

Hi-Fi Power Amplifier

Construction and Development Report

Discrete Class-AB Stereo Power Amplifier
with STM32-Based Speaker Protection

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Engineering Design and Construction Report

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Chapter 1

Introduction

Audio power amplifiers are a fundamental part of sound reproduction systems, providing the electrical power required to drive loudspeakers while preserving the characteristics of the original audio signal. The design of a high-quality power amplifier requires careful consideration of gain, output power, distortion, stability, thermal behaviour, power-supply quality, grounding, protection, and printed circuit board layout. Even when the amplifier topology itself is relatively simple, practical implementation has a significant influence on the final performance of the system.

The purpose of this project is the design and construction of a complete discrete-component Hi-Fi audio power amplifier system. The main amplifier stage is based on the principles of Rod Elliott's Project 3A power amplifier, while the overall implementation has been adapted and expanded to form a complete standalone amplifier. The design includes the main audio power amplifier, its power supply, signal input circuitry, loudspeaker connections, thermal management, and an independent protection and control system.

The output stage uses complementary bipolar power transistors, specifically the **2SC5200** NPN and **2SA1943** PNP devices. These transistors are commonly used in linear audio power amplifiers because of their high collector voltage capability, high current capability, and suitability for complementary Class-AB output stages. Adequate heatsinking is therefore an important part of the mechanical and thermal design of the amplifier.

The main amplifier is supplied from a symmetrical power supply derived from a **24–0–24 V AC toroidal transformer**. The nominal peak voltage of each 24 V secondary winding is

$$V_{\text{peak}} = 24\sqrt{2} \approx 33.94 \text{ V}.$$

After full-wave rectification and filtering, and allowing for the voltage drop across the rectifier diodes, the amplifier therefore operates from supply rails of approximately ± 33 to ± 34 V DC. The exact rail voltage depends on mains voltage, transformer regulation, load current, rectifier losses, and reservoir-capacitor ripple. Under load, the supply voltage will decrease slightly from its no-load value.

The selected 24–0–24 V AC transformer therefore provides a slightly lower supply voltage than the 25–0–25 V transformer recommended in the original Rod Elliott Project 3A design. This provides additional voltage and thermal margin when operating the amplifier

with 4 Ω loudspeaker loads while remaining close to the original design operating conditions. A symmetrical supply also allows the amplifier output to operate around ground potential without requiring a large output coupling capacitor.

In addition to the audio circuitry, a dedicated loudspeaker protection controller is incorporated into the project. The protection system is based on an **STM32G031K8** microcontroller and is electrically separated from the main audio signal path. Its purpose is to supervise important operating conditions such as DC voltage at the amplifier outputs, heatsink temperature, startup delay, and possible sensor faults. Loudspeaker relays are controlled by the microcontroller so that the speakers can be disconnected when an abnormal operating condition is detected. Status LEDs provide a simple visual indication of the operating state without requiring a display or complex user interface.

A major objective of the project is to achieve a practical implementation with low noise and good reliability. Particular attention is therefore given to the PCB layout. High-current power paths, low-level audio signal paths, grounding, supply decoupling, component positioning, transistor connections, and thermal considerations are treated separately where appropriate. Ground-current management is especially important because poorly designed return paths can introduce hum, noise, or unwanted coupling between the power stage and the sensitive input circuitry.

The project has been developed using **KiCad**, with separate schematic sections for the amplifier, power supply, and associated control circuitry. Component footprints, PCB layouts, manufacturing files, and bills of materials are developed as part of the complete design process. Components are selected according to their electrical ratings, physical dimensions, availability, and suitability for through-hole or PCB-mounted construction.

This report documents the complete development of the amplifier, beginning with the theoretical operation of the circuit and component selection and continuing through schematic design, PCB layout, power-supply design, thermal management, protection circuitry, manufacturing preparation, assembly, and testing. The design decisions made throughout the project are discussed together with their electrical and practical justification.

The final aim is to produce a robust, serviceable, and high-quality Hi-Fi power amplifier while gaining practical experience in analog electronics, power electronics, embedded control, PCB design, thermal design, protection systems, component selection, and electronic system integration.

Chapter 2

Project Objectives and Requirements

2.1 Project Objectives

The main objective of this project is the design and construction of a reliable, high-quality audio power amplifier suitable for driving conventional loudspeakers. The project is based on a discrete-component power amplifier topology and aims to combine good audio performance, electrical robustness, thermal stability, and practical construction techniques within a complete amplifier system.

A major objective of the project is to develop the amplifier as a complete engineering system rather than as an isolated power amplifier circuit. Therefore, the project includes the power amplifier stage, the power-supply circuitry, the input and preamplifier sections, the printed circuit board design, thermal management, grounding strategy, and a dedicated loudspeaker protection and monitoring system.

Particular attention is given to maintaining signal integrity throughout the system. Since audio circuits are sensitive to electromagnetic interference, ground loops, power-supply ripple, and coupling between high-current and low-level signal paths, the physical arrangement of the components and the PCB routing are considered an important part of the electrical design.

Another objective is to implement the system using components that are readily available and suitable for practical construction. Component selection therefore considers electrical ratings, tolerances, thermal characteristics, package types, PCB footprints, availability, and long-term reliability.

The project also aims to develop practical experience with the complete electronic design workflow. This includes schematic capture, component selection, footprint assignment, PCB layout, design-rule verification, generation of manufacturing files, bill-of-materials preparation, assembly planning, testing, and system verification.

2.2 Functional Requirements

The amplifier shall accept a low-level analogue audio signal and provide sufficient voltage and current amplification to drive a passive loudspeaker load. The amplifier must preserve the shape of the original audio waveform while introducing the minimum practical amount of unwanted noise and distortion.

The power amplifier shall operate from a symmetrical dual-rail DC power supply. The supply must provide sufficient current capability for the output stage while maintaining adequate voltage stability under changing audio loads. The power-supply section shall therefore include an appropriately rated mains transformer, bridge rectification, reservoir capacitors, fuses, and the necessary power distribution connections.

The amplifier output stage shall use complementary high-power bipolar transistors capable of safely handling the expected load current and power dissipation. In the final design, the output devices are the 2SC5200-O NPN and 2SA1943-O PNP power transistors. These devices shall be mounted on appropriate heatsinks to maintain their junction temperatures within safe operating limits.

The amplifier shall be capable of operating with typical loudspeaker loads while remaining electrically and thermally stable. Adequate provision must therefore be made for high-current PCB traces, suitable conductor widths, appropriate copper thickness, short output-current paths, and reliable high-current connectors.

2.3 Audio Performance Requirements

Low noise is an important design requirement because any unwanted signal generated inside the amplifier can become audible at the loudspeaker output. Low-level input signals must therefore be routed separately from high-current power and loudspeaker paths wherever practical.

The grounding system must be designed to minimise circulating ground currents and ground-loop-induced hum. Signal ground and high-current power ground paths shall be carefully managed and connected at appropriate common reference points. The PCB ground planes and copper zones shall be arranged so that power-stage return currents do not unnecessarily flow through sensitive input-stage ground paths.

The power-supply transformer, rectifier wiring, switching components, relay coils, digital electronics, and other potential sources of electromagnetic interference shall be positioned and routed so that their influence on the analogue audio circuitry is minimised.

The amplifier shall also maintain stable operation throughout the required audio frequency range and under normal loudspeaker loading conditions. PCB layout, component placement, local decoupling, grounding, and feedback-path routing must therefore be designed to reduce the possibility of unwanted oscillation or instability.

2.4 Protection and Monitoring Requirements

The amplifier shall include a dedicated loudspeaker protection system. Its primary purpose is to prevent potentially damaging operating conditions from being applied directly to the connected loudspeakers.

A microcontroller-based protection controller using an STM32 development board is incorporated into the project. The controller shall monitor important operating parameters and control the loudspeaker connection through a relay.

The protection system shall provide a power-on delay before connecting the loudspeakers. This prevents turn-on transients generated during amplifier and power-supply stabilisation from reaching the loudspeakers.

The system shall also monitor the DC voltage present at the amplifier outputs. During normal operation, the amplifier output consists primarily of an AC audio waveform centred close to 0 V DC. If a significant persistent DC component is detected, the protection controller shall disconnect the loudspeaker relay to reduce the risk of loudspeaker damage.

Temperature monitoring shall also be implemented using sensors associated with the amplifier's thermal system. If an excessive temperature is detected, the controller shall disconnect the loudspeakers and provide an appropriate visual indication.

Status LEDs shall provide information about the operating condition of the protection system. These indications include the initial loudspeaker connection delay, DC fault detection, temperature-related faults, and other relevant protection states.

2.5 Thermal Requirements

The output transistors dissipate a significant amount of power in the form of heat during amplifier operation. Effective thermal management is therefore necessary to maintain reliability and prevent semiconductor failure.

The power transistors shall be mechanically mounted to suitable heatsinks with appropriate thermal interfaces and electrical insulation where required. Their physical orientation on the PCB shall also allow practical attachment to the heatsinks without creating excessive mechanical stress on the transistor leads.

The enclosure shall permit sufficient airflow around the heatsinks and other heat-generating components. Additional forced-air cooling using fans may be employed where necessary to improve heat dissipation under sustained high-power operating conditions.

Components located near heat-generating devices shall be selected and positioned so that their rated operating temperatures are not exceeded. Temperature-sensitive components should not be placed unnecessarily close to power resistors, rectifiers, output transistors, or other significant heat sources.

2.6 PCB Design Requirements

The printed circuit boards shall be designed in KiCad using clearly defined schematics, verified component footprints, and appropriate electrical design rules. The PCB layout must reflect both the electrical and mechanical requirements of the amplifier.

High-current paths, particularly those associated with the power supply, output transistors, and loudspeaker connections, shall use sufficiently wide copper tracks or copper zones. Sensitive analogue signal traces shall be kept short and routed away from switching, rectification, relay, digital, and high-current paths.

Decoupling capacitors shall be positioned close to the circuits they are intended to stabilise. Power and ground routing shall be arranged to reduce impedance and prevent unwanted coupling between different functional sections.

The PCB shall include suitable mounting holes for secure mechanical installation inside the amplifier enclosure. Component spacing shall also allow practical soldering, inspection, maintenance, and replacement of components if required.

Silkscreen markings shall clearly identify important components, connectors, polarity-sensitive devices, power rails, input and output connections, and other information useful during assembly and troubleshooting.

2.7 Component Selection Requirements

Components shall be selected with appropriate safety margins relative to the maximum expected operating voltage, current, power dissipation, and temperature.

Resistors used in low-power signal-processing sections should generally have a tolerance of 1% to improve circuit consistency. Components subjected to greater power dissipation shall use suitable high-power resistor technologies and packages.

Electrolytic capacitors shall have voltage ratings comfortably above the maximum voltage that can appear across them. Their capacitance, ripple-current capability, temperature rating, and physical dimensions shall also be considered during selection.

Semiconductors shall be selected according to voltage rating, current rating, power dissipation, gain characteristics, package type, thermal performance, and availability. Their PCB footprints must correspond to the actual mechanical package of the selected component.

Connectors, switches, fuses, relay contacts, bridge rectifiers, and other power-related components shall be rated for the expected current and voltage of their respective circuits.

2.8 Safety Requirements

The amplifier contains mains-voltage circuitry and relatively high DC supply voltages. Electrical safety is therefore an essential design requirement.

The mains input section shall include appropriate fuse protection and a suitably rated power switch. Mains-voltage conductors must be physically separated from low-voltage

audio and control circuitry, and all exposed conductive parts that require protective earthing shall be connected appropriately.

The PCB and wiring layout shall maintain suitable clearance between mains-voltage and low-voltage sections. Connections carrying high current shall also be mechanically secure to minimise the risk of overheating or intermittent contact.

Reservoir capacitors can retain significant electrical energy after the amplifier has been disconnected from the mains supply. Appropriate discharge provisions and safe testing procedures must therefore be considered during assembly, measurement, and maintenance.

2.9 Mechanical and Construction Requirements

The complete system shall be suitable for installation inside a rigid amplifier enclosure. The enclosure must provide adequate space for the PCBs, transformer, heatsinks, cooling system, connectors, switches, wiring, and protection electronics.

The physical arrangement shall separate sensitive low-level audio circuitry from the mains transformer and high-current power-supply components as much as practical. Signal wiring should be kept short and may use shielded cable where appropriate.

The mechanical design shall allow convenient access to the input connectors, loudspeaker terminals, mains switch, fuses, status indicators, and any required service points.

2.10 Testing and Verification Requirements

The amplifier shall not be connected directly to a loudspeaker during the initial power-up procedure. Preliminary testing shall first verify the power-supply voltages, current consumption, amplifier DC offset, protection operation, and the absence of abnormal heating.

After basic electrical verification, the amplifier shall be tested using a controlled signal source and a suitable dummy load. Measurements shall be performed to verify correct waveform amplification, gain, output capability, frequency response, and general stability.

The protection system shall be tested independently by simulating the operating conditions that should trigger each protection state. These tests shall include the power-on delay, abnormal DC output detection, temperature fault detection, relay disconnection, and LED status indications.

Oscilloscope measurements shall be used where appropriate to observe the input waveform, output waveform, clipping behaviour, unwanted oscillations, switching transients, and power-supply behaviour.

2.11 Documentation Requirements

The project shall be documented sufficiently to allow the amplifier to be understood, reproduced, tested, and maintained. The documentation shall include schematics, PCB

layouts, component information, design decisions, calculations, protection-system operation, manufacturing files, and the bill of materials.

The final report shall describe both the theoretical operation of the amplifier and the practical engineering decisions made during its development. Particular emphasis shall be placed on explaining component selection, PCB layout, grounding, thermal design, protection circuitry, construction procedures, testing methodology, and any modifications made during the development process.

The resulting system should therefore represent not only a functional audio amplifier, but also a complete engineering project incorporating analogue electronics, power electronics, PCB design, embedded control, protection, thermal management, mechanical integration, testing, and technical documentation.

Chapter 3

Overall System Architecture

3.1 System Overview

The complete Hi-Fi power amplifier is composed of several interconnected subsystems, including two power amplifier channels, the main power supply, the loudspeaker protection controller, the audio input connections, and the grounding network. Figure 3.1 illustrates the overall electrical interconnection between these sections.

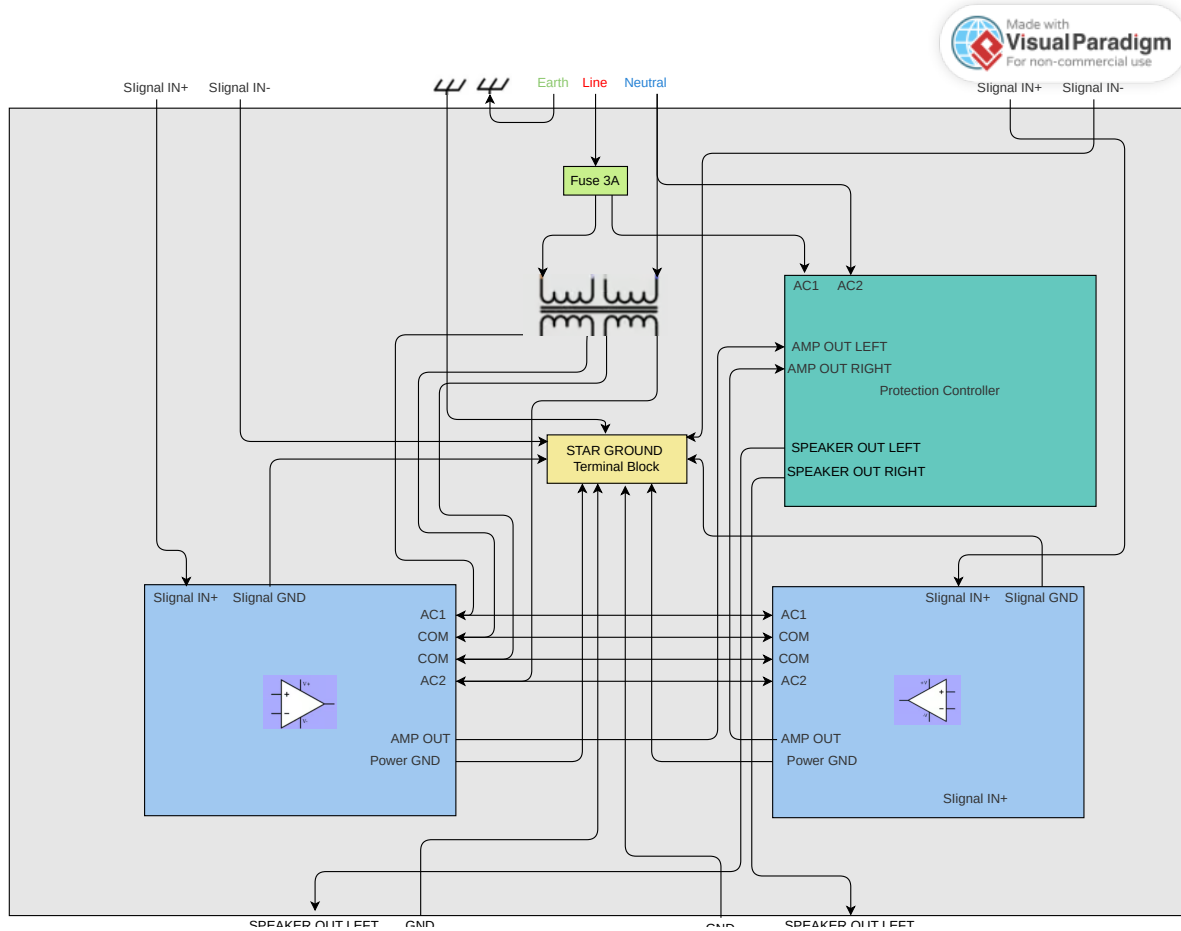


Figure 3.1: Overall system interconnection diagram of the stereo Hi-Fi power amplifier, showing the mains input, transformer, star-ground connection, left and right amplifier channels, signal inputs, and protection controller.

3.2 Main System Blocks

The system consists of two independent audio power amplifier channels supplied from a common symmetrical power supply. Each amplifier receives its own audio input signal and provides an amplified output signal for the corresponding loudspeaker channel.

A dedicated STM32-based protection controller supervises the amplifier outputs and thermal conditions. The controller operates independently from the analogue audio signal path and controls the loudspeaker connection when abnormal conditions are detected.

3.3 Power Distribution

Mains power enters the system through the protected mains input and fuse before being supplied to the main transformer. The transformer provides the required secondary AC voltages for the amplifier power supply and associated circuitry.

The two power amplifier channels share the main supply rails while their high-current

return paths are routed toward a common star-ground point.

3.4 Audio Signal Path

The left and right audio input signals are routed independently to their respective amplifier channels. The low-level signal paths are kept separate from the high-current power wiring in order to minimise unwanted coupling, noise, and hum.

3.5 Grounding Architecture

A central star-ground terminal is used as the principal reference point for the amplifier system. Signal-ground and power-ground connections are arranged so that large output-stage and power-supply currents do not unnecessarily flow through sensitive input-stage ground paths.

This grounding arrangement is intended to reduce circulating ground currents and minimise mains-frequency hum within the audio signal path.

3.6 Protection Controller Integration

The protection controller is connected to the relevant amplifier monitoring points while remaining outside the main analogue audio path. It monitors operating conditions such as amplifier output DC voltage and temperature and controls the loudspeaker relays accordingly.

In the event of an abnormal condition, the controller can disconnect the loudspeakers from the amplifier outputs to reduce the possibility of loudspeaker damage.

Chapter 4

Power Amplifier Design

4.1 Design Overview

The main audio amplification stage of this project is a discrete Class-AB power amplifier based on Rod Elliott's Project 3A topology [1]. The circuit was selected because it combines a relatively simple discrete architecture with good audio performance, high output-current capability, global negative feedback, adjustable quiescent current, and the ability to drive conventional $4\ \Omega$ and $8\ \Omega$ loudspeaker loads.

The complete amplifier system contains two identical power-amplifier channels for stereo operation. Each channel receives an independent audio input signal and drives one loudspeaker output. The two amplifier channels share the main symmetrical power supply but retain independent audio signal paths.

An important modification relative to the standard Project 3A power-supply recommendation is the transformer used in this project. The implemented amplifier uses a **24–0–24 V AC toroidal transformer**, rather than the 25–0–25 V AC transformer normally associated with the approximately $\pm 35\text{ V}$ Project 3A supply. Consequently, the actual amplifier supply rails are expected to be slightly below the nominal $\pm 35\text{ V}$ reference condition.

4.2 Power Amplifier Design Requirements

The main electrical characteristics of the implemented amplifier are summarised in Table 4.1.

Table 4.1: Main design parameters of the power amplifier.

Parameter	Design Value
Amplifier topology	Discrete Class-AB
Number of channels	2 (stereo)
Reference design	Rod Elliott Project 3A
Transformer secondary	24–0–24 V AC
Expected DC supply rails	Approximately ± 33 to ± 34 V
Loudspeaker impedance	4 Ω or 8 Ω
Output transistor, NPN	2SC5200-O
Output transistor, PNP	2SA1943-O
Output emitter resistors	0.33 Ω
Bias adjustment	2 k Ω trimmer
Nominal voltage gain	Approximately 23 V/V
Nominal voltage gain	Approximately 27.2 dB

The original Project 3A documentation specifies approximately 60 W into 8 Ω and 100 W into 4 Ω when operated from nominal ± 35 V supplies [1]. These values are used in this report as **reference performance figures** and are not presented as measured output-power values for the constructed amplifier.

Because the implemented supply voltage is slightly lower than the original ± 35 V reference supply, the maximum unclipped output power is expected to be somewhat lower. The exact available power depends on transformer regulation, mains voltage, rectifier losses, reservoir-capacitor ripple, loudspeaker impedance, transistor voltage drops, and thermal conditions. The actual maximum output power will therefore be established experimentally during final testing.

4.3 Power Amplifier Schematic

The final schematic of one amplifier channel is shown in Figure 4.1. The circuit consists entirely of discrete semiconductor devices and can be divided into several functional sections: the input network, differential input stage, current sink, voltage-amplification stage, bias-control network, driver stage, complementary-feedback power output stage, global negative-feedback network, and output-stability network.

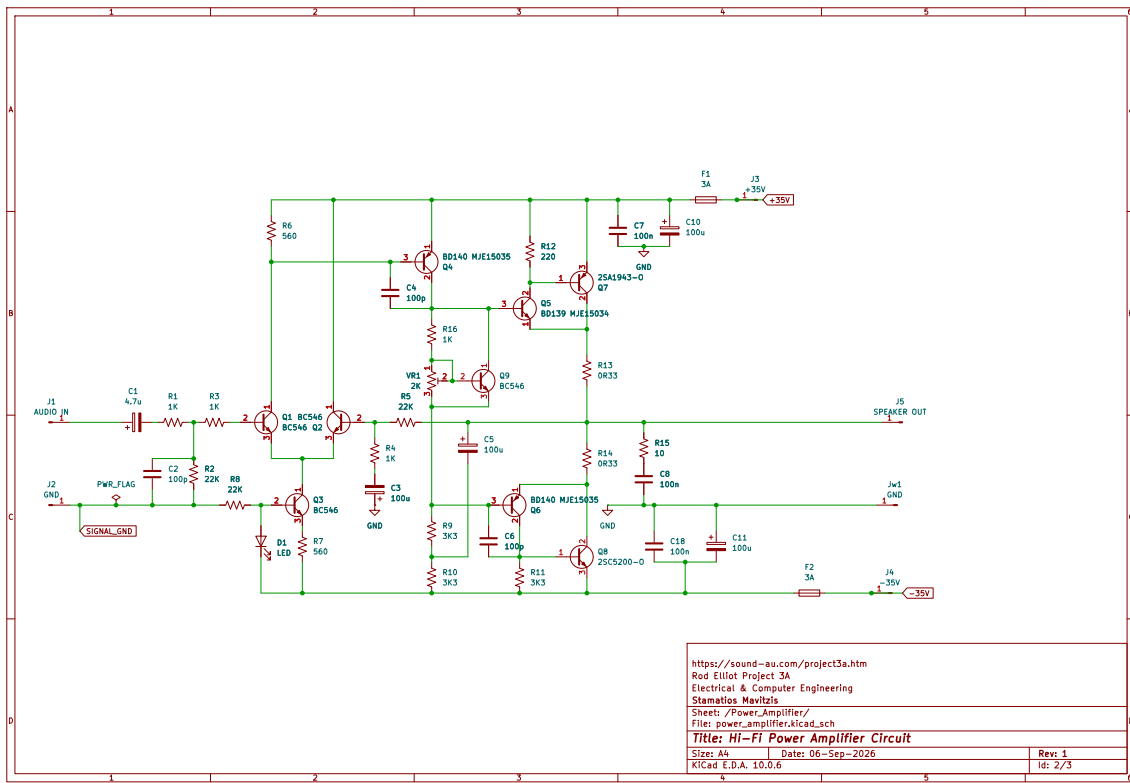


Figure 4.1: Final schematic of one power-amplifier channel.

A second identical circuit is used for the other stereo channel. The use of identical left and right amplifier stages simplifies construction and helps maintain similar electrical characteristics between channels.

4.4 Amplifier Architecture

The amplifier uses global negative feedback around several discrete transistor stages. At a system level, the signal path can be represented as

Audio Input → Differential Input Stage → Voltage Amplification → Driver Stage → Output Stage → I

A fraction of the loudspeaker output voltage is returned to the differential input stage through the feedback network. The amplifier therefore responds to the difference between the applied input signal and the feedback signal.

The differential error voltage may be represented as

$$V_{\text{error}} = V_{\text{input}} - V_{\text{feedback}}.$$

The high open-loop gain of the amplifier causes the circuit to reduce this error voltage, forcing the output waveform to follow the input waveform according to the closed-loop gain established by the feedback network.

Negative feedback provides several advantages, including improved linearity, reduced distortion, more predictable voltage gain, reduced output impedance, and improved control of the loudspeaker.

4.5 Input Network

The audio input is connected through connector J1. Capacitor C1 has a value of

$$C_1 = 4.7 \mu\text{F}$$

and provides AC coupling between the external audio source and the amplifier input.

The purpose of the input coupling capacitor is to block any DC component that may be present at the output of the preceding audio source. Preventing an external DC voltage from reaching the differential input stage helps preserve the required internal DC operating conditions.

The input network also contains R1 and R3, each with a value of 1 k Ω , together with R2=22 k Ω and a small 100 pF capacitor. These components provide input biasing, impedance definition, isolation, and high-frequency filtering.

The published Project 3A design has a nominal input impedance of approximately

$$Z_{\text{in}} \approx 24 \text{ k}\Omega$$

[1], making it suitable for connection to conventional preamplifiers and line-level audio sources.

4.6 Differential Input Stage

Transistors Q1 and Q2 are BC546 small-signal NPN transistors and form the differential input stage of the amplifier.

The differential stage compares the incoming audio signal with the feedback signal obtained from the amplifier output. Any difference between these two voltages is amplified by the following stages.

The use of a differential input stage is particularly appropriate in a feedback audio amplifier because it provides a direct method of comparing the desired output with the actual output. It also contributes to stable DC operation and low output offset.

4.7 Differential-Stage Current Sink

Transistor Q3, also a BC546 device, forms the current sink associated with the differential stage. A green LED, D1, is used as a voltage-reference component within this part of the circuit.

In this application the LED is not included primarily as a visual indicator. Its forward voltage is used to establish the operating conditions of the current sink. The Project 3A design requires a conventional green LED with a forward voltage of approximately 1.9–2.0 V at the operating current [1].

The current sink improves the operating consistency of the differential pair by reducing its dependence on changes in supply voltage. This contributes to stable amplifier operation and improved linearity.

4.8 Voltage-Amplification Stage

The signal produced by the differential input stage is applied to the voltage-amplification stage. Q4 provides most of the voltage gain required before the signal reaches the output drivers.

The voltage-amplification stage operates in Class-A and uses a bootstrap network formed principally by R9, R10, and C5. In the implemented design,

$$R_9 = R_{10} = 3.3 \text{ k}\Omega$$

and

$$C_5 = 100 \text{ }\mu\text{F}.$$

The bootstrap arrangement increases the effective AC load impedance seen by the voltage-amplification transistor and assists in obtaining sufficient voltage swing from the stage.

4.9 Driver and Output Stage

The output of the voltage-amplification stage cannot directly supply the current required by a low-impedance loudspeaker. Driver transistors are therefore placed between the voltage-amplification stage and the main output devices.

The implemented design provides BD139/MJE15034-type NPN and BD140/MJE15035-type PNP driver positions. These devices provide the current gain required to control the main power transistors while reducing the current demand placed on the preceding voltage-amplification circuitry.

The final output devices selected for the amplifier are

$$2\text{SC}5200\text{-O}$$

for the NPN output transistor and

$$2\text{SA}1943\text{-O}$$

for the complementary PNP output transistor.

These devices are high-power bipolar junction transistors intended for linear power-amplifier applications. Their high voltage, current, and power-handling capabilities make them suitable for the output-current requirements of this design.

4.10 Complementary Feedback Pair Topology

A characteristic feature of Project 3A is that the output stage uses *complementary feedback pairs*, also known as Sziklai pairs, rather than a conventional complementary Darlington output arrangement [1].

Each output transistor works together with its corresponding driver transistor as a locally feedback-controlled compound transistor stage. This arrangement provides high current gain while also introducing local negative feedback into the output stage.

The local feedback improves output-stage linearity and contributes to the overall low-distortion operation of the amplifier.

4.11 Class-AB Biasing

The output stage operates in Class-AB. A small standing current is maintained through the output stage when no audio signal is present. This quiescent current reduces the crossover distortion that would otherwise occur when the output waveform passes through zero.

The quiescent current is adjusted using VR1, which has a resistance of

$$VR_1 = 2 \text{ k}\Omega.$$

According to the Project 3A documentation, a normal quiescent-current range is approximately

$$30 \text{ mA} \leq I_Q \leq 75 \text{ mA},$$

with approximately 50 mA representing a practical nominal adjustment [1].

The final bias setting and the detailed thermal behaviour of the circuit are discussed separately in the chapter dedicated to biasing and thermal stability.

4.12 Output Emitter Resistors

The output stage includes R13 and R14, each having a resistance of

$$R_{13} = R_{14} = 0.33 \text{ }\Omega.$$

These low-value resistors provide local current feedback and improve thermal and bias stability. Elliott specifically recommends the 0.33 Ω value because it provides improved

bias stability compared with lower values [1].

The combined resistance used when checking the quiescent current is

$$R_{\text{total}} = R_{13} + R_{14} = 0.66 \, \Omega.$$

For a nominal quiescent current of 50 mA,

$$V_Q = I_Q R_{\text{total}}$$

and therefore

$$V_Q = 0.05 \times 0.66 = 0.033 \, \text{V}.$$

Thus,

$$\boxed{V_Q \approx 33 \, \text{mV}}$$

corresponds to approximately 50 mA of quiescent current when measured across the two $0.33 \, \Omega$ resistors.

At 75 mA,

$$V_Q = 0.075 \times 0.66 = 49.5 \, \text{mV}.$$

These values are consistent with the Project 3A bias-adjustment procedure [1].

4.13 Global Negative Feedback and Voltage Gain

The global feedback network determines the closed-loop voltage gain of the amplifier. The principal gain-setting resistors in the implemented circuit are

$$R_5 = 22 \, \text{k}\Omega$$

and

$$R_4 = 1 \, \text{k}\Omega.$$

The approximate closed-loop AC voltage gain is therefore

$$A_v = 1 + \frac{R_5}{R_4}.$$

Substitution of the implemented component values gives

$$A_v = 1 + \frac{22\,000}{1\,000} = 23.$$

The corresponding voltage gain in decibels is

$$A_{v,\text{dB}} = 20 \log_{10}(A_v),$$

therefore

$$A_{v,\text{dB}} = 20 \log_{10}(23) \approx 27.2 \text{ dB}.$$

Hence,

$$A_v \approx 23$$

and

$$A_{v,\text{dB}} \approx 27.2 \text{ dB}.$$

This calculated value agrees closely with the approximately 27 dB gain published for Project 3A [1].

4.14 Frequency Compensation

Because the amplifier contains several transistor stages enclosed by a global feedback loop, frequency compensation is necessary to maintain stability at high frequencies.

The implemented design includes small compensation capacitors with values of

$$C_4 = C_6 = 100 \text{ pF}.$$

These capacitors limit the high-frequency open-loop gain and reduce the risk of unwanted oscillation caused by accumulated phase shift around the feedback loop.

The Project 3A documentation specifically identifies C4 as an important stability component and states that its value should not be reduced [1].

4.15 Output Stability Network

The amplifier output contains a series RC network consisting of R15 and C8, with values

$$R_{15} = 10 \text{ } \Omega$$

and

$$C_8 = 100 \text{ nF}.$$

This arrangement is commonly referred to as a Zobel or Boucherot network. Its purpose is to provide the amplifier with a more predictable load impedance at high frequencies.

A loudspeaker is not a purely resistive load. Its impedance contains inductive and capacitive components that vary with frequency, while the speaker cable introduces additional parasitic reactance. The output RC network therefore contributes to stable operation when the amplifier is connected to a real loudspeaker system.

4.16 Local Supply Decoupling

Local decoupling is provided on both power-supply rails close to the amplifier circuitry.

On the positive rail, the circuit contains

$$C_7 = 100 \text{ nF}$$

and

$$C_{10} = 100 \text{ }\mu\text{F}.$$

Corresponding capacitors on the negative rail are

$$C_{18} = 100 \text{ nF}$$

and

$$C_{11} = 100 \text{ }\mu\text{F}.$$

The electrolytic capacitors provide local energy storage at lower frequencies, while the smaller 100 nF capacitors provide a low-impedance path for higher-frequency currents.

This local decoupling reduces the impedance of the supply rails at the amplifier and helps prevent unwanted interaction through the common power supply.

4.17 Implemented Supply Voltage

The implemented power amplifier does not use the 25–0–25 V AC transformer specified for the standard ± 35 V Project 3A supply. Instead, the transformer selected for this project has a secondary voltage of

$$24\text{--}0\text{--}24 \text{ V}_{\text{RMS}}.$$

The theoretical peak voltage of each 24 V_{RMS} half-secondary is

$$V_{\text{peak}} = V_{\text{RMS}}\sqrt{2}.$$

Substituting the transformer voltage gives

$$V_{\text{peak}} = 24\sqrt{2} = 33.94 \text{ V}.$$

Thus, before semiconductor losses, the theoretical split supply is approximately

$$\boxed{\pm 33.94 \text{ V}}.$$

In practice, the rectifier diodes introduce a forward-voltage drop, while the transformer secondary voltage varies with loading and mains voltage. Reservoir-capacitor ripple also causes the instantaneous DC voltage to change during operation.

The practical amplifier supply is therefore described in this report as approximately

$$\boxed{\pm 33\text{--}34 \text{ VDC}},$$

with the exact values to be verified experimentally at both no-load and loaded operating conditions.

The slightly lower supply voltage compared with the nominal $\pm 35 \text{ V}$ Project 3A configuration provides additional voltage and thermal margin when driving 4Ω loudspeakers. Elliott specifies that 4Ω operation should not be used with supply rails greater than approximately $\pm 35 \text{ V}$ [1].

4.18 Reference Output Power

The published Project 3A performance data specifies approximately

$$P_{\text{out}} \approx 60 \text{ W}$$

into an 8Ω load and

$$P_{\text{out}} \approx 100 \text{ W}$$

into a 4Ω load when operated from nominal $\pm 35 \text{ V}$ supply rails [1].

These values provide a useful reference for evaluating the present design. However, they should not be interpreted as guaranteed output-power values for the implemented amplifier because the present design uses slightly lower supply rails.

4.18.1 Reference 8 Ohm Operating Condition

For the published 60 W output into an 8Ω load, the RMS loudspeaker voltage is

$$V_{\text{RMS}} = \sqrt{P_{\text{out}} R_L}.$$

Therefore,

$$V_{\text{RMS}} = \sqrt{60 \times 8} = 21.91 \text{ V}.$$

The corresponding peak voltage is

$$V_{\text{peak}} = \sqrt{2}V_{\text{RMS}} \approx 30.98 \text{ V}.$$

The RMS load current is

$$I_{\text{RMS}} = \frac{V_{\text{RMS}}}{R_L} = \frac{21.91}{8} \approx 2.74 \text{ A},$$

while the peak current is

$$I_{\text{peak}} = \sqrt{2}I_{\text{RMS}} \approx 3.87 \text{ A}.$$

Thus, the Project 3A 60 W reference condition corresponds to

$$\boxed{V_{\text{RMS}} \approx 21.9 \text{ V}, \quad V_{\text{peak}} \approx 31.0 \text{ V}, \quad I_{\text{RMS}} \approx 2.74 \text{ A}, \quad I_{\text{peak}} \approx 3.87 \text{ A}}.$$

4.18.2 Reference 4 Ohm Operating Condition

For the published 100 W output into a 4 Ω load,

$$V_{\text{RMS}} = \sqrt{100 \times 4} = 20 \text{ V}.$$

The peak voltage is

$$V_{\text{peak}} = 20\sqrt{2} \approx 28.28 \text{ V}.$$

The RMS load current is

$$I_{\text{RMS}} = \frac{20}{4} = 5 \text{ A},$$

and therefore

$$I_{\text{peak}} = 5\sqrt{2} \approx 7.07 \text{ A}.$$

The corresponding reference values are therefore

$$\boxed{V_{\text{RMS}} = 20 \text{ V}, \quad V_{\text{peak}} \approx 28.3 \text{ V}, \quad I_{\text{RMS}} = 5 \text{ A}, \quad I_{\text{peak}} \approx 7.07 \text{ A}}.$$

These calculations demonstrate why operation into 4 Ω is significantly more demanding than operation into 8 Ω . The required voltage is similar, but the output current is considerably greater. This directly affects transistor dissipation, PCB conductor requirements, connector ratings, power-supply loading, and heatsink requirements.

4.19 Expected Output with the 24–0–24 V Transformer

No exact maximum output-power value is assigned to the implemented amplifier before testing. The actual clipping voltage depends on several parameters that cannot be obtained accurately from the nominal transformer voltage alone.

These parameters include transformer regulation, mains voltage, bridge rectifier voltage drop, reservoir-capacitor ripple, the instantaneous loudspeaker current, semiconductor voltage drops, and wiring resistance.

Since the implemented rails of approximately ± 33 – ± 34 V are slightly lower than the nominal ± 35 V rails associated with the published Project 3A 60 W and 100 W results, the maximum unclipped output power is expected to be slightly lower than those reference values.

The actual output capability will therefore be determined during testing by applying a sinusoidal input signal to the amplifier, connecting an appropriate dummy load, and increasing the input amplitude until the onset of output clipping is observed.

4.20 Approximate Input Sensitivity

The calculated voltage gain of approximately 23 can be combined with the Project 3A reference output voltages to estimate the required input level.

For the reference 60 W, 8 Ω condition,

$$V_{\text{in,RMS}} = \frac{V_{\text{out,RMS}}}{A_v} = \frac{21.91}{23} \approx 0.95 V_{\text{RMS}}.$$

For the reference 100 W, 4 Ω condition,

$$V_{\text{in,RMS}} = \frac{20}{23} \approx 0.87 V_{\text{RMS}}.$$

These values are theoretical estimates derived from the feedback resistor ratio. The actual input voltage required to reach clipping will be established experimentally.

4.21 Reference Electrical Performance

Table 4.2 summarises published electrical performance figures for the original Project 3A design. These values are included for comparison and are not claimed as measurements of the amplifier constructed in this project.

Table 4.2: Published Project 3A reference performance [1].

Parameter	Published Reference Value
Voltage gain	27 dB
Input impedance	24 k Ω
Frequency response	10 Hz–30 kHz (–1 dB), typical
THD	Approximately 0.04%, typical
Output power, ± 35 V, 8 Ω	Approximately 60 W
Output power, ± 35 V, 4 Ω	Approximately 100 W
Hum and noise	Approximately –73 dBV, unweighted
DC output offset	< 100 mV

The frequency response and distortion figures depend on the specific component values, PCB layout, supply quality, semiconductor characteristics, and test conditions. Since a custom PCB and different output transistor pair are used in this project, the published results must be treated as design-reference values until the completed amplifier is measured.

4.22 High-Current Design Considerations

The output-current calculations show that several amperes can flow through the output stage and loudspeaker path. The 4 Ω reference condition is particularly demanding, with a calculated current of

$$I_{\text{RMS}} = 5 \text{ A}$$

and

$$I_{\text{peak}} \approx 7.07 \text{ A}.$$

Consequently, the high-current paths of the amplifier require particular attention. These include the positive and negative supply connections, output transistor current paths, 0.33 Ω emitter resistors, loudspeaker output, reservoir-capacitor connections, and power-ground return.

High-current PCB conductors are therefore made wider than the small-signal tracks. Detailed PCB current handling, conductor width, copper distribution, and layout considerations are discussed separately in the PCB design chapters.

4.23 Grounding Considerations

The power amplifier contains both very small input signals and high-current output-stage currents. If these currents share inappropriate ground paths, voltage drops in the common conductor can be introduced into the input signal reference and become audible as hum or distortion.

For this reason, the design distinguishes between sensitive signal-ground paths and high-current power-ground paths. These return paths are directed towards a controlled common ground reference within the overall system.

The detailed star-ground arrangement, signal-ground routing, power-ground routing, and copper-zone implementation are discussed in the dedicated grounding chapters of this report.

4.24 Thermal Considerations

The output transistors dissipate significant power as heat during normal operation. This is particularly important with $4\ \Omega$ loads, where the output current is substantially higher.

The 2SC5200-O and 2SA1943-O output devices are therefore mounted so that they can be thermally coupled to substantial external heatsinks. The physical position of the power transistors near the PCB edge facilitates direct mechanical mounting to the heatsink while keeping the high-current electrical connections short.

Bias current also contributes to idle power dissipation. The relationship between bias, transistor temperature, heatsink thermal resistance, forced-air cooling, and safe operating temperature is examined in detail in the *Biasing and Thermal Stability* and *Heatsink Selection and Thermal Management* chapters.

4.25 Protection Considerations

The basic Project 3A power-amplifier circuit does not contain active output short-circuit protection [1]. A direct short circuit at the loudspeaker output while the amplifier is delivering significant signal power can therefore place destructive current stress on the output transistors.

The complete amplifier developed in this project includes an independent STM32-based loudspeaker protection controller. Its primary functions include power-on delay, DC-output monitoring, temperature monitoring, fault indication, and loudspeaker relay control.

The protection controller does not form part of the analogue audio amplification loop and is discussed separately in the dedicated protection system chapters.

4.26 Design Summary

The implemented amplifier is a two-channel discrete Class-AB design based on Rod Elliott's Project 3A. Each channel contains a differential BC546 input stage, current sink, voltage-amplification stage, adjustable Class-AB bias network, driver transistors, and a complementary-feedback output stage.

The final power transistors are the 2SC5200-O NPN and 2SA1943-O PNP devices. The output stage uses $0.33\ \Omega$ emitter resistors to improve bias and thermal stability. The global feedback network produces a calculated closed-loop gain of approximately

$$A_v \approx 23$$

or

$$A_{v,\text{dB}} \approx 27.2 \text{ dB}.$$

Unlike the original Project 3A $\pm 35 \text{ V}$ reference configuration, the implemented amplifier uses a 24–0–24 V AC toroidal transformer. The theoretical secondary peak voltage is

$$24\sqrt{2} \approx 33.94 \text{ V},$$

resulting in practical amplifier supply rails in the approximate ± 33 – $\pm 34 \text{ V}$ region after rectification and filtering.

This voltage remains below Elliott’s recommended maximum of approximately $\pm 35 \text{ V}$ for operation into 4Ω loads and provides a small additional electrical and thermal safety margin.

The published Project 3A output figures of approximately 60 W into 8Ω and 100 W into 4Ω are retained only as reference values for the nominal $\pm 35 \text{ V}$ version of the circuit. The exact maximum power, DC offset, distortion, frequency response, supply-rail voltage, and thermal behaviour of the implemented amplifier will be established during the testing and measurement phase.

Chapter 5

Power Amplifier Operating Principle

5.1 Overview of Operation

The power amplifier used in this project is a discrete Class-AB design based on Rod Elliott's Project 3A topology [1]. The circuit converts a relatively low-level audio input voltage into a substantially larger voltage and current capable of driving a passive loudspeaker.

Although the amplifier contains several transistor stages, its operation can be understood as a continuous closed-loop process. The input stage compares the incoming audio signal with a fraction of the amplifier output voltage. The resulting difference, or error signal, is amplified through the voltage-amplification and driver stages before controlling the high-power output transistors.

The overall signal flow may be represented as

Audio Input → Differential Input Stage → Voltage Amplification Stage → Bias and Driver Stage → Po

A global feedback path returns part of the output signal to the differential input stage:

Output → Feedback Network → Differential Input Stage.

The amplifier therefore continuously corrects its output so that the amplified waveform follows the input waveform with the required voltage gain.

5.2 Quiescent Operating Condition

Before considering an applied audio signal, it is useful to examine the amplifier under quiescent conditions. In this state, no significant AC input signal is present and the loudspeaker output should remain close to 0 V DC.

The amplifier is supplied from positive and negative DC power rails derived from the 24–0–24 V AC toroidal transformer. Following rectification and filtering, the practical

rail voltages are expected to be approximately ± 33 to ± 34 V, although their exact values depend on transformer regulation, mains voltage, loading, rectifier losses, and reservoir-capacitor ripple.

The use of a symmetrical power supply allows the loudspeaker output to be centred around ground. Consequently, no large series output coupling capacitor is required.

Even without an audio signal, a small current flows through the Class-AB output stage. This current is deliberately established by the bias-control circuit so that the upper and lower output sections are both slightly conductive around the zero crossing.

The purpose of this quiescent current is to reduce crossover distortion while avoiding the high idle dissipation that would occur if the output stage were operated in pure Class-A.

5.3 Input Signal Coupling

The audio signal enters the amplifier through connector J1 and first passes through coupling capacitor C1.

$$C_1 = 4.7 \mu\text{F}.$$

C1 blocks any DC voltage originating from the external audio source while allowing the wanted AC audio signal to pass into the amplifier.

The signal then passes through the input resistor network before reaching the base of transistor Q1. Resistors R1 and R3 provide series isolation, while R2 provides the required DC reference for the input.

The small capacitor C2 provides high-frequency filtering. Signals far above the required audio bandwidth are therefore attenuated before entering the high-gain section of the amplifier.

5.4 Differential Input Stage

Q1 and Q2 are BC546 NPN transistors and form the differential input pair.

The base of Q1 receives the incoming audio signal, while the base of Q2 receives the global feedback signal derived from the amplifier output.

The differential pair therefore responds to the difference between these two voltages:

$$V_{\text{error}} = V_{\text{input}} - V_{\text{feedback}}.$$

If the amplifier output is exactly equal to the value required by the input signal and feedback ratio, the differential error voltage becomes very small.

If the output is too low, the differential stage generates an error signal that drives the following stages in a direction that increases the output.

If the output is too high, the polarity of the error changes and the amplifier reduces its output.

This continuous correction process forms the basis of closed-loop amplifier operation.

5.5 Current Sink for the Differential Pair

The emitters of Q1 and Q2 share a common current sink based on Q3, which is also a BC546 transistor.

Q3 establishes a relatively stable total current for the differential input pair. Instead of allowing the current through Q1 and Q2 to vary strongly with supply voltage, the current sink causes the two input transistors to share an approximately controlled current.

The green LED D1 forms part of the reference network for Q3. Its forward voltage provides a relatively stable reference from which the current through Q3 and R7 is established.

When the amplifier is balanced and no significant error signal exists, the differential-pair current is approximately shared between Q1 and Q2.

When an audio signal creates a difference between the two base voltages, current is steered from one differential transistor toward the other. This variation produces the signal that controls the following voltage-amplification stage.

5.6 Voltage-Amplification Stage

Q4 is a BD140 PNP transistor and forms the principal voltage-amplification stage of the amplifier.

The differential input stage itself operates with relatively small voltage changes. Q4 converts these changes into the much larger voltage swing required to drive the bias and output circuitry.

Q4 operates in a Class-A region during normal amplifier operation. This is important because the voltage-amplification stage must reproduce both positive and negative portions of the audio waveform continuously before the signal is divided between the two sides of the output stage.

The voltage-amplification stage provides a large proportion of the amplifier's open-loop voltage gain. Global negative feedback then reduces this high open-loop gain to the controlled closed-loop gain required by the complete amplifier.

5.7 Bootstrap Operation

The voltage-amplification stage is assisted by the bootstrap network, including C5 and the associated resistors R9 and R10.

The bootstrap capacitor is connected so that one of its terminals follows the amplifier output waveform. As the output voltage changes, the voltage at the bootstrap network also changes.

This causes the load presented to part of the voltage-amplification stage to behave as

though it has a higher effective AC impedance than its physical resistance alone would suggest.

The result is an increased available voltage swing and reduced loading of the voltage-amplification stage. This is useful in a power amplifier because a large voltage excursion is required before the output stage can approach the positive and negative supply rails.

5.8 Bias Generation

The voltage produced by the voltage-amplification stage is applied across the bias-control network before reaching Q5 and Q6.

The bias-control circuit includes Q9, a BC546 transistor, together with VR1 and its associated resistors.

VR1 is a

$$2\text{ k}\Omega$$

adjustable resistor that allows the voltage between the upper and lower driver sections to be changed.

This voltage determines the standing current in the output stage. If the bias voltage is too small, both sides of the output stage become insufficiently conductive close to the zero crossing, producing crossover distortion.

If the bias voltage is too large, excessive idle current flows through the output stage, increasing transistor temperature and power dissipation.

The correct adjustment therefore establishes a small overlap in conduction between the upper and lower output sections.

5.9 Class-AB Operating Principle

The output stage operates in Class-AB rather than pure Class-A or Class-B.

In an ideal Class-B output stage, the upper output device conducts during one half of the waveform and the lower output device conducts during the opposite half. However, bipolar junction transistors require a finite base-emitter voltage before substantial conduction begins.

If both halves of a Class-B output stage were completely switched off at zero signal, a small dead region would exist around the zero crossing. This would create crossover distortion.

Class-AB operation avoids this condition by applying sufficient bias so that both output sections remain slightly conductive around zero output voltage.

At small signal levels, both sides contribute around the crossover region. As the required output magnitude increases, one output section progressively provides most of the load current while the opposite section reduces its conduction.

5.10 Upper Driver Stage

Q5 is a BD139 NPN transistor and forms the upper driver device.

Q5 does not directly provide the complete loudspeaker output current. Instead, it controls the high-power PNP transistor Q7.

The use of an intermediate driver transistor provides additional current gain, allowing the preceding voltage-amplification stage to operate at comparatively low current while the output transistor supplies several amperes to the loudspeaker when required.

5.11 Lower Driver Stage

Q6 is a BD140 PNP transistor and performs the corresponding driver function for the lower half of the output stage.

Q6 controls the NPN power transistor Q8 and allows the negative side of the output stage to sink the required loudspeaker current during the appropriate portion of the audio waveform.

5.12 Complementary Feedback Output Stage

The power output stage uses the complementary transistors

$$Q7 = 2SA1943-O$$

and

$$Q8 = 2SC5200-O.$$

Q7 is a PNP power transistor and Q8 is an NPN power transistor.

A significant characteristic of the Project 3A circuit is that these output devices are not used as a conventional complementary emitter-follower pair. Instead, each power transistor operates together with its corresponding driver transistor as a complementary feedback pair, commonly called a Sziklai pair [1].

The upper complementary feedback pair consists principally of

$$Q5 \text{ (BD139)} + Q7 \text{ (2SA1943-O)},$$

while the lower pair consists principally of

$$Q6 \text{ (BD140)} + Q8 \text{ (2SC5200-O)}.$$

Each pair behaves as a high-current compound device while incorporating local feedback around the driver and output transistor.

This local feedback improves linearity and reduces the effect of individual transistor gain variations. It also helps the main global feedback loop control the loudspeaker voltage accurately.

5.13 Positive Output Half-Cycle

Consider a positive-going input signal. The voltage at Q1 changes relative to the feedback voltage applied to Q2.

The differential input stage detects this difference and alters the drive applied to Q4.

Q4 changes the voltage across the bias and driver network so that the upper driver section becomes more conductive.

Q5 then controls Q7, the 2SA1943-O PNP power transistor, causing the upper complementary feedback pair to supply increasing current from the positive supply rail toward the amplifier output.

The loudspeaker output voltage therefore moves in the positive direction.

The corresponding current path may be represented conceptually as

$$+V \rightarrow Q7 \rightarrow R13 \rightarrow \text{Amplifier Output} \rightarrow \text{Loudspeaker} \rightarrow \text{Ground}.$$

As the output rises, part of this increasing voltage is returned to Q2 through the feedback network.

The feedback voltage therefore increases until the difference between the input and feedback signals becomes only the small error required to maintain the commanded output.

5.14 Negative Output Half-Cycle

During a negative-going input signal, the differential stage detects an error of the opposite polarity.

The operating condition of Q4 changes correspondingly, reducing drive to the upper output pair and increasing drive to the lower output pair.

Q6 controls Q8, the 2SC5200-O NPN output transistor, causing the lower complementary feedback pair to conduct more strongly.

Current can then flow from the loudspeaker output toward the negative supply rail, causing the amplifier output voltage to move below ground.

The corresponding current path may be represented conceptually as

$$\text{Ground} \rightarrow \text{Loudspeaker} \rightarrow \text{Amplifier Output} \rightarrow R14 \rightarrow Q8 \rightarrow -V.$$

As with the positive half-cycle, the changing output voltage is immediately returned to the differential stage through the global negative-feedback network.

The amplifier therefore does not independently generate positive and negative waveforms. Instead, the two halves operate together within the same feedback loop to reconstruct one continuous amplified waveform.

5.15 Operation Around the Zero Crossing

The transition between the positive and negative output stages is one of the most critical regions of Class-AB amplifier operation.

As a positive waveform approaches zero, conduction through the upper output pair decreases. At the same time, the lower output pair begins contributing increasing current.

The bias circuit ensures that the two sections overlap slightly rather than switching completely off and on at exactly zero current.

This smooth transfer of conduction is necessary to reduce crossover distortion.

Global negative feedback further suppresses any residual error occurring in the crossover region by correcting deviations between the required and actual output waveform.

5.16 Role of the 0.33 Ohm Output Resistors

R13 and R14 each have a value of

$$0.33 \, \Omega.$$

These resistors are located in the high-current output stage and develop a small voltage proportional to the output-stage current.

From Ohm's law,

$$V_R = IR.$$

An increase in transistor current therefore produces an increased voltage drop across the corresponding 0.33 Ω resistor.

This introduces local current feedback and helps stabilise the current in the output stage.

The resistors are particularly important for bias stability because they make the standing current less sensitive to changes in transistor gain and temperature.

5.17 Global Negative Feedback Operation

The loudspeaker output is connected back to Q2 through R5.

$$R_5 = 22 \, \text{k}\Omega.$$

The lower part of the feedback network includes R4:

$$R_4 = 1 \text{ k}\Omega.$$

For audio frequencies, the approximate closed-loop voltage gain is

$$A_v = 1 + \frac{R_5}{R_4}.$$

Substitution gives

$$A_v = 1 + \frac{22\,000}{1\,000} = 23.$$

Therefore, if the circuit is operating linearly, the output voltage attempts to follow

$$V_{\text{out}} \approx 23V_{\text{in}}$$

within the available output-voltage range.

The corresponding gain in decibels is

$$A_{v,\text{dB}} = 20 \log_{10}(23) \approx 27.2 \text{ dB}.$$

If nonlinear behaviour in the transistors causes the output to deviate from the required value, the feedback signal immediately changes the differential error voltage.

The amplifier then alters its internal drive so that the output error is reduced. This is one of the principal mechanisms by which distortion is kept low.

5.18 DC Feedback Behaviour

The feedback network also influences the DC operating condition of the amplifier.

Capacitor C3 is connected in the feedback network so that the AC closed-loop gain and DC behaviour are not identical.

At DC, the capacitor prevents the full audio-frequency gain from being applied to DC errors. The resulting DC feedback helps maintain the amplifier output close to ground potential.

This is important because the loudspeaker is directly coupled to the amplifier output. A significant DC voltage at the output would create a continuous current through the loudspeaker voice coil.

The independent STM32-based protection system used in this project provides an additional layer of protection by monitoring the output for abnormal DC voltage and disconnecting the loudspeaker if a persistent fault is detected.

5.19 Loudspeaker Current Delivery

The low-level stages of the amplifier primarily provide voltage gain, whereas the output stage provides the large current required by the loudspeaker.

For a resistive load, loudspeaker current is related to output voltage by

$$I_{\text{out}} = \frac{V_{\text{out}}}{R_L}.$$

For example, an output voltage of

$$20 \text{ V}_{\text{RMS}}$$

across a

$$4 \Omega$$

load requires

$$I_{\text{RMS}} = \frac{20}{4} = 5 \text{ A}.$$

The corresponding peak current is

$$I_{\text{peak}} = 5\sqrt{2} \approx 7.07 \text{ A}.$$

These current levels illustrate why the small-signal input transistors cannot drive the loudspeaker directly. The successive voltage-amplification, driver, and output stages provide the required increase in current capability.

5.20 Output Stability Network

The amplifier output is connected to a series RC network consisting of

$$R_{15} = 10 \Omega$$

and

$$C_8 = 100 \text{ nF}.$$

This is commonly known as a Zobel or Boucherot network.

Its purpose is primarily related to high-frequency stability rather than normal audio power transfer.

A loudspeaker is a reactive load whose impedance changes with frequency. Speaker cables can also introduce additional capacitance and inductance.

At high frequencies, the Zobel network provides the amplifier with a more predictable load and reduces the likelihood of instability when connected to a real loudspeaker.

5.21 Frequency Compensation

The amplifier also contains small compensation capacitors C4 and C6:

$$C_4 = C_6 = 100 \text{ pF}.$$

These capacitors limit the high-frequency open-loop response of the amplifier.

Without appropriate frequency compensation, the phase shift introduced by the several transistor stages could approach a condition in which the negative-feedback loop effectively becomes positive feedback at a sufficiently high frequency.

Such a condition could cause oscillation even when no audible instability is immediately apparent.

The compensation network therefore ensures that the loop gain falls to a safe level before excessive phase shift occurs.

5.22 Behaviour Near Maximum Output

As the input signal amplitude increases, the amplifier output voltage also increases according to the closed-loop gain until the internal transistor stages can no longer produce the required additional voltage.

The maximum clean voltage swing is less than the total supply-rail magnitude because voltage is lost across the output transistors, driver stages, emitter resistors, and other circuit elements.

When the requested output exceeds the available voltage swing, the waveform can no longer follow the input proportionally and clipping occurs.

During positive clipping, the output approaches its maximum available positive voltage and can no longer increase.

During negative clipping, the same effect occurs near the negative supply limit.

Clipping introduces significant nonlinear distortion and should therefore be avoided during normal Hi-Fi operation.

5.23 Power Flow Through the Amplifier

It is important to distinguish between the audio input signal and the power delivered to the loudspeaker.

The source connected to J1 does not provide the loudspeaker power. It provides only the signal information that determines the required output waveform.

The actual loudspeaker energy is obtained from the positive and negative DC power-supply rails.

The transistor stages continuously control the amount of supply current delivered to or removed from the loudspeaker so that the resulting output voltage follows the amplified input waveform.

The amplifier can therefore be regarded as using the low-power input signal to control a much larger flow of energy from the main power supply.

5.24 Complete Positive-to-Negative Signal Cycle

The complete operation of one sinusoidal cycle can now be summarised.

As the input signal initially rises above zero, the differential input stage detects the positive error and drives Q4 accordingly. The bias and driver network causes Q5 and Q7 to supply increasing positive current to the load.

The output voltage rises and the feedback voltage simultaneously increases. The feedback signal opposes the original input difference, maintaining the output at the value established by the closed-loop gain.

As the input waveform returns toward zero, conduction through the upper output pair falls. Around the zero crossing, the Class-AB bias arrangement allows a smooth transfer of load current from the upper pair to the lower pair.

When the input becomes negative, Q6 and Q8 increasingly conduct and the amplifier output moves below ground. Again, the feedback network continuously corrects the output voltage.

As the negative waveform returns to zero, current transfers back toward the upper section and the complete sequence repeats for the next audio cycle.

5.25 Operating Principle Summary

The Project 3A-based amplifier operates by combining a high-gain voltage amplifier with both local and global negative feedback.

The BC546 differential pair Q1 and Q2 compares the input signal with the feedback signal from the loudspeaker output. Q3 provides a controlled current for this input stage.

The resulting error signal controls Q4, the BD140 voltage-amplification transistor, which generates the larger voltage swing required by the output section.

Q9 and VR1 establish the Class-AB bias voltage required to maintain a small standing current through the output stage and reduce crossover distortion.

The upper output section consists of the BD139 driver Q5 and 2SA1943-O power transistor Q7, while the lower section consists of the BD140 driver Q6 and 2SC5200-O power transistor Q8.

These devices operate as complementary feedback pairs, allowing the low-current voltage-

amplification circuitry to control the several amperes required by the loudspeaker.

During positive output excursions, the upper output pair supplies current from the positive rail to the loudspeaker. During negative excursions, the lower pair conducts toward the negative rail.

Around the waveform zero crossing, both output sections remain slightly conductive as a result of the Class-AB bias arrangement, reducing crossover distortion.

Finally, global negative feedback continuously compares the actual output with the required output. Any difference generates a corrective signal at the differential input stage, producing a stable and accurately amplified representation of the original audio waveform.

Chapter 6

Power Transistor Selection

6.1 Introduction

The selection of the output power transistors is one of the most important decisions in the design of a discrete audio power amplifier. The output devices must withstand the full amplifier supply voltage, deliver several amperes of loudspeaker current, dissipate significant heat, and maintain suitable electrical characteristics throughout the audio frequency range.

For the amplifier developed in this project, the selected complementary output transistor pair is

2SC5200-O

for the NPN device and

2SA1943-O

for the PNP device.

These transistors are used as Q8 and Q7 respectively in each power-amplifier channel. They operate together with the BD139 and BD140 driver transistors as complementary feedback pairs within the Project 3A output-stage topology.

The 2SC5200 and 2SA1943 are complementary high-power bipolar junction transistors specifically intended for linear power applications, including audio amplifier output stages [4, 5]. Their voltage rating, current capability, power-dissipation capability, current gain, bandwidth, and mechanical package make them suitable for the operating requirements of the present amplifier.

6.2 Output Transistor Requirements

An audio power transistor must satisfy several requirements simultaneously. A suitable device cannot be selected only from its maximum collector-current rating or maximum power-dissipation figure.

The principal parameters considered in the transistor selection are:

- collector-emitter voltage capability,
- continuous and peak collector-current capability,
- safe operating area,
- collector power dissipation,
- DC current gain,
- current gain at high collector current,
- transition frequency,
- thermal characteristics,
- complementary matching,
- package type,
- heatsink compatibility,
- component availability and authenticity.

These characteristics must be considered together because a power transistor may remain below its absolute maximum current and voltage ratings while still being operated outside its safe operating area because of excessive simultaneous voltage, current, or junction temperature.

6.3 Selected Complementary Pair

The selected power transistor pair consists of the Toshiba 2SC5200 and 2SA1943. The 2SC5200 is an NPN power transistor, while the 2SA1943 is its complementary PNP device.

Table 6.1: Main specifications of the selected output transistors [4, 5].

Parameter	2SC5200-O	2SA1943-O
Polarity	NPN	PNP
Collector-emitter voltage, V_{CEO}	230 V	−230 V
Maximum collector current, I_C	15 A	−15 A
Maximum collector dissipation, P_C	150 W	150 W
DC current gain, O rank	80–160	80–160
Typical transition frequency, f_T	30 MHz	30 MHz
Package	TO-3P(L)	TO-3P(L)
Mounting	Through-hole	Through-hole

The absolute maximum ratings shown in Table 6.1 represent device limits under the conditions specified by the manufacturer. They must not be interpreted as conditions at which the transistor can operate continuously without thermal and safe-operating-area analysis.

6.4 Collector-Emitter Voltage Requirement

The amplifier is powered from a 24–0–24 V AC transformer. Following rectification and filtering, the practical DC supply is expected to be approximately

$$V_+ \approx +33\text{--}34 \text{ V}$$

and

$$V_- \approx -33\text{--}34 \text{ V}.$$

The approximate total rail-to-rail voltage is therefore

$$V_{\text{rail-to-rail}} \approx 68 \text{ V}$$

under the nominal high end of the expected DC supply range.

Both selected power devices have a collector-emitter voltage rating with a magnitude of

$$|V_{CEO}| = 230 \text{ V}.$$

Comparing the device voltage rating with the approximate total amplifier supply gives

$$\frac{230}{68} \approx 3.38.$$

The selected devices therefore provide a substantial voltage margin relative to the normal operating voltage of the amplifier.

This margin is desirable because the transistor voltage must not only accommodate the nominal power-supply voltage but also tolerate mains variation, transformer regulation, switching transients, reactive loudspeaker behaviour, and abnormal operating conditions.

6.5 Collector Current Requirement

The current requirement is more demanding than the voltage requirement, particularly when a 4 Ω loudspeaker is used.

For comparison, the original Project 3A reference operating condition of 100 W into 4 Ω requires an RMS load current of

$$I_{\text{RMS}} = \sqrt{\frac{P}{R_L}} = \sqrt{\frac{100}{4}} = 5 \text{ A}.$$

The corresponding peak current is

$$I_{\text{peak}} = \sqrt{2}I_{\text{RMS}},$$

therefore

$$I_{\text{peak}} = 5\sqrt{2} \approx 7.07 \text{ A.}$$

For the 60 W, 8 Ω Project 3A reference condition,

$$I_{\text{RMS}} = \sqrt{\frac{60}{8}} \approx 2.74 \text{ A,}$$

and

$$I_{\text{peak}} \approx 3.87 \text{ A.}$$

The 2SC5200 and 2SA1943 each have a maximum collector-current rating with a magnitude of

$$|I_C| = 15 \text{ A.}$$

Consequently, the devices provide considerable current capability relative to the expected loudspeaker-current requirement.

The implemented amplifier uses slightly lower supply rails than the nominal $\pm 35 \text{ V}$ Project 3A reference condition, so the actual maximum unclipped output current will depend on the measured rail voltage and clipping point. The 7.07 A value is therefore used as a demanding reference condition rather than a claimed measured current for the completed amplifier.

6.6 Current Gain

Current gain is an important output-transistor parameter because the driver stage must provide sufficient base current to produce the required collector current.

The DC current gain of a bipolar transistor is defined as

$$h_{FE} = \frac{I_C}{I_B}.$$

Rearranging this expression gives the required base current:

$$I_B = \frac{I_C}{h_{FE}}.$$

The selected devices use the **O gain rank**. According to the manufacturer's classification, the O rank corresponds to

$$80 \leq h_{FE} \leq 160$$

at the specified test condition of approximately 1 A collector current and 5 V collector-emitter voltage [4, 5].

Selecting the same gain classification for the complementary devices helps maintain broadly similar current-gain characteristics between the positive and negative sides of the output stage.

It is important, however, not to assume that h_{FE} remains constant at high current. Bipolar transistor gain normally decreases as collector current approaches the upper operating region. The transistor datasheets therefore also specify current-gain behaviour at several amperes of collector current.

This reduction in gain at high current is one reason why dedicated BD139 and BD140 driver transistors are used rather than attempting to drive the power transistors directly from the voltage-amplification stage.

6.7 Driver Current Requirement

The driver transistors provide the base current required by the main output devices.

As an illustrative example, if an output transistor were conducting

$$I_C = 7 \text{ A}$$

with an effective current gain of approximately

$$h_{FE} = 35,$$

the required base current would be approximately

$$I_B = \frac{7}{35} = 0.2 \text{ A}.$$

Thus,

$$\boxed{I_B \approx 200 \text{ mA}}$$

could be required under a demanding high-current operating condition.

This example demonstrates why the output transistor cannot be treated as a voltage-controlled device. Substantial base current can be necessary when the amplifier drives a low-impedance load near maximum output.

The BD139 and BD140 driver stage therefore provides an essential intermediate current-amplification function between the voltage-amplification stage and the 2SC5200-O/2SA1943-O power devices.

6.8 Transition Frequency

The selected transistors have a typical transition frequency of

$$f_T \approx 30 \text{ MHz}$$

under the manufacturer's specified test conditions [4, 5].

The transition frequency is the approximate frequency at which the small-signal current gain of the transistor falls to unity.

Since the audible frequency range extends only to approximately

$$20 \text{ kHz},$$

the transistor transition frequency is many orders of magnitude higher than the upper audible frequency.

This does not mean that the complete power amplifier has a bandwidth of 30 MHz. The overall amplifier bandwidth is intentionally limited by the circuit compensation and global feedback network to maintain adequate phase margin and stability.

A reasonably high transistor transition frequency is nevertheless beneficial because it allows the output devices to maintain useful current gain throughout the audio range and beyond it.

6.9 Complementary Electrical Characteristics

The 2SC5200 and 2SA1943 were designed as complementary devices. Their major electrical ratings are closely matched, including collector-emitter voltage, collector current, collector dissipation, current gain range, and transition frequency.

This is advantageous in a Class-AB power amplifier because the positive and negative halves of the waveform should behave as similarly as practical.

Perfect transistor matching is neither expected nor required because the Project 3A circuit incorporates both local complementary feedback and global negative feedback. Nevertheless, selecting a genuine complementary pair with similar electrical behaviour improves the symmetry of the output stage.

6.10 Suitability for the Complementary Feedback Pair

The Project 3A output stage does not use Q7 and Q8 as a conventional complementary emitter-follower pair. Instead, each output transistor operates with a driver transistor as a complementary feedback pair, or Sziklai pair [1].

In the implemented amplifier, the upper pair consists of

$$Q5 = \text{BD139}$$

and

$$Q7 = 2SA1943\text{-O},$$

while the lower pair consists of

$$Q6 = \text{BD140}$$

and

$$Q8 = 2SC5200\text{-O}.$$

The power transistors therefore provide the majority of the load-current capability, while the driver transistors control them through the local feedback arrangement.

The relatively high current capability and useful gain of the selected power transistors make them well suited to this topology.

6.11 Power Dissipation

The 2SC5200 and 2SA1943 are each specified with a maximum collector power dissipation of

$$P_C = 150 \text{ W}$$

under the manufacturer's specified case-temperature condition [4, 5].

This figure must not be interpreted as meaning that each transistor can continuously dissipate 150 W in the completed amplifier.

The maximum permissible transistor dissipation decreases as the device case temperature increases. The practical power capability is therefore strongly dependent on the complete thermal path from the semiconductor junction to the surrounding air.

This path can be represented as

$$T_J \rightarrow T_C \rightarrow T_H \rightarrow T_A,$$

where

$$T_J = \text{junction temperature,}$$

$$T_C = \text{transistor case temperature,}$$

$$T_H = \text{heatsink temperature,}$$

and

$$T_A = \text{ambient temperature.}$$

The thermal design must therefore consider the transistor junction-to-case thermal resistance, electrical insulation material, thermal interface compound, heatsink thermal resistance, airflow, ambient temperature, and the actual power dissipated by the output devices.

6.12 Safe Operating Area

The safe operating area, or SOA, is a more meaningful parameter than the individual maximum voltage, current, and power ratings when evaluating a linear power transistor.

The SOA defines combinations of collector current and collector-emitter voltage that the transistor can safely withstand for specified time intervals and thermal conditions.

A transistor having a 15 A collector-current rating and a 230 V collector-emitter rating cannot necessarily withstand

$$15 \text{ A}$$

and

$$230 \text{ V}$$

simultaneously.

Such a condition would correspond to an instantaneous transistor power of

$$P = VI = 230 \times 15 = 3450 \text{ W,}$$

which is clearly far beyond the thermal capability of the device.

The operating point must therefore remain inside the safe-operating-area boundary given by the transistor manufacturer.

This issue is particularly important in a Class-AB audio amplifier because the output transistor can simultaneously carry significant current while supporting a substantial collector-emitter voltage.

6.13 Why the 4 Ohm Load is More Demanding

The 4 Ω loudspeaker condition places greater stress on the output transistors than an 8 Ω load because the required current is approximately doubled for the same output voltage.

For example, at

$$V_{\text{out,RMS}} = 20 \text{ V},$$

the load current for 8Ω would be

$$I_{\text{RMS}} = \frac{20}{8} = 2.5 \text{ A},$$

while the same voltage across 4Ω requires

$$I_{\text{RMS}} = \frac{20}{4} = 5 \text{ A}.$$

The lower impedance therefore increases output-device current, emitter resistor dissipation, PCB conductor current, transformer loading, bridge rectifier current, and heatsink demand.

For this reason, the use of the slightly lower approximately ± 33 – ± 34 V supply produced by the selected 24–0–24 V transformer provides useful additional operating margin when driving 4Ω loudspeakers.

6.14 Output Emitter Resistors

Each power transistor operates with a

$$0.33 \Omega$$

output resistor.

These resistors provide a degree of local current feedback. As transistor current rises, the voltage developed across the resistor also rises:

$$V_R = IR.$$

At a current of 5 A, for example,

$$V_R = 5 \times 0.33 = 1.65 \text{ V}.$$

The instantaneous resistor power at this current would be

$$P_R = I^2 R = 5^2 \times 0.33 = 8.25 \text{ W}.$$

This example demonstrates that the output resistor experiences substantial instantaneous power during high-current signal operation. Actual average power depends on the waveform and conduction interval.

More importantly for transistor operation, the voltage developed by these resistors provides local negative feedback that opposes excessive increases in transistor current and contributes to improved bias and thermal stability.

6.15 TO-3P(L) Package Selection

Both selected output transistors are supplied in the TO-3P(L) through-hole power package.

The package is advantageous for this amplifier because it provides:

- a large physical surface suitable for heatsink mounting,
- mechanically robust through-hole leads,
- low-resistance high-current connections,
- convenient external heatsink attachment,
- straightforward PCB integration,
- a common package for both complementary devices.

The PCB layout positions Q7 and Q8 close to the edge of the board so that their bodies can face outward toward the heatsink.

This arrangement keeps the transistor leads relatively short while allowing the package to be mechanically clamped to a substantial external heatsink.

6.16 Thermal Interface and Heatsink Requirement

The electrical power that is not delivered to the loudspeaker must ultimately be dissipated as heat, primarily by the output transistors.

Consequently, Q7 and Q8 must never be operated at normal amplifier power without suitable heatsinking.

The thermal path between the transistor package and heatsink should have low thermal resistance. Depending on the mechanical construction, an electrical insulator may also be required between the transistor mounting surface and the heatsink.

Thermal compound is used to reduce the thermal resistance caused by microscopic air gaps between mating surfaces.

The detailed heatsink calculations, cooling fans, thermal monitoring, and temperature protection are treated separately in the thermal-management chapters of this report.

6.17 Comparison with the Original Project 3A Devices

Rod Elliott's Project 3A has been designed to operate with several suitable families of audio power transistors rather than depending on one unique device type [1].

The Project 3A documentation discusses devices including the MJL4281A and MJL4302A, MJL21193 and MJL21194, NJW3281 and NJW1302, and several alternative power transistor pairs.

This flexibility is possible partly because of the simplicity of the amplifier and its use of complementary feedback in the output stage.

For the present implementation, the 2SC5200-O and 2SA1943-O were selected because their characteristics satisfy the electrical requirements of the approximately ± 33 – ± 34 V amplifier supply while providing substantial voltage and current capability.

The use of the same complementary pair throughout both stereo channels also simplifies component sourcing, PCB assembly, thermal mounting, replacement, and maintenance.

6.18 Importance of Genuine Power Transistors

Power transistors used in an audio amplifier must be obtained from a reliable source. Devices carrying popular audio transistor part numbers are sometimes counterfeited or relabelled.

A counterfeit transistor may have a much smaller semiconductor die or significantly lower voltage, current, power, and safe-operating-area capability than the genuine component.

Such a component may operate correctly at low signal level and still fail catastrophically when the amplifier is driven into a demanding loudspeaker load.

For this reason, the output devices for this project are specified by their exact manufacturer part numbers and should be obtained from an authorised electronic-component distributor rather than from an unknown marketplace source.

6.19 Design Margin

A useful summary of the electrical margin can be obtained by comparing the main transistor ratings with representative amplifier requirements.

Table 6.2: Comparison between output-transistor capability and amplifier requirements.

Parameter	Amplifier Condition	Transistor Rating
Supply rail magnitude	≈ 33 – 34 V	$ V_{CEO} = 230$ V
Approx. rail-to-rail voltage	≈ 66 – 68 V	$ V_{CEO} = 230$ V
$4\ \Omega$ reference peak current	≈ 7.07 A	$ I_C = 15$ A
Gain classification	O rank selected	$h_{FE} = 80$ – 160 at specified test point
Audio frequency limit	≈ 20 kHz	$f_T \approx 30$ MHz typical

The table shows that the selected devices provide substantial nominal voltage and current margins.

However, these simple numerical comparisons do not replace safe-operating-area and thermal analysis. Reliable operation requires the voltage, current, power, junction tempera-

ture, heatsink performance, and duration of high-power operation to remain within safe limits simultaneously.

6.20 Selection Summary

The 2SC5200-O and 2SA1943-O complementary transistor pair was selected as the power output stage for the amplifier.

The principal reasons for their selection are their high collector-emitter voltage capability, high collector-current capability, useful current gain, wide bandwidth, complementary electrical characteristics, and mechanically suitable TO-3P(L) package.

Each transistor has a collector-emitter voltage rating with a magnitude of

$$230 \text{ V},$$

a maximum collector-current rating with a magnitude of

$$15 \text{ A},$$

and a maximum collector-dissipation rating of

$$150 \text{ W}$$

under the manufacturer's specified case-temperature conditions.

The selected O rank provides a specified low-current gain classification of

$$80 \leq h_{FE} \leq 160,$$

while the typical transition frequency is approximately

$$30 \text{ MHz}.$$

These ratings provide comfortable nominal margins relative to the approximately ± 33 – ± 34 V supply used in the implemented amplifier and the several-amperes output-current requirement associated with 4Ω and 8Ω loudspeakers.

The transistors are used together with the BD139 and BD140 driver devices in the complementary-feedback output topology of Project 3A. This arrangement provides the current amplification required to drive the loudspeaker while local and global negative feedback improve linearity and control.

The nominal ratings alone do not guarantee safe operation. Safe operating area, junction temperature, thermal resistance, heatsink performance, bias current, and load impedance remain critical design constraints. These considerations are addressed further in the chapters concerning biasing, thermal stability, heatsink selection, and thermal management.

Chapter 7

Biasing and Thermal Stability

7.1 Introduction

Biasing and thermal stability are critical aspects of the design of a Class-AB audio power amplifier. The output stage must maintain a sufficiently small standing current when no audio signal is present so that crossover distortion is minimised. At the same time, the quiescent current must remain low enough to prevent unnecessary transistor dissipation, excessive heatsink temperature, and the possibility of thermal instability.

The amplifier developed in this project is based on Rod Elliott's Project 3A topology and uses a complementary-feedback-pair output stage [1]. The main power transistors are the 2SA1943-O PNP device and the 2SC5200-O NPN device, driven by BD139 and BD140 transistors.

The amplifier incorporates an adjustable bias-control network based on Q9, a BC546 transistor, and the variable resistor VR1. This network allows the quiescent current to be adjusted during initial commissioning.

The thermal behaviour of this amplifier differs in an important way from that of many conventional complementary emitter-follower amplifiers. In the Project 3A complementary-feedback-pair topology, the driver transistors play an important role in determining the quiescent current. For this reason, the bias servo transistor must not be treated as though it were a conventional heatsink-mounted thermal compensation transistor.

7.2 Purpose of Output-Stage Biasing

A conventional Class-B output stage uses one transistor section to reproduce the positive portion of the output waveform and another transistor section to reproduce the negative portion.

A bipolar junction transistor does not begin conducting significant current until an appropriate base-emitter voltage has been established. If both halves of an output stage are completely turned off when the signal crosses zero, a short region exists in which neither side conducts adequately.

This produces crossover distortion around the zero crossing of the output waveform.

Class-AB operation reduces this effect by deliberately maintaining a small current through the output stage even when no audio signal is present. This current is referred to as the quiescent current and is denoted by

$$I_Q.$$

The bias circuit establishes a small overlap between the conduction of the positive and negative output stages. As a result, transfer of the loudspeaker current from one side of the amplifier to the other occurs more smoothly.

7.3 Quiescent Current

The Project 3A documentation recommends a quiescent-current range of approximately

$$30 \text{ mA} \leq I_Q \leq 75 \text{ mA},$$

with a current of approximately

$$I_Q \approx 50 \text{ mA}$$

representing a suitable nominal operating point [1].

The amplifier can operate at a lower quiescent current, but crossover distortion becomes increasingly visible as the current is reduced. Elliott indicates that approximately

$$20 \text{ mA}$$

should be regarded as a practical minimum for low-distortion operation [1].

Increasing the quiescent current substantially above the required value provides little further improvement in distortion while increasing power dissipation and heatsink temperature.

The bias setting is therefore a compromise between crossover performance and thermal dissipation.

7.4 Bias-Control Circuit

The adjustable bias network in the implemented amplifier is centred around Q9, a BC546 transistor, and VR1.

The value of the bias adjustment potentiometer is

$$VR_1 = 2 \text{ k}\Omega.$$

Changing the resistance of VR1 alters the voltage developed by the bias-control network. This changes the voltage applied between the upper and lower driver stages and consequently changes the standing current through the output stage.

If the bias voltage is too low, the quiescent current becomes insufficient and crossover distortion increases.

If the bias voltage is too high, excessive standing current flows through the driver and output transistors. This increases semiconductor dissipation and may eventually produce excessive junction temperature.

Correct adjustment of VR1 is therefore essential before the amplifier is operated at significant output power.

7.5 Complementary Feedback Pair Bias Behaviour

The Project 3A output stage uses complementary feedback pairs, also known as Sziklai pairs, rather than conventional complementary Darlington or emitter-follower output stages.

The upper complementary feedback pair consists of

$$Q5 = \text{BD139}$$

and

$$Q7 = \text{2SA1943-O},$$

while the lower pair consists of

$$Q6 = \text{BD140}$$

and

$$Q8 = \text{2SC5200-O}.$$

An important consequence of this topology is that the quiescent-current behaviour is strongly influenced by the driver transistors. The thermal compensation mechanism is therefore different from that of a conventional emitter-follower output stage.

Rod Elliott specifically notes that the quiescent current in Project 3A is controlled largely by the driver transistors and that mounting the bias servo transistor on the main heatsink produces excessive compensation [1].

For this reason, Q9 should remain thermally associated with the amplifier PCB rather than being mounted directly to the power-transistor heatsink.

7.6 Output Emitter Resistors and Bias Stability

The output stage contains two low-value power resistors,

$$R_{13} = R_{14} = 0.33 \, \Omega.$$

These resistors perform several important functions. They introduce local current feedback, improve bias stability, reduce sensitivity to transistor parameter variations, and provide a convenient method of measuring the quiescent current.

The voltage developed across a resistor is given by Ohm's law:

$$V = IR.$$

An increase in transistor current therefore produces a corresponding increase in voltage across the $0.33 \, \Omega$ resistor. This voltage opposes further increase in current through the local feedback action of the output stage.

The $0.33 \, \Omega$ value is deliberately used in the updated Project 3A design. Elliott recommends this value rather than the lower $0.22 \, \Omega$ value found in some earlier implementations because the higher resistance improves bias stability [1].

7.7 Measurement of Quiescent Current

The quiescent current can be determined by measuring the voltage across both $0.33 \, \Omega$ output resistors.

The total resistance between the relevant measurement points is

$$R_{\text{total}} = R_{13} + R_{14}.$$

Therefore,

$$R_{\text{total}} = 0.33 + 0.33 = 0.66 \, \Omega.$$

Using Ohm's law,

$$I_Q = \frac{V_{\text{measured}}}{0.66}.$$

For a target quiescent current of

$$I_Q = 50 \, \text{mA},$$

the expected voltage is

$$V_{\text{measured}} = 0.05 \times 0.66 = 0.033 \, \text{V}.$$

Thus,

$$V_{\text{measured}} \approx 33 \text{ mV}$$

corresponds to approximately

$$I_Q \approx 50 \text{ mA}.$$

Similarly, for

$$I_Q = 75 \text{ mA},$$

the expected voltage is

$$V_{\text{measured}} = 0.075 \times 0.66 = 0.0495 \text{ V},$$

or approximately

$$49.5 \text{ mV}.$$

Consequently, a measurement in the approximate range of

$$33\text{--}50 \text{ mV}$$

corresponds to a quiescent current of approximately

$$50\text{--}75 \text{ mA}.$$

7.8 Bias Measurement Relationship

A useful direct relationship between measured voltage and quiescent current is

$$I_Q = \frac{V_{\text{measured}}}{0.66}.$$

If the measured voltage is expressed in millivolts and current in milliamperes,

$$I_Q[\text{mA}] \approx 1.515 V_{\text{measured}}[\text{mV}].$$

Therefore, every millivolt measured across the combined emitter resistors corresponds to approximately

$$1.5 \text{ mA}$$

of quiescent current.

For example,

$$20 \text{ mV} \rightarrow 30.3 \text{ mA},$$

$$33 \text{ mV} \rightarrow 50.0 \text{ mA},$$

and

$$50 \text{ mV} \rightarrow 75.8 \text{ mA}.$$

7.9 Initial Bias Adjustment

Before initial power-up, VR1 must be positioned so that the amplifier starts with the minimum possible quiescent current.

For Project 3A, Elliott specifies that VR1 should be set to its maximum resistance before power is first applied [1].

This is an important safety requirement. If the potentiometer is initially set toward the high-current end of its adjustment range, excessive current may flow immediately after power-up and may damage the output transistors.

The initial bias-adjustment sequence is therefore:

1. Disconnect the loudspeaker load.
2. Set VR1 to maximum resistance before power is applied.
3. Power the amplifier using the controlled initial power-up procedure.
4. Verify that the DC output voltage is within a safe range.
5. Measure the voltage across the two 0.33Ω output resistors.
6. Slowly adjust VR1 until the required quiescent-current voltage is obtained.
7. Allow the amplifier to reach thermal equilibrium.
8. Measure the voltage again and readjust VR1 if necessary.

The final bias adjustment should not be considered complete until the amplifier temperature has stabilised.

7.10 Temperature Dependence of Semiconductor Junctions

The electrical characteristics of bipolar junction transistors vary with temperature. In particular, the base-emitter voltage required to produce a given collector current decreases as the semiconductor junction temperature increases.

This creates an important feedback mechanism in a power amplifier. An increase in transistor current causes additional power dissipation:

$$P_D = V_{CE}I_C.$$

The increased power raises the transistor junction temperature. As the junction temperature rises, transistor conduction characteristics change, potentially allowing additional current to flow.

If this process is not adequately controlled, a positive thermal feedback cycle can develop:

$$I_C \uparrow \Rightarrow P_D \uparrow \Rightarrow T_J \uparrow \Rightarrow I_C \uparrow.$$

This effect is known as thermal runaway.

7.11 Prevention of Thermal Runaway

Several features of the implemented amplifier help prevent thermal runaway.

First, the $0.33\ \Omega$ output resistors provide local current feedback. Increasing transistor current produces a greater resistor voltage drop, which tends to oppose further current increase.

Second, the complementary-feedback-pair output topology introduces local feedback through the driver and output transistors.

Third, the quiescent current is deliberately limited to a relatively small value.

Finally, the 2SC5200-O and 2SA1943-O power devices are mounted on substantial external heatsinks so that heat can be transferred away from the transistor junctions.

Thermal stability therefore results from the combined action of electrical feedback, correct bias setting, emitter resistance, transistor selection, and adequate heatsinking.

7.12 Quiescent Power Dissipation

Even when no audio signal is present, the quiescent current causes power to be drawn from the positive and negative supply rails.

The implemented amplifier is expected to operate from rails of approximately

$$V_S \approx 33\text{--}34\ \text{V}.$$

An approximate expression for the idle power associated with the Class-AB output-stage quiescent current is

$$P_Q \approx 2V_S I_Q.$$

Using a nominal rail magnitude of

$$V_S = 34 \text{ V}$$

and

$$I_Q = 50 \text{ mA},$$

the approximate idle power is

$$P_Q \approx 2 \times 34 \times 0.05.$$

Therefore,

$$P_Q \approx 3.4 \text{ W}$$

per amplifier channel, excluding the smaller consumption of the input, voltage-amplification, and other supporting stages.

At a quiescent current of

$$75 \text{ mA},$$

the corresponding value becomes

$$P_Q \approx 2 \times 34 \times 0.075 = 5.1 \text{ W}.$$

These calculations demonstrate why increasing the bias current unnecessarily causes additional heating even when no audio signal is present.

7.13 Bias Current and Crossover Distortion

Increasing quiescent current initially reduces crossover distortion because it increases the conduction overlap between the two halves of the Class-AB output stage.

However, the improvement does not continue indefinitely. According to the Project 3A design information, crossover distortion becomes very small at moderate quiescent current and increasing the current significantly above approximately 50 mA provides little further improvement [1].

The nominal 50 mA setting therefore provides a useful compromise:

Low $I_Q \Rightarrow$ lower idle heating but increased crossover distortion,

while

High $I_Q \Rightarrow$ reduced crossover effects but increased heating.

The selected bias point is intended to remain within the region in which crossover distortion is already low without imposing unnecessary thermal stress on the output stage.

7.14 Supply Voltage Influence on Bias

The quiescent current is not completely independent of supply voltage.

The voltage-amplification stage uses a bootstrapped current-loading arrangement rather than an ideal constant-current source. Consequently, changes in supply voltage can influence the current through the driver and bias circuitry.

The implemented amplifier uses a 24–0–24 V AC transformer and is expected to operate from approximately

$$\pm 33\text{--}\pm 34 \text{ VDC}$$

rather than the nominal ± 35 V supply used for the standard Project 3A reference configuration.

The bias must therefore be adjusted using the actual completed amplifier rather than assuming that a potentiometer position derived from another supply voltage will produce the correct quiescent current.

The adjustment should also be checked with the final transformer, rectifier, reservoir capacitors, and heatsink installed.

7.15 Thermal Stabilisation After Adjustment

The quiescent current should be adjusted initially while the amplifier is relatively cool and then checked again after the amplifier has warmed.

As the amplifier temperature changes, the semiconductor junction voltages also change. The measured quiescent current may therefore drift during the first several minutes after power is applied.

After the initial setting is made, the amplifier should remain powered while the output-transistor and heatsink temperatures are monitored.

Once the temperature has stabilised, the voltage across R13 and R14 should be measured again.

If necessary, VR1 can then be readjusted to restore the required quiescent current.

This process may require more than one adjustment because heatsink temperature and quiescent current are interdependent.

7.16 Power Transistor Heatsinking

The 2SA1943-O and 2SC5200-O output transistors must be mechanically attached to suitable heatsinks.

The purpose of the heatsink is to reduce the temperature difference between the transistor case and the surrounding air by providing a large surface area for heat transfer.

The complete thermal path can be represented by

$$T_J \rightarrow T_C \rightarrow T_H \rightarrow T_A,$$

where

T_J = semiconductor junction temperature,

T_C = transistor case temperature,

T_H = heatsink temperature,

and

T_A = ambient temperature.

Each stage of the thermal path has an associated thermal resistance.

In general,

$$T_J = T_A + P_D (R_{\theta JC} + R_{\theta CS} + R_{\theta SA}),$$

where

$R_{\theta JC}$ = junction-to-case thermal resistance,

$R_{\theta CS}$ = case-to-heatsink thermal resistance,

and

$R_{\theta SA}$ = heatsink-to-ambient thermal resistance.

This relationship demonstrates that safe junction temperature depends on both the transistor itself and the mechanical cooling arrangement.

7.17 Thermal Interface

The interface between the transistor and heatsink contributes additional thermal resistance.

Even apparently smooth metal surfaces contain microscopic imperfections that trap air when pressed together. Since air has poor thermal conductivity, these gaps increase the thermal resistance between the transistor and heatsink.

A thin layer of suitable thermal compound is therefore used to fill these microscopic voids and improve heat transfer.

Where electrical isolation from the heatsink is required, an appropriate electrically insulating but thermally conductive mounting interface must also be used.

The mounting hardware must provide sufficient mechanical pressure without damaging the transistor package.

7.18 Heatsink Requirement

Rod Elliott recommends substantial heatsinking for Project 3A and notes that there is effectively no disadvantage in using a heatsink larger than the minimum requirement [1].

For a pair of Project 3A amplifiers operated under normal Hi-Fi conditions and at the minimum recommended quiescent current, a heatsink thermal resistance in the region of $0.5\text{ }^{\circ}\text{C}/\text{W}$ is cited as a suitable reference [1].

This value should be regarded as a reference from the original amplifier rather than as a calculated specification for the mechanical assembly used in this project.

The required heatsink performance depends on the actual quiescent current, load impedance, listening level, enclosure airflow, ambient temperature, transistor mounting, and duration of high-power operation.

The final thermal design of this project therefore uses substantial heatsinks and incorporates active temperature monitoring as part of the protection system.

7.19 Driver Transistor Temperature

The BD139 and BD140 driver transistors dissipate less power than the main 2SA1943-O and 2SC5200-O output devices.

At supply voltages close to the Project 3A nominal $\pm 35\text{ V}$ value, the drivers do not normally require individual heatsinks during typical operation [1].

However, operation into $4\text{ }\Omega$ loads produces significantly higher output current. Under sustained high-power conditions, the temperature of Q5 and Q6 should therefore be checked during prototype testing.

If the driver transistors become excessively hot under the intended operating conditions, additional local heatsinking may be used.

7.20 Bias Servo Placement

The physical location of Q9 requires particular attention.

In many traditional Class-AB amplifiers, the bias transistor is intentionally thermally coupled to the output-transistor heatsink. This arrangement is used to reduce the bias voltage as the power devices become hotter.

That approach should not be applied directly to the Project 3A circuit.

Because the amplifier uses complementary feedback pairs, the driver transistors are primarily responsible for the thermal behaviour of the quiescent-current system. Elliott specifically states that mounting the bias servo transistor on the main heatsink results in excessive compensation [1].

Excessive compensation would reduce the quiescent current too strongly as the heatsink temperature rises and could reintroduce crossover distortion.

Therefore,

Q9 should not be mounted to the main power-transistor heatsink.

It should remain in the circuit-board environment intended by the Project 3A topology.

7.21 Detection of Inadequate Cooling

A correctly designed heatsink should eventually reach a stable temperature for a constant operating condition.

If the heatsink temperature continues increasing without approaching thermal equilibrium, the cooling system is inadequate for the power being dissipated.

A particularly serious condition exists if both heatsink temperature and quiescent current continue to rise together.

This behaviour indicates insufficient thermal stability and must not be ignored. Power should be removed and the cause investigated before further operation.

Possible causes include:

- insufficient heatsink surface area,
- inadequate airflow,
- excessive quiescent-current adjustment,
- poor transistor-to-heatsink thermal contact,
- incorrect transistor installation,
- incorrect component values,
- incorrect bias-circuit wiring,
- excessive ambient temperature.

7.22 Thermal Behaviour Under Load

Output-transistor heating is not determined only by loudspeaker output power.

The instantaneous transistor dissipation is approximately

$$P_D(t) = V_{CE}(t)I_C(t).$$

Consequently, transistor heating depends simultaneously on output current and the voltage remaining across the transistor.

At very small output power, current is low and the transistors dissipate only their quiescent power.

At intermediate output levels, a transistor may simultaneously carry significant current while supporting a substantial fraction of the supply voltage. This can produce considerable device dissipation.

Near clipping, more of the supply power is transferred to the load, although large peak currents still place significant electrical and thermal stress on the output stage.

For this reason, heatsink selection cannot be based solely on the rated loudspeaker output power.

7.23 Effect of Loudspeaker Impedance

A $4\ \Omega$ loudspeaker represents a more demanding thermal condition than an $8\ \Omega$ loudspeaker because it requires approximately twice the current for the same output voltage.

For example, an RMS output voltage of

$$20\ \text{V}$$

requires

$$I_{\text{RMS}} = \frac{20}{8} = 2.5\ \text{A}$$

for an $8\ \Omega$ load.

The same output voltage into $4\ \Omega$ requires

$$I_{\text{RMS}} = \frac{20}{4} = 5\ \text{A}.$$

The increased collector current places greater stress on the power transistors, drivers, emitter resistors, PCB tracks, power supply, and heatsinks.

This is another reason why the implemented 24–0–24 V transformer, producing approximately $\pm 33\text{--}\pm 34\ \text{VDC}$ rails, is advantageous for a system intended to support both $4\ \Omega$ and $8\ \Omega$ loudspeaker loads.

7.24 Cooling Fans

Forced-air cooling may be used to supplement the main heatsinks when the amplifier is operated for extended periods at high power.

Airflow reduces the effective heatsink-to-ambient thermal resistance by increasing convective heat transfer.

This can be particularly useful when:

- both stereo channels operate at high power simultaneously,
- 4 Ω loudspeakers are used,
- ambient temperature is high,
- the amplifier is installed in an enclosed chassis,
- prolonged test signals or demanding programme material are used.

The cooling fans form part of the overall mechanical and thermal design and are discussed in greater detail in the dedicated cooling chapter.

7.25 Temperature Monitoring and Protection

In addition to passive thermal design, the complete amplifier includes temperature monitoring as part of the STM32-based protection system.

The temperature sensors monitor the thermal condition of the amplifier and allow the controller to respond if a predefined excessive-temperature condition is reached.

If an overheating condition is detected, the protection system can disconnect the loudspeakers and activate the corresponding fault indication.

It is important to note that electronic temperature protection does not replace adequate heatsinking. The heatsink must be capable of maintaining safe transistor temperatures during normal operation.

The protection system provides an additional safety layer for abnormal or unexpected operating conditions.

7.26 Recommended Bias Verification Procedure

After construction, the following sequence should be used to verify stable bias operation:

1. Ensure that both output power transistors are correctly mounted to their heatsinks.
2. Confirm the orientation and values of all components in the bias and output stages.
3. Set VR1 to maximum resistance before applying power.

4. Perform the initial power-up without a loudspeaker connected.
5. Verify that no transistor or resistor shows abnormal rapid heating.
6. Measure the amplifier DC output voltage.
7. Measure the voltage across R13 and R14.
8. Slowly adjust VR1 toward the desired quiescent-current setting.
9. A value of approximately 33 mV corresponds to approximately 50 mA quiescent current.
10. Allow the amplifier and heatsink to warm until the temperature approaches equilibrium.
11. Recheck the measured bias voltage and readjust if required.
12. Continue monitoring temperature and bias current to confirm that neither increases continuously.
13. Repeat the procedure independently for the second stereo channel.

Only after stable bias and acceptable DC offset have been verified should the amplifier proceed to higher-power load testing.

7.27 Bias and Thermal Stability Summary

The Class-AB output stage requires a controlled quiescent current to minimise crossover distortion while limiting unnecessary heat generation.

In the implemented amplifier, Q9 and the 2 k Ω potentiometer VR1 provide adjustment of the output-stage bias.

A nominal quiescent current of approximately

$$50 \text{ mA}$$

is used as the preferred starting point, while the Project 3A documentation allows operation over an approximate range of 30–75 mA and identifies approximately 20 mA as a practical minimum for low-distortion operation.

The two 0.33 Ω output resistors provide both local current feedback and a convenient method of measuring the bias current. A voltage of approximately

$$33 \text{ mV}$$

across the combined 0.66 Ω resistance corresponds to approximately

$$50 \text{ mA}.$$

At the implemented approximately ± 34 V supply, a 50 mA output-stage quiescent current corresponds to approximately

$$\boxed{3.4 \text{ W}}$$

of idle output-stage supply power per channel, excluding the smaller power consumed by the remaining amplifier circuitry.

Thermal stability is improved by the $0.33 \text{ } \Omega$ emitter resistors, local feedback of the complementary-feedback-pair output stage, correct bias adjustment, and substantial heatsinking of the 2SA1943-O and 2SC5200-O power transistors.

Unlike many conventional Class-AB amplifiers, the Project 3A bias servo transistor Q9 should not be mounted directly on the main heatsink. The thermal behaviour of the complementary-feedback-pair topology is primarily influenced by the driver devices, and heatsink mounting of the bias servo would produce excessive thermal compensation.

After adjustment, the amplifier must be allowed to reach thermal equilibrium and the quiescent current must be checked again. A continuing increase in both temperature and bias current indicates an inadequate thermal condition and requires the amplifier to be switched off and corrected before operation continues.

Correct biasing, adequate heatsinking, controlled thermal interfaces, and continuous temperature supervision therefore work together to provide stable and reliable Class-AB operation.

Chapter 8

Main Power Supply

8.1 Introduction

The main power supply is responsible for providing the symmetrical DC supply rails required by the two power-amplifier channels. In a high-power audio amplifier, the power supply is a critical subsystem because almost all of the energy delivered to the loudspeaker originates from the DC supply rails.

The low-level audio input signal controls the amplifier transistors, but the actual electrical power delivered to the loudspeaker is supplied by the transformer, rectifier, and reservoir-capacitor system.

The power supply implemented in this project is an unregulated symmetrical linear supply. It consists of an external 24–0–24 V AC toroidal transformer, a full-wave bridge rectifier, large electrolytic reservoir capacitors, and high-frequency bypass capacitors.

The expected practical supply rails are approximately

$$V_+ \approx +33\text{--}34 \text{ VDC}$$

and

$$V_- \approx -33\text{--}34 \text{ VDC}.$$

The exact rail voltages depend on mains voltage, transformer regulation, rectifier voltage drop, capacitor ripple, amplifier loading, and wiring resistance.

8.2 Power-Supply Architecture

The overall power-conversion path of the amplifier can be represented as

Mains AC → Toroidal Transformer → 24–0–24 V AC → Bridge Rectifier → Reservoir Capacitors → Sy

The transformer performs two important functions. First, it provides galvanic isolation between the mains supply and the low-voltage amplifier circuitry. Second, it reduces the mains voltage to the required secondary voltage.

The bridge rectifier converts the transformer secondary AC waveform into rectified current. The reservoir capacitors then store energy and smooth the rectified waveform to produce the positive and negative DC supply rails.

The resulting supply provides three principal electrical potentials:

$$V_+, \quad 0 \text{ V}, \quad V_-.$$

The central 0 V node acts as the reference point for both amplifier channels.

8.3 Power-Supply Schematic

The main power-supply circuitry is shown on the second page of the combined power-amplifier schematic PDF. The first page contains the power-amplifier circuit, while the second page contains the rectifier and reservoir-capacitor section.

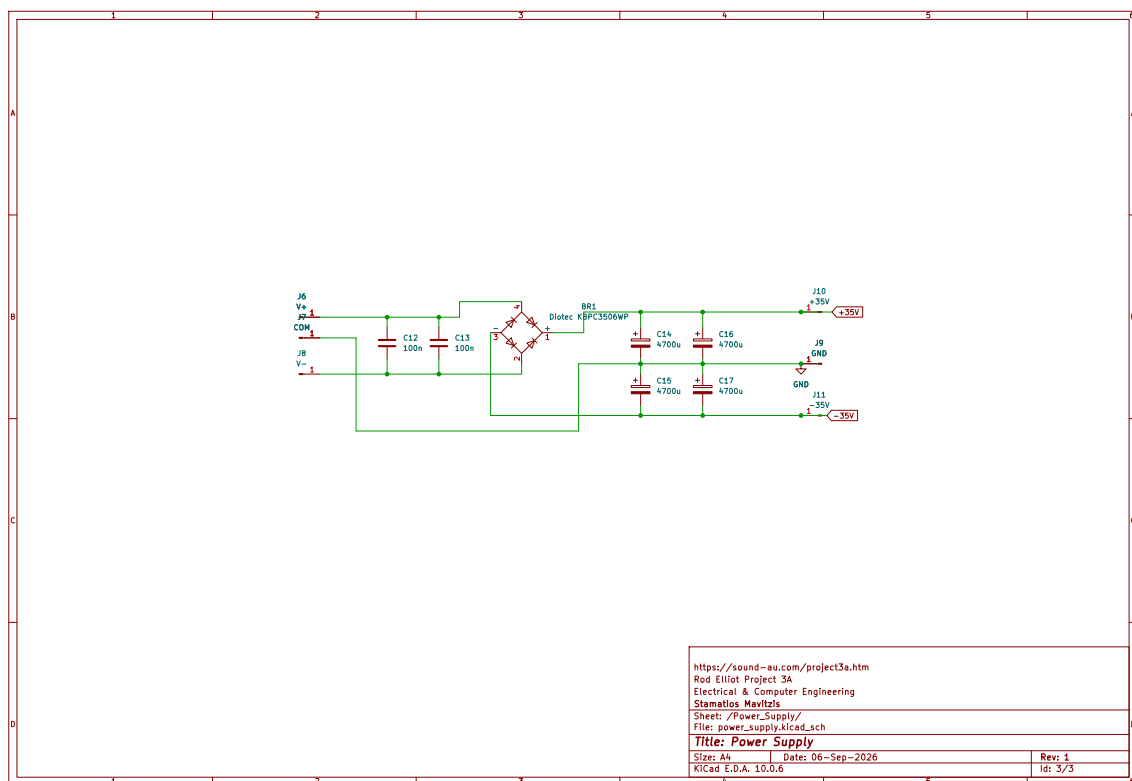


Figure 8.1: Main symmetrical power-supply schematic for the stereo power amplifier.

Figure 8.1 represents the circuitry following the toroidal transformer. The external 24–0–24 V AC transformer is not drawn on this schematic sheet.

The two outer ends of the transformer secondary and the centre tap are connected to the power-supply section through the corresponding external connections. The centre tap establishes the central reference point of the split supply.

The main rectifier is BR1, implemented using a Diotec KBPC3506WP bridge rectifier. Four $4700\ \mu\text{F}$ electrolytic capacitors provide the main energy storage, while C12 and C13 provide high-frequency bypassing.

8.4 24–0–24 V AC Transformer Interface

The toroidal transformer selected for this project has a centre-tapped secondary winding rated at

$$24\text{--}0\text{--}24\ \text{V}_{\text{RMS}}.$$

Each half of the secondary therefore provides approximately

$$24\ \text{V}_{\text{RMS}}$$

relative to the centre tap.

The voltage between the two outer secondary connections is nominally

$$48\ \text{V}_{\text{RMS}}.$$

The centre tap becomes the electrical reference between the positive and negative supply rails.

The selected transformer differs slightly from the 25–0–25 V AC transformer commonly used with the original Project 3A configuration. The resulting DC rails are therefore slightly lower than the nominal $\pm 35\ \text{V}$ rails associated with the original reference design.

8.5 Theoretical DC Rail Voltage

For a sinusoidal waveform, the relationship between RMS voltage and peak voltage is

$$V_{\text{peak}} = V_{\text{RMS}}\sqrt{2}.$$

For one $24\ \text{V}_{\text{RMS}}$ half-secondary,

$$V_{\text{peak}} = 24\sqrt{2}.$$

Therefore,

$$V_{\text{peak}} \approx 33.94\ \text{V}.$$

Before rectifier and wiring losses, the theoretical supply rails are therefore approximately

$$V_+ \approx +33.94 \text{ V}$$

and

$$V_- \approx -33.94 \text{ V}.$$

In practice, the actual DC voltage will not be exactly equal to this value. The conducting rectifier diodes introduce a forward-voltage drop, while the transformer secondary voltage itself changes with load.

Consequently, the nominal operating supply is described throughout this report as approximately

$$\boxed{\pm 33 - \pm 34 \text{ VDC}}.$$

At very light load, the measured voltage may be somewhat higher than this range because the unloaded secondary voltage of a real transformer is often higher than its rated full-load voltage.

8.6 Relation to the Project 3A Reference Supply

Rod Elliott's Project 3A is commonly operated from approximately $\pm 35 \text{ V}$ for operation with both 4Ω and 8Ω loudspeakers [1].

The present amplifier uses slightly lower supply rails because of the selected 24–0–24 V AC transformer.

The difference between the theoretical peak voltage of the implemented transformer and the nominal Project 3A rail voltage is approximately

$$35 - 33.94 = 1.06 \text{ V}$$

per rail.

This difference is relatively small, so the amplifier remains close to the intended electrical operating conditions of Project 3A.

The slightly reduced rail voltage also provides additional voltage, current, and thermal margin when the amplifier is used with 4Ω loudspeaker loads.

The published Project 3A output-power figures should therefore be regarded as reference values rather than guaranteed performance values for the present implementation.

8.7 Full-Wave Rectification

The transformer secondary provides an alternating voltage, whereas the amplifier requires fixed positive and negative DC supply rails.

BR1 converts the alternating secondary waveform into rectified current that charges the reservoir capacitors.

The implemented rectifier is a

KBPC3506WP.

A bridge rectifier contains four diodes arranged so that current is directed toward the correct supply polarity during both halves of the AC waveform.

Because both halves of the mains waveform are used, the circuit performs full-wave rectification.

For a mains frequency of

$$f_{\text{mains}} = 50 \text{ Hz},$$

the principal charging and ripple frequency after full-wave rectification is

$$f_{\text{ripple}} = 2f_{\text{mains}}.$$

Thus,

$$\boxed{f_{\text{ripple}} = 100 \text{ Hz}}.$$

The bridge rectifier is discussed in greater detail in the dedicated *Bridge Rectifier and Reservoir Capacitors* chapter.

8.8 Reservoir Capacitor Arrangement

The power supply uses four large electrolytic reservoir capacitors:

$$C_{14} = C_{15} = C_{16} = C_{17} = 4700 \mu\text{F}.$$

Two capacitors are used for the positive supply rail and two for the negative supply rail.

Because the two capacitors associated with each rail operate in parallel, the effective capacitance per rail is

$$C_{\text{rail}} = 4700 \mu\text{F} + 4700 \mu\text{F}.$$

Therefore,

$$\boxed{C_{\text{rail}} = 9400 \mu\text{F}}.$$

In farads,

$$C_{\text{rail}} = 9.4 \times 10^{-3} \text{ F}.$$

The complete power supply therefore contains a physical installed capacitance of

$$4 \times 4700 \text{ } \mu\text{F} = 18\,800 \text{ } \mu\text{F}.$$

However, because the supply is divided into positive and negative capacitor banks, the effective reservoir capacitance relevant to each individual rail is

$$9400 \text{ } \mu\text{F}.$$

8.9 Function of the Reservoir Capacitors

The bridge rectifier does not produce perfectly constant DC voltage. Its output follows the peaks of the rectified AC waveform.

The reservoir capacitors charge when the instantaneous rectifier voltage exceeds their existing voltage. Between charging intervals, they discharge into the amplifier load.

The capacitors therefore act as local energy-storage elements.

Their principal functions are:

- smoothing the full-wave rectified voltage,
- reducing the magnitude of the 100 Hz ripple,
- supplying short-duration current peaks,
- reducing the effective impedance of the main supply,
- supporting the amplifier during transient audio demands.

8.10 Stored Energy

The electrical energy stored in a capacitor is given by

$$E = \frac{1}{2}CV^2.$$

Using the effective capacitance per rail,

$$C = 9.4 \times 10^{-3} \text{ F},$$

and an approximate rail voltage of

$$V = 34 \text{ V},$$

the stored energy in one rail capacitor bank is

$$E_{\text{rail}} = \frac{1}{2} (9.4 \times 10^{-3}) (34)^2.$$

Therefore,

$$E_{\text{rail}} \approx 5.43 \text{ J}.$$

For both capacitor banks together,

$$E_{\text{total}} \approx 2E_{\text{rail}},$$

giving

$$\boxed{E_{\text{total}} \approx 10.86 \text{ J}}.$$

This energy is available to support short-duration amplifier current demands between transformer charging intervals.

8.11 Supply Ripple

The reservoir capacitors discharge slightly between successive charging pulses. This produces a periodic variation in rail voltage known as ripple.

A simplified approximation for the peak-to-peak capacitor discharge is

$$\Delta V \approx \frac{I}{f_{\text{ripple}} C},$$

where

I = average current supplied by the capacitor,

$$f_{\text{ripple}} = 100 \text{ Hz},$$

and

$$C = 9400 \text{ } \mu\text{F}.$$

As an illustrative example, if an average current of 2 A were drawn from one rail,

$$\Delta V \approx \frac{2}{100 \times 9.4 \times 10^{-3}}.$$

Thus,

$$\Delta V \approx 2.13 \text{ V}.$$

This value should be considered only an illustrative first-order estimate. An audio amplifier does not draw constant current from each rail, and the rectifier current occurs in relatively short charging pulses.

The true rail ripple must therefore be determined experimentally under defined amplifier load conditions.

8.12 Unregulated Supply Operation

The main power-amplifier supply is intentionally unregulated.

Providing linear regulation for a high-power stereo amplifier would require regulators capable of carrying several amperes while dissipating significant additional power.

This would increase complexity, heat generation, cost, and physical size.

The Project 3A amplifier is designed to operate directly from a conventional rectified and filtered supply, so regulation of the main power rails is not required.

An important consequence is that the supply voltage changes with load.

At light load, the transformer secondary voltage is comparatively high and the reservoir capacitors charge close to the available peak voltage.

As amplifier output current increases, voltage drops occur in the transformer windings, bridge rectifier, PCB tracks, wiring, and capacitor ESR.

The supply rails therefore decrease under heavy loading.

This effect can be represented qualitatively as

$$I_{\text{load}} \uparrow \Rightarrow V_{\text{rail}} \downarrow.$$

This reduction is commonly referred to as supply sag.

8.13 Stereo Power Distribution

The same main supply provides power to both the left and right power-amplifier channels.

The positive rail is distributed to both amplifier boards:

$$V_+ \rightarrow \begin{cases} \text{Left Power Amplifier} \\ \text{Right Power Amplifier} \end{cases}$$

while the negative rail is distributed in the same manner:

$$V_- \rightarrow \begin{cases} \text{Left Power Amplifier} \\ \text{Right Power Amplifier} \end{cases}.$$

Both channels also share the central 0 V power reference.

Because the supply is shared, simultaneous high-output signals in both channels place greater demand on the transformer and reservoir capacitors than operation of only one channel.

The transformer and rectifier system must therefore be rated for stereo operation rather than for only one amplifier channel.

8.14 Central Ground Reference

The centre point of the split supply forms the main 0 V power reference.

This node is associated with the transformer centre tap and the midpoint between the positive and negative reservoir-capacitor banks.

The three supply potentials are therefore

$$+V_S, \quad 0 \text{ V}, \quad -V_S.$$

Under normal operating conditions, the amplifier loudspeaker output is centred close to the 0 V reference.

The power-supply ground is also important to the overall star-ground architecture of the amplifier. Large rectifier charging currents and loudspeaker currents must not be allowed to produce significant voltage drops in the sensitive low-level signal-ground path.

For this reason, grounding is treated as a separate design problem and is discussed in detail in the dedicated grounding chapters.

8.15 High-Frequency Bypass Capacitors

In addition to the large electrolytic reservoir capacitors, the power-supply schematic contains

$$C_{12} = C_{13} = 100 \text{ nF}.$$

These smaller capacitors provide high-frequency bypassing.

Large electrolytic capacitors are highly effective for low-frequency energy storage but become less ideal as frequency increases because of parasitic inductance and equivalent series resistance.

The 100 nF capacitors have much lower impedance at high frequencies and therefore complement the large reservoir capacitors.

The combination provides effective power-supply decoupling over a wider frequency range.

8.16 Local Amplifier Decoupling

Further decoupling is provided locally on each power-amplifier channel.

The positive rail includes

$$C_7 = 100 \text{ nF}$$

and

$$C_{10} = 100 \text{ } \mu\text{F},$$

while the negative rail includes

$$C_{18} = 100 \text{ nF}$$

and

$$C_{11} = 100 \text{ } \mu\text{F}.$$

The main reservoir capacitors therefore provide bulk energy storage, while the smaller local capacitors reduce the supply impedance immediately adjacent to the active amplifier circuitry.

The arrangement can be represented as

Main Reservoir Capacitors → Power Distribution → Local Amplifier Decoupling → Active Circuitry.

This helps reduce interaction between the two amplifier channels and improves high-frequency stability.

8.17 Power-Supply Current Demand

The current required from the main power supply varies continuously with the audio waveform and loudspeaker impedance.

For a resistive load,

$$I_{\text{load}} = \frac{V_{\text{out}}}{R_L}.$$

For example, at

$$V_{\text{out,RMS}} = 20 \text{ V},$$

an 8 Ω loudspeaker requires

$$I_{\text{RMS}} = \frac{20}{8} = 2.5 \text{ A.}$$

The same output voltage into a 4Ω loudspeaker requires

$$I_{\text{RMS}} = \frac{20}{4} = 5 \text{ A.}$$

Consequently, the lower-impedance load places significantly greater demand on the transformer, bridge rectifier, reservoir capacitors, PCB copper, wiring, and amplifier output transistors.

8.18 Dynamic Audio Loads

Audio programme material is normally dynamic rather than continuous.

Music contains short high-amplitude peaks together with longer periods of lower average output power. During a short high-current transient, the reservoir capacitors can supply a significant fraction of the instantaneous current required by the amplifier.

The transformer subsequently replenishes the charge stored in the capacitors.

This energy flow can be represented conceptually as

Transformer $\rightarrow$ Rectifier $\rightarrow$ Reservoir Capacitors $\rightarrow$ Power Amplifier $\rightarrow$ Loudspeaker.

Adequate reservoir capacitance therefore contributes both to low ripple and to good transient current capability.

8.19 Rectifier Charging Currents

The current flowing through the bridge rectifier is not equal to the smooth average current drawn by the amplifier.

The reservoir capacitors discharge between charging intervals and are recharged only when the instantaneous transformer voltage becomes greater than the capacitor voltage plus the rectifier diode drops.

As a result, capacitor recharge current occurs in relatively short, high-current pulses around the peaks of the AC waveform.

These charging pulses influence the design of:

- the bridge rectifier,
- transformer secondary current rating,
- reservoir capacitor ripple-current rating,

- PCB track width,
- wiring cross-sectional area,
- connector ratings,
- ground-current routing.

They can also generate electrical and magnetic interference if the high-current rectifier loop is positioned too close to the sensitive audio input stage.

8.20 Relationship Between Supply Voltage and Output Power

The maximum unclipped output voltage of the amplifier is directly related to the available DC rail voltage.

The amplifier cannot produce an output voltage exactly equal to the supply rail because voltage is lost across the driver transistors, power transistors, output resistors, and other circuit elements.

For a sinusoidal output signal, loudspeaker power is

$$P_{\text{out}} = \frac{V_{\text{out,RMS}}^2}{R_L}.$$

Therefore, a reduction in the loaded supply-rail voltage reduces the maximum available output voltage and consequently reduces the maximum unclipped output power.

This is why the exact power capability of the implemented amplifier cannot be derived solely from the nominal 24–0–24 V transformer rating.

The final value will be established experimentally by measuring the maximum undistorted output voltage into known dummy loads.

8.21 Power-Supply Noise and Hum

The power supply is a potential source of unwanted noise within the audio system.

Important interference mechanisms include:

- transformer magnetic field coupling,
- 50 Hz mains-frequency coupling,
- 100 Hz rectifier ripple,
- high-current rectifier charging pulses,
- diode switching transients,

- voltage drops in shared ground conductors.

These effects must be prevented from entering the low-level audio signal path.

The transformer and high-current rectifier wiring should therefore be kept away from sensitive input circuitry where practical.

Similarly, high-current ground paths should be routed directly toward the main power ground rather than through signal-ground conductors.

The detailed grounding and noise-control techniques are discussed later in the dedicated grounding and noise-reduction chapters.

8.22 Power-Supply Safety

Although the amplifier supply voltage is considerably lower than the mains voltage, the reservoir capacitors can store enough electrical energy to produce significant current during an accidental short circuit.

The calculated stored energy of approximately

$$10.9 \text{ J}$$

across the two capacitor banks demonstrates that the supply must be treated with care even after the mains power has been removed.

Before servicing the amplifier, the positive and negative rail voltages must be measured and confirmed to have discharged to a safe level.

The mains side of the transformer requires substantially greater safety precautions because it is directly connected to hazardous mains voltage.

Mains fusing, protective earthing, insulation, conductor separation, switch rating, and mechanical safety are addressed in the dedicated mains-wiring and safety chapters.

8.23 Initial Power-Supply Verification

The main power supply should be verified before high-power amplifier testing is performed.

A suitable initial test sequence is:

1. Verify all transformer secondary wiring.
2. Confirm the centre-tap connection.
3. Measure approximately $24 \text{ V}_{\text{RMS}}$ from each outer secondary connection to the centre tap.
4. Measure approximately $48 \text{ V}_{\text{RMS}}$ between the two outer secondary connections.
5. Verify the bridge-rectifier connections.

6. Verify the polarity of all four electrolytic reservoir capacitors.
7. Confirm that no unintended short circuit exists between the positive rail, ground, and negative rail.
8. Apply power using the controlled initial power-up procedure.
9. Measure the positive rail relative to ground.
10. Measure the negative rail relative to ground.
11. Confirm that the two rail magnitudes are approximately symmetrical.
12. Check the transformer, bridge rectifier, and reservoir capacitors for abnormal heating.
13. Repeat the rail-voltage measurements after the amplifier channels are connected.
14. Observe the rail voltage during controlled load testing.

8.24 Measurements to be Recorded

The final constructed amplifier should record the actual supply voltages rather than relying exclusively on theoretical calculations.

Important measurements include

$$V_{+,idle},$$

$$V_{-,idle},$$

$$V_{+,load},$$

$$V_{-,load},$$

and the rail ripple voltage.

The symmetry of the split supply can be expressed as

$$\Delta V_{\text{sym}} = ||V_+| - |V_-||.$$

Ideally, this value should remain small.

Minor differences between the positive and negative rails are normal because of transformer tolerances, diode characteristics, amplifier loading, capacitor tolerances, wiring resistance, and measurement uncertainty.

8.25 Main Power Supply Summary

The main power supply used in this project is an unregulated symmetrical linear supply designed to power both channels of the stereo Class-AB amplifier.

An external toroidal transformer provides

$$\boxed{24-0-24 \text{ VAC}}.$$

The transformer itself is not shown on the power-supply schematic page. The second page of the combined power-amplifier schematic begins at the secondary-side power connections and contains the bridge rectifier, reservoir capacitors, bypass capacitors, and DC rail outputs.

The theoretical peak voltage of each 24 V secondary half is

$$24\sqrt{2} \approx 33.94 \text{ V}.$$

The amplifier is therefore expected to operate from practical DC rails in the approximate region of

$$\boxed{\pm 33 - \pm 34 \text{ VDC}}.$$

The exact voltages will vary with transformer regulation, mains voltage, rectifier losses, load current, and capacitor ripple.

A KBPC3506WP full-wave bridge rectifier is used to convert the transformer secondary voltage into the required positive and negative rectified supply.

Four 4700 μF electrolytic capacitors provide

$$\boxed{9400 \mu\text{F}}$$

of reservoir capacitance per rail.

At approximately 34 V, the two reservoir-capacitor banks together store approximately

$$\boxed{10.9 \text{ J}}$$

of electrical energy.

The supply also includes 100 nF high-frequency bypass capacitors, while each amplifier channel contains additional local rail decoupling.

The same main supply powers both stereo channels, making low-impedance power distribution and controlled grounding particularly important.

The implemented 24-0-24 V transformer produces slightly lower rails than the nominal $\pm 35 \text{ V}$ Project 3A reference condition. This retains operation close to the intended Project 3A supply range while providing a small additional electrical and thermal margin for 4 Ω loudspeaker operation.

The actual positive and negative rail voltages, supply sag, ripple, and rail symmetry will be measured during the final testing of the constructed amplifier.

Chapter 9

Toroidal Transformer Selection

9.1 Introduction

The mains transformer is one of the most important components of the power supply because it determines the basic voltage and current capability available to the power-amplifier stages. It must provide sufficient secondary voltage to obtain the required positive and negative DC supply rails while also having enough apparent-power capability to supply both stereo amplifier channels under demanding operating conditions.

For this project, a toroidal transformer with a centre-tapped secondary of

24–0–24 VAC

is used.

This differs slightly from the 25–0–25 V AC transformer commonly used with Rod Elliott’s Project 3A reference implementation [1]. The slightly lower secondary voltage was selected so that the implemented amplifier operates from DC rails slightly below the nominal ± 35 V Project 3A condition.

The resulting supply is particularly suitable for an amplifier intended to drive both 4 Ω and 8 Ω loudspeakers because it provides useful voltage and thermal margin for the output transistors.

9.2 Transformer Function

The transformer performs two fundamental functions within the amplifier system.

First, it provides galvanic isolation between the hazardous mains supply and the low-voltage amplifier circuitry.

Second, it reduces the mains voltage to the lower AC voltage required by the audio power supply.

The power-conversion sequence can be represented as

Mains AC $\rightarrow$ Toroidal Transformer $\rightarrow$ 24–0–24 VAC $\rightarrow$ Rectifier $\rightarrow$ Reservoir Capacitors $\rightarrow$ DC Supply

The transformer itself does not produce DC voltage. Its output remains an AC waveform and must therefore be rectified and filtered before it can supply the power-amplifier stages.

9.3 Centre-Tapped Secondary

The selected transformer provides a centre-tapped secondary winding described as

$$24\text{--}0\text{--}24 \text{ VAC.}$$

This means that each outer secondary terminal provides approximately

$$24 \text{ V}_{\text{RMS}}$$

relative to the centre tap.

The voltage measured between the two outer secondary terminals is therefore

$$V_{\text{outer}} = 24 + 24 = 48 \text{ V}_{\text{RMS}}.$$

The centre tap forms the central reference of the symmetrical power supply. After rectification and filtering, the three principal power nodes become

$$+V_S, \quad 0 \text{ V}, \quad -V_S.$$

This symmetrical supply arrangement is particularly appropriate for a direct-coupled audio amplifier because the loudspeaker output can operate around 0 V without requiring a large series output coupling capacitor.

9.4 Secondary Voltage Selection

The secondary voltage was selected according to the supply-voltage requirements of the Project 3A-based amplifier.

The relationship between sinusoidal RMS voltage and peak voltage is

$$V_{\text{peak}} = V_{\text{RMS}}\sqrt{2}.$$

For each 24 V_{RMS} secondary half,

$$V_{\text{peak}} = 24\sqrt{2}.$$

Therefore,

$$V_{\text{peak}} \approx 33.94 \text{ V}.$$

Before rectifier losses, each half of the transformer can therefore charge the corresponding reservoir capacitor bank toward approximately 33.94 V relative to the centre tap.

In practice, diode forward voltage, transformer winding resistance, mains variation, capacitor ripple, and amplifier loading modify the actual voltage. The power amplifier is therefore expected to operate from rails in the approximate region of

$$\pm 33 - \pm 34 \text{ VDC}.$$

The exact rail voltages will be measured after construction because the nominal transformer secondary voltage alone is not sufficient to predict the loaded DC supply with complete accuracy.

9.5 Comparison with a 25–0–25 V Transformer

Rod Elliott's Project 3A reference design is commonly associated with a 25–0–25 V AC transformer for operation at approximately $\pm 35 \text{ VDC}$ [1].

For comparison, a 25 V_{RMS} secondary has a theoretical peak voltage of

$$25\sqrt{2} \approx 35.36 \text{ V}.$$

The transformer used in this project produces

$$24\sqrt{2} \approx 33.94 \text{ V}.$$

The theoretical difference is therefore

$$35.36 - 33.94 = 1.42 \text{ V}$$

per rail before rectifier losses.

The percentage reduction in nominal secondary voltage is

$$\frac{25 - 24}{25} \times 100 = 4\%.$$

Thus, the implemented transformer produces only a modest reduction in supply voltage compared with the Project 3A reference configuration.

This is beneficial for the present design because Elliott recommends that approximately $\pm 35 \text{ V}$ should not be exceeded when operating Project 3A with a 4 Ω loudspeaker load [1].

9.6 Effect on Maximum Output Power

The slightly lower secondary voltage also results in a slightly lower maximum available amplifier output voltage.

The output power delivered to a resistive load is

$$P_{\text{out}} = \frac{V_{\text{out,RMS}}^2}{R_L}.$$

Consequently, maximum output power is influenced by the available supply voltage.

The published Project 3A reference figures of approximately

$$60 \text{ W}$$

into 8Ω and

$$100 \text{ W}$$

into 4Ω are associated with nominal $\pm 35 \text{ V}$ supply rails [1].

Because the implemented transformer produces slightly lower rails, the final amplifier may provide slightly less maximum unclipped output power.

This reduction is acceptable because the lower supply voltage also reduces the electrical and thermal stress applied to the power transistors, particularly during 4Ω operation.

The actual maximum output power will therefore be determined experimentally using a sinusoidal input signal and suitable dummy loads.

9.7 Transformer VA Rating

Transformer size is specified primarily by its apparent-power rating, expressed in volt-amperes (VA).

For a transformer,

$$S = V_{\text{RMS}} I_{\text{RMS}},$$

where S is the apparent power.

For a centre-tapped secondary, the complete secondary voltage is

$$48 \text{ V}_{\text{RMS}}.$$

A transformer in the approximate

$$\boxed{300 \text{ VA}}$$

class is appropriate for a stereo Project 3A implementation and is consistent with the transformer capacity recommended for a pair of amplifier channels [1].

If a 300 VA transformer is considered at its complete 48 V secondary voltage, its approximate secondary current rating is

$$I_{\text{secondary}} = \frac{S}{V}.$$

Therefore,

$$I_{\text{secondary}} = \frac{300}{48} = 6.25 \text{ A}.$$

Thus, a 300 VA, 24–0–24 V transformer corresponds approximately to

$$\boxed{24\text{--}0\text{--}24 \text{ V, } 6.25 \text{ A}}$$

for the series secondary winding.

Each 24 V half-secondary therefore represents approximately

$$24 \times 6.25 = 150 \text{ VA},$$

with the two halves together providing the complete 300 VA transformer rating.

9.8 Relationship Between Transformer VA and Audio Power

The transformer VA rating should not be assumed to be numerically identical to the loudspeaker output-power rating.

The amplifier is not perfectly efficient. Some of the electrical power drawn from the supply is dissipated as heat in the output transistors, driver transistors, emitter resistors, rectifier, transformer, and wiring.

In addition, the bridge rectifier and reservoir capacitors draw current from the transformer in short charging pulses rather than as an ideal sinusoidal current.

Therefore,

$$P_{\text{audio}} \neq S_{\text{transformer}}.$$

A transformer must have sufficient VA capability not only for the useful loudspeaker power but also for amplifier losses and the non-ideal current waveform of the rectifier-capacitor supply.

For a stereo amplifier, the transformer must also support both channels simultaneously.

9.9 Stereo Operation

The transformer supplies both left and right power-amplifier channels.

The complete power path is therefore

$$\text{Transformer} \rightarrow \text{Common DC Supply} \rightarrow \begin{cases} \text{Left Amplifier} \\ \text{Right Amplifier} \end{cases}.$$

When both channels reproduce high-amplitude signals simultaneously, the transformer experiences a greater current demand than when only one channel is active.

This is why the transformer is selected according to the requirements of the complete stereo system rather than one individual amplifier channel.

9.10 Why a Toroidal Transformer was Selected

A toroidal transformer was selected instead of a conventional laminated EI-core transformer.

Toroidal transformers offer several characteristics that are useful in an audio amplifier:

- compact physical dimensions for a given VA rating,
- relatively low weight compared with some conventional transformer constructions,
- good electrical efficiency,
- relatively good voltage regulation,
- comparatively low external magnetic leakage,
- convenient central mechanical mounting,
- low mechanical noise when correctly manufactured and operated.

The relatively low stray magnetic field is particularly valuable in audio equipment because the low-level input circuitry is susceptible to magnetic coupling from the mains transformer.

A reduction in stray magnetic field reduces the possibility of inducing a 50 Hz hum voltage into sensitive signal wiring and PCB tracks.

9.11 Transformer Regulation

The nominal 24 V secondary rating does not imply that exactly 24 V will be measured under all operating conditions.

Transformer secondary voltage normally decreases as load current increases because the windings have finite resistance and internal impedance.

At very light or zero load, the secondary voltage can therefore be somewhat higher than its rated value.

Transformer regulation may be expressed as

$$\text{Regulation} = \frac{V_{\text{no-load}} - V_{\text{full-load}}}{V_{\text{full-load}}} \times 100\%.$$

A transformer having good regulation exhibits a smaller change in secondary voltage between no-load and full-load operation.

This behaviour explains why the actual amplifier rails may exceed ± 34 V at idle even though the theoretical calculation using the nominal transformer voltage gives approximately ± 33.94 V before rectifier losses.

9.12 Voltage Sag Under Amplifier Load

When the amplifier delivers significant loudspeaker power, the transformer secondary current increases.

The secondary voltage then decreases because of transformer regulation. Additional voltage is lost across the bridge rectifier, wiring, PCB copper, and reservoir capacitors.

The amplifier rail voltage therefore falls under load:

$$I_{\text{load}} \uparrow \Rightarrow V_{\text{secondary}} \downarrow \Rightarrow V_{\text{DC}} \downarrow.$$

This is one of the principal reasons why maximum amplifier output power must be verified experimentally rather than being calculated only from the transformer nameplate voltage.

9.13 Secondary Current and 4 Ohm Operation

A $4\ \Omega$ loudspeaker creates the most demanding normal operating condition for the transformer because substantially more output current is required than with an $8\ \Omega$ load.

For example, an amplifier output of

$$20\ \text{V}_{\text{RMS}}$$

into $4\ \Omega$ requires

$$I_{\text{load,RMS}} = \frac{20}{4} = 5\ \text{A}.$$

The peak loudspeaker current is

$$I_{\text{load,peak}} = 5\sqrt{2} \approx 7.07\ \text{A}.$$

These currents do not correspond directly to the transformer RMS secondary current because a Class-AB amplifier and capacitor-input rectifier have different current waveforms. They nevertheless demonstrate why adequate transformer VA rating is essential for reliable low-impedance operation.

9.14 Inrush Current

A large toroidal transformer can draw a significant current pulse when first connected to the mains.

There are two principal causes of this effect.

First, the transformer core may initially be energised at an unfavourable point of the mains waveform, producing a temporary magnetising current much larger than the normal steady-state value.

Second, the main reservoir capacitors are initially discharged. Immediately after power is applied, they draw a large charging current through the transformer and bridge rectifier.

The initial current can therefore be considerably greater than the normal operating current.

This must be considered when selecting the mains switch, primary fuse, connectors, and wiring.

A soft-start circuit can also be incorporated if the measured inrush current of the completed amplifier is found to be excessive.

9.15 Magnetic Hum Considerations

Even though a toroidal transformer generally produces a relatively low stray magnetic field, it should still be positioned carefully inside the amplifier chassis.

The transformer should be kept as far as practical from:

- RCA input connectors,
- low-level input wiring,
- preamplifier circuitry,
- high-gain amplifier input stages,
- sensitive signal-ground conductors.

The high-current secondary and rectifier wiring should also be kept separate from low-level audio wiring.

If hum is observed during prototype testing, the physical orientation of the toroidal transformer can be investigated because rotation may alter the magnetic coupling between the transformer and nearby circuitry.

9.16 Mechanical Mounting

Toroidal transformers are normally mounted through their central opening using suitable insulating pads, washers, and mechanical clamping hardware.

The transformer must be mounted securely because a high-VA unit has considerable mass and must remain mechanically stable during transport and normal operation.

Excessive mounting pressure should be avoided because it may damage the transformer insulation or winding structure.

A particularly important rule is that the mounting system must not create a closed conductive loop passing through the centre of the toroidal core.

Such a loop would behave as a shorted secondary turn and could carry a very large current.

The central mounting bolt and surrounding metalwork must therefore be arranged so that they do not form a complete electrically conductive turn through the transformer aperture.

9.17 Transformer Position within the Chassis

The transformer should be mounted close to the mains input and main power-supply section where practical.

This reduces the length of mains-voltage wiring and the high-current secondary conductors.

At the same time, sufficient distance should be maintained between the transformer and the amplifier input circuitry.

The preferred physical organisation is therefore approximately

Mains Input → Transformer → Rectifier and Reservoir Capacitors → Power Amplifiers,

with the low-level audio inputs positioned away from the mains and transformer region.

9.18 Primary-Side Safety

The transformer primary is connected to hazardous mains voltage and must therefore be treated separately from the low-voltage secondary circuitry.

The primary wiring must use appropriately insulated conductors and must be kept physically separated from signal and low-voltage wiring.

The primary side must also be protected by an appropriately selected mains fuse and power switch.

The metal chassis is connected to protective earth independently of the audio signal grounding arrangement.

Protective earth must never depend on the PCB ground system or amplifier electronics for continuity.

Detailed mains wiring, fuse selection, protective-earthing requirements, and electrical clearances are discussed in the dedicated safety and mains-wiring chapters.

9.19 Secondary-Side Wiring

The three transformer secondary connections must be clearly identified before connection to the power-supply board.

They are

$$24\text{ V}, \quad 0\text{ V (centre tap)}, \quad 24\text{ V}.$$

Before connecting the transformer to the rectifier, the secondary voltages should be verified using a multimeter.

The expected measurements are approximately

$$24\text{ V}_{\text{RMS}}$$

between either outer connection and the centre tap, and

$$48\text{ V}_{\text{RMS}}$$

between the two outer connections.

These measurements also provide a useful method of confirming that the correct centre-tap conductor has been identified.

9.20 Initial Transformer Testing

The transformer should be verified independently before full amplifier commissioning.

A suitable sequence is:

1. Inspect the transformer and wiring for mechanical damage.
2. Verify the transformer primary voltage rating.
3. Confirm the identity of the primary and secondary conductors.
4. Verify continuity of the secondary windings with power removed.
5. Confirm the centre-tap conductor.
6. Secure the transformer mechanically inside the chassis.
7. Verify protective-earth bonding of the chassis separately.

8. Apply mains power using the controlled initial power-up procedure.
9. Measure each secondary half relative to the centre tap.
10. Measure the complete outer-to-outer secondary voltage.
11. Check for abnormal mechanical vibration, heating, or audible hum.
12. Disconnect power before connecting or modifying secondary wiring.

9.21 Expected Measurements

The nominal transformer measurements expected before connection to the main rectifier are

$$V_{AC1-CT} \approx 24 \text{ V}_{\text{RMS}},$$

$$V_{AC2-CT} \approx 24 \text{ V}_{\text{RMS}},$$

and

$$V_{AC1-AC2} \approx 48 \text{ V}_{\text{RMS}}.$$

The measured no-load values may be somewhat greater than the nominal ratings because of transformer regulation.

After connection to the rectifier and reservoir capacitors, the positive and negative DC rail voltages should then be measured separately relative to the centre-tap ground reference.

9.22 Selection Summary

A toroidal transformer with a

$$\boxed{24-0-24 \text{ VAC}}$$

centre-tapped secondary was selected for the stereo power amplifier.

Each half-secondary provides a theoretical peak voltage of

$$24\sqrt{2} \approx 33.94 \text{ V},$$

resulting in expected amplifier supply rails in the approximate region of

$$\boxed{\pm 33 - \pm 34 \text{ VDC}}$$

after rectification and filtering.

This is slightly lower than the nominal ± 35 V supply associated with the standard Project 3A configuration. The approximately 4% reduction in transformer secondary voltage provides additional electrical and thermal margin for operation with $4\ \Omega$ loudspeakers while remaining close to the intended Project 3A operating conditions.

For a stereo implementation, a transformer in the approximately 300 VA class provides suitable power capability. At a total secondary voltage of 48 V_{RMS}, a 300 VA rating corresponds to a nominal secondary current of approximately

$$\boxed{6.25\text{ A}}.$$

Toroidal construction was selected because of its compact dimensions, good efficiency, relatively good regulation, and comparatively low external magnetic field, all of which are advantageous in a high-quality audio amplifier.

Transformer location and wiring remain important despite the relatively low magnetic leakage of the toroidal construction. The device should be located away from sensitive audio input circuitry, while the secondary connections to the rectifier should be kept short and appropriately sized for the expected current.

Finally, the primary side of the transformer is a mains-voltage circuit and must be installed using appropriate fusing, insulation, protective earthing, mechanical mounting, and electrical separation. The exact secondary voltages, loaded voltage regulation, mechanical noise, and thermal behaviour will be verified during final amplifier testing.

Chapter 10

Bridge Rectifier and Reservoir Capacitors

10.1 Introduction

Following the 24–0–24 V AC toroidal transformer, the next stage of the main power supply consists of the bridge rectifier and reservoir-capacitor bank. These components convert the transformer secondary voltage into the positive and negative DC rails required by the two Class-AB power-amplifier channels.

The implemented rectifier is the Diotec KBPC3506WP single-phase bridge rectifier. The reservoir network consists of four 4700 μF electrolytic capacitors, with two capacitors connected in parallel for the positive rail and two connected in parallel for the negative rail.

The selected reservoir capacitors are 4700 μF , 63 V snap-in electrolytic devices. This voltage rating provides substantial margin above the expected amplifier rail voltage of approximately ± 33 – ± 34 VDC.

The complete rectifier and capacitor arrangement is shown in Figure 8.1, on the second page of the combined power-amplifier schematic.

10.2 Bridge Rectifier Selection

The bridge rectifier selected for the main power supply is

KBPC3506WP.

The KBPC3506WP is a high-current single-phase bridge rectifier designed for power-conversion applications. Its principal ratings are summarised in Table 10.1.

Table 10.1: Principal specifications of the KBPC3506WP bridge rectifier [6].

Parameter	Value
Rectifier configuration	Single-phase full-wave bridge
Repetitive peak reverse voltage, V_{RRM}	600 V
Maximum average forward current, I_{FAV}	35 A
Peak surge current at 50 Hz	375 A
Maximum junction temperature	150 °C
Typical package family	KBPC-WP
Mounting type	Through-hole / wire lead

These ratings are considerably greater than the normal voltage and current requirements of the amplifier power supply. This provides useful electrical margin and makes the device suitable for the large capacitor-charging current pulses produced by the reservoir network.

10.3 Full-Wave Bridge Operation

The bridge rectifier contains four internal diodes. During each half-cycle of the transformer secondary waveform, two of these diodes conduct while the other two are reverse biased.

During the following half-cycle, the opposite diode pair conducts. Current is therefore directed into the DC supply with the required polarity during both halves of the AC waveform.

The process may be represented conceptually as

$$\text{AC Secondary} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Full-Wave Rectified Voltage.}$$

Since both halves of the AC waveform are used, the rectified waveform has twice the mains frequency.

For the Greek mains frequency of

$$f_{\text{mains}} = 50 \text{ Hz},$$

the principal charging frequency is

$$f_{\text{ripple}} = 2f_{\text{mains}},$$

giving

$$\boxed{f_{\text{ripple}} = 100 \text{ Hz}}.$$

10.4 Bridge Connection to the Centre-Tapped Transformer

The transformer provides a

$$24\text{--}0\text{--}24 \text{ VAC}$$

centre-tapped secondary.

The two outer secondary connections are connected to the AC inputs of the bridge rectifier. The transformer centre tap is connected to the central 0 V reference between the positive and negative capacitor banks.

Consequently, the bridge produces a positive DC rail above the centre tap and a negative DC rail below the centre tap.

The resulting three power-supply nodes are

$$+V_S, \quad 0 \text{ V}, \quad -V_S.$$

This arrangement is appropriate for the direct-coupled Class-AB amplifier because it provides symmetrical positive and negative supplies relative to ground.

10.5 Secondary Peak Voltage

The complete transformer secondary voltage between the two outer connections is

$$V_{\text{secondary}} = 48 \text{ V}_{\text{RMS}}.$$

Its peak value is

$$V_{\text{secondary,peak}} = 48\sqrt{2}.$$

Therefore,

$$V_{\text{secondary,peak}} \approx 67.88 \text{ V}.$$

Each half of the centre-tapped winding produces

$$24\sqrt{2} \approx 33.94 \text{ V}$$

relative to the centre tap.

The reservoir capacitors therefore charge to voltages in the approximate region of the secondary peak voltage, reduced by rectifier and circuit losses.

10.6 Bridge Rectifier Voltage Margin

The KBPC3506WP has a repetitive peak reverse-voltage rating of

$$V_{\text{RRM}} = 600 \text{ V.}$$

The complete transformer secondary has a calculated peak voltage of only

$$67.88 \text{ V.}$$

A simple comparison gives a voltage-rating ratio of

$$M_V = \frac{600}{67.88} \approx 8.84.$$

Thus, the bridge rectifier has a reverse-voltage rating almost nine times the nominal secondary peak voltage.

This large margin is beneficial because the rectifier must also tolerate mains variation and electrical transients rather than only the ideal sinusoidal transformer voltage.

10.7 Bridge Rectifier Current Capability

The KBPC3506WP is rated for a maximum average forward current of

$$I_{\text{FAV}} = 35 \text{ A}$$

under the conditions specified by the manufacturer [6].

The 300 VA, 24–0–24 V transformer considered for the complete stereo amplifier has an approximate full-secondary current rating of

$$I_{\text{secondary}} = \frac{300}{48} = 6.25 \text{ A.}$$

The bridge current rating is therefore substantially greater than the nominal transformer secondary current.

However, the two quantities should not be interpreted as directly equivalent. A capacitor-input power supply draws narrow, high-amplitude current pulses from the transformer rather than a continuous sinusoidal current.

The large current rating of the bridge therefore provides useful margin for these charging pulses.

10.8 Surge Current Capability

When the amplifier is first powered, the reservoir capacitors are initially discharged.

Their initial voltage is approximately

$$V_C(0) \approx 0 \text{ V}.$$

The transformer and rectifier therefore experience a large charging-current pulse while the capacitors are brought toward their normal operating voltage.

The KBPC3506WP has a specified peak forward surge-current capability of approximately

$$\boxed{375 \text{ A}}$$

for the manufacturer's specified 50 Hz surge condition [6].

This high surge rating is useful in a power supply containing large reservoir capacitors. It does not imply that the amplifier normally draws hundreds of amperes; rather, it indicates that the rectifier is designed to withstand short-duration current surges significantly larger than its normal average current.

10.9 Rectifier Forward-Voltage Loss

A silicon rectifier diode is not an ideal conductor. When forward biased, a voltage develops across it.

The KBPC3506WP datasheet specifies a forward voltage of approximately

$$V_F = 1.1 \text{ V}$$

at a forward current of 17.5 A [6].

During bridge conduction, two diodes are active in the complete current path. The instantaneous bridge voltage drop may therefore be represented approximately by

$$V_{\text{bridge}} \approx 2V_F.$$

Using the datasheet test value gives

$$V_{\text{bridge}} \approx 2.2 \text{ V}.$$

The actual voltage drop during amplifier operation depends strongly on the instantaneous charging current and junction temperature and will not remain fixed at this value.

The bridge voltage drop is one reason why the final DC supply voltage is lower than the ideal secondary peak voltage.

10.10 Bridge Rectifier Power Dissipation

The forward-voltage drop across the conducting rectifier diodes results in power dissipation within BR1.

Instantaneous rectifier dissipation can be approximated by

$$P_{\text{BR}}(t) = V_{\text{bridge}}(t)I_{\text{BR}}(t).$$

Because the charging current occurs in pulses, the average thermal dissipation cannot be calculated accurately by simply multiplying the transformer RMS current by a fixed diode voltage.

The rectifier temperature must therefore be checked experimentally under sustained amplifier loading.

BR1 should also be positioned so that its heat does not unnecessarily warm the nearby electrolytic capacitors, since capacitor lifetime decreases as operating temperature increases.

10.11 Reservoir Capacitor Selection

The main reservoir network uses four aluminium electrolytic capacitors:

$$C_{14} = C_{15} = C_{16} = C_{17} = 4700 \mu\text{F}.$$

The selected capacitor type has the following principal characteristics:

$$C = 4700 \mu\text{F},$$

$$V_{\text{rated}} = 63 \text{ V},$$

with a snap-in through-hole package having approximately

$$25 \text{ mm}$$

body diameter and

$$10 \text{ mm}$$

lead spacing.

The selected part is

$$\boxed{380\text{LQ}472\text{M}063\text{J}032}$$

from the 380LQ aluminium-electrolytic capacitor family [7].

10.12 Capacitor Voltage Rating

The expected normal supply-rail magnitude is approximately

$$34 \text{ V.}$$

The reservoir capacitors are rated at

$$63 \text{ V.}$$

The voltage-rating ratio is therefore approximately

$$M_C = \frac{63}{34} \approx 1.85.$$

The capacitors therefore operate at only about

$$\frac{34}{63} \times 100 \approx 54\%$$

of their rated DC voltage under the nominal condition.

This margin is desirable because the rail voltage may rise above its nominal value when the amplifier is lightly loaded or when the mains supply voltage is high.

Using capacitors with a voltage rating only slightly above the nominal rail voltage would provide less tolerance for these variations.

10.13 Capacitor Arrangement

Two 4700 μF capacitors are connected in parallel on each supply rail.

For the positive rail,

$$C_{14} \parallel C_{16},$$

while the negative rail uses

$$C_{15} \parallel C_{17}.$$

For capacitors connected in parallel,

$$C_{\text{total}} = C_1 + C_2.$$

Therefore, the total reservoir capacitance per rail is

$$C_{\text{rail}} = 4700 + 4700 = 9400 \mu\text{F}.$$

Thus,

$$C_{\text{rail}} = 9400 \mu\text{F}.$$

Expressed in farads,

$$C_{\text{rail}} = 9.4 \times 10^{-3} \text{ F}.$$

The physical total capacitance installed across both rail banks is

$$4 \times 4700 = 18\,800 \mu\text{F},$$

but each supply rail independently has an effective reservoir capacitance of

$$9400 \mu\text{F}.$$

10.14 Electrolytic Capacitor Polarity

Correct capacitor polarity is essential because the reservoir devices are polarised aluminium electrolytic capacitors.

For the positive capacitor bank, the positive capacitor terminals are connected to the positive DC rail and the negative terminals are connected to the central ground reference.

Thus,

$$C_{14}, C_{16} : \quad (+) \rightarrow +V_S, \quad (-) \rightarrow 0 \text{ V}.$$

For the negative capacitor bank, the orientation is reversed relative to the rail labels. The positive capacitor terminals are connected to the central ground reference and the negative capacitor terminals are connected to the negative rail.

Therefore,

$$C_{15}, C_{17} : \quad (+) \rightarrow 0 \text{ V}, \quad (-) \rightarrow -V_S.$$

Reversing the polarity of an electrolytic reservoir capacitor can cause excessive leakage current, heating, gas generation, venting, and potentially catastrophic failure.

10.15 Energy Storage

The reservoir capacitors store energy that can be supplied rapidly to the power amplifier during transient current demands.

The energy stored in a capacitor is

$$E = \frac{1}{2}CV^2.$$

For a single 4700 μF capacitor operating at approximately 34 V,

$$E = \frac{1}{2}(4.7 \times 10^{-3})(34)^2.$$

Therefore,

$$E \approx 2.72 \text{ J}.$$

Since two capacitors are connected in parallel on each rail,

$$E_{\text{rail}} \approx 2 \times 2.72 = 5.44 \text{ J}.$$

The positive and negative capacitor banks together therefore store approximately

$$E_{\text{total}} \approx 10.9 \text{ J}.$$

Hence,

$$E_{\text{total}} \approx 10.9 \text{ J}$$

at an approximate rail magnitude of 34 V.

10.16 Capacitor Ripple Voltage

Between successive rectifier charging peaks, the reservoir capacitors supply current to the amplifier and their voltage decreases slightly.

A simplified first-order approximation for the resulting peak-to-peak ripple is

$$\Delta V \approx \frac{I_{\text{load}}}{f_{\text{ripple}}C}.$$

For the implemented supply,

$$f_{\text{ripple}} = 100 \text{ Hz}$$

and

$$C = 9400 \mu\text{F} = 9.4 \times 10^{-3} \text{ F}.$$

For an illustrative average rail current of 1 A,

$$\Delta V \approx \frac{1}{100(9.4 \times 10^{-3})} \approx 1.06 \text{ V}.$$

For 2 A,

$$\Delta V \approx 2.13 \text{ V},$$

and for 3 A,

$$\Delta V \approx 3.19 \text{ V}.$$

Table 10.2: Approximate theoretical rail ripple for different average rail currents using 9400 μF per rail.

Average Rail Current	Approximate Ripple
1 A	1.06 V _{pp}
2 A	2.13 V _{pp}
3 A	3.19 V _{pp}

These values are simplified theoretical estimates rather than predicted measurements of the completed amplifier.

The actual current drawn by a Class-AB audio amplifier is not constant, the two channels share the supply, and the capacitors are recharged by short, high-current rectifier pulses. Actual ripple must therefore be measured under defined load and signal conditions.

10.17 Capacitor Ripple-Current Capability

Reservoir capacitors must withstand not only DC voltage but also substantial AC ripple current.

The selected 4700 μF , 63 V capacitor has a specified low-frequency ripple-current capability of approximately

$$3.59 \text{ A}$$

at 120 Hz and the manufacturer's specified temperature condition [7].

Since two identical capacitors are used in parallel on each rail, the ripple current is shared between the two devices.

Under ideal equal-current-sharing conditions, the combined nominal capability would be approximately

$$I_{\text{ripple,total}} \approx 2 \times 3.59 = 7.18 \text{ A}.$$

In practice, exact current sharing depends on capacitor ESR, temperature, component tolerance, and PCB conductor resistance.

The parallel configuration therefore provides benefits not only in total capacitance but also in ripple-current capability.

10.18 Equivalent Series Resistance

A real capacitor has internal resistance known as equivalent series resistance or ESR.

The selected capacitor has an ESR of approximately

$$88 \text{ m}\Omega$$

at the specified 120 Hz test condition [7].

For two approximately identical capacitors connected in parallel, the ideal effective ESR becomes

$$R_{\text{ESR,parallel}} = \frac{R_{\text{ESR}}}{2}.$$

Thus,

$$R_{\text{ESR,parallel}} \approx \frac{88}{2} = 44 \text{ m}\Omega.$$

The actual effective value depends on component tolerance, temperature, age, frequency, and PCB resistance.

Lower ESR is advantageous because it reduces the instantaneous voltage drop during high-current transients and reduces internal capacitor heating.

10.19 Capacitor Heating

Ripple current flowing through capacitor ESR produces internal heat.

The approximate resistive power loss is

$$P_{\text{ESR}} = I_{\text{ripple,RMS}}^2 R_{\text{ESR}}.$$

Excessive ripple current therefore increases the internal temperature of the capacitor.

Higher operating temperature accelerates ageing of aluminium electrolytic capacitors and reduces their useful service life.

For this reason, adequate airflow should be maintained around the reservoir capacitors and they should not be positioned unnecessarily close to high-heat components such as the bridge rectifier, output transistors, or power resistors.

10.20 Parallel Capacitor Advantages

Using two 4700 μF capacitors per rail instead of a single 9400 μF device provides several practical advantages.

The total capacitance remains

$$9400 \mu\text{F},$$

while the two devices can share ripple current.

Parallel connection also reduces the effective ESR and distributes thermal dissipation between two components.

It can additionally provide a more convenient physical arrangement on the PCB than one very large capacitor.

The advantages can therefore be summarised as

$$\text{Parallel capacitors} \Rightarrow \begin{cases} \text{greater total capacitance,} \\ \text{greater ripple-current capability,} \\ \text{lower effective ESR,} \\ \text{distributed heating.} \end{cases}$$

10.21 Capacitor Charging Current

The reservoir capacitors are not charged continuously throughout the mains cycle.

After a charging peak, the capacitors supply current to the amplifier and their voltage gradually decreases.

The bridge diodes remain non-conductive until the instantaneous transformer secondary voltage once again becomes greater than the capacitor voltage plus the required diode forward voltage.

At that point, current flows into the capacitors over a comparatively short part of the AC cycle.

Consequently, the bridge and transformer experience charging-current pulses whose peak amplitude can be considerably greater than the average DC current drawn by the amplifier.

These pulses are an important consideration for rectifier rating, transformer rating, capacitor ripple-current rating, PCB track width, connector selection, and grounding.

10.22 Initial Capacitor Charging

When the amplifier is switched on, all four reservoir capacitors begin in a discharged or partially discharged state.

From the capacitor relationship

$$i_C = C \frac{dV_C}{dt},$$

a rapid increase in capacitor voltage requires a large charging current.

The combination of the large 9400 μF capacitance on each rail and the low impedance of the toroidal transformer can therefore produce a significant turn-on current pulse.

The bridge rectifier must tolerate this condition, while the mains fuse and power switch must be selected so that normal turn-on current does not cause unwanted operation.

The large turn-on current is one reason why inrush-current behaviour is an important consideration in a high-power audio amplifier.

10.23 High-Frequency Bypass Capacitors

The power-supply schematic also contains C12 and C13, each with a value of

$$C_{12} = C_{13} = 100 \text{ nF}.$$

These capacitors complement the large electrolytic reservoir capacitors at higher frequencies.

A large aluminium electrolytic capacitor is highly effective at storing low-frequency energy, but parasitic inductance and ESR cause its impedance to increase at sufficiently high frequency.

The smaller 100 nF capacitors provide a lower-impedance path for high-frequency components and transients.

The combination of large electrolytic and small bypass capacitance therefore provides effective supply filtering over a wider frequency range.

10.24 PCB Placement of the Reservoir Capacitors

The four reservoir capacitors are physically large snap-in components and are positioned within the high-current section of the PCB.

The connections between BR1 and C14–C17 should be kept short and wide because they carry the large pulsed charging currents produced by the rectifier.

Similarly, the connections from the capacitor banks to the amplifier supply rails should have low resistance.

The most important high-current loops include

$$\text{Transformer} \rightarrow \text{Bridge} \rightarrow \text{Reservoir Capacitors} \rightarrow \text{Transformer}$$

during capacitor charging, and

Reservoir Capacitors → Power Amplifier → Ground Return → Reservoir Capacitors

during amplifier operation.

Reducing the area and impedance of these loops helps reduce unwanted voltage drop, magnetic radiation, and interaction with sensitive low-level audio signals.

10.25 Ground Connection Between the Capacitor Banks

The electrical midpoint between the positive and negative reservoir-capacitor banks forms the main 0 V supply reference.

This point is also associated with the centre tap of the transformer secondary.

The capacitor midpoint is important because very large power-stage and rectifier currents return through this region.

The signal-ground system must therefore be arranged so that these currents do not create unwanted voltage drops in the low-level audio input reference.

The use of a controlled central ground or star-ground arrangement is discussed in detail in the dedicated grounding chapters.

10.26 Discharge After Power-Off

The reservoir capacitors retain stored energy after the amplifier has been disconnected from the mains supply.

Immediately after switch-off, the voltage may still be close to the normal rail voltage.

The stored energy is given by

$$E = \frac{1}{2}CV^2,$$

and the two capacitor banks can contain approximately 10.9 J at the nominal operating voltage.

Consequently, the absence of mains power does not immediately mean that the amplifier circuitry is electrically discharged.

Before servicing the circuit, the positive and negative rail voltages must be measured with a multimeter and confirmed to have fallen to a safe level.

10.27 Initial Verification

The bridge rectifier and capacitor bank should be carefully verified before the amplifier is connected to a loudspeaker.

The following checks should be performed:

1. Verify the pinout and orientation of BR1.
2. Confirm the two transformer outer secondary connections are connected to the bridge AC inputs.
3. Confirm the transformer centre tap is connected to the central supply reference.
4. Verify the polarity of C14 and C16 on the positive rail.
5. Verify the polarity of C15 and C17 on the negative rail.
6. Confirm that no short circuit exists between $+V_S$, ground, and $-V_S$.
7. Confirm that the reservoir capacitors have adequate voltage ratings.
8. Apply power using the controlled initial power-up procedure.
9. Measure the positive supply voltage relative to ground.
10. Measure the negative supply voltage relative to ground.
11. Verify that the positive and negative rail magnitudes are approximately symmetrical.
12. Check BR1 and the reservoir capacitors for abnormal heating.
13. Measure the rail ripple under load using an oscilloscope.

10.28 Expected Oscilloscope Waveforms

When observed directly at the transformer secondary, the expected waveform is approximately sinusoidal.

After rectification but before effective smoothing, the waveform consists of full-wave rectified sinusoidal peaks.

Across each reservoir capacitor bank, the voltage should appear as a mostly constant DC level with a smaller sawtooth-like ripple component.

The sequence can therefore be represented as

Sinusoidal AC $\rightarrow$ Full-Wave Rectified Waveform $\rightarrow$ DC with Ripple.

At higher amplifier output power, the capacitors discharge further between charging peaks, causing the ripple amplitude to increase.

10.29 Expected Measurements

The following parameters should be recorded during final testing:

$$V_{+,idle},$$

$$V_{-,idle},$$

$$V_{+,loaded},$$

$$V_{-,loaded},$$

$$V_{ripple,+},$$

and

$$V_{ripple,-}.$$

The bridge temperature and capacitor temperature should also be monitored during sustained output-power testing.

These measurements will allow the theoretical power-supply calculations to be compared with the behaviour of the physical implementation.

10.30 Design Summary

The main amplifier power supply uses a Diotec KBPC3506WP full-wave bridge rectifier together with four 4700 μF aluminium electrolytic reservoir capacitors.

The bridge rectifier is rated at

$$\boxed{600 \text{ V}}$$

repetitive peak reverse voltage and

$$\boxed{35 \text{ A}}$$

maximum average forward current, with a specified 50 Hz surge-current capability of approximately

$$\boxed{375 \text{ A}}.$$

These ratings provide substantial margin relative to the 24–0–24 V transformer and the approximately 6.25 A nominal secondary current associated with the 300 VA stereo transformer.

The complete 48 V_{RMS} transformer secondary has a theoretical peak voltage of only

$$48\sqrt{2} \approx 67.9 \text{ V},$$

which is considerably below the bridge rectifier's 600 V reverse voltage rating.

The four reservoir capacitors are arranged as two parallel capacitors on each supply rail, giving

$$\boxed{9400 \text{ } \mu\text{F per rail}}.$$

The selected capacitors are rated at

$$\boxed{63 \text{ V}},$$

providing substantial voltage margin above the approximately 33–34 V normal rail magnitude.

At approximately 34 V, the two capacitor banks together store approximately

$$\boxed{10.9 \text{ J}}$$

of electrical energy.

The parallel capacitor arrangement also increases ripple-current capability and reduces effective ESR compared with a single capacitor of the same individual type.

Together, the KBPC3506WP bridge rectifier and the four large reservoir capacitors provide the high-current rectification, filtering, and transient energy storage required by both channels of the Class-AB stereo amplifier.

Correct capacitor polarity, short high-current PCB paths, adequate thermal management of the rectifier, controlled grounding, and verification of rail voltage and ripple during testing are essential for reliable and low-noise operation.

Chapter 11

Preamplifier

11.1 Introduction

The preamplifier forms the low-level analogue signal-processing section of the complete Hi-Fi amplifier system. Its purpose is to accept signals from different audio sources, provide the required signal conditioning and voltage gain, allow the user to control the playback level and channel balance, and deliver a suitable line-level signal to the main power-amplifier stage.

Unlike the power amplifier, the preamplifier is not required to deliver large currents to a loudspeaker. Instead, its design priorities are low noise, low distortion, suitable input impedance, accurate gain, good channel matching, and immunity to interference from the high-current sections of the amplifier.

The preamplifier used in this project was originally developed as a separate analogue audio project and has subsequently been incorporated into the present Hi-Fi amplifier system. The design includes a dedicated phono stage for a turntable, additional line-level inputs, selectable signal routing, gain stages, balance control, volume control, and a regulated symmetrical power supply.

For integration with the present amplifier, the preamplifier is treated as a separate functional subsystem from the main power-amplifier boards and the STM32-based speaker-protection controller.

11.2 Role within the Complete Audio System

The main power amplifier requires a line-level input signal before it can deliver significant output power to the loudspeaker. The preamplifier therefore operates between the external audio source and the power amplifier.

The overall signal path can be represented as

Audio Source → Preamplifier → Power Amplifier → Speaker Protection → Loudspeaker.

For a conventional line-level source, the preamplifier mainly performs input selection, level

control, balance adjustment, and any required voltage gain.

For a turntable input, additional processing is necessary because the output from a moving-magnet phono cartridge is much smaller than a conventional line-level signal and has been recorded using RIAA frequency equalisation.

The preamplifier therefore contains a dedicated phono stage before the normal line-level signal-processing chain.

11.3 Preamplifier Architecture

The preamplifier can be divided into the following principal functional sections:

- phono input stage,
- line-level inputs,
- input selection,
- RIAA equalisation,
- first and second voltage-gain stages,
- gain selection,
- balance control,
- volume control,
- line-level output stage,
- regulated symmetrical power supply,
- local supply decoupling,
- star-ground network.

The functional signal flow is approximately

$$\left\{ \begin{array}{l} \text{Phono Input} \rightarrow \text{RIAA Phono Stage,} \\ \text{Line Inputs} \end{array} \right. \rightarrow \text{Input Selection} \rightarrow \text{Gain and Level Control} \rightarrow \text{Preamplifier}$$

The preamplifier output is then connected to the input of the main Class-AB power amplifier.

11.4 Audio Inputs

The preamplifier provides two general categories of analogue input: phono-level and line-level.

The implemented design contains one dedicated phono input intended for a moving-magnet turntable cartridge and three additional line-level inputs.

The line inputs are intended for sources that already provide signals at approximately line level, such as:

- CD players,
- digital-to-analogue converters,
- media streamers,
- external audio interfaces,
- tuners,
- other analogue source equipment.

These signals do not require RIAA equalisation and can therefore be routed directly toward the normal gain and level-control stages after input selection.

11.5 Need for a Dedicated Phono Stage

A turntable fitted with a moving-magnet cartridge cannot normally be connected directly to a standard line-level amplifier input.

The cartridge generates a comparatively small electrical signal, typically several millivolts, whereas conventional line-level sources operate at considerably higher signal amplitudes.

The phono signal therefore requires substantially more voltage gain before it can be processed in the same manner as the other inputs.

In addition, the frequency response recorded on a vinyl record is deliberately modified according to the RIAA recording characteristic. The phono preamplifier must apply the inverse characteristic during playback.

The phono stage therefore performs two essential functions:

Voltage Amplification

and

RIAA Equalisation.

11.6 RIAA Equalisation

Vinyl records are not recorded with a flat relationship between frequency and groove amplitude. Low frequencies are reduced and high frequencies are increased during recording.

This process reduces excessive groove displacement at low frequencies and improves the signal-to-noise performance of the recording at high frequencies.

During playback, the phono preamplifier applies the inverse response so that the original approximate frequency balance is restored.

The standard RIAA playback characteristic is based on the three principal time constants

$$\tau_1 = 3180 \mu\text{s},$$

$$\tau_2 = 318 \mu\text{s},$$

and

$$\tau_3 = 75 \mu\text{s}.$$

The corresponding break frequencies are obtained from

$$f = \frac{1}{2\pi\tau}.$$

These correspond approximately to

$$50.05 \text{ Hz},$$

$$500.5 \text{ Hz},$$

and

$$2122 \text{ Hz}.$$

The resistor-capacitor feedback network within the phono stage shapes the closed-loop gain as a function of frequency so that the required playback equalisation is obtained.

11.7 Phono Amplifier Circuit

The stereo phono stage is implemented using low-noise NE5532N operational amplifiers.

Separate left- and right-channel signal paths are used so that the two stereo channels remain electrically independent.

The NE5532 is particularly appropriate for low-level audio processing because it provides low input-noise voltage, low distortion, useful output-current capability, and wide small-signal bandwidth when operated within its intended conditions.

These characteristics are especially important in the phono stage because any noise generated at this point is subsequently amplified together with the wanted signal.

The phono stage uses frequency-dependent negative feedback around the operational amplifiers. The feedback network provides both the required voltage gain and the RIAA frequency response.

The signal path for each channel can therefore be represented as

Moving-Magnet Cartridge → Low-Noise Input Stage → RIAA Equalisation → Line-Level Phono Output

11.8 Stereo Phono Operation

The phono stage contains equivalent circuitry for the left and right audio channels.

Maintaining similar component values in the two channels is important because differences in resistor or capacitor values can result in unequal gain or frequency response.

Component tolerance is therefore particularly relevant within the RIAA network.

The phono signal levels are very small, so the phono wiring and PCB traces must also be kept away from the mains transformer, rectifier wiring, digital control electronics, relay coils, and other potential sources of electromagnetic interference.

11.9 Line-Level Signal Path

The three line-level inputs bypass the specialised RIAA processing required by the turntable input.

After input selection, the chosen source is passed to the gain and user-control stages.

The line-level signal path can be represented as

Line Input → Input Selector → Gain Stage → Balance and Volume Control → Output.

The phono stage ultimately enters the same general signal-processing chain after its voltage and frequency response have been corrected.

11.10 Gain Stages

The preamplifier incorporates dedicated voltage-gain stages following the input and phono sections.

The purpose of these stages is to ensure that the selected audio source is presented to the main power amplifier at an appropriate voltage level.

The existing preamplifier design contains first and second gain-stage sections. This arrangement allows the required total voltage amplification to be distributed between multiple stages instead of relying on one very high-gain amplifier.

Distributing gain can be advantageous because it allows signal level, headroom, noise, and user controls to be managed at suitable positions within the circuit.

11.11 Operational-Amplifier Gain

For a non-inverting operational-amplifier configuration, the closed-loop gain is determined by the feedback and ground resistors according to

$$A_v = 1 + \frac{R_f}{R_g},$$

where

R_f = feedback resistance

and

R_g = resistance between the inverting input and reference node.

The gain in decibels is then

$$A_{v,\text{dB}} = 20 \log_{10}(A_v).$$

These relationships allow the gain of the preamplifier to be selected according to the signal level required by the following power-amplifier stage.

11.12 Gain Selection

The existing preamplifier design incorporates selectable gain so that the signal amplification can be adjusted according to the connected source and required system level.

This is useful because different analogue sources can produce substantially different output amplitudes.

The purpose of gain selection is not simply to make the signal louder. Instead, the objective is to maintain an appropriate signal level throughout the preamplifier while retaining sufficient headroom to prevent clipping.

Too little gain would prevent the main power amplifier from reaching its required output level, whereas excessive gain would reduce the available headroom and make the system more sensitive to noise.

11.13 Volume Control

The volume control adjusts the amplitude of the signal presented to the power amplifier.

The power amplifier itself operates with a substantially fixed closed-loop gain. Therefore, normal listening-level adjustment is performed in the preamplifier rather than by changing the gain of the main power stage.

The volume control acts as an adjustable attenuator.

If the potentiometer position produces an attenuation factor k , where

$$0 \leq k \leq 1,$$

then the resulting output signal is approximately

$$V_{\text{out}} = kV_{\text{signal}}.$$

At the minimum setting,

$$k \rightarrow 0,$$

while at the maximum setting,

$$k \rightarrow 1.$$

The volume-control potentiometer should maintain good tracking between the left and right channels so that stereo balance does not change significantly with listening level.

11.14 Balance Control

A balance control is incorporated to allow the relative levels of the left and right channels to be adjusted.

With the balance control centred, the two channels should have approximately equal attenuation.

Moving the control toward one side progressively attenuates the opposite channel, allowing compensation for differences in loudspeaker position, listening position, source material, or room acoustics.

Under normal operation, the balance control is intended to remain near its central position.

11.15 Preamplifier Output

The preamplifier output is a line-level signal intended to drive the input of the main power amplifier.

The power amplifier used in this project has a voltage gain of approximately

$$A_v \approx 23$$

or

$$27.2 \text{ dB}.$$

Consequently, approximately one volt RMS at the power-amplifier input is sufficient to approach high output-power levels, depending on the loudspeaker load and the actual supply-rail voltage.

The preamplifier therefore does not need to generate a high-power output. It must instead provide a clean, low-distortion line-level voltage with adequate headroom.

11.16 Preamplifier Power Supply

The preamplifier uses a separate regulated symmetrical power supply rather than being connected directly to the approximately ± 33 – ± 34 V rails of the power amplifier.

The preamplifier supply provides

$$\boxed{+15 \text{ V}}$$

and

$$\boxed{-15 \text{ V}}$$

relative to the analogue ground reference.

Thus, the operational amplifiers operate from a total supply difference of

$$30 \text{ V}.$$

The symmetrical supply is suitable for audio operational amplifiers because the audio signal can remain centred around approximately 0 V without requiring a single-supply midpoint reference.

11.17 Voltage Regulation

The preamplifier power supply incorporates linear voltage regulation to provide stable low-noise rails.

A 7815 regulator is used for the positive supply:

$$V_{+, \text{preamp}} = +15 \text{ V},$$

while a 7915 regulator is used for the negative supply:

$$V_{-, \text{preamp}} = -15 \text{ V}.$$

Using regulated supplies is particularly beneficial in the preamplifier because the signals handled by the phono and line stages are much smaller than those present in the power amplifier.

Power-supply ripple or noise coupled into these low-level stages could otherwise become audible after subsequent amplification.

11.18 Preamplifier Supply Filtering

The preamplifier supply includes rectification, smoothing capacitors, regulators, and additional bypass capacitors.

The filtering chain can be represented as

AC Supply → Rectification → Bulk Filtering → Voltage Regulation → Local Decoupling.

This multistage approach reduces low-frequency ripple while also providing a low-impedance supply for the operational amplifiers over a wider frequency range.

11.19 Operational-Amplifier Decoupling

Local decoupling capacitors are positioned close to the operational-amplifier supply connections.

These components provide a short local current path for high-frequency transients and help prevent disturbances on the supply rails from coupling between amplifier stages.

The use of both bulk and small-value bypass capacitors is important because large electrolytic capacitors alone become less effective at high frequency due to their parasitic characteristics.

Proper local decoupling therefore contributes to:

- low noise,
- stable operational-amplifier operation,
- reduced inter-stage coupling,
- reduced susceptibility to supply transients,
- prevention of unwanted high-frequency oscillation.

11.20 Separation from the Main Power Supply

The preamplifier should not be powered directly from the unregulated high-current rails used by the main power amplifier.

The power-amplifier supply experiences large current variations, rectifier charging pulses, and rail-voltage movement as loudspeaker output power changes.

The preamplifier, particularly the phono input, processes signals several orders of magnitude smaller than the main power-stage signals.

Using its own regulated low-voltage supply therefore reduces the possibility that power-stage activity will modulate or contaminate the preamplifier signal.

11.21 Star-Ground Arrangement

The preamplifier uses a star-ground architecture.

Instead of allowing different signal and supply currents to flow through a long shared ground path, the individual sections are returned toward a common reference point.

Conceptually,

$$\left\{ \begin{array}{l} \text{Phono Ground} \\ \text{Line Input Ground} \\ \text{Gain Stage Ground} \\ \text{Output Ground} \\ \text{Power-Supply Ground} \end{array} \right. \rightarrow \text{Common Star Point.}$$

The objective is to reduce voltage differences caused by current flowing through shared ground impedances.

This is particularly important for the phono section because even a small ground-related interference voltage can become significant when amplified by the high-gain phono circuitry.

11.22 Grounding within the Complete Amplifier

The preamplifier analogue ground must also be integrated correctly with the grounding architecture of the complete stereo amplifier.

The low-level preamplifier ground should not be used as a return path for loudspeaker current, reservoir-capacitor charging current, relay current, fan current, or digital controller current.

These currents can create voltage drops that would appear directly in series with the wanted audio signal.

The preamplifier ground is therefore connected to the overall system ground at a controlled location rather than through arbitrary chassis or PCB paths.

11.23 Noise Considerations

Noise performance is particularly important in the preamplifier because signals at this point have not yet been amplified to the levels found in the power stage.

The most sensitive section is the phono input.

Important potential noise sources include:

- mains-transformer magnetic fields,
- 50 Hz electric-field coupling,
- 100 Hz rectifier ripple,
- power-amplifier ground currents,
- digital switching from the STM32 controller,
- relay-coil switching,
- cooling fans,
- long unshielded input wiring,
- poor grounding,
- operational-amplifier supply noise.

The physical installation must therefore keep the preamplifier, and especially the phono input, away from the mains transformer and high-current power wiring.

11.24 Signal Wiring

Low-level audio wiring between the RCA input connectors and preamplifier should be kept as short as practical.

Shielded cable may be used where the signal must travel a significant distance inside the amplifier enclosure.

The shield should be connected according to the chosen grounding strategy so that it provides electrostatic screening without creating unwanted ground loops.

Phono input wiring requires the greatest care because of the very small signal amplitude produced by the cartridge.

11.25 PCB Organisation

The preamplifier PCB is organised according to the functional sections of the circuit.

The existing board design separates the power-supply section, phono stage, and gain stages so that electrically noisy supply currents are kept away from the most sensitive signal-processing section.

The phono circuitry is positioned as a distinct functional region, while the first and second gain stages occupy their own section of the board.

This functional placement makes signal flow easier to follow while also supporting controlled grounding and shorter interconnections.

11.26 Integration into the KiCad Project

For the present complete amplifier project, the preamplifier is maintained as a separate schematic section within the KiCad project.

This separation is useful because the preamplifier, main power amplifier, power supply, and protection controller have significantly different electrical functions and layout requirements.

Keeping the preamplifier as an independent design section also simplifies maintenance, modification, PCB development, and documentation.

11.27 Signal-Level Compatibility with the Power Amplifier

The power amplifier has a calculated closed-loop voltage gain of approximately

$$A_v = 23.$$

As an example, a 1 V_{RMS} signal applied to its input would ideally attempt to produce

$$V_{\text{out,RMS}} = 23 \times 1 = 23 \text{ V}_{\text{RMS}},$$

provided that the supply voltage and output stage allow the required voltage swing.

This corresponds to a theoretical output power of

$$P = \frac{V_{\text{RMS}}^2}{R_L}.$$

The example demonstrates that only a conventional line-level voltage is required from the preamplifier to drive the main amplifier toward full output.

The main purpose of the preamplifier is therefore accurate signal preparation and control rather than high output voltage or current.

11.28 Headroom

Adequate signal headroom must be maintained throughout the preamplifier.

For an operational amplifier powered from

$$\pm 15 \text{ V},$$

the output cannot generally reach exactly

$$+15 \text{ V}$$

or

$$-15 \text{ V}.$$

Some voltage margin is required between the signal peak and the supply rails.

The gain distribution should therefore ensure that normal input sources do not cause any internal preamplifier stage to clip before the main power amplifier reaches its intended operating level.

11.29 Clipping Considerations

If the preamplifier is driven beyond its available voltage range, the output waveform clips.

Once clipping occurs, increasing the input voltage no longer produces a proportional increase in output voltage.

This introduces harmonic distortion before the signal reaches the main power amplifier.

Correct gain selection and volume-control positioning therefore ensure that

$$|V_{\text{signal,peak}}| < |V_{\text{available,peak}}|.$$

11.30 Channel Separation

The preamplifier uses independent left- and right-channel signal paths.

Electrical coupling between these paths would reduce stereo separation.

Channel separation is improved by maintaining independent signal routing, avoiding unnecessary shared impedances, providing proper supply decoupling, and controlling the common grounding network.

The layout should also avoid long parallel low-level traces where possible, particularly in high-gain parts of the phono circuit.

11.31 Relationship to the Protection Controller

The STM32 protection controller is not placed in the preamplifier audio signal path.

This is an intentional architectural decision.

The microcontroller operates digitally and controls protection functions such as loud-speaker relay operation, DC-fault monitoring, temperature monitoring, startup delay, and status indicators.

Keeping the digital control subsystem electrically separate from the low-level audio signal path reduces the possibility of digital switching noise being coupled into the preamplifier.

11.32 Testing of the Preamplifier

The preamplifier should be verified independently before being connected to the main power amplifier.

The initial testing procedure should include:

1. Verify the positive regulated supply voltage.
2. Verify the negative regulated supply voltage.
3. Confirm approximately +15 V and −15 V rails.
4. Check all operational amplifiers for abnormal heating.
5. Verify the absence of significant DC voltage at the preamplifier outputs.
6. Apply a low-level sinusoidal signal to each line input.
7. Confirm correct operation of the input selection system.
8. Verify the volume-control range.
9. Verify correct balance-control operation.
10. Verify any selectable gain functions.
11. Apply an appropriate low-level signal to the phono input.
12. Verify amplification of both left and right phono channels.
13. Measure the phono frequency response to assess the RIAA equalisation.
14. Check for 50 Hz and 100 Hz hum components.
15. Confirm adequate output level without visible clipping.
16. Only after successful independent testing, connect the preamplifier output to the power-amplifier input.

11.33 Frequency-Response Testing

The line-level signal path should ideally maintain an approximately flat response throughout the audible frequency range.

A sinusoidal input signal can be applied while frequency is swept through the required range.

The voltage gain at each frequency can be calculated as

$$A_v(f) = \frac{V_{\text{out}}(f)}{V_{\text{in}}(f)}.$$

In decibels,

$$A_{v,\text{dB}}(f) = 20 \log_{10} \left(\frac{V_{\text{out}}(f)}{V_{\text{in}}(f)} \right).$$

The phono input requires a different interpretation because its frequency response is intentionally non-flat according to the RIAA playback characteristic.

11.34 Noise Testing

Preamplifier noise should be measured with the appropriate input terminated and no intentional audio signal applied.

The output can then be observed using an oscilloscope or suitable audio measurement equipment.

Particular attention should be given to spectral components at

50 Hz

and

100 Hz,

which can indicate mains magnetic coupling or power-supply ripple.

If excessive hum is observed, the grounding arrangement, transformer position, input shielding, supply wiring, and PCB return paths should be investigated before changing the amplifier circuitry.

11.35 Preamplifier Design Summary

The preamplifier provides the low-level signal-conditioning and user-control functions required before the audio signal is applied to the main Class-AB power amplifier.

The implemented design provides a dedicated moving-magnet phono input together with three conventional line-level inputs.

The phono section uses low-noise NE5532N operational amplifiers and includes a frequency-dependent feedback network that provides the gain and RIAA playback equalisation required for vinyl-record reproduction.

After source selection, the audio signal passes through the main gain and control stages, where the required voltage amplification, volume adjustment, and stereo balance adjustment are performed.

The preamplifier operates from a regulated symmetrical supply of

$$\boxed{\pm 15 \text{ VDC}},$$

which is electrically separate from the approximately ± 33 – ± 34 V unregulated supply used by the main power amplifier.

Local decoupling capacitors are used close to the operational amplifiers to reduce supply impedance and improve high-frequency stability.

A star-ground architecture is used to minimise common ground impedances and reduce the possibility of mains hum and power-stage current entering the sensitive audio signal path.

Particular care is required in the phono section because its input signal is considerably smaller than a normal line-level signal and is consequently much more sensitive to noise, electromagnetic interference, grounding errors, and poor PCB routing.

The resulting preamplifier provides the interface between the system's external analogue audio sources and the main power-amplifier channels, allowing source selection, phono reproduction, signal amplification, level control, and channel-balance adjustment while maintaining the low-noise requirements of the complete Hi-Fi amplifier.

Chapter 12

STM32 Speaker Protection Controller

12.1 Introduction

The power amplifier developed in this project incorporates an independent microcontroller-based loudspeaker protection system. The purpose of this system is to supervise the operating condition of both power-amplifier channels and disconnect the loudspeakers whenever a potentially damaging condition is detected.

The controller is based on the STM32G031K8 microcontroller and forms a separate subsystem from the analogue audio amplification path. The microcontroller does not process the audio signal for reproduction. Instead, it observes selected electrical and thermal parameters and controls the loudspeaker relays according to the measured operating conditions.

The protection system performs several principal functions: delayed loudspeaker connection during startup, DC-offset monitoring of the amplifier outputs, temperature monitoring, loudspeaker relay control, fault-state management, and visual indication using status LEDs.

The protection controller therefore provides an additional safety layer between the power amplifiers and the connected loudspeakers.

12.2 Purpose of Loudspeaker Protection

Under normal operation, the output of each power amplifier consists of an AC audio waveform centred close to

0 VDC.

A loudspeaker is designed to reproduce an alternating signal. A significant continuous DC voltage applied to the loudspeaker can produce a large unidirectional current through its voice coil.

For a loudspeaker having an approximate resistance R_L , the DC current would be

$$I_{\text{DC}} = \frac{V_{\text{DC}}}{R_L}.$$

The corresponding electrical power dissipated in the loudspeaker is

$$P_{\text{DC}} = \frac{V_{\text{DC}}^2}{R_L}.$$

Unlike an audio waveform, this DC power does not contribute to useful acoustic output. Instead, it produces continuous heating of the loudspeaker voice coil and can force the cone away from its normal rest position.

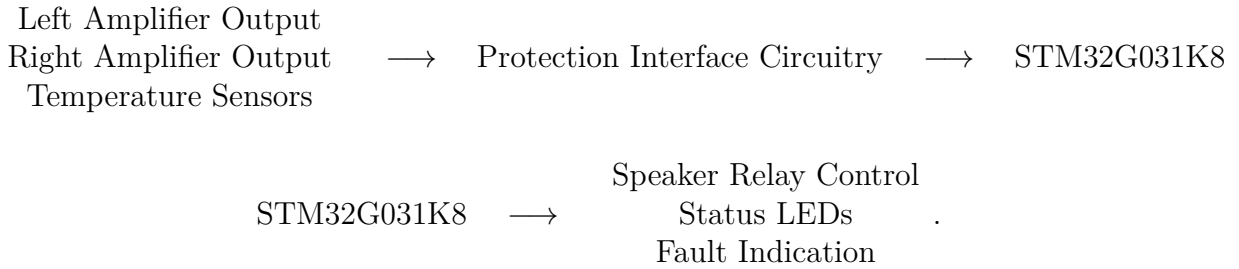
A severe amplifier failure can therefore damage a connected loudspeaker even if the power amplifier itself is protected by supply fuses.

The speaker-protection controller is designed to detect such abnormal conditions and electrically isolate the loudspeaker from the amplifier output.

12.3 System Architecture

The protection controller is implemented as a separate electronic subsystem that interfaces with both power-amplifier channels.

Its overall functional structure can be represented as



The analogue monitoring circuits condition the measured quantities so that they can be safely read by the microcontroller.

The STM32 then applies the programmed protection logic and determines whether the loudspeakers are permitted to remain connected.

12.4 STM32G031K8 Controller

The protection system uses an STM32G031K8 microcontroller [2]. During development, the device is used through the NUCLEO-G031K8 development platform [3].

The microcontroller provides the digital and analogue peripherals required for the protection system, including analogue-to-digital conversion, general purpose digital inputs and outputs, timers, and internal processing resources.

The analogue-to-digital converter is used to acquire conditioned monitoring signals such as amplifier-output DC level and temperature.

Digital GPIO outputs are used to control the loudspeaker relay driver and the status-indicator LEDs.

Internal timing resources allow the startup delay, fault confirmation times, sampling intervals, and other time-dependent protection functions to be implemented in firmware rather than using large analogue timing components.

12.5 Separation from the Audio Signal Path

A fundamental design requirement is that the STM32 protection controller remains outside the principal audio amplification path.

The normal signal path remains

Preamplifier → Power Amplifier → Speaker Relay → Loudspeaker.

The microcontroller monitors the amplifier but does not digitally process the audio signal.

This arrangement preserves the fully analogue character of the main audio path while still allowing digital supervision and protection.

Keeping the microcontroller outside the signal path also helps prevent digital switching activity from directly contributing to audio distortion or noise.

12.6 Power-On Loudspeaker Delay

When the amplifier is initially switched on, the loudspeakers are not connected immediately.

During startup, the main supply rails, preamplifier supply, bias conditions, and amplifier internal operating points require a short time to reach their normal values.

Temporary output transients may occur during this period. If the loudspeakers were connected immediately, these transients could be reproduced as an audible turn-on sound.

The protection controller therefore begins operation with the speaker relays in the disconnected state.

Conceptually, the startup sequence is

Power Applied → MCU Initialisation → Relays Disconnected → Startup Delay → Protection Checks →

The loudspeakers are connected only if no fault condition is detected when the startup interval has completed.

The precise timing strategy is discussed in the dedicated *Power-On Delay and Speaker Connection Sequence* chapter.

12.7 DC Offset Monitoring

Continuous monitoring of the power-amplifier outputs is one of the principal functions of the protection controller.

During normal operation, the audio waveform alternates above and below the ground reference and should have only a small DC component.

The monitoring circuitry conditions each amplifier output before it reaches the STM32 ADC input. The microcontroller can therefore evaluate the DC component without exposing its low-voltage input directly to the much larger power-amplifier output signal.

If a significant DC level remains present for a sufficient period, the condition is interpreted as an amplifier fault.

The controller then commands the speaker relay to disconnect.

The use of a confirmation interval is important because normal low-frequency audio signals can momentarily resemble a slowly changing DC voltage. The protection algorithm must therefore distinguish between legitimate programme material and a persistent abnormal DC condition.

The analogue interface circuit, ADC interpretation, threshold selection, and fault-confirmation method are treated in detail in the *DC Offset Detection Circuit* chapter.

12.8 Temperature Monitoring

The power output transistors can dissipate significant heat, particularly during prolonged high-power operation into a $4\ \Omega$ loudspeaker load.

Temperature sensors are therefore incorporated into the amplifier thermal system and connected to the protection controller.

The STM32 periodically evaluates the measured temperature and compares it with the protection criteria defined in the firmware.

During normal operation,

$$T < T_{\text{fault}},$$

and loudspeaker operation is permitted.

If the monitored temperature reaches an unsafe region,

$$T \geq T_{\text{fault}},$$

the controller enters an overtemperature fault state and disconnects the speaker relays.

A corresponding status indication is also provided so that the reason for the protection event can be identified.

The temperature sensor interface, measurement method, and thermal thresholds are discussed separately in the *Temperature Monitoring System* chapter.

12.9 Speaker Relay Interface

The microcontroller cannot connect a loudspeaker directly. Instead, it controls an electromechanical relay located in the loudspeaker output path.

The relay provides two distinct states:

Relay Energised $\Rightarrow$ Loudspeaker Connected

and

Relay De-energised $\Rightarrow$ Loudspeaker Disconnected.

A transistor driver stage is used between the STM32 GPIO output and the relay coil because the microcontroller output is not intended to supply the relay coil current directly.

The relay control stage therefore provides electrical current amplification between the low-power microcontroller output and the electromechanical relay.

The detailed relay-driver circuitry and contact arrangement are discussed in the dedicated *Speaker Relay Control* chapter.

12.10 Normally Disconnected Protection Philosophy

The control strategy is designed around the principle that the loudspeaker should remain disconnected unless the controller has confirmed that operation is safe.

Immediately after power-up, the relay is therefore inactive.

Only after the controller has completed its startup sequence and confirmed that no relevant fault is present does it activate the speaker relay.

This approach provides a useful fail-safe characteristic because loss of microcontroller control or loss of relay-drive power tends to return the system to the disconnected condition rather than forcing the loudspeaker to remain connected.

The preferred system condition is therefore

Unknown or Fault State $\Rightarrow$ Speaker Disconnected

12.11 Fault Detection

The protection controller evaluates several different operating conditions.

The principal fault categories in the present design are associated with abnormal amplifier-output DC voltage, excessive temperature, and invalid monitoring conditions.

The controller combines these conditions into an overall protection decision.

The relay can be considered enabled only when all required conditions are valid:

$$\text{RelayEnable} = \text{StartupComplete} \wedge \text{DCSafe} \wedge \text{TemperatureSafe} \wedge \text{SystemValid}.$$

If any required condition becomes false,

$$\text{RelayEnable} = 0,$$

and the loudspeaker is disconnected.

This Boolean representation illustrates an important property of the protection system: all protection conditions must indicate safe operation before the loudspeaker connection is permitted.

12.12 Protection States

The controller can be viewed as operating through several logical states.

Table 12.1: Principal operating states of the speaker-protection controller.

State	Speaker Relay	Purpose
Initialisation	Disconnected	Configure MCU and peripherals
Startup delay	Disconnected	Allow amplifier stabilisation
Normal operation	Connected	Continuous system monitoring
DC fault	Disconnected	Protect against amplifier DC output
Thermal fault	Disconnected	Protect against excessive temperature
System/sensor fault	Disconnected	Prevent operation with invalid monitoring

The detailed firmware state transitions are discussed later in the *Fault Detection and Protection Logic* and *STM32 Firmware Design* chapters.

12.13 Normal Operating Sequence

During a normal power-up sequence, the STM32 first configures the required GPIO, ADC, timer, and monitoring functions.

The loudspeaker relay remains disconnected during this process.

The controller then begins the startup timing interval while monitoring the required protection signals.

If the monitored conditions remain valid until the startup delay has completed, the controller activates the relay driver and connects the loudspeakers.

After connection, monitoring continues continuously.

A normal operating sequence can therefore be represented as

$$\boxed{\text{OFF} \rightarrow \text{INITIALISE} \rightarrow \text{DELAY} \rightarrow \text{CHECK} \rightarrow \text{CONNECTED}}.$$

The connected state is maintained only while all monitored conditions remain within the permitted operating range.

12.14 Fault Operating Sequence

If a protection condition becomes active during operation, the controller must remove the loudspeaker connection.

For example, a DC fault sequence can be represented as

Normal Operation → Abnormal DC Detected → Fault Confirmation → Relay Disconnected → Fault I

Similarly, an overheating sequence can be represented as

Normal Operation → Excessive Temperature → Thermal Fault → Relay Disconnected.

The exact recovery behaviour depends on the firmware implementation and is described in the later fault-logic chapter.

12.15 Status and Fault Indicators

Visual status indication is provided using LEDs connected to STM32 GPIO outputs.

These indicators allow the user to distinguish between normal amplifier operation and protection-related conditions without requiring a display.

The status indication system is intended to communicate conditions such as startup delay, normal speaker connection, DC-related faults, and temperature-related faults.

This is particularly useful during commissioning because the LEDs allow the internal protection state to be observed while measurements are performed.

The exact GPIO assignments and LED behaviour are documented separately in the *Status and Fault Indicator LEDs* chapter.

12.16 Analogue-to-Digital Conversion

Several protection quantities are analogue and must therefore be converted into digital values before they can be evaluated by the firmware.

For an ADC having a reference voltage V_{REF} and an N -bit resolution, the ideal relationship between input voltage and digital output is

$$N_{\text{ADC}} = \frac{V_{\text{ADC}}}{V_{\text{REF}}} (2^N - 1).$$

Conversely, the approximate measured ADC voltage is

$$V_{\text{ADC}} = \frac{N_{\text{ADC}}}{2^N - 1} V_{\text{REF}}.$$

The conditioned amplifier-output and temperature signals can therefore be converted into numerical values that are compared with firmware thresholds.

The conversion from ADC input voltage back to the original monitored physical quantity depends on the external resistor networks and sensor characteristics.

12.17 Sampling and Fault Confirmation

A protection controller should not normally react to every individual ADC sample as though it represents a genuine fault.

Electrical noise, relay switching, ADC quantisation, and rapidly changing audio signals can produce isolated samples outside a nominal range.

The firmware therefore evaluates measurements over time before deciding that a persistent protection condition exists.

Conceptually, the decision can be represented as

$$\text{Measured Threshold Exceeded} + \text{Persistence Requirement} \rightarrow \text{Confirmed Fault}.$$

This approach reduces nuisance relay operation while retaining protection against genuine sustained faults.

The exact filtering, sampling, and timing algorithm belongs to the firmware implementation and is therefore discussed in the dedicated firmware chapter.

12.18 Interaction with the Two Amplifier Channels

The complete audio system is stereo, so protection monitoring must account for both the left and right power-amplifier channels.

Each amplifier output is independently monitored for abnormal DC conditions.

A fault in either channel is considered relevant to loudspeaker protection.

The monitoring arrangement can therefore be represented as

$$\begin{cases} \text{DC}_L \\ \text{DC}_R \\ T_L \\ T_R \end{cases} \rightarrow \text{STM32 Protection Logic}.$$

This ensures that the protection system is not dependent on only one channel remaining healthy.

12.19 Interaction with Thermal Management

The electronic protection controller complements the passive thermal design of the amplifier.

The main defence against excessive transistor temperature remains correct heatsink selection, thermal mounting, airflow, and appropriate Class-AB bias.

The STM32 temperature-monitoring system provides an additional level of supervision for abnormal operating conditions.

It should therefore be regarded as

$$\text{Thermal Protection} = \text{Correct Thermal Design} + \text{Electronic Monitoring},$$

rather than as a replacement for adequate heatsinking.

12.20 Noise and Grounding Considerations

The STM32 and its digital circuitry generate switching currents that could potentially couple into the low-level analogue audio circuitry.

For this reason, the protection-controller PCB and wiring are treated separately from the sensitive preamplifier and amplifier input paths.

Particular care is required around:

- microcontroller digital supply currents,
- relay coil currents,
- LED currents,
- ADC sensing grounds,
- analogue amplifier signal ground.

Relay-coil and digital return currents should not share sensitive low-impedance audio-ground paths unnecessarily.

The detailed implementation is discussed later in the grounding and protection-controller PCB chapters.

12.21 Independent Controller Power Supply

The protection controller requires a low-voltage DC supply suitable for the STM32 and relay-control circuitry.

This supply is separate from the approximately ± 33 – ± 34 V rails used directly by the power-amplifier output stages.

The low-voltage protection supply must provide stable operation during startup, normal monitoring, and relay switching.

It must also provide appropriate decoupling for both the microcontroller and the relay driver circuitry.

The complete design of this supply is treated separately in the *Protection Controller Power Supply* chapter.

12.22 Firmware Role

The behaviour of the protection controller is determined primarily by the firmware executed by the STM32G031K8.

The firmware is responsible for peripheral initialisation, ADC acquisition, measurement interpretation, startup timing, fault confirmation, relay-state control, temperature supervision, and LED indication.

Separating these functions into firmware provides greater flexibility than a purely analogue protection circuit.

Thresholds, timing behaviour, fault confirmation, and status indication can be modified without redesigning the complete hardware.

The structure and implementation of the software are documented separately in the *STM32 Firmware Design* chapter, while the complete source code is included in the appendix.

12.23 Advantages of the Microcontroller-Based Approach

The STM32-based protection controller provides several advantages over a simple analogue relay-delay circuit.

A single controller can evaluate multiple protection conditions simultaneously, apply time-dependent decision logic, provide different visual fault indications, and implement a defined startup sequence.

The use of ADC measurements also means that protection decisions can be based on numerical signal levels rather than only on simple comparator states.

The system can therefore be expanded later to include additional monitoring or diagnostic functions without requiring major changes to the main audio amplifier.

12.24 Protection Philosophy

The controller is designed according to three principal protection rules.

First, the loudspeaker remains disconnected while the operating state of the amplifier is

unknown.

Second, connection is permitted only after the required startup checks have been completed successfully.

Third, a detected unsafe condition causes removal of the loudspeaker connection.

These rules can be expressed as

$$\boxed{\text{Unknown State} \Rightarrow \text{Disconnected}}$$

$$\boxed{\text{Confirmed Safe State} \Rightarrow \text{Connected}}$$

and

$$\boxed{\text{Confirmed Fault} \Rightarrow \text{Disconnected}}.$$

This behaviour forms the basis of the firmware state machine used by the protection controller.

12.25 Testing Requirements

The protection controller must be tested independently before valuable loudspeakers are connected to the amplifier.

Verification should confirm correct MCU startup, speaker-relay operation, power-on delay, left- and right-channel DC fault detection, thermal-fault detection, status LED behaviour, and correct disconnection of the loudspeakers when a protection condition is produced.

Faults should initially be simulated using controlled electrical test conditions rather than by deliberately damaging or overdriving the power amplifier.

After individual protection functions have been verified, the complete controller can be tested together with the two power-amplifier channels.

Detailed validation procedures are presented in the *Protection System Testing* chapter.

12.26 Design Summary

The speaker-protection controller is an independent microcontroller-based subsystem designed to protect the loudspeakers and supervise important operating conditions of the stereo power amplifier.

The controller is based on the STM32G031K8 microcontroller [2] and is developed using the NUCLEO-G031K8 platform [3].

The system monitors both amplifier outputs for abnormal DC conditions and also monitors the thermal condition of the amplifier.

At startup, the speaker relays remain disconnected while the amplifier power supplies and analogue circuitry stabilise. Once the startup interval has completed and the monitored quantities are confirmed to be within their permitted ranges, the controller enables the loudspeaker relays.

During normal operation, monitoring continues continuously. Detection of a persistent DC-output fault, excessive temperature, or other invalid protection condition causes the relay control system to disconnect the loudspeakers.

Status LEDs provide visual information regarding startup, normal operation, and fault conditions.

The microcontroller remains outside the main analogue audio path and therefore does not perform audio processing. Its function is supervisory and protective.

The use of firmware allows the protection behaviour to combine ADC measurements, timing, fault confirmation, relay control, and visual indication within a single controller while retaining flexibility for future modification.

The following chapters describe the individual parts of the controller in greater detail, including the STM32G031K8 microcontroller, DC-offset detection, temperature monitoring, relay control, startup timing, protection logic, indicator LEDs, controller power supply, and firmware implementation.

Chapter 13

STM32G031K8 Microcontroller

13.1 Introduction

The central processing element of the loudspeaker protection system is the STM32G031K8 microcontroller from STMicroelectronics [2]. The device is responsible for acquiring analogue monitoring signals, evaluating the operating condition of the two power-amplifier channels, controlling the loudspeaker relays, managing the power-on delay, and driving the status and fault indicator LEDs.

During development, the STM32G031K8 is used through the NUCLEO-G031K8 development board [3]. This board provides convenient access to the microcontroller pins together with an integrated ST-LINK programmer and debugger, allowing the protection firmware to be developed, programmed, tested, and modified without requiring a separate external programmer.

The microcontroller is used only for supervision and protection. It does not process the analogue audio signal that is sent to the loudspeakers. The main audio path therefore remains analogue, while the STM32 operates as a parallel monitoring and control subsystem.

13.2 Reasons for Selecting the STM32G031K8

The speaker protection controller does not require a high-performance microprocessor, but it does require reliable analogue measurement, deterministic timing, several GPIO connections, and sufficient processing capability to evaluate multiple protection conditions simultaneously.

The STM32G031K8 was selected because it provides these functions within a compact and relatively inexpensive microcontroller.

The principal characteristics relevant to this project include:

- 32-bit Arm Cortex-M0+ processor core,
- maximum CPU frequency of 64 MHz,

- 64 KB of embedded Flash memory,
- 8 KB of SRAM,
- integrated 12-bit analogue-to-digital converter,
- multiple GPIO pins,
- several hardware timers,
- watchdog timers,
- serial communication peripherals,
- low operating voltage and power consumption,
- integrated debugging and programming support when used with the NUCLEO-G031K8 board.

These resources are considerably greater than the minimum processing requirements of the protection system and therefore provide sufficient margin for additional firmware functions and future expansion.

13.3 Main Device Characteristics

The principal STM32G031K8 characteristics relevant to the protection controller are summarised in Table 13.1.

Table 13.1: Principal STM32G031K8 characteristics [2].

Parameter	STM32G031K8
Processor core	Arm Cortex-M0+
Architecture	32-bit RISC
Maximum CPU frequency	64 MHz
Flash memory	64 KB
SRAM	8 KB
ADC resolution	12 bit
ADC maximum conversion rate	up to 2.5 MS/s
Operating voltage	1.7–3.6 V
Hardware timers	Multiple 16-bit and 32-bit timers
Watchdog timers	Independent and window watchdog
Communication interfaces	USART, SPI, I ² C
Development platform	NUCLEO-G031K8

13.4 Arm Cortex-M0+ Processor Core

The STM32G031K8 is based on the 32-bit Arm Cortex-M0+ processor core.

The Cortex-M0+ architecture is designed for embedded control applications where low power consumption, compact implementation, and deterministic real-time behaviour are more important than the computational performance required by desktop or general-purpose computer systems.

The processor can operate at a clock frequency of up to

64 MHz

.

This processing capability is significantly greater than that required for the speaker-protection functions.

The controller primarily performs operations such as:

- reading ADC channels,
- comparing measurements with thresholds,
- calculating filtered or averaged values,
- maintaining software timers,
- controlling GPIO outputs,
- executing protection-state transitions,
- monitoring system validity.

These tasks require relatively little processing power, so the STM32G031K8 can perform them while retaining considerable computational margin.

13.5 Program Memory

The STM32G031K8 contains

64 KB

of embedded Flash memory.

The Flash memory stores the protection firmware and remains non-volatile when the amplifier is switched off.

The program stored in Flash includes the startup routine, peripheral configuration, ADC acquisition functions, protection logic, relay control, status indication, and other firmware functions.

The protection program is relatively small compared with the available 64 KB memory, providing considerable space for future firmware expansion.

Possible future functions could include:

- additional temperature sensors,

- fan-speed control,
- amplifier operating-time logging,
- serial diagnostics,
- additional fault histories,
- communication with a front-panel display,
- configurable protection thresholds.

13.6 SRAM

The microcontroller contains

8 KB

of SRAM.

SRAM is used for temporary variables and data while the firmware is running.

In the present protection application, RAM can be used to store:

- ADC samples,
- filtered voltage values,
- temperature measurements,
- timer counters,
- protection flags,
- relay states,
- LED states,
- state-machine variables.

The memory requirement of these functions is relatively small compared with the available SRAM.

13.7 Analogue-to-Digital Converter

One of the most important reasons for selecting the STM32G031K8 is its integrated 12-bit analogue-to-digital converter.

The ADC allows continuously varying analogue quantities to be converted into numerical values that can be processed by the firmware.

In this project, these analogue quantities include principally:

- left-channel amplifier DC monitoring signal,
- right-channel amplifier DC monitoring signal,
- left-channel temperature signal,
- right-channel temperature signal.

Additional analogue monitoring channels can also be incorporated in future versions if required.

13.8 ADC Resolution

For a 12-bit ADC, the number of possible digital output codes is

$$2^{12} = 4096.$$

The ideal ADC code therefore ranges from

$$0$$

to

$$4095.$$

For an analogue reference voltage V_{REF} , the ideal conversion is

$$N_{\text{ADC}} = \frac{V_{\text{ADC}}}{V_{\text{REF}}}(4095).$$

Conversely,

$$V_{\text{ADC}} = \frac{N_{\text{ADC}}}{4095} V_{\text{REF}}.$$

If the ADC reference is approximately

$$V_{\text{REF}} = 3.3 \text{ V},$$

the ideal voltage represented by one least-significant bit is

$$V_{\text{LSB}} = \frac{3.3}{4095}.$$

Therefore,

$$\boxed{V_{\text{LSB}} \approx 0.806 \text{ mV}}.$$

This represents the ideal ADC quantisation step at a 3.3 V reference.

The actual measurement accuracy also depends on ADC characteristics, reference accuracy, noise, PCB layout, resistor tolerances, and the analogue signal-conditioning circuit.

13.9 ADC Input-Voltage Requirement

The STM32 ADC cannot be connected directly to the power-amplifier output.

The main power amplifier operates from approximately

$$\pm 33\text{--}\pm 34\text{ V}$$

supply rails, while the STM32 operates from a low-voltage supply in the approximately 3.3 V region.

The power-amplifier output can also swing both positively and negatively, whereas a normal STM32 ADC input must remain within the voltage range permitted by the microcontroller.

Consequently,

$$V_{\text{amplifier}} \not\rightarrow V_{\text{ADC}} \text{ directly}$$

and an analogue conditioning circuit is required between the two systems.

The conditioning network performs the required attenuation, level shifting, filtering, and protection before the signal reaches the ADC input.

The detailed interface is described in the *DC Offset Detection Circuit* chapter.

13.10 ADC Signal Conditioning

The conditioned ADC voltage can generally be expressed in the form

$$V_{\text{ADC}} = aV_{\text{AMP}} + V_{\text{OFFSET}},$$

where

$$a$$

represents the attenuation factor and

$$V_{\text{OFFSET}}$$

represents any DC offset introduced to move a bipolar monitored signal into the unipolar ADC input range.

The firmware can then mathematically reverse this relationship:

$$V_{\text{AMP}} = \frac{V_{\text{ADC}} - V_{\text{OFFSET}}}{a}.$$

This allows the STM32 to estimate the actual amplifier-output DC voltage from the ADC measurement.

13.11 ADC Sampling Rate

The integrated ADC is capable of conversion rates substantially higher than those required by the speaker-protection system.

The STM32G031 family ADC can operate at conversion rates of up to approximately

$$2.5 \text{ MS/s}$$

under the appropriate operating conditions [2].

The protection controller does not require measurements at megasample rates. DC offset and temperature change relatively slowly compared with the audio waveform.

A significantly lower firmware sampling rate can therefore be used, reducing processing requirements while still allowing faults to be detected rapidly.

The exact sampling frequency and digital filtering strategy are defined by the firmware and are discussed in the dedicated firmware chapter.

13.12 GPIO Resources

General-purpose input/output pins provide the digital connection between the microcontroller and the remaining protection circuitry.

GPIO outputs are used for functions such as:

- loudspeaker relay control,
- normal-operation indication,
- startup-delay indication,
- DC-fault indication,
- overtemperature indication,
- other status or diagnostic indicators.

GPIO inputs can also be used for switches, diagnostic inputs, additional fault signals, or future expansion.

The number of available GPIO pins is more than sufficient for the present protection-controller requirements.

13.13 GPIO Output Limitations

Although STM32 GPIO pins can provide digital control signals, they are not intended to directly drive high-current loads.

A loudspeaker relay coil requires substantially more current than should be drawn directly from an MCU output.

The required relationship is therefore

$$\text{STM32 GPIO} \rightarrow \text{Transistor Driver} \rightarrow \text{Relay Coil.}$$

The GPIO controls the base current of an external transistor driver, while the transistor provides the larger current required by the relay coil.

In this project, a BC337 transistor is used as the relay-coil driver.

The relay interface therefore electrically separates the low-current microcontroller output function from the higher-current electromechanical load.

13.14 Relay Driver Control

The firmware controls the relay driver using a digital GPIO signal.

Conceptually,

$$GPIO = 1 \Rightarrow \text{Relay Driver Active}$$

and

$$GPIO = 0 \Rightarrow \text{Relay Driver Inactive,}$$

subject to the final transistor-driver polarity.

The protection firmware is designed so that the relay is not enabled until the system has passed the required startup and safety checks.

If a protection fault occurs, the firmware removes the relay-enable signal.

The transistor driver and relay circuit are discussed separately in the *Speaker Relay Control* chapter.

13.15 Status LED Control

Several GPIO outputs are dedicated to visual status indicators.

The LEDs communicate the operating state of the protection controller without requiring an LCD or other complex user interface.

Each indicator is controlled through an appropriate current-limiting resistor.

The general connection is

STM32 GPIO → Current-Limiting Resistor → LED.

The firmware can therefore independently enable the required indicator according to the current state of the protection system.

The precise indicator assignments are described in the *Status and Fault Indicator LEDs* chapter.

13.16 Hardware Timers

The STM32G031K8 contains several hardware timers that can be used for time-dependent protection functions.

This is particularly useful for implementing the loudspeaker startup delay.

Rather than relying on a large RC analogue timing network, the delay can be generated digitally.

A timer can also provide regular events for:

- ADC sampling,
- temperature measurements,
- DC-fault confirmation,
- LED timing,
- relay timing,
- periodic system diagnostics.

This produces repeatable timing that is not strongly affected by analogue component tolerances.

13.17 Power-On Delay Timing

When the amplifier is powered, the STM32 begins execution but keeps the speaker relay disabled.

A timer or firmware time base measures the required startup interval.

The basic process is

$$t = 0 \Rightarrow \text{Relay OFF}$$

followed by

$$0 < t < t_{\text{delay}} \Rightarrow \text{Relay OFF}.$$

Once

$$t \geq t_{\text{delay}},$$

the firmware evaluates the monitored protection conditions.

The relay is connected only if the system is confirmed to be safe.

13.18 Watchdog Functions

The STM32G031K8 includes hardware watchdog resources [2].

A watchdog timer can be used to detect abnormal firmware execution.

During correct program operation, the firmware periodically services the watchdog.

If the program becomes trapped, stops executing correctly, or fails to service the watchdog within the permitted interval, the microcontroller can automatically reset.

This feature is useful in a protection controller because the system should not remain indefinitely in an uncontrolled software state.

A robust implementation should ensure that an MCU reset results in the speaker relays returning to the disconnected state until the startup protection sequence has been completed again.

13.19 Reset Behaviour

The reset state of the microcontroller is important because the loudspeakers must not be connected accidentally while the MCU is starting.

At reset, the relay-control output should therefore remain in a condition that keeps the relay driver inactive.

The required philosophy is

$$\boxed{\text{MCU Reset} \Rightarrow \text{Speaker Relay OFF}}.$$

Only after GPIO initialisation and successful protection checks should the firmware intentionally activate the loudspeaker relay.

This contributes to the fail-safe behaviour of the complete protection system.

13.20 Brownout and Supply Monitoring

The STM32G031K8 includes internal power-management and reset functions, including power-on reset and programmable brownout reset [2].

A brownout condition occurs when the microcontroller supply voltage falls below the range required for reliable operation.

Allowing the processor to continue executing at an inadequate supply voltage could result in unpredictable behaviour.

The internal reset system therefore provides additional protection against incorrect operation during supply startup or collapse.

This is particularly relevant when the amplifier is switched off because the controller supply and the main amplifier rails may not fall at exactly the same rate.

13.21 Clock System

The STM32G031K8 contains internal clock resources that allow the processor and peripherals to operate without requiring a complex external oscillator circuit.

The system clock controls the CPU, timers, ADC timing, and other peripherals.

For the present application, absolute clock accuracy at the level required for precision communication or instrumentation is not necessary.

The important requirement is that timer intervals and ADC acquisition remain sufficiently stable and repeatable for the protection functions.

13.22 Communication Interfaces

The STM32G031K8 also includes communication peripherals such as USART, SPI, and I²C interfaces [2].

These interfaces are not essential to the basic speaker-protection function, but they provide useful possibilities for future development.

For example, a future version could use serial communication to transmit

- measured amplifier DC offset,
- heatsink temperature,
- current protection state,
- fault history,
- firmware diagnostic information.

An I²C interface could also be used for an external display, digital temperature sensor, or other peripheral.

13.23 Operating Voltage

The STM32G031K8 itself is designed to operate from a supply voltage in the range

$$1.7 \text{ V} \leq V_{DD} \leq 3.6 \text{ V}$$

[2].

In the protection controller, the MCU operates in the approximately 3.3 V logic domain.

This is substantially lower than the main amplifier supply voltage of approximately

$$\pm 33 - \pm 34 \text{ V}.$$

Direct electrical connection between the power-amplifier nodes and MCU pins is therefore not permitted unless the signals have first been appropriately conditioned.

13.24 Protection of MCU Inputs

The protection-interface circuits must ensure that the voltage at every microcontroller input remains within the permitted range.

For an ADC input, the normal signal must satisfy approximately

$$0 \leq V_{\text{ADC}} \leq V_{DDA}.$$

This requirement is particularly important for amplifier-output sensing because the original monitored signal is bipolar.

Resistors, clamping elements, filtering components, and level-shifting circuitry are therefore used before the MCU input where required.

The analogue input-protection design is described in greater detail in the chapter concerning DC-offset measurement.

13.25 NUCLEO-G031K8 Development Board

The STM32G031K8 is used during development through the NUCLEO-G031K8 board [3].

The Nucleo-32 platform simplifies prototype development by providing the microcontroller together with supporting circuitry and accessible header connections.

Important board features include:

- onboard STM32G031K8 microcontroller,
- integrated ST-LINK programmer/debugger,

- USB programming connection,
- reset push-button,
- user LED,
- accessible expansion connectors,
- compatibility with the STM32Cube development ecosystem.

The integrated ST-LINK interface allows the firmware to be programmed and debugged directly from the development computer.

13.26 Use of ST-LINK

ST-LINK provides the programming and debugging interface between the development computer and the STM32 microcontroller.

During firmware development it allows:

- firmware downloading,
- processor reset,
- breakpoint insertion,
- single-step execution,
- variable inspection,
- peripheral-register inspection,
- debugging of protection logic.

These features are particularly useful during the development of safety functions because the internal state of the controller can be observed while simulated faults are applied.

13.27 STM32CubeIDE Development

Firmware for the protection controller can be developed using the STMicroelectronics STM32Cube software ecosystem.

Peripheral configuration is defined for functions such as

- system clock,
- GPIO,
- ADC,

- timers,
- interrupts,
- watchdog operation.

The application firmware then implements the higher-level protection functions.

Keeping peripheral configuration separate from protection logic makes the software easier to understand, test, and maintain.

13.28 Role in the Protection State Machine

The STM32 executes the logical state machine that determines whether the speaker relay may be energised.

A simplified set of states is

INITIALISATION,

STARTUP DELAY,

NORMAL OPERATION,

DC FAULT,

THERMAL FAULT,

and

SYSTEM FAULT.

The microcontroller continuously evaluates the current measurements and determines whether a state transition is necessary.

For example,

NORMAL + DC Fault Detected $\rightarrow$ DC FAULT.

A transition into a fault state immediately causes the relay-control output to move to the loudspeaker-disconnected condition.

13.29 System-Level Fail-Safe Behaviour

The microcontroller is incorporated into a system in which the default condition is intended to be loudspeaker disconnection.

This principle is important because the controller itself cannot be assumed to be permanently operational under every possible fault condition.

The intended behaviour is therefore

$$\text{No MCU Control} \Rightarrow \text{Relay De-energised} \Rightarrow \text{Speaker Disconnected.}$$

Similarly,

$$\text{MCU Reset} \Rightarrow \text{Speaker Disconnected,}$$

and

$$\text{Protection Fault} \Rightarrow \text{Speaker Disconnected.}$$

Only a confirmed safe system state permits loudspeaker connection.

13.30 Why a Microcontroller is Suitable for this Application

The use of a microcontroller allows several functions that would otherwise require multiple analogue comparators, timing networks, logic gates, and latches to be combined into one programmable device.

The STM32 can simultaneously monitor multiple analogue signals, apply different thresholds, implement delays, filter measurements, identify different fault types, and provide corresponding visual indication.

The protection behaviour can also be modified in firmware without redesigning the complete PCB.

For example, changing the DC-fault persistence time can be implemented by changing a firmware parameter rather than replacing a timing capacitor or resistor.

The same applies to temperature limits, LED behaviour, relay delays, and sampling intervals.

13.31 Advantages over a Simple Analogue Protection Circuit

A conventional analogue speaker-protection circuit can provide reliable operation, but the microcontroller-based implementation offers additional flexibility.

The principal advantages in the present project include:

- simultaneous monitoring of both stereo channels,
- programmable DC fault thresholds,
- programmable temperature thresholds,
- configurable startup delay,
- digital fault confirmation,
- multiple independent status indicators,
- straightforward modification of protection behaviour,
- potential for future diagnostics and communications.

The use of software does, however, require careful firmware design so that the controller always returns to a safe state following reset or abnormal operation.

13.32 Isolation from High-Current Circuits

The MCU and ADC circuitry process low-level electrical signals and must be protected from the high-current environment of the main amplifier.

The PCB layout should therefore keep sensitive MCU analogue connections away from:

- transformer secondary wiring,
- bridge-rectifier current paths,
- reservoir-capacitor charging loops,
- speaker output current paths,
- relay-coil current paths,
- fan wiring.

Particular attention should also be given to grounding. High-current return paths should not flow through the same PCB conductors used as the ADC reference ground.

Otherwise, ground voltage drops could appear as measurement errors at the ADC.

13.33 Decoupling

Digital microcontrollers generate short current pulses as internal logic changes state.

Local power-supply decoupling is therefore required close to the STM32 supply connections.

A small ceramic capacitor provides a local low-impedance current source for high-frequency switching activity, while larger capacitance provides lower-frequency energy storage.

The purpose of local decoupling is to reduce

- supply-voltage transients,
- digital switching noise,
- interaction between MCU and analogue circuits,
- susceptibility to external interference.

Correct decoupling is especially important because the ADC shares the same general electrical environment as the digital core.

13.34 Testing of the Microcontroller Subsystem

The STM32 subsystem should be tested independently before it is allowed to control a loudspeaker connected to the amplifier.

The verification procedure should include:

1. Verify the MCU supply voltage.
2. Confirm correct firmware programming through ST-LINK.
3. Confirm that the MCU starts correctly following power application.
4. Verify that the speaker relay remains disconnected immediately after reset.
5. Verify all required GPIO outputs.
6. Verify each status LED independently.
7. Confirm operation of the relay-driver GPIO.
8. Measure the ADC reference and supply voltage.
9. Apply known test voltages to each ADC monitoring input.
10. Confirm that the calculated ADC values correspond to the applied voltages.
11. Verify temperature-channel acquisition.
12. Verify left- and right-channel DC monitoring.

13. Confirm timer operation.
14. Confirm startup-delay behaviour.
15. Simulate individual fault conditions.
16. Verify that every relevant fault forces the loudspeaker relay to the disconnected condition.
17. Reset or power-cycle the controller and confirm that the safe startup sequence is repeated.

13.35 Future Expansion

The STM32G031K8 provides additional resources that are not required by the initial protection implementation.

These resources allow future versions of the system to incorporate additional functions without necessarily replacing the microcontroller.

Possible future additions include:

- automatic fan-speed control,
- amplifier temperature display,
- DC-offset display,
- serial diagnostic output,
- protection-event logging,
- configurable protection thresholds,
- power-button control,
- additional sensor monitoring,
- remote system monitoring.

The selected microcontroller therefore provides sufficient capability for the current design while retaining useful expansion potential.

13.36 Design Summary

The STM32G031K8 is the central processing device of the speaker-protection controller. It is based on a 32-bit Arm Cortex-M0+ processor operating at frequencies up to

64 MHz,

and provides

64 KB

of Flash memory together with

8 KB

of SRAM [2].

The integrated 12-bit ADC provides

4096

possible conversion levels and is used to measure conditioned amplifier-output and temperature signals.

With an ideal 3.3 V reference, the theoretical ADC resolution is approximately

0.806 mV/LSB.

The ADC inputs cannot be connected directly to the approximately ± 33 – ± 34 V power-amplifier circuitry. Dedicated analogue signal-conditioning networks are therefore used to attenuate, shift, filter, and protect the monitored signals before they reach the microcontroller.

GPIO outputs control the BC337 relay driver and the protection-status LEDs, while hardware timers provide accurate timing for startup delay, ADC sampling, fault confirmation, and other time-dependent functions.

Watchdog and reset resources provide additional means of ensuring that abnormal firmware execution does not leave the protection controller indefinitely in an uncontrolled state.

The STM32 is developed using the NUCLEO-G031K8 platform, which provides an integrated ST-LINK programmer/debugger and convenient access to the microcontroller I/O connections [3].

The microcontroller remains completely outside the principal analogue audio path. Its purpose is supervisory: it measures the operating condition of the amplifier, executes the programmed protection logic, controls the loudspeaker relays, and communicates system status through visual indicators.

The STM32G031K8 therefore provides a compact and flexible platform for implementing the complete startup, monitoring, protection, and fault-control requirements of the stereo power amplifier.

Chapter 14

DC Offset Detection Circuit

14.1 Introduction

One of the principal functions of the STM32-based speaker protection system is the detection of abnormal DC voltage at the outputs of the left and right power-amplifier channels.

During normal operation, the power-amplifier output consists primarily of an alternating audio waveform centred close to

0 VDC.

A small DC offset is unavoidable in a practical direct-coupled power amplifier, but a large sustained DC voltage can indicate a serious amplifier fault. Possible causes include failure of an output transistor, failure of a driver stage, loss of one power-supply rail, or another condition that forces the amplifier output toward one of the supply rails.

If such a voltage were applied continuously to a loudspeaker, a large unidirectional current could flow through its voice coil. The purpose of the DC detection circuit is therefore to monitor both amplifier outputs and provide the STM32G031K8 with safe analogue signals from which dangerous DC conditions can be identified.

14.2 Need for Signal Conditioning

The STM32 ADC cannot be connected directly to the power-amplifier output.

The amplifier operates from supply rails of approximately

± 33 – ± 34 V,

and its output signal can therefore contain both positive and negative voltages of several tens of volts.

The STM32G031K8, in contrast, operates in the approximately 3.3 V logic domain and its ADC input must remain within its permitted low-voltage range.

Direct connection would therefore create two major problems:

- the amplifier-output magnitude can greatly exceed the ADC input voltage range;
- negative amplifier-output voltage cannot be applied directly to the ADC input.

The protection circuit therefore performs four functions before the signal reaches the STM32:

1. attenuation of the amplifier-output voltage;
2. positive DC biasing of the sensed signal;
3. low-pass filtering to reject normal audio;
4. current limiting and voltage clamping for ADC protection.

14.3 Stereo Sensing Architecture

Two identical DC sensing circuits are used, one for each amplifier channel.

The left-channel circuit monitors the left amplifier output and provides the signal

ADC_L

to STM32 pin

PA2.

The right-channel circuit provides

ADC_R

to STM32 pin

PA3.

The architecture can therefore be represented as

Left Amplifier Output → DC Sensing Network → PA2/ADC_L

and

Right Amplifier Output → DC Sensing Network → PA3/ADC_R.

Using identical component values for both channels provides approximately the same voltage scaling, filtering, and fault thresholds.

14.4 Final Component Values

The final sensing network used for each amplifier channel is based on the component values shown in Table 14.1.

Table 14.1: DC sensing circuit component values per amplifier channel.

Component	Value	Function
R1	150 k Ω , 1%	Amplifier-output attenuation and current limiting
R2	10 k Ω , 1%	Upper part of the 3.3 V bias network
R3	10 k Ω , 1%	Lower part of the bias network
R4	4.7 k Ω	ADC and clamp-diode current limiting
C1	4.7 μ F	Main audio-rejection low-pass filter
C_{HF}	100 nF optional	High-frequency bypass filtering
D1, D2	Clamp diodes	ADC overvoltage and undervoltage protection

The 4.7 μ F value is used as the principal filtering capacitor. A 100 nF capacitor alone would provide insufficient attenuation of low-frequency audio and bass signals.

14.5 Bias Network

R2 and R3 create a positive bias voltage for the sensing circuit.

Both resistors have the same value:

$$R_2 = R_3 = 10 \text{ k}\Omega.$$

If no other circuit were connected to their midpoint, the ideal voltage would be

$$V_{\text{BIAS}} = 3.3 \frac{R_3}{R_2 + R_3}.$$

Therefore,

$$V_{\text{BIAS}} = 3.3 \frac{10}{10 + 10} = 1.65 \text{ V}.$$

However, R1 also connects the node to the amplifier output. Consequently, the actual nominal centre voltage with the amplifier output at 0 V is slightly lower than 1.65 V.

14.6 Derivation of the Sensing Transfer Function

The sensing node is connected to three voltage sources through three resistors:

- amplifier output through R1;

- +3.3 V through R2;
- ground through R3.

Ignoring the very small DC input current of the STM32 ADC, the sensing-node voltage can be written as

$$V_{\text{SENSE}} = \frac{\frac{V_{\text{AMP}}}{R_1} + \frac{3.3}{R_2} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}.$$

Substituting

$$R_1 = 150 \text{ k}\Omega,$$

$$R_2 = R_3 = 10 \text{ k}\Omega,$$

gives approximately

$$V_{\text{ADC}} \approx 1.597 + 0.03226V_{\text{AMP}}.$$

Therefore,

$$V_{\text{ADC}} \approx 1.597 \text{ V} + 0.03226V_{\text{AMP}}$$

where V_{AMP} is the DC component of the amplifier output.

The ADC input therefore changes by approximately

$$32.26 \text{ mV}$$

for every

$$1 \text{ V}$$

change at the power-amplifier output.

14.7 Healthy Zero-Output Condition

For a healthy amplifier with approximately zero DC voltage,

$$V_{\text{AMP}} = 0 \text{ V}.$$

The expected ADC voltage is therefore

$$V_{\text{ADC}} = 1.597 + 0.03226(0).$$

Hence,

$$\boxed{V_{\text{ADC}} \approx 1.597 \text{ V}}$$

represents the nominal healthy centre value.

The protection firmware therefore does not expect an ADC voltage close to zero during normal operation. Instead, it expects a value near 1.6 V.

14.8 Detection of Positive DC Offset

A positive DC voltage at the amplifier output causes the ADC voltage to rise above the nominal 1.597 V centre.

For example, with

$$V_{\text{AMP}} = +1 \text{ V},$$

the ADC voltage becomes

$$V_{\text{ADC}} = 1.597 + 0.03226(1),$$

giving

$$V_{\text{ADC}} \approx 1.629 \text{ V}.$$

For a more severe

$$+10 \text{ V}$$

fault,

$$V_{\text{ADC}} = 1.597 + 0.03226(10) \approx 1.920 \text{ V}.$$

The STM32 therefore detects positive DC faults as an upward displacement from the healthy centre voltage.

14.9 Detection of Negative DC Offset

A negative DC voltage causes the sensed voltage to move below the centre value.

For example,

$$V_{\text{AMP}} = -1 \text{ V}$$

produces

$$V_{\text{ADC}} = 1.597 - 0.03226,$$

giving approximately

$$V_{\text{ADC}} \approx 1.565 \text{ V}.$$

Similarly,

$$V_{\text{AMP}} = -10 \text{ V}$$

produces approximately

$$V_{\text{ADC}} \approx 1.274 \text{ V}.$$

The positive centre bias therefore allows both positive and negative amplifier faults to be represented using positive voltages that can be safely measured by the STM32 ADC.

14.10 Full-Rail Fault Conditions

The implemented amplifier operates from approximately ± 33 – ± 34 VDC rails.

If a severe fault causes the speaker output to move close to the positive +34 V rail,

$$V_{\text{ADC}} = 1.597 + 0.03226(34).$$

Therefore,

$$\boxed{V_{\text{ADC}} \approx 2.694 \text{ V}}$$

for an approximate +34 V output fault.

For a corresponding negative-rail fault,

$$V_{\text{AMP}} = -34 \text{ V},$$

giving

$$V_{\text{ADC}} = 1.597 - 0.03226(34)$$

and therefore

$$V_{\text{ADC}} \approx 0.500 \text{ V.}$$

Both values remain within the intended normal ADC measurement range, allowing severe positive and negative amplifier faults to be identified without requiring the protection diodes to conduct during normal expected fault conditions.

14.11 Useful Voltage Scaling

Table 14.2: Approximate relationship between amplifier DC output and STM32 ADC voltage.

Amplifier DC	ADC Voltage	Interpretation
0 V	1.597 V	Healthy centre
+0.5 V	1.613 V	Small positive offset
−0.5 V	1.581 V	Small negative offset
+1 V	1.629 V	Positive fault threshold region
−1 V	1.565 V	Negative fault threshold region
+5 V	1.758 V	Clear positive fault
−5 V	1.436 V	Clear negative fault
+10 V	1.920 V	Serious positive fault
−10 V	1.274 V	Serious negative fault
+34 V	2.694 V	Approximate positive rail fault
−34 V	0.500 V	Approximate negative rail fault

14.12 Conversion Back to Amplifier DC Voltage

The sensing equation can be rearranged so that the firmware can estimate the original amplifier-output DC voltage.

Starting with

$$V_{\text{ADC}} = 1.597 + 0.03226V_{\text{AMP}},$$

the estimated amplifier DC voltage is

$$V_{\text{AMP}} \approx \frac{V_{\text{ADC}} - 1.597}{0.03226}.$$

This equation allows the firmware to express the measured fault quantity in terms of approximate volts at the actual amplifier output rather than only in ADC counts.

14.13 ADC Resolution and DC Measurement

The STM32G031K8 contains a 12-bit ADC. Assuming an approximate 3.3 V ADC reference, the ideal ADC step size is

$$V_{\text{LSB}} = \frac{3.3}{4095} \approx 0.806 \text{ mV}.$$

A 1 V change at the amplifier output produces approximately

$$32.26 \text{ mV}$$

change at the ADC.

This corresponds ideally to approximately

$$\frac{32.26}{0.806} \approx 40$$

ADC counts.

A 0.5 V amplifier-output change therefore corresponds to approximately 20 ADC counts.

This demonstrates that the 12-bit ADC provides sufficient nominal resolution to detect DC offsets considerably smaller than a full supply-rail fault.

The actual measurement uncertainty will also depend on resistor tolerances, ADC reference variation, PCB noise, grounding, and ADC accuracy.

14.14 Low-Pass Filtering

Normal audio must not be interpreted as a DC fault.

The loudspeaker output may contain large AC voltages, particularly at high listening levels. A simple instantaneous ADC threshold would therefore cause false protection events whenever the audio waveform moved sufficiently above or below zero.

C1 is therefore used to create a low-pass filter that strongly attenuates the normal audio waveform while preserving slowly changing and sustained DC components.

The final main filter capacitor is

$$\boxed{C_1 = 4.7 \text{ } \mu\text{F}}.$$

14.15 Approximate Filter Cutoff Frequency

The Thevenin resistance of R1, R2 and R3 seen from the sensing node is

$$R_{\text{TH}} = R_1 \parallel R_2 \parallel R_3.$$

Therefore,

$$R_{\text{TH}} = 150 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel 10 \text{ k}\Omega \approx 4.84 \text{ k}\Omega.$$

Including the series resistor R4 gives an approximate resistance seen by C1 of

$$R_{\text{FILTER}} \approx 4.84 + 4.7 = 9.54 \text{ k}\Omega.$$

The approximate first-order cutoff frequency is therefore

$$f_c = \frac{1}{2\pi R_{\text{FILTER}} C_1}.$$

Substituting

$$R_{\text{FILTER}} = 9.54 \text{ k}\Omega$$

and

$$C_1 = 4.7 \text{ }\mu\text{F},$$

gives

$$f_c \approx 3.55 \text{ Hz}.$$

Thus,

$$\boxed{f_c \approx 3.5 \text{ Hz}}.$$

This low cutoff frequency strongly reduces normal audio content, including low-frequency bass, while still allowing a sustained DC fault to move the ADC voltage away from its normal centre.

14.16 Optional High-Frequency Capacitor

An optional

$$100 \text{ nF}$$

ceramic capacitor may be placed in parallel with the main $4.7 \text{ }\mu\text{F}$ capacitor.

Its main purpose is high-frequency noise suppression rather than determining the low-frequency DC-detection response.

The 100 nF capacitor should therefore not replace the $4.7 \text{ }\mu\text{F}$ main filter capacitor.

14.17 Behaviour with Normal Audio

A normal music waveform alternates above and below 0 V.

Before filtering, the sensed ADC signal would therefore move both above and below its approximately 1.597 V centre.

The analogue low-pass network suppresses most of this rapidly changing audio component.

Digital averaging in the firmware can then further reduce the remaining AC variation.

The protection system is therefore intended to respond primarily to the slow or average DC component rather than to instantaneous audio voltage.

14.18 Behaviour with Low-Frequency Bass

Strong low-frequency programme material presents an important protection challenge because a large bass waveform can remain on one side of zero for a relatively long fraction of each cycle.

The 4.7 μF main filtering capacitor was selected specifically to provide substantially better rejection of low-frequency audio than the earlier 100 nF value.

The analogue low-pass filter is combined with digital filtering and a persistence requirement so that normal 20–50 Hz programme material does not cause unnecessary loud-speaker disconnection.

14.19 ADC Protection Resistor

R4 is connected in series between the biased sensing node and the STM32 ADC input.

$$R_4 = 4.7 \text{ k}\Omega.$$

Its principal purpose is to limit current during abnormal conditions, especially if the ADC node reaches the conduction voltage of one of the external protection diodes.

R4 therefore provides an additional layer of protection between the high-voltage amplifier environment and the low-voltage STM32 input.

14.20 Clamp Diodes

External clamp diodes are connected from the ADC node toward the 0 V and 3.3 V rails.

Under normal operation, these diodes should remain non-conductive.

If an extreme positive voltage attempts to raise the ADC node beyond the permitted upper range, the upper clamp diode provides a current path toward the positive low-voltage rail.

If an extreme negative voltage attempts to pull the ADC node below ground, the lower clamp diode conducts toward ground.

R4 limits the resulting current.

The protection chain can therefore be represented as

Amplifier Output $\rightarrow R_1 \rightarrow$ Biased Divider $\rightarrow R_4 \rightarrow$ Clamp Network $\rightarrow$ STM32 ADC.

The clamp diodes are intended as a backup protection mechanism. The normal resistor network is designed so that expected amplifier voltages remain within the ADC measurement range without continuous clamp conduction.

14.21 Input Resistor Protection Function

The large

150 k Ω

value of R1 also limits current from the power amplifier into the protection controller.

This is particularly useful during abnormal conditions or if the microcontroller is not powered while the amplifier output remains active.

The current-limiting action of R1 therefore supplements the protection provided by R4 and the clamp diodes.

14.22 Recommended Fault Threshold Concept

The final protection thresholds should be calibrated and validated on the assembled hardware rather than being assumed solely from theoretical component values.

A useful initial decision concept is:

Table 14.3: Recommended conceptual DC fault regions.

Amplifier DC Offset	Interpretation	Controller Action
$ V_{DC} < 0.5 \text{ V}$	Normal region	Continue monitoring
0.5–1.0 V	Suspicious region	Continue filtering and timing
$ V_{DC} > 1.0 \text{ V}$	Potential DC fault	Trip after required persistence

The controller should not disconnect the loudspeakers because of a single ADC sample outside the nominal region.

The decision should combine

Analogue Filtering + Digital Filtering + Persistence Time

before declaring a DC fault.

14.23 Fault Response

If either the left or right amplifier output develops a confirmed dangerous DC condition, both speaker relays are disconnected.

The protection response is therefore

DC Fault on Left OR Right Channel

↓

RELAY_L_CTRL OFF + RELAY_R_CTRL OFF.

The DC fault indicator is then activated.

Disconnecting both channels avoids leaving the amplifier system operating in a partially failed state when one channel has developed a serious fault.

14.24 DC Fault During Startup

The speaker relays remain de-energised when the protection controller first starts.

During the startup delay, the STM32 monitors both DC sensing channels together with the other protection inputs.

If a DC fault is already present at startup, the speaker relays are never energised.

The sequence can therefore be represented as

Power On → Relays OFF → DC Check

followed by either

Safe → Continue Startup Sequence

or

DC Fault → Relays Remain OFF.

14.25 Loss of One Amplifier Supply Rail

The DC detector does not directly measure the positive and negative power-supply rails.

Nevertheless, failure of one amplifier rail can cause the amplifier output to move strongly toward the remaining rail.

The DC sensing network detects the resulting output offset.

This means that many power-stage and supply faults are detected indirectly by monitoring the quantity that is most important for loudspeaker safety: the actual amplifier output voltage.

14.26 Amplifier Clipping

Normal symmetrical clipping should not automatically be interpreted as a DC fault.

Although a clipped waveform is distorted, its average value can remain close to zero if the clipping is approximately symmetrical.

The analogue low-pass filter and firmware averaging therefore allow the ADC centre value to remain near its normal value.

Clipping may still be undesirable for audio and thermal reasons, but it is a different condition from sustained DC offset.

14.27 Ground Reference

The DC sensing circuit measures amplifier output voltage relative to the controller ground reference.

A defined connection between the amplifier reference ground and the protection-controller measurement ground is therefore essential.

The controller ground must be incorporated into the overall star-ground strategy so that high loudspeaker, rectifier, relay, or power-stage currents do not create significant voltage errors in the ADC measurement reference.

Incorrect grounding could cause

$$V_{\text{ADC,error}} = V_{\text{true}} + V_{\text{ground offset}}.$$

Grounding of the protection controller is therefore part of the measurement accuracy as well as the overall amplifier noise design.

14.28 Sensing Circuit Limitation

One limitation of the present sensing topology is that an open connection between the amplifier output and R1 cannot necessarily be detected.

If the sensing connection is interrupted, R2 and R3 still bias the ADC node near its normal centre voltage.

The controller may therefore interpret an open sensing connection as an approximately zero-DC amplifier output.

In this design the sensing connection is implemented as a permanent internal PCB connection rather than a removable external measurement lead, which reduces the practical

likelihood of this failure mode.

Nevertheless, this limitation should be recognised when evaluating the overall protection architecture.

14.29 Component Tolerance and Calibration

The calculated centre voltage of approximately

$$1.597 \text{ V}$$

is a nominal theoretical value.

The actual assembled circuit will be influenced by:

- R1, R2 and R3 tolerance;
- 3.3 V regulator accuracy;
- ADC reference variation;
- PCB ground voltage differences;
- ADC offset and gain error;
- temperature variation.

R1, R2 and R3 are therefore specified as 1% resistors.

The actual healthy centre value should also be measured on the completed PCB and used when validating the firmware thresholds.

14.30 PCB Placement Considerations

The ADC protection components should be positioned close to the STM32 input pins where practical.

In particular, R4 and the clamp diodes should be located near the ADC node so that abnormal current is limited before reaching the microcontroller.

C1 should also be placed close to the sensing and ADC circuitry.

The amplifier-output sensing trace should be kept away from digital switching, relay-coil, transformer, and high-current rectifier paths to reduce coupled noise.

14.31 Validation Procedure

The DC sensing circuit must be verified before expensive loudspeakers are connected to the amplifier.

A suitable test procedure is:

1. Power the protection controller with the speaker relays disabled.
2. Confirm the 3.3 V controller supply.
3. Apply approximately 0 V to each simulated amplifier output input.
4. Verify an ADC sensing voltage close to 1.6 V.
5. Apply a small controlled positive DC voltage and confirm that the ADC voltage rises.
6. Apply a small controlled negative DC voltage and confirm that the ADC voltage falls.
7. Compare measured values with the theoretical transfer equation.
8. Verify the left and right sensing channels separately.
9. Confirm that normal AC test signals are sufficiently attenuated by the low-pass filtering.
10. Inject a controlled persistent DC fault and verify relay disconnection.
11. Verify that a fault on either channel disconnects both relays.
12. Confirm correct DC fault LED operation.
13. Perform protection validation using a dummy load rather than a valuable loud-speaker.

14.32 Design Summary

The DC offset detection circuit provides a safe interface between the high-voltage power-amplifier outputs and the low-voltage STM32G031K8 ADC inputs.

Each stereo channel uses

$$R_1 = 150 \text{ k}\Omega,$$

$$R_2 = R_3 = 10 \text{ k}\Omega,$$

$$R_4 = 4.7 \text{ k}\Omega,$$

and

$$C_1 = 4.7 \text{ }\mu\text{F},$$

with an optional 100 nF high-frequency bypass capacitor.

The resistor network attenuates the amplifier output while biasing the healthy zero-output condition to approximately

$$1.597 \text{ V}.$$

The approximate transfer characteristic is

$$V_{\text{ADC}} \approx 1.597 + 0.03226V_{\text{AMP}}.$$

Consequently, both positive and negative amplifier DC faults can be measured using only positive ADC voltages.

With the actual amplifier supply of approximately $\pm 34 \text{ V}$, a positive full-rail output fault produces approximately

$$2.694 \text{ V}$$

at the ADC, while a negative full-rail fault produces approximately

$$0.500 \text{ V}.$$

The $4.7 \mu\text{F}$ filter capacitor and approximately $9.54 \text{ k}\Omega$ effective source resistance produce a first-order cutoff frequency close to

$$3.5 \text{ Hz},$$

strongly reducing normal audio while preserving sustained DC information.

R4 and the external clamp diodes provide additional ADC protection during abnormal voltage conditions, while R1 limits current from the high-voltage amplifier output.

The protection algorithm should combine analogue filtering, digital filtering, and a persistence timer rather than responding to individual ADC samples.

A confirmed serious DC fault on either amplifier channel causes both loudspeaker relays to be de-energised, thereby isolating both loudspeakers from the amplifier.

The resulting circuit provides the STM32 with a simple, symmetrical, and protected method of detecting positive and negative DC faults while remaining electrically compatible with the approximately 3.3 V ADC domain.

Chapter 15

Temperature Monitoring System

15.1 Introduction

Thermal monitoring forms an important part of the loudspeaker protection and supervision system developed for the power amplifier. The Class-AB output stages can dissipate significant power as heat, particularly during prolonged high-power operation and when driving low-impedance $4\ \Omega$ loudspeaker loads.

Although the primary thermal protection of the output transistors is provided by correct heatsink sizing, thermal mounting, airflow, and bias adjustment, an electronic temperature-monitoring system provides an additional layer of protection against abnormal operating conditions.

Two independent temperature sensors are used in the amplifier, one for the left-channel heatsink and one for the right-channel heatsink. Each sensor is connected to the STM32G031K8 protection controller through an analogue voltage-divider network.

The microcontroller periodically measures both temperatures. If either heatsink exceeds the configured safe operating limit, the controller de-energises the loudspeaker relays and prevents continued loudspeaker operation until the thermal condition has been resolved.

15.2 Temperature Monitoring Architecture

The stereo amplifier uses one temperature-monitoring channel for each power-amplifier heatsink.

The overall architecture can be represented as

Left Heatsink $\rightarrow$ NTC Sensor $\rightarrow$ Voltage Divider $\rightarrow$ STM32 ADC

and

Right Heatsink $\rightarrow$ NTC Sensor $\rightarrow$ Voltage Divider $\rightarrow$ STM32 ADC.

The two channels are evaluated independently by the firmware.

The resulting temperature information is used for three principal purposes:

- detection of excessive heatsink temperature;
- detection of abnormal temperature-sensor conditions;
- supervision of thermal behaviour during amplifier operation.

15.3 Selected Temperature Sensor

The selected temperature sensor is the TDK

B57703M0103A047

NTC thermistor [8].

Two devices are used, one mechanically attached to each amplifier heatsink.

The principal characteristics of the selected sensor are summarised in Table 15.1.

Table 15.1: Principal characteristics of the B57703M0103A047 NTC sensor [8].

Parameter	Value
Sensor type	NTC thermistor
Nominal resistance, R_{25}	10 k Ω
Rated temperature	25 °C
Resistance tolerance	$\pm 3\%$
$B_{25/100}$ value	3988 K
B-value tolerance	$\pm 1\%$
Maximum power rating at 25°C	150 mW
Operating-temperature range	−25 to 125 °C
Mechanical construction	Ring-terminal surface sensor
Intended application	Surface / heatsink temperature measurement

The ring-terminal construction is particularly suitable for this application because the sensor can be mechanically fixed directly to the heatsink rather than being suspended in the surrounding air.

15.4 NTC Operating Principle

NTC stands for *Negative Temperature Coefficient*. The electrical resistance of an NTC thermistor decreases as its temperature increases.

Therefore,

$$T \uparrow \Rightarrow R_{\text{NTC}} \downarrow.$$

At the rated temperature of

$$25\text{ }^{\circ}\text{C},$$

the selected thermistor has a nominal resistance of

$$\boxed{R_{25} = 10\text{ k}\Omega}.$$

As the heatsink becomes hotter, the thermistor resistance progressively falls. The STM32 cannot measure resistance directly, so the thermistor is combined with a fixed resistor to form a voltage divider.

15.5 Resistance–Temperature Relationship

The resistance of an NTC thermistor can be approximated using the Beta equation

$$R(T) = R_0 \exp \left[B \left(\frac{1}{T} - \frac{1}{T_0} \right) \right],$$

where

$$R(T) = \text{thermistor resistance at temperature } T,$$

$$R_0 = \text{thermistor resistance at reference temperature } T_0,$$

and

$$B = \text{thermistor Beta constant}.$$

Temperature must be expressed in kelvin.

For the selected sensor,

$$R_0 = 10\text{ k}\Omega,$$

$$T_0 = 25 + 273.15 = 298.15\text{ K},$$

and

$$B = 3988\text{ K}.$$

Consequently,

$$R(T) = 10\,000 \exp \left[3988 \left(\frac{1}{T} - \frac{1}{298.15} \right) \right].$$

15.6 Thermistor Resistance at Elevated Temperature

The negative temperature coefficient causes the resistance to decrease substantially as the heatsink warms.

Approximate values obtained from the Beta relationship are shown in Table 15.2.

Table 15.2: Approximate resistance of the 10 k Ω , 3988 K NTC thermistor at selected temperatures.

Temperature	Approximate NTC Resistance
25 °C	10.0 k Ω
40 °C	$\approx$ 5.3 k Ω
50 °C	$\approx$ 3.6 k Ω
60 °C	$\approx$ 2.5 k Ω
70 °C	$\approx$ 1.8 k Ω
80 °C	$\approx$ 1.3 k Ω
90 °C	$\approx$ 0.93 k Ω

These values demonstrate that a relatively large and easily measurable resistance change occurs over the expected heatsink operating-temperature range.

15.7 Voltage-Divider Interface

Each 10 k Ω NTC thermistor is used together with a fixed

10 k Ω

resistor to form an analogue voltage divider.

The divider converts the temperature-dependent thermistor resistance into a temperature-dependent voltage that can be measured by the STM32 ADC.

For a generic divider containing the thermistor and a fixed resistor R_F , the node voltage is determined by the relative values of the two resistances.

If the fixed resistor is connected to the supply and the NTC is connected toward ground, the ADC voltage is

$$V_{\text{ADC}} = V_{\text{SUPPLY}} \frac{R_{\text{NTC}}}{R_F + R_{\text{NTC}}}.$$

In this orientation, increasing temperature causes the NTC resistance and therefore the ADC voltage to decrease.

If the physical divider orientation is reversed, the ADC voltage rises with temperature instead. The temperature calculation remains equivalent once the actual circuit orientation is taken into account in firmware.

15.8 Mid-Scale Operation at Room Temperature

The use of a 10 k Ω fixed resistor is appropriate because it matches the nominal thermistor resistance at 25 °C.

For

$$R_F = R_{\text{NTC}} = 10 \text{ k}\Omega,$$

the voltage divider produces approximately one half of the supply voltage.

With a 3.3 V ADC supply,

$$V_{\text{ADC},25} = 3.3 \frac{10}{10 + 10}.$$

Therefore,

$$\boxed{V_{\text{ADC},25} \approx 1.65 \text{ V}}$$

for the divider orientation in which the thermistor forms the lower leg.

This places the room-temperature measurement close to the centre of the ADC range and provides useful measurement range in both directions.

15.9 Temperature Conversion from ADC Measurement

The STM32 first converts the measured ADC code into voltage.

For the 12-bit ADC,

$$V_{\text{ADC}} = \frac{N_{\text{ADC}}}{4095} V_{\text{REF}}.$$

Once the divider voltage is known, the thermistor resistance can be calculated.

For the divider orientation in which R_F is connected to the supply and the thermistor is connected to ground,

$$R_{\text{NTC}} = R_F \frac{V_{\text{ADC}}}{V_{\text{SUPPLY}} - V_{\text{ADC}}}.$$

The calculated resistance can then be converted to temperature by rearranging the Beta equation:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \ln \left(\frac{R_{\text{NTC}}}{R_0} \right).$$

Thus,

$$T = \left[\frac{1}{T_0} + \frac{1}{B} \ln \left(\frac{R_{\text{NTC}}}{R_0} \right) \right]^{-1}.$$

The resulting value is in kelvin and is converted to degrees Celsius using

$$T_{\circ C} = T_K - 273.15.$$

15.10 ADC Resolution

The STM32G031K8 provides a 12-bit ADC.

For an approximate reference voltage of

$$3.3 \text{ V},$$

the ideal ADC voltage resolution is

$$V_{\text{LSB}} = \frac{3.3}{4095}.$$

Therefore,

$$V_{\text{LSB}} \approx 0.806 \text{ mV}.$$

The large resistance change of the NTC over the relevant thermal range therefore produces a voltage change that can be resolved comfortably by the ADC.

The practical temperature accuracy is determined not only by ADC resolution but also by thermistor tolerance, Beta tolerance, fixed-resistor tolerance, reference-voltage variation, thermal contact, and sensor placement.

15.11 Analogue Filtering

Each NTC measurement channel includes local analogue filtering using a

$$100 \text{ nF}$$

capacitor.

Temperature changes occur relatively slowly, so high-frequency electrical content at the ADC input does not contain useful thermal information.

The capacitor therefore helps reduce interference from

- relay switching,
- digital activity,

- amplifier output currents,
- fan wiring,
- rectifier switching,
- external electromagnetic interference.

Additional digital averaging can also be applied in firmware because rapid sample-to-sample changes should not correspond to genuine heatsink temperature changes.

15.12 Thermal Response Time

The measured heatsink temperature cannot change instantaneously.

Several thermal time constants are involved:

Transistor Junction $\rightarrow$ Transistor Case $\rightarrow$ Heatsink $\rightarrow$ Temperature Sensor.

The sensor therefore measures the temperature of the heatsink surface rather than the instantaneous semiconductor junction temperature.

This distinction is important. A rapid transistor junction-temperature rise may occur before the entire heatsink reaches the same temperature.

The temperature-monitoring system is therefore an additional protective and supervisory layer and does not replace correct semiconductor safe-operating-area design or proper heatsinking.

15.13 Sensor Mounting

The B57703M0103A047 uses a ring-terminal mechanical construction intended for surface-temperature measurement.

Each sensor should be firmly attached to the heatsink of the corresponding amplifier channel.

The preferred mounting location is close to the power output transistors while remaining mechanically secure and electrically safe.

The sensor should not simply measure the temperature of the internal chassis air because the heatsink surface provides a more direct indication of output stage thermal loading.

Good thermal contact is important because poor contact introduces additional thermal delay and causes the measured temperature to underestimate the actual heatsink temperature.

15.14 Left and Right Channel Monitoring

The two amplifier channels are monitored independently.

The firmware therefore maintains two measured temperatures:

$$T_L$$

for the left heatsink and

$$T_R$$

for the right heatsink.

Normal operation requires both channels to remain within the permitted temperature region.

The protection condition can be represented as

$$T_L < T_{\text{trip}}$$

and

$$T_R < T_{\text{trip}}.$$

If either condition is violated, a thermal fault is declared.

15.15 Overtemperature Threshold

The final thermal trip temperature is implemented as a firmware parameter,

$$T_{\text{trip}}.$$

A fixed numerical shutdown temperature is not assumed before the physical amplifier has been thermally characterised.

The final value should be selected after measuring:

- output-transistor case temperature,
- heatsink temperature,
- sensor temperature,
- ambient temperature,
- bias stability,
- behaviour under 4 Ω and 8 Ω loading,
- operation with and without forced-air cooling.

This allows the programmed threshold to be based on the actual thermal behaviour of the constructed amplifier rather than an arbitrary assumed temperature.

15.16 Overtemperature Fault Response

If either measured heatsink temperature exceeds the configured trip temperature,

$$T_L \geq T_{\text{trip}}$$

or

$$T_R \geq T_{\text{trip}},$$

the STM32 enters the overtemperature protection state.

The protection sequence is

Excessive Temperature $\rightarrow$ Thermal Fault Confirmed $\rightarrow$ Speaker Relays OFF.

Both loudspeakers are disconnected rather than allowing the unaffected channel to continue operating in a partially faulted amplifier system.

The appropriate fault indication is also activated.

15.17 Temperature Hysteresis

A single temperature threshold for both disconnection and reconnection can cause repeated relay switching if the heatsink temperature remains close to the threshold.

For example, the amplifier could disconnect at the trip temperature, cool slightly, reconnect, immediately heat again, and disconnect repeatedly.

To prevent this behaviour, thermal hysteresis should be implemented.

Two temperatures can therefore be defined:

$$T_{\text{trip}}$$

and

$$T_{\text{reset}},$$

with

$$T_{\text{reset}} < T_{\text{trip}}.$$

The amplifier disconnects when

$$T \geq T_{\text{trip}},$$

but is not considered thermally recovered until

$$T \leq T_{\text{reset}}.$$

This provides a deliberate cooling margin before normal operation can resume.

15.18 Persistence and Digital Filtering

Temperature is a slowly changing physical quantity, so a single unusual ADC sample should not immediately be interpreted as genuine overheating.

The firmware can apply averaging or another low-pass digital filter to the ADC measurements.

A thermal fault can then be confirmed only after the calculated temperature has remained beyond the permitted region for a defined interval.

The decision process can be represented as

ADC Samples → Digital Filtering → Temperature Calculation → Threshold Comparison → Fault Dec.

15.19 Temperature-Sensor Fault Detection

The protection controller also supervises the validity of the temperature sensors themselves.

Possible sensor-related faults include:

- open thermistor connection,
- short-circuited thermistor,
- broken cable,
- disconnected terminal,
- ADC value outside the physically plausible range.

An open- or short-circuit condition drives the divider output toward an extreme value determined by the divider orientation.

The firmware can therefore define a valid ADC range corresponding to plausible thermistor resistances and temperatures.

A reading outside this range is treated as a sensor fault rather than simply as an unusual temperature.

15.20 Open-Circuit Sensor

If the thermistor or its wiring becomes open circuit, the voltage-divider node is no longer controlled by the expected thermistor resistance.

The remaining fixed resistor tends to pull the ADC input toward one of the supply rails, depending on the physical divider orientation.

The resulting ADC value should lie outside the normal temperature measurement region.

The controller can therefore identify the condition as

TEMP SENSOR FAIL

.

The speaker relays should remain disconnected when sensor integrity cannot be confirmed.

15.21 Short-Circuit Sensor

A short circuit across the thermistor produces the opposite extreme divider condition.

Again, the ADC reading approaches one end of its available range rather than a normal temperature-dependent value.

The firmware can distinguish this from normal heatsink operation by checking whether the calculated resistance and temperature are physically plausible.

Both open- and short-circuit sensor conditions should therefore be treated as protection faults.

15.22 Sensor Fault Indicators

Separate status indicators are provided for the two temperature channels.

The protection controller includes

TEMP SENSOR 1 FAIL

and

TEMP SENSOR 2 FAIL

indications.

The corresponding front-panel outputs are assigned to STM32 GPIOs PB8 and PB9 respectively.

These indicators allow a failed temperature-monitoring channel to be distinguished from a DC-output fault or normal startup delay.

15.23 Startup Temperature Check

Temperature monitoring begins before the loudspeaker relays are enabled.

During startup, the controller verifies that both temperature channels produce valid readings.

The sequence is therefore

Power Applied $\rightarrow$ MCU Initialisation $\rightarrow$ Temperature Sensors Checked $\rightarrow$ Startup Delay $\rightarrow$ Speaker C

If either temperature sensor is invalid or an overtemperature condition is already present, the relays remain de-energised.

15.24 Interaction with Speaker Protection

Temperature monitoring forms one input to the overall protection state machine.

Speaker connection is permitted only when

$$\text{TemperatureSafe}_L = 1$$

and

$$\text{TemperatureSafe}_R = 1.$$

Together with the other protection conditions, the relay-enable decision can be expressed conceptually as

$$\text{RelayEnable} = \text{DCSafe} \wedge \text{TemperatureSafe} \wedge \text{SensorsValid} \wedge \text{StartupComplete}.$$

A thermal or temperature-sensor failure therefore removes permission for the speaker relays to remain energised.

15.25 Relationship to Cooling Fans

The amplifier incorporates forced-air cooling in addition to the passive heatsinks.

The temperature measurements provide useful information for evaluating the effectiveness of this cooling system during testing.

The thermal response can be measured under conditions such as

Natural Convection

and

Forced-Air Cooling.

A future firmware version could also use the measured temperature to control fan operation automatically.

For example, different temperature regions could be used for

Fans OFF,

Fans ON,

and

Thermal Shutdown.

This is an optional future extension rather than a requirement of the present protection architecture.

15.26 Difference Between Heatsink and Junction Temperature

The NTC sensor measures heatsink surface temperature and does not directly measure the internal junction temperature of the 2SC5200-O and 2SA1943-O power transistors.

In general,

$$T_J > T_C > T_H$$

while the amplifier is dissipating significant power, where

T_J = transistor junction temperature,

T_C = transistor case temperature,

and

T_H = heatsink temperature.

The exact temperature differences depend on transistor dissipation and the thermal resistances of the transistor package, insulating interface, mounting system, and heatsink.

Consequently, the firmware trip temperature must not simply be set equal to the maximum permissible semiconductor junction temperature.

A significant safety margin is required because the heatsink sensor responds more slowly and measures a cooler point than the transistor junction.

15.27 Importance During 4 Ohm Operation

Thermal monitoring is particularly important when the amplifier drives a

$$4\ \Omega$$

load.

The lower load impedance requires substantially greater output current than an

$$8\ \Omega$$

load for a comparable output voltage.

Higher output current increases stress and dissipation in the output stage and therefore increases heatsink temperature.

The temperature-monitoring system provides an additional means of detecting conditions in which sustained output power, inadequate airflow, or abnormal bias causes the heatsink temperature to approach an unsafe region.

15.28 Sensor Self-Heating

Current flowing through an NTC thermistor produces a small amount of internal power dissipation.

This effect can cause the sensor to heat itself slightly and introduce measurement error.

For this reason, the divider current should remain small.

With resistances in the approximately 10 k Ω region and a 3.3 V measurement supply, the divider current is small and sensor self-heating is correspondingly limited.

The selected sensor has a maximum power rating of

$$150\ \text{mW}$$

at 25 °C under the manufacturer's specified conditions [8], while the sensing divider operates at a much lower power level.

15.29 Measurement Accuracy

The calculated temperature should not be interpreted as an ideal exact value.

Sources of uncertainty include:

- thermistor resistance tolerance,
- Beta-value tolerance,

- fixed-resistor tolerance,
- ADC gain and offset error,
- supply/reference-voltage tolerance,
- PCB ground error,
- sensor mounting pressure,
- thermal interface quality,
- position of the sensor on the heatsink.

For thermal protection, repeatable and conservative threshold behaviour is more important than achieving laboratory-grade temperature measurement accuracy.

15.30 Calibration

The temperature channels should be checked against a known temperature measurement during prototype testing.

A reference thermometer or thermocouple can be placed close to the NTC sensor while the heatsink temperature is varied.

The STM32 calculated temperature can then be compared with the independent measurement.

If required, a calibration offset or correction can be incorporated into the firmware.

15.31 Validation Procedure

The temperature-monitoring system should be tested independently before being relied upon for amplifier protection.

A suitable validation procedure is:

1. Verify both 10 k Ω NTC sensors with a multimeter at room temperature.
2. Confirm that the resistance is approximately 10 k Ω near 25 °C.
3. Warm each sensor carefully and verify that its resistance decreases.
4. Check the corresponding ADC voltage for each temperature channel.
5. Confirm that the calculated temperature changes in the correct direction.
6. Compare the STM32 temperature reading with an independent thermometer.
7. Disconnect TEMP1 and verify that the controller identifies a sensor fault.
8. Repeat the open-circuit test for TEMP2.

9. Simulate a short-circuit sensor condition and verify that it is detected as invalid.
10. Apply controlled heating to one heatsink.
11. Verify correct detection of the programmed overtemperature condition.
12. Confirm that both loudspeaker relays are de-energised during a thermal fault.
13. Confirm correct temperature-fault LED indication.
14. Allow the heatsink to cool and verify the programmed hysteresis and recovery behaviour.

15.32 Thermal Testing of the Complete Amplifier

After individual sensor validation, the complete amplifier should be tested under realistic thermal loading.

Relevant tests include:

- idle operation;
- moderate output power into $8\ \Omega$;
- sustained operation into $8\ \Omega$;
- controlled operation into $4\ \Omega$;
- operation with forced-air cooling;
- operation with fans disabled for comparison;
- elevated ambient-temperature testing where practical.

The temperatures of the left and right heatsinks should be recorded together with amplifier output power and ambient temperature.

These measurements will provide the experimental basis for selecting the final firmware thermal threshold.

15.33 Design Summary

The temperature-monitoring system provides two independent thermal measurement channels, one for each power-amplifier heatsink.

Each channel uses a TDK

B57703M0103A047

ring-terminal NTC thermistor.

The selected sensor has a nominal resistance of

$$10 \text{ k}\Omega$$

at

$$25 \text{ }^\circ\text{C},$$

and a Beta value of

$$B_{25/100} = 3988 \text{ K}.$$

As heatsink temperature rises, thermistor resistance decreases.

Each sensor is combined with a $10 \text{ k}\Omega$ fixed resistor to form a voltage divider whose output is measured using an STM32 ADC channel.

A 100 nF capacitor provides local filtering because heatsink temperature changes much more slowly than electrical switching noise.

The STM32 converts the measured divider voltage to thermistor resistance and then estimates temperature using the NTC resistance–temperature relationship.

The left and right temperatures are evaluated independently. If either channel exceeds the configured thermal limit,

$$T \geq T_{\text{trip}},$$

the protection controller de-energises both loudspeaker relays.

The controller also detects abnormal sensor conditions such as open or short-circuited thermistors and prevents speaker connection when valid temperature supervision is unavailable.

The final trip and recovery temperatures are not assumed before hardware testing. They will be established after the completed amplifier, heatsinks, fans, output transistors, and sensor mounting arrangement have been thermally characterised.

The temperature-monitoring system therefore complements the amplifier’s passive thermal design by providing continuous electronic supervision of both heatsinks and allowing the STM32 protection controller to disconnect the loudspeakers when abnormal thermal conditions are detected.

Chapter 16

Speaker Relay Control

16.1 Introduction

The loudspeaker relays form the final switching element of the STM32-based protection system. Their purpose is to physically connect or disconnect the left and right loudspeakers from the corresponding power amplifier outputs.

Under normal operation, the relays remain energised and the amplifier outputs are connected to the loudspeakers. During startup, reset, loss of controller power, DC fault, thermal fault, or another protection condition, the relays are de-energised and the loudspeakers are disconnected.

Two independent Omron G2RL-1A-E-DC12 relays are used, with one relay assigned to each audio channel. Each relay has a 12 V DC coil and a normally-open single-pole contact arrangement suitable for switching the loudspeaker output path [9].

The STM32G031K8 cannot drive the relay coils directly because its GPIO pins are designed for low-current logic signals. Each relay therefore uses an external BC337 NPN transistor as a low-side coil driver.

16.2 Relay Selection

The selected speaker relay is

G2RL-1A-E-DC12

from Omron.

Two relays are used:

Relay_L

for the left channel and

Relay_R

for the right channel.

The relay was selected because it provides a 12 V DC coil, high-current contacts, through-hole PCB mounting, and a normally-open contact arrangement.

Table 16.1: Principal relay characteristics [9].

Parameter	Value
Relay model	G2RL-1A-E-DC12
Coil voltage	12 VDC
Contact arrangement	SPST-NO
Rated coil current	approximately 33.3 mA
Coil resistance	approximately 360 Ω
Coil power	approximately 400 mW
Contact current class	16 A
Number used	2

16.3 Relay Position in the Audio Path

The relay contact is placed in series between each power-amplifier output and the corresponding loudspeaker terminal.

For the left channel,

Left Amplifier Output $\rightarrow$ Relay_L $\rightarrow$ Left Loudspeaker.

For the right channel,

Right Amplifier Output $\rightarrow$ Relay_R $\rightarrow$ Right Loudspeaker.

When the relay coil is energised, the normally-open contact closes and the audio signal is allowed to pass to the loudspeaker.

When the relay coil is de-energised, the contact opens and interrupts the speaker-current path.

16.4 Normally-Open Protection Philosophy

A normally-open relay contact is advantageous for a protection system because the loudspeaker remains disconnected when no relay power is applied.

The default hardware condition is therefore

$$\boxed{\text{Relay Coil OFF} \Rightarrow \text{Speaker Disconnected}}.$$

The loudspeaker is connected only when the STM32 actively determines that the system is operating safely.

This provides a fail-safe characteristic for conditions such as

- initial power-up,
- controller reset,
- loss of STM32 power,
- firmware watchdog reset,
- DC output fault,
- excessive temperature,
- temperature-sensor fault.

Consequently,

Unknown or Fault Condition $\Rightarrow$ Relay OFF.

16.5 Need for a Relay Driver Transistor

The relay coil requires substantially more current than should be supplied directly by an STM32 GPIO output.

The 12 V relay coil requires approximately

$$I_{\text{coil}} \approx 33.3 \text{ mA}.$$

The STM32 therefore provides only a logic-level control signal. A separate BC337 transistor supplies the actual relay current.

The control chain is

STM32 GPIO $\rightarrow$ Base Resistor $\rightarrow$ BC337 $\rightarrow$ Relay Coil.

The transistor operates primarily as a switch rather than as a linear amplifier.

16.6 BC337 Relay Driver

The selected driver transistor is

BC33716BU.

The BC337 is an NPN bipolar transistor in a TO-92 through-hole package [10].

The device has a continuous collector-current capability substantially greater than the approximately 33.3 mA required by one relay coil.

The basic low-side switching arrangement is

$$+12\text{ V} \rightarrow \text{Relay Coil} \rightarrow \text{BC337 Collector} \rightarrow \text{BC337 Emitter} \rightarrow \text{GND}.$$

The emitter of the BC337 is connected to controller ground, while the collector is connected to the low side of the relay coil.

The high side of the relay coil is connected directly to the regulated +12 V protection-controller supply.

16.7 BC337 Current Capability

The BC337 has a specified continuous collector-current capability of

$$I_C = 800\text{ mA}$$

[10].

The relay requires only approximately

$$I_{\text{coil}} = 33.3\text{ mA}.$$

The ratio is therefore

$$\frac{800}{33.3} \approx 24.$$

The transistor current rating is therefore approximately 24 times greater than the normal relay-coil current.

This provides substantial current margin for the switching application.

16.8 Base Resistor

A resistor is placed between the STM32 GPIO output and the BC337 base.

The selected value is

$$R_B = 1\text{ k}\Omega.$$

The resistor limits the current drawn from the GPIO pin and establishes the base drive of the transistor.

Using an approximate GPIO high voltage of

$$V_{\text{GPIO}} = 3.3\text{ V}$$

and an approximate base-emitter voltage of

$$V_{BE} \approx 0.7 \text{ V},$$

the base current is approximately

$$I_B = \frac{V_{\text{GPIO}} - V_{BE}}{R_B}.$$

Therefore,

$$I_B = \frac{3.3 - 0.7}{1000}.$$

Thus,

$$\boxed{I_B \approx 2.6 \text{ mA}}.$$

This provides strong base drive for switching the approximately 33 mA relay-coil current.

16.9 Forced Current Gain

The effective forced current gain required when the relay is energised can be estimated from

$$\beta_{\text{forced}} = \frac{I_C}{I_B}.$$

Using

$$I_C \approx 33.3 \text{ mA}$$

and

$$I_B \approx 2.6 \text{ mA},$$

gives

$$\beta_{\text{forced}} \approx \frac{33.3}{2.6} \approx 12.8.$$

Thus,

$$\boxed{\beta_{\text{forced}} \approx 13}.$$

This is intentionally much lower than the normal small-signal transistor gain and is appropriate when the BC337 is used as a saturated switch.

16.10 Base Pulldown Resistor

A

$$10 \text{ k}\Omega$$

resistor is connected between the BC337 base and ground.

This resistor ensures that the transistor remains off when the STM32 GPIO is in a high-impedance state, such as during reset or initial microcontroller configuration.

Without the pulldown resistor, the transistor base could float and electrical noise might cause partial or unintended relay activation.

The desired reset behaviour is therefore

$$\text{GPIO High Impedance} \rightarrow \text{Base Pulled Low} \rightarrow \text{BC337 OFF} \rightarrow \text{Relay OFF}.$$

The pulldown resistor is therefore an important part of the hardware fail-safe behaviour.

16.11 Relay Coil Current

The nominal relay-coil current can also be verified from Ohm's law.

For a coil resistance of approximately

$$R_{\text{coil}} = 360 \text{ }\Omega$$

and a coil voltage of

$$V_{\text{coil}} = 12 \text{ V},$$

the current is

$$I_{\text{coil}} = \frac{V_{\text{coil}}}{R_{\text{coil}}}.$$

Therefore,

$$I_{\text{coil}} = \frac{12}{360} = 0.0333 \text{ A}.$$

Hence,

$$I_{\text{coil}} \approx 33.3 \text{ mA}.$$

16.12 Relay Coil Power

The approximate electrical power consumed by one relay coil is

$$P_{\text{coil}} = V_{\text{coil}} I_{\text{coil}}.$$

Therefore,

$$P_{\text{coil}} = 12 \times 0.0333.$$

Thus,

$$P_{\text{coil}} \approx 0.40 \text{ W}.$$

With both stereo relays energised, the total relay-coil consumption is approximately

$$P_{\text{relay, total}} \approx 0.8 \text{ W}.$$

This load must be included when considering the power requirement of the 12 V protection-controller supply.

16.13 Inductive Flyback

The relay coil is an inductive component.

While current flows through the coil, energy is stored in its magnetic field.

The stored energy can be represented by

$$E_L = \frac{1}{2} L I^2.$$

When the BC337 is suddenly switched off, the coil current cannot fall instantaneously.

From the inductor relationship,

$$V_L = L \frac{dI}{dt},$$

a rapid reduction in current can produce a large voltage across the coil.

Without protection, this voltage could exceed the transistor's allowable collector-emitter voltage and generate significant electromagnetic interference.

16.14 Flyback Diode

A 1N4007 diode is connected directly across each relay coil.

The selected device is

$$\boxed{1N4007\text{-E3/54}}.$$

During normal relay operation, the diode is reverse biased and does not conduct.

The diode orientation is

$$\boxed{\text{Cathode} \rightarrow +12 \text{ V}}$$

and

$$\boxed{\text{Anode} \rightarrow \text{BC337 Collector} / \text{Coil Low Side}}.$$

When the transistor switches off, the coil voltage reverses and forward biases the diode.

The remaining coil current then circulates through the diode and coil rather than generating a large voltage spike.

The current path during turn-off becomes

$$\text{Relay Coil} \rightarrow \text{Flyback Diode} \rightarrow \text{Relay Coil}.$$

The stored magnetic energy is then dissipated gradually in the coil resistance and diode.

16.15 Importance of Flyback-Diode Orientation

Correct diode orientation is essential.

If the diode were installed in the opposite direction, it would become forward biased whenever the 12 V relay supply was applied.

This would effectively place a very low resistance path across the relay supply and could damage the diode, regulator, wiring, or PCB.

The diode must therefore remain reverse biased while the relay is normally energised.

16.16 Relay Driver Circuit

The complete relay-control circuit for each channel can be represented conceptually as

$$+12 \text{ V} \rightarrow \text{Relay Coil} \rightarrow Q_{\text{DRV}} \rightarrow \text{GND},$$

where

$$Q_{\text{DRV}} = \text{BC337}.$$

The MCU side is

STM32 GPIO $\rightarrow$ 1 k Ω $\rightarrow$ BC337 Base.

A

10 k Ω

resistor connects the base to ground, while the 1N4007 diode is connected across the coil. Thus, each relay driver consists of

- one Omron G2RL-1A-E-DC12 relay;
- one BC33716BU transistor;
- one 1 k Ω base resistor;
- one 10 k Ω base pulldown resistor;
- one 1N4007-E3/54 flyback diode.

16.17 Relay ON Operation

When the protection controller determines that amplifier operation is safe, the corresponding STM32 relay-control GPIO is driven high.

The sequence is

GPIO HIGH $\rightarrow$ Base Current $\rightarrow$ BC337 ON $\rightarrow$ Coil Current $\rightarrow$ Relay Energised.

When energised, the normally-open relay contact closes and connects the amplifier output to the corresponding loudspeaker.

The system therefore enters the state

Protection Healthy $\Rightarrow$ Relay ON $\Rightarrow$ Speaker Connected

16.18 Relay OFF Operation

When the MCU removes the GPIO drive, the BC337 base current falls to zero.

The base pulldown resistor ensures that the transistor remains off.

The relay coil current then falls through the flyback-diode path and the relay returns to its normally-open mechanical state.

The sequence is therefore

GPIO LOW $\rightarrow$ BC337 OFF $\rightarrow$ Relay Coil De-energised $\rightarrow$ Relay Contact Open $\rightarrow$ Speaker Disconnect

16.19 Power-On Behaviour

The loudspeaker relays must not energise immediately when power is first applied.

At startup, the STM32 GPIO outputs are configured so that both BC337 driver transistors remain off.

The relay state is therefore

OFF

during MCU initialisation and the programmed startup delay.

Only after the controller has verified that the DC sensing channels, temperature sensors, and other required protection conditions are valid may the relay-control outputs be activated.

This prevents amplifier startup transients from reaching the loudspeakers.

16.20 Fault Response

During normal operation, the protection firmware continuously monitors the two amplifier channels.

If a confirmed protection fault occurs, relay drive is removed.

For a serious DC fault, the selected system behaviour is to de-energise both speaker relays even if the problem is detected on only one amplifier channel.

The sequence is

$$\text{DC Fault} \rightarrow \text{RELAY_L_CTRL} = 0$$

and

$$\text{RELAY_R_CTRL} = 0.$$

The same general disconnected condition is used for serious thermal or monitoring faults.

This prevents the system from continuing to operate in a partially failed state.

16.21 MCU Reset or Power Loss

The relay-control hardware is intentionally designed so that loss of MCU control causes the loudspeakers to disconnect.

If the STM32 resets,

$$\text{GPIO inactive} \rightarrow \text{BC337 OFF} \rightarrow \text{Relay OFF}.$$

If the STM32 loses power,

No GPIO Drive $\rightarrow$ Relay OFF.

If the watchdog resets the microcontroller,

MCU Reset $\rightarrow$ Relay OFF.

The speakers therefore remain connected only while the controller is actively operating and has confirmed that the system is healthy.

16.22 Relay Contact Current

The relay contacts carry the power-amplifier loudspeaker current.

For a $4\ \Omega$ load at the Project 3A reference condition of approximately 100 W,

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5\ \text{A},$$

while the peak current is

$$I_{\text{peak}} = 5\sqrt{2} \approx 7.07\ \text{A}.$$

The selected relay belongs to the 16 A contact-current class, providing useful current margin relative to the expected loudspeaker current.

The audio current rating nevertheless depends on the actual switching conditions, contact temperature, PCB track resistance, connector resistance, and contact ageing.

16.23 Contact Resistance

Any mechanical relay contact has a small but finite resistance.

The resulting voltage drop is

$$V_{\text{contact}} = I_{\text{speaker}} R_{\text{contact}},$$

while the power dissipated at the contact is

$$P_{\text{contact}} = I_{\text{speaker}}^2 R_{\text{contact}}.$$

For this reason, the speaker relay must use low-resistance power contacts and the PCB tracks connected to the relay should also be sized for the expected speaker current.

The relay should not be connected through narrow signal-level PCB traces.

16.24 Relay Placement in the PCB Layout

The speaker relays should be located so that the high-current audio path between the power amplifier, relay contacts, and speaker terminals remains short.

The relay coil-control traces carry relatively small current and should be treated separately from the speaker-current path.

The PCB therefore contains two conceptually distinct relay circuits:

Low-Current Control Side

and

High-Current Contact Side.

The control side contains the STM32, base resistor, BC337, coil, and flyback diode.

The contact side contains the amplifier output and loudspeaker connection.

16.25 Grounding Considerations

Relay coil current should return through the protection-controller power ground rather than through sensitive analogue signal-ground conductors.

Although the coil current is only several tens of milliamperes, switching it produces transient current changes.

Incorrect grounding could allow these transients to couple into the ADC measurement reference or audio signal path.

The relay-driver ground should therefore be integrated into the planned controller and system star-ground architecture.

16.26 Flyback Diode and Release Time

The flyback diode greatly reduces the voltage generated when the relay coil is switched off.

One consequence is that the coil current decays more slowly than it would if the inductive voltage were allowed to rise to a larger controlled value.

This can slightly increase the mechanical release time of the relay.

For the present protection system, the simplicity and transistor protection provided by the 1N4007 flyback diode are considered appropriate.

The actual speaker-disconnection time should nevertheless be verified during protection-system testing.

16.27 Avoiding Relay Chatter

The firmware should not repeatedly energise and de-energise the relays when a measured parameter remains close to a protection threshold.

Such behaviour is known as relay chatter.

The protection algorithms therefore use appropriate persistence, filtering, and hysteresis before allowing relay-state transitions.

For example, after an overtemperature fault, the relay should not immediately reconnect as soon as the measured temperature falls by a very small amount.

The detailed timing and state logic are discussed in the *Fault Detection and Protection Logic* chapter.

16.28 Audio Switching Considerations

The loudspeaker relay should normally change state when the power-amplifier output is close to a safe condition.

Disconnecting a large audio current can produce contact arcing and mechanical stress.

The principal purpose of the protection relay is therefore fault isolation, startup muting, and speaker protection rather than frequent audio switching.

The relay is expected to remain continuously energised during normal listening and to operate only during startup, shutdown, or a protection event.

16.29 Relay Contact Wear

Mechanical relay contacts have a finite lifetime.

Contact wear can be accelerated by

- high switching current,
- switching during amplifier clipping,
- repeated fault cycling,
- DC current,
- contact arcing,
- contamination,
- excessive operating temperature.

The protection firmware should therefore avoid unnecessary relay operation.

The relay should not be used as a normal volume muting device if repeated switching is not required.

16.30 Relay Failure Considerations

Although the relay is a protection component, the relay itself can also fail.

Possible failure modes include

- open coil,
- welded contacts,
- increased contact resistance,
- mechanical sticking,
- damaged driver transistor,
- failed flyback diode,
- broken PCB connection.

The normally-open arrangement provides useful protection against several failure modes because loss of coil power or an open driver path leaves the speaker disconnected.

A welded contact, however, would prevent electronic disconnection of that channel. This illustrates why the relay is one part of the complete protection system rather than an absolute guarantee against every possible fault.

16.31 Testing the Relay Driver

The relay-driver circuit should be verified before connecting valuable loudspeakers.

A suitable initial procedure is:

1. Verify the 12 V relay supply.
2. Confirm the orientation of both BC337 transistors.
3. Verify the 1 k Ω base resistors.
4. Verify the 10 k Ω base pulldown resistors.
5. Confirm the polarity of both 1N4007 flyback diodes.
6. Verify that the relay coils are approximately 360 Ω .
7. Power the controller with the speaker outputs disconnected.
8. Confirm that both relays remain OFF during MCU reset.
9. Command the left relay ON and verify correct operation.
10. Command the right relay ON and verify correct operation.

11. Measure the coil voltage while energised.
12. Verify that the BC337 transistors do not show abnormal heating.
13. Remove the relay command and verify reliable release.
14. Simulate a DC fault and confirm that both relays open.
15. Simulate an overtemperature condition and confirm relay disconnection.

16.32 Testing the High-Current Contact Path

After the low-current driver circuitry has been verified, the relay contact path should be tested independently.

Continuity should be absent when the relay is de-energised and present only when the coil is deliberately energised.

The resistance of the complete high-current path should be checked for abnormal connections before a loudspeaker is attached.

Initial amplifier protection testing should be performed with a dummy load rather than an expensive loudspeaker.

16.33 Design Summary

The stereo speaker-protection system uses two Omron

G2RL-1A-E-DC12

relays, with one device controlling each loudspeaker channel.

Each relay has a 12 V coil with an approximate resistance of

$360\ \Omega$,

requiring approximately

$33.3\ \text{mA}$

and consuming approximately

$0.40\ \text{W}$.

The STM32 does not supply this current directly. Each relay is driven by a

BC33716BU

NPN transistor configured as a low-side switch.

A

$$1\text{ k}\Omega$$

base resistor limits GPIO current and provides approximately

$$2.6\text{ mA}$$

of transistor base drive from a nominal 3.3 V GPIO output.

A

$$10\text{ k}\Omega$$

base pulldown resistor ensures that the BC337 remains switched off during MCU reset or when the GPIO is in a high-impedance state.

A

$$1\text{N4007}$$

flyback diode is connected across each relay coil, with its cathode connected to +12 V and its anode connected to the transistor collector.

The diode suppresses the inductive voltage generated when the relay is de-energised and protects the BC337 and surrounding electronics.

The use of normally-open relay contacts gives the protection system a fail-safe default condition:

$$\boxed{\text{No Active Relay Drive} \Rightarrow \text{Speaker Disconnected}}.$$

During normal listening, the STM32 actively energises both relays only after the startup delay has completed and all monitored protection conditions are valid.

A confirmed DC fault, thermal fault, sensor fault, MCU reset, or loss of controller power removes relay drive and disconnects the loudspeakers.

The relay stage therefore provides the physical isolation mechanism that allows the low-power STM32 protection logic to control the high-current connection between the power amplifiers and the loudspeakers.

Chapter 17

Power-On Delay and Speaker Connection Sequence

17.1 Introduction

The loudspeakers are not connected to the power-amplifier outputs immediately when the amplifier is switched on. Instead, the STM32-based protection controller maintains both speaker relays in the disconnected state while the power supplies, analogue amplifier stages, bias conditions, sensing circuits, and microcontroller itself are allowed to stabilise.

After this initial delay, the controller verifies that the monitored operating conditions are safe. Only then are the left and right speaker relays energised.

The sequence therefore provides two important functions. First, it prevents turn-on transients from reaching the loudspeakers. Second, it prevents the loudspeakers from being connected if a fault is already present when the amplifier is powered.

17.2 Reason for the Power-On Delay

When mains power is first applied, the complete amplifier does not instantaneously reach its normal steady-state operating condition.

The main reservoir capacitors begin charging, the positive and negative power rails rise toward their normal values, the power-amplifier transistor stages establish their DC operating points, and the regulated low-voltage supplies for the control electronics stabilise.

During this period, temporary voltage disturbances may appear at the amplifier outputs.

If a loudspeaker were connected during these transients, the resulting output could be reproduced as an audible turn-on “thump” or, under an abnormal condition, could apply an undesirable voltage to the loudspeaker.

The protection controller therefore intentionally delays speaker connection until the system has reached a valid operating state.

17.3 Default Relay State

The speaker relays used in the protection system have normally-open contacts.

Therefore, with no coil current,

$$\boxed{\text{Relay De-energised} \Rightarrow \text{Speaker Disconnected}}$$

This is the default condition during startup.

The BC337 relay-driver transistors are also provided with base pulldown resistors. Consequently, while the STM32 GPIO pins are still unconfigured or in a high-impedance reset state, the relay drivers remain switched off.

The hardware therefore supports the required fail-safe startup condition even before the firmware begins normal execution.

17.4 Initial Power Application

When the amplifier is switched on, the overall system begins from the condition

$$t = 0.$$

At this moment, both speaker relays are de-energised:

$$\text{Relay}_L = 0$$

and

$$\text{Relay}_R = 0.$$

The loudspeakers are therefore electrically isolated from the two power-amplifier outputs.

At the same time, the main amplifier supply and protection-controller supply begin rising toward their normal operating voltages.

17.5 Microcontroller Reset and Initialisation

The STM32G031K8 remains in its reset or startup condition until its supply voltage reaches a level suitable for reliable operation.

Once the microcontroller begins execution, the firmware configures the peripherals required by the protection system.

These include the GPIO outputs, analogue-to-digital converter channels, timing functions, status indicators, and relay-control outputs.

During this entire process, the relay-control outputs remain in the speaker-disconnected state.

The required behaviour is

$$\boxed{\text{MCU Initialisation} \Rightarrow \text{Speaker Relays OFF}}$$

17.6 Beginning of the Startup Delay

After the microcontroller has completed its essential initialisation, the power-on delay interval begins.

The delay duration is represented by

$$t_{\text{delay}}.$$

For the interval

$$0 < t < t_{\text{delay}},$$

both speaker relays remain de-energised.

The delay allows the analogue amplifier and sensing circuitry additional time to settle before the loudspeaker connection is permitted.

17.7 Monitoring During the Delay

The power-on delay is not treated simply as a period during which the microcontroller waits without performing any useful operation.

During this interval, the protection controller can already begin monitoring the important system quantities.

These include the left and right amplifier DC sensing channels and the two temperature-monitoring channels.

The system therefore evaluates quantities such as

$$V_{\text{DC,L}},$$

$$V_{\text{DC,R}},$$

$$T_L,$$

and

$$T_R.$$

This means that a fault present at startup can be detected before the loudspeaker is ever connected.

17.8 Safe Startup Conditions

Completion of the time delay alone is not sufficient to energise the speaker relays.

At the end of the delay, all required protection conditions must indicate safe operation.

Conceptually, the relay-enable condition can be written as

$$\text{RelayEnable} = \text{DelayComplete} \wedge \text{DCSafe}_L \wedge \text{DCSafe}_R \wedge \text{TemperatureSafe}_L \wedge \text{TemperatureSafe}_R \wedge \text{SensorsValid}$$

Only when the complete expression evaluates as true are the loudspeakers permitted to connect.

17.9 DC Verification Before Connection

The left and right amplifier outputs are monitored independently during startup.

The DC sensing circuits convert each amplifier-output DC level into a low-voltage signal suitable for the STM32 ADC.

For a healthy amplifier, both outputs should remain close to

$$0 \text{ VDC.}$$

If a persistent abnormal DC condition is detected on either channel, the startup sequence is prevented from reaching the relay-connected state.

The result is

$$\text{DC Fault During Startup} \Rightarrow \text{Relays Remain OFF.}$$

This is particularly important because a failed amplifier could otherwise be connected to the loudspeaker immediately after the startup delay expires.

17.10 Temperature Verification Before Connection

Both heatsink temperature-monitoring channels are also checked during startup.

The controller verifies that both sensors return physically valid measurements and that neither channel indicates an excessive temperature.

Although a cold amplifier would normally be expected at initial power-on, this check is still important because it also detects possible sensor wiring or measurement faults.

Therefore,

Temperature Sensor Fault $\Rightarrow$ Relays Remain OFF

and

Overtemperature Condition $\Rightarrow$ Relays Remain OFF.

17.11 End of the Power-On Delay

Once

$$t \geq t_{\text{delay}},$$

the controller evaluates whether all required conditions have remained valid.

If the system is healthy,

$$\text{RelayEnable} = 1.$$

The firmware then drives the relay-control GPIO outputs to activate the BC337 relay-driver transistors.

Current begins flowing through the 12 V relay coils and the normally-open speaker contacts close.

The sequence is therefore

Delay Complete $\rightarrow$ Safety Conditions Valid $\rightarrow$ Relay Drivers ON $\rightarrow$ Speaker Relays Energised $\rightarrow$ Speaker

17.12 Left and Right Relay Connection

The stereo protection system contains one relay for each power-amplifier channel.

Once the common startup validation has been completed successfully, both channels are enabled:

$$\text{Relay}_L = 1$$

and

$$\text{Relay}_R = 1.$$

The audio paths then become

Left Power Amplifier → Left Speaker Relay → Left Loudspeaker

and

Right Power Amplifier → Right Speaker Relay → Right Loudspeaker.

Although the two relays use independent driver circuits, the normal stereo operating state requires both protection channels to be healthy.

17.13 Complete Startup Sequence

The complete speaker-connection process can be represented as

POWER ON → MCU RESET → INITIALISATION → RELAYS OFF → STARTUP DELAY → SAFE

provided that no protection condition is detected.

If any required condition is invalid, the sequence instead becomes

POWER ON → MONITORING → FAULT DETECTED → RELAYS REMAIN OFF.

17.14 Startup State Diagram

The logical startup sequence may be represented as the following state progression:

OFF → INITIALISE → DELAY → VALIDATE → NORMAL.

The transition from the validation state to normal operation is permitted only if all required protection conditions are valid.

Otherwise, the controller enters the appropriate fault state:

VALIDATE → FAULT.

17.15 Startup Delay Timing

The delay is generated by the STM32 timing system rather than by a large analogue RC timing network.

A firmware counter or hardware timer provides the required time base.

If the system uses a periodic timing interval

$$\Delta t,$$

and the required startup delay is

$$t_{\text{delay}},$$

then the corresponding number of timing events is

$$N_{\text{delay}} = \frac{t_{\text{delay}}}{\Delta t}.$$

Using a digital delay provides repeatable timing and makes the startup period easy to modify in firmware without changing PCB components.

17.16 Why the Delay Should Not Be Excessively Short

If the startup delay is too short, the speaker relays may connect before the power-amplifier output has fully stabilised.

This can allow turn-on transients to reach the loudspeaker.

The delay should therefore exceed the normal settling period of the main amplifier, preamplifier, power supplies, and sensing circuitry.

The final value should be validated experimentally by observing the amplifier output voltage with an oscilloscope during startup.

17.17 Why the Delay Should Not Be Excessively Long

An unnecessarily long delay does not normally damage the system, but it makes the amplifier slower to become usable after power-on.

The objective is therefore not to select the longest possible delay.

The required design condition is

$$t_{\text{delay}} > t_{\text{stabilisation}}$$

with a reasonable safety margin.

The final delay should therefore be based on measured startup behaviour rather than an arbitrary value.

17.18 Normal Operation After Connection

The power-on delay is only the initial part of the protection sequence.

Once the relays have closed, the controller continues monitoring the amplifier continuously. The normal operating state can be represented as

$$\text{Relay}_L = 1, \quad \text{Relay}_R = 1$$

while

$$\text{DCSafe} = 1,$$

$$\text{TemperatureSafe} = 1,$$

and

$$\text{SensorsValid} = 1.$$

The relay state therefore remains conditional rather than becoming permanent after startup.

17.19 Fault Occurring After Speaker Connection

If a confirmed fault develops after the relays have closed, the STM32 removes the relay-drive signals.

For example,

$$\text{Normal Operation} + \text{Confirmed DC Fault} \rightarrow \text{Relays OFF}.$$

Similarly,

$$\text{Normal Operation} + \text{Overtemperature} \rightarrow \text{Relays OFF}.$$

The firmware command to disconnect can occur immediately after fault confirmation, although the physical relay contacts require a finite mechanical release time.

17.20 Relay Release Time

An electromechanical relay cannot change state instantaneously.

When the BC337 driver is switched off, current stored in the relay-coil inductance circulates through the 1N4007 flyback diode and gradually decays.

Once the magnetic field becomes insufficient to hold the relay armature, the contact opens. Therefore,

$$t_{\text{disconnect}} = t_{\text{firmware}} + t_{\text{coil}} + t_{\text{mechanical}}.$$

The firmware response is only one component of the total loudspeaker disconnection time. The complete disconnection delay should therefore be measured during final protection-system testing.

17.21 Effect of the Flyback Diode on Release

The 1N4007 flyback diode protects the BC337 driver from the large inductive voltage that would otherwise appear when coil current is interrupted.

However, clamping the coil voltage to a relatively low value also causes the current to decay more gradually.

This can slightly increase relay release time.

For this amplifier, the simplicity and reliability of the conventional flyback-diode arrangement are considered appropriate, but the resulting release time should be included in the protection validation.

17.22 Power-Off Behaviour

The speaker relay should also return to its disconnected state when the amplifier is switched off.

As controller power disappears, the relay-driver transistors lose their active base drive and the coils become de-energised.

The normally-open contacts then disconnect the loudspeakers.

The desired behaviour is

Controller Power Lost $\Rightarrow$ Relay OFF $\Rightarrow$ Speaker Disconnected
--

This hardware behaviour is useful because it does not rely solely on firmware executing a shutdown routine successfully.

17.23 MCU Reset During Operation

If the STM32 resets unexpectedly while the amplifier is operating, the GPIO outputs return to their reset configuration.

The BC337 base pulldown resistors then ensure that the relay drivers remain off.

Therefore,

MCU Reset $\Rightarrow$ Relay OFF

After the microcontroller restarts, the complete startup sequence must be performed again.

The loudspeakers are not reconnected until the startup delay and safety checks have again been completed successfully.

17.24 Watchdog Reset Behaviour

The same behaviour applies if the microcontroller watchdog detects abnormal firmware execution and resets the STM32.

A watchdog event therefore produces the sequence

Firmware Fault → MCU Reset → Relays OFF → Startup Sequence Restarted.

This prevents the system from leaving the loudspeakers connected while the protection firmware is no longer executing correctly.

17.25 Startup Fault Example: DC Output

Consider a fault in which one output transistor fails before the amplifier is powered.

When the system starts, the affected amplifier output may move toward one of the supply rails.

The DC sensing circuit detects the abnormal average output voltage during the startup delay.

The protection controller therefore obtains

$$\text{DCSafe} = 0.$$

At the end of the delay,

$$\text{RelayEnable} = 0.$$

The speakers are therefore never connected to the faulty amplifier.

17.26 Startup Fault Example: Temperature Sensor

A disconnected or short-circuited heatsink temperature sensor can also be detected before speaker connection.

If the measured sensor voltage lies outside the physically valid region,

$$\text{SensorsValid} = 0.$$

The startup sequence is then prevented from entering the normal connected state.

The relays remain de-energised while the appropriate fault indicator informs the user of the invalid monitoring condition.

17.27 Protection Against Turn-On Thump

One of the audible benefits of the delay circuit is the suppression of turn-on thump.

A temporary output disturbance may occur while the amplifier's internal DC operating conditions settle.

Because the speaker relay is open,

$$V_{\text{AMP,transient}}$$

is present only at the amplifier side of the relay and is not transferred to the loudspeaker.

Once the amplifier output has settled close to its normal DC operating point, the relay is permitted to close.

17.28 Connection Sequence and Audio Path

Before relay activation, the signal path is

Power Amplifier → OPEN RELAY Loudspeaker Disconnected.

After successful startup validation,

Power Amplifier → CLOSED RELAY → Loudspeaker.

The protection system therefore changes only the physical connection between the amplifier output and speaker. It does not modify or digitally process the audio waveform.

17.29 Relay Status Indication

The protection system uses status LEDs to indicate its operating condition.

During the startup delay, the system can indicate that the loudspeakers are intentionally disconnected while the controller waits for stabilisation.

After successful connection, the normal operating indication confirms that the protection controller has enabled the speaker relays.

If the sequence is interrupted by a fault, the corresponding fault LED provides information about the cause.

17.30 Testing the Startup Sequence

The startup sequence should be tested before valuable loudspeakers are connected.

A suitable validation procedure is:

1. Disconnect the loudspeakers and use suitable test equipment or a dummy load.
2. Apply power to the complete amplifier.
3. Confirm that both speaker relays remain de-energised immediately after power-on.
4. Verify correct STM32 initialisation.
5. Measure the amplifier output during the startup interval.
6. Record the time required for the amplifier DC output to stabilise.
7. Confirm that the programmed delay exceeds the measured stabilisation time.
8. Confirm that both DC sensing channels indicate a healthy condition.
9. Confirm that both temperature channels are valid.
10. Verify that both relays energise only after the complete startup delay.
11. Simulate a DC fault during startup and confirm that the relays remain off.
12. Simulate a temperature-sensor fault and confirm that speaker connection is inhibited.
13. Allow normal speaker connection and then simulate a fault.
14. Measure the time between confirmed fault detection and physical relay contact opening.
15. Reset the STM32 and verify that both relays immediately return to the disconnected state.
16. Confirm that the complete delay sequence is repeated after the reset.

17.31 Determination of the Final Delay

The final value of

$$t_{\text{delay}}$$

should be established experimentally.

During prototype testing, the power-amplifier output should be observed with an oscilloscope from the instant mains power is applied.

The measured settling time should then be compared with the programmed relay delay.

The selected value should satisfy

$$t_{\text{delay}} > t_{\text{maximum observed settling}}$$

with suitable margin for component tolerance, temperature, mains variation, and normal differences between the left and right channels.

The final report can then replace the symbolic value with the measured and implemented startup delay.

17.32 Design Summary

The power-on delay prevents the loudspeakers from being connected while the amplifier is still reaching its normal operating condition.

Immediately after power is applied, both normally-open loudspeaker relays are de-energised:

$$\boxed{\text{Relay}_L = 0, \quad \text{Relay}_R = 0.}$$

The STM32G031K8 then initialises its ADC channels, GPIO outputs, timers, protection variables, and monitoring functions.

The relays remain disconnected for a firmware-defined period

$$\boxed{t_{\text{delay}}}.$$

During this interval, the controller already monitors both amplifier DC outputs and both heatsink temperature channels.

At the end of the delay, the speakers are connected only if the complete protection condition is satisfied:

$$\boxed{\text{RelayEnable} = \text{DelayComplete} \wedge \text{DCSafe}_L \wedge \text{DCSafe}_R \wedge \text{TemperatureSafe}_L \wedge \text{TemperatureSafe}_R \wedge \text{SensorSafe}_L \wedge \text{SensorSafe}_R}$$

If all conditions are valid, the STM32 activates the BC337 relay-driver transistors, energising both 12 V speaker relays and connecting the loudspeakers to the amplifier outputs.

If any fault is present during startup, the relays remain de-energised and the loudspeakers are never exposed to the abnormal amplifier condition.

After connection, monitoring continues continuously. A confirmed fault, controller reset, watchdog reset, or loss of control power removes relay drive and returns the system to its default fail-safe condition:

$$\boxed{\text{Relay OFF} \Rightarrow \text{Speaker Disconnected.}}$$

The final value of the startup delay will be determined experimentally by measuring the settling behaviour of the completed power amplifier and choosing a delay greater than the maximum observed stabilisation interval with appropriate safety margin.

Chapter 18

Fault Detection and Protection Logic

18.1 Introduction

The STM32-based protection controller continuously evaluates the electrical and thermal condition of the stereo power amplifier and determines whether the loudspeakers may remain connected.

The protection logic combines information from several monitoring subsystems, including the left and right amplifier DC sensing circuits, the two heatsink temperature sensors, the startup timer, and the validity of the sensing hardware.

The controller follows a fail-safe operating principle. Loudspeaker connection is permitted only when all required operating conditions have been confirmed as safe.

The fundamental rule is therefore

$$\boxed{\text{All Required Conditions Safe} \Rightarrow \text{Speakers Connected}}$$

while

$$\boxed{\text{Any Confirmed Protection Fault} \Rightarrow \text{Speakers Disconnected}}.$$

The protection system is designed so that loss of valid control, controller reset, abnormal sensor behaviour, excessive temperature, or dangerous DC output causes the speaker relays to return to their normally-open state.

18.2 Protection Objectives

The protection logic has several principal objectives:

- prevent speaker connection during amplifier startup;
- detect dangerous positive or negative DC voltage at either amplifier output;
- detect excessive heatsink temperature;

- detect invalid temperature-sensor conditions;
- disconnect both loudspeakers when a serious fault is confirmed;
- prevent nuisance relay operation caused by short electrical disturbances;
- provide a defined recovery path following a fault;
- maintain a safe relay state during MCU reset or loss of control.

The protection firmware must therefore distinguish between normal dynamic audio behaviour and abnormal conditions that persist long enough to represent a genuine fault.

18.3 Monitored Variables

The controller monitors the following principal quantities:

$$V_{\text{DC,L}}$$

for the left-channel amplifier DC output,

$$V_{\text{DC,R}}$$

for the right-channel amplifier DC output,

$$T_L$$

for the left-channel heatsink temperature, and

$$T_R$$

for the right-channel heatsink temperature.

In addition, the controller evaluates the validity of the temperature-sensor signals and the completion state of the startup delay.

The complete protection decision therefore depends on both measured analogue quantities and internal controller state.

18.4 Protection Conditions

The principal logical protection conditions are defined as

$$\text{DCSafe}_L,$$

$$\text{DCSafe}_R,$$

TemperatureSafe_L,

TemperatureSafe_R,

SensorValid_L,

SensorValid_R,

and

StartupComplete.

The loudspeaker relays may be energised only when all of these conditions are true.

The relay permission can therefore be represented logically as

$$\text{RelayEnable} = \text{StartupComplete} \wedge \text{DCSafe}_L \wedge \text{DCSafe}_R \wedge \text{TemperatureSafe}_L \wedge \text{TemperatureSafe}_R \wedge \text{SensorValid}_L \wedge \text{SensorValid}_R$$

Thus,

$$\boxed{\text{RelayEnable} = 1}$$

only when every required safety condition is satisfied.

18.5 Fail-Safe Logic

The speaker-protection system is based on a fail-safe philosophy.

The relays use normally-open contacts, so the loudspeakers are physically disconnected when the relay coils are not energised.

Therefore,

$$\boxed{\text{No Relay Drive} \Rightarrow \text{Speaker Disconnected}}.$$

The firmware must actively maintain the relay-enable condition during normal operation.

Any state in which controller operation cannot be confirmed should result in relay de-energisation.

Examples include:

- MCU reset;
- watchdog reset;

- controller power loss;
- startup state;
- DC fault;
- overtemperature;
- invalid temperature sensor;
- undefined or unexpected protection state.

18.6 Protection State Machine

The protection firmware can be represented as a finite-state machine.

The principal states are:

- INITIALISATION;
- STARTUP_DELAY;
- NORMAL;
- DC_FAULT;
- THERMAL_FAULT;
- SENSOR_FAULT.

The system begins in the initialisation state after power-on or reset.

The normal progression is

INITIALISATION $\rightarrow$ STARTUP_DELAY $\rightarrow$ NORMAL.

A detected fault causes a transition from the normal operating state into the appropriate fault state.

For example,

NORMAL + DCFault $\rightarrow$ DC_FAULT,

and

NORMAL + Overtemperature $\rightarrow$ THERMAL_FAULT.

18.7 Protection State Summary

Table 18.1: Principal protection-controller states.

State	Relay Condition	Function
INITIALISATION	OFF	Configure MCU and protection peripherals
STARTUP_DELAY	OFF	Allow amplifier and supplies to stabilise
NORMAL	ON	Continuous protection monitoring
DC_FAULT	OFF	Dangerous amplifier DC detected
THERMAL_FAULT	OFF	Excessive heatsink temperature detected
SENSOR_FAULT	OFF	Temperature measurement invalid

18.8 Initialisation Logic

Immediately after power is applied, the controller enters the **INITIALISATION** state.

During this state,

$$\text{Relay}_L = 0$$

and

$$\text{Relay}_R = 0.$$

The STM32 configures the required GPIO pins, ADC channels, timers, internal variables, and status indicators.

Protection measurements are not considered valid until the corresponding peripherals have been correctly initialised.

Once initialisation is complete, the controller enters the startup-delay state.

18.9 Startup Protection Logic

During the **STARTUP_DELAY** state, both loudspeakers remain disconnected.

The controller nevertheless begins measuring both amplifier DC outputs and both temperature channels.

The startup state therefore provides an opportunity to identify a fault before any speaker connection occurs.

The transition to normal operation is allowed only when

$$t \geq t_{\text{delay}}$$

and all protection conditions are safe.

Therefore,

$$\text{STARTUP_DELAY} + \text{AllSafe} \rightarrow \text{NORMAL}.$$

If a fault exists,

$$\text{STARTUP_DELAY} + \text{Fault} \rightarrow \text{FAULT STATE}.$$

18.10 DC Fault Detection Logic

The left and right power-amplifier outputs are independently monitored for persistent DC voltage.

A DC fault condition can be expressed as

$$|V_{\text{DC}}| \geq V_{\text{DC,trip}},$$

where

$$V_{\text{DC,trip}}$$

is the selected firmware trip threshold.

Separate conditions are evaluated for the two amplifier channels:

$$|V_{\text{DC,L}}| \geq V_{\text{DC,trip}}$$

and

$$|V_{\text{DC,R}}| \geq V_{\text{DC,trip}}.$$

Either condition can initiate a DC fault.

The combined condition is therefore

$$\text{DCFault} = \text{DCFault}_L \vee \text{DCFault}_R.$$

A confirmed fault on either stereo channel causes both loudspeaker relays to be switched off.

18.11 Positive and Negative DC Faults

The DC detector must respond to both positive and negative output faults.

A positive failure condition may produce

$$V_{\text{DC}} > +V_{\text{DC},\text{trip}},$$

while a negative failure condition may produce

$$V_{\text{DC}} < -V_{\text{DC},\text{trip}}.$$

The protection firmware therefore operates on the magnitude of the estimated DC output:

$$|V_{\text{DC}}|.$$

This allows identical protection logic to be used for failures toward either the positive or negative amplifier supply rail.

18.12 DC Persistence Requirement

A single ADC sample outside the normal range should not immediately be declared a DC fault.

The amplifier output normally contains an AC audio waveform that repeatedly moves far above and below 0 V.

Although the analogue DC sensing network strongly attenuates audio frequencies, residual low-frequency variation and measurement noise may remain.

The firmware therefore requires the abnormal DC condition to persist for a defined confirmation period.

This period is represented as

$$t_{\text{DC},\text{confirm}}.$$

The protection condition can therefore be expressed as

$$|V_{\text{DC}}| \geq V_{\text{DC},\text{trip}}$$

continuously or sufficiently consistently for

$$t \geq t_{\text{DC},\text{confirm}}.$$

Only then is

$$\text{DCFault} = 1.$$

18.13 Digital Filtering of DC Measurements

The analogue filter is supplemented by digital processing.

A simple moving average of N samples can be represented as

$$\bar{V}_{\text{DC}} = \frac{1}{N} \sum_{k=1}^N V_{\text{DC}}(k).$$

The averaged value contains less high-frequency noise than an individual sample.

The protection decision can then be based on

$$|\bar{V}_{\text{DC}}|$$

rather than an individual instantaneous ADC conversion.

The exact digital filter implementation belongs to the firmware design and can be modified without changing the sensing hardware.

18.14 Thermal Fault Detection Logic

The controller independently calculates the left and right heatsink temperatures.

An overtemperature fault occurs when

$$T_L \geq T_{\text{trip}}$$

or

$$T_R \geq T_{\text{trip}},$$

where

$$T_{\text{trip}}$$

is the programmed thermal trip temperature.

The combined thermal fault can therefore be expressed as

$$\text{ThermalFault} = (T_L \geq T_{\text{trip}}) \vee (T_R \geq T_{\text{trip}}).$$

If either channel exceeds the permitted temperature, both speaker relays are de-energised.

18.15 Thermal Hysteresis

Thermal hysteresis prevents the relays from repeatedly switching on and off when the heatsink temperature remains close to the trip point.

Two separate temperatures are used:

$$T_{\text{trip}}$$

and

$$T_{\text{reset}},$$

with

$$T_{\text{reset}} < T_{\text{trip}}.$$

The thermal fault is entered when

$$T \geq T_{\text{trip}}.$$

Thermal recovery is not considered until

$$T \leq T_{\text{reset}}.$$

The difference

$$\Delta T_{\text{hyst}} = T_{\text{trip}} - T_{\text{reset}}$$

defines the temperature hysteresis.

18.16 Temperature Sensor Fault Logic

The protection controller must also verify that the temperature measurements themselves are valid.

An open or short-circuited NTC thermistor produces an ADC value near one of the extreme ends of the divider range.

The firmware can therefore define a valid sensor interval:

$$V_{\text{TEMP,min}} < V_{\text{TEMP}} < V_{\text{TEMP,max}}.$$

If the ADC reading lies outside the valid interval,

$$\text{SensorValid} = 0.$$

The two channels are checked independently:

$$\text{SensorValid}_L$$

and

$$\text{SensorValid}_R.$$

A failure of either sensor prevents or removes loudspeaker connection.

18.17 Sensor Plausibility

The measured sensor value should also correspond to a physically plausible temperature.

For example, a calculated value far outside the operating range of the NTC sensor should not be interpreted as a genuine heatsink temperature.

Instead, it should be classified as a measurement or sensor fault.

The logical condition is

$$T_{\min,\text{valid}} < T_{\text{measured}} < T_{\max,\text{valid}}.$$

Values outside this interval cause

$$\text{SensorValid} = 0.$$

18.18 Combined Fault Logic

The controller combines the individual fault conditions into an overall system fault.

The combined protection state may be represented as

$$\text{SystemFault} = \text{DCFault} \vee \text{ThermalFault} \vee \text{SensorFault}.$$

Therefore,

$$\boxed{\text{SystemFault} = 1 \Rightarrow \text{RelayEnable} = 0}.$$

This logic ensures that every monitored protection category has the ability to disconnect the loudspeakers.

18.19 Relay Disconnection Logic

When a confirmed fault occurs, the STM32 immediately removes the relay-enable commands.

The digital control signals become

$$\text{RELAY_L_CTRL} = 0$$

and

$$\text{RELAY_R_CTRL} = 0.$$

The BC337 driver transistors then switch off and the 12 V relay coils become de-energised.

The normally-open contacts return to the disconnected state.

Thus,

$$\boxed{\text{Confirmed Fault} \rightarrow \text{Both Speaker Relays OFF}}.$$

18.20 Why Both Channels are Disconnected

The system uses a common protection response for serious faults.

If a DC or thermal fault occurs in either channel, both loudspeaker relays are opened.

This approach prevents the amplifier from continuing to operate in a partially failed stereo condition.

The protection relation is therefore

$$\text{Fault}_L \vee \text{Fault}_R \Rightarrow \text{Relay}_L = 0 \wedge \text{Relay}_R = 0.$$

18.21 Fault Priority

Protection takes priority over normal speaker connection.

Consequently, even if the startup delay has completed, a fault condition overrides the relay-enable request.

The logical priority can be represented as

$$\text{Fault} > \text{NormalOperation}.$$

In practical firmware terms, relay activation should occur only after the fault conditions have been evaluated.

A suitable logical structure is

$$\text{Check Faults} \rightarrow \begin{cases} \text{Fault Present} & \rightarrow \text{Relay OFF} \\ \text{No Fault} & \rightarrow \text{Evaluate Relay Permission} \end{cases}$$

18.22 Fault Indication

The protection controller provides visual indication of the reason for a speaker-disconnection event.

Different LEDs can identify conditions such as

- startup delay;
- DC fault;
- overtemperature;
- temperature sensor 1 fault;
- temperature sensor 2 fault;
- normal operation.

The indication logic is separate from the fundamental relay protection action.

A failure of an indicator LED must not prevent the loudspeaker relay from being disconnected.

18.23 Fault Recovery Philosophy

A fault should not result in immediate loudspeaker reconnection as soon as a single measurement returns to the safe region.

The controller must verify that the fault condition has genuinely disappeared.

A generic recovery sequence is

FAULT → Measurements Return Safe → Recovery Validation → Reconnect Permission.

The precise recovery behaviour can differ according to fault type.

For example, a thermal fault requires the heatsink to cool below the lower hysteresis temperature before recovery is considered.

A DC fault may require the output to remain inside the safe range for a defined recovery interval before the startup connection process is repeated.

18.24 DC Fault Recovery

Following a DC protection event, the controller should continue monitoring the amplifier output with the loudspeaker disconnected.

Recovery should be considered only when

$$|V_{\text{DC,L}}| < V_{\text{DC,safe}}$$

and

$$|V_{\text{DC,R}}| < V_{\text{DC,safe}}$$

for a sufficiently long period.

A recovery confirmation time may therefore be defined as

$$t_{\text{DC,recovery}}.$$

This prevents a rapidly changing or intermittent amplifier fault from causing repeated relay cycling.

18.25 Thermal Fault Recovery

After an overtemperature fault, the amplifier must cool before speaker reconnection can be considered.

The thermal recovery condition is

$$T_L < T_{\text{reset}}$$

and

$$T_R < T_{\text{reset}}.$$

This ensures that both amplifier channels have returned to an acceptable thermal region before normal operation resumes.

18.26 Sensor Fault Recovery

A sensor fault is considered cleared only when the affected ADC measurement returns to the valid physical range.

The controller should also verify the measurement for multiple samples rather than treating one apparently valid sample as sufficient proof that the sensor has recovered.

This is particularly important for intermittent connectors or wiring faults.

18.27 Relay Chatter Prevention

Rapid repeated relay operation must be avoided.

Relay chatter can increase contact wear and produce repeated speaker connection transients.

The protection logic therefore uses a combination of

Filtering + Persistence + Hysteresis + Recovery Timing.

This ensures that relay state changes correspond to meaningful system conditions rather than ADC noise or values located directly on a threshold.

18.28 MCU Reset Logic

If the STM32 resets unexpectedly, both relay-control GPIO outputs return to their safe inactive condition.

The BC337 base pulldown resistors ensure that the relay drivers remain off.

Therefore,

MCU Reset $\Rightarrow$ Both Relays OFF.

After reset, the system must repeat the complete initialisation and power-on delay sequence before the loudspeakers can reconnect.

18.29 Watchdog Protection

The STM32 watchdog can provide additional supervision of firmware execution.

During correct operation, the firmware periodically refreshes the watchdog.

If the program becomes trapped or fails to execute correctly, the watchdog can reset the microcontroller.

The resulting protection sequence is

Firmware Failure $\rightarrow$ Watchdog Reset $\rightarrow$ Relay OFF $\rightarrow$ Startup Sequence Restarted.

This helps prevent the speakers from remaining connected indefinitely if the protection program stops functioning correctly.

18.30 Loss of Controller Power

The speaker-protection system does not rely on the microcontroller being powered in order to disconnect the speakers.

If the protection-controller supply fails,

$$V_{\text{MCU}} \rightarrow 0,$$

the GPIO outputs can no longer drive the BC337 transistors.

The relay coils therefore become de-energised.

Consequently,

Controller Power Loss $\Rightarrow$ Speaker Disconnected.

18.31 Fault Logic during Power-Off

When the complete amplifier is switched off, the controller supply and main power-amplifier supply rails begin to collapse.

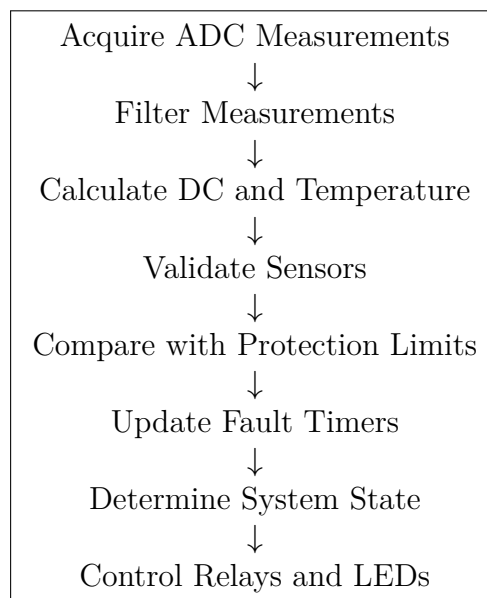
The protection hardware is arranged so that loss of active relay drive opens the speaker contacts.

Thus, the loudspeaker connection does not depend on the controller completing a complex software shutdown sequence.

The default mechanical state of the relays provides the required protection.

18.32 Protection Logic Flow

The repeated protection cycle during normal operation can be represented conceptually as



This sequence repeats continuously while the protection controller is operating.

18.33 Conceptual Firmware Logic

The overall behaviour may be represented in simplified pseudocode form as

```

Read DC_left
Read DC_right
Read TEMP_left
Read TEMP_right

Filter measurements

Check temperature sensor validity

if sensor fault:
    relays OFF
    enter SENSOR_FAULT

else if confirmed DC fault:
    relays OFF
    enter DC_FAULT

else if confirmed overtemperature:
    relays OFF
    enter THERMAL_FAULT

else if startup delay not complete:
    relays OFF
    remain in STARTUP_DELAY

else:
    relays ON
    enter NORMAL

```

The final firmware implementation may use separate functions and state variables, but the protection priority remains equivalent to this structure.

18.34 Protection Threshold Calibration

The final numerical values used by the protection firmware should be established during prototype testing.

These parameters include

$$V_{\text{DC},\text{trip}},$$

$$t_{\text{DC},\text{confirm}},$$

$$T_{\text{trip}},$$

$$T_{\text{reset}},$$

and any recovery timing intervals.

Using configurable firmware constants allows these values to be refined without changing the PCB.

18.35 Avoiding False DC Trips

Low-frequency audio signals represent one of the most important potential sources of false DC detection.

The protection strategy therefore uses several layers of rejection:

Analogue Low-Pass Filter

followed by

Digital Filtering

followed by

Persistence Timing.

A genuine DC fault remains present continuously, whereas normal audio changes polarity around zero.

This difference allows the protection logic to distinguish a sustained dangerous output condition from normal music content.

18.36 Avoiding False Thermal Trips

Temperature measurements can also contain electrical noise even though the physical heatsink temperature changes slowly.

Analogue filtering and software averaging are therefore used before the temperature threshold is evaluated.

The use of thermal hysteresis further ensures that small ADC variations around the thermal limit do not repeatedly operate the speaker relays.

18.37 Fault Logging and Diagnostics

The microcontroller architecture allows the detected fault type to be stored internally while the system remains powered.

This information can be used to control the corresponding status indicator and simplify troubleshooting.

For example, an internal variable can represent

FAULT_NONE,

FAULT_DC,

FAULT_THERMAL,

or

FAULT_SENSOR.

Future firmware versions could extend this mechanism to maintain a fault history or transmit diagnostic information through a serial interface.

18.38 Testing the Protection Logic

Each fault condition must be tested deliberately under controlled conditions before the protection system is trusted with valuable loudspeakers.

A suitable validation procedure includes:

1. Power the system with the loudspeakers disconnected.
2. Verify that the relays remain OFF during startup.
3. Confirm normal relay connection after all conditions are safe.
4. Inject a controlled positive DC sensing fault.
5. Confirm that the DC fault is detected only after the programmed persistence requirement.
6. Repeat the test using a negative DC fault.
7. Test the left and right DC channels independently.
8. Verify that a fault on either channel opens both speaker relays.
9. Heat the left temperature sensor in a controlled manner.
10. Confirm thermal fault detection.
11. Repeat the thermal test on the right channel.
12. Disconnect each temperature sensor and verify sensor-fault detection.

13. Simulate a short-circuited temperature sensor.
14. Verify that sensor faults prevent relay connection.
15. Confirm thermal hysteresis during cooling.
16. Reset the STM32 and verify immediate speaker disconnection.
17. Verify that the startup sequence is repeated after reset.
18. Test controller power loss and confirm relay release.
19. Confirm correct status LED indication for every fault category.

18.39 Protection Logic Summary

The STM32 protection controller uses a fail-safe decision structure in which speaker connection is permitted only when every required safety condition is valid.

The overall relay condition is represented by

$$\text{RelayEnable} = \text{StartupComplete} \wedge \text{DCSafe}_L \wedge \text{DCSafe}_R \wedge \text{TemperatureSafe}_L \wedge \text{TemperatureSafe}_R \wedge \text{SensorValid}$$

The protection logic monitors both amplifier outputs for sustained positive or negative DC voltage, both heatsinks for excessive temperature, and both temperature channels for sensor failure.

A DC fault is confirmed only when the measured DC component exceeds the selected threshold for a sufficient persistence interval. This prevents normal low-frequency audio signals or individual ADC disturbances from causing unnecessary relay operation.

A thermal fault occurs when either heatsink reaches the programmed trip temperature. A separate lower recovery temperature provides hysteresis and prevents relay chatter near the thermal limit.

Open, short-circuited, or otherwise implausible temperature-sensor readings are treated as faults because reliable thermal supervision can no longer be guaranteed.

Any confirmed serious fault produces

$$\text{Relay}_L = 0, \quad \text{Relay}_R = 0$$

and therefore disconnects both loudspeakers.

The same safe condition occurs during MCU reset, watchdog reset, controller power loss, and initial startup.

Filtering, fault persistence, hysteresis, and recovery validation are used to ensure that the protection system responds rapidly to genuine faults while remaining immune to normal audio signals and short electrical disturbances.

The numerical DC thresholds, persistence intervals, thermal shutdown temperature, thermal recovery temperature, and other firmware timing parameters will be finalised during testing of the completed protection controller and power amplifier.

Chapter 19

Status and Fault Indicator LEDs

19.1 Introduction

The STM32-based speaker protection controller incorporates a simple visual indication system that allows the operating state of the protection circuitry to be identified without requiring a display or external diagnostic equipment.

Four front-panel indicator LEDs are controlled directly by GPIO outputs of the STM32G031K8. Three red LEDs indicate protection-related faults, while one yellow LED indicates that the loudspeakers are intentionally disconnected.

The implemented indicators are:

- Temperature Sensor 1 Failure;
- Temperature Sensor 2 Failure;
- DC Output Fault;
- Wait / Speakers Disconnected.

The indicator system is intended to provide clear information during startup, normal operation, fault diagnosis, testing, and maintenance.

19.2 Indicator LED Assignment

The final LED assignments are shown in Table 19.1.

Table 19.1: STM32 protection-controller LED assignments.

LED	Function	STM32 Pin	Colour
LED1	TEMP SENSOR 1 FAIL	PB8 / D8	Red
LED2	TEMP SENSOR 2 FAIL	PB9 / D10	Red
LED3	DC FAULT	PB5 / D11	Red
LED4	WAIT / SPEAKERS DISCONNECTED	PB4 / D12	Yellow

The three red indicators use the WP1503IT 5 mm LED, while the yellow WAIT indicator uses the WP7113SYT 5 mm LED.

19.3 Electrical Connection

Each LED is driven directly from an STM32 GPIO output through a $470\ \Omega$ current-limiting resistor.

The connection for each indicator is

$$\text{STM32 GPIO} \rightarrow 470\ \Omega \rightarrow \text{LED} \rightarrow \text{GND}.$$

The individual connections are therefore

$$\text{PB8} \rightarrow 470\ \Omega \rightarrow \text{LED1} \rightarrow \text{GND},$$

$$\text{PB9} \rightarrow 470\ \Omega \rightarrow \text{LED2} \rightarrow \text{GND},$$

$$\text{PB5} \rightarrow 470\ \Omega \rightarrow \text{LED3} \rightarrow \text{GND},$$

and

$$\text{PB4} \rightarrow 470\ \Omega \rightarrow \text{LED4} \rightarrow \text{GND}.$$

The LEDs therefore use an active-high control arrangement.

19.4 Active-High Operation

Because the LED anode is driven from the STM32 through the current-limiting resistor and the cathode is connected to ground, a logic-high GPIO output causes current to flow through the LED.

Therefore,

$$\boxed{\text{GPIO} = 1 \Rightarrow \text{LED ON}}$$

and

$$\boxed{\text{GPIO} = 0 \Rightarrow \text{LED OFF.}}$$

This provides simple firmware control because each indicator can be activated or deactivated directly by changing the corresponding GPIO output state.

19.5 Current-Limiting Resistors

A light-emitting diode must not be connected directly between an STM32 GPIO and ground because its current would otherwise be limited primarily by the internal resistance of the GPIO and LED.

A resistor is therefore connected in series with every indicator.

The selected value is

$$R_{\text{LED}} = 470 \, \Omega.$$

The LED current can be estimated from

$$I_{\text{LED}} = \frac{V_{\text{GPIO}} - V_F}{R_{\text{LED}}},$$

where

$$V_{\text{GPIO}}$$

is the logic-high output voltage and

$$V_F$$

is the LED forward voltage.

19.6 Approximate LED Current

Using a nominal STM32 logic voltage of

$$V_{\text{GPIO}} \approx 3.3 \, \text{V}$$

and an illustrative LED forward voltage near

$$V_F \approx 2.0 \, \text{V},$$

the current is approximately

$$I_{\text{LED}} = \frac{3.3 - 2.0}{470}.$$

Therefore,

$$I_{\text{LED}} \approx 2.77 \, \text{mA}.$$

Thus, the selected resistor produces only a few milliamperes of indicator current.

This is sufficient for visible status indication while keeping the GPIO load small.

The exact current depends on the forward voltage of the individual LED and the actual STM32 output voltage.

19.7 LED Resistor Dissipation

The approximate power dissipated in one current-limiting resistor is

$$P_R = I_{\text{LED}}^2 R.$$

For an illustrative current of

$$2.77 \text{ mA},$$

the resistor power is

$$P_R = (2.77 \times 10^{-3})^2 \times 470.$$

Therefore,

$$P_R \approx 3.6 \text{ mW}.$$

This is far below the rating of a normal through-hole resistor.

19.8 TEMP SENSOR 1 FAIL Indicator

LED1 is a red fault indicator assigned to

PB8

and identifies a failure in the first temperature-sensing channel.

The indication is labelled

TEMP SENSOR 1 FAIL

.

The first temperature channel monitors the corresponding amplifier heatsink using the 10 k Ω NTC thermistor described in the temperature-monitoring chapter.

The indicator is activated when the firmware determines that the temperature measurement is invalid.

Examples include:

- open-circuit NTC sensor;

- disconnected sensor cable;
- short-circuited sensor;
- ADC value outside the valid physical range;
- calculated temperature outside the plausible sensor range.

The logical indication is therefore

$$\text{SensorValid}_1 = 0 \Rightarrow \text{LED1} = 1.$$

19.9 TEMP SENSOR 2 FAIL Indicator

LED2 is the corresponding red fault indicator for the second temperature channel.

It is connected to

PB9

and is labelled

TEMP SENSOR 2 FAIL

.

Its behaviour is equivalent to that of LED1:

$$\text{SensorValid}_2 = 0 \Rightarrow \text{LED2} = 1.$$

Separate sensor-failure indicators are useful because they identify which heatsink-monitoring channel requires investigation.

19.10 Difference Between Sensor Failure and Overtemperature

A temperature-sensor failure and an overtemperature condition are two different faults.

A sensor failure means that the controller cannot obtain a trustworthy temperature measurement.

An overtemperature condition means that the sensor is operating correctly but the measured heatsink temperature has exceeded the permitted operating limit.

Therefore,

$$\text{Sensor Failure} \neq \text{Overtemperature}.$$

LED1 and LED2 specifically identify sensor integrity failures.

In the implemented four-LED indicator system, no separate dedicated overtemperature LED is provided. An overtemperature condition causes the speaker relays to open, so the system enters the speakers-disconnected state represented by the yellow WAIT / SPEAKERS DISCONNECTED indicator.

19.11 DC FAULT Indicator

LED3 is a red fault indicator connected to

PB5

.

It is labelled

DC FAULT

.

The indicator is activated when the protection firmware confirms dangerous DC voltage at either power-amplifier output.

The DC fault condition is derived from the two independent sensing channels:

$$\text{DCFault} = \text{DCFault}_L \vee \text{DCFault}_R.$$

Therefore,

$$\text{DCFault} = 1 \Rightarrow \text{LED3} = 1.$$

A serious DC fault also causes both loudspeaker relays to be de-energised.

The protection response is therefore

$$\text{DC Fault} \rightarrow \begin{cases} \text{Both speaker relays OFF} \\ \text{DC FAULT LED ON} \end{cases}$$

regardless of whether the fault originated in the left or right amplifier channel.

19.12 WAIT / SPEAKERS DISCONNECTED Indicator

LED4 is a yellow status indicator connected to

PB4

.

It is labelled

WAIT / SPEAKERS DISCONNECTED

Unlike the three red LEDs, this indicator does not necessarily represent an electrical failure.

Its main purpose is to indicate that the speaker relays are not currently energised.

The indicator is therefore active during the normal startup delay.

The startup behaviour is

Power Applied $\rightarrow$ WAIT LED ON $\rightarrow$ Speakers Disconnected.

After the startup delay has completed and all protection conditions have been verified,

All Conditions Safe $\rightarrow$ Relays ON $\rightarrow$ WAIT LED OFF.

19.13 WAIT Indicator During Fault Conditions

The yellow indicator can also represent the general physical condition that the loudspeakers are disconnected.

For example,

DC Fault $\Rightarrow$ Relays OFF

and therefore the system is in a speaker-disconnected state.

Similarly,

Thermal Fault $\Rightarrow$ Relays OFF.

The red fault indicators communicate the specific diagnostic information where a dedicated indicator exists, while the yellow LED communicates that normal speaker connection has not been permitted.

19.14 Startup LED Sequence

At initial power application, the loudspeaker relays are intentionally de-energised.

The expected visible sequence is therefore

Power ON

$\Downarrow$

WAIT LED ON



DC and Temperature Checks

followed, if no fault is detected, by

Speaker Relays ON

and

WAIT LED OFF.

This gives the user direct confirmation that the delay before loudspeaker connection is intentional rather than a relay malfunction.

19.15 DC Fault Indication Sequence

If a confirmed DC fault occurs during normal operation, the sequence is

Normal Operation → DC Fault Detected → Both Relays OFF → DC FAULT LED ON.

The yellow speaker-disconnected indication may also be active according to the final firmware state representation.

This allows the user to distinguish between a normal startup wait condition and a speaker disconnection caused by a dangerous DC-output fault.

19.16 Sensor Failure Indication Sequence

If temperature sensor 1 fails,

TEMP1 Invalid → TEMP SENSOR 1 FAIL LED ON → Speaker Relays OFF.

If temperature sensor 2 fails,

TEMP2 Invalid → TEMP SENSOR 2 FAIL LED ON → Speaker Relays OFF.

The independent indicators make troubleshooting easier because the failed measurement channel is identified immediately.

19.17 Indicator State Table

Table 19.2: Expected LED behaviour for the principal protection states.

System State	TEMP1	TEMP2	DC Fault	WAIT
Startup delay	OFF	OFF	OFF	ON
Normal operation	OFF	OFF	OFF	OFF
TEMP1 sensor failure	ON	OFF	OFF	ON
TEMP2 sensor failure	OFF	ON	OFF	ON
DC fault	OFF	OFF	ON	ON
Overtemperature	OFF	OFF	OFF	ON
MCU reset / relay disabled	OFF	OFF	OFF	ON or hardware default state

The precise LED behaviour during MCU reset depends on when firmware control becomes active. The essential hardware requirement is that the speaker relays remain de-energised during reset.

19.18 Normal Operating State

During normal listening, all monitored parameters are within their valid ranges and the loudspeaker relays are energised.

The expected fault-indicator condition is therefore

$$\text{LED1} = 0,$$

$$\text{LED2} = 0,$$

$$\text{LED3} = 0,$$

and

$$\text{LED4} = 0.$$

Thus, the absence of illuminated protection LEDs represents normal connected operation.

19.19 GPIO Initialisation

The LED GPIO outputs should be configured during STM32 startup before normal protection operation begins.

The three red fault LEDs should initially remain off unless a corresponding fault has been confirmed.

The yellow WAIT indicator is then activated during the startup delay.

The resulting initialization concept is

Reset $\rightarrow$ GPIO Configuration $\rightarrow$ Fault LEDs OFF $\rightarrow$ WAIT LED ON.

Once startup validation is completed successfully, the WAIT indicator is switched off as the speaker relays are energised.

19.20 Firmware Representation

The LED control can be directly associated with internal firmware status flags.

For example,

$$\text{LED1} = \neg \text{SensorValid}_1,$$

$$\text{LED2} = \neg \text{SensorValid}_2,$$

$$\text{LED3} = \text{DCFault},$$

and approximately

$$\text{LED4} = \neg \text{SpeakersConnected}.$$

This representation makes the visual indicators a direct reflection of the protection state.

19.21 LEDs and Protection Independence

The LEDs provide information only. They do not perform the protection function themselves.

The protection decision must therefore remain independent of the physical condition of an indicator LED.

For example, if the DC fault LED becomes open circuit,

$$\text{DCFault} = 1$$

must still produce

$$\text{Relay}_L = 0$$

and

$$\text{Relay}_R = 0.$$

The indication system is therefore diagnostic rather than safety-critical.

19.22 Failure of an Indicator LED

Possible LED-related failure modes include:

- open LED;
- reversed LED polarity;
- open current-limiting resistor;
- incorrect GPIO assignment;
- damaged PCB trace;
- shorted LED;
- firmware configuration error.

An LED failure should not interfere with ADC measurement, relay operation, or the fundamental protection logic.

19.23 Front-Panel Colour Selection

Red is used for the three fault indicators because it provides an immediate visual association with abnormal operating conditions.

Therefore,

$$\text{Red} \Rightarrow \text{Fault} / \text{Service Attention}.$$

Yellow is used for the WAIT / SPEAKERS DISCONNECTED condition because this state is not necessarily a failure.

During normal startup, for example, the yellow LED simply informs the user that the speaker relays are intentionally being held open.

Thus,

$$\text{Yellow} \Rightarrow \text{Wait} / \text{Speaker Connection Inhibited}.$$

19.24 Physical Front-Panel Arrangement

The four LEDs should be arranged so that the corresponding labels are clearly visible from the normal operating position.

A logical ordering is

TEMP1 TEMP2 DC FAULT WAIT.

Clear labelling is important because a red temperature-sensor failure indicator has a different meaning from the red DC fault indicator.

The front-panel chapter later in this report addresses the mechanical placement and appearance of these indicators in greater detail.

19.25 PCB Routing Considerations

The LED circuits carry only a few milliamperes and therefore do not require wide high-current PCB traces.

Nevertheless, their digital switching traces should not be routed through sensitive analogue ADC regions unnecessarily.

Particular care should be taken to prevent indicator return current from sharing a sensitive analogue measurement path.

The protection controller ground system should therefore provide a controlled return path for LED current.

19.26 Testing the Indicator System

All status LEDs should be tested independently before complete protection testing is performed.

A suitable procedure is:

1. Verify the orientation of all four LEDs.
2. Verify each $470\ \Omega$ series resistor.
3. Confirm continuity between each STM32 GPIO and the corresponding resistor.
4. Power the protection controller without connecting valuable loudspeakers.
5. Drive PB8 high and confirm TEMP SENSOR 1 FAIL indication.
6. Drive PB9 high and confirm TEMP SENSOR 2 FAIL indication.
7. Drive PB5 high and confirm DC FAULT indication.
8. Drive PB4 high and confirm WAIT indication.

9. Confirm that each LED switches off when its GPIO is driven low.
10. Verify that the WAIT indicator operates during the complete startup delay.
11. Simulate a TEMP1 sensor failure and verify LED1.
12. Simulate a TEMP2 sensor failure and verify LED2.
13. Simulate a confirmed DC fault and verify LED3.
14. Confirm that a protection fault also causes the loudspeaker relays to disconnect independently of the LED operation.

19.27 Indicator Logic Summary

Table 19.3: Final protection-controller indicator configuration.

Function	Pin	Colour	Resistor	LED Type
TEMP SENSOR 1 FAIL	PB8 / D8	Red	470 Ω	WP1503IT
TEMP SENSOR 2 FAIL	PB9 / D10	Red	470 Ω	WP1503IT
DC FAULT	PB5 / D11	Red	470 Ω	WP1503IT
WAIT / SPEAKERS DISCONNECTED	PB4 / D12	Yellow	470 Ω	WP7113SYT

19.28 Design Summary

The STM32 speaker protection controller uses four front-panel LEDs to provide simple visual information about the state of the amplifier protection system.

Three

red 5 mm LEDs

are used for fault indication:

TEMP SENSOR 1 FAIL, TEMP SENSOR 2 FAIL, DC FAULT.

One

yellow 5 mm LED

is used for

WAIT / SPEAKERS DISCONNECTED.

The final STM32 assignments are

$$\boxed{\text{PB8} \rightarrow \text{TEMP1 FAIL}},$$
$$\boxed{\text{PB9} \rightarrow \text{TEMP2 FAIL}},$$
$$\boxed{\text{PB5} \rightarrow \text{DC FAULT}},$$

and

$$\boxed{\text{PB4} \rightarrow \text{WAIT}}.$$

Each LED is connected through a

$$\boxed{470\ \Omega}$$

current-limiting resistor between the corresponding GPIO and ground.

The indicators operate in an active-high configuration:

$$\boxed{\text{GPIO HIGH} \Rightarrow \text{LED ON}}.$$

The TEMP1 and TEMP2 red LEDs identify invalid temperature-sensor channels, while the DC FAULT LED identifies a confirmed dangerous DC condition at one of the amplifier outputs.

The yellow WAIT indicator identifies the condition in which the loudspeakers are intentionally disconnected. It is active during the startup delay and can also indicate the relay-disconnected state during protection events.

The indicator system is diagnostic rather than safety-critical. Relay disconnection and amplifier protection must continue to function even if an individual LED fails.

Together, the four indicators provide a simple and easily interpreted user interface for startup, fault diagnosis, protection-system testing, and normal amplifier operation without requiring a digital display.

Chapter 20

Protection Controller Power Supply

20.1 Introduction

The STM32-based speaker protection controller requires a low-voltage power supply that is electrically separate from the high-power symmetrical supply used by the Class-AB power amplifiers.

The main power-amplifier stages operate from approximately

$$\pm 33 - \pm 34 \text{ VDC},$$

whereas the protection relays require a nominal

$$12 \text{ VDC}$$

supply and the STM32G031K8 operates in the low-voltage digital domain.

For this reason, a dedicated linear power supply is incorporated into the protection-controller PCB. The supply converts the 230 V AC mains input into an isolated 12 V AC secondary voltage, rectifies and filters this voltage, and then regulates it to a stable

$$\boxed{+12 \text{ VDC}}.$$

The regulated 12 V rail is used principally to power the two loudspeaker relay coils and the NUCLEO-G031K8 development board through its VIN supply input. The Nucleo board subsequently generates the lower-voltage rails required by the STM32 microcontroller and its associated logic circuitry.

20.2 Power-Supply Requirements

The protection-controller power supply must satisfy several requirements.

It must provide:

- galvanic isolation from the 230 V AC mains supply;

- sufficient current for both speaker-relay coils;
- sufficient current for the NUCLEO-G031K8 board;
- a stable 12 V relay supply;
- low electrical noise for reliable ADC measurements;
- sufficient filtering to prevent relay switching from disturbing the microcontroller;
- safe physical separation between mains and low-voltage circuitry.

The protection controller does not require a large power supply compared with the audio output stages. Nevertheless, reliable controller operation is essential because the same supply powers the circuitry responsible for loudspeaker disconnection during abnormal conditions.

20.3 Power-Supply Architecture

The protection-controller supply follows the conventional linear power-supply architecture

230 V AC $\rightarrow$ Isolation Transformer $\rightarrow$ 12 V AC $\rightarrow$ Bridge Rectifier $\rightarrow$ Reservoir Capacitor $\rightarrow$ 12 V Regulator

Using the selected components, the actual chain is

230 VAC $\rightarrow$ FS24-250-C2 $\rightarrow$ 12 VAC $\rightarrow$ W04G $\rightarrow$ 2200 μ F $\rightarrow$ L7812CV $\rightarrow$ 12 VDC.

The regulated output is then distributed to

$$\begin{cases} \text{Left speaker relay coil,} \\ \text{Right speaker relay coil,} \\ \text{NUCLEO-G031K8 VIN.} \end{cases}$$

20.4 Selected Power-Supply Components

Table 20.1: Principal components of the protection-controller power supply.

Component	Selected Part	Function
Transformer	FS24-250-C2	Mains isolation and voltage reduction
Bridge rectifier	W04G-E4/51	Full-wave rectification
Main reservoir capacitor	2200 μ F/25 V	Raw DC smoothing
Voltage regulator	L7812CV	Regulated +12 V output
Output capacitor	100 μ F/25 V	12 V rail energy storage and filtering
Bypass capacitors	100 nF	High-frequency decoupling

20.5 FS24-250-C2 Transformer

The protection-controller supply uses the

FS24-250-C2

PCB-mount power transformer.

The transformer is rated at approximately

6 VA

and contains two 12 V secondary windings.

For the present application, the two secondary windings are connected in parallel rather than in series.

This produces a nominal output voltage of

12 V_{RMS}

while increasing the available current compared with a single secondary winding.

With the two windings correctly paralleled, the approximate secondary rating is

$12\text{ V}, 0.5\text{ A}$.

This corresponds to

$$12 \times 0.5 = 6\text{ VA},$$

which agrees with the transformer rating.

20.6 Secondary Winding Phasing

The two 12 V secondary windings must be connected with the correct relative phase before they are paralleled.

For parallel operation, equivalent winding starts must be connected together and equivalent winding ends must be connected together.

Conceptually,

$$S_{1,\text{start}} \rightarrow S_{2,\text{start}}$$

and

$$S_{1,\text{end}} \rightarrow S_{2,\text{end}}.$$

The resulting pair of connections then forms the 12 V AC input to the bridge rectifier.

The windings must never be paralleled with opposite polarity.

If one winding is reversed relative to the other, the two secondary voltages oppose each other and an extremely high circulating current may flow.

Therefore,

Secondary phasing must be verified before parallel connection.

The transformer manufacturer's pinout must be followed during PCB assembly.

20.7 Galvanic Isolation

The transformer provides galvanic isolation between the mains supply and the protection electronics.

There is no intentional conductive connection between the primary and secondary windings.

The isolation boundary can therefore be represented as

230 V Mains | Transformer Isolation | Low-Voltage Protection Circuitry.

This isolation is essential because the STM32, relay drivers, temperature sensors, and DC sensing circuits are all intended to operate on the isolated low-voltage side of the system.

20.8 Theoretical Secondary Peak Voltage

The transformer secondary voltage is specified as an RMS quantity.

For a sinusoidal voltage,

$$V_{\text{peak}} = V_{\text{RMS}}\sqrt{2}.$$

With

$$V_{\text{RMS}} = 12 \text{ V},$$

the theoretical secondary peak voltage is

$$V_{\text{peak}} = 12\sqrt{2}.$$

Therefore,

$$V_{\text{peak}} \approx 16.97 \text{ V}.$$

This is the ideal peak secondary voltage before bridge-rectifier losses are considered.

20.9 Bridge Rectifier

The selected bridge rectifier is the

$$\boxed{\text{W04G-E4/51}}.$$

The device performs full-wave rectification of the 12 V AC transformer secondary.

During each half-cycle of the AC waveform, two internal bridge diodes conduct.

The approximate raw DC peak voltage is therefore

$$V_{\text{RAW,peak}} \approx V_{\text{secondary,peak}} - 2V_D,$$

where

$$V_D$$

is the forward voltage of one conducting rectifier diode.

Using an illustrative silicon diode drop of approximately

$$V_D \approx 0.7 \text{ V},$$

the raw peak voltage would be approximately

$$V_{\text{RAW,peak}} \approx 16.97 - 1.4.$$

Thus,

$$\boxed{V_{\text{RAW,peak}} \approx 15.6 \text{ V}}$$

under this simplified condition.

The actual value varies with transformer regulation, load current, mains voltage, and rectifier forward voltage.

20.10 Full-Wave Ripple Frequency

The bridge uses both halves of the AC waveform.

For a 50 Hz mains supply,

$$f_{\text{mains}} = 50 \text{ Hz}.$$

After full-wave rectification, the capacitor charging frequency becomes

$$f_{\text{ripple}} = 2f_{\text{mains}}.$$

Therefore,

$$f_{\text{ripple}} = 100 \text{ Hz.}$$

20.11 Main Smoothing Capacitor

The rectified waveform is smoothed using a

$$2200 \mu\text{F}/25 \text{ V}$$

electrolytic capacitor.

The selected part is

EEU-FC1E222.

The capacitor charges near the peaks of the rectified waveform and supplies the load between charging intervals.

The result is a raw DC voltage with a smaller superimposed ripple rather than a full-wave rectified waveform.

20.12 Approximate Raw DC Ripple

A simplified approximation for the reservoir-capacitor ripple is

$$\Delta V \approx \frac{I_{\text{load}}}{f_{\text{ripple}} C}.$$

For

$$f_{\text{ripple}} = 100 \text{ Hz}$$

and

$$C = 2200 \mu\text{F} = 2.2 \times 10^{-3} \text{ F},$$

the relationship becomes

$$\Delta V \approx \frac{I_{\text{load}}}{100(2.2 \times 10^{-3})}.$$

Therefore,

$$\Delta V \approx 4.55 I_{\text{load}}$$

with ΔV in volts when I_{load} is expressed in amperes.

For example, the two relay coils alone require approximately

$$I_{\text{relay,total}} = 2(33.3 \text{ mA}) \approx 66.6 \text{ mA}.$$

Ignoring the additional Nucleo and control-circuit current, the ripple caused by the relay load alone would be approximately

$$\Delta V \approx 4.55(0.0666) \approx 0.30 \text{ V}.$$

The actual ripple will be greater because the Nucleo board, LEDs, sensing circuits, and other protection electronics also consume current.

20.13 L7812CV Voltage Regulator

The raw rectified and filtered voltage is applied to an

$$\text{L7812CV}$$

linear regulator.

The regulator provides a nominal output voltage of

$$12 \text{ VDC}.$$

The purpose of the regulator is to remove much of the remaining supply variation and provide a more stable voltage for the relay coils and controller.

The supply chain therefore becomes

$$15\text{--}16 \text{ VDC approximately} \rightarrow \text{L7812CV} \rightarrow 12 \text{ VDC}.$$

20.14 Regulator Input Headroom

A linear regulator requires its input voltage to remain above its regulated output by a sufficient margin.

The input condition may be represented as

$$V_{\text{IN,min}} > V_{\text{OUT}} + V_{\text{headroom}}.$$

For the present supply,

$$V_{\text{OUT}} = 12 \text{ V}.$$

The nominal rectified voltage of approximately 15–16 V provides several volts of regulator headroom under normal operating conditions.

However, the lowest instantaneous regulator input voltage occurs at the bottom of the reservoir-capacitor ripple waveform.

Consequently, regulator performance must be verified under the maximum actual controller load rather than from the transformer no-load voltage alone.

20.15 Regulator Power Dissipation

Because the L7812CV is a linear regulator, the difference between its input and output voltage is dissipated as heat.

The approximate regulator dissipation is

$$P_{\text{REG}} = (V_{\text{IN}} - V_{\text{OUT}})I_{\text{load}}.$$

For example, if the regulator input is approximately

$$15.5 \text{ V},$$

then

$$P_{\text{REG}} \approx (15.5 - 12)I_{\text{load}}.$$

Thus,

$$P_{\text{REG}} \approx 3.5I_{\text{load}}.$$

The exact power dissipation depends on the total protection-controller current and should be checked during final testing.

The L7812CV temperature should therefore be monitored during prolonged operation with both speaker relays energised.

20.16 Output Filtering

The regulated 12 V rail includes an additional

$100 \mu\text{F}/25 \text{ V}$

electrolytic capacitor.

The selected component is

EEU-FR1E101.

This capacitor provides local energy storage on the regulated rail and helps reduce voltage movement when the relay load changes.

It is particularly useful when both relay coils are switched simultaneously.

20.17 100 nF Bypass Capacitors

The protection-controller supply also uses

100 nF

ceramic bypass capacitors.

These capacitors provide low impedance at higher frequencies where the large electrolytic capacitors are less effective because of parasitic inductance and equivalent series resistance.

The power-supply design includes three 100 nF capacitors for regulator and supply bypassing.

The combined filtering philosophy is therefore

$2200\ \mu\text{F} \rightarrow$ Bulk Raw-DC Filtering,

$100\ \mu\text{F} \rightarrow$ Regulated-Rail Energy Storage,

and

$100\ \text{nF} \rightarrow$ High-Frequency Bypassing.

20.18 12 V Power Distribution

The regulated 12 V output is distributed to the two principal load groups of the protection controller.

The first group consists of the two Omron speaker-relay coils:

$$+12\ \text{V} \rightarrow \begin{cases} \text{Relay}_L, \\ \text{Relay}_R. \end{cases}$$

The second output feeds the NUCLEO-G031K8 through its VIN supply input:

$+12\ \text{V} \rightarrow$ NUCLEO VIN.

The Nucleo board then provides the regulated low-voltage logic supply required by the STM32G031K8 and the associated 3.3 V sensing and control circuits.

The 12 V rail is therefore not connected directly to any STM32 GPIO or ADC pin.

20.19 Relay Supply Requirement

Each G2RL-1A-E-DC12 relay requires approximately

$$33.3 \text{ mA}$$

at its nominal 12 V coil voltage.

With both relays energised,

$$I_{\text{relay,total}} = 2 \times 33.3 = 66.6 \text{ mA}.$$

The total relay-coil power is approximately

$$P_{\text{relay}} = 12 \times 0.0666.$$

Therefore,

$$P_{\text{relay}} \approx 0.80 \text{ W}.$$

The transformer and regulator must additionally supply the Nucleo board and the remaining protection circuitry.

20.20 Transformer Current Margin

With the two transformer secondaries connected correctly in parallel, the nominal secondary capability is approximately

$$500 \text{ mA}.$$

The two relay coils require only

$$66.6 \text{ mA},$$

leaving the remainder of the transformer capability available for the Nucleo board and associated monitoring circuits.

However, the relationship between transformer RMS current and available DC load current is not one-to-one in a bridge-rectifier and capacitor-input supply.

The bridge draws current in short charging pulses near the peaks of the AC waveform.

Consequently, final supply capability should be confirmed by measuring the transformer secondary voltage, raw DC voltage, regulated 12 V rail, and component temperatures under the complete operating load.

20.21 Power-Up Behaviour

When mains power is first applied, the 2200 μF reservoir capacitor is initially discharged. A charging current therefore flows through the transformer and bridge rectifier until the capacitor approaches its normal operating voltage.

The L7812CV output rises as sufficient input voltage becomes available.

The Nucleo board then begins its own power-on and reset sequence.

Importantly, the speaker relays remain de-energised during this period because the STM32 has not yet completed the protection startup sequence.

The overall startup chain is therefore

Mains Applied $\rightarrow$ 12 V AC Generated $\rightarrow$ Raw DC Rises $\rightarrow$ 12 V Regulation Established $\rightarrow$ STM32 St

20.22 Effect of Relay Switching on the Supply

When the two speaker relays are energised, approximately

$$66.6 \text{ mA}$$

of additional current is suddenly applied to the 12 V supply.

Without adequate decoupling, this current step could produce a temporary voltage reduction.

The 100 μF output capacitor and local ceramic capacitors help reduce this disturbance.

This is important because the relay coils and microcontroller share the same regulated 12 V source.

The controller supply must remain sufficiently stable that energising the relays does not cause an STM32 reset or corrupt ADC measurements.

20.23 Separation from the Main Amplifier Supply

The protection controller is not powered directly from the approximately $\pm 33\text{--}\pm 34 \text{ V}$ main amplifier rails.

Using a separate transformer and regulator has several advantages.

It avoids the need to drop a very large voltage from the main amplifier supply, provides independent isolation, and reduces the coupling of high-current audio supply variations into the protection electronics.

The main power amplifier can draw several amperes and its rail voltage changes with audio output power.

In contrast, the controller requires a comparatively small and stable low-voltage supply.

The separate supply therefore helps maintain reliable microcontroller and ADC operation.

20.24 Power-Supply Noise Considerations

Although the controller does not process the main audio signal, supply noise can still affect its ADC measurements.

Potential noise sources include:

- bridge-rectifier charging pulses;
- relay coil switching;
- digital switching from the STM32;
- mains-frequency magnetic coupling;
- high-current power-amplifier wiring.

The power-supply capacitors, local MCU decoupling, controlled grounding, and physical PCB separation help reduce these effects.

In particular, relay-coil return current should not flow through the same sensitive ground path used as the ADC measurement reference.

20.25 Grounding

The protection-controller supply establishes a local low-voltage ground reference.

This ground is used by the Nucleo board, relay-driver transistors, status LEDs, temperature-sensing circuits, and DC sensing circuits.

Because the DC detection circuit measures the power-amplifier output relative to ground, the protection-controller ground must ultimately have a defined relationship with the amplifier signal reference.

This connection should be made according to the overall grounding strategy rather than through multiple uncontrolled chassis or PCB paths.

The detailed connection between signal ground, protection ground, power ground, and protective earth is discussed separately in the grounding chapters.

20.26 Mains and Low-Voltage PCB Separation

The FS24-250-C2 transformer is mounted on the protection-controller PCB, which means that hazardous mains voltage is physically present on part of that PCB.

The primary side must therefore be clearly separated from the isolated secondary and low-voltage electronics.

The board should contain a clearly defined isolation boundary:

MAINS / PRIMARY AREA ISOLATION BOUNDARY LOW-VOLTAGE AREA.

The mains side contains the transformer primary connections and associated mains conductors.

The isolated side contains:

- transformer secondaries;
- bridge rectifier;
- smoothing capacitors;
- L7812CV regulator;
- Nucleo board;
- relay drivers;
- relays;
- ADC sensing circuits.

Low-voltage copper must not unnecessarily cross the primary-side area.

20.27 Creepage and Clearance

Adequate creepage and clearance distances must be maintained between mains conductors and isolated low-voltage conductors.

Clearance refers to the shortest distance through air between two conductive parts.

Creepage refers to the shortest distance measured along the surface of an insulating material.

The PCB layout must therefore provide deliberate physical spacing between the 230 V AC primary circuit and the protection-controller low-voltage circuitry.

The exact safety requirements depend on the applicable insulation category, pollution degree, PCB material, system construction, and relevant electrical safety standard.

These requirements are treated in greater detail in the *Safety Considerations* chapter.

20.28 Capacitor Voltage Ratings

The main smoothing capacitor is rated at

$$25 \text{ V},$$

while its normal operating voltage is expected to be approximately 15–16 V DC.

The 100 μF output capacitor is also rated at

$$25 \text{ V},$$

while operating from the regulated 12 V rail.

The capacitor voltage ratings therefore provide useful margin above the normal operating voltages.

Nevertheless, the raw no-load voltage should be measured on the completed hardware to confirm that the smoothing capacitor remains comfortably below its rated voltage under mains and transformer regulation conditions.

20.29 Protection-Supply Failure Behaviour

The protection system is designed so that loss of its supply moves the loudspeakers toward the safe disconnected state.

If the 12 V rail disappears,

$$V_{12} \rightarrow 0,$$

the relay coils cannot remain energised.

Therefore,

Protection Supply Failure $\Rightarrow$ Relay Coils OFF $\Rightarrow$ Speakers Disconnected.
--

This is an important fail-safe property because the speaker connection does not remain closed merely because the microcontroller has lost power.

20.30 Failure of the 12 V Regulator

The L7812CV regulator represents an important component of the controller supply and should be verified during commissioning.

Possible abnormal conditions include:

- no output voltage;

- output voltage below 12 V;
- excessive regulator temperature;
- instability caused by incorrect bypassing;
- input voltage falling below the required regulation headroom.

If the regulated voltage becomes too low to operate the relay coils or Nucleo board, the protection system naturally tends toward the speaker-disconnected state.

20.31 Expected Test Points

Several voltage measurements should be recorded during commissioning.

The principal test points are

$$V_{\text{SEC}},$$

the transformer secondary RMS voltage,

$$V_{\text{RAW}},$$

the rectified and smoothed voltage before the regulator,

$$V_{12},$$

the regulated 12 V output, and

$$V_{3.3},$$

the logic supply used by the STM32 and sensing circuitry.

Expected nominal values are approximately

$$V_{\text{SEC}} \approx 12 \text{ V}_{\text{RMS}},$$

$$V_{\text{RAW}} \approx 15\text{--}16 \text{ VDC}$$

under appropriate loading conditions,

$$V_{12} \approx 12 \text{ VDC},$$

and

$$V_{3.3} \approx 3.3 \text{ VDC}.$$

20.32 Initial Power-Supply Verification

The protection-controller supply should be verified before the Nucleo board and loud-speaker relays are relied upon for normal operation.

A suitable test procedure is:

1. Inspect the PCB for solder bridges and incorrect component orientation.
2. Verify the primary-side wiring before applying mains power.
3. Confirm the transformer secondary pin configuration.
4. Confirm that the two 12 V secondary windings are connected with the correct phase before paralleling them.
5. Verify the orientation of the W04G bridge rectifier.
6. Verify the polarity of the 2200 μF smoothing capacitor.
7. Verify the L7812CV pinout and orientation.
8. Verify the polarity of the 100 μF output capacitor.
9. Verify installation of the 100 nF bypass capacitors.
10. Apply mains power using the controlled initial power-up procedure.
11. Measure approximately 12 V_{RMS} at the transformer secondary.
12. Measure the raw rectified voltage across the 2200 μF capacitor.
13. Confirm approximately 12 VDC at the regulator output.
14. Verify the Nucleo logic rail.
15. Confirm that the Nucleo starts correctly.
16. Energise each relay independently.
17. Energise both relays simultaneously and verify that the 12 V rail remains stable.
18. Check the transformer, bridge rectifier, L7812CV, and capacitors for abnormal heating.
19. Observe the raw and regulated rails with an oscilloscope to evaluate ripple and switching disturbances.

20.33 Measurements Under Maximum Controller Load

The supply should also be tested with both relay coils energised and the protection controller operating normally.

The following quantities should be recorded:

$$V_{\text{SEC,load}},$$

$$V_{\text{RAW,load}},$$

$$V_{12,\text{load}},$$

$$I_{\text{controller}},$$

and

$$\Delta V_{\text{RAW,ripple}}.$$

The regulator case temperature should also be measured after sufficient time has been allowed for thermal equilibrium.

These measurements confirm that adequate regulator headroom remains available under the real controller load.

20.34 Design Summary

The STM32 speaker-protection controller is powered by an independent isolated linear supply rather than from the high-voltage power-amplifier rails.

The supply uses an

$$\boxed{\text{FS24-250-C2}}$$

6 VA PCB transformer configured with its two 12 V secondary windings correctly connected in parallel.

The resulting secondary supply is approximately

$$\boxed{12 \text{ V}_{\text{RMS}}, 500 \text{ mA}}.$$

A W04G-E4/51 full-wave bridge rectifier converts the AC voltage to rectified DC.

The theoretical secondary peak voltage is

$$12\sqrt{2} \approx 16.97 \text{ V},$$

while the practical raw DC voltage after bridge losses is expected to be in the approximate

$$15\text{--}16\text{ V}$$

region under suitable operating conditions.

A

$$2200\text{ }\mu\text{F}/25\text{ V}$$

capacitor provides the principal raw-DC smoothing.

The raw voltage is then regulated using an

$$\text{L7812CV}$$

linear regulator to produce the required

$$+12\text{ VDC}$$

controller rail.

A $100\text{ }\mu\text{F}/25\text{ V}$ output capacitor and several 100 nF bypass capacitors provide additional low- and high-frequency decoupling.

The regulated 12 V rail supplies the two G2RL-1A-E-DC12 speaker-relay coils and the NUCLEO-G031K8 VIN input. The Nucleo board then provides the low-voltage logic supply required by the STM32G031K8 and associated sensing circuits.

The two relay coils require approximately

$$66.6\text{ mA}$$

together when energised, corresponding to approximately

$$0.8\text{ W}$$

of 12 V relay-coil power.

The supply is designed so that a loss of controller power automatically removes relay-coil current. Because the loudspeaker relays use normally-open contacts, this gives the fail-safe behaviour

$$\text{Protection Supply Lost} \Rightarrow \text{Relays OFF} \Rightarrow \text{Speakers Disconnected.}$$

Because the FS24-250-C2 is mounted directly on the protection-controller PCB, the PCB contains both hazardous mains voltage and isolated low-voltage electronics. A clearly defined mains isolation boundary, appropriate creepage and clearance, safe primary wiring, and correct transformer-secondary phasing are therefore essential parts of the final design.

The transformer secondary voltage, raw DC voltage, regulated 12 V output, 3.3 V logic supply, ripple, controller current, and regulator temperature will be verified experimentally during final protection-system testing.

Chapter 21

PCB Design

21.1 Introduction

Printed circuit board design is a critical part of the practical implementation of the Hi-Fi amplifier because the electrical performance of the complete system depends not only on the schematic but also on component placement, conductor geometry, grounding, high-current routing, thermal considerations, and physical separation between different functional sections.

Two principal PCBs were developed in KiCad for this project:

- the power-amplifier and main power-supply PCB;
- the STM32-based speaker-protection-controller PCB.

The two boards perform significantly different functions and therefore require different layout priorities. The power-amplifier PCB contains both sensitive analogue circuitry and several-ampere output-current paths, while the protection-controller PCB combines low-voltage analogue sensing, digital control, loudspeaker relays, and an isolated mains-powered auxiliary supply.

Both layouts were therefore developed by dividing the circuitry into functional regions and routing each region according to its electrical and mechanical requirements.

21.2 PCB Design Objectives

The principal PCB design objectives were:

- maintain short and low-impedance high-current paths;
- separate low-level audio signals from high-current power paths;
- provide controlled ground return paths;
- minimise unnecessary coupling between circuit sections;

- position decoupling capacitors close to the circuits they support;
- provide suitable track widths for different current levels;
- provide mechanically practical positions for power transistors, relays, connectors, and large capacitors;
- provide external access to all required power and signal connections;
- provide mounting holes for secure installation inside the chassis;
- maintain separation between mains-voltage and low-voltage circuitry on the protection-controller board;
- produce layouts suitable for standard Gerber and drill-file manufacturing.

Particular attention was given to the difference between low-level signal currents and high-current power currents. These should not unnecessarily share the same narrow PCB conductors or ground-return paths.

21.3 Board Overview

The principal physical characteristics extracted from the final KiCad PCB files are summarised in Table 21.1.

Table 21.1: Main physical characteristics of the two PCB designs.

Parameter	Power Amplifier PCB	Protection PCB
Board width	136 mm	121 mm
Board height	124.5 mm	182.25 mm
PCB thickness	1.6 mm	1.6 mm
Copper layers	2	2
Copper layers used	F.Cu, B.Cu	F.Cu, B.Cu
Ground plane	B.Cu	B.Cu
Mounting holes	4 × 5 mm	4 × 3.2 mm M3
Principal routed widths	0.4, 1.0, 2.5 mm	1.0, 2.4 mm

Both boards use a conventional two-layer construction with front and back copper layers and a nominal board thickness of

$$\boxed{1.6 \text{ mm}}.$$

The copper weight is a fabrication parameter and is not defined directly by the uploaded KiCad PCB files. Consequently, a specific copper weight is not assumed in this chapter.

21.4 Two-Layer PCB Structure

The PCB stack used for both boards can be represented in simplified form as

Top Silkscreen

Top Solder Mask

F.Cu

FR-4 Substrate

B.Cu

Bottom Solder Mask.

The two copper layers provide sufficient routing flexibility while retaining a simple and economical PCB construction.

The front copper layer is used extensively for component-to-component routing, while the rear copper layer also provides routing and carries the principal ground copper zone.

Layer changes are made through plated vias where required.

21.5 Power Amplifier PCB Architecture

The power-amplifier PCB has an overall rectangular outline of

136 mm × 124.5 mm.

The board integrates the analogue power-amplifier circuit together with the main bridge-rectifier and reservoir-capacitor section.

The component placement follows the functional signal and power flow of the circuit.

The low-level audio input connections are positioned toward the left-hand side of the board. The small-signal transistor stages occupy the central portion of the amplifier area, while the driver and power-output components progress toward the opposite side.

The selected 2SA1943-O and 2SC5200-O output devices are positioned very close to the outer edge of the PCB. This allows the TO-3P power packages to be mounted mechanically to an external heatsink without requiring long high-current connections.

The arrangement therefore follows the approximate physical signal flow

Audio Input → Small-Signal Stage → Driver Stage → Power Output Stage.

This helps prevent high-current output wiring from unnecessarily passing through the sensitive input region.

21.6 Main Power-Supply Region

The same PCB also incorporates the high-current main power-supply section.

The power-supply region contains the KBPC3506WP bridge rectifier and four large

4700 μF

reservoir capacitors.

The capacitors use large snap-in footprints with a body diameter of approximately

25 mm.

These physically large components are grouped separately from the sensitive input transistor circuitry.

The main supply connectors are also located near the power-supply section, reducing the distance travelled by transformer and reservoir-capacitor currents.

This arrangement is important because rectifier charging currents contain large current pulses that should not share long PCB paths with the small audio input signal.

21.7 Power Transistor Placement

The final power devices Q7 and Q8 use vertical TO-3P footprints and are positioned along the board edge.

The output transistor locations are arranged so that the transistor bodies can face outward toward the external heatsink.

This provides several advantages:

- short electrical connections between the drivers and output devices;
- short high-current connections to the emitter resistors and loudspeaker output;
- straightforward heatsink attachment;
- reduced mechanical stress compared with extending the devices far from the PCB;
- improved accessibility during assembly.

The power resistors and driver transistors are positioned in the same general output-stage region.

21.8 Power Amplifier Routing Widths

The final power-amplifier PCB uses three principal routed conductor widths:

0.4 mm ,

1.0 mm ,

and

2.5 mm .

The 0.4 mm routing is used only in a limited small-signal portion of the design.

A large proportion of the amplifier routing uses

1.0 mm

tracks, providing considerably wider conductors than those normally required for very low-current logic circuitry.

The widest

2.5 mm

tracks are used on selected power and high-current paths, including sections of the supply and amplifier-output network.

This deliberate variation in track width prevents the complete PCB from being routed using a single conductor size regardless of current requirement.

21.9 Protection Controller PCB Architecture

The STM32 speaker-protection-controller PCB has an overall size of

$121\text{ mm} \times 182.25\text{ mm}$.

The board combines several different functional regions:

- auxiliary mains transformer and power supply;
- bridge rectifier and 12 V regulation;
- NUCLEO-G031K8 controller;
- left and right DC-offset sensing circuits;
- left and right temperature-monitoring inputs;

- relay-driver transistors;
- two loudspeaker relays;
- amplifier-input and speaker-output connectors;
- push buttons;
- fault and status indicator LEDs.

The physical board arrangement separates these functions according to their electrical requirements.

21.10 Protection Controller Functional Placement

The transformer occupies the upper-left/central portion of the protection PCB. The bridge rectifier, reservoir capacitor, and 12 V regulator are grouped in the upper power-supply region.

The NUCLEO-G031K8 module is positioned in the central-right portion of the board, allowing relatively short connections to the ADC sensing circuitry, relay drivers, control switches, and indicator LEDs.

The two speaker relays are positioned toward the lower-left area, close to the amplifier-output and loudspeaker-output wiring.

This arrangement reduces the length of the high-current speaker path through the board.

The status LEDs are positioned together in the lower-right region, providing a convenient physical grouping for connection to the front-panel indication system.

The temperature-sensor connectors are positioned near the right-hand edge, allowing the external NTC sensor wiring to leave the PCB without crossing the high-current relay region.

21.11 Protection Controller Routing Widths

The protection PCB uses two principal routed track widths:

1.0 mm

and

2.4 mm.

The 1.0 mm routing is used for the majority of the low-voltage power, sensing, GPIO, LED, and control connections.

The wider

2.4 mm

tracks are used on the high-current loudspeaker paths between the amplifier input connectors, relay contacts, and speaker output connectors.

The speaker path is therefore physically distinguished from the microcontroller and sensing circuitry.

21.12 Speaker Relay Current Paths

Each protection relay is located between an amplifier-output input connector and the corresponding speaker-output connector.

The current path for the left channel is approximately

$$\text{AMP_OUT_L} \rightarrow K_1 \rightarrow \text{SPEAKER_OUT_L},$$

while the right channel uses

$$\text{AMP_OUT_R} \rightarrow K_2 \rightarrow \text{SPEAKER_OUT_R}.$$

These routes use the wider 2.4 mm copper tracks rather than the normal controller routing width.

This is appropriate because the relay contacts can carry several amperes of loudspeaker current, whereas the MCU GPIO and sensor circuits operate at much smaller currents.

21.13 Ground Planes

Both final PCB files contain a copper ground zone on the bottom copper layer.

For the power-amplifier PCB, the zone is assigned to

GND

on

B.Cu

The protection-controller PCB similarly uses a bottom-layer zone named

GROUND PLANE

connected to the GND net.

The zone clearance defined in both PCB files is

0.5 mm,

with a minimum copper-zone thickness of

0.25 mm.

The ground zones reduce the impedance of ground connections and provide shorter current-return paths than would be obtained using only narrow tracks.

Their detailed current-return behaviour and relationship to signal ground and power ground are discussed separately in the dedicated grounding and ground plane chapters.

21.14 Layer Changes and Vias

The layouts were routed with relatively few layer transitions.

The power-amplifier PCB contains seven through vias. The routed vias use a diameter of approximately

1.0 mm

with a drill of

0.8 mm.

The protection-controller PCB contains only two routing vias in the final file.

These consist of one

0.6 mm

via with a

0.3 mm

drill and one

0.8 mm

via with a

0.4 mm

drill.

The small number of vias reflects the predominantly through-hole construction and the effort to keep important current paths direct.

21.15 Through-Hole Construction

Both boards make extensive use of through-hole components.

The power-amplifier PCB includes through-hole transistors, axial resistors, electrolytic capacitors, fuse holders, the bridge rectifier, large snap-in reservoir capacitors, and solder-wire connection pads.

The protection PCB similarly contains a through-hole transformer, bridge rectifier, TO-220 regulator, relays, BC337 transistors, axial resistors, capacitors, terminal blocks, switches, LEDs, and the Nucleo module.

Through-hole construction is appropriate for this project because many of the components are mechanically large or carry significant current.

It also simplifies manual assembly, inspection, replacement, and troubleshooting.

21.16 External Connections

The layouts use large solder-wire footprints for several external high-current connections.

On the power-amplifier board these include:

- audio input and signal ground;
- speaker output;
- power-supply connections;
- main ground connections.

The protection-controller PCB includes dedicated connections for:

- transformer and AC power connections;
- left amplifier output;
- right amplifier output;
- left loudspeaker output;
- right loudspeaker output;
- temperature sensor 1;
- temperature sensor 2.

Placing external connectors near the relevant circuit sections reduces the need for long PCB routes and crossing external wires.

21.17 Mechanical Mounting

Both boards include four mechanical mounting holes.

The power-amplifier board uses

$$4 \times 5 \text{ mm}$$

mounting holes positioned close to the board corners.

The protection-controller PCB uses four

$$3.2 \text{ mm}$$

holes intended for M3 mounting hardware.

The use of four mounting points prevents the comparatively large boards from flexing significantly during connector insertion, component servicing, or transport.

This is particularly important for the power-amplifier PCB because it contains large reservoir capacitors, fuse holders, a bridge rectifier, and heatsink-mounted power-transistor connections.

21.18 Mains and Low-Voltage Separation

The protection-controller PCB requires additional safety consideration because the auxiliary transformer is mounted directly on the PCB.

The transformer primary and mains-input connections are physically located on one side of the transformer, while the transformer secondary connections, bridge rectifier, regulator, Nucleo board, relays, and sensing electronics are located on the isolated low-voltage side.

The required layout philosophy is therefore

$$\boxed{\text{Mains Region} \mid \text{Transformer Isolation} \mid \text{Low-Voltage Region}}.$$

High-voltage and low-voltage copper should not be routed arbitrarily across this isolation boundary.

The exact creepage and clearance requirements depend on the final fabrication and safety standard and are discussed separately in the *Safety Considerations* chapter.

21.19 Analogue and Digital Separation

The protection-controller PCB contains both analogue sensing circuits and digital micro-controller signals.

The DC-offset and thermistor sensing circuits must provide stable voltages to the STM32 ADC, while the relay drivers, push buttons, LEDs, and MCU generate digital switching currents.

The component placement therefore keeps the sensing networks physically organised around their corresponding ADC inputs rather than routing the analogue signals repeatedly through the relay-current region.

The relay coil and LED current paths return through the controller power system rather than being intentionally routed through the analogue sensing nodes.

21.20 Power and Decoupling Placement

Power-supply and decoupling capacitors are placed near the sections that they support.

On the power-amplifier board, the large reservoir capacitors are positioned close to the bridge rectifier and main supply distribution region.

The amplifier itself also contains local rail-decoupling capacitors close to the active stages.

On the protection board, the 2200 μF reservoir capacitor, rectifier, regulator, and regulator bypass capacitors are grouped within the auxiliary supply region.

This reduces unnecessary loop area between the rectifier, smoothing capacitors, regulator, and loads.

21.21 PCB Design Verification

KiCad provides electrical and physical verification tools that should be used before PCB manufacture.

Important checks include:

- confirmation that all required nets are routed;
- clearance checking between copper objects;
- verification of footprint orientation;
- verification of pad and drill dimensions;
- inspection of polarity-sensitive components;
- verification of board-edge clearance;
- inspection of copper-zone connections;
- confirmation of mounting-hole locations;
- verification of high-current track widths;
- inspection of mains-to-low-voltage separation.

Particular attention must be given to custom footprints because an electrically correct schematic can still produce an unusable PCB if the physical pin numbering or dimensions are incorrect.

21.22 Manufacturing Preparation

After PCB layout and verification, the KiCad design can be converted into the standard files required by a PCB manufacturer.

These include Gerber files for the copper, solder-mask, silkscreen, and board outline layers together with separate plated and non-plated drill information where required.

The board outline is defined on the KiCad

`Edge.Cuts`

layer.

Generating the drill files in addition to the Gerber layers is essential because the boards contain many through-hole components and mechanical mounting holes.

The manufacturing package should therefore contain both the Gerber layer data and the corresponding drill files.

21.23 Relationship to the Following PCB Chapters

This chapter describes the overall PCB design methodology used throughout the project. More detailed analysis is provided in the following chapters.

The *Power Amplifier PCB Layout* chapter examines the physical layout of the analogue power stage and main supply.

The *Protection Controller PCB Layout* chapter examines the STM32, relay, sensing, and auxiliary power sections.

The *PCB Track Width and Current Handling* chapter considers conductor current capability in greater detail.

The *Grounding Strategy*, *Audio Signal Ground and Power Ground*, and *Copper Zones and Ground Planes* chapters examine the return-current and noise-control aspects of the layouts.

21.24 PCB Design Summary

Two principal two-layer PCBs were developed in KiCad for the complete Hi-Fi amplifier system.

The power-amplifier PCB measures

136 mm × 124.5 mm

and combines the discrete Class-AB amplifier with the main bridge-rectifier and reservoir-capacitor section.

Its routing uses actual track widths of

$0.4\text{ mm}, 1.0\text{ mm}, 2.5\text{ mm}$,

with the widest conductors used on selected high-current and power paths.

The 2SA1943-O and 2SC5200-O power transistors are located close to the board edge to simplify external heatsink mounting.

The protection-controller PCB measures

$121\text{ mm} \times 182.25\text{ mm}$

and integrates the auxiliary power supply, NUCLEO-G031K8, DC sensing, temperature sensing, relay drivers, loudspeaker relays, controls, and indicators.

Its routing uses

1.0 mm

tracks for the majority of the controller circuitry and

2.4 mm

tracks for the high-current loudspeaker relay paths.

Both PCBs use a nominal

1.6 mm

board thickness and employ both front and back copper layers.

A GND copper zone is implemented on the bottom layer of each PCB, with a defined clearance of

0.5 mm

and a minimum zone thickness of

0.25 mm .

The layouts use relatively few routing vias and extensive through-hole construction, helping maintain direct current paths and simplifying prototype assembly and servicing.

Mechanical integration was also considered during layout. The amplifier PCB contains four 5 mm mounting holes, while the protection board contains four 3.2 mm M3 mounting holes.

The resulting PCB architecture separates sensitive analogue circuitry, high-current amplifier paths, power-supply circuitry, digital control, relay switching, and mains-voltage circuitry according to their individual electrical requirements.

The PCB design therefore forms an integral part of the amplifier engineering rather than simply providing mechanical interconnection between schematic components.

Chapter 22

Power Amplifier PCB Layout

22.1 Introduction

The printed circuit board layout is an important part of the practical implementation of an audio power amplifier. Although the schematic defines the electrical connections between components, the PCB determines the actual conductor lengths, current-return paths, parasitic resistance and inductance, component spacing, thermal arrangement, and interaction between low-level signal circuitry and high-current power circuitry.

The power-amplifier PCB developed for this project was designed in KiCad and combines the discrete Class-AB amplifier circuit with the principal main power-supply components on a single two-layer printed circuit board.

The final layout was organised so that the low-level input circuitry, voltage amplification stage, bias network, driver stage, output stage, bridge rectifier, reservoir capacitors, rail fuses, and external connections form clearly identifiable functional regions.

Particular attention was given to the physical separation of sensitive audio circuitry from the high-current power-supply paths, the location of the output transistors for heatsink attachment, the width of high-current conductors, and the mechanical accessibility of the principal service components.

22.2 PCB Dimensions and Construction

The final power-amplifier PCB has a rectangular outline with dimensions

$$136 \text{ mm} \times 124.5 \text{ mm}$$

and a nominal PCB thickness of

$$1.6 \text{ mm}.$$

A two-layer PCB construction is used, consisting of

F.Cu

and

B.Cu.

The front copper layer is used extensively for point-to-point routing between the amplifier components, while the bottom copper layer provides additional routing capability and also contains the principal ground copper zone.

The final KiCad layout is shown in Figure 22.1.

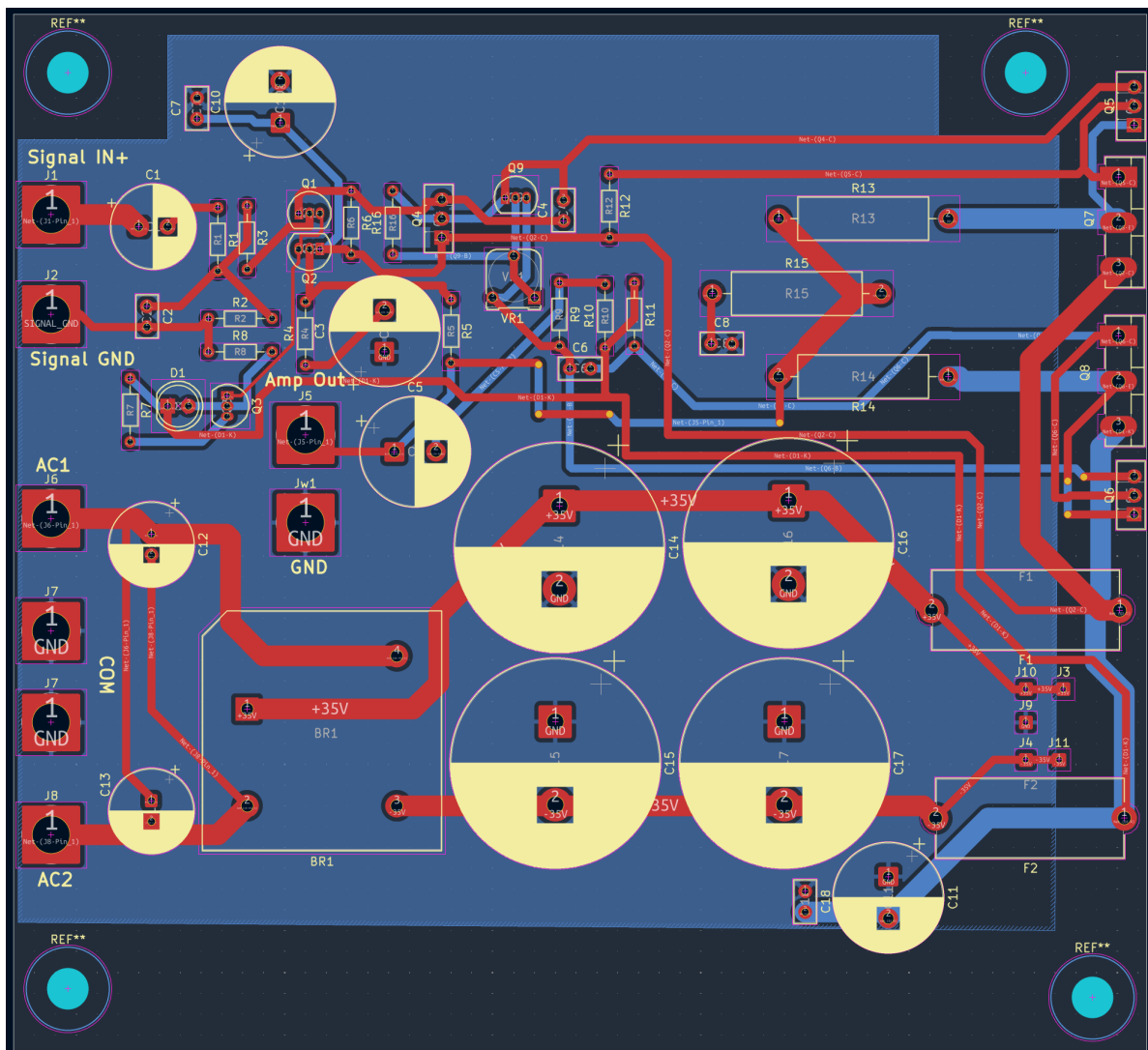


Figure 22.1: Final KiCad PCB layout of the Project 3A-derived power amplifier and main power-supply section. The figure shows the low-level amplifier stages, driver and output stages, rail fuses, bridge rectifier, reservoir capacitors, copper routing, ground plane, mounting holes, and external wiring connections.

As shown in Figure 22.1, the low-level input circuitry is concentrated toward the upper-left area of the PCB. The differential input stage, voltage-amplification stage, bias network,

and driver circuitry then progress toward the right-hand side of the board.

The power output transistors Q7 and Q8 are positioned directly along the right-hand PCB edge. This arrangement permits the devices to face outward toward an external heatsink while maintaining relatively short electrical connections to the driver circuitry and emitter resistors.

The four large reservoir capacitors occupy the lower-central part of the PCB, while the bridge rectifier is positioned toward the lower-left section. The two supply-rail fuses are positioned toward the right-hand edge, where they remain accessible during servicing.

The wide copper tracks visible in Figure 22.1 correspond to the principal high-current supply and output paths, while narrower conductors are used in the small-signal amplifier circuitry. The bottom copper layer and broad ground zone provide additional routing and low-impedance current-return paths.

22.3 Functional Partitioning of the PCB

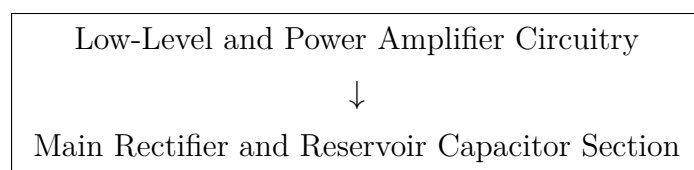
The PCB can be divided into two principal functional regions:

- the discrete power-amplifier circuit;
- the bridge-rectifier and reservoir-capacitor power-supply circuit.

The upper section of the board primarily contains the amplifier signal path, while the lower section is dominated by the main power-supply components.

This arrangement is intentional because the bridge rectifier and reservoir capacitors carry high-amplitude charging pulses that should not be routed through the same physical area as the sensitive input circuitry.

The basic physical organisation is therefore



22.4 Signal Flow Through the PCB

The component placement approximately follows the electrical progression of the amplifier:

Audio Input → Differential Input Stage → Voltage Amplification Stage → Driver Stage → Power Out

This physical progression reduces unnecessary crossing between the sensitive input signal and the high-current output conductors.

The audio input is located at the left-hand side of the board, while the driver and output transistors are located toward the right-hand side.

22.5 Audio Input Region

The audio signal enters the PCB through J1, identified on the board as

Signal IN+.

The corresponding signal-ground connection is provided through J2, labelled

Signal GND.

These connections are positioned at the outer edge of the PCB so that the external input wiring does not need to cross the power-supply or output-stage regions.

The input coupling capacitor C1 and the surrounding bias resistors are positioned close to the audio-input connector.

Keeping the low-level input path short reduces its susceptibility to noise coupled from the main power-supply and output-current conductors.

22.6 Differential Input Stage

The amplifier input stage contains Q1 and Q2, which form the differential input pair.

As visible in Figure 22.1, the two devices are positioned close together in the upper-left amplifier region.

Their associated resistors and capacitors are also concentrated around the same section.

This compact placement minimises unnecessary conductor length around the high-gain input stage.

Q3 is located nearby as part of the following small-signal circuitry.

22.7 Voltage Amplification Stage

The voltage-amplification stage occupies the central section between the differential input stage and the driver circuitry.

Q4 and its associated components are positioned so that the transition toward the driver transistors remains direct.

The high-frequency compensation capacitors

$$C_4 = 100 \text{ pF}$$

and

$$C_6 = 100 \text{ pF}$$

are located inside this active amplifier region.

Because these components influence the high-frequency behaviour and stability of the amplifier, short PCB connections are desirable.

22.8 Bias Adjustment Circuit

The bias-control circuit is positioned near the centre of the amplifier section.

The adjustment is performed using

$$VR1 = 2 \text{ k}\Omega.$$

The trimmer is accessible from the component side of the PCB so that the quiescent current can be adjusted during commissioning.

The bias transistor Q9 is positioned within the amplifier circuitry but is not placed directly against the main output heatsink.

This follows the thermal behaviour of the complementary-feedback output stage used in the Project 3A-derived circuit.

22.9 Driver Stage Placement

The driver transistors Q5 and Q6 are positioned immediately before the main output devices in the physical signal path.

Their location near the right-hand side of the board reduces the length of the connections between the driver transistors and the bases of Q7 and Q8.

Short driver-to-output connections are desirable because unnecessary inductance in this part of the circuit can adversely affect high-frequency behaviour.

22.10 Power Output Transistor Placement

The output stage uses the complementary power devices Q7 and Q8.

In the selected amplifier implementation these correspond to the complementary power-transistor pair

$$2SA1943-O$$

and

2SC5200-O

.

Both devices use vertical TO-3P through-hole footprints.

As clearly visible in Figure 22.1, the transistors are positioned directly along the right-hand edge of the PCB.

The mechanical arrangement is therefore

PCB → Output Transistor → Thermal Interface → External Heatsink.

This arrangement avoids the need for long flying wires between the PCB and the output transistors.

22.11 Advantages of Edge-Mounted Output Devices

Positioning Q7 and Q8 along the PCB edge provides several advantages:

- short electrical connections to the driver stage;
- short high-current paths through the output stage;
- straightforward external heatsink mounting;
- reduced heat transfer directly into the PCB;
- easier transistor replacement during servicing;
- improved mechanical accessibility.

The heatsink therefore carries the primary thermal load of the output devices, while the PCB mainly provides the required electrical interconnection.

22.12 Emitter Resistor Placement

The power emitter resistors are

$$R_{13} = R_{14} = 0.33 \, \Omega.$$

As shown in Figure 22.1, R13 and R14 are positioned immediately before the output transistor region.

These resistors carry the high loudspeaker current and therefore use physically larger footprints than the small-signal resistors.

Their relatively large spacing also helps prevent unnecessary heating of nearby small components.

22.13 Speaker Output Routing

The amplifier output connection is labelled

Amp Out

on the PCB.

This node carries the full loudspeaker current.

The associated routing therefore uses substantially wider copper than the small-signal input and voltage-amplification stages.

The current-handling capability of the selected output conductor is analysed separately in the *PCB Track Width and Current Handling* chapter.

22.14 Zobel Network Placement

The output network contains

$$R_{15} = 10 \, \Omega$$

and

$$C_8 = 100 \, \text{nF}.$$

These components form the Zobel network and are positioned in the output section close to the high-current stage.

Their placement keeps the high-frequency output-current path local to the output region rather than routing it back through the small-signal section.

22.15 Main Power-Supply Region

The main rectifier and reservoir-capacitor section is integrated onto the same PCB as the amplifier.

The principal components are the

KBPC3506WP

bridge rectifier and four

4700 μF

reservoir capacitors.

The bridge rectifier BR1 is positioned in the lower-left section of the board.

The four large reservoir capacitors occupy the lower-central area and are clearly visible in Figure 22.1.

22.16 Reservoir Capacitor Arrangement

The reservoir capacitors are

$$C_{14} = C_{15} = C_{16} = C_{17} = 4700 \mu\text{F}.$$

Two capacitors operate in parallel on each power rail.

The resulting nominal capacitance per rail is therefore

$$C_{\text{rail}} = 4700 + 4700.$$

Thus,

$$C_{\text{rail}} = 9400 \mu\text{F}.$$

The capacitors are grouped close to the bridge rectifier and main power-distribution paths. This reduces the physical area of the rectifier charging-current loops.

22.17 Rectifier Current Path

The transformer secondary enters the PCB through the connections labelled

$$\boxed{\text{AC1}}$$

and

$$\boxed{\text{AC2}}.$$

The centre connection is brought into the common ground region.

The bridge rectifier converts the transformer secondary voltage into positive and negative DC supply rails.

The high-current rectifier paths are physically concentrated around BR1 and the four reservoir capacitors.

This avoids routing the large charging-current pulses through the sensitive input-amplifier area.

22.18 Power Rail Distribution

The external transformer used in this project has a nominal secondary voltage of

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}}.$$

The corresponding practical DC rails are expected to be approximately

$$\boxed{\pm 33\text{--}\pm 34 \text{ VDC}}$$

depending on mains voltage, transformer regulation, rectifier losses, and load current.

The current PCB image still contains the historical Project 3A labels

$$+35\text{V}$$

and

$$-35\text{V}.$$

Within this project these labels represent the functional positive and negative amplifier supply rails,

$$V_+$$

and

$$V_-,$$

rather than an exact measured operating voltage of $\pm 35 \text{ V}$.

For the final PCB revision, replacing the historical labels with

$$\boxed{V_+}$$

and

$$\boxed{V_-}$$

would make the PCB notation fully consistent with the actual 24–0–24 V transformer used in the amplifier.

22.19 Rail Fuse Placement

Two PCB-mounted 5 mm × 20 mm fuse holders are included in the amplifier design.

The rail fuse value is

$$\boxed{3 \text{ A}}.$$

F1 protects the positive amplifier supply path and F2 protects the negative supply path.

The arrangement is therefore

$$V_+ \rightarrow F_1 \rightarrow \text{Positive Amplifier Rail}$$

and

$$V_- \rightarrow F_2 \rightarrow \text{Negative Amplifier Rail}.$$

As visible in Figure 22.1, both fuse holders are positioned near the right-hand edge of the board, improving accessibility for inspection and replacement.

22.20 Local Power-Supply Decoupling

Local rail-decoupling capacitors are included around the amplifier circuitry in addition to the four large main reservoir capacitors.

The relevant components include

$$C_{10} = 100 \mu\text{F},$$

$$C_{11} = 100 \mu\text{F},$$

$$C_7 = 100 \text{ nF},$$

and

$$C_{18} = 100 \text{ nF}.$$

The electrolytic capacitors provide local energy storage, while the smaller 100 nF capacitors provide a low-impedance path at higher frequencies.

This reduces the dependence of the amplifier stages on long PCB paths back to the main reservoir capacitors.

22.21 Track Width Selection

Three principal track widths are present in the final PCB layout:

0.4 mm, 1.0 mm, 2.5 mm.

The narrowest 0.4 mm tracks are limited to low-current small-signal circuitry.

A significant proportion of the amplifier uses

1.0 mm

routing.

The high-current supply and output regions use tracks up to

2.5 mm.

This distinction is visible in Figure 22.1, where the supply and output paths are noticeably wider than the input-stage routing.

The quantitative current-carrying capability of the selected widths is analysed separately in the *PCB Track Width and Current Handling* chapter.

22.22 Front and Bottom Copper Layers

The front copper layer carries a large proportion of the direct component interconnections.

The bottom copper layer provides additional routing where required and also contains the main ground copper zone.

The two-layer approach makes it possible to maintain relatively direct electrical paths while avoiding excessive track crossings.

Layer transitions are used selectively.

The final amplifier PCB contains approximately

7

routing vias.

22.23 Ground Copper Zone

A large copper zone connected to

GND

is implemented on the bottom copper layer.

The KiCad zone uses a clearance of

0.5 mm

from unrelated copper and a minimum zone thickness of

0.25 mm.

The broad ground area is clearly visible in Figure 22.1.

The purpose of the ground zone is to provide low-impedance current-return paths and reduce the need for long individual ground tracks.

The relationship between signal ground, power ground, loudspeaker-current return, and the system star-ground point is discussed in greater detail in the dedicated grounding chapters.

22.24 Through-Hole Construction

The PCB uses predominantly through-hole components.

These include:

- BC546 small-signal transistors;
- BD139 and BD140 driver devices;
- TO-3P output transistors;
- axial resistors;
- power resistors;
- electrolytic capacitors;
- film and ceramic capacitors;
- bias trimmer;
- fuse holders;
- bridge rectifier;
- snap-in reservoir capacitors;
- external wire connections.

Through-hole construction is suitable for the present amplifier because many components are mechanically large, dissipate significant heat, or carry substantial current.

It also simplifies manual prototype assembly, inspection, and replacement.

22.25 Thermal Layout Considerations

Thermal behaviour was considered when arranging the board.

The dominant heat-generating devices are Q7 and Q8.

Their position at the right-hand PCB edge allows them to be connected directly to an external heatsink.

The bridge rectifier can also dissipate noticeable power under high-output conditions and is positioned away from the differential input pair.

The large emitter resistors R13 and R14 are also given sufficient physical space.

The thermally sensitive low-level input components are therefore not deliberately concentrated around the main heat-producing devices.

22.26 Mechanical Mounting

Four mechanical mounting holes are provided in the PCB.

Each hole has a nominal diameter of

5 mm.

As shown in Figure 22.1, the holes are positioned close to the four corners of the board.

This provides stable four-point mechanical support.

The mounting arrangement is particularly useful because the PCB contains several large components, including the reservoir capacitors, bridge rectifier, fuse holders, and power-stage circuitry.

22.27 External Wiring Connections

The principal external connections are positioned around the perimeter of the PCB.

The visible labels include

Signal IN+,

Signal GND,

Amp Out,

AC1,

AC2,

GND,

and

COM.

Placing these connection points near the edge simplifies chassis wiring and reduces the need for external wires to cross directly over the active amplifier circuitry.

22.28 Serviceability

The layout was also developed with practical servicing in mind.

The bias trimmer remains accessible from the component side of the PCB.

The two supply fuses can be reached from the board edge.

The power transistors remain accessible for heatsink attachment and replacement.

The large electrolytic capacitors and their polarity markings remain visually accessible during assembly.

The predominantly through-hole construction further simplifies component replacement during prototype development.

22.29 PCB Layout Verification

Before manufacture, the PCB should be checked using KiCad design-rule tools and careful manual inspection.

Particular attention should be given to:

- Q7 and Q8 footprint pin assignments;
- BD139 and BD140 footprint orientation;
- electrolytic capacitor polarity;
- bridge-rectifier pin arrangement;
- positive and negative supply routing;
- rail fuse connectivity;
- speaker-output routing;
- signal-ground connections;

- ground-zone connectivity;
- mounting-hole clearance;
- board-edge clearance;
- remaining unrouted connections.

Correct output-transistor footprint verification is particularly important because an incorrect base, collector, or emitter assignment could result in immediate amplifier failure during first power-up.

22.30 Manufacturing Output

After the PCB layout has been verified, KiCad is used to generate the files required by the PCB manufacturer.

The manufacturing package includes Gerber data for the required layers, including:

- F.Cu;
- B.Cu;
- front solder mask;
- back solder mask;
- front silkscreen;
- Edge.Cuts.

Drill files are also required because the design contains numerous plated through-holes as well as the four mechanical mounting holes.

The complete manufacturing package therefore consists of both Gerber and drill information.

22.31 Design Summary

The final power-amplifier PCB integrates the discrete Class-AB amplifier and the principal main power-supply components onto a single two-layer board.

The final board measures

$$136 \text{ mm} \times 124.5 \text{ mm}$$

and has a nominal thickness of

$$1.6 \text{ mm}.$$

The completed layout is shown in Figure 22.1.

The component arrangement approximately follows the electrical signal flow, with the sensitive input circuitry concentrated toward the left-hand side and the driver and output circuitry progressing toward the right-hand side.

The power output transistors Q7 and Q8 are positioned directly at the PCB edge to allow attachment to an external heatsink.

The lower portion of the board contains the KBPC3506WP bridge rectifier and four

4700 μ F

reservoir capacitors, giving a nominal

9400 μ F

of reservoir capacitance per supply rail.

The positive and negative amplifier rails are protected by individual

3 A

fuses.

The PCB uses conductor widths of

0.4 mm, 1.0 mm, 2.5 mm

according to the current level and purpose of the corresponding circuit region.

A broad GND copper zone is provided on the bottom layer with a clearance of

0.5 mm

and a minimum copper-zone thickness of

0.25 mm.

Four 5 mm mounting holes provide mechanical attachment to the amplifier chassis.

The PCB therefore combines electrical, thermal, mechanical, grounding, high-current, and serviceability considerations in a practical implementation of the Project 3A-derived Hi-Fi power amplifier.

Chapter 23

Protection Controller PCB Layout

23.1 Introduction

The loudspeaker protection controller combines analogue sensing, digital processing, relay control, user indication, and an isolated auxiliary power supply on a single printed circuit board. Because these functions operate at different voltage, current, and noise levels, their physical arrangement on the PCB is an important part of the overall protection-system design.

The PCB was designed in KiCad around the NUCLEO-G031K8 development board and contains the two DC-offset sensing channels, two temperature-monitoring channels, two loudspeaker relay drivers, two power relays, status and fault indicators, user-control push buttons, and the dedicated controller power supply.

The layout was organised into functional regions so that the high-current speaker paths, relay switching circuitry, analogue ADC sensing networks, microcontroller circuitry, and power-supply section remain physically identifiable and can be routed according to their individual requirements.

23.2 PCB Dimensions and Construction

The final protection-controller PCB has overall dimensions of

$$121 \text{ mm} \times 182.25 \text{ mm}$$

with a nominal PCB thickness of

$$1.6 \text{ mm}.$$

The board is implemented using two copper layers:

$$\text{F.Cu}$$

and

B.Cu.

The front layer is used for a large proportion of the point-to-point routing, while the bottom layer provides additional routing and supports the principal ground copper zone.

The complete protection-controller PCB layout is shown in Figure 23.1.

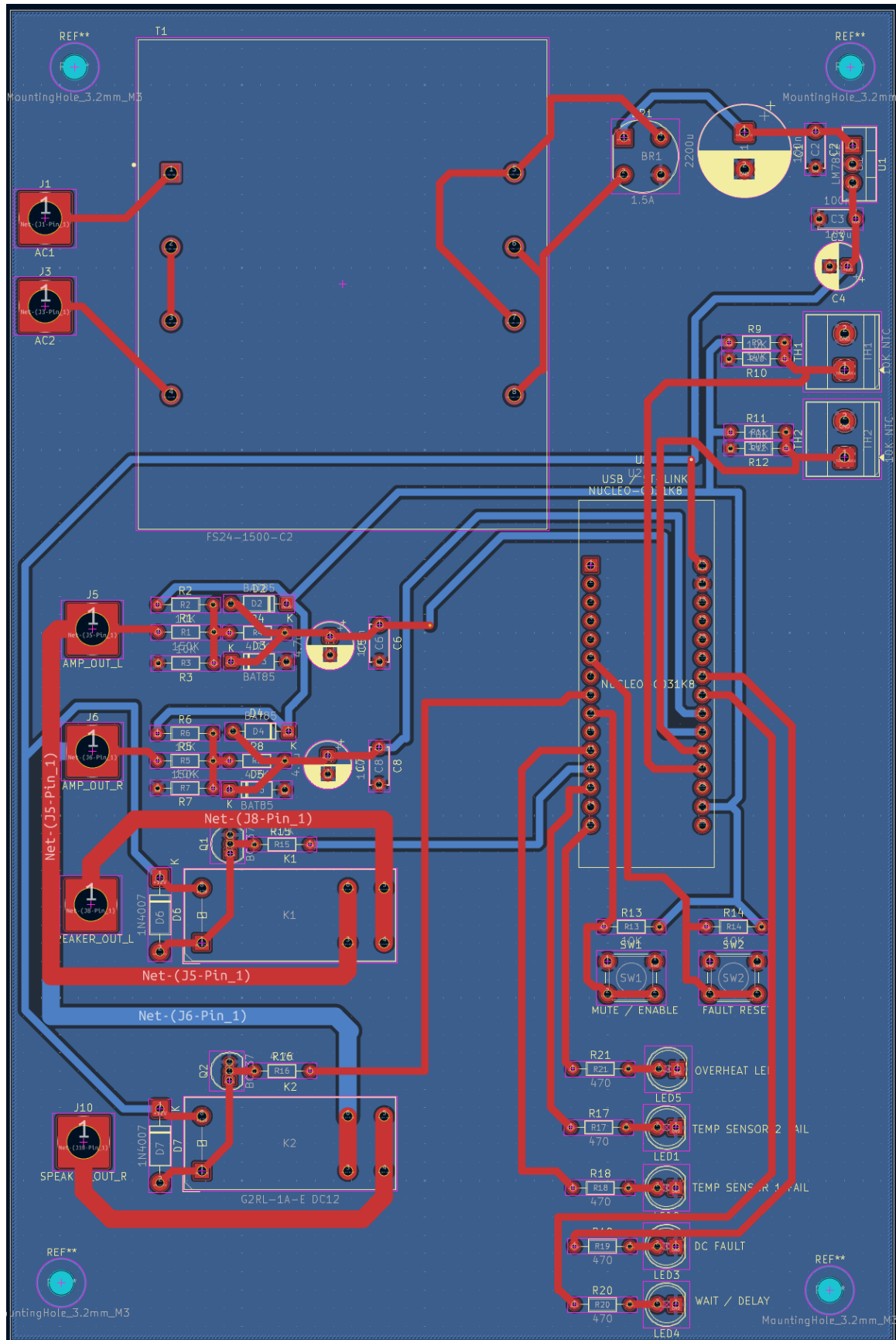


Figure 23.1: Final KiCad PCB layout of the STM32-based loudspeaker protection controller, including the auxiliary power supply, NUCLEO-G031K8, DC-offset sensing circuits, temperature-monitoring connections, relay drivers, loudspeaker relays, control buttons, status indicators, copper routing, and ground plane.

As shown in Figure 23.1, the PCB is substantially taller than the power-amplifier board because it combines the complete protection electronics and its dedicated power supply on one board.

23.3 Functional Organisation

The PCB can be divided into several principal functional regions:

- auxiliary isolated power supply;
- STM32 microcontroller and Nucleo development board;
- left-channel DC-offset sensing circuit;
- right-channel DC-offset sensing circuit;
- left- and right-channel temperature-monitoring interfaces;
- relay-driver circuitry;
- left and right loudspeaker relays;
- amplifier-input and speaker-output connections;
- user-control push buttons;
- status and fault indicator LEDs.

Separating these regions physically reduces unnecessary interaction between high-current relay paths, analogue ADC measurements, and digital microcontroller signals.

23.4 Auxiliary Power-Supply Region

The upper portion of the PCB contains the dedicated protection-controller power supply.

The principal components in this region are:

- PCB-mounted isolation transformer T1;
- bridge rectifier BR1;
- 2200 μF reservoir capacitor C1;
- 12 V linear regulator U1;
- 100 μF output capacitor;
- local 100 nF bypass capacitors.

The transformer is positioned toward the upper-left/central region of the board, while the rectifier, reservoir capacitor, regulator, and output filtering are grouped toward the upper-right.

This provides a short physical progression through the power-supply stages:

Transformer → Bridge Rectifier → Reservoir Capacitor → 12 V Regulator

.

Keeping these components close together reduces unnecessary loop area in the rectifier and filtering paths.

23.5 Power-Supply Distribution

The regulated

+12 V

rail is distributed from the upper power-supply region to the Nucleo board and the two relay coils.

The NUCLEO-G031K8 subsequently provides the low-voltage logic supply used by the microcontroller and sensing circuitry.

The layout therefore contains several different supply domains, including

+12 V,

+3.3 V,

and

GND.

These supply nets are routed independently from the high-current loudspeaker contact paths.

23.6 NUCLEO-G031K8 Placement

The NUCLEO-G031K8 module, designated U2, is positioned in the central-right portion of the PCB.

This location provides relatively direct connections to all major controller functions while avoiding placement directly within the high-current relay contact paths.

The Nucleo module interfaces with:

- the two DC sensing channels;

- the two temperature ADC channels;
- the two relay-control outputs;
- the status and fault LEDs;
- the MUTE / ENABLE push button;
- the FAULT RESET push button;
- the 12 V input and 3.3 V logic supply.

The central position therefore reduces the maximum trace length required between the microcontroller and the distributed protection circuits.

23.7 DC Offset Sensing Circuit Placement

The two DC-offset sensing networks are grouped together in the central-left region of the PCB.

The left and right amplifier-output connections enter through the dedicated connectors

AMP_OUT_L

and

AMP_OUT_R.

Each sensing channel contains the resistor network, clamp diodes, series ADC resistor, and analogue filtering components required to reduce the high-voltage amplifier output to a safe STM32 measurement range.

The two channels are laid out in a similar physical form. This helps maintain consistent electrical behaviour between the left and right sensing circuits.

The ADC networks are positioned between the amplifier-output connections and the Nucleo board, thereby avoiding unnecessarily long analogue sensing traces.

23.8 DC Sensing Components

The principal resistor values used in each DC sensing channel are

150 k Ω ,

10 k Ω ,

10 k Ω ,

and

4.7 k Ω .

The sensing circuits also include the

4.7 μ F

main low-pass capacitors and

100 nF

high-frequency bypass capacitors.

BAT85 clamp diodes are present in the final PCB layout to provide additional protection for the ADC inputs.

The duplicated physical arrangement of the two channels makes inspection, assembly, and troubleshooting easier.

23.9 Temperature Monitoring Connections

The two temperature-sensor connections are provided through TH1 and TH2.

Both use two-position terminal blocks located close to the right-hand edge of the PCB.

The two associated measurement nets are

TEMP1_ADC

and

TEMP2_ADC.

Placing the terminal blocks near the PCB edge allows the external heatsink NTC wiring to enter the controller without crossing the relay or mains-supply regions.

The temperature-divider resistors are located between these connectors and the Nucleo board, producing a compact analogue signal path.

23.10 Relay Driver Placement

Two independent BC337 transistor stages are used to drive the loudspeaker relays.

Q1 controls the first relay and Q2 controls the second relay.

The driver transistors are positioned immediately adjacent to their corresponding relay circuitry rather than next to the microcontroller.

This arrangement reduces the length of the relay-coil current paths.

Each driver section also contains its corresponding

1 k Ω

base resistor,

10 k Ω

base pulldown resistor, and

1N4007

flyback diode.

The flyback diodes are placed close to the relay and transistor circuits so that the inductive turn-off current remains local to the driver region.

23.11 Speaker Relay Placement

The protection controller uses two

G2RL-1A-E DC12

speaker relays, designated K1 and K2.

Both relays are positioned in the lower-left half of the PCB.

K1 controls the left loudspeaker path and K2 controls the right loudspeaker path.

Locating the relays close to the amplifier and speaker connectors allows the high-current audio path to remain physically short.

23.12 Left Speaker Current Path

The left amplifier output enters the board through

AMP_OUT_L.

The high-current signal is routed to relay K1 and then to

SPEAKER_OUT_L.

The physical path is therefore

AMP_OUT_L $\rightarrow$ K₁ $\rightarrow$ SPEAKER_OUT_L.

This path is routed using the wider copper traces used on the protection PCB.

23.13 Right Speaker Current Path

The second channel is arranged similarly:

$$\boxed{\text{AMP_OUT_R} \rightarrow K_2 \rightarrow \text{SPEAKER_OUT_R}}.$$

The physical symmetry of the two relay sections simplifies the stereo wiring and helps maintain similar electrical resistance in the left and right output paths.

23.14 High-Current Relay Track Width

Two principal routed track widths are present in the final protection PCB:

$$\boxed{1.0 \text{ mm}}$$

and

$$\boxed{2.4 \text{ mm}}.$$

The majority of the control, ADC, power-supply, LED, and GPIO routing uses

$$1.0 \text{ mm}$$

tracks.

The high-current loudspeaker paths use

$$\boxed{2.4 \text{ mm}}$$

tracks.

Inspection of the final KiCad design shows that the 2.4 mm routing is concentrated specifically on the nets connecting the amplifier-output connectors, relay contacts, and speaker-output connectors.

This prevents the several-ampere loudspeaker current from being routed through the same conductor geometry used by the ADC and logic circuitry.

23.15 Analogue and Digital Routing

The controller contains both sensitive analogue measurements and digital switching signals.

The ADC sensing networks are routed toward the Nucleo analogue inputs, while relay-control, LED-control, and push-button nets form the digital-control portion of the PCB.

The layout avoids deliberately routing the high-current speaker paths through the central ADC sensing area.

This is important because voltage developed across a shared conductor by relay or loud-speaker current could otherwise appear as an error in the low-voltage ADC measurements.

23.16 Status and Fault Indicator Region

The final PCB contains five 5 mm status and fault LEDs grouped vertically near the lower-right side of the board.

The indicators present in the actual PCB design are:

- OVERHEAT;
- TEMP SENSOR 2 FAIL;
- TEMP SENSOR 1 FAIL;
- DC FAULT;
- WAIT / DELAY.

Each LED uses a

470 Ω

series resistor.

Grouping the indicators together simplifies wiring to the front-panel region and allows the protection state to be interpreted visually.

23.17 User-Control Push Buttons

Two PCB-mounted push buttons are included near the lower-central/right area of the board.

SW1 is labelled

MUTE / ENABLE

while SW2 is labelled

FAULT RESET

Their location close to the Nucleo module keeps the digital control traces short while separating the switches from the high-current speaker connections.

23.18 Ground Plane

A large ground copper zone named

GROUND PLANE

is present on the bottom copper layer.

The zone is connected to

GND

and extends over most of the board area.

The KiCad zone is configured with a clearance of

0.5 mm

and a minimum copper thickness of

0.25 mm.

The purpose of the ground plane is to provide a broad, low-impedance reference for the microcontroller, ADC sensing circuits, LEDs, relay-driver circuitry, and low-voltage power supply.

The detailed treatment of return-current paths is discussed separately in the *Grounding Strategy* and *Copper Zones and Ground Planes* chapters.

23.19 Routing Vias

Only two routing vias are present in the final PCB layout.

One via has a nominal diameter of

0.6 mm

with a

0.3 mm

drill.

The second has a nominal diameter of

0.8 mm

with a

0.4 mm

drill.

The small number of vias results from the extensive use of through-hole components and the relatively direct routing between functional sections.

Reducing unnecessary layer transitions also simplifies visual inspection of the prototype PCB.

23.20 Through-Hole Construction

The protection-controller PCB uses predominantly through-hole components.

These include:

- the power transformer;
- bridge rectifier;
- LM7812 regulator;
- electrolytic capacitors;
- resistors;
- BAT85 clamp diodes;
- 1N4007 flyback diodes;
- BC337 relay-driver transistors;
- G2RL speaker relays;
- NTC terminal blocks;
- push buttons;
- 5 mm indicator LEDs;
- NUCLEO-G031K8 module;
- amplifier and speaker wiring pads.

This construction style is appropriate for a prototype protection controller because the board can be assembled manually and individual components remain accessible for measurement and replacement.

23.21 Mechanical Mounting

Four mechanical mounting holes are included near the corners of the PCB.

Each hole has a nominal diameter of

3.2 mm

and is intended for

M3

mounting hardware.

The four-point attachment arrangement provides mechanical stability for the relatively large

121 mm × 182.25 mm

board.

This is particularly important because the PCB contains relatively heavy components such as the mains transformer and two electromechanical relays.

23.22 External Connection Placement

The external amplifier and speaker connections are concentrated near the left-hand board edge.

The relevant connections are:

AMP_OUT_L,

AMP_OUT_R,

SPEAKER_OUT_L,

and

SPEAKER_OUT_R.

The NTC sensor connectors are positioned separately near the right-hand edge.

This physical arrangement helps prevent the low-current temperature-sensor wiring from becoming mixed with the high-current speaker wiring.

23.23 Mains and Low-Voltage Regions

Unlike the power-amplifier PCB, the protection-controller PCB contains a PCB-mounted mains isolation transformer.

The transformer primary section and incoming AC wiring therefore constitute a hazardous-voltage region of the PCB.

The rectified secondary-side supply, regulator, Nucleo board, ADC circuits, relays, and sensing electronics operate on the isolated low-voltage side.

The intended physical structure is therefore

Mains Primary Isolation Transformer Low-Voltage Electronics

The transformer itself provides galvanic isolation between the two electrical domains.

23.24 Mains Isolation Requirement

Because the PCB contains mains-voltage circuitry, the physical copper layout around the transformer primary requires special attention.

Adequate creepage and clearance must be maintained between primary-side copper and all isolated low-voltage conductors.

The final manufacturing revision should therefore be checked specifically for

- primary-to-secondary copper clearance;
- primary-to-ground-plane clearance;
- solder-mask and board-edge spacing;
- transformer pin spacing;
- separation of incoming mains wiring;
- removal of unnecessary low-voltage copper beneath or close to the primary-side region.

The applicable minimum distances must be determined from the final safety classification and manufacturing requirements rather than from ordinary low-voltage PCB design rules.

23.25 Prototype Ground-Plane Consideration

The current KiCad file uses a bottom-layer GND zone covering almost the entire board, with the normal zone clearance set to

0.5 mm.

This is suitable as an ordinary low-voltage copper-zone rule but should not be assumed to define adequate mains isolation.

Before the board is released for fabrication, a dedicated copper keepout or equivalent isolation region should therefore be applied around the mains-primary section where necessary.

This allows the low-voltage ground plane to remain useful throughout the controller while maintaining deliberate separation from hazardous mains conductors.

23.26 Component Placement and Serviceability

The physical layout also considers access during assembly and maintenance.

The Nucleo module remains accessible for firmware programming and USB/ST-LINK connection.

The relay and driver circuits are positioned where they can be probed during protection testing.

The two push buttons and five status indicators remain easily identifiable, and the NTC and speaker connections are located near board edges.

The transformer, rectifier, regulator, and reservoir capacitor are grouped together, making power-supply troubleshooting straightforward.

23.27 Silkscreen and Identification

The PCB silkscreen identifies the major components and protection functions.

Important markings include the relay references, Nucleo reference, temperature sensor connections, amplifier outputs, speaker outputs, push-button functions, and LED status descriptions.

Clear silkscreen information is particularly valuable on a protection controller because testing frequently requires individual sensor inputs, relay channels, or status outputs to be located quickly.

23.28 PCB Layout Verification

Before manufacturing the protection-controller PCB, both automatic design-rule checking and manual inspection should be performed.

Particular attention should be given to:

- relay footprint pin numbering;
- BC337 transistor pinout;
- 1N4007 flyback-diode polarity;

- BAT85 clamp-diode orientation;
- electrolytic capacitor polarity;
- bridge-rectifier orientation;
- LM7812 pinout;
- Nucleo header orientation;
- ADC sensing connections;
- left/right channel identification;
- relay high-current track continuity;
- ground-plane connectivity;
- transformer primary and secondary isolation;
- mains-to-low-voltage creepage and clearance;
- mounting-hole clearance;
- absence of unrouted connections.

23.29 Manufacturing Output

After the layout has been verified, the PCB manufacturing package is produced from KiCad.

The required Gerber layers include:

- F.Cu;
- B.Cu;
- front solder mask;
- back solder mask;
- front silkscreen;
- Edge.Cuts.

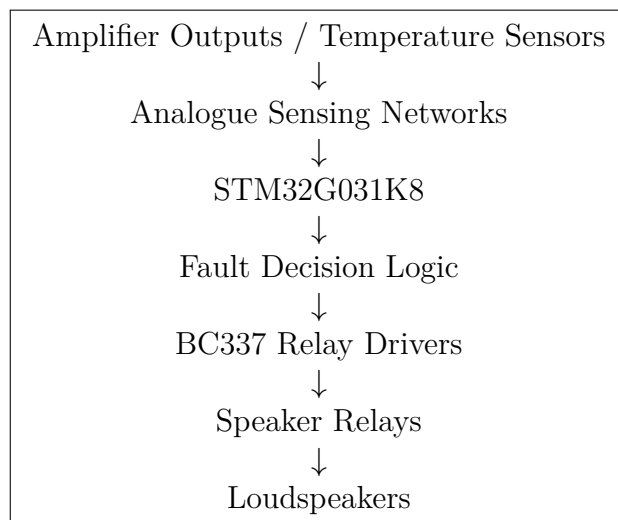
Separate drill information is also required because the PCB contains many plated through-hole components and four mechanical M3 mounting holes.

The Gerber and drill files together form the manufacturing data required to fabricate the protection-controller board.

23.30 Relationship to the Protection Architecture

The physical PCB layout reflects the functional architecture of the protection system.

The signal progression can be represented as



This progression is reflected physically on the PCB by keeping the sensing, processing, driver, and switching sections identifiable and relatively direct.

23.31 Design Summary

The STM32 loudspeaker protection controller is implemented on a two-layer KiCad PCB measuring

121 mm × 182.25 mm

with a nominal board thickness of

1.6 mm.

The completed layout is shown in Figure 23.1.

The upper region contains the dedicated auxiliary power supply, including the PCB-mounted isolation transformer, bridge rectifier, reservoir capacitor, 12 V regulator, and local filtering components.

The NUCLEO-G031K8 is positioned centrally so that the analogue sensing, relay-control, LED, and push-button connections can be routed efficiently.

The two DC-offset sensing circuits are grouped in the central-left region, while the two NTC temperature-sensor connectors are located close to the right-hand board edge.

Two BC337 transistor drivers control two Omron G2RL-1A-E DC12 speaker relays.

The relays are positioned close to the amplifier and loudspeaker connections to minimise the high-current contact path.

The normal protection-controller routing uses

1.0 mm

tracks, while the high-current speaker paths use

2.4 mm

tracks.

A large GND copper plane is implemented on the bottom layer with a nominal zone clearance of

0.5 mm

and a minimum zone thickness of

0.25 mm.

Only two routing vias are required in the final design, reflecting the predominantly through-hole construction and direct physical organisation of the board.

Five fault and status LEDs and two control push buttons provide the local user interface.

Four

3.2 mm

M3 mounting holes provide mechanical attachment to the chassis.

Because the protection PCB also contains the mains-powered transformer, the final manufacturing revision must additionally ensure deliberate primary-to-low-voltage isolation and appropriate creepage and clearance around the transformer primary section.

The resulting layout integrates analogue sensing, embedded control, high-current relay switching, user indication, and an isolated power supply into a single serviceable protection-controller PCB while maintaining clear functional separation between the different circuit regions.

Chapter 24

PCB Track Width and Current Handling

24.1 Introduction

PCB conductor width is an important design consideration in a power amplifier because several parts of the circuit carry currents that are much larger than those normally encountered in low-power analogue or digital electronics.

The audio input and microcontroller circuits operate at relatively small currents, whereas the amplifier supply rails, output transistors, loudspeaker connections, bridge rectifier, reservoir capacitors, and protection-relay contacts can carry currents of several amperes.

If a PCB conductor is too narrow for the current passing through it, several undesirable effects can occur:

- excessive conductor temperature rise;
- increased voltage drop;
- unnecessary power dissipation;
- localised heating around pads and vias;
- reduced efficiency;
- possible damage to the PCB under severe overload conditions.

For this reason, the PCBs developed for this project use different conductor widths according to the electrical function of each connection.

All PCBs in the project are specified using

2 oz copper

rather than the common 1 oz copper thickness.

24.2 Copper Weight

PCB copper thickness is commonly specified using copper weight per unit area.

A 1 oz copper layer corresponds approximately to

$$35 \mu\text{m},$$

while the 2 oz copper specified for this project corresponds approximately to

$$\boxed{70 \mu\text{m}}.$$

Therefore,

$$t_{\text{Cu}} \approx 0.070 \text{ mm}.$$

Doubling the copper thickness approximately doubles the conductor cross-sectional area for a given track width.

This reduces both electrical resistance and conductor heating compared with an otherwise identical 1 oz PCB.

24.3 Conductor Cross-Section

For a PCB track of width w and copper thickness t , the approximate copper cross-sectional area is

$$A = wt.$$

For the 2 oz boards used in this project,

$$t = 0.070 \text{ mm}.$$

For example, a

$$2.5 \text{ mm}$$

track therefore has an approximate copper cross-section of

$$A = 2.5 \times 0.070.$$

Thus,

$$\boxed{A \approx 0.175 \text{ mm}^2}.$$

Similarly, a 1.0 mm track has an approximate cross-section of

$A = 1.0 \times 0.070 = 0.070 \text{ mm}^2.$

24.4 Track Widths Used in the Project

The final power-amplifier PCB contains three principal routed track widths:

$0.4 \text{ mm}, \quad 1.0 \text{ mm}, \quad 2.5 \text{ mm}.$

The STM32 speaker-protection controller uses two principal track widths:

$1.0 \text{ mm} \quad \text{and} \quad 2.4 \text{ mm}.$

The width selection reflects the electrical purpose of the corresponding circuit.

Table 24.1: Principal PCB conductor widths used in the project.

PCB	Track Width	Typical Function
Power amplifier	0.4 mm	Low-current small-signal routing
Power amplifier	1.0 mm	General amplifier routing
Power amplifier	2.5 mm	Supply and high-current output paths
Protection controller	1.0 mm	ADC, GPIO, 12 V and control routing
Protection controller	2.4 mm	Loudspeaker relay contact paths

24.5 Electrical Resistance of a PCB Track

The resistance of a copper conductor is calculated from

$R = \rho \frac{L}{A},$

where

ρ

is the resistivity of copper,

L

is the conductor length, and

A

is its cross-sectional area.

For copper at approximately room temperature,

$$\rho \approx 1.724 \times 10^{-8} \Omega\text{m}.$$

Since

$$A = wt,$$

the track resistance becomes

$$R = \rho \frac{L}{wt}.$$

24.6 Resistance of the Selected Track Widths

Using 2 oz copper with an approximate thickness of 70 μm , the calculated resistances of the track widths used in the project are shown in Table 24.2.

Table 24.2: Approximate conductor resistance for 2 oz copper.

Track Width	Cross-Section	Resistance per 100 mm
0.4 mm	0.028 mm ²	61.6 m Ω
1.0 mm	0.070 mm ²	24.6 m Ω
2.4 mm	0.168 mm ²	10.3 m Ω
2.5 mm	0.175 mm ²	9.85 m Ω

The difference between the 1.0 mm and 2.5 mm tracks is therefore significant.

For the same physical length, the 2.5 mm conductor has only about

$$\frac{9.85}{24.6} \approx 0.40$$

of the resistance of the 1.0 mm track.

24.7 Voltage Drop

The voltage drop produced by a PCB conductor is

$$V_{\text{drop}} = IR.$$

For example, a 100 mm section of 2.5 mm wide, 2 oz copper has an approximate resistance of

$$9.85 \text{ m}\Omega.$$

At a current of

$$5 \text{ A},$$

the voltage drop would be approximately

$$V_{\text{drop}} = 5(9.85 \times 10^{-3}).$$

Therefore,

$$V_{\text{drop}} \approx 49 \text{ mV}.$$

A similar 2.4 mm track would produce approximately

$$51 \text{ mV}$$

of voltage drop over 100 mm at 5 A.

24.8 Copper Power Dissipation

Track power dissipation is given by

$$P_{\text{track}} = I^2 R.$$

For the same 100 mm section of 2.5 mm conductor carrying 5 A,

$$P = 5^2(9.85 \times 10^{-3}).$$

Therefore,

$$P \approx 0.246 \text{ W}.$$

For the 2.4 mm protection-controller speaker path,

$$P \approx 5^2(10.3 \times 10^{-3}),$$

giving approximately

$$P \approx 0.26 \text{ W}$$

per 100 mm of conductor.

24.9 Approximate Current Capability

The current capability of a PCB conductor depends on several factors, including:

- copper thickness;
- conductor width;
- conductor length;
- ambient temperature;
- permitted temperature rise;
- nearby copper area;
- airflow;
- solder coating;
- enclosure temperature.

For an approximate engineering estimate, the conventional external-conductor relationship associated with IPC PCB design guidance can be expressed as

$$I = k(\Delta T)^{0.44} A^{0.725},$$

where

$$I$$

is current in amperes,

$$\Delta T$$

is the permitted conductor temperature rise in degrees Celsius,

$$A$$

is copper cross-sectional area in square mils, and for an external conductor

$$k \approx 0.048.$$

This relationship is used here only as a design estimate. Actual PCB temperature depends on the complete board geometry and should ultimately be verified experimentally.

24.10 Estimated Current Capability of the Project Tracks

Applying the approximate calculation to the 2 oz conductors used in this project gives the results shown in Table 24.3.

Table 24.3: Approximate external-track current capability for 2 oz copper.

Width	Approx. Current at 10°C Rise	Approx. Current at 20°C Rise
0.4 mm	2.0 A	2.8 A
1.0 mm	4.0 A	5.4 A
2.4 mm	7.5 A	10.1 A
2.5 mm	7.7 A	10.4 A

These values show why the wide 2.4–2.5 mm tracks are used for the loudspeaker and major power paths.

24.11 Power Amplifier Output Current

The loudspeaker-output track must be sized according to the load current rather than according to the amplifier supply voltage alone.

The RMS load current is

$$I_{\text{RMS}} = \sqrt{\frac{P}{R_L}}.$$

Using the Project 3A reference operating condition of approximately

$$100 \text{ W}$$

into

$$4 \, \Omega,$$

the corresponding RMS current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}}.$$

Therefore,

$$I_{\text{RMS}} = 5 \text{ A}.$$

For a sinusoidal waveform, the peak output current is

$$I_{\text{peak}} = \sqrt{2}I_{\text{RMS}},$$

giving

$$I_{\text{peak}} \approx 7.07 \text{ A.}$$

These figures correspond to the published reference operating condition and are not presented as a measured output of the completed amplifier.

Because the actual project uses approximately $\pm 33\text{--}\pm 34$ V rails rather than nominal ± 35 V rails, the final measured output power may be somewhat lower.

24.12 8 Ohm Load Current

For the Project 3A reference condition of approximately

$$60 \text{ W}$$

into

$$8 \Omega,$$

the RMS load current is

$$I_{\text{RMS}} = \sqrt{\frac{60}{8}}.$$

Therefore,

$$I_{\text{RMS}} \approx 2.74 \text{ A.}$$

The corresponding peak current is

$$I_{\text{peak}} = 2.74\sqrt{2} \approx 3.87 \text{ A.}$$

The 4Ω operating condition is therefore the more demanding condition for loudspeaker-path current handling.

24.13 Power Amplifier High-Current Tracks

The widest tracks on the amplifier PCB are

$$2.5 \text{ mm.}$$

Inspection of the final KiCad layout shows that these tracks are used on important high-current nets including sections of:

- the positive supply rail;
- the negative supply rail;
- amplifier output routing;
- output-stage emitter paths;
- power-supply distribution.

With 2 oz copper, the approximate 10°C-rise estimate for this width is

$$7.7 \text{ A.}$$

This exceeds the

$$5 \text{ A}$$

RMS current corresponding to the 100 W/4 Ω reference condition.

24.14 Amplifier Rail Current

The positive and negative amplifier supply paths are additionally protected by

$$\boxed{3 \text{ A}}$$

rail fuses.

The 2.5 mm supply tracks therefore have substantial thermal margin relative to the nominal fuse current.

Even the approximate 10°C-rise current estimate of

$$7.7 \text{ A}$$

is considerably greater than the

$$3 \text{ A}$$

fuse rating.

The fuse therefore becomes the intended protective element before the main wide PCB conductor approaches its calculated continuous-current limit.

24.15 General 1.0 mm Amplifier Routing

The majority of the amplifier PCB uses

1.0 mm

routing.

For 2 oz copper, this width has an estimated continuous capability of approximately

4.0 A

for a 10°C conductor rise and approximately

5.4 A

for a 20°C rise.

This is more than adequate for the small-signal and driver-stage circuitry, whose currents are substantially below the full loudspeaker current.

24.16 0.4 mm Small-Signal Track

The narrowest routed track on the amplifier PCB is

0.4 mm .

Inspection of the KiCad layout shows that this width is confined to a small-signal transistor connection rather than a loudspeaker or supply-current path.

At 2 oz copper its cross-sectional area is approximately

0.028 mm^2 .

Its calculated current capability is far greater than the actual current expected in the corresponding small-signal stage.

Its use therefore does not represent a high-current limitation.

24.17 Protection Controller Speaker Paths

The protection-controller PCB uses

2.4 mm

tracks specifically on the high-current loudspeaker paths.

The relevant paths connect the amplifier outputs to the relay contacts and then to the loudspeaker terminals.

The left channel is

$$\text{AMP_OUT_L} \rightarrow K_1 \rightarrow \text{SPEAKER_OUT_L},$$

while the right channel is

$$\text{AMP_OUT_R} \rightarrow K_2 \rightarrow \text{SPEAKER_OUT_R}.$$

At 2 oz copper, the approximate 10°C-rise current estimate for a

$$2.4 \text{ mm}$$

external track is

$$\boxed{7.5 \text{ A.}}$$

The approximate 20°C-rise estimate is

$$\boxed{10.1 \text{ A.}}$$

24.18 Suitability for 4 Ohm Speaker Operation

The protection-controller speaker path must carry essentially the same current as the amplifier output because the relay is connected in series with the loudspeaker.

At the 100 W/4 Ω reference condition,

$$I_{\text{RMS}} = 5 \text{ A.}$$

This is below the approximate

$$7.5 \text{ A}$$

10°C-rise estimate of the 2.4 mm 2 oz track.

The selected width therefore provides useful thermal margin for the expected speaker-current path.

24.19 1.0 mm Protection-Controller Routing

The remaining protection-controller routing uses mainly

1.0 mm

tracks.

These conductors carry signals and supply currents associated with:

- the STM32 ADC inputs;
- 3.3 V logic;
- the 12 V controller supply;
- relay coils;
- BC337 base-drive circuits;
- LEDs;
- switches;
- temperature sensors.

These currents are much smaller than the loudspeaker current.

For example, each speaker-relay coil requires only approximately

33.3 mA,

which is negligible relative to the several-ampere capability of a 1.0 mm 2 oz conductor.

24.20 Track Neck-Downs

Track width should not be evaluated only from the widest visible portion of a net.

A conductor may temporarily narrow when entering a transistor pad, relay pad, fuse holder, terminal, or other through-hole component.

The electrical current must pass through the narrowest section of the complete path.

Inspection of the final PCB files shows that some nets which predominantly use 2.4 or 2.5 mm routing also contain short 1.0 mm sections around component connections.

These local reductions in width should therefore be kept as short as practical.

For 2 oz copper, a 1.0 mm external track still has an approximate current estimate of

4.0 A

for a 10°C rise and

5.4 A

for a 20°C rise.

A very short neck-down therefore has much lower total resistance than a long 1.0 mm conductor, but unnecessary narrow sections should still be avoided on the highest-current paths.

24.21 Importance of Track Length

Track width is not the only factor determining voltage drop.

Resistance is directly proportional to conductor length:

$$R \propto L.$$

Therefore, a short 2.4 mm speaker track is preferable to an unnecessarily long route of the same width.

This was one reason for positioning the speaker relays close to the amplifier and speaker output connectors on the protection-controller PCB.

Similarly, the amplifier output transistors, emitter resistors, fuses, and output connection were positioned so that the principal high-current paths remain relatively direct.

24.22 Peak Current and RMS Heating

The loudspeaker current is an AC waveform rather than a continuous DC current.

For a sinusoidal output,

$$I_{\text{peak}} = \sqrt{2}I_{\text{RMS}}.$$

However, conductor heating is primarily related to

$$I_{\text{RMS}}^2 R.$$

Therefore, the

$$5 \text{ A}$$

RMS current corresponding to 100 W/4 Ω is the more relevant value for estimating average PCB copper heating.

The corresponding

$$7.07 \text{ A}$$

peak current remains important for component, connector, relay-contact, and semiconductor current capability.

24.23 Ground Plane Current Handling

Both principal PCBs use broad bottom-layer GND copper zones.

Unlike a narrow track, a large ground plane has a much greater effective cross-sectional width.

Its resistance is therefore considerably lower than that of a long narrow ground trace.

This is beneficial for both electrical performance and current handling.

However, the effective current path through a plane can still be restricted by narrow sections, thermal-relief spokes, connector pads, or local copper geometry.

The ground-plane architecture and return-current distribution are discussed in greater detail in the dedicated *Copper Zones and Ground Planes* chapter.

24.24 Thermal Relief Connections

Pads connected to a copper plane are often attached using thermal-relief spokes.

These spokes make soldering easier by limiting the rate at which the plane removes heat from the pad.

For low-current signal connections this is normally advantageous.

For high-current connections, however, narrow thermal-relief spokes can become a local resistance bottleneck.

High-current power and ground pads should therefore be reviewed carefully to ensure that their connection to the surrounding copper area is sufficiently robust.

24.25 Via Current Handling

A via also has finite resistance and current capability.

Its performance depends on:

- finished hole diameter;
- copper plating thickness;
- via length;
- number of parallel vias.

The power-amplifier PCB contains only a small number of routing vias, while the protection-controller PCB contains only two.

For high-current routes, maintaining the conductor on a single copper layer is preferable where practical.

If a multi-ampere path must change layers, multiple suitably sized vias can be placed in parallel rather than forcing the complete current through one small via.

24.26 Pads, Connectors and Relay Contacts

The PCB conductor is only one part of the complete high-current path.

For example, the loudspeaker current on the protection board passes through

PCB Track → Relay Pad → Relay Contact → PCB Track → Output Connection.

The complete current capability is therefore limited by the weakest part of this chain.

Connector contacts, solder joints, relay contacts, fuse holders, external wire gauge, and PCB pads must all be rated for the required current.

Using a wide PCB track does not compensate for an undersized connector or poor solder joint.

24.27 External Wiring

The external wires connected to the high-current PCB tracks must also be selected appropriately.

This includes:

- transformer secondary wiring;
- supply-rail wiring;
- loudspeaker output wiring;
- star-ground connections;
- connections between the amplifier and protection controller.

The external wire should not have substantially greater resistance than the PCB conductor that it connects.

24.28 Effect of 2 oz Copper

The decision to manufacture all boards using 2 oz copper provides significant benefits for this amplifier project.

For the same conductor width, 2 oz copper approximately halves the resistance compared with 1 oz copper.

As a result,

$$V_{\text{drop}}$$

is reduced,

$$I^2R$$

loss is reduced, and conductor temperature rise is reduced.

This is particularly advantageous for the speaker output and amplifier rail paths, where currents of several amperes are possible.

24.29 Why Wider Tracks Are Still Required with 2 oz Copper

Using thicker copper does not eliminate the requirement for wide high-current tracks.

A narrow conductor still has substantially greater resistance than a wide conductor of the same copper thickness.

For example, over the same length,

$$R_{1.0\text{ mm}} \approx 24.6\text{ m}\Omega/100\text{ mm},$$

whereas

$$R_{2.5\text{ mm}} \approx 9.85\text{ m}\Omega/100\text{ mm}.$$

The wider conductor therefore provides lower voltage drop and lower heating even though both use the same 2 oz copper.

24.30 Design Margin

PCB traces should not normally be designed to operate continuously at their estimated maximum thermal capability.

Useful current margin is required to account for:

- elevated chassis temperature;
- manufacturing copper-thickness tolerance;
- local neck-downs;
- nearby heat-producing components;
- restricted airflow;
- long-duration high-power operation;
- possible overloads.

The use of 2 oz copper together with 2.4–2.5 mm high-current tracks therefore provides useful margin compared with selecting conductor width only from the theoretical expected current.

24.31 Practical Validation

The calculations in this chapter are design estimates and should be confirmed experimentally after the PCBs are manufactured.

During high-power testing, the following should be examined:

- temperature of the high-current PCB tracks;
- temperature of relay contacts and pads;
- fuse-holder temperature;
- connector temperature;
- voltage drop between amplifier and speaker terminals;
- voltage drop along the amplifier supply paths;
- solder-joint temperature;
- temperature around narrow track transitions.

An infrared thermometer or thermal camera can be useful for identifying local hot spots.

Voltage-drop measurements can also be made directly with a multimeter while the amplifier is operating into a suitable dummy load.

24.32 Current Handling Summary

Table 24.4: Summary of the principal conductor choices used in the project.

Width	Copper	Approx. 10°C Current	Application
0.4 mm	2 oz	2.0 A	Small-signal amplifier
1.0 mm	2 oz	4.0 A	General PCB routing
2.4 mm	2 oz	7.5 A	Protection speaker path
2.5 mm	2 oz	7.7 A	Amplifier power/output paths

24.33 Design Summary

All PCBs used in the amplifier project are specified using

2 oz copper

with an approximate copper thickness of

70 μm .

The power-amplifier PCB uses three main conductor widths:

0.4 mm, 1.0 mm, 2.5 mm.

The 0.4 mm conductor is restricted to low-current small-signal routing, while 1.0 mm is used for general amplifier interconnections.

The principal high-current supply and loudspeaker-output paths use

2.5 mm

copper tracks.

The STM32 protection-controller PCB uses

1.0 mm

routing for its ADC, logic, relay-coil, LED and low-voltage power circuits, while the loudspeaker relay contact paths use

2.4 mm

tracks.

Using an approximate external-conductor thermal calculation, the 2.5 mm power-amplifier track has an estimated continuous current capability of approximately

7.7 A

for a 10°C temperature rise.

The 2.4 mm protection-controller speaker path has a corresponding estimate of approximately

7.5 A.

These values exceed the approximately

5 A_{RMS}

speaker current associated with the 100 W/4 Ω Project 3A reference operating point.

The corresponding peak speaker current is approximately

$$7.07 \text{ A.}$$

The 2.5 mm amplifier supply routing also provides substantial margin relative to the installed

$$3 \text{ A}$$

rail fuses.

The calculations further show that track length, neck-downs, vias, thermal reliefs, connector contacts, fuse holders, relay contacts, solder joints, and external wiring must also be considered. The current capability of the complete electrical path is determined by its weakest element rather than by the widest section of copper alone.

The combination of 2 oz copper, wide high-current tracks, short loudspeaker paths, broad ground planes, limited via use, and appropriate external connectors therefore provides a suitable basis for the high-current requirements of the amplifier and speaker-protection system.

Final verification will be performed experimentally during load and thermal testing of the completed hardware, where track temperature, voltage drop, and connector heating can be measured under realistic amplifier operating conditions.

Chapter 25

Grounding Strategy

25.1 Introduction

Grounding is a critical part of the practical design of a high-power audio amplifier. Although several circuit nodes may all be represented by the same electrical ground symbol in a schematic, the physical conductors connecting those nodes have finite resistance and inductance.

Consequently, currents flowing through one ground path can produce small voltages that become visible to other parts of the circuit. In an audio amplifier, this can result in audible mains-frequency hum, power-supply noise, channel interaction, instability, or unwanted modulation of the low-level input signal.

The grounding strategy adopted for this project therefore separates the different types of return current according to their electrical function and brings them together at a controlled common reference point.

The fundamental grounding principle is

Sensitive Signal Returns $\neq$ High-Current Power Returns
--

until both are intentionally connected at the common system reference.

25.2 Grounding Objectives

The principal objectives of the grounding system are:

- minimise circulating ground currents;
- prevent loudspeaker current from flowing through input-signal ground;
- prevent rectifier charging current from flowing through sensitive analogue ground paths;
- maintain a common reference for both amplifier channels;

- provide a stable reference for the STM32 protection controller;
- reduce mains-frequency hum and power-supply ripple coupling;
- avoid unnecessary ground loops;
- maintain a clear distinction between signal ground, power ground, chassis, and protective earth.

25.3 Central Star-Ground Architecture

The complete stereo amplifier uses a central star-ground architecture.

The general concept is shown in Figure 3.1, where the principal ground returns from the amplifier subsystems converge at a common point.

The star-ground concept can be represented as



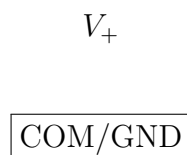
The purpose of the star connection is not simply to connect all ground wires together. Its purpose is to control where different ground currents are allowed to combine.

Each major subsystem is provided with a defined return path to the common reference rather than allowing several high-current sections to share arbitrary sections of signal-ground wiring.

25.4 Main Power-Supply Reference

The amplifier uses a symmetrical power supply with positive and negative supply rails referenced to a common midpoint.

The supply can be represented as



The common midpoint of the rectifier and reservoir-capacitor network forms the principal zero-volt reference of the amplifier power system.

This node carries substantial current because both amplifier channels return their load and supply currents toward the common power-supply reference.

For this reason, the common supply reference must be treated as a high-current node rather than as a small-signal ground connection.

25.5 Ground Current Categories

The ground currents present in the complete amplifier can be divided into several functional categories.

Table 25.1: Principal ground-current categories in the amplifier system.

Ground Type	Typical Current Level	Examples
Signal ground	Very low	RCA input, feedback, preamplifier
Amplifier circuit ground	Low to moderate	Input and voltage-amplification stages
Power ground	High	Output stage, supply rails
Speaker return	High	Loudspeaker current return
Rectifier / reservoir return	Pulsed high current	Bridge and filter capacitors
Protection-controller ground	Low to moderate	STM32, ADC, LEDs, relay drivers
Protective earth	Fault current only	Metal chassis safety bond

The purpose of the grounding architecture is to prevent the high-current categories from sharing unnecessary impedance with the low-level categories.

25.6 Signal Ground

Signal ground is the reference used by the low-level audio input and related small-signal circuitry.

The amplifier PCB includes a separate input connection identified as

Signal GND.

The voltage at this point directly affects the voltage seen by the input stage.

For example, if a high speaker current were allowed to flow through a section of conductor shared with the input ground, the resulting voltage would be

$$V_{\text{error}} = I_{\text{power}}R_{\text{shared}}.$$

Even when

$$R_{\text{shared}}$$

is very small, the large value of

$$I_{\text{power}}$$

can produce a voltage comparable with a low-level audio signal.

Signal-ground conductors are therefore prevented from carrying substantial output-stage or rectifier current.

25.7 Power Ground

Power ground carries the return currents associated with the high-power sections of the amplifier.

These include:

- output-stage current;
- loudspeaker return current;
- reservoir-capacitor current;
- rectifier charging current;
- amplifier supply-current return.

These currents are significantly larger than the currents present in the input stage.

Power-ground conductors must therefore be short, low resistance, and capable of carrying several amperes without producing significant common-impedance voltage.

25.8 Speaker Ground Classification

The loudspeaker return is treated as a high-current power-ground connection, not as a signal-ground connection.

The speaker current is the same current delivered by the amplifier output stage and can therefore reach several amperes.

For a $4\ \Omega$ load at the Project 3A reference operating point of approximately 100 W,

$$I_{\text{speaker,RMS}} = \sqrt{\frac{100}{4}} = 5\ \text{A}.$$

The corresponding peak current is approximately

$$I_{\text{speaker,peak}} = 5\sqrt{2} \approx 7.07\ \text{A}.$$

This current must not be returned through the low-level audio input-ground connection.

The required principle is therefore

Speaker Return $\rightarrow$ Power Ground / Star Ground

rather than

Speaker Return $\rightarrow$ Signal Input Ground.

25.9 Rectifier and Reservoir-Capacitor Ground

The bridge rectifier and reservoir capacitors produce one of the most electrically demanding ground-current paths in the complete amplifier.

The transformer and bridge do not deliver perfectly continuous current into the filter capacitors.

Instead, the capacitors are recharged in relatively short current pulses near the peaks of the rectified AC waveform.

These charging pulses can have considerably greater instantaneous amplitude than the average amplifier supply current.

They must therefore remain local to the power-supply section and its common connection as far as practical.

The grounding strategy is intended to prevent these currents from flowing through

Signal GND

or through the low-level preamplifier return wiring.

25.10 Left and Right Channel Ground Returns

The stereo amplifier consists of two power-amplifier channels sharing the same main power supply.

The left and right channels therefore require a common electrical reference, but their high-current return paths should remain independent until they reach the central ground region.

Conceptually,

Left Amplifier Return $\rightarrow$ Star Ground

and

Right Amplifier Return $\rightarrow$ Star Ground.

The alternative arrangement

Left Amplifier → Right Amplifier → Power Ground

is avoided because the current of one channel could then develop a voltage in the ground connection used by the other channel.

This could result in undesirable inter-channel coupling.

25.11 Common-Impedance Coupling

One of the principal reasons for star grounding is reduction of common-impedance coupling.

Consider two circuit sections sharing a ground conductor having resistance

$$R_G.$$

If one circuit draws current

$$I_1,$$

then the shared conductor develops a voltage

$$V_G = I_1 R_G.$$

The second circuit sees this voltage as a change in its own ground reference.

Thus, even though both nodes are labelled “GND” in the schematic, they are not at exactly the same instantaneous potential.

Reducing shared conductor impedance and controlling where the currents merge therefore improves audio performance.

25.12 Protection Controller Ground

The STM32 protection controller has its own low-voltage power supply and local ground plane.

However, the protection system must measure the power-amplifier outputs relative to the same electrical reference used by the amplifiers.

The DC sensing relationship is meaningful only when the protection-controller ADC ground and the amplifier reference ground have a defined connection.

Therefore,

Protection Controller GND → System Ground Reference

is required.

The connection should be deliberate and should not be duplicated through several unrelated wires.

Multiple independent ground connections between the protection controller and amplifier could create an additional loop through which relay, digital, or power-supply current could circulate.

25.13 DC Sensing and Ground Accuracy

The DC protection circuit determines amplifier output voltage relative to the STM32 ADC reference.

The measured value can be expressed generally as

$$V_{\text{measured}} = V_{\text{output}} - V_{\text{ADC GND}}.$$

Any difference between the amplifier ground and ADC ground therefore appears as measurement error.

For example, if the controller ground differs from the amplifier ground by

$$V_{\text{GND,error}},$$

then approximately

$$V_{\text{measured}} = V_{\text{actual}} + V_{\text{GND,error}}.$$

Ground-reference integrity is therefore important not only for audio noise but also for accurate loudspeaker-protection operation.

25.14 PCB Ground Planes

Both main PCB designs incorporate broad ground copper areas.

The power-amplifier PCB contains a GND copper zone on the bottom layer, while the protection-controller PCB contains its own bottom-layer GND plane.

The use of large copper areas reduces the resistance of local ground paths compared with long narrow tracks.

All boards in the project use

$$\boxed{2 \text{ oz copper}},$$

providing approximately

$$70 \mu\text{m}$$

of copper thickness.

The thicker copper further reduces ground resistance and improves high-current handling.

The detailed geometry of the ground zones is discussed separately in the *Copper Zones and Ground Planes* chapter.

25.15 Ground Plane Is Not Automatically a Star Ground

A large copper plane does not automatically guarantee an ideal grounding system.

Current through a plane follows a physical return path determined by the location of the source and load.

If a high-current output-stage return is forced to travel through the same region of copper used by the input circuitry, common-impedance coupling can still occur even though both connections are made to a large plane.

The placement of ground connections on the plane is therefore as important as the presence of the plane itself.

25.16 Local Decoupling and Ground Return

Local decoupling capacitors provide a short high-frequency current path between the supply rails and the local amplifier reference.

The power-amplifier PCB includes local decoupling such as

$$100 \text{ nF}$$

and

$$100 \mu\text{F}$$

capacitors on the positive and negative rails.

The high-frequency current loop should therefore remain local:

$$\text{Active Stage} \rightarrow \text{Decoupling Capacitor} \rightarrow \text{Local Ground / Rail Reference}.$$

This reduces the amount of high-frequency current that must flow through long system-ground connections.

25.17 Preamplifier Ground

The preamplifier processes substantially smaller signals than the main power-amplifier stages.

Its ground return therefore belongs to the sensitive signal-ground category.

Preamplifier ground should not be connected in series with the speaker or power-amplifier current return.

The preferred system concept is

Preamplifier Ground $\rightarrow$ Signal Ground Reference $\rightarrow$ Controlled Common Ground.

This is particularly important for the phono input, where the audio signal is significantly smaller than a normal line-level signal.

25.18 RCA Input Grounding

The RCA input shield carries the input-signal reference.

It should therefore be connected to the amplifier signal-ground network rather than being used as a return for power-stage current.

The RCA input wiring should remain physically separated from

- transformer secondary wiring;
- mains wiring;
- bridge-rectifier wiring;
- speaker output wiring;
- relay-coil wiring.

The detailed implementation of the RCA connectors is discussed separately in the *RCA Audio Inputs* chapter.

25.19 Chassis and Protective Earth

Protective earth and audio ground perform different functions.

Protective earth is a safety conductor and must provide a reliable low-impedance connection between the metal chassis and the mains protective earth conductor.

Therefore,

Protective Earth $\rightarrow$ Metal Chassis

is a safety connection.

Protective earth must not be used as the normal loudspeaker-current return or audio-signal return.

Similarly, the chassis should not acquire multiple accidental audio-ground connections through RCA sockets, PCB mounting hardware, heatsinks, or other mechanical parts.

25.20 Audio Ground to Chassis Connection

The relationship between the audio zero-volt system and the metal chassis must be intentional.

If the audio ground is bonded to the chassis at several different points, the chassis can become part of a circulating ground-current loop.

The design principle is therefore

One Controlled Audio-Ground / Chassis Relationship

rather than multiple accidental connections.

The exact final hardware implementation of the audio-ground-to-chassis connection will be verified during system integration and hum testing.

No specific ground-lift network is assumed here unless such a network is deliberately added to the final hardware.

25.21 Protective Earth Must Not Be Switched

The protective-earth connection is a safety function and must remain connected regardless of the operating state of the amplifier.

It should not be interrupted by the mains power switch, fuse, speaker relay, or microcontroller.

The basic safety arrangement is

IEC Protective Earth → Dedicated Chassis Bond.

Detailed mains and safety requirements are discussed later in the *Mains Wiring, Switches and Fuses* and *Safety Considerations* chapters.

25.22 Avoiding Ground Loops

A ground loop occurs when two circuit points are connected by more than one conductive path.

If magnetic fields or voltage differences exist between those paths, a circulating current can flow.

For example, a possible unwanted loop could be formed by

Signal Cable Ground → Amplifier Ground → Chassis → External Equipment Ground → Signal Cable

The star-ground strategy reduces the number of uncontrolled parallel ground paths within the amplifier itself.

25.23 Ground Wiring Geometry

High-current ground wiring should be kept short and should use sufficiently large conductors.

The resistance of a ground connection is

$$R = \rho \frac{L}{A}.$$

Reducing length

L

and increasing conductor cross-sectional area

A

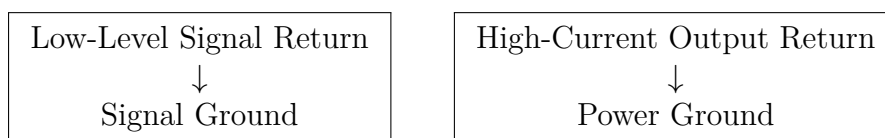
both reduce the resulting ground voltage.

For this reason, speaker returns, amplifier power returns, and the central supply-common connection should use heavier wiring than low-current control and signal-ground conductors.

25.24 Physical Separation of Ground Paths

Where practical, high-current and low-current ground conductors should remain physically separate until they reach their intended common point.

The design philosophy can therefore be summarised as



followed by

Signal Ground + Power Ground $\rightarrow$ Star Ground.

25.25 Stereo Grounding

The same concept is applied independently to both stereo channels.

The system should avoid routing one channel's ground current through the other channel.

Thus,

Left Channel Return $\rightarrow$ Star Ground

and

Right Channel Return $\rightarrow$ Star Ground.

This reduces the possibility of one channel modulating the electrical reference of the other channel when the two channels reproduce different audio signals.

25.26 Grounding of Relay Circuits

The speaker-relay coils are controlled by BC337 transistor drivers located on the protection-controller PCB.

The coil current is approximately tens of milliamperes and is switched on and off digitally.

The relay-driver return is therefore routed through the protection-controller power ground rather than through the amplifier's sensitive signal-ground connection.

The flyback diodes also keep the relay turn-off current local to the coil and driver circuit.

This reduces the possibility that relay switching generates a transient in the amplifier input-ground path.

25.27 Digital Ground Currents

The STM32 microcontroller, indicator LEDs, push buttons, and associated logic generate switching currents.

Although these currents are small compared with the loudspeaker current, they contain relatively fast edges.

They should therefore remain within the controller ground system rather than sharing long return paths with the analogue audio input.

The protection controller communicates with the power amplifier primarily through sensing and relay functions, not by inserting digital circuitry into the analogue audio signal path.

25.28 Grounding Verification

The grounding system should be verified before final amplifier operation.

The following checks should be performed:

1. Verify continuity between the intended star-ground connections.

2. Verify that the left and right amplifier grounds have low-resistance paths to the common reference.
3. Verify that speaker returns do not pass through the RCA input-ground wiring.
4. Verify that the preamplifier signal ground is not used as a power-current return.
5. Verify the connection between protection-controller ground and the amplifier reference.
6. Check for unintended chassis-ground connections through mounting hardware.
7. Check whether RCA connector bodies are intentionally isolated from or bonded to the chassis.
8. Verify that protective earth is securely bonded to the chassis.
9. Verify that protective earth is not used as an audio-current return.
10. Measure resistance between intended ground points before applying power.

25.29 Grounding Validation During Operation

After initial electrical verification, the amplifier should be tested for ground-related noise.

Useful tests include operation with:

- inputs shorted to signal ground;
- no loudspeaker load;
- an $8\ \Omega$ dummy load;
- a $4\ \Omega$ dummy load;
- one amplifier channel operating;
- both channels operating;
- the protection relays energised;
- the preamplifier connected.

The output should be observed with an oscilloscope and, where practical, the residual AC noise should also be measured using an RMS voltmeter or audio measurement system.

Particular attention should be given to

50 Hz

and

100 Hz

components because these frequencies can indicate mains coupling or full-wave-rectifier ripple respectively.

25.30 Ground Potential Measurements

A useful practical test is to measure the AC voltage between different points that are nominally labelled ground.

For example, measurements can be made between:

- signal ground and star ground;
- left amplifier ground and star ground;
- right amplifier ground and star ground;
- protection-controller ground and star ground;
- chassis and audio ground.

The measured voltages should remain small under normal operating conditions.

A significant signal between two nominal ground points can indicate excessive shared impedance or an unsuitable current-return path.

25.31 Relationship to the Following Chapters

This chapter defines the overall grounding philosophy of the complete amplifier.

The next chapter, *Audio Signal Ground and Power Ground*, examines the distinction between low-level and high-current return paths in greater detail.

The chapter *Copper Zones and Ground Planes* then describes how the PCB copper areas implement low-impedance local ground connections.

Finally, the *Noise and Hum Reduction* chapter examines the broader measures used to reduce electromagnetic coupling, mains hum, and other unwanted signals.

25.32 Design Summary

The complete Hi-Fi amplifier uses a controlled grounding architecture based on a central star-ground reference.

The fundamental principle is that sensitive audio signal currents and large power currents should not share unnecessary ground impedance.

The system therefore distinguishes between

Signal Ground

and

Power Ground

before both are brought to an appropriate common reference.

The left and right power-amplifier channels have independent high-current return paths toward the central star-ground region rather than one amplifier returning through the other.

The loudspeaker return is treated as

high-current power ground

because it can carry several amperes during high-power operation.

The rectifier and reservoir-capacitor ground connections are also treated as high-current paths because the capacitor charging current occurs in large pulses.

The RCA input and preamplifier grounds belong to the low-level signal-ground system and should not carry loudspeaker or rectifier current.

The STM32 protection controller uses its own local ground plane but requires a defined connection to the amplifier system ground so that the DC sensing circuits measure the amplifier outputs relative to the correct reference.

Both principal PCBs use broad bottom-layer ground copper and

2 oz copper

to provide low-resistance local return paths.

Protective earth remains conceptually separate from the audio-current return system. It is bonded to the metal chassis for electrical safety and is not used as the normal loudspeaker or signal return.

The relationship between audio ground and chassis must be deliberate and should not be duplicated through several accidental mechanical connections.

The resulting grounding architecture is intended to minimise common-impedance coupling, circulating currents, mains-frequency hum, rectifier noise, inter-channel interaction, and ground-reference errors within the protection system.

Final verification will be performed during system integration by checking ground continuity, unintended chassis connections, output hum, ground potential differences, and noise behaviour under both unloaded and high-current amplifier operation.

Chapter 26

Audio Signal Ground and Power Ground

26.1 Introduction

Although the terms *signal ground* and *power ground* both represent nominally zero-volt reference nodes, they perform different functions within a high-power audio amplifier.

Signal ground carries the return currents associated with low-level audio signals and sensitive analogue circuitry. Power ground carries substantially larger currents associated with the output stage, loudspeaker load, power supply, rectifier, and reservoir capacitors.

Connecting these currents without considering their physical return paths can cause high-current voltage drops to become superimposed on the low-level audio reference.

For this reason, the amplifier developed in this project distinguishes physically between

Audio Signal Ground

and

Power Ground

even though both ultimately share the same electrical zero-volt reference.

26.2 Why Ground Is Not an Ideal Zero-Volt Node

In an ideal schematic, every point labelled GND is assumed to have exactly the same voltage.

A real PCB track or wire, however, has finite resistance and inductance.

If current I flows through a ground conductor having resistance R_G , the voltage developed across that conductor is

$$V_G = IR_G.$$

Therefore, two physically separated points labelled GND can have slightly different instantaneous voltages.

For high-current circuits, even a very small ground resistance can become important.

For example, if a ground connection has a resistance of only

$$10 \text{ m}\Omega$$

and carries

$$5 \text{ A},$$

the resulting voltage is

$$V_G = 5 \times 0.010 = 0.050 \text{ V}.$$

Thus,

$$V_G = 50 \text{ mV}.$$

A 50 mV unwanted ground voltage is small compared with the approximately $\pm 33\text{--}\pm 34 \text{ V}$ amplifier supply rails, but it is significant when compared with a low-level audio signal.

26.3 Audio Signal Ground

Audio signal ground is the low-current reference used by the input and small-signal sections of the amplifier.

The power-amplifier PCB contains a dedicated external connection identified as

$$\text{Signal GND}.$$

This connection accompanies the audio input and forms the reference against which the amplifier interprets the incoming audio voltage.

The input relationship can be represented as

$$V_{\text{IN}} = V_{\text{Signal}} - V_{\text{Signal GND}}.$$

Any unwanted voltage appearing on the signal-ground conductor therefore appears directly as part of the amplifier input.

For this reason, signal ground must be protected from large supply, loudspeaker, and rectifier currents.

26.4 Circuits Referenced to Signal Ground

The low-level signal-ground system principally includes the returns associated with:

- RCA audio inputs;
- preamplifier outputs;
- input coupling networks;
- differential input stages;
- feedback networks;
- low-level small-signal circuitry.

These circuits carry comparatively small currents.

Their grounding priority is therefore not high-current capability but maintaining a clean and stable voltage reference.

26.5 RCA Input Ground

The RCA connector provides both the audio signal and its corresponding return reference.

The centre conductor carries the audio signal, while the outer connection is associated with the audio signal ground.

The basic input arrangement is therefore

RCA Centre → Audio Input
RCA Shield → Signal Ground

The RCA ground connection should not be used as a return path for loudspeaker current, rectifier current, relay current, or other high-current loads.

Otherwise, the voltage generated by those currents would become part of the amplifier input reference.

26.6 Preamplifier Ground

The preamplifier also belongs to the low-level audio domain.

This is particularly important for the phono section, where the input signal can be substantially smaller than the normal line-level inputs.

The preamplifier output and its corresponding ground therefore connect to the power amplifier through the signal-input system.

Conceptually,

Preamplifier Output $\rightarrow$ Power Amplifier Input

while

Preamplifier Signal Ground $\rightarrow$ Power Amplifier Signal Ground.

The preamplifier ground should not be placed in series with the speaker or main power-supply return path.

26.7 Power Ground

Power ground is the return system used by circuits that carry comparatively large currents. These include:

- the output stage;
- the loudspeaker return;
- the main reservoir capacitors;
- bridge-rectifier charging currents;
- positive and negative rail return currents;
- other high-current power connections.

The power-ground conductors must therefore have low electrical resistance and sufficient current capability.

All boards in the project use

2 oz copper

,

which provides approximately

$70\ \mu\text{m}$

of copper thickness and reduces the resistance of the high-current return paths.

26.8 Speaker Ground Is Power Ground

One particularly important design decision is the classification of the loudspeaker return connection.

Although the loudspeaker return is sometimes casually referred to simply as “speaker ground”, it must be treated electrically as

Power Ground

rather than signal ground.

The loudspeaker return carries the same AC current that is delivered through the amplifier output terminal.

For a load resistance R_L and output power P ,

$$I_{\text{RMS}} = \sqrt{\frac{P}{R_L}}.$$

At the Project 3A reference condition of approximately

$$100 \text{ W}$$

into

$$4 \Omega,$$

the RMS loudspeaker current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5 \text{ A}.$$

The corresponding peak current is

$$I_{\text{PEAK}} = 5\sqrt{2} \approx 7.07 \text{ A}.$$

Therefore, the speaker return can carry currents thousands of times greater than those present in the amplifier input circuitry.

26.9 Incorrect Shared Ground Path

An undesirable arrangement would allow the speaker-return current and input signal current to share part of the same conductor.

For example,

Speaker Return $\rightarrow$ R_{shared} $\rightarrow$ Signal Ground.

The speaker current would then produce

$$V_{\text{error}} = I_{\text{speaker}} R_{\text{shared}}.$$

This unwanted voltage would appear at the input-stage ground reference.

The amplifier could then amplify a component of its own output-current return, creating distortion, hum, or instability.

26.10 Preferred Ground Arrangement

The preferred arrangement separates the two current paths:

Audio Input Return → Signal Ground

and

Speaker Return → Power Ground.

The two systems are then connected at a controlled common reference:

Signal Ground → Common / Star Ground ← Power Ground.

This prevents the high-current speaker return from flowing through the low-level signal-ground conductor.

26.11 Main Supply Common Point

The amplifier uses a symmetrical supply derived from a 24–0–24 V AC transformer.

After rectification and filtering, the supply provides approximately

$$V_+ \approx +33 - +34 \text{ V}$$

and

$$V_- \approx -33 - -34 \text{ V}.$$

The midpoint of the positive and negative reservoir-capacitor network forms the principal power-supply common reference.

This is a high-current node because it is associated with the return currents of the power-amplifier channels and reservoir capacitors.

It therefore forms an important part of the system's star-ground architecture.

26.12 Reservoir Capacitor Ground Currents

The reservoir capacitors carry substantial charging and discharging currents.

The four main capacitors provide

9400 μF

of nominal capacitance per supply rail.

During operation, the bridge rectifier replenishes these capacitors during limited portions of the AC waveform.

The resulting charging currents occur in pulses rather than as a smooth DC current.

These pulsed currents must therefore be prevented from flowing through the audio signal-ground path.

The preferred current loop is approximately

Transformer $\rightarrow$ Bridge Rectifier $\rightarrow$ Reservoir Capacitors $\rightarrow$ Supply Common.

The sensitive input-ground connection should remain outside this charging loop.

26.13 Output Stage Return Current

The power output stage also produces substantial ground currents as it delivers energy to the loudspeaker.

These currents flow through the power devices, emitter resistors, supply rails, loudspeaker, and power-ground network.

The output-stage return therefore belongs to the same general high-current ground system as the loudspeaker return and main supply common.

It should not use the low-current signal-ground conductor as its return path.

26.14 Left and Right Amplifier Channels

The stereo system contains two independent power-amplifier channels.

The left and right channel grounds ultimately share a common reference, but their high-current paths should remain independent until reaching the controlled common-ground region.

The preferred structure is

Left Power Ground $\rightarrow$ Star Ground

and

Right Power Ground $\rightarrow$ Star Ground.

The arrangement

Left Power Ground → Right Amplifier Ground → Star Ground

would be less desirable because current produced by the left channel could generate a voltage in the ground reference used by the right channel.

26.15 Channel Interaction Through Ground Impedance

The mechanism by which one amplifier channel can influence the other can be demonstrated using a shared ground impedance.

Suppose both channels share a conductor having impedance Z_G .

A current I_L from the left amplifier produces

$$V_G = I_L Z_G.$$

This voltage becomes part of the reference seen by the right amplifier.

Consequently, the left-channel output signal can appear indirectly in the right channel through the common ground impedance.

Independent high-current return paths therefore help reduce stereo inter-channel coupling.

26.16 Feedback Ground Reference

The amplifier's global negative-feedback network is especially sensitive to ground-reference errors because it determines the closed-loop gain and output correction.

The nominal closed-loop voltage gain of the amplifier is approximately

$$A_v = 1 + \frac{R_5}{R_4} = 1 + \frac{22 \text{ k}\Omega}{1 \text{ k}\Omega} \approx 23.$$

If the feedback-network ground reference contains voltage produced by high-current output paths, that unwanted voltage becomes part of the feedback signal.

The feedback reference should therefore be associated with a clean amplifier reference rather than an arbitrary high-current ground path.

26.17 Protection Controller Ground

The STM32 speaker-protection controller has its own local power supply and ground plane.

Nevertheless, its DC-offset sensing circuits measure the power-amplifier outputs relative to the amplifier ground reference.

The ADC therefore requires a defined relationship between

ADC GND

and the amplifier system ground.

The protection-controller ground is consequently connected to the system reference at a controlled point.

26.18 Importance for DC Offset Measurement

The DC sensing circuit does not measure an absolute voltage. It measures the difference between the amplifier output and the controller ground reference.

Conceptually,

$$V_{\text{measured}} = V_{\text{AMP}} - V_{\text{Controller GND}}.$$

If the controller ground contains a ground offset

$$V_{\text{GND,error}},$$

the measured value becomes

$$V_{\text{measured}} = V_{\text{AMP}} - V_{\text{GND,error}}.$$

Ground-reference errors can therefore affect the apparent DC offset.

The protection-controller ground connection must consequently be stable and must not carry large loudspeaker return currents.

26.19 Relay Driver Ground

The speaker relays are operated by BC337 transistor drivers.

Each relay coil draws approximately

$$33.3 \text{ mA}.$$

The relay-driver current returns through the protection-controller power ground.

It is not routed through the audio signal-ground connection.

This distinction reduces the possibility of relay switching producing a transient in the low-level audio reference.

26.20 Digital Ground Currents

The STM32, indicator LEDs, control buttons, and other digital circuits generate switching currents.

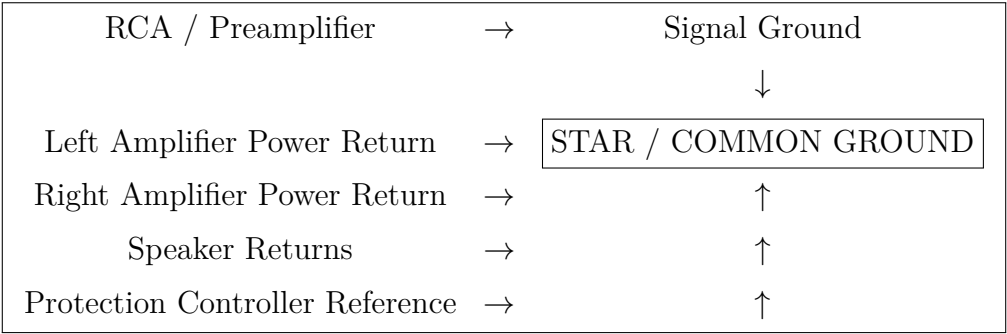
These currents are much smaller than the loudspeaker current but contain higher-frequency components because of the relatively fast logic transitions.

They therefore remain primarily within the protection-controller local ground system.

The protection electronics are not inserted directly into the analogue signal path.

26.21 Ground Connection Hierarchy

The overall ground architecture can be represented conceptually as



The diagram is conceptual rather than a statement that all conductors must be physically identical.

The important feature is that high-current returns do not flow through the sensitive signal-ground conductors before reaching the common reference.

26.22 Signal Ground on the Power Amplifier PCB

The final power-amplifier PCB visibly distinguishes the incoming



connection from the main high-current ground and common power connections.

The signal-ground connection is placed close to the audio input in the small-signal region of the PCB.

The main power and common connections are located separately within the power-supply and output-current regions.

This physical separation helps maintain the intended distinction between small-signal and high-current return currents.

26.23 Effect of 2 oz Copper

All project PCBs use

2 oz copper

.

The increased copper thickness reduces the impedance of both signal and power ground conductors.

This is particularly useful for power ground because currents of several amperes are possible.

However, thicker copper does not eliminate the need for correct ground-current routing.

Even a low-resistance conductor can generate unwanted voltage when large currents flow through a section shared with a sensitive circuit.

Therefore,

Low Resistance + Correct Return Path

are both required.

26.24 Ground Plane Considerations

Broad bottom-layer ground zones are used on the principal PCBs.

These reduce local ground resistance and provide short return paths.

However, the existence of a ground plane does not remove the distinction between signal and power currents.

Current still flows through particular regions of the plane between its source and return.

A high-current path should therefore not be arranged so that it crosses the same region of ground copper used as the reference for the amplifier input.

The physical copper-zone implementation is analysed in detail in the *Copper Zones and Ground Planes* chapter.

26.25 RCA Connector and Chassis

The RCA ground is part of the audio signal-ground network.

Its relationship with the metal chassis must therefore be deliberate.

If the RCA connector body is electrically connected to the chassis while the signal ground is also connected to the chassis elsewhere, a second ground path may be created.

This can produce a ground loop.

The final mechanical implementation should therefore verify whether the RCA connector is isolated from or intentionally bonded to the chassis.

26.26 Protective Earth Is Not Signal Ground

The mains protective-earth conductor performs a safety function and must not be confused with either signal ground or power ground.

The protective-earth conductor is connected securely to the metal chassis.

Thus,

$$\boxed{\text{Protective Earth} \rightarrow \text{Metal Chassis}}.$$

It is not intended to carry normal loudspeaker, amplifier, or input-signal current.

The relationship between the audio ground and chassis is a separate design consideration.

26.27 Incorrect Grounding Examples

Several ground arrangements should be avoided.

The following are undesirable:

- connecting the loudspeaker return through the RCA ground conductor;
- allowing bridge-rectifier charging current to flow through the input ground;
- connecting one amplifier's power ground through the second amplifier;
- using the preamplifier ground as a main power return;
- providing several uncontrolled connections between audio ground and chassis;
- allowing relay-coil current to return through an ADC reference path;
- using protective earth as the normal amplifier-current return.

26.28 Correct Grounding Principles

The preferred principles used in this project can be summarised as follows:

1. Keep low-level audio returns separate from high-current power returns.
2. Treat speaker return as a power-ground connection.
3. Keep rectifier and reservoir-capacitor charging loops local.
4. Give the left and right amplifier channels independent return paths toward the common ground.
5. Reference the protection controller to the amplifier ground through a defined connection.

6. Keep relay and digital return currents within the protection controller where practical.
7. Use broad copper areas and sufficiently heavy wiring for high-current ground paths.
8. Bring the different functional grounds together at a controlled common reference.

26.29 Grounding Verification Before Power-Up

The ground system should be checked before initial power-up.

A suitable procedure is:

1. Verify continuity between the intended amplifier ground points.
2. Confirm continuity between the power-supply common point and the intended star-ground point.
3. Verify the left amplifier power-ground connection.
4. Verify the right amplifier power-ground connection.
5. Check that the speaker returns connect to the high-current ground system.
6. Confirm that the RCA input ground is not being used as a speaker return.
7. Verify the preamplifier signal-ground connection.
8. Check the protection-controller ground reference.
9. Verify that no unintended PCB mounting hardware connects signal ground to the chassis.
10. Verify the intended protective-earth-to-chassis connection.

26.30 Grounding Verification During Operation

After the electrical connections have been verified, the amplifier should be checked for ground-related noise under operating conditions.

Useful measurements include:

- output noise with the input shorted to signal ground;
- output noise with the preamplifier connected;
- output noise with the speaker relays energised;
- AC voltage between signal ground and star ground;
- AC voltage between power ground and star ground;

- behaviour with one amplifier channel operating;
- behaviour with both amplifier channels operating;
- behaviour under $8\ \Omega$ load;
- behaviour under controlled $4\ \Omega$ loading.

If significant unwanted voltage is measured between nominal ground points, the associated conductor and current path should be investigated.

26.31 Hum Diagnosis

Ground-related problems frequently appear as low-frequency hum at the amplifier output. A strong component near

50 Hz

can indicate coupling from the mains system or transformer.

A strong component near

100 Hz

can indicate coupling from the full-wave rectifier and reservoir-capacitor charging system.

These observations can help determine whether the unwanted signal is entering through signal-ground wiring, power-ground wiring, magnetic coupling, or power-supply ripple.

26.32 Relationship to the Star-Ground Strategy

The distinction between audio signal ground and power ground forms the practical basis of the star-ground architecture described in the previous chapter.

The star-ground point is where the otherwise separated functional return paths are intentionally referenced together.

Thus,

Signal Ground + Left Power Ground + Right Power Ground + Supply Common + Protection Reference

The objective is not to electrically isolate these grounds from one another, but to control the paths taken by their currents before they reach the common reference.

26.33 Design Summary

The amplifier system distinguishes physically between low-level

Audio Signal Ground

and high-current

Power Ground

The audio signal-ground network carries the return currents associated with the RCA input, preamplifier, differential input stage, feedback network, and other sensitive analogue circuitry.

The power-ground system carries the much larger currents associated with the output stage, loudspeaker, main supply, bridge rectifier, and reservoir capacitors.

The loudspeaker return is specifically treated as a

power-ground connection

because the current can reach several amperes during high-power operation.

At the Project 3A reference condition of 100 W into 4 Ω , the loudspeaker current is approximately

$$5 \text{ A}_{\text{RMS}}$$

with peaks of approximately

$$7.07 \text{ A}.$$

These high currents must not flow through the RCA or low-level signal-ground conductors.

The left and right amplifier channels are given controlled high-current return paths toward the common system reference so that one channel does not use the other channel's ground path.

The bridge rectifier and reservoir-capacitor charging currents are similarly kept within the high-current power-supply ground system.

The STM32 protection controller uses its own local ground plane but is referenced deliberately to the amplifier system ground because accurate DC offset measurement depends on a stable common reference.

All project PCBs use

2 oz copper

which reduces conductor resistance, but correct return-current routing remains essential because even very small shared impedances can generate significant ground voltages when several amperes flow.

The signal and power grounds therefore remain functionally separated until they meet at the controlled common/star-ground reference.

This arrangement reduces common-impedance coupling, mains hum, rectifier noise, inter-channel interaction, and measurement errors while preserving a common electrical reference throughout the complete stereo amplifier system.

Chapter 27

Copper Zones and Ground Planes

27.1 Introduction

Copper zones and ground planes are used throughout the PCB designs to provide low-impedance ground connections, improve current-return paths, reduce the length of individual ground traces, and make efficient use of otherwise unused copper area.

Both the power-amplifier PCB and the STM32 protection-controller PCB use a large copper zone connected to GND on the bottom copper layer.

The copper planes complement the grounding strategy described in the previous chapters. The star-ground architecture defines how the principal subsystem grounds are interconnected at system level, while the PCB ground planes provide low-impedance local connections between components within each individual board.

The important distinction is that a copper plane does not replace the need for controlled grounding. Instead,

System Grounding Strategy + Local Ground Plane

are used together.

27.2 Copper Zones in KiCad

A copper zone in KiCad is an area assigned to a particular electrical net. When the zone is filled, KiCad automatically places copper throughout the defined region while maintaining the required clearance from tracks, pads, vias, and zones belonging to other nets.

For the present project, the principal copper zones are assigned to the ground network.

The main purposes of these zones are:

- reducing ground-path impedance;
- providing broad current-return paths;

- reducing the number of individually routed ground tracks;
- improving local supply-decoupling return paths;
- providing a stable voltage reference for analogue circuitry;
- providing a common reference for digital and protection circuitry;
- reducing unnecessary loop area.

27.3 Copper Weight

All PCBs in this project are manufactured using

$$\boxed{2 \text{ oz copper}}.$$

This corresponds to an approximate copper thickness of

$$\boxed{t_{\text{Cu}} \approx 70 \text{ } \mu\text{m}}.$$

Compared with a conventional 1 oz PCB, the increased copper thickness reduces the electrical resistance of both routed tracks and large copper planes.

The use of 2 oz copper is particularly beneficial in this project because the power amplifier and loudspeaker paths can carry currents of several amperes.

27.4 Ground Plane Location

Both principal PCB designs use their main ground copper zone on the bottom copper layer,

$$\boxed{\text{B.Cu}}.$$

The front copper layer is used extensively for component interconnection and signal routing, while the bottom layer provides additional routing and a broad ground reference.

This two-layer arrangement can be represented as

$$\boxed{\begin{array}{c} \text{F.Cu} \rightarrow \text{Signal and Power Routing} \\ \text{PCB Substrate} \\ \text{B.Cu} \rightarrow \text{Routing} + \text{Ground Plane} \end{array}}$$

27.5 Power Amplifier Ground Zone

The power-amplifier PCB contains a copper zone assigned directly to the net

GND

.

The zone is located on

B.Cu

and covers a substantial part of the available PCB area.

The zone follows the irregular outline required by the amplifier board and surrounds the numerous through-hole components while maintaining clearance from conductors belonging to the positive supply, negative supply, amplifier output, and small-signal nets.

The ground plane can be seen in Figure 22.1.

Its purpose is to provide a broad local reference for the amplifier circuitry while reducing reliance on long individual ground tracks.

27.6 Power Amplifier Zone Parameters

The ground-zone parameters defined in the final KiCad amplifier PCB are shown in Table 27.1.

Table 27.1: Power-amplifier ground-zone parameters.

Parameter	Value
Net	GND
Copper layer	B.Cu
Copper weight	2 oz
Approximate copper thickness	70 μm
Zone clearance	0.5 mm
Minimum zone thickness	0.25 mm
Thermal-relief gap	0.5 mm
Thermal-relief bridge width	0.5 mm

27.7 Protection Controller Ground Plane

The protection-controller PCB also contains a large copper zone on the bottom layer.

The zone is named

GROUND PLANE

in the KiCad file and is electrically assigned to

GND.

Unlike the more irregular amplifier-board zone boundary, the protection controller uses a rectangular zone extending across essentially the board outline.

The plane provides the local reference for:

- the NUCLEO-G031K8;
- STM32 ADC measurements;
- DC-offset sensing circuits;
- temperature-sensing circuits;
- BC337 relay drivers;
- indicator LEDs;
- control switches;
- the low-voltage auxiliary power supply.

The ground plane can be seen in Figure 23.1.

27.8 Protection Controller Zone Parameters

The final protection-controller PCB uses the zone parameters listed in Table 27.2.

Table 27.2: Protection-controller ground-plane parameters.

Parameter	Value
Zone name	GROUND PLANE
Electrical net	GND
Copper layer	B.Cu
Copper weight	2 oz
Approximate copper thickness	70 μm
Zone clearance	0.5 mm
Minimum zone thickness	0.25 mm
Thermal-relief gap	0.5 mm
Thermal-relief bridge width	0.5 mm

27.9 Zone Clearance

Both PCB designs use a normal copper-zone clearance of

0.5 mm .

This means that the GND plane is automatically separated from copper belonging to unrelated nets by the specified distance.

For example, the ground zone maintains clearance from:

- positive power rails;
- negative power rails;
- amplifier-output tracks;
- audio-signal tracks;
- STM32 signal nets;
- relay-contact paths.

The clearance prevents accidental electrical connection between the ground plane and unrelated conductors.

27.10 Minimum Zone Thickness

The minimum copper-zone thickness used in both designs is

0.25 mm .

This parameter controls the minimum width of narrow copper sections that KiCad is permitted to generate when filling the zone.

Very small or narrow pieces of copper are generally undesirable because they may provide little electrical benefit and can complicate PCB fabrication.

The minimum-thickness rule therefore helps prevent the ground plane from forming unnecessarily fine copper features around densely spaced pads and tracks.

27.11 Thermal Relief Connections

A through-hole pad connected directly to a large 2 oz copper plane can be difficult to solder because the surrounding copper rapidly conducts heat away from the pad.

Thermal reliefs reduce this effect by connecting the pad to the plane through a limited number of narrow copper bridges.

The ground-zone configuration used in both PCBs specifies a thermal-relief gap of

0.5 mm

and a thermal bridge width of

$$\boxed{0.5 \text{ mm}}.$$

The concept can be represented as

$$\text{Ground Plane} \rightarrow \text{Thermal Spokes} \rightarrow \text{Ground Pad}.$$

This arrangement simplifies manual soldering while still providing an electrical connection to the ground plane.

27.12 Thermal Relief and High-Current Connections

Although thermal reliefs are useful during assembly, they must be considered carefully for high-current ground connections.

The current flowing from a pad into a plane must pass through the thermal bridges. Therefore, these bridges can form a local restriction even when the surrounding plane is very large.

For low-current signal and control circuitry, this is normally insignificant.

For high-current nodes such as

- power-supply common;
- speaker return;
- amplifier output-stage ground;
- reservoir-capacitor ground;

the thermal-relief geometry should be inspected carefully.

Where necessary, a direct plane connection or wider thermal spokes may be preferable, provided that the resulting pad remains practical to solder.

27.13 Ground Plane Resistance

A principal advantage of a ground plane is its low electrical resistance.

The resistance of a copper conductor is

$$R = \rho \frac{L}{A},$$

where A is the conductor cross-sectional area.

For a normal track,

$$A = wt.$$

A ground plane effectively has a much larger available width w than an ordinary routed conductor.

Consequently,

$$R_{\text{plane}} < R_{\text{narrow track}}$$

for comparable current paths.

The 2 oz copper thickness further reduces the resistance of the plane.

27.14 Ground Plane and Return Current

A common misconception is that current spreads uniformly throughout an entire ground plane.

In practice, current follows a path determined by the physical locations of the source and return connections, conductor geometry, and signal frequency.

The existence of a large ground plane therefore does not mean that the placement of ground connections can be ignored.

For example, a speaker-return connection placed next to a sensitive input ground can still cause undesirable common-impedance effects if the current paths overlap in a critical region of the plane.

The ground plane must therefore be considered together with the star-ground and signal-ground strategy described previously.

27.15 Ground Plane and Audio Signal Return

The low-level audio input requires a stable local ground reference.

The ground plane provides a low-impedance path for the input and small-signal circuits, but the high-current output and power-supply connections should be located so that their principal return currents do not unnecessarily cross the same plane region used by the input stage.

The desired arrangement is conceptually

Input Circuit $\rightarrow$ Local Ground Plane $\rightarrow$ Controlled Ground Reference
--

without forcing the path to carry substantial loudspeaker or rectifier current.

27.16 Ground Plane and Power Return

High-current ground connections benefit considerably from the wide copper area available in the ground plane.

However, the current should enter the plane close to its intended high-current ground region.

The power-amplifier return path should therefore remain associated with the power stage and supply common rather than being deliberately routed toward the audio input before reaching ground.

This preserves the distinction between signal ground and power ground even though both are connected electrically.

27.17 Local Decoupling Return Paths

Ground planes are particularly useful in conjunction with local decoupling capacitors.

A local decoupling capacitor provides a short transient-current loop between a circuit and its supply reference.

The desired high-frequency loop is

Active Circuit $\rightarrow$ Decoupling Capacitor $\rightarrow$ Ground Plane $\rightarrow$ Active Circuit.

Reducing this loop area lowers parasitic inductance and prevents high-frequency current from unnecessarily travelling through long power-supply paths.

27.18 Power Amplifier Decoupling

The power-amplifier PCB includes local decoupling capacitors in addition to the large reservoir capacitors.

These include

$$C_7 = 100 \text{ nF},$$

$$C_{18} = 100 \text{ nF},$$

$$C_{10} = 100 \text{ }\mu\text{F},$$

and

$$C_{11} = 100 \text{ }\mu\text{F}.$$

The local ground plane helps provide a low-impedance return for the high-frequency currents associated with these components.

27.19 Protection Controller Analogue Reference

The protection-controller ground plane is particularly important for ADC measurements.

The STM32 measures the left and right amplifier output voltages and the two temperature-sensor voltages relative to its local ground reference.

Any unwanted voltage appearing between the sensing networks and the MCU ground can become an ADC measurement error.

A broad ground plane therefore provides a lower-impedance reference than a collection of long narrow tracks.

27.20 Digital and Relay Return Currents

The same protection PCB also contains digital and electromechanical loads.

These include:

- the STM32 microcontroller;
- indicator LEDs;
- control switches;
- BC337 relay drivers;
- two relay coils.

The ground plane provides a common return structure, but component placement and routing are used to keep the larger relay-driver return currents from unnecessarily flowing through the analogue ADC sensing region.

27.21 Plane Interruptions

Tracks, pads, vias, holes, and clearance boundaries interrupt the continuity of a copper zone.

A plane that appears visually large can therefore contain narrow regions or “necks” through which current must pass.

These constrictions are important because the local resistance is determined by the narrowest effective copper path.

During PCB inspection, particular attention should therefore be given to:

- narrow copper channels;
- isolated copper sections;
- copper trapped between rows of pads;
- ground paths around large footprints;
- thermal-relief spokes;
- plane regions cut by high-current tracks.

27.22 Copper Islands

A copper island is a portion of filled copper that is not usefully connected to the intended electrical network.

Such islands provide little or no grounding benefit and can create unwanted capacitive coupling.

After zone filling, the PCB should therefore be visually inspected for isolated or poorly connected copper areas.

Ground copper should form electrically useful return paths rather than simply maximising the visible amount of copper on the board.

27.23 Copper Zone Filling

In KiCad, the copper zones should be refilled whenever tracks, footprints, board outlines, or zone boundaries are modified.

The standard zone-fill command recalculates the copper geometry according to the current PCB layout.

The resulting filled zone should then be checked for:

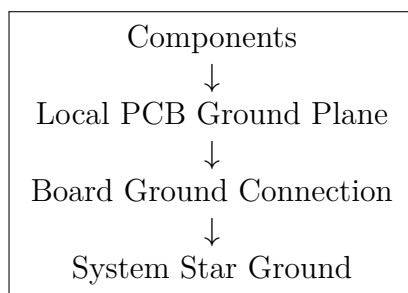
- correct net assignment;
- correct clearance;
- complete pad connectivity;
- absence of unintended copper islands;
- adequate width through restricted regions;
- correct thermal-relief connections.

27.24 Relationship to the Star Ground

The PCB ground plane should not be interpreted as replacing the central star-ground architecture.

The local PCB plane connects components within one board efficiently, while the star-ground strategy controls how separate subsystems are connected at system level.

The architecture can therefore be represented as



This hierarchy maintains low local impedance without allowing arbitrary parallel ground paths between separate subsystems.

27.25 Mains Isolation on the Protection PCB

The protection-controller PCB requires an additional consideration because it contains a PCB-mounted mains transformer.

The ground plane belongs to the isolated low-voltage electronics and must not reduce the required separation between the hazardous mains-primary circuitry and the isolated secondary circuitry.

The normal zone clearance used elsewhere on the board is

0.5 mm.

This ordinary PCB routing clearance must not automatically be interpreted as an adequate mains creepage or clearance distance.

The primary-side mains region must instead be treated as a dedicated high-voltage isolation region.

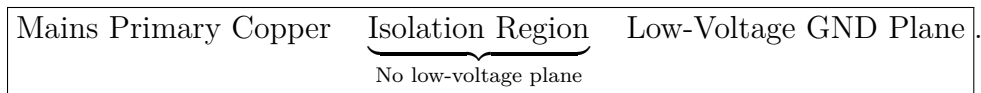
27.26 Ground Plane Keepout Around the Mains Region

The current protection-controller KiCad design defines the bottom GND plane across essentially the complete rectangular board area.

Before final fabrication, the region around the mains-primary connections of the PCB transformer should therefore be reviewed carefully.

Where required, a dedicated copper keepout should prevent the low-voltage GND plane from extending into the required primary-to-secondary isolation region.

The desired principle is



The final isolation distance must be selected according to the applicable safety requirements and PCB manufacturing process rather than the ordinary 0.5 mm zone rule.

27.27 Ground Plane Around Mounting Holes

The relationship between copper planes and mounting holes must also be considered during mechanical integration.

If a mounting hole is intended to remain electrically isolated from the chassis, adequate copper clearance should be maintained around the associated screw, washer, and standoff.

Conversely, if a mounting position is deliberately intended to establish a chassis connection, that connection should be explicitly designed rather than created accidentally by exposed copper or mounting hardware.

This is particularly important because the audio-ground-to-chassis relationship should remain controlled.

27.28 Solder Mask and Copper Planes

The copper plane is normally covered by solder mask except at exposed pads and other deliberately uncovered features.

The solder mask reduces the possibility of accidental short circuits and protects the large copper area from contamination and mechanical contact.

Solder mask, however, should not be considered an electrical safety barrier that replaces the required copper clearance around mains-voltage circuitry.

27.29 Ground Plane and Heat Distribution

A large copper area also spreads heat more effectively than a narrow track.

This can be beneficial around moderate-current ground connections because the 2 oz plane provides a relatively large thermal mass.

The same property explains why pads connected directly to the plane can be more difficult to solder and why thermal reliefs are useful during manual assembly.

27.30 Ground Plane Verification

Before PCB fabrication, the ground planes should be checked using both KiCad and manual inspection.

The following checks are recommended:

1. Refill all copper zones.
2. Confirm that each ground zone is assigned to the correct GND net.
3. Verify that all intended ground pads are connected.
4. Check for isolated copper islands.
5. Inspect narrow plane sections and bottlenecks.
6. Verify thermal-relief connections.
7. Check high-current ground pads for sufficient copper connection.
8. Verify ground clearance from positive and negative supply rails.
9. Confirm clearance from amplifier-output tracks.
10. Check the relationship between PCB ground and mounting holes.
11. On the protection board, verify the mains-primary keepout and isolation region.
12. Run the KiCad Design Rule Checker before generating Gerber files.

27.31 Manufacturing Verification

The Gerber files should also be inspected after export because the Gerber representation is the actual data supplied to the PCB manufacturer.

The copper Gerbers should confirm:

- correct ground-plane shape;
- correct plane clearances;
- correct thermal reliefs;
- absence of unintended isolated copper;
- correct board-edge clearance;
- correct mains isolation region;
- correct pad and via connections.

This provides an additional verification stage between the KiCad project and physical PCB fabrication.

27.32 Relationship to Noise Performance

The use of broad ground planes reduces local ground impedance and can therefore contribute to lower noise.

However, noise performance depends on more than simply filling unused board area with copper.

The location of the input circuitry, transformer, rectifier, digital electronics, relay currents, grounding points, and external wiring all affect the final result.

These wider system-level issues are examined in the following *Noise and Hum Reduction* chapter.

27.33 Design Summary

Both principal PCBs in the amplifier project use large GND copper zones on the

B.Cu

layer.

The power-amplifier zone is named

GND,

while the protection-controller zone is named

GROUND PLANE

and is electrically connected to the GND net.

All boards use

2 oz copper,

corresponding to an approximate copper thickness of

70 μm .

Both ground-zone definitions use a normal clearance of

0.5 mm,

a minimum zone thickness of

0.25 mm,

a thermal-relief gap of

0.5 mm ,

and a thermal-relief bridge width of

0.5 mm .

The broad copper areas reduce local ground resistance, provide efficient current-return paths, simplify ground routing, and support the local decoupling networks.

The ground planes do not replace the star-ground strategy. Instead, each PCB uses its local plane to connect its components efficiently before the board ground is connected to the controlled system ground reference.

The location of high-current connections remains important because loudspeaker, output-stage, and rectifier currents should not be allowed to create significant voltage in the portion of the plane used as the low-level audio reference.

Thermal reliefs improve solderability but must be reviewed at high-current ground pads because narrow thermal spokes can form a local current restriction.

The protection-controller PCB requires additional attention because its PCB-mounted transformer introduces a mains-primary region. The normal 0.5 mm copper-zone clearance is an ordinary PCB design rule and must not be interpreted as the required mains isolation distance.

A dedicated primary-side copper keepout or equivalent isolation region should therefore be verified in the final protection-controller PCB before manufacturing.

The combination of broad 2 oz ground planes, controlled ground connections, appropriate zone clearances, local decoupling, and system-level star grounding provides a low-impedance ground structure suitable for the analogue amplifier and digital protection-controller sections.

Chapter 28

Noise and Hum Reduction

28.1 Introduction

Noise and hum performance are important design considerations in a Hi-Fi power amplifier because unwanted electrical signals can become audible even when their amplitude is small compared with the normal amplifier output.

The complete amplifier developed in this project contains several potential noise sources, including the mains transformer, bridge rectifier, reservoir capacitors, high-current output stage, preamplifier, STM32 protection controller, loudspeaker relays, and cooling system.

Noise reduction therefore cannot be achieved by a single component or PCB feature. It depends on the combined design of the grounding system, PCB layout, wiring arrangement, power-supply filtering, physical component placement, shielding, and mechanical construction.

The principal objective is to prevent unwanted signals from entering the sensitive audio path and to prevent high-current circuits from disturbing the low-level ground reference.

28.2 Principal Sources of Noise

The principal potential noise sources within the amplifier system are:

- the 230 V AC mains wiring;
- the main 24–0–24 V AC toroidal transformer;
- magnetic fields produced by transformer current;
- bridge-rectifier switching and capacitor charging currents;
- residual power-supply ripple;
- shared ground impedance;
- high-current loudspeaker return paths;

- preamplifier and phono-stage noise;
- digital switching currents from the STM32 controller;
- loudspeaker relay switching;
- relay-coil transients;
- cooling-fan motors;
- electromagnetic coupling between internal cables;
- external ground loops through connected audio equipment.

Each source requires a different mitigation method.

28.3 Difference Between Noise and Hum

The terms *noise* and *hum* are related but do not describe exactly the same phenomenon. Hum normally refers to low-frequency unwanted signals associated with the mains power system.

For the Greek mains supply frequency of

$$50 \text{ Hz},$$

magnetic or electrical mains coupling can produce a strong component at

$$50 \text{ Hz}.$$

After full-wave rectification, the reservoir capacitors are recharged twice during every mains cycle.

Therefore, power-supply ripple and rectifier charging effects commonly occur at

$$100 \text{ Hz}.$$

These frequencies provide useful diagnostic information when investigating the completed amplifier.

28.4 50 Hz Hum

A strong unwanted component close to

$$50 \text{ Hz}$$

can indicate direct coupling from the mains system.

Possible causes include:

- transformer magnetic-field coupling;
- mains wiring positioned close to the audio input;
- ground loops through connected equipment;
- inadequate shielding;
- unsuitable chassis-ground connections.

The presence of a dominant 50 Hz component does not identify the cause automatically, but it provides an important starting point for diagnosis.

28.5 100 Hz Hum

A dominant component at approximately

100 Hz

is more commonly associated with the rectified power supply.

Possible causes include:

- insufficient reservoir capacitance;
- rectifier charging-current coupling;
- shared power and signal ground impedance;
- excessive supply ripple;
- poor connection of the reservoir-capacitor common point;
- high-current return paths passing through sensitive circuitry.

Separating the diagnosis of 50 Hz and 100 Hz components is therefore useful during amplifier testing.

28.6 Main Toroidal Transformer

The main amplifier supply uses a

24-0-24 V_{AC}

toroidal transformer.

Toroidal transformers generally produce less external magnetic leakage than many conventional laminated-core transformers, but the external magnetic field is not zero.

The transformer should therefore be located as far as practically possible from:

- the RCA input connectors;
- the preamplifier;
- the phono input circuitry;
- the small-signal section of the power-amplifier PCBs;
- sensitive analogue interconnection cables.

Physical separation is particularly important for the phono preamplifier because its input signals are significantly smaller than normal line-level signals.

28.7 Transformer Orientation

The orientation of a transformer inside the chassis can influence the amount of magnetic coupling into nearby circuits.

If hum is detected during prototype testing, one useful diagnostic method is to change the physical orientation of the transformer while monitoring the amplifier output.

A reduction in hum after rotating or repositioning the transformer indicates that magnetic coupling is contributing to the problem.

The final transformer orientation should therefore be selected experimentally during system integration rather than assumed solely from mechanical convenience.

28.8 Mains Wiring

The 230 V mains wiring should be treated as a potential interference source.

Mains conductors should:

- remain physically separated from audio input wiring;
- follow short and direct routes;
- remain close to the chassis where practical;
- avoid crossing the preamplifier and input stages;
- enter and leave the power switch and fuse region without unnecessary loops.

The line and neutral conductors should be routed together so that the magnetic fields generated by their currents partially cancel.

Twisting paired AC conductors is useful because it reduces the effective loop area and therefore decreases radiated magnetic field.

28.9 Transformer Secondary Wiring

The transformer secondary wiring also carries relatively large AC current.

Although its voltage is much lower than the mains voltage, it can still create magnetic coupling into nearby audio circuitry.

The secondary wires should therefore be kept short and routed directly to the bridge-rectifier input.

The AC secondary conductors should be routed together rather than separated across different areas of the chassis.

Where practical, twisting the relevant secondary conductors can reduce the area of the AC current loop.

28.10 Bridge Rectifier Charging Currents

The KBPC3506WP bridge rectifier supplies the four main reservoir capacitors.

The current flowing through the rectifier is not a smooth sinusoidal current.

The reservoir capacitors are recharged during short intervals close to the peaks of the rectified waveform.

Consequently, the charging current can have a relatively high peak amplitude.

These high-current pulses can generate noise through:

- conductor resistance;
- parasitic inductance;
- shared ground impedance;
- electromagnetic coupling.

The bridge rectifier, reservoir capacitors, and associated ground connection are therefore grouped closely on the amplifier PCB.

28.11 Reservoir Capacitors

The main supply contains four

4700 μ F

reservoir capacitors.

Two capacitors operate in parallel on each rail, providing

9400 μ F

of nominal reservoir capacitance per supply rail.

The reservoir capacitors reduce the rectified supply ripple by storing energy between rectifier conduction intervals.

However, increasing capacitance alone does not guarantee low hum.

The capacitor-current return path and physical connection to the system ground are equally important because large charging currents can still produce unwanted voltages across shared conductors.

28.12 Power-Supply Loop Area

The physical loop containing the transformer secondary, bridge rectifier, and reservoir capacitors should be kept compact.

The desired arrangement is approximately

Transformer → Bridge → Reservoir Capacitors → Transformer / Common Return.

Reducing the loop area reduces its ability to radiate magnetic fields and reduces the conductor inductance associated with the charging pulses.

28.13 Star Grounding

One of the principal techniques used to reduce hum is the star-ground architecture described in the previous grounding chapters.

The large currents associated with the loudspeakers, output stages, and reservoir capacitors should not flow through the same conductors used by the audio input.

The system therefore separates

Signal Ground

and

Power Ground

before they meet at a controlled common reference.

The basic principle is

Low-Level Signal Return → Star Ground ← High-Current Power Return.

This reduces common-impedance coupling between the output stage and the sensitive amplifier input.

28.14 Speaker Return Current

The loudspeaker return is treated as a high-current power-ground connection.

At the Project 3A reference condition of approximately

$$100 \text{ W}$$

into

$$4 \Omega,$$

the loudspeaker current is

$$I_{\text{RMS}} = 5 \text{ A}.$$

A current of this magnitude should not flow through any part of the RCA or preamplifier ground path.

Otherwise, even milliohms of common conductor resistance can create an unwanted voltage in the input reference.

28.15 Ground Planes

Both major PCBs use broad bottom-layer GND copper zones.

All boards are manufactured using

$$\boxed{2 \text{ oz copper}},$$

providing an approximate copper thickness of

$$70 \mu\text{m}.$$

The broad copper areas reduce local ground resistance and provide short current-return paths.

However, a ground plane is not by itself sufficient to guarantee low noise.

High-current connections must still be positioned so that their return paths do not unnecessarily cross the sensitive input-ground region.

28.16 Local Supply Decoupling

Local supply decoupling is used to prevent high-frequency current from travelling unnecessarily through long supply and ground conductors.

The power amplifier contains both

100 μF

electrolytic capacitors and

100 nF

high-frequency decoupling capacitors close to the active circuitry.

The small-value capacitors are particularly useful at frequencies where the large reservoir electrolytics have increased impedance due to parasitic inductance.

The desired local current path is approximately

Amplifier Stage $\rightarrow$ Local Decoupling $\rightarrow$ Local Return

.

28.17 Input Stage Placement

The small-signal input section of the power-amplifier PCB is positioned away from the bridge rectifier and main power-supply section.

This reduces the possibility of coupling from the high-current rectifier and reservoir-capacitor paths into the differential input stage.

The RCA/input wiring should similarly connect directly to this low-level region rather than crossing the main supply or output stage.

28.18 Audio Signal Wiring

Internal audio wiring should be kept as short as practical.

The following signals require particular attention:

- RCA input to preamplifier;
- preamplifier output to power amplifier;
- selector and volume-control wiring;
- phono-stage input wiring.

These cables should be kept away from mains conductors, transformer secondaries, relay coils, and loudspeaker wiring.

28.19 Shielded Audio Cable

Shielded cable is appropriate for low-level analogue interconnections, particularly between the RCA connectors and preamplifier or between physically separated signal-processing stages.

The shield provides electrostatic screening from nearby electric fields.

The shield connection should form part of the controlled signal-ground system and should not create an additional uncontrolled chassis or ground-loop path.

28.20 RCA Connector Placement

The RCA connectors should be positioned away from the mains transformer and high-current AC wiring.

Rear-panel placement should be selected so that the signal cable can travel directly toward the preamplifier or amplifier input rather than crossing the power-supply section.

The RCA ground connection must also be considered carefully with respect to the metal chassis.

An unintended chassis connection at the RCA socket can create a second ground path and potentially form a loop with another grounded audio source.

28.21 Preamplifier Noise Considerations

The preamplifier is more sensitive to noise than the main power amplifier because it processes smaller signals.

This is especially true for the phono input.

Noise-reduction measures for the preamplifier therefore include:

- locating it away from the mains transformer;
- using short low-level input wiring;
- using shielded cable where appropriate;
- maintaining a clean signal-ground reference;
- providing local supply decoupling;
- separating low-level audio tracks from high-current wiring.

28.22 Preamplifier Power Supply

The preamplifier operates from regulated low-voltage supplies rather than directly from the main amplifier rails.

Regulation of the preamplifier supply helps reduce the amount of power-supply ripple reaching the low-level analogue circuitry.

Local decoupling capacitors should also be positioned close to the operational amplifiers to reduce high-frequency supply impedance.

28.23 Protection Controller Noise Sources

The STM32 protection controller contains digital electronics that can generate switching noise.

Potential sources include:

- the STM32 clock and GPIO transitions;
- status LEDs;
- relay drivers;
- relay coils;
- the auxiliary rectifier and regulator.

The protection circuitry is therefore maintained outside the direct analogue audio signal path.

The controller monitors the amplifier outputs and controls speaker relays but does not process the audio waveform digitally.

28.24 Protection Controller Ground

The protection controller uses a local bottom-layer ground plane.

The ADC sensing circuits require a stable ground reference because the measured voltages are interpreted relative to the STM32 ground.

Relay and LED return currents should therefore not be deliberately routed through the most sensitive analogue sensing nodes.

The PCB layout keeps the analogue sensing networks physically identifiable from the relay-current region.

28.25 Relay Switching Noise

The protection system contains two 12 V loudspeaker relays.

Each relay coil is driven by a BC337 transistor.

When current through a relay coil is interrupted, the inductance attempts to maintain the current and can generate a high-voltage transient.

A 1N4007 flyback diode is connected across each relay coil.

During normal energised operation the diode is reverse biased.

When the transistor switches off, the diode provides a path for the coil current:

$$\boxed{\text{Relay Coil} \rightarrow \text{Flyback Diode} \rightarrow \text{Relay Coil.}}$$

This limits the voltage transient and reduces conducted switching noise.

28.26 Relay Contact Placement

The speaker relays are positioned close to the amplifier and speaker connections on the protection PCB.

This keeps the high-current audio path relatively short and reduces both conductor resistance and loop area.

The relay-coil driver circuit is electrically separate from the relay contact audio path, preventing the digital control current from flowing directly through the loudspeaker signal conductor.

28.27 DC Sensing Filter

The protection controller's DC-offset sensing network incorporates analogue low-pass filtering.

The approximate effective resistance associated with the ADC filter is

$$R_{\text{eq}} \approx 9.54 \text{ k}\Omega,$$

with

$$C = 4.7 \text{ }\mu\text{F}.$$

The corresponding approximate cutoff frequency is

$$f_c = \frac{1}{2\pi RC} \approx 3.55 \text{ Hz}.$$

This deliberately attenuates the normal audio waveform and high-frequency noise before the STM32 evaluates the DC component.

The filter therefore improves protection-system noise immunity while allowing persistent DC faults to be detected.

28.28 Digital Filtering

The firmware should not make a protection decision from a single ADC sample.

Analogue filtering is complemented by digital averaging and persistence checking.

This helps prevent relay disconnection due to short-duration noise spikes or normal audio transients.

The final thresholds and timing constants will be established during prototype testing.

28.29 Cooling Fans

Cooling fans represent another possible electrical and acoustic noise source.

A fan motor can produce:

- electrical switching noise;
- magnetic-field interference;
- conducted supply noise;
- mechanical vibration;
- audible airflow noise.

Fan wiring should therefore be kept away from the audio input and preamplifier wiring.

Where possible, the fan current should not share a sensitive signal-ground return path.

The physical mounting of the fans should also avoid transmitting excessive mechanical vibration into the chassis.

28.30 High-Current Loudspeaker Wiring

The loudspeaker output wiring carries several amperes during high-power operation.

These conductors should therefore be kept away from low-level signal wiring.

Where the left and right speaker conductors run through the chassis, each channel's output and return wiring should be kept close together.

This reduces the effective magnetic loop area associated with the speaker current.

28.31 Separation of Signal and Power Cables

Internal chassis wiring can be grouped into three broad categories:

Low-Level Audio

High-Current Power / Speaker

and

Mains AC

These groups should not be bundled together.

The preferred physical arrangement is to route them through different parts of the chassis whenever practical.

If signal and power wires must cross, a near-perpendicular crossing is preferable to a long parallel run because the coupling length is reduced.

28.32 Chassis Shielding

A conductive metal chassis provides useful electrostatic shielding for the internal audio circuitry.

The chassis is bonded to protective earth for electrical safety.

The relationship between chassis and audio ground must remain controlled so that the shielding benefit is obtained without introducing unnecessary ground loops.

The chassis is therefore treated as a safety and shielding structure rather than as an arbitrary current-return conductor.

28.33 PCB Track Routing

The PCB layouts use different track widths according to current level.

The power-amplifier PCB uses

0.4 mm, 1.0 mm, 2.5 mm

routing.

The protection controller uses

1.0 mm

routing for most control functions and

2.4 mm

routing for the high-current loudspeaker relay paths.

The use of wider tracks reduces resistance and therefore reduces unwanted voltage developed by high currents.

28.34 Effect of 2 oz Copper

All project PCBs use

2 oz copper

.

The approximately 70 μm copper thickness reduces conductor resistance compared with conventional 1 oz construction.

Lower conductor resistance reduces

$$V_{\text{noise}} = IR$$

generated by currents flowing through shared copper.

However, thicker copper does not replace correct grounding and routing.

The return-current topology remains more important than simply increasing copper thickness.

28.35 Component Placement

The physical component arrangement contributes directly to noise reduction.

The amplifier PCB places the sensitive input stage toward one side of the board, while the bridge rectifier, large reservoir capacitors, fuses, driver stage, and output devices occupy separate functional regions.

The protection-controller PCB similarly groups the ADC sensing networks, microcontroller, relay drivers, relays, and auxiliary power supply into recognisable regions.

This physical organisation reduces unnecessary crossing between unrelated high-current and low-level circuits.

28.36 Avoiding Large Current Loops

Any current travels in a complete loop.

The magnetic field associated with the loop increases with the effective enclosed area.

For this reason, conductor pairs carrying equal and opposite currents should be kept physically close where practical.

Examples include:

- mains line and neutral;
- transformer secondary conductors;
- loudspeaker output and return;
- low-level audio signal and its ground reference.

28.37 Mechanical Vibration

Noise in an amplifier is not exclusively electrical.

The toroidal transformer can produce mechanical vibration, particularly if the mains waveform contains DC or if the mounting system is excessively rigid.

Cooling fans and large transformers can also transmit vibration through the chassis.

Mechanical noise should therefore be considered separately from electrical hum during final system evaluation.

28.38 Transformer Mounting

The toroidal transformer should be mounted securely using appropriate mechanical hardware.

The mounting arrangement must not create an unintended shorted turn around the transformer core.

Only the intended transformer mounting hardware should pass through the central opening.

Mechanical isolation material may also be used where appropriate to reduce transmission of transformer vibration to the chassis.

28.39 Initial Noise Testing

Noise performance should be evaluated systematically after construction.

The first test should be performed with the amplifier inputs shorted directly to their corresponding signal-ground connections.

This removes external source equipment as a possible noise source.

The amplifier output can then be monitored with an oscilloscope and suitable AC measurement equipment.

28.40 Recommended Test Sequence

A useful diagnostic sequence is:

1. Power the amplifier with the audio inputs shorted to signal ground.
2. Measure the output noise with no loudspeaker connected.
3. Observe the waveform and frequency spectrum where available.
4. Determine whether the dominant component is near 50 Hz or 100 Hz.

5. Connect the protection controller and repeat the measurement.
6. Energise the speaker relays and check for additional switching or steady-state noise.
7. Connect the preamplifier and repeat the measurement.
8. Connect an $8\ \Omega$ dummy load.
9. Repeat the test under controlled output power.
10. Where safe and appropriate, repeat selected tests using a $4\ \Omega$ dummy load.

28.41 Diagnostic Interpretation

Table 28.1: Examples of possible noise symptoms and likely areas to investigate.

Observed Symptom	Possible Area to Investigate
Strong 50 Hz hum	Transformer coupling, mains wiring, chassis/ground loop
Strong 100 Hz hum	Rectifier current, reservoir capacitors, supply ground, ripple
Hum changes when RCA connected	External ground loop or RCA grounding
Noise increases with volume	Input or preamplifier section
Noise independent of volume	Power amplifier or power-supply section
Click when relay switches	Relay driver, coil suppression, grounding
Noise when fans operate	Fan motor supply, wiring or mechanical vibration
Different hum between channels	Channel grounding, wiring position or component placement

28.42 Oscilloscope Investigation

An oscilloscope can be used to observe unwanted signals at different points in the amplifier.

Useful measurement locations include:

- amplifier output;
- signal input;
- positive supply rail;
- negative supply rail;

- star-ground reference;
- signal ground relative to star ground;
- protection-controller ground;
- preamplifier output.

Care must be taken when using earth-referenced laboratory oscilloscopes because the probe ground is normally connected to protective earth.

Incorrect probe-ground placement can create a short circuit or unintended ground path.

28.43 Noise Measurement with Input Shorted

Shorting the amplifier input to its local signal ground is a useful method of measuring the amplifier's own residual noise.

If the output becomes quiet when the input is shorted but becomes noisy when an external source is connected, the source, cable, or inter-equipment ground connection should be investigated.

If the hum remains with the input shorted, the cause is more likely to be inside the amplifier.

28.44 Channel-by-Channel Testing

Because the system is stereo, each channel should first be investigated independently.

Testing one channel at a time makes it easier to determine whether an unwanted signal originates from:

- one specific PCB;
- shared power-supply wiring;
- the transformer;
- the common grounding network;
- interaction between both channels.

After each channel has been verified independently, both channels should be operated simultaneously.

28.45 Final Chassis Verification

Noise testing should be repeated after final installation in the chassis.

A circuit that performs quietly on an open workbench can behave differently after all transformers, cables, fans, front-panel controls, RCA sockets, and ground connections are installed.

The final chassis arrangement therefore forms part of the electrical design, not merely the mechanical packaging.

28.46 Noise-Reduction Measures Used in the Project

The principal noise- and hum-reduction measures incorporated into the project can be summarised as follows:

- central star-ground architecture;
- separation of signal ground and power ground;
- speaker return treated as high-current power ground;
- broad bottom-layer ground planes;
- 2 oz PCB copper;
- wide high-current PCB tracks;
- physically separated low-level and power circuitry;
- grouped bridge rectifier and reservoir capacitors;
- local amplifier decoupling capacitors;
- short driver and output-stage connections;
- external heatsink positioning of power devices;
- regulated preamplifier supply;
- shielded low-level audio wiring where required;
- physical separation of audio and mains wiring;
- compact transformer-secondary wiring;
- flyback diodes across loudspeaker relay coils;
- analogue and digital filtering of protection measurements;
- separation of digital protection circuitry from the direct audio path;
- deliberate audio-ground/chassis relationship.

28.47 Future Experimental Verification

The effectiveness of the noise-reduction strategy will be verified after the complete amplifier has been assembled.

The following values and observations will be recorded during testing:

- residual output noise with inputs shorted;
- dominant hum frequency;
- output noise with the preamplifier connected;
- left-channel and right-channel noise separately;
- noise with both channels operating simultaneously;
- influence of transformer position and orientation;
- influence of protection-controller operation;
- relay switching transient amplitude;
- influence of cooling-fan operation;
- ground potential differences between major ground points.

No measured noise or hum values are claimed at this stage because these tests require the completed physical amplifier.

28.48 Design Summary

Noise and hum reduction in the amplifier is achieved through the combined design of the power supply, PCB layouts, grounding system, internal wiring, component placement, shielding, and protection circuitry.

The principal low-frequency interference mechanisms are expected around

50 Hz

from direct mains coupling and around

100 Hz

from full-wave rectification and reservoir-capacitor charging.

The 24–0–24 V toroidal transformer is therefore positioned away from sensitive input and preamplifier circuitry, while mains and transformer secondary wiring are kept physically separated from the audio signal paths.

The KBPC3506WP bridge rectifier and four 4700 μF reservoir capacitors are grouped closely so that the high-current charging loops remain compact.

A star-ground architecture prevents loudspeaker and rectifier currents from sharing unnecessary impedance with the RCA and preamplifier signal grounds.

Broad bottom-layer ground zones and

2 oz copper

further reduce local conductor resistance.

The power-amplifier PCBs use local 100 nF and 100 μF rail-decoupling capacitors to keep high-frequency supply currents local to the amplifier stages.

Low-level audio cables are routed separately from mains, transformer, loudspeaker, relay, and fan wiring, with shielded cable used where appropriate.

The STM32 protection controller remains outside the direct audio path. Its relay coils are driven through BC337 transistors and protected by 1N4007 flyback diodes, reducing inductive switching transients.

The DC sensing system also combines analogue low-pass filtering with digital averaging and persistence checking so that normal audio signals and brief noise spikes do not cause unnecessary relay disconnection.

The final noise performance will be evaluated experimentally after complete system integration. Measurements with the inputs shorted, individual and simultaneous channel operation, dummy loads, relay operation, fan operation, and frequency analysis of the residual output will be used to identify any remaining hum or noise mechanisms.

The resulting strategy is intended to minimise mains hum, rectifier noise, common-impedance coupling, digital switching interference, relay transients, and magnetic coupling while maintaining a clean low-level audio reference throughout the complete Hi-Fi amplifier system.

Chapter 29

Component Selection

29.1 Introduction

Component selection is an important part of the development of the complete Hi-Fi amplifier because the electrical performance, reliability, thermal behaviour, serviceability, manufacturability, and safety of the final system depend on the characteristics of the individual components.

The project contains several different electronic subsystems, each with different component requirements:

- two discrete Class-AB power-amplifier channels;
- the main symmetrical power supply;
- the audio preamplifier;
- the STM32-based speaker-protection controller;
- the protection-controller auxiliary power supply;
- loudspeaker switching and temperature monitoring;
- chassis-mounted connectors, fuses, switches, heatsinks, and cooling hardware.

Components were therefore not selected only according to their nominal electrical value. The selection process also considered voltage rating, current capability, power dissipation, tolerance, thermal behaviour, package type, PCB compatibility, availability, and suitability for through-hole prototype construction.

29.2 General Selection Criteria

The principal criteria used throughout the project were:

- adequate voltage margin;

- adequate continuous and peak current capability;
- suitable power-dissipation capability;
- appropriate thermal characteristics;
- appropriate tolerance for the circuit function;
- suitable frequency performance;
- compatibility with the intended PCB footprint;
- practical through-hole assembly;
- mechanical robustness;
- availability from established distributors;
- suitability for long-term servicing and replacement.

A component was therefore considered suitable only when both its electrical and mechanical characteristics matched the requirements of the corresponding circuit.

29.3 Component Rating Margin

Components should not normally be operated continuously at their absolute maximum ratings.

For example, if a component has a maximum voltage rating

$$V_{\text{MAX}},$$

the normal operating voltage

$$V_{\text{OP}}$$

should satisfy

$$V_{\text{OP}} < V_{\text{MAX}}$$

with an appropriate engineering margin.

The same principle applies to current and power:

$$I_{\text{OP}} < I_{\text{MAX}}$$

and

$$P_{\text{OP}} < P_{\text{MAX}}.$$

This margin is particularly important in the present amplifier because mains variation, loudspeaker impedance, thermal conditions, startup transients, and music peaks can all cause instantaneous operating conditions to differ from their nominal values.

29.4 Power Amplifier Semiconductor Selection

The discrete power-amplifier stage is derived from Rod Elliott's Project 3A topology [1]. Different semiconductor classes are used according to the voltage, current, and thermal requirements of each stage.

The principal transistor selection is summarised in Table 29.1.

Table 29.1: Principal transistor selection for one power-amplifier channel.

Reference	Selected Device	Function
Q1, Q2	BC546	Differential input stage
Q3	BC546	Differential-stage current sink
Q4	BD140	Voltage-amplification stage
Q5	BD139	Upper driver
Q6	BD140	Lower driver
Q7	2SA1943-O	PNP power-output transistor
Q8	2SC5200-O	NPN power-output transistor
Q9	BC546	Bias-control circuit

29.5 BC546 Small-Signal Transistors

BC546 NPN transistors are used in the differential input, current-sink, and bias-control sections.

These locations require relatively low current but benefit from a small-signal transistor with adequate voltage capability for operation in the amplifier's approximately

$$\pm 33\text{--}\pm 34\text{ V}$$

supply environment.

Using the same transistor family in several small-signal positions also simplifies component sourcing, PCB assembly, and servicing.

For Q1 and Q2, electrical similarity between the two devices is desirable because the pair forms the differential input stage.

29.6 BD139 and BD140 Driver Devices

The driver stage uses the complementary medium-power devices

BD139

and

BD140.

These transistors provide greater current capability than the BC546 small-signal devices and form the interface between the voltage-amplification stage and the main output transistors.

Their use is appropriate because the driver stage must deliver sufficient base current to the output devices while remaining capable of dissipating the associated power.

The BD139 and BD140 are also available in practical through-hole packages, which is consistent with the construction philosophy of the amplifier.

29.7 Output Power Transistors

The selected main output devices are

2SA1943-O

for the PNP transistor and

2SC5200-O

for the NPN transistor.

These devices are used as Q7 and Q8 respectively and operate with the BD139 and BD140 drivers as complementary feedback pairs.

The 2SC5200 and 2SA1943 families are intended for high-power linear applications such as audio amplifier output stages [4, 5].

Their selection provides appropriate margin for the operating voltage and several-ampere loudspeaker current required by the amplifier.

The selected O gain rank provides a specified current-gain classification of

$$80 \leq h_{FE} \leq 160.$$

The devices are supplied in TO-3P packages, which also suit the mechanical design because the transistors can be mounted along the PCB edge and attached directly to external heatsinks.

29.8 Complementary Output Pair

Using complementary NPN and PNP output transistors provides the symmetrical positive and negative current capability required by the Class-AB output stage.

The positive and negative output sections are arranged as

$$\text{BD139} + 2\text{SA1943-O}$$

and

$$\text{BD140} + 2\text{SC5200-O}.$$

This topology provides the current gain required to drive $4\ \Omega$ and $8\ \Omega$ loudspeaker loads while the preceding amplifier stages operate at considerably lower current.

29.9 Power Resistor Selection

Not all resistors in the amplifier experience the same dissipation.

Most small-signal resistors carry relatively little current and can therefore use conventional through-hole resistor packages.

The output-stage emitter resistors,

$$R_{13} = R_{14} = 0.33\ \Omega,$$

carry the loudspeaker current and require physically larger power-rated components.

Their selection must provide adequate power capability as well as low resistance.

The instantaneous dissipation is determined by

$$P = I^2 R.$$

For example, a current of $5\ \text{A}$ through $0.33\ \Omega$ would correspond instantaneously to

$$P = 5^2(0.33) = 8.25\ \text{W}.$$

The actual dissipation depends on the waveform and current sharing within the output stage, but the calculation demonstrates why these positions require power resistors rather than ordinary low-power components.

29.10 Resistor Tolerance

Precision is particularly important in the amplifier feedback and bias networks and in the protection-controller sensing circuits.

The project therefore primarily uses

$$\boxed{1\%}$$

resistors in positions where resistance accuracy affects gain, voltage scaling, ADC interpretation, or channel matching.

For example, the closed-loop amplifier gain depends on the ratio of

$$R_5 = 22 \text{ k}\Omega$$

and

$$R_4 = 1 \text{ k}\Omega.$$

The nominal gain is

$$A_v = 1 + \frac{R_5}{R_4} \approx 23.$$

Using reasonably accurate resistor values helps maintain predictable channel gain.

29.11 Bias Adjustment Component

The output-stage bias is adjusted using

$$\boxed{VR1 = 2 \text{ k}\Omega}.$$

A trimmer potentiometer is required rather than a fixed resistor because the quiescent current depends on transistor characteristics and therefore must be adjusted during amplifier commissioning.

The selected component must provide stable adjustment and must remain mechanically accessible after PCB assembly.

29.12 Amplifier Capacitor Selection

Different capacitor technologies and values are used according to the required electrical function.

The amplifier contains:

- input coupling capacitors;
- bootstrap capacitors;
- local rail-decoupling capacitors;

- compensation capacitors;
- Zobel-network capacitors;
- large power-supply reservoir capacitors.

Small-value capacitors such as the

100 pF

compensation components are used where high-frequency behaviour is important.

The local rail decoupling combines

100 nF

capacitors with

100 μ F

electrolytic capacitors.

This combination provides low impedance over a broader frequency range than either capacitor value alone.

29.13 Main Reservoir Capacitors

The main power supply uses four

4700 μ F

large electrolytic capacitors.

Two operate in parallel on each rail, producing

9400 μ F

of nominal reservoir capacitance per supply rail.

Large reservoir capacitors were selected because the power amplifier requires substantial short-duration current during audio peaks.

Their voltage rating must also provide sufficient margin above the expected rail voltage of approximately

33–34 VDC.

In addition to capacitance and voltage rating, suitable reservoir capacitors must have appropriate ripple-current capability and physical dimensions compatible with the PCB footprints.

29.14 Main Bridge Rectifier

The selected main bridge rectifier is

$$\boxed{\text{KBPC3506WP}}.$$

The device is rated substantially above the normal current and voltage requirements of the amplifier supply, providing useful margin for the high peak charging currents associated with the reservoir capacitors [6].

A mechanically large bridge package is also advantageous because it provides robust terminals and is well suited to the high-current power-supply region of the PCB.

29.15 Main Transformer

The main amplifier supply uses a toroidal transformer with a secondary voltage of

$$\boxed{24-0-24 \text{ V}_{\text{AC}}}.$$

The theoretical unloaded secondary peak is

$$V_{\text{peak}} = 24\sqrt{2} \approx 33.94 \text{ V}.$$

After rectifier losses and taking transformer regulation and mains variation into account, this is appropriate for obtaining the approximately

$$\pm 33 - \pm 34 \text{ VDC}$$

rail region required by the implemented amplifier.

A toroidal transformer was selected because it provides the required symmetrical high-current supply in a compact form and generally offers low external magnetic leakage when installed correctly.

29.16 Power-Supply Fuse Selection

Each amplifier rail incorporates a

$$\boxed{3 \text{ A}}$$

fuse.

The fuse rating was selected so that the device can carry the normal amplifier supply current while providing protection against sustained abnormal high-current conditions.

The fuse is not intended to replace semiconductor protection or correct thermal design, but provides an additional level of protection for the amplifier and power supply.

29.17 Preamplifier Operational Amplifiers

The audio preamplifier uses

NE5532N

dual operational amplifiers.

This device is appropriate for audio applications because it combines low noise, low distortion, useful output-current capability, and compatibility with symmetrical supply operation.

The dual-amplifier package also allows two operational-amplifier stages to be implemented using a single integrated circuit, reducing component count.

The selection is particularly suitable for line-level amplification and active audio filtering.

29.18 Preamplifier Supply Regulation

The preamplifier uses regulated

$\pm 15\text{ V}$

supply rails.

The use of conventional positive and negative linear regulators provides a stable supply for the NE5532 circuitry and reduces the transfer of power-supply ripple into the low-level audio stages.

Linear regulation is appropriate in this application because the preamplifier current requirement is relatively small compared with the main power amplifier.

29.19 Microcontroller Selection

The speaker-protection system is based on the

STM32G031K8

microcontroller and is developed using the

NUCLEO-G031K8

development board [2, 3].

The STM32G031K8 provides the required combination of:

- multiple GPIO pins;

- 12-bit analogue-to-digital conversion;
- timers;
- watchdog capability;
- sufficient Flash memory;
- sufficient SRAM;
- low-voltage operation;
- integrated development and debugging support through the Nucleo platform.

The 12-bit ADC is particularly important because the protection controller must monitor the DC level at both amplifier outputs and the voltages produced by two temperature-sensor networks.

29.20 Why a Microcontroller Was Selected

A microcontroller-based protection system provides greater flexibility than a purely analogue protection circuit.

The firmware can implement:

- startup delay;
- persistent DC-fault detection;
- temperature monitoring;
- sensor-fault detection;
- relay sequencing;
- fault indication;
- hysteresis;
- digital averaging;
- future diagnostic extensions.

This flexibility was a principal reason for selecting the STM32 solution for the protection controller.

29.21 Speaker Relay Selection

The selected loudspeaker relays are

Omron G2RL-1A-E-DC12

devices [9].

One relay is used for each stereo channel.

The relays use

12 VDC

coils and normally-open contacts.

The normally-open configuration is particularly appropriate for a protection system because loss of controller power or relay drive naturally disconnects the loudspeaker.

The contacts are also designed for currents substantially greater than the small-signal switching levels of ordinary signal relays, making the component suitable for the several-ampere loudspeaker path.

29.22 Relay Driver Transistors

The STM32 GPIO outputs cannot directly supply the current required by the 12 V relay coils.

Each relay is therefore driven by an

BC337

NPN transistor [10].

The selected BC337 provides considerably greater collector-current capability than required by the approximately

33.3 mA

relay coil.

A transistor driver also isolates the microcontroller GPIO from the 12 V relay supply.

29.23 Relay Flyback Diodes

Each relay coil is protected by a

1N4007

flyback diode.

When the relay driver switches off, the coil inductance attempts to maintain the current. Without a suppression path, this could produce a high collector voltage.

The flyback diode provides the current path

$$\text{Relay Coil} \rightarrow 1\text{N4007} \rightarrow \text{Relay Coil},$$

thereby limiting the transient and protecting the BC337 driver and surrounding electronics.

29.24 ADC Protection Diodes

The amplifier outputs can reach tens of volts, while the STM32 ADC operates from the 3.3 V logic domain.

The DC sensing network therefore includes dedicated input scaling together with Schottky clamp diodes.

BAT85 Schottky devices are used as the external ADC clamps in the final protection circuit.

Schottky diodes are appropriate because of their relatively low forward voltage, allowing them to begin limiting an abnormal ADC input before a conventional silicon diode would conduct as strongly.

These external clamps provide secondary protection; the resistor-divider and series resistance remain the primary means of limiting voltage and current.

29.25 Temperature Sensor Selection

Each output-stage heatsink is monitored using a

B57703M0103A047

NTC temperature sensor [8].

The selected component has a nominal resistance of

10 k Ω

at

25°C

and a nominal beta value of approximately

$$B_{25/100} = 3988 \text{ K}.$$

A major mechanical advantage of this sensor is its ring-terminal construction, which allows it to be mounted directly to the heatsink.

The sensor therefore measures the heatsink temperature more reliably than an unmounted bead thermistor located elsewhere inside the chassis.

29.26 Temperature Divider Resistors

Each 10 k Ω NTC is used with a fixed 10 k Ω resistor to form a voltage divider for the STM32 ADC.

Using equal nominal values provides approximately mid-supply voltage around room temperature.

The use of a simple passive divider provides a reliable and inexpensive interface between the resistive sensor and the microcontroller.

29.27 Protection Controller Auxiliary Supply

The protection electronics use an independent isolated low-voltage supply.

The power-supply architecture is

$$230 \text{ VAC} \rightarrow \text{Isolation Transformer} \rightarrow 12 \text{ VAC} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Reservoir Capacitor} \rightarrow 7812 \rightarrow 12 \text{ VDC}$$

A separate supply was selected rather than deriving the controller voltage directly from the main $\pm 33\text{--}\pm 34 \text{ V}$ amplifier rails.

This simplifies regulation, maintains isolation, and prevents the controller load from becoming directly dependent on the high-power amplifier supply.

29.28 Protection Supply Rectifier

A compact W04G-family full-wave bridge rectifier is used in the protection controller power supply.

This component is appropriate because the auxiliary supply current is much smaller than the main amplifier current and therefore does not require a large high-current bridge such as the KBPC3506WP.

The bridge package also integrates all four rectifier diodes into a single through-hole component, simplifying PCB assembly.

29.29 Protection Supply Reservoir Capacitor

A

$$\boxed{2200 \mu\text{F}}$$

electrolytic capacitor is used as the principal reservoir capacitor in the controller supply. For 50 Hz mains operation, full-wave rectification produces a charging frequency of approximately

$$100 \text{ Hz.}$$

The approximate ripple relationship is

$$\Delta V \approx \frac{I}{fC}.$$

The selected capacitance provides substantial filtering for the relatively modest current required by the Nucleo board, two relay coils, LEDs, and sensing circuitry.

29.30 12 V Regulator

The auxiliary supply uses an

$$\boxed{\text{L7812}}$$

linear regulator.

The regulated 12 V rail supplies the loudspeaker relay coils and the Nucleo input supply.

A linear regulator was selected because:

- the required current is moderate;
- the circuit is simple;
- switching-regulator noise is avoided;
- the input voltage after rectification provides suitable headroom.

Regulator power dissipation must nevertheless be verified because

$$P_{\text{REG}} = (V_{\text{IN}} - V_{\text{OUT}})I.$$

29.31 LED Selection

Through-hole status LEDs are used for the protection-controller indications.

The PCB provides separate visual indication for fault and operating states, including temperature-related faults, DC fault, startup/wait state, and overtemperature status.

Each LED is operated with a series resistor of

$$470\ \Omega.$$

Using modest LED current reduces the load on the STM32 GPIO outputs while remaining sufficient for clear visual indication on the amplifier front panel.

29.32 Connector Selection

Different connector types are used according to signal level and current.

Low-level audio inputs use RCA connectors, while the loudspeaker connections require mechanically stronger, low-resistance output terminals.

The main criteria for high-current connectors are:

- sufficient current rating;
- low contact resistance;
- secure mechanical retention;
- compatibility with the intended wire gauge;
- suitable chassis or PCB mounting.

Sensor and low-current controller connections can use smaller terminal blocks because their current requirements are much lower.

29.33 Fuse Selection

Different fuses are used according to the part of the system being protected.

The amplifier rails use PCB-mounted 5 mm × 20 mm fuses, while the mains input requires its own appropriately rated primary protection.

Fuse selection must consider:

- normal operating current;
- startup inrush current;
- transformer characteristics;

- fault current;
- conductor and component ratings.

The fuse is therefore selected as part of the complete protection system rather than simply choosing the smallest current rating that exceeds the normal current.

29.34 Heatsink Selection Considerations

The output transistors require external heatsinks because the transistor packages alone cannot safely dissipate the heat generated during high-power operation.

Heatsink selection depends on:

- transistor dissipation;
- thermal resistance from junction to case;
- thermal interface resistance;
- heatsink-to-ambient thermal resistance;
- ambient chassis temperature;
- airflow.

The TO-3P package of the selected 2SA1943-O and 2SC5200-O devices simplifies attachment to substantial external heatsinks.

The detailed thermal calculations are treated separately in the *Heatsink Selection and Thermal Management* chapter.

29.35 Cooling Fan Selection

Cooling fans are included in the mechanical thermal-management concept.

Fan selection must consider not only airflow but also:

- supply voltage;
- current consumption;
- acoustic noise;
- mechanical vibration;
- physical dimensions;
- bearing type;
- expected lifetime.

Because this is a Hi-Fi amplifier, excessive acoustic fan noise would be undesirable even if thermal performance were improved.

29.36 Through-Hole Construction

A major selection criterion throughout the project is compatibility with through-hole construction.

The amplifier and protection PCBs contain many THT components, including:

- transistors;
- power transistors;
- resistors;
- capacitors;
- bridge rectifiers;
- fuse holders;
- relays;
- regulator;
- diodes;
- terminal blocks;
- LEDs;
- push buttons.

Through-hole construction simplifies manual assembly and is particularly appropriate for components that carry significant current or experience mechanical stress.

29.37 PCB Footprint Compatibility

Electrical compatibility alone is not sufficient when selecting a component.

The physical package must also match the PCB footprint.

The following features were therefore checked during component selection:

- pin spacing;
- lead diameter;
- pin numbering;
- transistor pin order;
- body dimensions;
- capacitor diameter and lead pitch;

- relay pin arrangement;
- terminal-block pitch;
- bridge-rectifier pin layout;
- mounting-hole geometry.

This is particularly important for transistors and relays because devices with similar electrical specifications can have different physical pinouts.

29.38 Component Availability

Availability was also considered during the design process.

A theoretically ideal component is not practical if it is obsolete, unavailable, or subject to excessive lead time.

Where alternatives were considered, preference was given to parts that could be sourced from established electronics distributors and that were available in suitable through-hole packages.

Availability should be checked again immediately before the final order because distributor stock changes with time.

29.39 Counterfeit Risk

High-power audio transistors are components for which source quality is particularly important.

The output devices should therefore be obtained from an established authorised distributor rather than from an unknown marketplace source.

Counterfeit or incorrectly specified output transistors could have substantially lower safe-operating-area, voltage, or current capability than expected and could fail under amplifier load.

29.40 Mechanical Component Selection

Several non-electronic parts also form part of the component-selection process.

These include:

- heatsinks;
- fans;
- RCA connectors;

- loudspeaker binding posts;
- mains switch;
- IEC mains connection;
- fuse holders;
- chassis spacers;
- transistor mounting hardware;
- wire terminals and spade connectors.

Their mechanical dimensions must be coordinated with the PCB layout and final chassis design.

29.41 Component Selection Summary

Table 29.2: Representative major components selected for the amplifier system.

Subsystem	Component	Selected Device / Value
Power amplifier	Small-signal transistor	BC546
Power amplifier	Driver transistors	BD139 / BD140
Power amplifier	Output transistors	2SA1943-O / 2SC5200-O
Power amplifier	Emitter resistors	0.33 Ω power resistors
Main supply	Transformer	24–0–24 V AC toroidal
Main supply	Bridge rectifier	KBPC3506WP
Main supply	Reservoir capacitance	$2 \times 4700 \mu\text{F}$ per rail
Preamplifier	Operational amplifier	NE5532N
Preamplifier	Supply	Regulated ± 15 V
Protection	MCU	STM32G031K8
Protection	Development platform	NUCLEO-G031K8
Protection	Speaker relay	G2RL-1A-E-DC12
Protection	Relay driver	BC337
Protection	Flyback diode	1N4007
Protection	ADC clamp diode	BAT85
Protection	Temperature sensor	B57703M0103A047
Protection PSU	Reservoir capacitor	2200 μF
Protection PSU	Regulator	L7812

29.42 Relationship to the Bill of Materials

This chapter describes the engineering reasoning behind the major component choices.

The following *Bill of Materials* chapter provides the detailed component inventory required for construction.

The BOM therefore contains the exact quantities and component designations, while the present chapter explains why the principal types of devices were chosen.

29.43 Design Summary

Component selection for the amplifier was performed according to electrical, thermal, mechanical, and practical requirements rather than nominal value alone.

The power-amplifier small-signal stages use BC546 transistors, while BD139 and BD140 devices provide the greater current capability required by the driver stage.

The main power-output devices are the complementary

2SA1943-O

and

2SC5200-O

transistors, selected for their suitability for high-power linear audio operation and their practical TO-3P packages.

The main symmetrical supply uses a 24–0–24 V AC toroidal transformer, a KBPC3506WP bridge rectifier, and four 4700 μ F reservoir capacitors.

The preamplifier uses NE5532N operational amplifiers operating from regulated ± 15 V rails.

The protection controller uses the STM32G031K8 microcontroller on the NUCLEO-G031K8 platform because it provides the required ADC, GPIO, timer, and firmware capabilities.

Omron G2RL-1A-E-DC12 normally-open relays provide loudspeaker isolation, while BC337 transistor stages provide sufficient relay-coil drive from the STM32 control outputs.

1N4007 diodes suppress relay-coil transients and BAT85 Schottky diodes provide additional protection at the ADC sensing inputs.

B57703M0103A047 10 k Ω ring-terminal NTC sensors were selected because they can be mechanically attached directly to the two power-transistor heatsinks.

The complete design makes extensive use of through-hole components, which is appropriate for manual prototype assembly, high-current circuitry, and future maintenance.

Component voltage, current, power, thermal, tolerance, package, footprint, availability, and mechanical characteristics were considered together so that each selected part is compatible with both the electrical design and the physical implementation of the complete Hi-Fi amplifier.

Chapter 30

Bill of Materials

30.1 Introduction

The Bill of Materials (BOM) provides the complete component inventory required to construct the electronic system. It forms the link between the schematic design, PCB layout, component procurement, assembly process, and final documentation.

The complete amplifier consists of several different subsystems and the BOM is therefore divided into:

- two discrete power-amplifier channels;
- one shared main power supply;
- the audio preamplifier;
- one STM32 speaker-protection controller;
- the protection-controller auxiliary power supply;
- external electromechanical and mechanical components.

The quantities shown in this chapter represent the components required by the electrical design rather than spare procurement quantities. Additional parts may be ordered for prototype development, component matching, replacement, or future servicing.

30.2 BOM Organisation

The BOM is organised according to circuit function rather than manufacturer or supplier. For each item, the important information includes:

- reference designator;
- component type;

- nominal value or device;
- quantity;
- electrical rating where important;
- package or construction type.

Manufacturer part numbers and distributor order codes can change as components are replaced or stock availability changes. These procurement-specific details are therefore maintained separately in the detailed manufacturing BOM.

30.3 Power Amplifier Channel BOM

The stereo system contains two identical power-amplifier channels. Therefore, the quantity required for the complete stereo amplifier is twice the quantity required for one amplifier channel.

30.3.1 Power Amplifier Resistors

Table 30.1: Resistor BOM for the two power-amplifier channels.

References per Channel	Value	Qty./Channel	Stereo Qty.
R1, R3, R4, R16	1 k Ω	4	8
R2, R5, R8	22 k Ω	3	6
R6, R7	560 Ω	2	4
R9, R10, R11	3.3 k Ω	3	6
R12	220 Ω	1	2
R13, R14	0.33 Ω	2	4
R15	10 Ω	1	2

The low-power signal resistors are primarily 1% components. R13 and R14 are power emitter resistors and therefore require substantially greater power handling than the ordinary small-signal resistors.

R15 forms part of the output Zobel network and is also selected as a higher-power component.

30.3.2 Power Amplifier Capacitors

Table 30.2: Capacitor BOM for the two power-amplifier channels.

References per Channel	Value	Qty./Channel	Stereo Qty.
C1	4.7 μ F	1	2

References per Channel	Value	Qty./Channel	Stereo Qty.
C2, C4, C6	100 pF	3	6
C3, C5, C10, C11	100 μ F	4	8
C7, C8, C18	100 nF	3	6

The small 100 pF capacitors are used in frequency-compensation positions, while the 100 nF capacitors provide high-frequency decoupling and output-network functions.

The electrolytic capacitors provide coupling, bootstrap, and local power-supply energy storage.

30.3.3 Power Amplifier Semiconductors

Table 30.3: Semiconductor BOM for the two power-amplifier channels.

References per Channel	Device	Qty./Channel	Stereo Qty.
Q1, Q2, Q3, Q9	BC546	4	8
Q4, Q6	BD140	2	4
Q5	BD139	1	2
Q7	2SA1943-O	1	2
Q8	2SC5200-O	1	2
D1	5 mm LED	1	2

The current design uses the complementary 2SA1943-O and 2SC5200-O devices as the main output transistors.

The older alternative NJW0302G/NJW0281G devices considered during development are not part of the current amplifier PCB configuration.

30.3.4 Bias and Fuse Components

Table 30.4: Bias-adjustment and rail-protection components.

Reference	Component	Qty./Channel	Stereo Qty.
VR1	2 k Ω trimmer	1	2
F1	3 A rail fuse	1	2
F2	3 A rail fuse	1	2

Each amplifier channel therefore includes two independent 3 A rail fuses and one bias-adjustment trimmer.

30.4 Main Power Supply BOM

The two amplifier channels are powered from the common symmetrical main power supply.

Table 30.5: Main stereo power-supply BOM.

Reference	Component	Specification	Qty.
T _{MAIN}	Toroidal transformer	24–0–24 V AC, approximately 300 VA	1
BR1	Bridge rectifier	KBPC3506WP, 35 A, 600 V	1
C14–C17	Reservoir capacitors	4700 μ F each	4
C12, C13	HF bypass capacitors	100 nF	2

The four 4700 μ F capacitors provide a nominal capacitance of

$$2 \times 4700 \mu\text{F} = \boxed{9400 \mu\text{F}}$$

per supply rail.

The main transformer produces

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}},$$

which results in amplifier rails in the approximate

$$\boxed{\pm 33\text{--}\pm 34 \text{ VDC}}$$

region after rectification and filtering under practical operating conditions.

30.5 Protection Controller BOM

The speaker-protection controller is implemented as one independent PCB for the complete stereo amplifier.

30.5.1 Protection Controller Resistors

Table 30.6: Protection-controller resistor BOM.

References	Value	Function	Qty.
R1, R5	150 k Ω	DC sensing input	2
R2, R3, R6, R7, R9–R14	10 k Ω	DC sensing, temperature and relay circuits	10
R4, R8, R15, R16	4.7 k Ω	ADC and control networks	4
R17–R21	470 Ω	LED current limiting	5

30.5.2 Protection Controller Capacitors

Table 30.7: Protection-controller capacitor BOM.

References	Value	Function	Qty.
C1	2200 μF	Auxiliary PSU reservoir	1
C4	100 μF	Regulator output filtering	1
C5, C7	4.7 μF	DC sensing low-pass filters	2
C2, C3, C6, C8	100 nF	HF filtering / bypass	4

30.5.3 Protection Controller Diodes and Transistors

Table 30.8: Protection-controller semiconductor BOM.

References	Device	Function	Qty.
Q1, Q2	BC337	Relay-coil drivers	2
D2–D5	BAT85	ADC input clamps	4
D6, D7	1N4007	Relay flyback suppression	2

30.5.4 Microcontroller and Relay Components

Table 30.9: Major protection-controller components.

Reference	Component	Specification	Qty.
U2	MCU development board	NUCLEO-G031K8	1
U1	Linear regulator	LM7812	1
K1, K2	Speaker relays	G2RL-1A-E-DC12	2
BR1	Auxiliary bridge rectifier	1.5 A class	1
TH1, TH2	Temperature sensors	10 k Ω NTC	2
SW1	Push button	MUTE / ENABLE	1
SW2	Push button	FAULT RESET	1

30.5.5 Status Indicator LEDs

The final protection PCB contains five individual status indicators.

Table 30.10: Protection-controller status LEDs.

Reference	Indication	Qty.
LED1	TEMP SENSOR 2 FAIL	1
LED2	TEMP SENSOR 1 FAIL	1
LED3	DC FAULT	1
LED4	WAIT / DELAY	1
LED5	OVERHEAT	1

Each indicator uses a $470\ \Omega$ series resistor.

30.5.6 Protection Controller Power Transformer

The auxiliary protection supply uses a small PCB-mounted isolation transformer to generate the low-voltage AC supply required by the relay and controller electronics.

The final transformer part number must be identical in the schematic, PCB, BOM, and procurement documentation before fabrication.

The required electrical function is

$$230\ \text{V}_{\text{AC}} \rightarrow 12\ \text{V}_{\text{AC}} \rightarrow \text{Rectifier} \rightarrow 7812\ \text{regulator} \rightarrow 12\ \text{VDC}.$$

The PCB transformer selection should therefore be verified as part of the final BOM revision before component ordering.

30.6 Preamplifier BOM

The preamplifier is maintained as a separate schematic subsystem and therefore has its own detailed BOM.

The principal active components include:

Table 30.11: Principal preamplifier components.

Component Type	Selected Device	Function
Dual operational amplifier	NE5532N	Audio amplification and filtering
Positive regulator	7815	+15 V supply regulation
Negative regulator	7915	−15 V supply regulation
Input connectors	RCA	Phono and line inputs
Potentiometers	Audio taper / balance	Volume and balance control

The complete passive-component inventory for the preamplifier is generated directly from its final KiCad schematic so that the BOM remains synchronised with the implemented circuit.

30.7 System-Level Electromechanical Components

The complete amplifier also requires several components that do not belong to a single PCB.

Table 30.12: Principal system-level electromechanical components.

Component	Application	Quantity
Large power-transistor heatsink	One per amplifier channel	2
AC cooling fan	Forced-air cooling	4
RCA connectors	Audio input connections	As required
Gold-plated binding posts	Loudspeaker connections	2 channel sets
Mains inlet	230 V AC input	1
Mains power switch	Main amplifier control	1
Primary fuse holder	Mains protection	1
PCB standoffs and screws	Mechanical assembly	As required
Heatsink mounting hardware	Output transistor mounting	As required
Internal high-current wiring	Supply and speaker wiring	As required
Shielded audio cable	Low-level signal wiring	As required

30.8 PCB Manufacturing Specification

The PCB itself also forms part of the BOM because its construction parameters affect the current-carrying capability and mechanical compatibility of the system.

Table 30.13: Principal PCB manufacturing requirements.

Parameter	Specification
PCB type	Double-sided FR-4
Nominal thickness	1.6 mm
Copper weight	2 oz
Approximate copper thickness	70 μm
Power-amplifier board	136 $\times$ 124.5 mm
Protection-controller board	121 $\times$ 182.25 mm
Construction	Predominantly through-hole

30.9 Design BOM and Procurement BOM

Two different forms of BOM are useful during the project.

The *design BOM* describes the components actually required by the electrical design.

The *procurement BOM* additionally contains:

- manufacturer part number;
- distributor part number;
- supplier;
- stock status;

- quantity to order;
- spare quantity;
- package information.

These quantities do not necessarily match because additional spare components may deliberately be purchased.

For example, inexpensive resistors, small-signal transistors, diodes, LEDs, and fuses can be ordered in quantities greater than the exact design requirement.

30.10 BOM Revision Control

The BOM must remain synchronised with the schematic and PCB revision.

This is particularly important when a component is replaced during development.

A valid component substitution may require changes to:

- component value;
- manufacturer part number;
- package;
- PCB footprint;
- pin assignment;
- 3D model;
- procurement information.

For this reason, the final KiCad schematic and PCB files are treated as the primary design reference when producing the manufacturing BOM.

Older ordering spreadsheets should not be used without first confirming that their component selections still correspond to the current PCB revision.

30.11 BOM Verification Procedure

Before component ordering, the final BOM should be checked using the following procedure:

1. Generate the latest BOM directly from the KiCad project.
2. Confirm every reference designator against the schematic.
3. Verify component values.

4. Verify voltage and current ratings.
5. Verify resistor power ratings.
6. Verify capacitor voltage ratings and polarity.
7. Verify transistor pinouts.
8. Verify relay coil voltage and footprint.
9. Verify bridge-rectifier pin arrangement.
10. Verify transformer voltage and VA rating.
11. Verify PCB footprint dimensions against manufacturer datasheets.
12. Verify distributor stock before ordering.
13. Confirm that obsolete component alternatives have been removed.
14. Compare the assembled PCB quantities with the procurement quantity.

30.12 Spare Components

For prototype construction it is useful to order additional inexpensive components beyond the exact BOM quantity.

This is particularly appropriate for:

- resistors;
- small capacitors;
- small-signal transistors;
- diodes;
- LEDs;
- fuses;
- connectors.

Power transistors may also be purchased as additional matched or replacement devices where practical.

The spare quantity is a procurement decision and is therefore not included in the electrical design quantities listed in the previous tables.

30.13 Relationship to the Appendix BOM

The tables in this chapter provide an engineering-level summary of the parts used in the complete system.

The full manufacturing BOM is included separately in the appendix and repository.

That version contains the detailed manufacturer and distributor information required for component ordering and PCB assembly.

Separating the engineering BOM from the procurement BOM prevents the main report from becoming dependent on temporary supplier stock conditions or distributor catalogue numbers.

30.14 Design Summary

The complete Bill of Materials includes components for two discrete power-amplifier channels, the shared symmetrical main power supply, the audio preamplifier, the STM32 speaker-protection controller, and the mechanical and external wiring system.

Each amplifier channel uses BC546 small-signal transistors, BD139/BD140 drivers, and the complementary

2SA1943-O

and

2SC5200-O

power-output devices.

The shared main supply contains a 24–0–24 V AC toroidal transformer, KBPC3506WP bridge rectifier, and four 4700 μ F reservoir capacitors.

The protection controller uses a NUCLEO-G031K8, two BC337 relay drivers, two G2RL-1A-E-DC12 speaker relays, two 10 k Ω temperature sensors, four BAT85 ADC clamp diodes, two 1N4007 flyback diodes, and five status LEDs.

The system also requires two large output-stage heatsinks, four cooling fans, audio connectors, speaker terminals, mains switching and protection hardware, internal wiring, mounting hardware, and a suitable metal chassis.

All project PCBs are specified using

2 oz copper

and a nominal thickness of

1.6 mm.

The BOM is maintained under revision control because component substitutions can affect both the electrical design and PCB footprint.

The final procurement BOM must therefore be generated from and checked against the latest KiCad schematic and PCB revision before any manufacturing order is placed.

Chapter 31

KiCad Symbols and Footprints

31.1 Introduction

KiCad represents an electronic component using two closely related but different objects: the *schematic symbol* and the *PCB footprint*. Correct selection of both is essential because the schematic defines the electrical operation of the circuit, while the footprint defines how the physical component is mounted and connected on the printed circuit board.

A component can therefore be electrically correct in the schematic but still produce an unusable PCB if its footprint has an incorrect pin arrangement, lead spacing, package size, or mounting geometry.

For the present amplifier project, symbol and footprint verification was particularly important because the design contains a wide range of component types, including:

- small-signal transistors;
- medium-power driver transistors;
- TO-3P output transistors;
- large electrolytic capacitors;
- bridge rectifiers;
- PCB relays;
- the NUCLEO-G031K8 development board;
- PCB-mounted transformers;
- fuse holders;
- terminal blocks;
- LEDs;
- push buttons;
- mounting holes.

The symbol-to-footprint relationship was therefore treated as part of the engineering verification process rather than as a purely graphical step.

31.2 Difference Between a Symbol and a Footprint

A KiCad schematic symbol represents the electrical behaviour and logical connections of a component.

For example, an NPN transistor symbol identifies three electrical terminals:

Base	Collector	Emitter
------	-----------	---------

The symbol does not by itself define the physical position of these terminals on the real component.

The PCB footprint defines:

- pad positions;
- pad numbers;
- pad dimensions;
- drill diameters;
- component outline;
- courtyard;
- fabrication information;
- silkscreen;
- optional 3D model.

The relationship can therefore be represented as

Schematic Symbol → Footprint Assignment → Physical Component
--

31.3 Pin Number Consistency

The most important requirement when assigning a footprint is that the symbol pin numbers correspond exactly to the footprint pad numbers.

For example, if a transistor symbol uses

$$1 = B, \quad 2 = C, \quad 3 = E,$$

then footprint pads 1, 2, and 3 must connect to the same physical transistor terminals.

If the physical component instead has a different pin sequence, an electrically correct schematic can become an incorrectly wired PCB.

This type of error is particularly dangerous in the power-amplifier output stage because an incorrect base, collector, or emitter connection can cause immediate device failure during initial power-up.

31.4 Symbol Selection

The first stage of component placement in KiCad is selecting an appropriate schematic symbol.

Where suitable symbols already existed in the standard KiCad libraries, they were used directly.

Typical examples include:

- resistors;
- capacitors;
- diodes;
- bipolar transistors;
- LEDs;
- bridge rectifiers;
- voltage regulators;
- connectors;
- switches.

The graphical appearance of the symbol is less important than its electrical pin numbering and meaning.

A symbol should therefore always be checked against the manufacturer datasheet before it is accepted for the final design.

31.5 Footprint Selection

After the schematic has been completed, a physical footprint is assigned to every component that will be mounted on a PCB.

The selected footprint must match:

- package family;
- number of pins;

- pin spacing;
- pin numbering;
- body width;
- body length;
- lead diameter;
- mounting orientation;
- required mechanical clearance.

For large components, datasheet mechanical drawings are particularly important because components with similar electrical ratings may have substantially different physical dimensions.

31.6 Through-Hole Construction

The amplifier project uses predominantly through-hole components.

This influenced the footprint-selection process because the PCB requires physical drill holes rather than only surface-mount copper pads.

Through-hole construction was appropriate for this project because it offers:

- straightforward manual assembly;
- strong mechanical retention;
- easy component replacement;
- convenient probing during development;
- good compatibility with high-power components;
- practical mounting of large capacitors and relays.

The majority of the footprints therefore contain plated through-hole pads.

31.7 Resistor Footprints

Standard axial through-hole footprints are used for most resistors.

Different physical sizes are required according to resistor power rating.

A normal low-power resistor requires only a relatively small body and lead spacing, while components such as the

$$R_{13} = R_{14} = 0.33 \, \Omega$$

output-stage emitter resistors require substantially larger footprints.

The footprint must therefore be selected according to the actual resistor body rather than resistance value alone.

31.8 Capacitor Footprints

Several different capacitor packages are used in the project.

These include:

- small ceramic capacitors;
- film capacitors;
- radial electrolytic capacitors;
- large snap-in reservoir capacitors.

For electrolytic capacitors, both the body diameter and lead spacing must be verified.

Polarity marking is also important.

The positive pad and negative pad on the PCB must agree with the capacitor symbol and with the actual device orientation.

31.9 Reservoir Capacitor Footprints

The main power-amplifier PCB contains four large

4700 μ F

reservoir capacitors.

These components occupy a substantial portion of the lower power-supply region of the PCB.

Their footprints must accommodate:

- the capacitor body diameter;
- snap-in lead spacing;
- lead diameter;
- polarity;
- mechanical clearance between adjacent capacitors.

The physical body diameter is particularly important because an incorrect footprint could prevent the four capacitors from being installed side by side on the completed PCB.

31.10 Small-Signal Transistor Footprints

The amplifier uses BC546 small-signal transistors in several locations.

Although the electrical transistor symbol is simple, the physical pin order must always be checked.

A transistor footprint that fits mechanically but uses a different pad order is not interchangeable without changing the symbol pin mapping or PCB connections.

The final footprint must therefore correspond to the exact manufacturer package used for the purchased device.

31.11 BD139 and BD140 Footprints

The BD139 and BD140 devices used in the driver and voltage-amplification stages have larger packages than the BC546 devices.

Their footprint must match the actual lead arrangement and body dimensions.

Because the flat face of the component also establishes the transistor's physical orientation, the silkscreen outline should clearly show the intended installation direction.

Incorrect rotation or use of a footprint with a different pin sequence could connect the transistor incorrectly even if the PCB visually appears correct.

31.12 Output Transistor Footprints

The principal output transistors are

2SA1943-O

and

2SC5200-O

.

The final amplifier PCB uses vertical through-hole TO-3P style footprints for these devices.

The footprints are positioned directly along the right-hand edge of the PCB so that the transistor bodies can face outward toward the external heatsink.

This footprint placement serves both electrical and mechanical purposes.

Electrically, it keeps the driver-to-output and emitter-resistor connections short.

Mechanically, it permits

PCB → Power Transistor → Thermal Interface → Heatsink

without requiring long flying wires.

31.13 Output Transistor Pin Verification

The power-transistor footprint was treated as a critical verification item.

Before final manufacture, the following must agree:

- schematic symbol pin numbers;
- PCB footprint pad numbers;
- manufacturer datasheet pin assignment;
- physical transistor orientation;
- heatsink mounting direction.

A discrepancy in any one of these items can produce an incorrectly connected output stage.

31.14 Bridge Rectifier Footprint

The main power supply uses the

KBPC3506WP

bridge rectifier.

Unlike a small four-diode bridge, this is a physically large, high-current component.

Its footprint must match the physical arrangement of the four terminals and the body dimensions.

Particular attention is required because the four pins represent

+

−

and two

~

AC terminals.

A footprint with the correct physical spacing but an incorrect pad assignment would reverse or incorrectly route the power-supply connections.

31.15 Fuse Holder Footprints

The power-amplifier PCB contains two

$$5 \times 20 \text{ mm}$$

fuse holders.

The footprint must provide appropriate spacing for the fuse-holder terminals and sufficient surrounding clearance for insertion and removal of the fuse.

Mechanical accessibility is particularly important because the fuses are serviceable components.

Their footprints are therefore positioned close to the PCB edge.

31.16 Speaker Relay Footprints

The protection controller uses two

G2RL-1A-E-DC12

speaker relays.

Relay footprints require particularly careful verification because devices in the same relay family can have different contact arrangements.

The footprint must correctly represent:

- coil terminals;
- common contact;
- normally-open contact;
- any unused contact positions;
- physical pin spacing.

The G2RL-1A relay must therefore not be assumed to have an identical footprint to every other G2RL-family variant.

The manufacturer's mechanical drawing is the final reference for pad positions and contact numbering.

31.17 Relay Driver Footprints

The two relay drivers use BC337 NPN transistors.

The footprint must correspond to the selected TO-92 package and manufacturer pin arrangement.

The same principle applies here as for the amplifier transistors: package shape alone is not sufficient to guarantee correct pin mapping.

31.18 Diode Footprints

The protection controller uses several through-hole diodes, including:

- 1N4007 relay flyback diodes;
- BAT85 ADC clamp diodes.

Axial diode footprints are selected according to body size and lead spacing.

Diode orientation must also be visible clearly on the silkscreen because an incorrectly installed diode can prevent the protection circuit from operating correctly.

The PCB marking should make the cathode orientation clear.

31.19 LED Footprints

The protection-controller PCB uses five through-hole status LEDs.

The final board includes indications for:

- TEMP SENSOR 1 FAIL;
- TEMP SENSOR 2 FAIL;
- DC FAULT;
- WAIT / DELAY;
- OVERHEAT.

The LEDs use 5 mm through-hole packages.

The footprint must correctly identify

Anode

and

Cathode

and should contain a clear polarity indication on the silkscreen.

31.20 Push-Button Footprints

The protection board contains two PCB-mounted push buttons used for

MUTE / ENABLE

and

FAULT RESET.

Tactile push buttons often contain four physical pins even though they act electrically as a single-pole switch.

Two pins on one side of the component may already be internally connected, as may the two pins on the opposite side.

This internal connection must be understood before interpreting the PCB ratsnest or assigning the footprint.

The footprint therefore has to match both the mechanical pin pattern and the internal switch connectivity.

31.21 NUCLEO-G031K8 Footprint

The protection controller is based on the

NUCLEO-G031K8

development board.

Unlike a conventional integrated circuit, the Nucleo board is a complete development module containing the STM32G031K8 microcontroller, ST-LINK interface, regulators, headers, and supporting circuitry.

The PCB footprint must therefore represent the external Nucleo header connections and the mechanical dimensions of the entire module.

Important considerations include:

- header spacing;
- header pin numbering;
- module outline;
- USB/ST-LINK accessibility;
- mechanical clearance around the board.

The footprint was positioned so that the development board remains accessible for programming and debugging after installation.

31.22 Transformer Footprint

The protection-controller PCB contains a PCB-mounted isolation transformer.

Its footprint is especially important because the component is both electrically and mechanically large.

The footprint must match:

- primary pin locations;
- secondary pin locations;
- body dimensions;
- pin diameter;
- primary-to-secondary spacing;
- component orientation.

Because this transformer operates from mains voltage, the footprint also affects the available creepage and clearance between the primary and isolated low-voltage circuitry.

The exact final transformer model must therefore remain consistent between the schematic, footprint, PCB, BOM, and purchased part.

31.23 Voltage Regulator Footprint

The protection power supply uses an L7812 linear regulator.

The selected through-hole footprint must match the device package and the manufacturer pin order.

For a three-terminal regulator, the important connections are

IN, GND, OUT.

A footprint selected only because it physically resembles the regulator could still be incorrect if its pad numbering does not correspond to the symbol.

31.24 Terminal Block Footprints

Terminal blocks are used where external wires connect directly to the PCBs.

Examples include:

- temperature sensors;
- amplifier outputs;

- speaker outputs;
- auxiliary power connections.

The most important mechanical parameter is the terminal pitch.

The footprint must also provide sufficient space for the body of the terminal block and for screwdriver access after the PCB has been installed in the chassis.

31.25 Mounting-Hole Footprints

Mechanical mounting holes are represented in KiCad using dedicated mounting hole footprints.

The power-amplifier PCB contains four

5 mm

mounting holes.

The protection-controller PCB contains four

3.2 mm

mounting holes intended for M3 hardware.

The mounting-hole footprint determines not only the drill diameter but also the mechanical keepout and clearance around the fixing hardware.

Where a mounting hole is intended to remain electrically isolated from the chassis, it must not accidentally create a ground connection through exposed copper or plated hardware.

31.26 Footprint Courtyards

A footprint courtyard defines the approximate mechanical area required by a component and its assembly process.

Courtyards are useful for detecting components that are placed too close together.

This is especially important for:

- reservoir capacitors;
- fuse holders;
- relays;
- large bridge rectifiers;
- transformers;

- the Nucleo module.

Correct courtyard dimensions improve the reliability of component-placement checks and reduce the risk of mechanical interference during assembly.

31.27 Silkscreen Information

The footprint also defines the graphical information printed on the PCB silkscreen.

Useful silkscreen information includes:

- component outline;
- reference designator;
- transistor orientation;
- capacitor polarity;
- diode cathode marking;
- connector pin identification;
- relay orientation.

Clear silkscreen markings reduce assembly errors and make later servicing easier.

31.28 3D Models

KiCad footprints can optionally reference 3D component models.

A 3D model is not required for the electrical operation of the PCB and is not required to generate valid Gerber files.

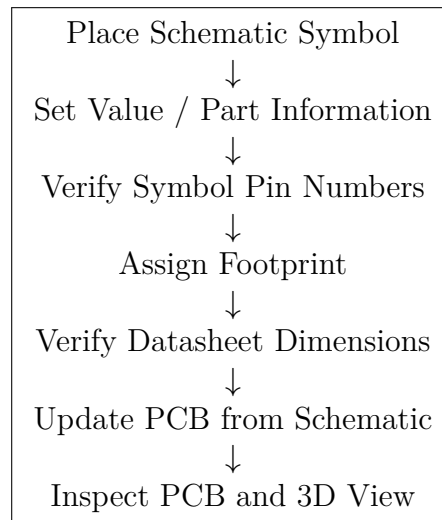
However, it is useful for checking:

- component orientation;
- approximate component height;
- mechanical interference;
- connector accessibility;
- heatsink positioning;
- chassis clearances;
- overall board presentation.

Therefore, 3D models were treated as useful mechanical documentation rather than as a mandatory electrical-design element.

31.29 Footprint Assignment Workflow

A typical component workflow in KiCad is



This sequence helps ensure that the physical PCB remains synchronized with the schematic.

31.30 Assign Footprints Tool

KiCad's *Assign Footprints* tool provides the connection between schematic symbols and physical footprints.

The workflow allows the designer to inspect each schematic component and choose an appropriate footprint from the available libraries.

Footprint filters can reduce the list of available packages, but the final choice still requires manual verification against the physical component.

Automatic filtering should therefore not replace datasheet checking.

31.31 Updating the PCB from the Schematic

After footprint assignment or a schematic modification, the PCB must be updated from the schematic.

This transfers:

- new components;
- removed components;
- changed net connections;
- updated footprint assignments;

- reference designators.

The schematic should remain the principal source of electrical connectivity.

Changes should therefore not be made independently in the PCB when the same change should also exist in the schematic.

31.32 Footprint Replacement

During the development process, a component may be replaced by an alternative device.

When this occurs, electrical similarity alone is not sufficient.

The replacement must be checked for:

- package compatibility;
- pinout compatibility;
- lead pitch;
- body dimensions;
- mounting orientation;
- electrical rating.

If the physical package changes, the footprint must be updated before the PCB can be considered final.

31.33 Importance of Datasheet Verification

The manufacturer datasheet is the primary reference for footprint verification.

The relevant mechanical drawing normally provides:

- maximum body dimensions;
- minimum body dimensions;
- pin spacing;
- lead width;
- lead thickness;
- package orientation;
- pin identification;
- mounting-hole position where applicable.

These values should be compared directly with the KiCad footprint rather than assuming that a similarly named library footprint is correct.

31.34 Common Footprint Errors

Several common footprint errors can occur during PCB development.

Table 31.1: Typical footprint errors and their consequences.

Error	Possible Consequence
Incorrect transistor pin order	Device connected incorrectly or destroyed
Incorrect capacitor polarity	Electrolytic capacitor failure
Wrong capacitor diameter	Component does not physically fit
Wrong relay pin arrangement	Coil or speaker contacts incorrectly connected
Incorrect terminal pitch	Connector cannot be installed
Incorrect drill diameter	Component lead does not fit through PCB
Incorrect bridge pin numbering	Power supply incorrectly rectified
Incorrect mounting-hole size	PCB cannot be installed mechanically
Incorrect silkscreen orientation	Increased assembly-error risk

31.35 Electrical Rule Check and Footprints

KiCad's Electrical Rules Checker verifies many schematic connectivity problems, but it cannot guarantee that the physical footprint corresponds to the purchased component.

For example, ERC may report no errors even if a transistor footprint has physically incorrect pad ordering, provided the schematic and footprint pad numbers are internally consistent.

Physical component verification is therefore still required.

31.36 Design Rule Check

The PCB Design Rule Checker verifies physical layout properties such as:

- copper clearance;
- track-to-pad clearance;
- unconnected nets;
- overlapping courtyards;
- board-edge clearance;
- via rules;

- zone connectivity.

DRC complements footprint verification but does not replace comparison with the manufacturer mechanical drawing.

31.37 Critical Components Requiring Manual Verification

The following components in the amplifier project require particularly careful manual footprint verification:

- 2SA1943-O;
- 2SC5200-O;
- BD139;
- BD140;
- BC546;
- BC337;
- KBPC3506WP;
- large 4700 μ F reservoir capacitors;
- G2RL-1A-E-DC12 speaker relays;
- PCB-mounted protection transformer;
- NUCLEO-G031K8;
- fuse holders;
- external terminal blocks.

These parts are critical either because an incorrect footprint could cause electrical failure or because their mechanical dimensions strongly influence the PCB layout.

31.38 Pre-Manufacturing Footprint Verification

Before Gerber generation, the following verification procedure should be completed:

1. Confirm that every schematic component requiring PCB assembly has a footprint.
2. Check all transistor pin assignments against the selected manufacturer datasheets.
3. Confirm output-transistor package and orientation.

4. Check bridge-rectifier terminal numbering.
5. Confirm electrolytic capacitor polarity.
6. Verify reservoir-capacitor body diameter and lead spacing.
7. Verify relay pin arrangement and coil/contact numbering.
8. Verify transformer primary and secondary pin positions.
9. Confirm regulator pin numbering.
10. Confirm Nucleo header positions and orientation.
11. Verify terminal-block pitch.
12. Verify fuse-holder spacing.
13. Check LED and diode polarity markings.
14. Confirm all mounting-hole diameters.
15. Run ERC and DRC.
16. Inspect the PCB using KiCad's 3D Viewer.

31.39 Footprint Verification Using Printed PCB View

A useful additional verification technique is to print the PCB at

1 : 1

scale.

Physical components can then be placed over the printed footprint positions to confirm approximate lead spacing and body size before the PCB is manufactured.

This method is especially useful for:

- large capacitors;
- relays;
- fuse holders;
- terminal blocks;
- bridge rectifiers;
- transformers.

A paper check cannot verify the electrical pinout, but it provides an effective additional mechanical check.

31.40 Symbols and Footprints in the Repository

The project repository separates design files from reusable library data.

The library structure includes dedicated directories for

Symbols

and

Footprints.

This organisation allows any non-standard resources used by the project to be stored together with the design rather than relying entirely on the local KiCad installation.

The detailed creation, organisation, and GitHub storage of these resources is described in the following *Custom KiCad Libraries* chapter.

31.41 Reproducibility

Correct management of symbols and footprints is important not only during design but also for long-term reproducibility.

If a PCB project references a footprint that exists only on the original designer's computer, another user may be unable to open or modify the project correctly.

Keeping project-specific symbols and footprints with the repository therefore helps ensure that the design can be reconstructed on another computer in the future.

31.42 Design Summary

KiCad uses schematic symbols to represent the logical electrical function of a component and footprints to represent its physical implementation on the PCB.

The connection between the two is established through matching pin and pad numbers.

Correct footprint assignment is particularly important in this project because the design contains a mixture of small-signal devices, large power transistors, relays, transformers, reservoir capacitors, connectors, and mechanical components.

The predominantly through-hole construction improves manual assembly and serviceability but requires correct drill sizes, pad geometry, lead spacing, and component orientation.

The 2SA1943-O and 2SC5200-O output devices use vertical TO-3P footprints positioned along the amplifier PCB edge to permit direct mounting to external heatsinks.

The KBPC3506WP bridge rectifier, four large reservoir capacitors, G2RL speaker relays, NUCLEO-G031K8 module, PCB transformer, fuse holders, terminal blocks, and mounting holes similarly require careful mechanical verification.

Transistor and relay footprints receive particular attention because incorrect pad numbering can produce a PCB that appears correct visually while being electrically unusable.

Datasheet mechanical drawings are therefore used as the primary reference when checking package dimensions and pin assignments.

ERC and DRC provide important automated verification but cannot confirm that a selected footprint physically matches the purchased part.

Final verification therefore combines:

Datasheet Check + Symbol Check + Footprint Check + DRC + 3D Inspection
--

before manufacturing files are generated.

This process reduces the risk of incorrect pin assignments, mechanical interference, assembly errors, and component incompatibility, and helps ensure that the final KiCad PCB corresponds directly to the components selected in the Bill of Materials.

Chapter 32

Custom KiCad Libraries

32.1 Introduction

Although KiCad provides extensive standard libraries for commonly used electronic components, some parts used in the amplifier project require project-specific symbols, footprints, or mechanical representations.

Custom KiCad libraries were therefore incorporated into the project so that components not adequately represented by the standard libraries could be defined accurately and stored together with the rest of the design.

The use of project-specific libraries has several advantages:

- exact compatibility with the selected physical components;
- control over pin numbering and pad geometry;
- independence from changes to global KiCad libraries;
- easier transfer of the project between computers;
- improved long-term reproducibility;
- easier version control using GitHub.

The custom library files are therefore treated as part of the engineering design rather than as temporary local KiCad configuration.

32.2 Library Structure Used in the Project

The GitHub repository contains a dedicated library directory organised into separate categories for schematic and PCB resources.

The general structure is


```
Libraries/
|-- Symbols/
|-- Footprints/
'-- 3D_Models/
```

Each category has a different purpose.

Table 32.1: Organisation of the project-specific KiCad libraries.

Directory	Purpose
Symbols	Custom schematic symbol libraries
Footprints	Custom PCB footprint libraries
3D_Models	Optional mechanical 3D component models

This separation keeps the reusable library resources independent from the schematic, PCB, Gerber, and documentation files.

32.3 Custom Symbol Libraries

Custom schematic symbols are stored using KiCad symbol-library files with the extension

.kicad_sym

A custom symbol defines the logical representation of a component in the schematic.

Typical symbol information includes:

- component graphical representation;
- pin numbers;
- pin names;
- electrical pin types;
- default component value;
- manufacturer information where required;
- optional footprint association.

The most important requirement is that the symbol pin numbers correspond correctly to the physical component.

32.4 When a Custom Symbol Is Required

A custom symbol is useful when:

- the required component does not exist in the standard KiCad libraries;
- the standard symbol uses unsuitable pin numbering;
- a complete module must be represented rather than a single integrated circuit;
- additional pin labels improve schematic readability;
- a specific manufacturer component must be represented exactly.

A custom symbol should not be created unnecessarily when an existing KiCad symbol already represents the component correctly.

32.5 Creating a Custom Symbol

Custom symbols are created using the KiCad *Symbol Editor*.

The general procedure is:

1. Create or open a project-specific symbol library.
2. Create a new symbol.
3. Define the component name and reference prefix.
4. Draw the graphical representation.
5. Add all required electrical pins.
6. Assign the correct pin numbers and names.
7. Define the electrical pin types.
8. Compare the result against the manufacturer datasheet.
9. Save the symbol to the project library.

The manufacturer datasheet remains the primary reference for pin numbering.

32.6 Pin Numbering

The pin numbering of a custom symbol must match the pad numbering of its assigned PCB footprint.

The required relationship is

$$\boxed{\text{Symbol Pin Number} = \text{Footprint Pad Number}}.$$

For example, if a component symbol defines

Pin 1

as a particular electrical terminal, footprint pad 1 must correspond to that same physical terminal.

An error at this stage can produce a PCB that passes normal connectivity checking while connecting the real component incorrectly.

32.7 Electrical Pin Types

KiCad also assigns an electrical type to each schematic pin.

Examples include:

- input;
- output;
- bidirectional;
- passive;
- power input;
- power output;
- open collector;
- no connection.

Correct pin types improve the effectiveness of the Electrical Rules Checker (ERC).

Incorrect pin-type selection can create unnecessary ERC warnings or, more importantly, prevent a genuine electrical connection error from being identified.

32.8 Custom Footprint Libraries

KiCad custom footprints are normally grouped inside directories ending in

`.pretty`.

For example, a custom footprint library may have a structure similar to

```
Footprints/  
'-- Power_Amplifier.pretty/  
    |-- Custom_Component_1.kicad_mod  
    |-- Custom_Component_2.kicad_mod  
    '-- Custom_Component_3.kicad_mod
```

The `.pretty` item is therefore a directory rather than an individual footprint file.

Each footprint stored inside the directory normally uses the extension

`.kicad_mod`.

32.9 Information Contained in a Footprint

A footprint defines the physical PCB implementation of the component.

The main information includes:

- pad locations;
- pad numbers;
- pad dimensions;
- plated or non-plated drill dimensions;
- component body outline;
- front silkscreen;
- fabrication layer information;
- courtyard;
- component reference and value text;
- optional 3D-model association.

For through-hole components, correct hole diameter and pad size are especially important.

32.10 Creating a Custom Footprint

Custom footprints are created using the KiCad *Footprint Editor*.

A typical procedure is:

1. Create a project-specific `.pretty` footprint library.
2. Create a new footprint.
3. Define the footprint origin.
4. Place the required pads.
5. Assign pad numbers.
6. Set pad and drill dimensions.
7. Draw the component body on the fabrication layer.
8. Add the required silkscreen outline.
9. Define the courtyard.
10. Compare dimensions against the manufacturer mechanical drawing.
11. Add a 3D model if one is available.
12. Save the footprint in the project library.

32.11 Datasheet-Based Footprint Design

Custom footprints are created from the component manufacturer's mechanical drawing rather than by visually estimating dimensions.

The relevant datasheet parameters include:

- pin-to-pin spacing;
- row spacing;
- lead diameter;
- lead width and thickness;
- component body dimensions;
- mounting-hole locations;
- maximum package dimensions;
- pin numbering.

Where dimensional tolerances are provided, the footprint should allow sufficient mechanical margin for normal manufacturing variation.

32.12 Pad Diameter and Drill Size

For a through-hole component, the finished PCB hole must be larger than the maximum physical lead diameter.

Conceptually,

$$D_{\text{hole}} > D_{\text{lead}}.$$

A manufacturing allowance is required so that component insertion remains practical after the holes have been plated.

The copper pad must also remain sufficiently larger than the drilled hole to provide an adequate annular ring.

Consequently,

$$D_{\text{pad}} > D_{\text{hole}}.$$

Both parameters are therefore checked against the physical component and PCB manufacturer capabilities.

32.13 Courtyard Definition

The footprint courtyard defines the approximate mechanical space required by the assembled component.

Correct courtyard geometry is useful for detecting component interference during PCB placement.

This is particularly valuable for large project components such as:

- reservoir capacitors;
- bridge rectifiers;
- relays;
- fuse holders;
- transformers;
- the Nucleo development board;
- large connectors.

32.14 Silkscreen Design

Custom footprints also provide an opportunity to improve the PCB silkscreen.

Useful information includes:

- component outline;
- pin 1 indication;
- transistor orientation;
- diode polarity;
- capacitor polarity;
- connector function;
- relay orientation.

Silkscreen information should remain readable without extending over exposed pads.

32.15 Custom Footprints in the Amplifier Project

Custom or manually verified footprints are especially important for the mechanically significant components in the amplifier system.

These include components such as:

- power transistors;
- large reservoir capacitors;
- bridge rectifiers;
- relay packages;
- PCB-mounted transformers;
- fuse holders;
- terminal blocks;
- development-board headers;
- mechanical connectors.

For each of these components, the physical package was considered together with the electrical characteristics.

32.16 Project-Specific Library Tables

KiCad uses library tables to determine where symbols and footprints are located.

The two relevant project-level files are generally

sym-lib-table

for schematic symbols and

fp-lib-table

for PCB footprints.

These files can contain references to the custom libraries stored inside the project repository.

Using project-specific library tables reduces dependence on global KiCad configuration.

32.17 Project Libraries Versus Global Libraries

KiCad libraries can be configured globally or specifically for an individual project.

A global library is available to many projects on the same computer.

A project library belongs specifically to one project and can be stored directly inside its repository.

For custom amplifier components, project-level libraries are advantageous because the required files can travel with the project.

The dependency becomes

Repository → Project Library → KiCad Design

rather than

Local Computer Configuration → KiCad Design.

32.18 Use of Relative Paths

Library references should use project-relative paths where practical.

An absolute path such as

C:/Users/User/Documents/KiCad/MyLibrary.pretty

may work on the original design computer but will generally fail when the repository is opened on another machine.

A project-relative structure is preferable because the library remains located relative to the project directory.

This significantly improves portability.

32.19 Recommended Repository Organisation

A suitable repository organisation for the project is

```
Power-Amplifier/  
|  
|-- KiCad/  
|   |-- Schematics/  
|   '-- PCB/  
|  
|-- Libraries/  
|   |-- Symbols/  
|   |-- Footprints/  
|   '-- 3D_Models/  
|  
|-- Gerber/  
|-- Drill/  
|-- Documentation/  
|-- BOM/  
|-- Firmware/  
'-- README.md
```

The exact directory names can vary, but symbols, footprints, and optional 3D models should remain identifiable and separate from generated manufacturing outputs.

32.20 Why Library Files Are Stored in GitHub

The custom KiCad libraries are stored in the GitHub repository so that the complete hardware design can be reproduced from the repository alone.

Without the custom libraries, another user might be able to open the schematic or PCB but could encounter missing symbols, missing footprints, or unresolved 3D-model references.

Storing the libraries with the project provides several benefits:

- backup of custom design work;
- version history;
- portability between computers;
- easier collaboration;

- reproducible PCB revisions;
- preservation of discontinued or modified footprints.

32.21 Version Control

Custom library files are included in version control together with the schematics and PCB layouts.

If a footprint is changed, the modification can therefore be associated with the PCB revision that uses it.

For example,

Footprint Revision $\leftrightarrow$ PCB Revision

can be tracked through the repository history.

This is useful because a footprint modification after a board has already been manufactured may otherwise make it difficult to reproduce the earlier board revision.

32.22 Library Naming Conventions

Clear naming conventions are useful when several custom components are stored in the same library.

A useful footprint name can contain information such as:

- manufacturer or device family;
- package;
- pin count;
- mounting type;
- important mechanical dimensions.

Names should remain descriptive enough that a footprint can be identified without opening it.

Generic names such as

`Footprint1`

or

`New_Component`

should be avoided in the final project.

32.23 Symbol Naming

Custom symbols should similarly use names that correspond closely to the actual device or functional module.

This makes the schematic easier to interpret and helps maintain consistency between:

- schematic;
- BOM;
- datasheet;
- footprint;
- procurement information.

32.24 3D Model Library

The repository also contains a dedicated directory for

`3D_Models`.

Three-dimensional models are optional because they are not required for electrical connectivity or PCB fabrication.

However, they are useful for verifying:

- component orientation;
- body dimensions;
- vertical clearance;
- connector accessibility;
- heatsink alignment;
- chassis integration;
- overall assembly appearance.

32.25 Common 3D Model Formats

Typical 3D model formats used with KiCad include

`.step`

and

`.wrl`.

STEP models are particularly useful for mechanical CAD integration because they provide accurate three-dimensional geometry.

When a suitable manufacturer model exists, it can be referenced from the corresponding footprint.

32.26 3D Models Are Not Required for Manufacturing

A missing 3D model does not prevent the PCB from being manufactured.

The manufacturing outputs are derived from the copper, solder-mask, silkscreen, board-outline, and drill data.

Therefore,

Symbol + Footprint

are electrically important, while

3D Model

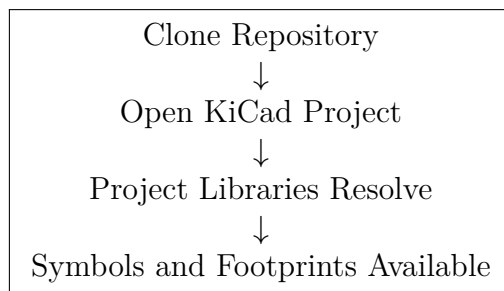
is primarily a mechanical and visual aid.

This distinction is useful when prioritising project-library development.

32.27 Library Portability

A major objective of the custom-library structure is to ensure that the project can be opened on another computer without modifying the original desktop KiCad installation.

The ideal workflow is:



This approach is considerably more reliable than requiring the user to install each custom footprint manually into a global KiCad library.

32.28 Avoiding Broken Library References

A common problem when transferring KiCad projects between computers is a broken library path.

Typical causes include:

- absolute local file paths;
- renamed folders;
- missing `.pretty` directories;
- missing `.kicad_sym` files;
- missing project library-table entries;
- 3D models stored outside the repository.

Keeping all project-specific libraries within a controlled directory structure greatly reduces these problems.

32.29 Library Verification

Before final PCB manufacture, custom libraries should be checked in the same way as the main design.

For each custom symbol, the following should be verified:

- pin number;
- pin name;
- electrical pin type;
- graphical orientation;
- reference prefix.

For each custom footprint, the following should be verified:

- pad number;
- pad position;
- pad size;
- drill diameter;
- pin spacing;

- body outline;
- courtyard;
- silkscreen;
- polarity or orientation marking.

32.30 Symbol-to-Footprint Verification

The final verification step is confirming that the custom symbol and footprint represent the same physical device.

The relationship should satisfy

Datasheet → Symbol Pins → Footprint Pads → Physical Component.
--

Any disagreement between these four stages must be corrected before the PCB is manufactured.

32.31 High-Risk Custom Components

Not all library errors have the same consequence.

The components requiring the most careful verification are those where a library error could damage the hardware or prevent assembly.

Examples include:

- power transistors;
- driver transistors;
- bridge rectifiers;
- voltage regulators;
- relays;
- mains transformers;
- polarised electrolytic capacitors;
- multi-pin development modules.

These components should be checked directly against the manufacturer's datasheet before Gerber generation.

32.32 Library Changes During Development

Component selection can change during development.

A replacement component may require:

- a different symbol;
- modified pin numbering;
- a new footprint;
- a different courtyard;
- an updated 3D model;
- a PCB layout change.

The schematic, PCB, library, and BOM should therefore be updated together.

Changing only the component name in the BOM is insufficient if the physical package has also changed.

32.33 Example of Component Replacement

The power-output transistor selection illustrates the importance of library revision control.

Different power-transistor families may provide similar electrical functions while using different physical packages or pin arrangements.

Whenever an output transistor is replaced, the designer must verify:

Electrical Rating + Pinout + Package + Footprint + Heatsink Geometry.

The present PCB revision is therefore documented using the specific output devices and footprints actually implemented in the KiCad design rather than older component alternatives considered during development.

32.34 ERC and DRC

Custom library verification is complemented by KiCad's automatic checking tools.

The Electrical Rules Checker can identify schematic problems related to pin types and connectivity.

The Design Rule Checker can identify PCB issues such as:

- clearance violations;
- unconnected pads;

- courtyard overlaps;
- board-edge violations;
- track-width violations;
- zone-connectivity errors.

However, neither ERC nor DRC can determine whether a custom footprint has been drawn from the wrong mechanical dimensions.

Manual datasheet verification remains essential.

32.35 3D Viewer Verification

KiCad's 3D Viewer provides an additional verification stage after footprint assignment.

The 3D representation can reveal:

- reversed components;
- unexpected body orientation;
- overlapping components;
- connector-access problems;
- excessive component height;
- heatsink interference.

The 3D view does not replace dimensional verification, but it is an effective final visual inspection tool.

32.36 Repository Reproducibility

A professional hardware repository should contain sufficient design information for the project to be reconstructed without relying on files that exist only on the original designer's computer.

For this reason, the repository stores the project-specific KiCad libraries together with the electronic design.

The intended reproducibility chain is

GitHub Repository → KiCad Project → Custom Libraries → PCB Manufacturing Files.

This also improves long-term maintainability if the project is opened several years after its initial development.

32.37 Recommended Library Maintenance Procedure

The following procedure is used when adding or modifying a project-specific component:

1. Obtain the official manufacturer datasheet.
2. Confirm the exact manufacturer part number.
3. Determine whether a suitable standard KiCad symbol already exists.
4. Determine whether a suitable standard footprint already exists.
5. Create a custom symbol only where required.
6. Create or modify a footprint using the datasheet mechanical drawing.
7. Verify all symbol pin numbers.
8. Verify all footprint pad numbers.
9. Confirm symbol-to-footprint correspondence.
10. Add a 3D model if useful and available.
11. Add the library files to the project repository.
12. Run ERC and DRC.
13. Inspect the component in the PCB and 3D Viewer.
14. Commit the library modification together with the associated design change.

32.38 Relationship to Manufacturing

Custom libraries are not supplied directly to the PCB manufacturer.

The manufacturer receives the generated PCB fabrication outputs, such as

- Gerber files;
- plated-through-hole drill files;
- non-plated drill information where required.

Nevertheless, any error in a custom footprint is transferred directly into these manufacturing files.

Library verification must therefore occur before Gerber and drill generation.

32.39 Relationship to the Bill of Materials

The custom-library entries should remain consistent with the Bill of Materials.

The BOM identifies the actual physical component, while the footprint defines the geometry into which that component must fit.

The relationship can be written as

BOM Part Number ↔ Datasheet ↔ KiCad Footprint.

If the BOM part changes, the footprint must be rechecked even if the replacement component has the same electrical value.

32.40 Design Summary

Custom KiCad libraries are used in this project to maintain accurate, project-specific representations of components that are not adequately covered by the standard KiCad libraries or that require additional verification.

The repository separates library resources into

Symbols

,

Footprints

,

and

3D Models

.

Custom schematic symbols are stored in

.kicad_sym

library files, while PCB footprint libraries are stored as

.pretty

directories containing individual

.kicad_mod

footprints.

Project-level

sym-lib-table

and

fp-lib-table

files can be used to associate these libraries with the KiCad project.

The custom libraries are stored inside the GitHub repository so that the schematic and PCB do not depend on library files existing only on the original development computer.

Footprints are created from official manufacturer mechanical drawings, with particular attention to pad numbering, drill diameter, lead spacing, body dimensions, courtyard, orientation, and polarity.

Optional STEP or WRL models can be associated with footprints to improve mechanical inspection using KiCad's 3D Viewer.

The most important verification relationship is

Datasheet → Symbol → Footprint → Physical Component.
--

Any inconsistency in this chain can result in an electrically incorrect or mechanically unusable PCB.

The final library workflow therefore combines datasheet verification, custom symbol and footprint management, ERC, DRC, 3D inspection, repository version control, and BOM cross-checking.

This library structure makes the amplifier project more portable, reproducible, maintainable, and suitable for future modification or independent reconstruction on another KiCad installation.

Chapter 33

Connectors and External Wiring

33.1 Introduction

The complete amplifier contains several independent electronic subsystems that must be interconnected electrically after the individual PCBs have been assembled. These connections include low-level audio signals, loudspeaker outputs, power-supply rails, ground connections, temperature-sensor wiring, protection signals, and mains-related wiring.

The electrical requirements of these connections differ considerably. A low-level RCA input may carry only a small audio voltage, while the loudspeaker output can carry several amperes. The wiring method, connector type, conductor size, routing, and shielding must therefore be selected according to the electrical function.

The internal wiring strategy used in the project separates the principal wiring categories into

Low-Level Audio

Low-Voltage Control

High-Current Power / Speaker

and

Mains AC.

These groups should remain physically separated wherever practical.

33.2 RCA Audio Inputs

The external analogue audio inputs use RCA connectors. RCA connectors are appropriate for unbalanced consumer and Hi-Fi audio systems and provide a simple connection consisting of a centre signal terminal and an outer ground connection.

The basic connection is

RCA Centre Pin → Audio Signal
RCA Outer Contact → Signal Ground

The RCA ground belongs to the low-level audio signal-ground system rather than the high-current power-ground system.

The RCA wiring should therefore not be used to carry loudspeaker, rectifier, relay, or power-supply return current.

33.3 RCA Connector Placement

The RCA connectors should be located on the rear panel and positioned away from the mains input, mains switch, transformer primary wiring, and main toroidal transformer.

The internal cable run from each RCA connector toward the preamplifier or power-amplifier input should be kept short.

Where the low-level signal must travel a significant distance inside the chassis, shielded audio cable should be used.

The cable shield should form part of the controlled signal-ground network and must not create an unintended second chassis-ground connection.

33.4 RCA Connector Insulation from the Chassis

The relationship between the RCA outer contact and the metal chassis requires careful consideration.

If the RCA body is electrically bonded to the chassis while the audio ground is also connected to the chassis at another location, an unintended parallel ground path can be formed.

This may create a ground loop.

The final mechanical arrangement should therefore ensure that the RCA connector is either intentionally isolated from the chassis or deliberately bonded according to the final grounding architecture.

33.5 Preamplifier to Power-Amplifier Wiring

The preamplifier output is connected to the two power-amplifier inputs using low-level signal wiring.

The preferred structure for each channel is

Preamplifier Output → Shielded Signal Cable → Power Amplifier Input

with the corresponding signal-ground reference travelling with the signal.

These cables should be kept away from:

- mains wiring;
- transformer secondary wiring;
- loudspeaker wiring;
- relay-coil wiring;
- cooling-fan wiring.

33.6 Speaker Output Terminals

The loudspeaker outputs require connectors capable of carrying substantially greater current than the RCA input connectors.

Gold-plated speaker terminals are used for the external left and right loudspeaker connections.

Each stereo channel requires two electrical connections:

Speaker Output

and

Speaker Return

.

The speaker return is treated as a high-current power-ground connection and must not be returned through the RCA signal-ground wiring.

33.7 Speaker Current

The required connector and wire current capability can be estimated from

$$P = I_{\text{RMS}}^2 R_L.$$

Therefore,

$$I_{\text{RMS}} = \sqrt{\frac{P}{R_L}}.$$

For the Project 3A reference condition of approximately

100 W

into

$$4\ \Omega,$$

the current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5\ \text{A}.$$

The corresponding sine-wave peak is

$$I_{\text{PEAK}} = 5\sqrt{2} \approx 7.07\ \text{A}.$$

The speaker terminals, relay contacts, PCB tracks, internal wiring, and solder joints must therefore all be suitable for several amperes.

33.8 Protection Relay Wiring

The loudspeaker signal does not pass directly from the power-amplifier PCB to the external loudspeaker terminal.

Instead, it passes through the speaker-protection controller.

For the left channel,

$$\boxed{\text{Left Amplifier Output} \rightarrow K_1 \rightarrow \text{Left Speaker Terminal}}$$

and for the right channel,

$$\boxed{\text{Right Amplifier Output} \rightarrow K_2 \rightarrow \text{Right Speaker Terminal.}}$$

This permits the STM32 protection controller to disconnect both loudspeakers during startup or when a fault is detected.

33.9 Speaker Wiring Geometry

The output and return conductors of each loudspeaker channel should be routed close together.

Keeping the two conductors close reduces the effective loop area and therefore reduces magnetic-field radiation.

The left and right loudspeaker wiring should also remain separated from the low-level RCA and preamplifier wiring.

33.10 Temperature Sensor Wiring

Two NTC temperature sensors are mounted on the power-transistor heatsinks.

Their wiring connects to the protection-controller inputs

TEMP1

and

TEMP2.

These are low-current analogue sensing signals and should therefore be routed away from the loudspeaker and mains wiring.

The two conductors associated with each sensor should be routed together to reduce noise pickup.

33.11 Power-Amplifier Supply Wiring

The main transformer provides a

24-0-24 V_{AC}

secondary supply.

The transformer secondary wiring connects to the amplifier power-supply rectifier and reservoir-capacitor section.

These conductors carry significant AC current and should therefore:

- use sufficiently large wire;
- remain short;
- be routed together;
- remain away from the RCA input;
- remain away from the preamplifier.

33.12 Ground Interconnections

The system contains signal ground, amplifier power ground, speaker returns, protection-controller ground, supply common, chassis, and protective earth.

The wiring should preserve the grounding hierarchy described previously.

In particular,

Speaker Return → Power Ground

and not

Speaker Return → RCA Signal Ground.

The principal subsystem grounds are connected through the controlled star-ground architecture.

33.13 Wire Identification

Clear wire identification is recommended during assembly.

Colour coding can be used consistently for:

- positive supply rail;
- negative supply rail;
- ground/common;
- left speaker;
- right speaker;
- mains line;
- mains neutral;
- protective earth.

The exact colours should remain consistent throughout the amplifier.

33.14 Strain Relief

External cables and internal high-current wires should not place mechanical stress directly on solder joints.

Appropriate cable ties, clamps, grommets, and chassis strain relief should be used where required.

This is particularly important for the mains cable and heavy loudspeaker or transformer wiring.

33.15 Design Summary

The amplifier uses separate wiring strategies for low-level audio, low-voltage control, high-current loudspeaker and power connections, and mains AC wiring.

RCA inputs carry the low-level audio signals and their signal-ground references, while gold-plated speaker terminals provide the high-current output connections.

The loudspeaker output passes through the two protection relays before reaching the external terminals.

Signal wiring is kept short and separated from transformer, mains, relay, fan, and speaker wiring.

Speaker return is treated as power ground, while RCA and preamplifier returns remain within the signal-ground system.

This external wiring architecture supports low noise, reliable current handling, serviceability, and correct operation of the loudspeaker-protection system.

Chapter 34

Mains Wiring and Electrical Safety

34.1 Introduction

The amplifier operates from the 230 V AC mains supply and therefore contains hazardous voltages capable of causing serious electric shock, fire, or damage if incorrectly constructed.

Mains wiring must consequently be treated differently from all low-voltage audio and control circuitry.

The mains system includes the external AC inlet, primary fuse, power switch, main amplifier transformer, protection-controller transformer, and protective earth connection.

34.2 Mains Input

The mains connection provides three conductors:

Line

Neutral

and

Protective Earth.

The line and neutral conductors provide operating power to the transformers, while protective earth is used exclusively for electrical safety.

34.3 Protective Earth

The protective-earth conductor must be bonded securely to the metal chassis.

The required safety connection is

Protective Earth → Dedicated Chassis Bond.

This connection must remain permanent and should not depend on PCB mounting screws, RCA connectors, signal wiring, or other removable components.

The protective-earth conductor should never be interrupted by the power switch or normal control circuitry.

34.4 Mains Power Switch

The amplifier includes a chassis-mounted mains power switch.

The switch must be appropriately rated for operation at 230 V AC and for the transformer primary current.

The switch forms part of the hazardous-voltage wiring region and should be positioned and insulated accordingly.

34.5 Mains Fuse

A mains fuse is required in addition to the 3 A amplifier rail fuses.

The mains fuse protects against sustained primary-side overcurrent conditions.

Its final rating must be selected according to:

- transformer VA rating;
- 230 V mains voltage;
- startup inrush current;
- fuse time-current characteristic;
- mains cable and switch ratings.

The final primary fuse value should therefore be confirmed from the actual transformer specification rather than inferred only from the secondary rail fuses.

34.6 Amplifier Rail Fuses

Each power-amplifier channel contains two 5 mm × 20 mm rail fuses.

The positive and negative rails are each protected by

3 A

fuses.

The rail protection is therefore

$$V_+ \rightarrow F_1 \rightarrow \text{Amplifier}$$

and

$$V_- \rightarrow F_2 \rightarrow \text{Amplifier}.$$

These fuses protect the amplifier supply paths but do not replace the mains primary fuse.

34.7 Mains Wire Routing

Mains conductors should be routed through a dedicated part of the chassis.

They should be physically separated from:

- RCA connectors;
- preamplifier wiring;
- STM32 ADC wiring;
- temperature-sensor wiring;
- low-level signal circuitry.

The line and neutral conductors should be kept close together so that their magnetic fields partially cancel.

34.8 Insulation and Mechanical Protection

Mains wiring should use insulation rated for the applied voltage and expected temperature.

Wires passing through metal panels require suitable bushings or grommets to prevent the chassis from damaging the insulation.

Exposed mains terminals should be avoided wherever possible.

Heat-shrink insulation or appropriate terminal covers should be used where necessary.

34.9 Primary and Secondary Separation

Mains-primary wiring and isolated low-voltage wiring should remain physically separated.

This applies both to chassis wiring and to the protection-controller PCB, which contains its own PCB-mounted transformer.

The normal

0.5 mm

PCB zone clearance used for ordinary low-voltage routing is not a mains-safety specification.

A dedicated primary-to-secondary isolation region must therefore be maintained around the protection-controller transformer.

34.10 Creepage and Clearance

Two related distances are important around hazardous-voltage circuitry.

Clearance is the shortest distance through air between conductive parts.

Creepage is the shortest distance along the surface of an insulating material.

The required values depend on factors such as:

- working voltage;
- insulation category;
- pollution degree;
- PCB material;
- applicable safety standard.

The final mains layout must therefore be checked against the relevant safety requirements before fabrication.

34.11 Mains Transformer Mounting

The 24–0–24 V toroidal transformer must be secured mechanically to the chassis.

Its mounting hardware must not create a conductive shorted turn around the toroidal core.

The transformer should also be positioned away from the RCA input and preamplifier circuitry to reduce magnetic coupling.

34.12 Protection Controller Transformer

The protection-controller PCB contains a second transformer that provides an isolated low-voltage supply for the STM32 and relay circuitry.

This transformer creates a mains-primary region directly on the PCB.

The primary connections therefore require the same attention to insulation, clearance, creepage, and accessibility as other mains wiring.

34.13 Chassis as a Safety Barrier

The enclosure provides mechanical protection and helps prevent accidental contact with hazardous wiring.

The chassis should therefore be fully closed during normal operation.

Openings should not permit direct access to live mains connections.

34.14 Protective Earth and Audio Ground

Protective earth and audio ground have different functions.

Protective earth is primarily a safety connection.

Signal and power grounds are electrical references used by the audio system.

The relationship between audio ground and chassis must therefore be intentional rather than created accidentally through multiple mounting points.

34.15 Safety Before Servicing

Before internal servicing:

1. disconnect the amplifier from mains power;
2. verify that the power is physically removed;
3. allow reservoir capacitors to discharge;
4. confirm the supply voltages with a meter;
5. avoid assuming that the power switch alone guarantees isolation.

The main reservoir capacitors can retain electrical energy after the amplifier has been switched off.

34.16 Pre-Power-Up Safety Inspection

Before applying mains power for the first time, the following should be checked:

- correct mains line and neutral connections;

- secure protective-earth connection;
- correct fuse installation;
- switch wiring;
- absence of exposed conductors;
- transformer primary wiring;
- transformer secondary wiring;
- PCB primary-to-secondary separation;
- chassis strain relief;
- absence of accidental shorts.

34.17 Design Summary

The amplifier operates from 230 V AC and therefore requires deliberate mains-safety design.

Protective earth is bonded directly to the metal chassis, while primary fusing and mains switching are provided independently from the low-voltage amplifier rail fuses.

Mains wiring is physically separated from signal circuitry, protected against mechanical damage, and kept outside sensitive audio regions.

The protection-controller PCB requires additional mains isolation around its PCB-mounted transformer.

The final system must be inspected for correct creepage, clearance, insulation, fusing, protective-earth continuity, and safe enclosure before operation.

Chapter 35

Heatsink and Thermal Management

35.1 Introduction

Thermal management is essential in a Class-AB power amplifier because a significant fraction of the electrical energy supplied to the output stage is converted into heat.

The two power-amplifier channels therefore use dedicated heatsinks and temperature monitoring.

Four cooling fans are also included in the mechanical cooling concept to improve airflow around the power stages and internal components.

35.2 Principal Heat Sources

The main heat-producing components are:

- Q7 and Q8 power-output transistors;
- Q5 and Q6 driver transistors;
- output emitter resistors;
- main bridge rectifier;
- protection-controller linear regulator;
- transformers;
- reservoir capacitors exposed to internal ambient temperature.

The output transistors are the most important thermal design elements because they can dissipate substantial power under sustained amplifier loading.

35.3 Output Transistor Mounting

The main output devices are the complementary

2SA1943-O

and

2SC5200-O

Both are positioned directly along the outer edge of the amplifier PCB.

This permits the mechanical arrangement

PCB → Output Transistor → Thermal Interface → External Heatsink.

The edge position reduces the need for long transistor wiring and prevents the heatsink from interfering with the central PCB component area.

35.4 Two Independent Heatsinks

The stereo amplifier uses two large heatsinks, one for each amplifier channel.

This provides several advantages:

- independent thermal path for each channel;
- simple mechanical arrangement;
- independent temperature measurement;
- reduced thermal interaction between channels;
- straightforward transistor mounting.

35.5 Thermal Resistance

Thermal behaviour can be represented using thermal resistance in the same way that electrical voltage drop is represented using electrical resistance.

The approximate temperature rise is

$$\Delta T = P_D \theta,$$

where

$$P_D$$

is the dissipated power and

$$\theta$$

is thermal resistance.

For the complete transistor-to-air thermal path,

$$\theta_{JA} = \theta_{JC} + \theta_{CS} + \theta_{SA},$$

where:

θ_{JC} = junction-to-case thermal resistance,

θ_{CS} = case-to-heatsink thermal resistance,

and

θ_{SA} = heatsink-to-ambient thermal resistance.

35.6 Junction Temperature

The estimated transistor junction temperature can be expressed as

$$T_J = T_A + P_D(\theta_{JC} + \theta_{CS} + \theta_{SA}).$$

The heatsink should be selected so that the transistor junction remains comfortably below its maximum rating under the worst expected operating condition.

A conservative thermal design is preferable to operating continuously close to the absolute maximum junction temperature.

35.7 Thermal Interface Material

The mating surfaces of the transistor and heatsink are not perfectly flat.

A thermal interface material is therefore used to reduce the thermal resistance between the device and heatsink.

Depending on the final mechanical arrangement, the interface may also provide electrical insulation where required.

The transistor mounting hardware must provide adequate pressure without mechanically damaging the package.

35.8 Electrical Isolation

The electrical relationship between the output-transistor mounting surface and the heatsink must be verified from the transistor package.

If the transistor mounting surface is electrically connected to an active terminal, suitable electrically insulating thermal pads and insulated mounting hardware must be used whenever both devices share a conductive heatsink.

Continuity should be checked with a multimeter before power is applied.

35.9 Heatsink Temperature Monitoring

Each heatsink is monitored using a

B57703M0103A047

10 k Ω NTC sensor.

The ring-terminal construction allows the sensor to be mechanically attached directly to the heatsink.

One sensor is assigned to each amplifier channel.

35.10 Temperature Measurement

The NTC resistance changes with temperature.

The approximate beta relationship is

$$R(T) = R_{25} \exp \left[B \left(\frac{1}{T} - \frac{1}{T_{25}} \right) \right].$$

For the selected sensor,

$$R_{25} = 10 \text{ k}\Omega$$

and

$$B_{25/100} \approx 3988 \text{ K}.$$

The STM32 converts the corresponding divider voltage into an estimated heatsink temperature.

35.11 Thermal Protection

If either heatsink exceeds the final protection threshold

$$T_{\text{trip}},$$

the protection controller removes speaker relay drive.

The required logic is approximately

$$T_L < T_{\text{trip}}$$

and

$$T_R < T_{\text{trip}}$$

for normal operation.

Temperature hysteresis is used so that the relay does not repeatedly switch near the trip temperature.

35.12 Cooling Fans

Four AC cooling fans are included in the thermal-management system.

The fans are used to increase airflow through the amplifier chassis and around the heatsinks.

Forced airflow decreases the effective heatsink-to-ambient thermal resistance and can therefore reduce transistor operating temperature.

35.13 Fan Placement

Fan placement should create a deliberate airflow path rather than simply circulating hot air inside the enclosure.

The preferred concept is

Cool Air In → Heatsink / Power Stage → Warm Air Out.

The chassis should include suitable inlet and outlet ventilation areas.

35.14 Fan Electrical Noise

Cooling fans can produce both electrical and acoustic noise.

Fan wiring should therefore remain away from:

- RCA wiring;
- preamplifier inputs;

- phono circuitry;
- STM32 ADC sensing wires.

The fan current should also not share sensitive signal-ground conductors.

35.15 Fan Acoustic Noise

Because the project is intended as a Hi-Fi amplifier, cooling performance must be balanced against audible fan noise.

Excessive fan speed, bearing noise, turbulence, or mechanical vibration can reduce the subjective quality of the completed amplifier.

Fan mounting should therefore avoid transmitting unnecessary vibration into the chassis.

35.16 Natural and Forced Convection

The heatsinks should remain capable of dissipating heat through natural convection even when airflow is reduced.

The cooling fans should therefore improve thermal margin rather than compensate for an fundamentally undersized heatsink.

35.17 Thermal Testing

The final thermal performance cannot be confirmed until the physical amplifier has been assembled.

Testing should be performed with:

- no signal;
- moderate output power;
- high output power;
- 8 Ω dummy load;
- controlled 4 Ω dummy load;
- one channel operating;
- both channels operating;
- fans operating;
- fans disabled temporarily for comparison where safe.

35.18 Thermal Measurements

The following temperatures should be recorded:

- left heatsink;
- right heatsink;
- bridge rectifier;
- protection regulator;
- chassis internal air;
- transformer where practical.

Measurements can be compared with the NTC values reported by the STM32.

35.19 Design Summary

The amplifier uses two large heatsinks, one per channel, with the 2SA1943-O and 2SC5200-O output devices mounted along the PCB edge.

Each heatsink is monitored by a 10 k Ω ring-terminal NTC sensor connected to the STM32 protection controller.

Four AC cooling fans provide additional forced airflow.

The final thermal design combines suitable heatsink thermal resistance, correct transistor mounting, thermal interface material, heatsink temperature monitoring, chassis ventilation, and fan-assisted airflow.

Final heatsink suitability and protection thresholds will be confirmed through physical thermal testing rather than assumed from calculation alone.

Chapter 36

Mechanical and Chassis Design

36.1 Introduction

The mechanical design integrates the amplifier electronics into a single enclosure while providing adequate support, cooling, accessibility, electrical safety, and electromagnetic separation.

The chassis must accommodate:

- two power-amplifier PCBs;
- the STM32 protection-controller PCB;
- the preamplifier;
- the main toroidal transformer;
- two large heatsinks;
- four cooling fans;
- front-panel controls and indicators;
- rear-panel audio and loudspeaker connectors;
- mains input hardware;
- internal wiring.

36.2 Chassis Architecture

The chassis interior should be organised according to electrical function.

A suitable arrangement separates:

Mains / Transformer Region

Power-Amplifier Region

Low-Level Audio Region

and

Protection / Control Region

.

This reduces interference and makes assembly and servicing easier.

36.3 Power-Amplifier PCB Mounting

Each power-amplifier PCB measures

$136\text{ mm} \times 124.5\text{ mm}$.

Four

5 mm

mounting holes are provided.

The boards should be supported using suitable standoffs so that the underside copper and solder joints cannot contact the chassis.

36.4 Protection Controller PCB Mounting

The protection-controller PCB measures

$121\text{ mm} \times 182.25\text{ mm}$.

It uses four

3.2 mm

M3 mounting holes.

Adequate clearance must be maintained beneath the board, particularly because the PCB contains mains-primary circuitry around the onboard transformer.

36.5 Main Transformer Placement

The main 24–0–24 V toroidal transformer is one of the heaviest components in the amplifier.

It should be mounted directly to a mechanically strong chassis surface.

The transformer should be positioned away from the preamplifier and RCA input region to reduce magnetic coupling.

36.6 Heatsink Placement

The two amplifier boards position their output transistors at the PCB edge.

The mechanical chassis design should therefore position each amplifier board close to its corresponding heatsink.

This allows the output devices to connect mechanically to the heatsink while remaining electrically connected directly to the PCB.

36.7 Airflow

The internal arrangement should permit a clear path for airflow.

Large objects such as the transformer, PCBs, and wiring harnesses should not completely obstruct the fan airflow.

Ventilation openings should allow cool air to enter and heated air to leave the enclosure.

36.8 Front Panel Design

The front panel provides the primary user interface.

It should present only the controls and indicators needed during normal operation.

The exact cosmetic arrangement can be finalised after the electrical and mechanical locations of the components are established.

36.9 Front Panel Controls

The front panel may accommodate controls associated with:

- main power;
- volume;
- balance;

- source selection;
- mute / enable;
- protection reset.

The exact control set depends on the final preamplifier and chassis implementation.

36.10 Protection Indicators

The protection-controller PCB currently contains five status and fault indicators:

- TEMP SENSOR 1 FAIL;
- TEMP SENSOR 2 FAIL;
- DC FAULT;
- WAIT / DELAY;
- OVERHEAT.

These indicators can be brought to the front panel so that the protection state is visible without opening the chassis.

36.11 WAIT / DELAY Indicator

The WAIT / DELAY indicator is active during the startup period while the speakers remain disconnected.

It can also be used to indicate that the speaker relays are not currently energised.

36.12 DC Fault Indicator

The DC fault indicator warns that the protection controller has detected a persistent abnormal DC level at one of the amplifier outputs.

Under this condition both speaker relays are de-energised.

36.13 Temperature Sensor Fault Indicators

The two temperature-sensor fault indicators identify failures of the temperature-monitoring channels.

They represent sensor-integrity faults rather than simply a high heatsink temperature.

36.14 Overheat Indicator

The dedicated OVERHEAT indicator identifies a thermal protection event.

This distinction allows the user to differentiate between

Sensor Failure

and

Actual Overtemperature.

36.15 Rear Panel

The rear panel contains the external connections that do not normally require continuous user adjustment.

These include:

- RCA audio inputs;
- left speaker output;
- right speaker output;
- mains input;
- fuse access where appropriate.

36.16 Rear Panel Separation

The mains input should be positioned away from the RCA input region.

Similarly, speaker terminals should not force high-current wiring to cross the low-level audio section of the chassis.

36.17 Internal Wiring Channels

Internal cable routing should be planned before final assembly.

Separate physical routes should be provided for:

- mains wiring;
- transformer secondary wiring;
- speaker wiring;

- low-level audio;
- sensor and control wiring.

36.18 Serviceability

Components requiring adjustment or replacement should remain accessible.

These include:

- amplifier bias trimmers;
- rail fuses;
- Nucleo programming interface;
- speaker relays;
- power transistors;
- mains fuse.

36.19 Mechanical Fasteners

PCB standoffs, heatsink screws, transformer mounting hardware, fan screws, and panel hardware should be selected so that they remain mechanically secure during transport and normal use.

Thread-locking methods or locking washers can be used where appropriate.

36.20 Sharp Edges and Cable Protection

Internal cables should not pass directly over sharp metal edges.

Cable openings through chassis panels should use grommets or protected bushings.

This requirement is particularly important for mains wiring.

36.21 Design Summary

The chassis is designed to integrate the amplifier, protection electronics, preamplifier, transformers, heatsinks, cooling fans, connectors, and control interface while maintaining separation between sensitive audio circuitry and high-current or mains wiring.

The two amplifier boards are positioned close to their dedicated heatsinks, while the main transformer is kept away from the low-level audio section.

Five protection-status LEDs are incorporated into the user-interface concept.

The rear panel provides the RCA, loudspeaker, and mains connections.

The final mechanical design therefore combines electrical performance, cooling, safety, user accessibility, and serviceability.

Chapter 37

Construction Procedure

37.1 Introduction

The amplifier should be assembled in a controlled sequence so that each subsystem can be inspected before it becomes integrated into the complete chassis.

Constructing and testing individual sections first reduces the risk of multiple simultaneous faults and simplifies troubleshooting.

37.2 Recommended Assembly Sequence

A suitable construction sequence is:

1. inspect all manufactured PCBs;
2. verify component values and footprints;
3. assemble the power-amplifier PCBs;
4. assemble the protection-controller PCB;
5. assemble the preamplifier;
6. inspect all solder joints;
7. mount the power transistors and heatsinks;
8. install PCBs in the chassis;
9. install transformers;
10. install front- and rear-panel hardware;
11. complete low-voltage wiring;
12. complete signal wiring;
13. complete loudspeaker wiring;

14. complete mains wiring;
15. perform continuity and resistance checks;
16. perform first power-up procedures.

37.3 PCB Inspection Before Assembly

Before components are installed, each PCB should be inspected for:

- manufacturing defects;
- incomplete drill holes;
- damaged copper;
- incorrect board outline;
- shorted pads;
- incorrect solder mask;
- mechanical damage.

37.4 Component Sorting

Components should be organised by value and circuit before soldering begins.

This reduces the risk of installing an incorrect resistor, capacitor, diode, or transistor.

Polarity-sensitive and orientation-sensitive components should be identified before installation.

37.5 Assembly Order

For through-hole PCBs, assembly normally begins with the lowest-profile components.

A practical order is:

1. resistors;
2. small diodes;
3. small capacitors;
4. transistor devices;
5. IC sockets or modules;
6. larger capacitors;

7. terminal blocks;
8. relays;
9. fuse holders;
10. bridge rectifiers;
11. large reservoir capacitors.

37.6 Resistor Verification

Resistor values should be verified before installation using either colour coding or a multimeter.

Particular attention should be given to the power-amplifier feedback and bias networks and the protection-controller ADC divider resistors.

37.7 Polarised Components

Electrolytic capacitors, LEDs, rectifier diodes, flyback diodes, and several semiconductor devices require correct orientation.

Before soldering, polarity should be confirmed against:

- schematic;
- PCB silkscreen;
- component datasheet.

37.8 Transistor Installation

All transistor pinouts should be checked before soldering.

This is especially important for:

- BC546;
- BC337;
- BD139;
- BD140;
- 2SA1943-O;
- 2SC5200-O.

A transistor should never be installed solely by assuming that the flat face or package shape determines the correct pin order.

37.9 Large Capacitor Installation

The four main

4700 μF

reservoir capacitors must be installed with correct polarity.

Their physical size means they should also be seated securely against the PCB without placing unnecessary mechanical force on the leads.

37.10 Relay Installation

The two G2RL-1A-E-DC12 relays should be installed according to the footprint orientation.

The coil and contact pins should be checked for continuity before the system is powered.

37.11 Nucleo Installation

The NUCLEO-G031K8 module should be installed only after the protection PCB has been inspected for shorts.

Its orientation must match the PCB footprint so that the GPIO, ADC, supply, and ground connections correspond correctly.

37.12 Solder Joint Inspection

After assembly, every PCB should be inspected visually.

The inspection should look for:

- solder bridges;
- insufficient solder;
- cold joints;
- unsoldered pins;
- damaged pads;
- reversed components;
- incorrect component values.

37.13 Continuity Testing

Before applying power, resistance and continuity checks should be performed.

Important checks include:

- positive rail to ground;
- negative rail to ground;
- positive rail to negative rail;
- speaker output to ground;
- mains primary isolation;
- protective-earth continuity;
- relay contact continuity.

A very low unexpected resistance may indicate a short circuit that should be resolved before power-up.

37.14 Heatsink Assembly

The two output transistors of each channel are mechanically attached to their corresponding heatsink.

The mounting surfaces should be clean and suitable thermal interface material should be installed.

Where electrical isolation is required, insulating pads and bushings must be installed correctly.

The transistor-to-heatsink insulation should be checked electrically before power is applied.

37.15 Temperature Sensor Installation

Each NTC ring sensor should be attached firmly to its corresponding heatsink.

The sensor should be positioned near the power transistors so that it responds to their thermal behaviour.

The sensor wiring should be secured mechanically and routed away from high current and mains wiring.

37.16 PCB Installation

The amplifier and protection boards should be mounted using appropriate standoffs.

The PCB underside must not contact the metal chassis.

The mounting hardware should also not accidentally create unwanted ground-to-chassis connections.

37.17 Transformer Installation

The main toroidal transformer should be mounted securely before final wiring.

Its primary and secondary conductors should remain identifiable and physically separated.

The transformer mounting arrangement should also avoid creating a conductive shorted turn around the core.

37.18 Panel Hardware Installation

The RCA connectors, speaker terminals, mains inlet, switch, front-panel controls, LEDs, and push buttons should be mounted before final internal wiring.

This allows wire lengths to be determined from the actual physical positions.

37.19 Signal Wiring

Low-level audio wiring should be completed using short, controlled routes.

Where required, shielded cable should be used.

Signal wires should remain separated from mains, transformer, loudspeaker, fan, and relay wiring.

37.20 Speaker Wiring

The power-amplifier outputs connect first to the protection-controller relay inputs.

The relay outputs then connect to the external speaker terminals.

The high-current connections should use appropriately sized conductors and secure solder or terminal connections.

37.21 Ground Wiring

The grounding network should be constructed according to the star-ground architecture described earlier.

The installer should verify that:

- speaker returns use power ground;
- RCA returns use signal ground;
- left and right amplifier returns are not daisy chained;
- protection ground has one defined system reference;
- protective earth is connected directly to chassis.

37.22 Mains Wiring

Mains wiring should be completed only after all low-voltage assembly has been verified.

Line, neutral, and protective earth should remain clearly identifiable.

The protective-earth connection should be installed first and verified before the mains circuit is energised.

37.23 Cable Management

After wiring is complete, cables should be secured so that they cannot move into fans, heatsinks, sharp chassis edges, or exposed terminals.

Signal and mains wiring should not be bundled together.

37.24 Pre-Power-Up Inspection

Before the first application of mains power, the complete system should be checked systematically.

The inspection should confirm:

- correct component polarity;
- correct transistor orientation;
- correct fuse values;
- no visible solder bridges;
- no PCB-to-chassis shorts;

- correct ground connections;
- secure protective earth;
- correct transformer wiring;
- correct speaker relay wiring;
- correct temperature-sensor wiring;
- correct heatsink mounting;
- correct mains insulation.

37.25 First Power-Up

The first power-up should be performed without loudspeakers connected.

A controlled current-limited or protected startup method should be used where available.

The supply rails should first be verified before attempting high-power operation.

The expected amplifier rails are approximately

$$\pm 33 - \pm 34 \text{ VDC}$$

with the 24–0–24 V AC transformer, subject to mains voltage and load.

37.26 Bias Adjustment

The amplifier quiescent current is adjusted using VR1.

The adjustment should be performed only after verifying that the amplifier has no major DC fault.

The exact procedure is covered separately in the *Amplifier Bias Adjustment* chapter.

37.27 Protection System Verification

Before loudspeakers are connected, the protection system should be checked for:

- startup delay;
- relay operation;
- DC fault detection;
- temperature sensing;
- sensor-fault detection;

- overtemperature indication;
- relay disconnection.

37.28 Dummy Load Testing

Initial amplifier output testing should use suitable dummy loads rather than valuable loudspeakers.

An 8 Ω load provides a suitable initial condition before controlled 4 Ω testing.

37.29 Final Assembly

Only after the electrical and thermal behaviour has been verified should the complete enclosure be closed.

A final inspection should ensure that no wire has moved into a fan, heatsink, sharp edge, or hazardous-voltage region.

37.30 Construction Summary

The amplifier is assembled subsystem by subsystem rather than wiring the entire system before testing.

The PCB assemblies are inspected before installation, semiconductor and polarised-component orientation is checked, the power transistors are mounted to their heatsinks, and the NTC sensors are attached close to the output devices.

Signal, speaker, ground, control, and mains wiring are installed according to their different electrical requirements.

The complete system is subjected to continuity, isolation, grounding, and mechanical checks before the first application of mains power.

Initial electrical tests are performed without loudspeakers, followed by bias, protection, dummy-load, thermal, and full-system testing.

This staged construction approach reduces the risk of damaging the power amplifier, protection controller, loudspeakers, or power-supply components and provides a systematic path from individual PCB assembly to the completed stereo Hi-Fi amplifier.

Chapter 38

Initial Power-Up Procedure

38.1 Introduction

The initial power-up is one of the most critical stages in the construction of the complete amplifier. An assembly error that is harmless while the circuit is unpowered can result in excessive current, semiconductor failure, damaged loudspeakers, or overheating immediately after power is applied.

For this reason, the complete system should not be powered for the first time with all subsystems connected and with loudspeakers attached.

Instead, the amplifier should be commissioned progressively. The main power supply, protection controller, individual power-amplifier channels, and preamplifier should be verified separately before the complete system is interconnected.

The recommended philosophy is

Inspect → Measure → Power Gradually → Verify → Integrate

No measured results are presented in this chapter because the procedure describes the intended commissioning process that will be followed after the physical amplifier has been assembled.

38.2 Safety During Initial Power-Up

The amplifier contains 230 V AC mains circuitry and must therefore be treated as hazardous equipment during commissioning.

The first mains-powered tests should be carried out only after the chassis protective-earth connection, mains fuse, switch wiring, transformer primary wiring, and insulation have been verified.

Before touching or modifying any internal connection, the amplifier should be disconnected from the mains supply and the main reservoir capacitors should be allowed to discharge.

Residual supply voltage should be confirmed using a multimeter before work is continued.

The protective-earth connection should remain installed throughout testing.

38.3 Initial Power-Up Sequence

The complete commissioning procedure is divided into the following stages:

1. mechanical and visual inspection;
2. unpowered continuity measurements;
3. main power-supply verification;
4. protection-controller verification;
5. first power-up of each amplifier channel individually;
6. DC output verification;
7. amplifier bias adjustment;
8. protection-system integration;
9. low-level signal testing;
10. dummy-load testing;
11. thermal testing;
12. final loudspeaker connection.

Each stage should be completed successfully before proceeding to the next.

38.4 Visual Inspection Before Power-Up

Every PCB should first be inspected visually under good lighting.

The inspection should confirm:

- correct component values;
- correct transistor orientation;
- correct diode orientation;
- correct LED polarity;
- correct electrolytic-capacitor polarity;
- correct bridge-rectifier orientation;
- correct relay orientation;

- correct Nucleo orientation;
- absence of solder bridges;
- absence of unsoldered component pins;
- absence of damaged PCB tracks or pads.

Special attention should be given to the output transistors

2SA1943-O

and

2SC5200-O,

because incorrect pin assignment or installation could result in immediate output-stage failure.

38.5 Mechanical Inspection

The mechanical assembly should also be inspected before electrical power is applied.

The following should be checked:

- PCBs are securely mounted on standoffs;
- PCB solder joints cannot contact the metal chassis;
- power transistors are securely attached to the heatsinks;
- thermal interface material is installed correctly;
- heatsink isolation is correct where required;
- temperature sensors are secured to the heatsinks;
- transformer mounting is mechanically secure;
- no wires can contact cooling fans;
- no wires pass over sharp chassis edges;
- strain relief is fitted where required.

38.6 Protective-Earth Verification

Before mains power is applied, continuity should be verified between the mains protective-earth conductor and the metal chassis.

The required connection is

$$\boxed{\text{Protective Earth} \rightarrow \text{Metal Chassis}}.$$

This connection must be secure and independent of audio signal-ground wiring.

Protective earth should not rely on PCB mounting screws or signal connectors for continuity.

38.7 Unpowered Resistance Measurements

A multimeter should be used to check for obvious short circuits before energising the amplifier.

Useful resistance checks include measurements between:

- positive supply rail and ground;
- negative supply rail and ground;
- positive rail and negative rail;
- amplifier output and ground;
- amplifier output and each supply rail;
- 12 V protection supply and ground;
- 3.3 V logic supply and ground.

A low resistance is not automatically a fault because capacitors and semiconductor junctions can affect the reading. However, an unexpected near short should be investigated before power is applied.

38.8 Reservoir Capacitor Polarity

The polarity of the four

$$4700 \mu\text{F}$$

main reservoir capacitors should be checked carefully.

The positive-rail capacitors and negative-rail capacitors are not installed with identical polarity relative to the common ground point.

Incorrect electrolytic-capacitor polarity can result in capacitor failure when the supply is energised.

38.9 Fuse Verification

The correct fuses should be installed before power-up.

Each amplifier channel uses

3 A

fuses in the positive and negative supply rails.

The mains input also requires its independently selected primary fuse.

A fuse should never be replaced by a wire or significantly oversized device during commissioning.

38.10 Bias Trimmer Initial Position

The amplifier bias control

$VR1$

should be placed in the safe minimum-bias position before the initial power-up.

The correct direction must be determined from the implemented circuit rather than by assuming that clockwise or counter-clockwise rotation always reduces the bias.

This minimises the risk of excessive quiescent current through the output stage during the first energisation.

38.11 No Loudspeakers During Initial Testing

Loudspeakers should not be connected during the first power-up.

Until the output DC offset and protection operation have been verified, a circuit fault could place a large DC voltage across the loudspeaker.

The initial configuration should therefore be

Speaker Outputs Unloaded

.

Dummy loads should only be added during a later stage after correct no-load operation has been confirmed.

38.12 Input Condition

The amplifier inputs should initially be connected to their local signal-ground reference. This provides a defined zero-input condition and prevents the high-impedance input from picking up environmental noise.

Thus,

$$V_{\text{IN}} = 0 \text{ V}$$

during the first amplifier power-up.

38.13 Current-Limited Initial Energisation

Where suitable commissioning equipment is available, the first energisation should be performed using a method that limits fault current.

The objective is to prevent a wiring or component fault from immediately delivering the full available transformer power to the circuit.

A current-limited commissioning arrangement can therefore provide an additional level of protection during initial testing.

If abnormal current consumption, unexpected heating, smoke, or an unusual transformer sound occurs, power should be removed immediately and the circuit should be inspected before another test is attempted.

38.14 Main Power-Supply Verification

The main power supply should be checked before full amplifier operation is attempted.

The main transformer secondary is

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}}.$$

After rectification and filtering, the expected unloaded or lightly loaded rails are approximately

$$V_+ \approx +33\text{--} + 34 \text{ V}$$

and

$$V_- \approx -33\text{--} - 34 \text{ V}.$$

The exact values depend on mains voltage, transformer regulation, rectifier voltage drop, and circuit loading.

38.15 Checking Rail Symmetry

The positive and negative rail magnitudes should be reasonably symmetrical.

A useful comparison is

$$|V_+| \approx |V_-|.$$

A substantial difference between the two rails may indicate:

- incorrect reservoir-capacitor wiring;
- rectifier wiring error;
- excessive load on one rail;
- incorrect ground/common connection;
- component failure.

The cause should be identified before proceeding.

38.16 Protection Controller Power-Up

The protection controller should also be verified independently before it is relied upon to protect the loudspeakers.

Its auxiliary supply is based on the sequence

$$230 \text{ V}_{\text{AC}} \rightarrow \text{Isolation Transformer} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Filter} \rightarrow 7812 \rightarrow 12 \text{ VDC}.$$

The regulated supply should first be confirmed before the relay-control functions are tested.

38.17 Protection Supply Measurements

The following supply points should be verified:

$$\boxed{12 \text{ V}}$$

for the relay and controller supply system, and the appropriate logic supply used by the STM32/Nucleo system.

An incorrect controller supply should be corrected before the Nucleo board, relays, or ADC sensing circuits are relied upon.

38.18 STM32 Startup

After the protection-controller supply has been verified, the STM32 firmware should be allowed to start.

The expected initial state is a safe state in which the loudspeaker relays are not immediately energised.

The startup logic should therefore satisfy

Power Applied → WAIT State → Fault Checks → Relay Enable

.

38.19 Startup Delay Verification

The WAIT / DELAY indicator should show that the controller is intentionally keeping the loudspeakers disconnected during the initial startup interval.

After the programmed delay, the controller should energise the relays only if:

- no persistent DC fault is detected;
- no overtemperature condition exists;
- both temperature sensors are valid;
- the controller itself is operating normally.

38.20 Relay Verification

The two G2RL-1A-E-DC12 relays should be verified before loudspeakers are connected.

The relay sequence should be

Safe Conditions → BC337 Driver ON → Relay Energised

.

Relay de-energisation should disconnect the loudspeaker output.

The normally-open contact configuration ensures that loss of control power results in speaker disconnection.

38.21 Powering One Amplifier Channel at a Time

The two amplifier channels should initially be tested independently.

Testing one channel at a time has several advantages:

- a fault is easier to locate;

- overall current is reduced;
- rail behaviour is easier to interpret;
- channel-to-channel wiring errors are eliminated from the first test.

After the first amplifier has been verified, the same procedure should be repeated for the second channel.

38.22 Immediate Checks After Energisation

Immediately after an amplifier channel is energised, the following should be observed:

- supply-rail voltage;
- excessive current indication;
- output DC voltage;
- unexpected heating;
- unusual smell or smoke;
- abnormal transformer noise.

If any abnormal condition appears, power should be removed immediately.

38.23 Output DC Offset

Before any loudspeaker or dummy load is connected, the DC voltage at the amplifier output should be measured relative to the amplifier ground.

Ideally,

$$V_{\text{OUT,DC}} \approx 0 \text{ V.}$$

A small residual offset is expected in a real discrete amplifier because of transistor mismatch and component tolerances.

However, a large DC voltage indicates an amplifier fault and the output should not be connected to a loudspeaker.

38.24 Comparing Both Channels

After both channels have been powered individually, their DC behaviour should be compared.

Measurements should include:

$$V_{\text{OUT,L,DC}}$$

and

$$V_{\text{OUT,R,DC}}.$$

A substantial difference between the channels may indicate a component or assembly difference that requires investigation.

38.25 Quiescent Current and Bias

Once the DC output and supply rails are confirmed, the Class-AB quiescent current can be adjusted using VR1.

The purpose of the bias adjustment is to establish sufficient standing current to minimise crossover distortion without producing excessive idle dissipation.

The bias should be adjusted progressively while monitoring the relevant voltage or current according to the amplifier commissioning method.

The adjustment should not be performed by estimating heatsink temperature or by turning the trimmer until the amplifier “sounds correct.”

38.26 Bias Thermal Stabilisation

After the initial bias setting, the amplifier should be allowed to warm up at idle.

The quiescent condition should then be measured again.

This is necessary because transistor junction voltages and therefore bias current vary with temperature.

A stable amplifier should not exhibit uncontrolled increasing quiescent current as the output stage warms.

38.27 Initial Temperature Monitoring

During the first extended idle test, the temperature of the output transistors and heatsinks should be observed.

The STM32 temperature measurements can also be compared with an independent temperature measurement where practical.

Unexpected rapid heating at zero signal is an indication that the bias or output stage should be investigated.

38.28 Protection DC Sensing Verification

After each amplifier channel is known to operate correctly, its output can be connected to the corresponding protection-controller sensing input.

Under normal conditions, the controller should interpret the amplifier output as safe.

The ADC reading should remain within the normal region defined by the firmware.

The protection controller should not repeatedly connect and disconnect the relays during normal zero-signal operation.

38.29 Protection Fault Verification

The protection logic should be deliberately tested before valuable loudspeakers are connected.

The test should confirm the controller response to:

- DC output fault;
- overtemperature condition;
- temperature-sensor failure;
- controller startup;
- power removal.

Fault simulation should be performed using controlled test conditions rather than by creating dangerous high-energy faults in the power amplifier.

38.30 Sensor Fault Verification

The two temperature-sensor channels should be checked individually.

The controller should distinguish between a normal temperature measurement and an invalid sensor condition.

The corresponding

TEMP SENSOR 1 FAIL

and

TEMP SENSOR 2 FAIL

indicators should identify the affected sensing channel.

38.31 Overtemperature Logic Verification

The OVERHEAT protection state should also be verified independently from the sensor-failure state.

The logic should distinguish

Valid Sensor + Excessive Temperature

from

Invalid Sensor Reading

In either condition, the loudspeaker relays should move to the safe state if required by the firmware.

38.32 Low-Level Signal Test

After correct idle operation has been confirmed, a low-amplitude sine wave can be applied to the amplifier input.

The signal should initially be kept small so that amplifier gain and waveform quality can be checked without substantial output-stage dissipation.

The output should be observed using suitable measurement equipment.

The expected relationship is approximately

$$V_{\text{OUT}} = A_v V_{\text{IN}},$$

where the nominal closed-loop gain is approximately

$$A_v \approx 23.$$

38.33 Oscilloscope Verification

An oscilloscope can be used to verify that the amplifier output follows the input waveform correctly.

The initial check should look for:

- correct waveform polarity;
- expected voltage gain;
- absence of unexpected DC shift;
- absence of visible oscillation;
- absence of severe distortion.

Particular care is required when using an earth-referenced oscilloscope because the probe ground may be electrically connected to protective earth.

38.34 No-Load Frequency Test

Before high-power testing, the amplifier can be examined at low amplitude over a limited range of audio frequencies.

This confirms basic small-signal operation before significant thermal stress is introduced.

High-frequency tests should be approached cautiously because oscillation or improper loading can produce high transistor dissipation even when the signal is not audible.

38.35 Dummy Load Connection

After correct no-load operation has been confirmed, a suitable non-inductive dummy load can be connected.

An

$$8\ \Omega$$

load is preferable for the first loaded test.

The dummy load must have sufficient power capability for the intended test.

Ordinary small resistors must not be used for high-power amplifier testing.

38.36 Gradual Output Power Increase

The signal amplitude should be increased progressively rather than applying full power immediately.

During this process, the following should be monitored:

- output waveform;
- supply-rail voltage;

- amplifier current;
- heatsink temperature;
- output transistor temperature;
- protection-controller status.

38.37 Rail Sag Under Load

The supply rails will normally decrease in magnitude as output power increases.

This occurs because of transformer regulation, rectifier losses, wiring resistance, and reservoir-capacitor ripple.

Therefore,

$$|V_{\text{rail,loaded}}| < |V_{\text{rail,no-load}}|$$

is expected.

The measured loaded values should later be included in the experimental results chapter.

38.38 Testing into 4 Ohms

Controlled

$$4\ \Omega$$

testing should only be performed after satisfactory operation into

$$8\ \Omega$$

has been confirmed.

The $4\ \Omega$ load approximately doubles the current required for a given output voltage compared with an $8\ \Omega$ load.

This significantly increases stress on:

- output transistors;
- driver transistors;
- power supply;
- PCB tracks;
- speaker relays;

- wiring;
- heatsinks.

38.39 Thermal Soak Test

A short high-power measurement does not fully verify the thermal design.

The amplifier should therefore eventually be operated for a longer period at a controlled load and output level.

The heatsink temperature should be allowed to approach a stable value.

The test can verify:

- adequacy of heatsink size;
- effectiveness of cooling fans;
- stability of amplifier bias;
- accuracy of temperature sensing;
- protection threshold operation.

38.40 Fan Verification

The four cooling fans should be checked for correct orientation and unobstructed airflow.

The airflow direction should support the intended path

Cool Air → Heatsinks → Warm Air Exhaust

.

Fan operation should not introduce unacceptable electrical noise into the audio output.

38.41 Preamplifier Integration

The preamplifier should be added only after both power-amplifier channels have been shown to operate correctly from a direct test source.

Its regulated supply should first be confirmed.

The preamplifier output can then be connected to the power-amplifier inputs and the complete signal path tested at low volume.

38.42 Noise and Hum Check

With the system fully assembled but without loudspeakers, residual output noise should be examined.

The input can first be shorted to signal ground.

A dominant component near

50 Hz

may indicate mains coupling, while a component near

100 Hz

may indicate rectifier or supply-ripple coupling.

Any significant hum should be investigated before final loudspeaker testing.

38.43 Relay Switching Noise

The output should also be observed when the loudspeaker relays connect and disconnect.

The 1N4007 flyback diodes should suppress the relay-coil voltage transients, while the grounding arrangement should prevent the driver currents from producing significant disturbances in the audio path.

38.44 Both Channels Operating

After each amplifier channel has been validated independently, both channels should be operated simultaneously.

This test is important because the channels share the main transformer and power supply.

Simultaneous operation allows evaluation of:

- shared supply loading;
- rail sag;
- transformer heating;
- inter-channel ground interaction;
- total chassis thermal load;
- cooling performance.

38.45 Final Loudspeaker Connection

Loudspeakers should be connected only after all of the following have been confirmed:

- correct positive and negative supply rails;
- acceptable output DC offset;
- stable amplifier bias;
- correct relay startup delay;
- correct DC protection operation;
- correct overtemperature protection;
- valid temperature-sensor operation;
- absence of abnormal heating;
- absence of excessive hum or oscillation.

Only at this stage should the amplifier be considered ready for an initial listening test.

38.46 Measurements to Record

The commissioning process should be documented rather than treated only as a pass/fail procedure.

The following values should be recorded for each channel:

- positive supply voltage;
- negative supply voltage;
- output DC offset;
- quiescent current or equivalent bias measurement;
- left heatsink temperature;
- right heatsink temperature;
- protection-controller supply voltage;
- startup-delay behaviour;
- relay state;
- noise or hum observations.

These results can later be included in the experimental validation section of the report.

38.47 Suggested Commissioning Record

Table 38.1: Suggested initial power-up measurement record.

Measurement	Left Channel	Right Channel
Positive rail	To be measured	To be measured
Negative rail	To be measured	To be measured
Output DC offset	To be measured	To be measured
Bias measurement	To be measured	To be measured
Idle heatsink temperature	To be measured	To be measured
Loaded heatsink temperature	To be measured	To be measured
Relay operation	To be verified	To be verified
DC protection	To be verified	To be verified
Thermal protection	To be verified	To be verified

38.48 Fault Response During Commissioning

If an abnormal condition is detected, continued power application should not be used as a troubleshooting method.

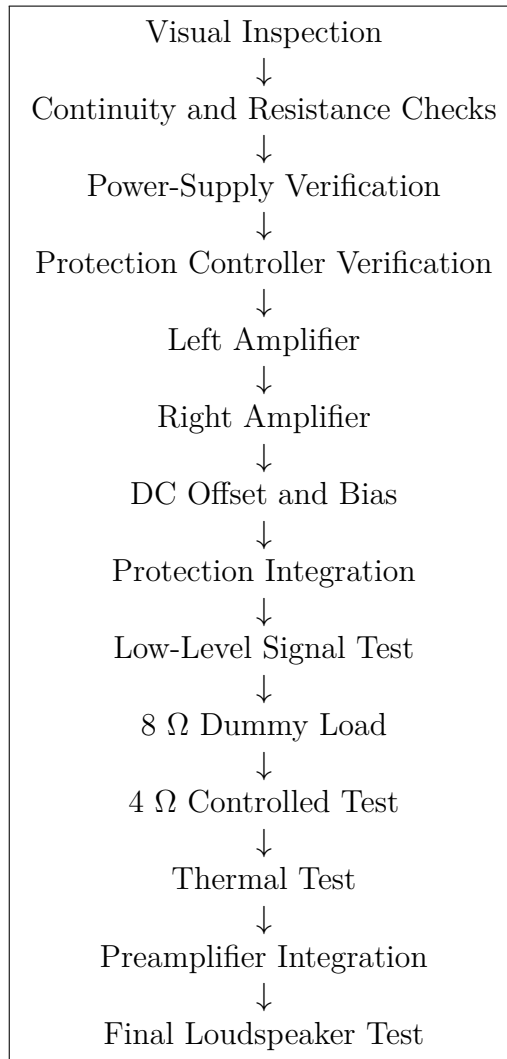
Power should be removed and the relevant subsystem should be inspected.

Examples requiring immediate shutdown include:

- excessive current consumption;
- rapidly heating output transistors;
- significant DC at the amplifier output;
- incorrect rail voltage;
- smoke or unusual smell;
- repeated fuse failure;
- transformer overheating or abnormal noise;
- unstable bias;
- visible oscillation.

38.49 Commissioning Flow

The complete procedure can be summarised as



38.50 Design Summary

The initial power-up is performed progressively rather than energising the complete stereo amplifier at once.

The procedure begins with visual, mechanical, continuity, grounding, polarity, and fuse checks.

The main 24-0-24 V AC supply is then verified, with expected amplifier rails in the approximate

$$\pm 33 - \pm 34 \text{ V}$$

region under light loading.

The protection-controller 12 V supply and STM32 operation are verified separately before the controller is relied upon for loudspeaker protection.

Each power-amplifier channel is then commissioned independently with its input referenced to signal ground and without a loudspeaker connected.

Output DC offset, supply rails, idle current, and transistor temperature are checked before the Class-AB bias is adjusted.

After thermal stabilisation, the protection sensing and speaker-relay functions are integrated and tested.

A low-level signal test is performed before connecting a dummy load. Testing begins with an $8\ \Omega$ load and progresses to controlled $4\ \Omega$ operation only after correct behaviour has been established.

Both amplifier channels are subsequently operated together so that shared power-supply loading and total thermal behaviour can be evaluated.

The preamplifier is integrated only after the two power-amplifier channels have been validated independently.

Loudspeakers are connected only after acceptable DC offset, stable bias, correct protection operation, satisfactory thermal behaviour, and absence of significant hum or oscillation have been confirmed.

All commissioning measurements will be recorded and later used as part of the experimental verification of the completed Hi-Fi amplifier.

Chapter 39

Amplifier Bias Adjustment

39.1 Introduction

The output stage of the power amplifier operates in Class-AB and therefore requires a controlled quiescent current when no audio signal is present.

This current is adjusted using the variable resistor

$$\boxed{VR1 = 2 \text{ k}\Omega}.$$

Correct bias adjustment is important because insufficient quiescent current increases crossover distortion, while excessive quiescent current increases idle power dissipation and output-transistor temperature.

The bias adjustment is therefore performed during initial commissioning after the supply rails and output DC offset have been verified, but before the amplifier is operated at significant output power.

The adjustment procedure described in this chapter is performed independently for the left and right amplifier channels.

39.2 Purpose of the Bias Adjustment

In a Class-B output stage, the positive and negative output devices alternate conduction around the zero crossing of the waveform.

If both sides are completely turned off around this region, a short interval exists during which neither output section conducts adequately.

This produces crossover distortion.

Class-AB operation reduces this effect by maintaining a small standing current through the output stage even when

$$V_{\text{IN}} = 0.$$

This standing current is the quiescent current,

$$I_Q.$$

The objective of the bias adjustment is therefore to obtain sufficient

$$I_Q$$

to minimise crossover distortion without causing unnecessary thermal dissipation.

39.3 Bias Circuit

The adjustable bias network is based around Q9, a BC546 transistor, together with VR1.

Changing the resistance of VR1 changes the voltage developed by the bias network and consequently changes the standing current through the driver and output stage.

The output stage consists of complementary feedback pairs.

The upper pair is formed by

$$Q5 = \text{BD139}$$

and

$$Q7 = \text{2SA1943-O},$$

while the lower pair is formed by

$$Q6 = \text{BD140}$$

and

$$Q8 = \text{2SC5200-O}.$$

The bias behaviour of this complementary-feedback-pair topology differs from that of a conventional complementary emitter-follower output stage, and the adjustment should therefore follow the requirements of the Project 3A circuit rather than a generic amplifier procedure [1].

39.4 Recommended Quiescent Current

The Project 3A design information gives a useful quiescent-current operating range of approximately

$$30 \text{ mA} \leq I_Q \leq 75 \text{ mA},$$

with approximately

$$I_Q \approx 50 \text{ mA}$$

providing an appropriate nominal operating point [1].

A considerably lower current reduces idle power dissipation but progressively increases crossover distortion.

A substantially higher current provides little additional distortion benefit while increasing output-stage heating.

The nominal adjustment target used for commissioning is therefore

$$I_Q = 50 \text{ mA}.$$

39.5 Emitter Resistors

The output stage contains two power emitter resistors,

$$R_{13} = 0.33 \text{ } \Omega$$

and

$$R_{14} = 0.33 \text{ } \Omega.$$

The bias current can be determined from the voltage developed across these resistors.

When the voltage is measured across both resistors, their total resistance is

$$R_{\text{BIAS}} = R_{13} + R_{14}.$$

Therefore,

$$R_{\text{BIAS}} = 0.33 + 0.33 = 0.66 \text{ } \Omega.$$

39.6 Bias Voltage and Current Relationship

Using Ohm's law,

$$V = IR,$$

the quiescent current can be calculated from

$$I_Q = \frac{V_{\text{BIAS}}}{R_{13} + R_{14}}.$$

Since

$$R_{13} + R_{14} = 0.66 \, \Omega,$$

the expression becomes

$$I_Q = \frac{V_{\text{BIAS}}}{0.66}.$$

For the nominal target current of

$$I_Q = 50 \, \text{mA},$$

the corresponding voltage is

$$V_{\text{BIAS}} = I_Q R.$$

Therefore,

$$V_{\text{BIAS}} = 0.050 \times 0.66 = 0.033 \, \text{V}.$$

Thus, the nominal bias target is approximately

$$V_{\text{BIAS}} \approx 33 \, \text{mV}$$

measured across the two $0.33 \, \Omega$ resistors.

39.7 Bias Reference Values

Table 39.1 shows several useful bias voltage and current relationships.

Table 39.1: Quiescent current corresponding to voltage across $R_{13} + R_{14} = 0.66 \, \Omega$.

Measured Voltage	Approximate Quiescent Current
13.2 mV	20 mA
20 mV	30.3 mA
33 mV	50.0 mA
50 mV	75.8 mA

The table demonstrates that very small voltage changes correspond to significant changes in quiescent current.

A multimeter with reliable millivolt resolution is therefore required.

39.8 Required Test Equipment

The bias-adjustment procedure requires only a small amount of test equipment.

The principal equipment is:

- digital multimeter with millivolt resolution;
- insulated adjustment tool for VR1;
- suitable controlled initial power-up arrangement;
- temperature measurement equipment where available.

An oscilloscope is not required for the basic bias-current adjustment, although it can later be used to evaluate crossover distortion and amplifier stability.

39.9 Conditions Before Bias Adjustment

Bias adjustment should not be attempted immediately after construction without first completing the initial electrical checks.

Before adjusting VR1, the following conditions should have been confirmed:

1. The PCB has been inspected for assembly errors.
2. The 2SA1943-O and 2SC5200-O devices are mounted correctly.
3. The BD139 and BD140 driver orientation has been verified.
4. The positive and negative supply rails have been measured.
5. The output DC offset is within a safe region.
6. No output-stage component is heating abnormally.
7. The amplifier input is connected to signal ground.
8. No loudspeaker is connected.

39.10 Initial VR1 Position

Before the amplifier is powered for the first time, VR1 must be placed in the minimum-current starting condition.

For the Project 3A circuit, VR1 should initially be set to its

maximum resistance

position [1].

This is an important commissioning precaution because an incorrect initial setting could produce excessive standing current immediately after power is applied.

The resistance of VR1 should preferably be confirmed with a multimeter before energising the amplifier rather than relying only on the mechanical direction of rotation.

39.11 Amplifier Conditions During Adjustment

Bias adjustment is performed with no audio input signal.

The preferred conditions are therefore

$$V_{\text{IN}} = 0$$

and

$$P_{\text{OUT}} = 0.$$

The audio input should be referenced to signal ground and the loudspeaker should remain disconnected.

Under these conditions, the measured current represents the amplifier's quiescent operating current rather than load current.

39.12 Multimeter Connection

The multimeter should be configured for DC millivolt measurement.

The measurement is made across the two output-stage emitter resistors R13 and R14.

The target measurement is therefore

$$V_{\text{BIAS}} = V_{R13+R14}.$$

The test leads should be positioned securely before adjusting VR1.

Accidental probe movement around the output transistor circuitry should be avoided because the nearby conductors include the high-current supply rails and amplifier output.

39.13 Initial Adjustment Procedure

The recommended adjustment sequence for each amplifier channel is:

1. Disconnect the loudspeaker.
2. Ensure that no input signal is applied.

3. Connect the amplifier input to signal ground.
4. Set VR1 to maximum resistance before applying power.
5. Connect the multimeter across R13 and R14.
6. Select the DC millivolt measurement range.
7. Apply power using the controlled initial power-up procedure.
8. Confirm that the amplifier output DC voltage remains within a safe region.
9. Observe the initial voltage across R13 and R14.
10. Slowly adjust VR1 while observing the multimeter.
11. Increase the bias until the measured voltage approaches

33 mV .

12. Stop adjusting and monitor the value as the amplifier warms.

39.14 Importance of Slow Adjustment

VR1 should be adjusted slowly.

The bias network affects semiconductor current, while transistor and heatsink temperatures respond more slowly than the electrical adjustment.

Rapid adjustment can therefore overshoot the required current before the thermal response becomes obvious.

A useful principle is

Adjust → Observe → Allow Stabilisation → Adjust Again if Required.

39.15 Thermal Stabilisation

The first 33 mV setting should not automatically be treated as the final adjustment.

Bipolar transistor characteristics change as the semiconductor temperature changes.

After the initial bias setting, the amplifier should remain powered at idle while the output devices and heatsink gradually warm.

The measured voltage across R13 and R14 should be monitored during this period.

The adjustment should not be considered final until the measured bias becomes reasonably stable.

39.16 Final Bias Readjustment

After the amplifier reaches a stable idle temperature, the bias voltage should be checked again.

If the value has moved significantly from the nominal

$$33 \text{ mV},$$

VR1 should be readjusted gradually.

The sequence may need to be repeated more than once:

$$\boxed{\text{Adjust} \rightarrow \text{Warm Up} \rightarrow \text{Measure} \rightarrow \text{Readjust.}}$$

This process ensures that the final setting represents the warm operating condition rather than only the cold startup condition.

39.17 Temperature and Bias Drift

A small amount of bias drift during warm-up is expected.

However, continuously increasing current accompanied by rapidly rising heatsink temperature is not normal.

The positive thermal feedback mechanism can be represented as

$$I_Q \uparrow \Rightarrow P_D \uparrow \Rightarrow T_J \uparrow \Rightarrow I_Q \uparrow.$$

If the measured bias rises continuously rather than approaching a stable value, power should be removed and the bias network, transistor installation, and thermal arrangement should be investigated.

39.18 Idle Dissipation

Increasing the quiescent current increases the power consumed by the amplifier even when no audio signal is present.

The approximate relationship is

$$P_Q \propto V_{\text{rail}} I_Q.$$

For this reason, selecting a quiescent current far above the value needed to minimise crossover distortion is undesirable.

It increases:

- transistor junction temperature;

- heatsink temperature;
- power-supply loading;
- long-term thermal stress.

39.19 Bias Current and Crossover Distortion

If the quiescent current is too low, the positive and negative output sections do not overlap sufficiently around the zero crossing.

The resulting crossover region can appear as a small discontinuity in the output waveform. Increasing I_Q initially reduces this distortion.

According to the Project 3A design information, increasing the current significantly beyond approximately

$$50 \text{ mA}$$

provides little additional improvement [1].

The nominal setting is therefore a compromise between

Low Crossover Distortion

and

Reasonable Idle Dissipation

39.20 Effect of the Actual Supply Voltage

The implemented amplifier uses a

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}}$$

transformer.

The expected DC rails are approximately

$\pm 33\text{--}\pm 34 \text{ V}$

under light practical loading.

This is slightly below the nominal

$$\pm 35 \text{ V}$$

reference supply commonly associated with Project 3A.

The final bias must therefore be established by measuring the actual completed amplifier rather than copying a mechanical VR1 position from another build.

The adjustment should be made with the final transformer, bridge rectifier, reservoir capacitors, output transistors, and heatsink installed.

39.21 Left and Right Channel Adjustment

The two amplifier channels should be adjusted independently.

The procedure is first completed for one channel and then repeated for the second.

The target is

$$V_{\text{BIAS,L}} \approx 33 \text{ mV}$$

and

$$V_{\text{BIAS,R}} \approx 33 \text{ mV}.$$

Small differences may occur because of normal component and transistor tolerances.

The goal is not to obtain identical potentiometer positions but to obtain the required electrical quiescent current in each channel.

39.22 Do Not Match VR1 Positions Mechanically

The left and right VR1 potentiometers should not be adjusted simply to the same physical position.

The relationship between potentiometer position and actual bias current depends on component tolerances and semiconductor characteristics.

Therefore,

Bias is set by measurement, not potentiometer position.

39.23 Post-Adjustment DC Offset Check

After the bias has been adjusted, the amplifier output DC offset should be measured again.

The desired condition remains

$$V_{\text{OUT,DC}} \approx 0 \text{ V}.$$

A significant change in output offset after adjustment may indicate a problem that should be investigated before proceeding to loaded testing.

39.24 Post-Adjustment Temperature Check

Both heatsinks should be monitored after the bias setting has stabilised.

At zero input and with no load, the output stage should not exhibit rapid or uncontrolled heating.

The two NTC sensors connected to the STM32 protection controller provide an additional method of monitoring the left and right heatsink temperatures.

Their readings can later be compared with an independent temperature measurement.

39.25 Bias Verification With an Oscilloscope

After the basic current adjustment is complete, crossover behaviour can be examined using a low-level sine-wave signal and an oscilloscope.

The waveform should be observed close to the zero-crossing region.

A pronounced discontinuity around zero can indicate insufficient output-stage bias.

However, the oscilloscope waveform should be treated as a verification tool, not as a replacement for the direct quiescent-current measurement.

39.26 Conditions Requiring Immediate Shutdown

Power should be removed immediately if any of the following occurs during bias adjustment:

- rapidly increasing bias voltage;
- excessive heatsink temperature;
- substantial DC voltage at the amplifier output;
- smoke or unusual smell;
- abnormal transformer noise;
- repeated fuse failure;
- unstable or uncontrollable VR1 response;
- sudden large change in supply current.

The fault should be identified before another adjustment is attempted.

39.27 Bias Adjustment Record

The final bias settings should be recorded for both amplifier channels.

Table 39.2: Suggested amplifier bias commissioning record.

Measurement	Left Channel	Right Channel
Cold bias voltage	To be measured	To be measured
Cold calculated I_Q	To be calculated	To be calculated
Stabilised bias voltage	To be measured	To be measured
Stabilised I_Q	To be calculated	To be calculated
Output DC offset	To be measured	To be measured
Idle heatsink temperature	To be measured	To be measured
Positive rail voltage	To be measured	To be measured
Negative rail voltage	To be measured	To be measured

39.28 Calculating the Measured Quiescent Current

The measured bias voltage can be converted directly into quiescent current.

For example, if a completed channel produces

$$V_{\text{BIAS}} = 31 \text{ mV},$$

then

$$I_Q = \frac{31 \times 10^{-3}}{0.66}.$$

Therefore,

$$I_Q \approx 46.97 \text{ mA}.$$

Similarly, a measured bias voltage of

$$35 \text{ mV}$$

would correspond to

$$I_Q = \frac{35 \times 10^{-3}}{0.66} \approx 53.0 \text{ mA}.$$

This demonstrates why recording the measured voltage is useful even when the target is expressed primarily in terms of quiescent current.

39.29 Final Acceptance Criteria

The bias adjustment can be considered satisfactory when:

- the quiescent current is within the intended Project 3A operating region;
- the nominal setting is close to 50 mA;
- the value remains stable after thermal warm-up;
- the output DC offset remains acceptable;
- neither amplifier channel exhibits abnormal heating;
- the supply rails remain normal;
- the bias can be adjusted smoothly using VR1.

Only after these conditions have been verified should the amplifier proceed to higher-power dummy-load testing.

39.30 Relationship to Later Measurements

Correct bias adjustment must be completed before meaningful distortion, frequency-response, or output-power measurements are performed.

If the amplifier is significantly under-biased, crossover distortion can artificially increase the measured distortion.

If it is over-biased, excessive thermal dissipation can alter the operating temperature and reduce the safety margin of the output stage.

Bias adjustment therefore forms part of the commissioning process rather than being considered an optional final tuning step.

39.31 Design Summary

The Project 3A-based amplifier uses an adjustable Class-AB bias circuit based around Q9 and the

2 k Ω

VR1 potentiometer.

Before the first power-up, VR1 is set to its maximum-resistance position so that the output stage begins at minimum quiescent current.

The output-stage bias is then measured using the voltage across

$$R_{13} + R_{14} = 0.33 + 0.33 = 0.66 \, \Omega.$$

The relationship between the measured voltage and quiescent current is

$$I_Q = \frac{V_{\text{BIAS}}}{0.66}.$$

The nominal Project 3A setting is approximately

$$I_Q = 50 \, \text{mA},$$

which corresponds to approximately

$$V_{\text{BIAS}} = 33 \, \text{mV}$$

across the two emitter resistors.

The useful Project 3A quiescent-current region is approximately

$$30 \, \text{mA} \leq I_Q \leq 75 \, \text{mA},$$

with substantially lower bias increasing crossover distortion and excessive bias increasing idle dissipation.

Each stereo channel is adjusted independently with no loudspeaker connected, no input signal applied, and the amplifier input referenced to signal ground.

After the initial adjustment, the amplifier is allowed to reach a stable idle temperature. The voltage across the emitter resistors is then measured again and VR1 is readjusted if necessary.

The final bias setting is determined by electrical measurement rather than by the physical position of the potentiometer.

The completed bias adjustment will be documented for both amplifier channels, including the measured bias voltage, calculated quiescent current, output DC offset, rail voltages, and heatsink temperature.

After stable bias operation has been verified, the amplifier can proceed to dummy-load, distortion, output-power, and thermal testing.

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Chapter 40

Testing and Measurements

After completion of the schematic design, PCB layout, firmware development, and assembly of the power amplifier and protection controller, a systematic testing procedure is required to verify the electrical performance, stability, thermal behaviour, and safety of the complete system. Testing is particularly important in a high-power audio amplifier because faults such as excessive DC offset, oscillation, overheating, incorrect biasing, or protection-system malfunction can potentially damage both the amplifier and the connected loudspeaker.

The testing procedure was therefore divided into several stages. Initial tests were carried out without a loudspeaker connected to the output. Where possible, a resistive dummy load was used instead of a loudspeaker so that the amplifier could be operated at high output power without risking damage to an actual loudspeaker.

The main parameters investigated during testing were:

- DC voltage at the loudspeaker output;
- small-signal and large-signal frequency response;
- maximum available output voltage and output power;
- waveform clipping behaviour;
- total harmonic distortion;
- heatsink and transistor temperature;
- operation of the STM32-based loudspeaker protection system;
- power-on delay operation;
- DC fault detection;
- over-temperature protection;
- relay disconnection and reconnection behaviour;
- general stability of the amplifier under load.

40.1 Test Equipment

The main equipment required for the measurements includes a digital multimeter, oscilloscope, signal generator, resistive dummy load, laboratory thermometer or temperature measurement system, and the STM32 protection controller developed as part of this project.

A sine-wave signal was used for most AC measurements because it provides a well-defined waveform from which gain, clipping, distortion, frequency response, and maximum output voltage can be determined.

For high-power measurements, an appropriately rated non-inductive dummy-load resistor should be connected to the amplifier output. Depending on the test configuration, typical load values are:

$$R_L = 8 \, \Omega \quad (40.1)$$

or, where required,

$$R_L = 4 \, \Omega. \quad (40.2)$$

The power rating of the dummy load must be significantly greater than the expected amplifier output power because a large amount of electrical energy is converted directly into heat during testing.

40.2 Initial Power-Up Procedure

Before performing full-power measurements, the amplifier was visually inspected for solder bridges, incorrect transistor orientation, reversed electrolytic capacitors, connector errors, and possible short circuits between the positive and negative supply rails.

The amplifier was initially powered without a loudspeaker connected. The main supply rails were measured and verified before the loudspeaker protection relay was permitted to connect the output.

For the amplifier developed in this project, a 25 – 0 – 25 VAC toroidal transformer is used. After bridge rectification and reservoir filtering, the resulting unloaded DC supply rails are approximately:

$$V_{DC} \approx V_{AC}\sqrt{2} \quad (40.3)$$

which gives approximately:

$$V_{DC} \approx 25\sqrt{2} \approx 35.4 \, \text{V}. \quad (40.4)$$

The actual voltage is slightly lower because of the forward voltage drop across the bridge rectifier and the loading of the power supply. Under amplifier load, the supply voltage decreases further due to transformer regulation and ripple voltage.

Only after the supply rails, DC offset, protection controller, and basic amplifier operation had been verified was the system operated at increasing output power.

40.3 Measurement Safety

Special care is required during these measurements because the system contains both mains voltage and high-current DC power rails. The mains-side circuitry must never be modified or probed while the amplifier is energised unless suitable isolation and safety procedures are used.

The amplifier output should also never be short-circuited during high-power testing. Oscilloscope ground connections must be made carefully because most conventional oscilloscopes have an earth-referenced ground connection.

The testing process therefore followed a progressive approach, beginning with low-voltage and no-load measurements before moving to high-power operation.

Chapter 41

DC Offset Testing

One of the most important initial measurements of a direct-coupled audio power amplifier is the DC voltage present at the loudspeaker output. Ideally, the average DC output voltage should be approximately 0 V. A large DC component can cause a continuous current through the loudspeaker voice coil and may result in overheating or permanent loudspeaker damage.

41.1 Measurement Procedure

The amplifier input was first shorted to signal ground so that no AC input signal was applied. No loudspeaker was connected during the initial test.

A digital multimeter was configured to measure DC voltage and connected between the amplifier output and signal ground. The DC output voltage was then recorded after the amplifier had completed its initial power-on stabilisation period.

The measurement was repeated after the amplifier had warmed up in order to determine whether thermal changes in the transistor stages significantly affected the output offset.

Table 41.1: DC output offset measurements.

Operating Condition	Measured DC Offset
Initial power-up	<i>[Insert measured value]</i> mV
After warm-up	<i>[Insert measured value]</i> mV

41.2 Importance of Low DC Offset

The DC current through a loudspeaker caused by an output offset can be estimated using Ohm's law:

$$I_{DC} = \frac{V_{DC}}{R_L}. \quad (41.1)$$

For example, if an 8 Ω loudspeaker were exposed to a 1 V DC offset,

$$I_{DC} = \frac{1}{8} = 0.125 \text{ A.} \quad (41.2)$$

The corresponding power dissipated continuously in the voice coil would be:

$$P_{DC} = \frac{V_{DC}^2}{R_L}. \quad (41.3)$$

Although a small offset in the millivolt range is normally harmless, a fault in the power amplifier output stage could place several volts or potentially a large fraction of the power-supply voltage directly across the loudspeaker.

41.3 Interaction with the Protection Controller

The STM32 speaker protection controller developed for this project continuously monitors the amplifier output for abnormal DC voltage. Under normal conditions, the measured offset should remain below the configured protection threshold and the loudspeaker relay remains energised.

If an excessive positive or negative DC voltage is detected for longer than the firmware qualification time, the microcontroller identifies a fault condition and immediately disconnects the loudspeaker relay.

This hardware and firmware protection provides an additional layer of safety in case the amplifier output stage fails or becomes significantly unbalanced.

Chapter 42

Frequency Response Testing

Frequency-response testing was performed to determine whether the amplifier maintains approximately constant voltage gain throughout the intended audio-frequency range. An audio power amplifier should reproduce signals from approximately 20 Hz to 20 kHz without significant attenuation or amplification variation.

42.1 Measurement Method

A sinusoidal input signal was applied to the amplifier using a signal generator. The input level was kept sufficiently low to ensure that the output stage remained well below clipping throughout the measurement.

The input amplitude was maintained constant while the signal frequency was varied across the audio range. The amplifier output voltage was measured using an oscilloscope or true-RMS digital measurement instrument.

Suggested measurement frequencies include:

$$20, 50, 100, 500, 1000, 5000, 10000, 20000 \text{ Hz.} \quad (42.1)$$

Additional points may be recorded above and below the audible range in order to estimate the amplifier's lower and upper cutoff frequencies.

42.2 Voltage Gain

The amplifier voltage gain is calculated from:

$$A_v = \frac{V_{out}}{V_{in}}. \quad (42.2)$$

The corresponding gain in decibels is:

$$A_{v,dB} = 20 \log_{10} \left(\frac{V_{out}}{V_{in}} \right). \quad (42.3)$$

The gain at 1 kHz may be used as the reference value for the frequency-response measurement.

42.3 Frequency-Response Measurements

Table 42.1: Measured amplifier frequency response.

Frequency	Output Voltage	Relative Gain
20 Hz	[Value]	[Value] dB
50 Hz	[Value]	[Value] dB
100 Hz	[Value]	[Value] dB
500 Hz	[Value]	[Value] dB
1 kHz	[Value]	0 dB
5 kHz	[Value]	[Value] dB
10 kHz	[Value]	[Value] dB
20 kHz	[Value]	[Value] dB

42.4 Bandwidth

The conventional bandwidth of an amplifier is determined from the frequencies at which its gain decreases by 3 dB relative to the mid-band value.

A -3 dB reduction corresponds to an output voltage equal to approximately 0.707 times the reference voltage:

$$V_{-3dB} = 0.707V_{\text{reference}}. \quad (42.4)$$

The lower and upper cutoff frequencies can therefore be identified experimentally from the measured response.

Figure 42.1: Measured frequency response of the completed power amplifier.

The final measured curve should demonstrate that the amplifier maintains a substantially flat response throughout the audible frequency range.

Chapter 43

Output Power Testing

Output-power testing determines the maximum continuous sinusoidal power that the amplifier can deliver to a specified load before significant waveform clipping occurs.

43.1 Test Configuration

For this measurement, a resistive dummy load was connected to the amplifier output instead of a loudspeaker. A sinusoidal input signal with a frequency of 1 kHz was applied to the amplifier.

The input signal amplitude was gradually increased while observing the output waveform on an oscilloscope. The highest output level immediately before visible clipping was recorded.

43.2 RMS Output Voltage

For a sinusoidal waveform, the RMS voltage is related to the peak voltage by:

$$V_{RMS} = \frac{V_{peak}}{\sqrt{2}}. \quad (43.1)$$

If the measurement is made using the peak-to-peak output voltage,

$$V_{RMS} = \frac{V_{pp}}{2\sqrt{2}}. \quad (43.2)$$

43.3 Output Power Calculation

The continuous output power delivered to a resistive load is:

$$P_{out} = \frac{V_{RMS}^2}{R_L}. \quad (43.3)$$

For example, an output of 25 V_{RMS} across an 8 Ω load would correspond to:

$$P_{out} = \frac{25^2}{8} = 78.125 \text{ W.} \quad (43.4)$$

The corresponding peak output voltage would be:

$$V_{peak} = 25\sqrt{2} \approx 35.36 \text{ V.} \quad (43.5)$$

The actual maximum power of the completed amplifier depends on the loaded supply voltage, transistor voltage drops, output-stage current capability, heatsink temperature, and the selected load impedance.

43.4 Measured Results

Table 43.1: Measured maximum amplifier output power.

Load	V_{RMS}	V_{pp}	Output Power
8 Ω	[Value] V	[Value] V	[Value] W
4 Ω	[Value] V	[Value] V	[Value] W

43.5 Clipping Behaviour

As the input voltage is increased beyond the linear operating region, the output waveform eventually approaches the available positive and negative supply rails. The tops and bottoms of the sine wave then begin to flatten, indicating clipping.

The maximum useful continuous output power is therefore defined using the largest approximately undistorted sine wave rather than the absolute largest voltage that can be obtained from the amplifier.

Figure 43.1: Amplifier output waveform immediately below clipping.

Chapter 44

Distortion Testing

Distortion testing evaluates the extent to which the waveform produced by the amplifier differs from the original input waveform. Ideally, an amplifier should increase only the amplitude of the input signal without modifying its waveform.

44.1 Total Harmonic Distortion

A practical audio amplifier generates small harmonic components due to transistor non-linearity, crossover behaviour, finite open-loop gain, bias conditions, and other circuit effects.

Total harmonic distortion can be expressed as:

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots + V_n^2}}{V_1}, \quad (44.1)$$

where V_1 represents the RMS amplitude of the fundamental frequency and V_2 , V_3 , and subsequent terms represent the RMS amplitudes of the harmonic components.

The result is normally expressed as a percentage:

$$THD(\%) = 100 \times THD. \quad (44.2)$$

44.2 Measurement Procedure

A low-distortion 1 kHz sine wave was applied to the amplifier input. The amplifier was operated at several output levels ranging from low power to near its maximum unclipped output.

The output signal can be analysed using an audio analyser, digital oscilloscope with FFT capability, or computer-based audio measurement system.

A frequency spectrum obtained from the output waveform allows the amplitude of the fundamental and harmonic components to be identified.

Table 44.1: Amplifier distortion measurements.

Output Power	Test Frequency	Measured THD
[Value] W	1 kHz	[Value] %
[Value] W	1 kHz	[Value] %
Maximum unclipped power	1 kHz	[Value] %

44.3 Distortion Near Clipping

THD normally remains relatively low while the amplifier operates within its linear region. As the output approaches the supply-rail limits, clipping begins and the harmonic content increases rapidly.

For this reason, output-power and distortion measurements should be considered together. A quoted output-power value is only meaningful when the corresponding distortion level remains acceptable.

Figure 44.1: FFT spectrum of the amplifier output during distortion testing.

Chapter 45

Thermal Testing

Thermal testing is essential for a Class-AB power amplifier because a significant fraction of the electrical input power is dissipated as heat in the output transistors. Excessive junction temperature reduces semiconductor reliability and can eventually cause permanent device failure.

45.1 Heat Generation in the Output Stage

The complementary output transistor stage used in the amplifier conducts substantial current when driving low-impedance loads. The difference between the instantaneous supply voltage and output voltage appears across the power transistors and results in power dissipation.

The transistor junction temperature may be approximated by the thermal-resistance relationship:

$$T_J = T_A + P_D (R_{\theta JC} + R_{\theta CS} + R_{\theta SA}), \quad (45.1)$$

where:

- T_J is the transistor junction temperature;
- T_A is the ambient temperature;
- P_D is the power dissipated by the transistor;
- $R_{\theta JC}$ is the junction-to-case thermal resistance;
- $R_{\theta CS}$ is the case-to-heatsink thermal resistance;
- $R_{\theta SA}$ is the heatsink-to-ambient thermal resistance.

45.2 Thermal Test Procedure

The amplifier was operated continuously into a dummy load while an audio-frequency sine wave was applied. Temperature was monitored at the heatsink near the output transistors.

The temperature was recorded periodically until either thermal equilibrium was reached or the test was stopped because the temperature approached the configured protection threshold.

Table 45.1: Example thermal-test measurement table.

Time	Heatsink Temperature	Operating Condition
0 min	[Value] °C	Start
5 min	[Value] °C	Continuous load
10 min	[Value] °C	Continuous load
15 min	[Value] °C	Continuous load
20 min	[Value] °C	Continuous load

45.3 Temperature Monitoring

The protection-controller PCB includes a temperature-monitoring system connected to the STM32G031K8 microcontroller. The temperature-sensing circuit is used to detect excessive heatsink temperature independently of the audio signal.

The firmware continuously evaluates the measured temperature. For the final protection configuration used in this project, a critical over-temperature threshold of approximately:

$$T_{\text{FAULT}} = 120^{\circ}\text{C} \quad (45.2)$$

is used.

When this limit is reached, the controller places the amplifier in a thermal-fault state and opens the loudspeaker relay. The loudspeaker is therefore disconnected before continued overheating can expose it to abnormal amplifier behaviour.

45.4 Cooling System

Large heatsinks are used for the output transistors to transfer heat from the semiconductor packages to the surrounding air. Forced-air cooling can additionally be provided by the fans included in the mechanical design.

The heatsink must provide good thermal contact with the power transistors. Thermal compound or suitable thermal interface material should be applied between the transistor and heatsink where appropriate. Electrical insulation must also be provided if required by the transistor package and mounting arrangement.

Figure 45.1: Measured heatsink temperature during continuous operation.

Chapter 46

Protection System Testing

The loudspeaker protection system is a critical part of the complete amplifier because it provides isolation between the power-amplifier output and the loudspeaker whenever unsafe operating conditions are detected.

The protection controller designed in this project is based on the STM32G031K8 microcontroller. Its firmware monitors the amplifier operating condition and controls a loudspeaker relay.

46.1 Power-On Delay Test

During power-up, amplifier internal voltages require a short time to stabilise. Connecting the loudspeaker immediately may result in an audible transient or “pop”.

The firmware therefore keeps the loudspeaker relay disconnected during the initial power-on period.

The power-on test consisted of energising the complete amplifier while monitoring the relay-control signal and amplifier output using an oscilloscope.

The expected sequence is:

1. power is applied to the amplifier;
2. the STM32 controller starts;
3. the loudspeaker relay remains open;
4. the controller monitors the DC-offset and temperature inputs;
5. the configured delay period expires;
6. if no fault is detected, the relay is energised;
7. the loudspeaker output is connected.

Table 46.1: Power-on protection test.

Parameter	Measured Result
Relay activation delay	[Value] s
Output stable before connection	Yes / No
Correct status indication	Yes / No

46.2 DC Fault Protection Test

The DC protection function was tested by applying a controlled test voltage to the DC-detection input of the protection controller rather than intentionally creating a destructive fault in the power amplifier.

Both positive and negative fault conditions should be tested because an amplifier output-stage failure may place either polarity of the supply voltage on the loudspeaker output.

The controller should disconnect the relay once the measured DC level remains outside the permitted range for the configured qualification period.

Table 46.2: DC protection test results.

Test Condition	Relay State	Result
Normal output	Connected	Pass / Fail
Positive DC fault	Disconnected	Pass / Fail
Negative DC fault	Disconnected	Pass / Fail

46.3 Over-Temperature Protection Test

The thermal-protection function was tested by increasing the temperature detected by the sensor or by electrically simulating the corresponding sensor voltage.

Under normal temperature conditions, the relay should remain connected. When the measured temperature reaches the configured critical limit of approximately 120°C, the firmware should enter the over-temperature fault state.

The required response is:

1. detect that the temperature limit has been exceeded;
2. disable the loudspeaker relay;
3. illuminate or activate the corresponding fault indication;
4. prevent loudspeaker reconnection while the fault remains present.

Table 46.3: Thermal protection test results.

Temperature Condition	Relay State	Result
Normal temperature	Connected	Pass / Fail
Near warning threshold	[State]	Pass / Fail
$\geq 120^{\circ}\text{C}$	Disconnected	Pass / Fail

46.4 Fault Indicator Testing

The status LEDs were checked during each operating condition. The LED indications provide immediate visual information concerning the controller state and simplify both normal operation and troubleshooting.

The following operating states should be verified:

- controller powered;
- normal amplifier operation;
- startup delay;
- DC-offset fault;
- over-temperature fault;
- loudspeaker relay disconnected.

46.5 Relay Fail-Safe Operation

The relay is used as the final physical disconnection element between the amplifier and loudspeaker. The control strategy should ensure that the loudspeaker is disconnected whenever the microcontroller detects an abnormal condition.

The relay was therefore repeatedly activated and deactivated while monitoring continuity through the loudspeaker-output path. This verified correct relay-driver operation and confirmed that the firmware state matched the physical relay state.

46.6 Protection-System Test Summary

Table 46.4: Summary of protection-system verification.

Protection Function	Result
Power-on delay	Pass / Fail
Positive DC fault detection	Pass / Fail
Negative DC fault detection	Pass / Fail
Over-temperature detection	Pass / Fail
Relay disconnection	Pass / Fail
Status LED operation	Pass / Fail
Fault recovery	Pass / Fail

Chapter 47

Troubleshooting and Design Corrections

The development of the complete power amplifier required several iterations involving the schematic, PCB layout, component selection, firmware, and mechanical arrangement. Troubleshooting was therefore treated as part of the engineering process rather than only as a final repair stage.

47.1 Schematic Verification

Before PCB fabrication, the power-amplifier schematic and protection-controller schematic were checked using the KiCad Electrical Rules Checker. Particular attention was given to power connections, transistor pin assignments, component values, connector orientation, and ground connections.

Warnings caused by duplicated references, inconsistent component values, or unconnected pins were investigated individually rather than automatically ignored.

47.2 PCB Footprint Verification

A major design check involved confirming that the selected physical components matched the assigned KiCad footprints.

This was especially important for:

- electrolytic capacitors;
- high-power resistors;
- output transistors;
- bridge rectifier;
- relay;

- connectors;
- fuse holders;
- temperature sensor;
- STM32 microcontroller;
- mounting holes.

Component pitch, lead spacing, body dimensions, pad size, pin numbering, and package orientation were compared against manufacturer drawings and the PCB footprint.

47.3 Output Transistor Selection

The output transistor selection was revised during the development process as component availability and package compatibility were investigated.

The final output stage uses the complementary Toshiba transistor pair:

- 2SC5200-O – NPN power transistor;
- 2SA1943-O – PNP power transistor.

These devices use high-power through-hole packages suitable for mounting directly to the external heatsinks.

The PCB transistor orientation was also considered mechanically so that the transistor mounting surfaces face the heatsink and can be secured without unnecessarily long connections.

47.4 High-Current PCB Routing

High-current paths were reviewed separately from low-level signal paths. The power-supply connections, output-transistor current paths, loudspeaker output, and ground returns require significantly wider copper tracks than small-signal circuitry.

Where possible, high-current tracks were shortened and widened. Copper pours were also used for ground distribution where appropriate.

The PCB design was evaluated for fabrication using 1 oz/ft² copper. Track widths were selected according to the expected continuous and transient currents, with particular attention given to the output stage and power-supply connections.

47.5 Grounding Corrections

Grounding is especially important in a power amplifier because the same circuit contains both millivolt-level input signals and several-ampere output currents.

If high-current loudspeaker or reservoir-capacitor return currents share the same narrow PCB path as the input signal reference, voltage drops across the copper can be introduced into the audio path and produce hum, distortion, or instability.

The layout was therefore organised so that small-signal ground, power ground, and loudspeaker return currents are controlled and joined at an appropriate common reference point.

47.6 Power-Supply Connections

The amplifier power supply was designed around a $25 - 0 - 25$ VAC toroidal transformer followed by a high-current bridge rectifier and large reservoir capacitors.

The PCB power nets were clearly identified according to their function, including the positive rail, negative rail, and ground reference. Particular care was taken to prevent confusion between AC transformer connections and the rectified DC supply rails.

47.7 Mechanical and Thermal Corrections

The placement of high-power transistors was reviewed together with the heatsink arrangement. The final orientation allows the devices to be mechanically coupled to the heatsinks while keeping the PCB interconnections short.

Mounting holes were incorporated into the PCB to permit secure installation inside the amplifier enclosure.

Cooling airflow was also considered because the project includes forced-air cooling in addition to passive heatsinks.

47.8 Protection Controller Firmware Debugging

The STM32 protection firmware was developed as several source and header files rather than placing all functionality directly in `main.c`. This provides better modularity and makes individual functions easier to test and modify.

The firmware is divided into modules responsible for:

- protection-state logic;
- sensor acquisition;
- relay control;
- GPIO interfaces;
- hardware pin definitions;
- configurable protection thresholds.

During debugging, firmware behaviour was verified under normal operation and simulated fault conditions. Particular attention was given to avoiding false relay activation during startup and ensuring that a detected fault always results in loudspeaker disconnection.

47.9 Temperature Cut-Off Correction

The over-temperature protection threshold was configured so that the loudspeaker is disconnected when the measured temperature reaches approximately:

$$120^{\circ}\text{C}. \quad (47.1)$$

The relevant firmware comparison and fault state were verified to ensure that the relay is opened rather than merely displaying an over-temperature indication.

47.10 Gerber and Manufacturing File Verification

Before manufacturing, the PCB production files were generated directly from KiCad. The manufacturing package contains both the Gerber layer files and the NC drill files.

The Gerber files define the copper, solder mask, silkscreen, and board outline, while the drill files define plated and non-plated holes.

The generated files were inspected using the KiCad Gerber viewer before being archived and uploaded to the project repository.

47.11 Repository and Documentation Corrections

The GitHub repository was organised to separate the main categories of project data. Dedicated directories were created for:

- KiCad source files;
- schematics;
- PCB layouts;
- fabrication files;
- component libraries;
- bill-of-materials files;
- STM32 firmware;
- project documentation.

This organisation ensures that the complete design can be reconstructed and reviewed without relying on files stored only on the original development computer.

47.12 Final Verification

After each design correction, the affected part of the system was tested again. A modification was therefore not considered complete simply because the schematic or firmware compiled successfully; its effect on the real system also had to be verified.

The final troubleshooting process included:

1. visual inspection;
2. continuity and short-circuit checks;
3. power-rail verification;
4. DC-offset measurement;
5. small-signal amplifier testing;
6. protection-controller testing;
7. thermal testing;
8. dummy-load testing;
9. high-power testing;
10. final loudspeaker testing.

This progressive verification approach reduces the probability that a single assembly or design error can damage the amplifier, protection electronics, or connected loudspeaker.

Chapter 48

Future Improvements

48.1 Introduction

The present project provides a complete design for a stereo Hi-Fi power amplifier consisting of two discrete Class-AB power-amplifier channels, a shared linear power supply, an audio preamplifier, an STM32-based loudspeaker protection controller, temperature monitoring, relay-based speaker disconnection, and a complete mechanical and grounding concept.

Although the current design satisfies the principal objectives of the project, several improvements could be investigated in future revisions. These improvements are not required for the operation of the present amplifier, but could increase performance, reliability, usability, protection capability, manufacturability, and diagnostic functionality.

The possible improvements can be divided into the following areas:

- power-amplifier optimisation;
- power-supply improvement;
- protection-system expansion;
- firmware development;
- thermal-management improvement;
- noise and electromagnetic compatibility;
- preamplifier development;
- user-interface improvement;
- PCB refinement;
- mechanical and chassis development;
- monitoring and diagnostic capability;
- experimental characterisation.

48.2 Further Power-Amplifier Optimisation

The present power-amplifier topology is based on the proven Elliott Project 3A architecture. A future revision could investigate whether additional optimisation of the discrete amplifier stage could improve measurable performance while preserving the simplicity and reliability of the original design.

Possible areas of investigation include:

- improved matching of the differential input transistors;
- tighter tolerance components in the feedback network;
- optimisation of the bias network;
- improved thermal coupling of the bias-control transistor;
- optimisation of compensation components;
- investigation of alternative driver devices;
- detailed safe-operating-area analysis of the output transistors.

Any circuit modification should be evaluated carefully because improving one parameter, such as bandwidth, can reduce stability margin or increase susceptibility to oscillation.

48.3 Input Transistor Matching

The differential input stage strongly influences the DC offset and small-signal behaviour of the amplifier.

In a future build, Q1 and Q2 could be selected from a larger group of BC546 transistors and matched according to parameters such as current gain and base-emitter voltage.

Improved matching could reduce the natural imbalance of the differential input stage and potentially reduce the output DC offset.

The effectiveness of this modification should be evaluated experimentally rather than assumed.

48.4 Automatic or Assisted Bias Monitoring

The present amplifier bias is adjusted manually using VR1 and verified from the voltage across the output-stage emitter resistors.

A future protection-controller revision could monitor the amplifier idle condition indirectly and provide a maintenance warning if abnormal thermal behaviour is detected.

A more advanced design could include dedicated low-voltage differential measurement circuitry across the emitter resistors.

This could allow the microcontroller to estimate quiescent current using

$$I_Q = \frac{V_R}{R_E}.$$

Such a system could provide diagnostic information but should not automatically control the analogue bias unless the additional control circuit had been carefully designed and validated.

48.5 Power-Supply Soft Start

The main amplifier uses a relatively high-power toroidal transformer and large reservoir capacitance.

At switch-on, the transformer and discharged capacitors can produce a high inrush current.

A future revision could include a dedicated soft-start circuit.

A typical soft-start architecture could use

Mains → Current-Limiting Element → Transformer

during the first part of startup, followed by a relay or other appropriately rated switching element that bypasses the limiter after a controlled delay.

This could reduce stress on:

- the mains switch;
- primary fuse;
- transformer;
- rectifier;
- reservoir capacitors.

48.6 Improved Mains Filtering

A future chassis revision could incorporate an appropriately rated mains EMI filter at the power entry point.

The purpose would be to reduce both conducted interference entering the amplifier and unwanted high-frequency interference returning to the mains.

Any mains filter would need to be implemented using components specifically rated and approved for direct connection to the AC mains supply.

48.7 Separate Channel Power Supplies

The present stereo amplifier uses a shared main power supply.

A more ambitious future version could investigate dual-mono construction, in which the left and right channels use separate transformers and reservoir capacitor banks.

The architecture would become

Left Transformer → Left PSU → Left Amplifier
Right Transformer → Right PSU → Right Amplifier

Potential advantages include reduced channel interaction and improved supply independence.

However, this would substantially increase cost, weight, chassis size, and power-supply complexity.

48.8 Protection Controller Expansion

The current STM32 protection controller monitors amplifier DC offset and heatsink temperature and controls the two loudspeaker relays.

The microcontroller architecture makes it possible to add further protection functions in a future hardware revision.

Possible additional monitoring functions include:

- amplifier supply-rail monitoring;
- undervoltage detection;
- abnormal power-supply asymmetry detection;
- additional internal temperature sensors;
- fan-failure detection;
- relay-state monitoring;
- amplifier current monitoring;
- power-on event logging.

48.9 Supply-Rail Monitoring

Additional ADC channels could be used to monitor the positive and negative amplifier supply rails through appropriately designed scaling and protection networks.

The controller could then compare

$$|V_+|$$

and

$$|V_-|.$$

A substantial difference could indicate a fuse failure, rectifier problem, wiring fault, or abnormal loading of one rail.

This would extend the controller from speaker protection toward broader amplifier health monitoring.

48.10 Output Current Monitoring

A future protection system could also incorporate amplifier output-current monitoring.

This could be implemented using an appropriately selected current-sensing method designed to introduce very little additional impedance into the loudspeaker path.

The controller could then identify abnormal sustained current conditions or possible short-circuit events.

Such a system would require careful analogue design because the speaker path carries several amperes and must remain suitable for high-quality audio operation.

48.11 Relay Contact Monitoring

The present controller commands the speaker relays but does not directly confirm the mechanical state of the contacts.

A future design could include relay-state feedback or diagnostic monitoring.

This could help identify conditions such as:

- relay coil failure;
- driver transistor failure;
- contact failure;
- unexpected relay disconnection.

48.12 Protection Firmware Improvements

The STM32 firmware provides a significant opportunity for future development without changing the basic amplifier circuitry.

Possible firmware improvements include:

- configurable startup delay;
- configurable DC-fault persistence time;
- adjustable temperature thresholds;
- improved digital filtering;
- event counters;
- fault-history storage;
- startup self-test;
- relay-cycle counting;
- temperature logging;
- serial diagnostic output.

Configuration constants should remain bounded to safe operating ranges so that firmware modification cannot accidentally disable essential protection.

48.13 Fault History

Future firmware could record the reason for each protection event.

For example, the controller could maintain counters for

$$N_{\text{DC}}, \quad N_{\text{OT}}, \quad N_{\text{SENSOR}},$$

representing DC faults, overtemperature events, and sensor failures.

Such information would be useful during troubleshooting because intermittent faults may disappear before the amplifier can be inspected.

48.14 Non-Volatile Storage

Selected diagnostic information could be stored in non-volatile memory.

The amplifier could therefore retain the previous fault condition even after the mains supply has been removed.

On the next startup, the controller could report the previous shutdown reason before clearing or updating the stored information.

48.15 Serial Diagnostic Interface

A future firmware version could expose diagnostic information through a UART or USB serial connection.

This could allow a computer to display:

- left heatsink temperature;
- right heatsink temperature;
- left DC offset;
- right DC offset;
- relay status;
- current protection state;
- previous fault;
- supply voltages where additional sensing is implemented.

Such functionality would be particularly useful during development and maintenance.

48.16 Improved Temperature Measurement

The current design uses two 10 k Ω NTC sensors attached directly to the heatsinks.

Future improvement could include calibration of each sensor channel against a known temperature reference.

The firmware could then compensate for:

- NTC tolerance;
- divider resistor tolerance;
- ADC reference tolerance;
- sensor mounting differences.

This would improve absolute temperature accuracy.

48.17 Adaptive Fan Control

The current thermal design includes four AC cooling fans.

A future revision could replace these with suitable low-voltage DC fans and allow the STM32 to control fan speed according to heatsink temperature.

A possible control strategy would be

$$T < T_1 \Rightarrow \text{Fans Off or Minimum Speed,}$$

$$T_1 \leq T < T_2 \Rightarrow \text{Medium Cooling,}$$

and

$$T \geq T_2 \Rightarrow \text{Maximum Cooling.}$$

This could reduce acoustic noise during low-power listening while preserving maximum airflow when the amplifier is operated heavily.

48.18 Fan Failure Detection

If future fans provide tachometer outputs, the protection controller could monitor actual fan rotation.

A requested fan speed without a valid tachometer signal could then be treated as a cooling-system fault.

The amplifier could respond by issuing a warning or disconnecting the speakers if the heatsink temperature continued to rise.

48.19 Improved User Interface

The present protection system uses discrete status LEDs.

A future version could incorporate a small display to provide more detailed information without requiring a large number of separate indicators.

Possible displayed information includes:

- left heatsink temperature;
- right heatsink temperature;
- startup countdown;
- speaker relay state;
- DC protection status;

- previous fault;
- system operating time.

48.20 Retaining Independent Fault LEDs

Even if a display were added, the most important fault states could continue to use dedicated LEDs.

A dedicated indicator can be interpreted immediately and does not depend on a display menu or software screen.

The display should therefore supplement rather than necessarily replace the essential protection indications.

48.21 Remote Control

A future preamplifier and controller revision could provide remote control for functions such as:

- mute;
- input selection;
- standby;
- volume, if an electronically controlled volume stage is introduced.

The loudspeaker safety functions should remain independent from any optional remote-control functionality.

48.22 Balanced Audio Input

The current amplifier uses conventional unbalanced RCA inputs.

A future version could investigate the addition of balanced audio inputs using XLR connectors and a suitable differential input receiver.

A balanced interconnection can provide greater rejection of common-mode noise over long cables.

The existing RCA inputs could still be retained so that both consumer and balanced sources are supported.

48.23 Preamplifier Development

The preamplifier could be expanded independently from the power-amplifier stage.

Possible improvements include:

- improved input switching;
- additional line-level inputs;
- optimised phono-stage noise performance;
- selectable phono loading;
- improved channel tracking of the volume control;
- balanced output stage;
- relay-based signal routing.

48.24 Motorised or Electronic Volume Control

The conventional analogue volume control could eventually be replaced by a high-quality motorised potentiometer or electronically controlled analogue attenuator.

This could improve remote-control capability while maintaining an analogue signal path.

Any electronic volume-control solution would need to be evaluated for noise, distortion, channel matching, and maximum signal level.

48.25 Further PCB Integration

The present design intentionally uses separate functional PCBs.

A future revision could improve the interconnection system without necessarily placing the entire amplifier onto one board.

Possible improvements include:

- dedicated board-to-board connectors;
- labelled cable harnesses;
- keyed connectors;
- separate signal and power connector groups;
- test points for commissioning;
- improved silkscreen documentation.

48.26 Dedicated Test Points

Future PCB revisions could provide clearly labelled test points for important measurements.

Useful test points include:

$$V_+, \quad V_-, \quad \text{GND}, \quad V_{\text{OUT}}, \quad V_{\text{BIAS}}.$$

The protection controller could similarly provide test points for

$$12 \text{ V}, \quad 3.3 \text{ V}, \quad \text{DC_LEFT}, \quad \text{DC_RIGHT}, \quad \text{TEMP1}, \quad \text{TEMP2}.$$

This would make commissioning and maintenance safer and more repeatable.

48.27 Further PCB Ground Optimisation

The ground planes and star-ground architecture can be evaluated experimentally once the physical amplifier has been constructed.

If measurements reveal significant ground-potential differences or hum, a future PCB revision could modify the ground-plane geometry and connection points according to the measured current-return behaviour.

Such modifications should be driven by measurements rather than by assuming that a visually larger copper area automatically improves performance.

48.28 Reduction of Track Neck-Downs

The present PCB routing uses wide tracks for the principal high-current paths, but some connections necessarily narrow near component pads.

Future PCB revisions could investigate larger pads, revised component placement, or alternative routing to reduce these local bottlenecks where mechanically practical.

This could further reduce voltage drop and local heating in the highest-current paths.

48.29 Four-Layer PCB Option

The current designs use two-layer 2 oz copper PCBs.

A future low-level or controller PCB could potentially use a multilayer construction to provide more continuous ground and power reference planes.

For the main power amplifier, however, the advantages of a four-layer board would need to be weighed against the greater manufacturing cost and the relatively simple through-hole topology.

48.30 Improved EMC Layout

After prototype measurements, additional electromagnetic compatibility improvements could be considered.

Possible modifications include:

- improved cable routing;
- further separation of digital and analogue wiring;
- revised transformer orientation;
- improved chassis shielding;
- ferrite suppression on selected non-audio cables;
- filtered external control connections.

Any component added directly to the audio path should be evaluated for its effect on frequency response and distortion.

48.31 Improved Mechanical Modularity

The chassis could be redesigned so that major subsystems can be removed individually without dismantling the entire amplifier.

For example, separate mounting assemblies could be used for:

- left amplifier and heatsink;
- right amplifier and heatsink;
- protection controller;
- preamplifier;
- transformer assembly.

This would simplify servicing and future upgrades.

48.32 Improved Internal Cable Harnesses

A future construction could replace individual point-to-point wires with documented cable harnesses.

Each harness could use defined connectors and wire colours.

This would improve:

- assembly repeatability;
- serviceability;
- wiring consistency;
- fault diagnosis;
- mechanical appearance.

48.33 Chassis Temperature Monitoring

In addition to the two heatsink sensors, a future controller could include an ambient temperature sensor inside the chassis.

The controller would then have access to

$$T_{\text{HS,L}}, \quad T_{\text{HS,R}}, \quad T_{\text{AMBIENT}}.$$

This would make it possible to evaluate the heatsink temperature rise above the internal ambient temperature,

$$\Delta T_{\text{HS}} = T_{\text{HS}} - T_{\text{AMBIENT}}.$$

This quantity is useful when evaluating cooling performance.

48.34 Additional Thermal Testing

The completed amplifier could be characterised under a wider range of thermal conditions. Future experiments could investigate:

- extended idle operation;
- continuous moderate-power operation;
- high-power short-duration operation;
- one-channel versus two-channel loading;
- fan-on and fan-off behaviour;
- 8 Ω and controlled 4 Ω loading;
- elevated ambient temperature.

The resulting data could be used to refine fan placement and thermal trip thresholds.

48.35 Expanded Audio Measurements

A major future development is complete experimental characterisation of the finished amplifier.

Measurements could include:

- maximum unclipped output power;
- frequency response;
- total harmonic distortion;
- THD+N;
- signal-to-noise ratio;
- residual output noise;
- channel separation;
- channel gain matching;
- DC offset;
- clipping behaviour;
- load regulation;
- power-supply ripple.

48.36 THD+N Measurement

A useful future measurement would be total harmonic distortion plus noise,

$$\text{THD} + \text{N}.$$

This could be evaluated as a function of:

- output power;
- frequency;
- loudspeaker load;
- amplifier temperature.

This would provide a quantitative indication of the completed amplifier's audio performance rather than relying only on waveform observation or subjective listening.

48.37 Stereo Crosstalk Measurement

Because both amplifier channels share the main power supply and chassis, future testing should quantify stereo channel separation.

A signal can be applied to one channel while the residual signal appearing at the output of the un-driven channel is measured.

This would provide direct information about the effectiveness of the grounding, power-supply, and physical layout strategies.

48.38 Protection Response Characterisation

The protection controller can also be tested quantitatively.

Future measurements could determine:

- DC fault detection threshold;
- DC fault response time;
- startup delay accuracy;
- temperature trip point;
- temperature reset point;
- temperature hysteresis;
- relay release time;
- behaviour during power loss.

These measured values could then be compared directly with the firmware design parameters.

48.39 Long-Term Reliability Testing

Long-duration operation would provide useful information that cannot be obtained during short bench tests.

Future reliability testing could monitor:

- heatsink temperature drift;
- quiescent-current stability;
- relay operation;
- fan reliability;

- connector temperature;
- transformer temperature;
- output DC offset over time.

48.40 Automatic Self-Test

A future firmware revision could perform a limited startup self-test before energising the loudspeaker relays.

For example, the controller could verify:

ADC channels valid,

temperature sensors valid,

DC outputs within range,

and, if additional hardware is available,

supply rails within range.

The relays would remain disconnected if any required condition failed.

48.41 Manufacturing and Documentation Improvements

Future revisions could also improve the project as a reproducible engineering design.

Possible improvements include:

- revision-controlled manufacturing BOM;
- complete assembly drawings;
- cable-harness drawings;
- chassis CAD files;
- PCB assembly photographs;
- labelled test-point documentation;
- automated firmware build instructions;
- formal release versions on GitHub.

48.42 Automated Production Testing

If multiple amplifier units were ever constructed, a repeatable production test procedure could be developed.

The test sequence could automatically or semi-automatically record:

$$V_+, \quad V_-, \quad V_{\text{OFFSET}}, \quad I_Q, \quad T_{\text{HS}},$$

followed by low-power functional verification.

This would improve consistency between separate amplifier builds.

48.43 Possible Development Priorities

Not all proposed improvements provide the same benefit relative to their complexity.

A practical future development order would be:

1. complete physical construction and validation of the present design;
2. measure electrical and thermal performance;
3. optimise grounding, wiring, and cooling using measured results;
4. add supply-rail and diagnostic monitoring to the STM32 controller;
5. improve firmware logging and fault reporting;
6. investigate quieter temperature-controlled cooling;
7. add soft-start capability if measured inrush justifies it;
8. consider user-interface and preamplifier enhancements;
9. consider larger architectural changes such as balanced inputs or dual-mono power supplies only after the baseline design is fully characterised.

48.44 Importance of Baseline Validation

The most important prerequisite for future improvement is complete characterisation of the present amplifier.

A modification cannot be judged objectively unless the performance of the original implementation is known.

The baseline design should therefore first be characterised in terms of

$$\boxed{\text{Power} + \text{Distortion} + \text{Noise} + \text{Temperature} + \text{Protection Performance}}.$$

Only then can a future revision demonstrate quantitatively whether a proposed change provides a real improvement.

48.45 Design Summary

The current amplifier provides a complete foundation for a high-quality stereo Class-AB amplifier with independent speaker protection, temperature monitoring, preamplification, and structured PCB and mechanical design.

Future improvements can therefore focus primarily on refinement rather than replacing the entire architecture.

Potential analogue improvements include input transistor matching, improved bias monitoring, refined grounding, and additional audio-performance optimisation.

The main power supply could be extended with soft-start and improved mains filtering, while a substantially more complex future version could investigate dual-mono power supplies.

The STM32 protection controller provides the largest opportunity for functional expansion. Additional rail monitoring, current sensing, fan monitoring, diagnostic logging, fault-history storage, and serial communication could be implemented in future hardware and firmware revisions.

The present four AC cooling fans could eventually be replaced by controllable low-voltage fans so that airflow can respond dynamically to heatsink temperature while reducing acoustic noise during low-power operation.

User-interface improvements could include a small diagnostic display, while the existing dedicated fault indicators could be retained for immediate fault identification.

Further PCB revisions could provide dedicated test points, improved connector organisation, reduced high-current track neck-downs, revised ground geometry, and more modular cable connections.

Mechanical improvements could increase serviceability through modular heatsink, PCB, and transformer assemblies and documented cable harnesses.

Most importantly, the completed amplifier should first undergo comprehensive experimental characterisation, including output power, frequency response, THD+N, signal-to-noise ratio, crosstalk, DC offset, power-supply ripple, thermal behaviour, and protection response.

The results of these measurements would provide the quantitative baseline required to determine which future modifications produce genuine improvements in performance, reliability, safety, and usability.

Chapter 49

Final System Integration

49.1 Introduction

Final system integration represents the stage at which the independently designed and assembled subsystems are combined into the complete stereo Hi-Fi amplifier.

The final system contains:

- two discrete Class-AB power-amplifier channels;
- the shared symmetrical main power supply;
- the audio preamplifier;
- the STM32-based speaker-protection controller;
- left and right speaker relays;
- two heatsinks with temperature sensors;
- the cooling system;
- RCA input connections;
- loudspeaker output terminals;
- mains input, switching, fusing, and protective earth;
- the front-panel controls and protection indicators.

The objective of system integration is not simply to connect these subsystems electrically. The complete assembly must preserve the grounding strategy, signal integrity, thermal design, current-carrying capability, protection functions, and mains-safety principles established during the earlier design stages.

The integration process can therefore be represented as

Independent Subsystems → Controlled Interconnection → Integrated Verification → Complete Amplifier
--

The procedures described in this chapter represent the intended final integration process. Final measured performance will be documented only after the physical system has been assembled and experimentally verified.

49.2 Integrated System Architecture

The complete signal path of the stereo amplifier can be represented as

Audio Source → RCA Input → Preamplifier → Power Amplifier → Protection Relay → Loudspeaker

The left and right channels follow equivalent signal paths but remain electrically independent through the majority of the audio chain.

They share the main power supply, chassis, protection controller, and mains input system.

The protection controller does not form part of the normal analogue audio signal-processing path. Its main interaction with the loudspeaker signal is through the relay contacts and the high-impedance DC-sensing networks.

49.3 Integration Sequence

The subsystems should be integrated progressively rather than connecting the complete amplifier in a single step.

A suitable final integration sequence is:

1. verify the main power supply;
2. verify the protection-controller supply;
3. commission the left power-amplifier channel;
4. commission the right power-amplifier channel;
5. adjust the bias of both channels;
6. verify the protection controller independently;
7. connect the amplifier outputs to the protection sensing circuits;
8. connect the amplifier outputs through the speaker relays;
9. integrate the temperature sensors;
10. connect the preamplifier;
11. complete the front-panel wiring;
12. complete the cooling system;

13. complete the chassis ground and protective-earth connections;
14. perform complete-system electrical checks;
15. perform integrated low-level testing;
16. perform dummy-load and thermal testing;
17. connect loudspeakers only after all previous checks are satisfactory.

49.4 Main Power-Supply Integration

The two power-amplifier channels share the main symmetrical supply.

The supply begins with the

$$24-0-24 \text{ V}_{\text{AC}}$$

toroidal transformer.

The secondary voltage is full-wave rectified by the KBPC3506WP bridge and filtered by four

$$4700 \text{ } \mu\text{F}$$

reservoir capacitors.

The nominal reservoir capacitance per rail is therefore

$$C_{\text{rail}} = 2 \times 4700 \text{ } \mu\text{F} = 9400 \text{ } \mu\text{F}.$$

The expected supply region is approximately

$$V_+ \approx +33 - +34 \text{ V}$$

and

$$V_- \approx -33 - -34 \text{ V}$$

under light practical loading.

The exact measured values will depend on mains voltage, transformer regulation, bridge-rectifier voltage drop, reservoir ripple, and amplifier loading.

49.5 Power Distribution to Both Channels

The left and right amplifier channels should receive their supply connections from the common power-supply region using independent high-current paths.

The desired distribution is conceptually

Main PSU → Left Amplifier
Main PSU → Right Amplifier

rather than

PSU → Left Amplifier → Right Amplifier.

This reduces common wiring impedance between the channels and helps prevent one channel's load current from modulating the supply reference of the other.

49.6 Amplifier Rail Fuses

Each amplifier channel contains independent positive- and negative-rail protection.

The two supply rails use

3 A

fuses.

The integrated supply path for each rail is therefore of the form

Main Supply Rail → Fuse → Amplifier Circuit.

The rail fuses remain separate from the primary mains fuse and perform a different protection function.

49.7 Power-Amplifier Integration

The two power-amplifier channels are based on the same discrete Project 3A architecture and are integrated as a stereo pair.

Each channel uses:

BC546

small-signal devices,

BD139/BD140

driver devices, and the complementary output pair

2SA1943-O

and

$$\boxed{2SC5200-O}.$$

The two amplifier PCBs should be installed so that the output transistors face their corresponding heatsinks.

49.8 Bias Verification Before Integration

Each amplifier channel should have its bias adjusted before the stereo system is subjected to significant load.

With

$$R_{13} = R_{14} = 0.33 \, \Omega,$$

the total resistance used for the bias measurement is

$$R_{13} + R_{14} = 0.66 \, \Omega.$$

The nominal target quiescent current is approximately

$$I_Q \approx 50 \, \text{mA},$$

corresponding to

$$V_{\text{BIAS}} = I_Q(0.66) \approx 33 \, \text{mV}.$$

Both channels should be thermally stabilised and checked independently before final stereo integration.

49.9 Preamplifier Integration

The preamplifier provides the interface between the external audio sources and the two power-amplifier channels.

The principal audio path becomes

$$\text{RCA Inputs} \rightarrow \text{Input Selection} \rightarrow \text{Preamplifier} \rightarrow \text{Power Amplifiers}.$$

The low-level output of the preamplifier should be connected to each amplifier using short signal cables with the appropriate signal-ground reference.

Where the physical distance is significant, shielded audio cable should be used.

49.10 Preamplifier Power Integration

The preamplifier uses regulated low-voltage supply rails.

These rails must remain electrically separated from the high-current loudspeaker and main amplifier supply paths except at the intended system ground reference.

The preamplifier supply wiring should be routed away from:

- transformer primary wiring;
- high-current transformer secondary wiring;
- speaker wiring;
- relay-coil wiring;
- fan wiring.

49.11 Protection Controller Integration

The STM32 protection controller supervises both amplifier channels.

Its functions include:

- startup speaker delay;
- left-channel DC monitoring;
- right-channel DC monitoring;
- left-heatsink temperature monitoring;
- right-heatsink temperature monitoring;
- temperature-sensor fault detection;
- overtemperature protection;
- speaker relay control;
- status indication.

The controller is based on the STM32G031K8 microcontroller using the NUCLEO-G031K8 platform.

49.12 Protection Controller Power

The controller uses an independent isolated low-voltage supply.

The architecture is

$230\text{ V}_{\text{AC}} \rightarrow \text{Auxiliary Transformer} \rightarrow \text{Rectifier} \rightarrow \text{Filter} \rightarrow 7812 \rightarrow 12\text{ VDC}.$

This independent supply permits the protection system to operate without being directly powered from the main high-voltage amplifier rails.

49.13 DC-Sense Integration

Each amplifier output is connected to a dedicated protection-controller DC-sensing network.

The sensing connection must be high impedance so that it does not significantly load the audio output.

The architecture is

$\text{Amplifier Output} \rightarrow \text{Attenuation / Filtering} \rightarrow \text{ADC Protection} \rightarrow \text{STM32 ADC}.$

The filtering intentionally rejects the normal audio waveform while allowing a persistent DC component to be detected.

49.14 Speaker Relay Integration

The loudspeaker outputs pass through the two Omron G2RL-1A-E-DC12 normally-open relays.

For the left channel,

$\text{Left Amplifier} \rightarrow K_1 \rightarrow \text{Left Speaker}$

and for the right channel,

$\text{Right Amplifier} \rightarrow K_2 \rightarrow \text{Right Speaker}.$

The relay coils are controlled by BC337 transistor drivers.

1N4007 flyback diodes suppress the inductive voltage generated when the relay coils are switched off.

49.15 Fail-Safe Relay State

The relays use normally-open contacts.

Therefore, when relay drive is absent,

Speaker Disconnected

This is advantageous because a controller power failure, driver failure, or intentional protection event naturally places the speaker connection in the open state.

49.16 Temperature Sensor Integration

Each amplifier heatsink uses one 10 k Ω NTC sensor.

The left sensor is associated with the left amplifier heatsink and the right sensor with the right amplifier heatsink.

The integration can be represented as

Left Heatsink $\rightarrow$ NTC_L $\rightarrow$ STM32
 Right Heatsink $\rightarrow$ NTC_R $\rightarrow$ STM32

The sensors should be mounted close to the power output devices so that they respond effectively to heatsink heating.

49.17 Cooling-System Integration

The complete thermal system combines passive heatsinking, forced airflow, and electronic temperature monitoring.

The intended relationship is

Output Transistors $\rightarrow$ Heatsinks $\rightarrow$ Forced Airflow

with the protection controller continuously monitoring the heatsink temperatures.

The four cooling fans should be arranged so that they support an intentional airflow path rather than simply moving air randomly within the chassis.

49.18 Thermal and Protection Coordination

The thermal system has two complementary functions.

The heatsinks and fans attempt to maintain acceptable operating temperature during normal operation.

The STM32 protection system provides a second level of defence by disconnecting the loudspeakers if an abnormal temperature condition is detected.

Thus,

Cooling = Normal Thermal Management

while

Overtemperature Trip = Fault Protection.

49.19 Grounding Integration

Grounding becomes particularly important during final system integration because all independent boards are now interconnected.

The final system contains several different return-current categories:

- low-level audio signal return;
- preamplifier return;
- amplifier power return;
- loudspeaker return;
- rectifier and reservoir return;
- protection-controller ground;
- chassis;
- protective earth.

These connections should not be treated as mechanically interchangeable simply because they may all eventually reference approximately the same electrical potential.

49.20 Star-Ground Integration

The main high-current return paths should converge at a controlled common ground region.

The desired principle is

Left Amplifier Return
 Right Amplifier Return
 Speaker Returns
 Power-Supply Common } → Central Ground Region

rather than connecting the channels sequentially.

This minimises common-impedance coupling between the high-current paths.

49.21 Signal-Ground Integration

The RCA and preamplifier grounds belong primarily to the low-level signal domain.

They should not carry:

- loudspeaker current;
- reservoir-capacitor charging current;
- relay-coil current;
- cooling-fan current.

The low-level signal ground should join the wider system grounding arrangement at a controlled point.

49.22 Speaker Return Integration

The speaker return is part of the high-current amplifier output circuit.

For example, at the Project 3A reference condition of

$$100 \text{ W}$$

into

$$4 \Omega,$$

the loudspeaker current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5 \text{ A}.$$

The corresponding peak current is

$$I_{\text{PEAK}} = 5\sqrt{2} \approx 7.07 \text{ A}.$$

The speaker return therefore cannot be treated as a small-signal ground connection.

49.23 Chassis and Protective-Earth Integration

The metal enclosure must be bonded securely to protective earth.

The required safety connection is

Protective Earth → Dedicated Chassis Bond.

Protective earth is not used as the normal audio current return.

The audio-ground-to-chassis relationship must remain controlled so that the final mechanical assembly does not create unintended additional ground paths.

49.24 RCA Connector Integration

The RCA connectors are located in the low-level audio region of the rear panel.

Their centre terminals carry the audio signals, while the outer conductors provide the corresponding signal-ground references.

If the RCA connector bodies are metal, their relationship to the chassis must be defined intentionally so that an accidental second signal-ground-to-chassis connection is not created.

49.25 Speaker Terminal Integration

The left and right speaker terminals are connected to the corresponding protection-relay outputs.

High-current conductors should be used between:

Power Amplifier → Protection Relay → Speaker Terminal.

The output and return conductors of each channel should be kept close together to minimise loop area.

49.26 Internal Wiring Separation

During final integration, internal wiring should be separated into functional groups.

The principal groups are:

- mains wiring;
- transformer secondary wiring;
- DC power wiring;

- speaker wiring;
- low-level audio wiring;
- temperature and control wiring;
- fan wiring.

Low-level audio wiring should remain physically separated from the high-current and mains groups.

49.27 Mains Wiring Integration

The mains system includes:

- mains inlet;
- primary fuse;
- power switch;
- main transformer primary;
- protection transformer primary;
- protective-earth conductor.

This wiring should remain within a dedicated chassis region and should not cross the low-level input or preamplifier area unless unavoidable.

49.28 Transformer Placement

The main toroidal transformer should be located away from the most sensitive analogue sections.

Its final orientation may be adjusted during prototype testing if magnetic hum coupling is detected.

The physical placement should also consider:

- mechanical weight;
- ventilation;
- secondary wire length;
- distance from RCA wiring;
- distance from phono circuitry where applicable.

49.29 Front-Panel Integration

The front panel provides the main user interface to the completed amplifier.

Depending on the final preamplifier implementation, the front panel can include:

- power control;
- source selection;
- volume control;
- balance control;
- MUTE / ENABLE control;
- FAULT RESET control;
- protection-status indicators.

49.30 Protection Indicators

The protection controller provides five status indicators:

- TEMP SENSOR 1 FAIL;
- TEMP SENSOR 2 FAIL;
- DC FAULT;
- WAIT / DELAY;
- OVERHEAT.

These indicators provide immediate information about the controller state and are especially useful during initial system integration.

49.31 WAIT / DELAY Integration

When the amplifier is first powered, the WAIT / DELAY state should be active and the speaker relays should remain open.

The intended sequence is

Power On → WAIT → Protection Checks → Relay Connection

provided that no fault condition is present.

49.32 DC Fault Integration

If persistent abnormal DC is detected at either amplifier output, both speaker relays should move to the safe state.

The expected system response is

DC Fault → Relay Drive Removed → Speakers Disconnected

.

This function should be confirmed before any valuable loudspeaker is connected.

49.33 Overtemperature Integration

An overtemperature condition should similarly cause loudspeaker disconnection.

The controller should differentiate between a valid high-temperature measurement and an invalid temperature-sensor condition.

This allows the user to distinguish

OVERHEAT

from

TEMP SENSOR FAIL

.

49.34 Power-Off Behaviour

The behaviour of the complete amplifier during power removal is also important.

The protection relays should release as the controller supply disappears so that the loudspeakers are disconnected rather than remaining connected during uncontrolled supply decay.

Power-off behaviour should be examined experimentally because different subsystems can lose voltage at different rates.

49.35 Pre-Integration Electrical Checks

Before connecting all subsystems together, the following should have been verified independently:

- main positive rail;
- main negative rail;

- protection-controller 12 V supply;
- STM32 operation;
- left amplifier output offset;
- right amplifier output offset;
- left amplifier bias;
- right amplifier bias;
- left temperature sensor;
- right temperature sensor;
- relay operation.

49.36 Integrated Continuity Checks

After all interconnections are installed but before mains power is applied, a new continuity check should be performed.

This is necessary because the integration wiring can introduce faults that did not exist when the boards were tested separately.

Important checks include:

- positive rail to ground;
- negative rail to ground;
- positive to negative rail;
- each speaker output to ground;
- left/right speaker separation;
- RCA grounds to intended signal ground;
- chassis to protective earth;
- absence of unintended chassis-to-signal-ground paths;
- relay contact routing.

49.37 Integrated First Power-Up

The first complete-system power-up should still be performed without loudspeakers connected.

The expected sequence is:

1. mains power is applied;
2. the main amplifier rails rise;
3. the protection controller starts;
4. WAIT / DELAY remains active;
5. DC and temperature conditions are evaluated;
6. the relays energise only if the system is safe;
7. the amplifier remains at zero-signal idle.

49.38 Integrated Rail Measurements

After the complete system is powered, the main rail voltages should be checked again.

The expected condition is approximately

$$V_+ \approx +33 - +34 \text{ V}$$

and

$$V_- \approx -33 - -34 \text{ V}$$

under light load.

The values may differ slightly from isolated PSU measurements because the full system now draws current.

49.39 Integrated DC Offset Check

The amplifier output DC offset should be checked again after the preamplifier, protection sensing, and grounding systems have been integrated.

This confirms that no new ground or signal connection has altered the amplifier operating point.

The desired condition remains

$$V_{\text{OUT,DC}} \approx 0 \text{ V.}$$

49.40 Integrated Bias Check

The final installation can alter heatsink temperature and airflow.

For this reason, the Class-AB bias of both channels should be checked again after the amplifier is mounted in the chassis.

The target remains approximately

$$V_{\text{BIAS}} \approx 33 \text{ mV}$$

across

$$R_{13} + R_{14} = 0.66 \text{ } \Omega.$$

49.41 Low-Level Audio Integration Test

After correct zero-signal operation is confirmed, a low-amplitude audio test signal can be applied through the normal RCA and preamplifier signal path.

The test therefore verifies the complete route

$$\text{RCA} \rightarrow \text{Preamplifier} \rightarrow \text{Power Amplifier} \rightarrow \text{Relay} \rightarrow \text{Output}.$$

The test should begin at low amplitude so that unexpected behaviour does not subject the output stage to unnecessary stress.

49.42 Channel Verification

The left and right channels should be tested separately before stereo operation is evaluated.

The checks should confirm:

- correct left/right routing;
- correct gain;
- absence of channel reversal;
- correct volume-control behaviour;
- correct relay operation;
- absence of unexpected crosstalk.

49.43 Noise and Hum Verification

Final integration can introduce noise that was absent when individual PCBs were tested on the bench.

With the input shorted to signal ground, the output should be checked for residual noise and hum.

A strong component near

50 Hz

can suggest mains or magnetic coupling, while a strong component near

100 Hz

can suggest full-wave rectifier or reservoir-charging coupling.

These frequencies provide diagnostic clues rather than absolute proof of a particular fault.

49.44 Ground-Loop Investigation

If hum increases after all subsystems are interconnected, the final chassis assembly should be checked for unintended parallel ground connections.

Potential locations include:

- RCA connector bodies;
- PCB mounting hardware;
- preamplifier chassis contact;
- protection-board mounting hardware;
- shield connections;
- external equipment grounds.

The fault should be identified by tracing return paths rather than randomly disconnecting protective earth.

49.45 Cooling Verification

The fans should be operated with the complete chassis configuration in place.

The airflow should be checked to ensure that:

- cool air reaches both amplifier heatsinks;

- hot air has a clear exit path;
- internal cables do not obstruct airflow;
- fans do not create excessive mechanical vibration;
- fan wiring does not introduce audible interference.

49.46 Integrated Thermal Test

The thermal behaviour of the complete system should be evaluated after the chassis is closed or configured as it will be during normal use.

This is important because open-bench thermal behaviour can differ substantially from enclosed operation.

The integrated thermal test should monitor:

$$T_{\text{HS,L}}, \quad T_{\text{HS,R}},$$

as well as internal ambient temperature where possible.

49.47 Dummy-Load Integration Test

Before loudspeakers are used, the complete signal and protection paths should be evaluated with suitable dummy loads.

Initial loaded testing should begin with

$$8 \, \Omega$$

loads.

Only after satisfactory operation should controlled

$$4 \, \Omega$$

testing be performed.

The dummy loads must be rated for the electrical power applied during the test.

49.48 Both Channels Under Load

Testing both channels simultaneously is an important final integration test because both channels share the main transformer and reservoir capacitors.

This test evaluates:

- rail sag;
- transformer loading;
- reservoir ripple;
- channel interaction;
- chassis thermal loading;
- fan effectiveness;
- grounding behaviour.

49.49 Protection Testing in the Integrated System

The speaker-protection system should be verified again after complete system integration.

The final tests should confirm:

- startup relay delay;
- DC fault detection;
- relay disconnection during DC fault;
- temperature sensing;
- overtemperature response;
- temperature-sensor fault response;
- relay release at power-off;
- correct indicator operation.

Fault tests should be performed using controlled test conditions rather than by deliberately producing destructive faults in the power amplifier.

49.50 Front-Panel Functional Check

All front-panel controls and indicators should be checked after integration.

The user interface should be consistent and unambiguous.

A final check should confirm the function of:

- power control;
- volume control;

- source selection;
- balance control where implemented;
- MUTE / ENABLE;
- FAULT RESET;
- WAIT / DELAY indicator;
- DC FAULT indicator;
- OVERHEAT indicator;
- both temperature-sensor fault indicators.

49.51 Rear-Panel Functional Check

The rear-panel connections should also be inspected.

The final assembly should clearly identify:

- left and right RCA inputs;
- left speaker output;
- right speaker output;
- speaker polarity;
- mains input;
- primary fuse where externally accessible.

Connectors should be mechanically secure and should not rotate or move when cables are inserted or removed.

49.52 Cable Management

After the electrical tests are complete, internal wiring should be secured in its final position.

The cable management should ensure that:

- no wire can contact a fan blade;
- no wire rests against a hot heatsink;
- no wire passes across a sharp chassis edge;
- mains and low-level audio cables remain separated;
- sensor wiring remains away from high-current conductors;
- serviceable components remain accessible.

49.53 Final Mechanical Inspection

Before the chassis is closed, all mechanical fasteners should be checked.

This includes:

- PCB standoffs;
- transformer mounting;
- heatsink mounting;
- fan screws;
- power-transistor mounting;
- NTC mounting;
- RCA connectors;
- speaker terminals;
- front-panel controls;
- protective-earth hardware.

49.54 Final Electrical Safety Inspection

The final system should not be considered complete until the mains-related construction has been inspected.

The inspection should verify:

- secure protective-earth bond;
- correct mains fuse;
- correct line and neutral routing;
- insulated mains terminals;
- strain relief;
- primary-to-secondary separation;
- adequate mechanical protection;
- absence of exposed hazardous conductors.

49.55 Final Integration Measurement Record

The final integration process should be documented using measured values.

Table 49.1: Suggested final system integration measurement record.

Parameter	Left Channel	Right Channel
Positive rail voltage	To be measured	To be measured
Negative rail voltage	To be measured	To be measured
Output DC offset	To be measured	To be measured
Stabilised bias voltage	To be measured	To be measured
Quiescent current	To be calculated	To be calculated
Idle heatsink temperature	To be measured	To be measured
Loaded heatsink temperature	To be measured	To be measured
Relay operation	To be verified	To be verified
DC protection	To be verified	To be verified
Thermal protection	To be verified	To be verified

49.56 System-Level Acceptance Criteria

The final integrated amplifier can proceed to normal listening tests when the following conditions have been confirmed:

- both amplifier supply rails are correct;
- both amplifier channels have acceptable DC offset;
- both channels have stable Class-AB bias;
- no output-stage component exhibits abnormal heating;
- the preamplifier operates correctly;
- the protection-controller supply is correct;
- startup relay delay operates correctly;
- DC protection disconnects the loudspeakers correctly;
- both temperature sensors operate correctly;
- overtemperature protection operates correctly;
- the speaker relays connect and disconnect correctly;
- no excessive hum or oscillation is present;
- cooling airflow is satisfactory;
- protective earth is secure;
- all mains connections are mechanically and electrically safe.

49.57 Relationship to Experimental Validation

Final integration is not equivalent to complete experimental validation.

Successful integration confirms that the individual subsystems operate correctly together.

Subsequent measurements are still required to determine the actual performance of the completed amplifier.

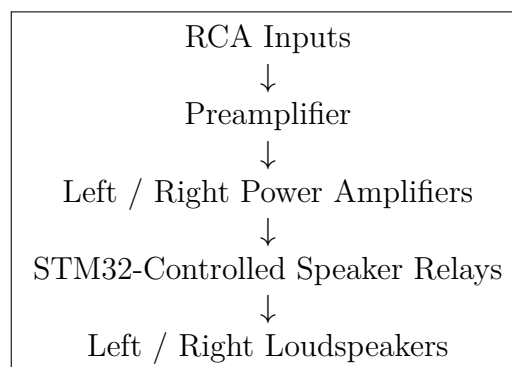
These measurements may include:

- maximum unclipped output power;
- frequency response;
- total harmonic distortion;
- THD+N;
- signal-to-noise ratio;
- residual noise;
- stereo crosstalk;
- supply ripple;
- thermal behaviour;
- protection response time.

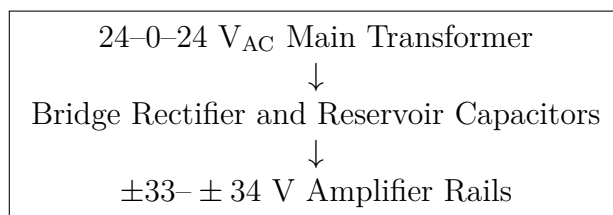
The measured values should be reported separately so that calculated, manufacturer-specified, and experimentally verified quantities remain clearly distinguished.

49.58 Final System Architecture Summary

After integration, the complete amplifier can be summarised as



while the supporting system consists of



and an independent protection-controller supply.

49.59 Design Summary

Final system integration combines the two Class-AB power-amplifier channels, main power supply, preamplifier, STM32 protection controller, speaker relays, temperature sensors, cooling system, connectors, grounding network, front panel, and mains wiring into the complete stereo amplifier.

The integration is performed progressively so that each subsystem remains independently verifiable before the next subsystem is connected.

The shared main power supply provides approximately

$\pm 33 - \pm 34 \text{ V}$

to both power-amplifier channels from the 24–0–24 V AC toroidal transformer.

The two amplifier channels retain separate high-current supply and return paths where practical, while the preamplifier and RCA connections remain within the low-level signal domain.

The STM32 protection controller monitors both amplifier outputs and both heatsink temperatures and controls the two normally-open G2RL speaker relays.

The loudspeaker path therefore becomes

Amplifier → Protection Relay → Speaker Terminal.
--

Two 10 k Ω NTC sensors provide independent thermal monitoring of the left and right heatsinks, while the four cooling fans provide forced airflow.

The grounding architecture preserves the distinction between signal ground, high-current power return, speaker return, chassis, and protective earth.

Final integration testing includes supply-rail measurement, DC-offset verification, bias rechecking, protection testing, low-level signal testing, noise and hum investigation, dummy-load operation, two-channel loading, thermal testing, and complete front-panel and rear-panel verification.

The system is considered ready for final experimental characterisation only after both channels operate correctly together, all protection functions have been verified, thermal behaviour is satisfactory, mains construction is safe, and no significant integration-related noise or grounding problems are observed.

This stage transforms the individual electronic subsystems into a single coordinated stereo Hi-Fi amplifier and provides the foundation for the final performance measurements and evaluation of the completed design.

Chapter 50

Conclusion

50.1 Project Overview

This project presented the complete design and development of a stereo Hi-Fi power amplifier based on a discrete Class-AB architecture derived from Rod Elliott's Project 3A.

The work extended considerably beyond the power-amplifier circuit itself and developed the amplifier as a complete system consisting of:

- two independent power-amplifier channels;
- a shared symmetrical linear power supply;
- an audio preamplifier;
- an STM32-based speaker-protection controller;
- DC fault detection;
- heatsink temperature monitoring;
- relay-based loudspeaker disconnection;
- thermal management and cooling;
- PCB design and manufacturing preparation;
- grounding and noise-control strategies;
- mechanical and chassis integration;
- mains wiring and electrical-safety considerations.

The result is therefore not simply an implementation of an existing amplifier schematic, but a complete engineering design in which analogue electronics, digital control, power electronics, PCB design, firmware, thermal management, mechanical design, and safety requirements are integrated into a single system.

50.2 Power-Amplifier Design

The power-amplifier section is based on a discrete Class-AB topology using a differential input stage, voltage-amplification stage, driver stage, and complementary high-power output devices.

The implemented output stage uses the complementary transistor pair

$$\boxed{2\text{SA1943-O}}$$

and

$$\boxed{2\text{SC5200-O}},$$

with BD139 and BD140 devices used in the driver circuitry.

The use of discrete devices allows the internal operation of the amplifier to be studied in detail and provides direct control over biasing, compensation, power-device selection, thermal design, and PCB layout.

The nominal closed-loop voltage gain of the amplifier is approximately

$$A_v = 1 + \frac{22 \text{ k}\Omega}{1 \text{ k}\Omega} \approx 23.$$

The design is intended for conventional $4 \text{ }\Omega$ and $8 \text{ }\Omega$ loudspeaker loads, subject to final verification of output power, thermal behaviour, and safe operating conditions on the completed hardware.

50.3 Power-Supply Design

The two power-amplifier channels are supplied from a common linear power supply based on a

$$\boxed{24\text{--}0\text{--}24 \text{ V}_{\text{AC}}}$$

toroidal transformer.

The AC supply is rectified by a KBPC3506WP bridge rectifier and filtered using four

$$4700 \text{ }\mu\text{F}$$

reservoir capacitors.

Two capacitors are effectively used per rail, producing a nominal reservoir capacitance of

$$2 \times 4700 \text{ }\mu\text{F} = \boxed{9400 \text{ }\mu\text{F}}$$

per rail.

The expected amplifier supply voltage is approximately

$$\boxed{\pm 33 - \pm 34 \text{ VDC}},$$

depending on mains voltage, transformer regulation, rectifier losses, and amplifier loading.

The final rail voltages will be measured during commissioning rather than assumed from calculation alone.

50.4 Speaker Protection System

One of the main extensions developed for this project is the independent speaker-protection controller.

The protection system is based on the STM32G031K8 microcontroller using the NUCLEO-G031K8 development platform.

The controller monitors both amplifier channels and is responsible for:

- startup speaker delay;
- left-channel DC fault detection;
- right-channel DC fault detection;
- left-heatsink temperature monitoring;
- right-heatsink temperature monitoring;
- temperature-sensor fault detection;
- overtemperature protection;
- speaker-relay control;
- visual fault and status indication.

The two loudspeaker channels are switched using Omron G2RL-1A-E-DC12 normally-open relays.

This means that the default unpowered state is

$$\boxed{\text{Speaker Disconnected}},$$

which provides a useful fail-safe characteristic.

BC337 transistors provide the relay-coil drive, while 1N4007 diodes suppress the inductive voltage generated when the relay coils are switched off.

50.5 DC Fault Detection

The amplifier outputs are monitored by dedicated attenuating and filtering networks connected to the STM32 ADC.

The protection controller is therefore able to distinguish a persistent DC component from the normal AC audio signal.

This allows the loudspeakers to be disconnected if an amplifier fault causes a dangerous DC voltage to appear at the output.

The final protection thresholds and response times will be verified experimentally during commissioning.

50.6 Thermal Protection

Each amplifier channel uses a dedicated heatsink monitored by a 10 k Ω NTC temperature sensor.

The selected sensors are mechanically attached to the heatsinks so that the controller can monitor the thermal state of both output stages independently.

The thermal-management system therefore consists of three levels:

Heatsink + Forced Airflow + Electronic Thermal Protection.

Four cooling fans are included in the mechanical cooling concept.

The final effectiveness of the heatsinks and fans will be established through thermal testing under idle, moderate-power, and higher-power operating conditions.

50.7 Bias Adjustment

The Class-AB operating point is adjusted independently for each amplifier channel using VR1.

The output stage contains two

$$0.33 \, \Omega$$

emitter resistors.

The bias current can therefore be determined from the voltage measured across

$$R_{13} + R_{14} = 0.66 \, \Omega.$$

For a nominal quiescent current of approximately

$$I_Q = 50 \, \text{mA},$$

the expected voltage is

$$V_{\text{BIAS}} = 0.050 \times 0.66 \approx \boxed{33 \text{ mV}}.$$

This provides a practical and repeatable method for adjusting the Class-AB operating point during commissioning.

50.8 PCB Development

Dedicated KiCad PCBs were developed for both the power-amplifier circuitry and the protection controller.

The power-amplifier PCB has dimensions of approximately

$$136 \text{ mm} \times 124.5 \text{ mm},$$

while the protection-controller PCB measures approximately

$$121 \text{ mm} \times 182.25 \text{ mm}.$$

Both boards use

$$\boxed{2 \text{ oz copper}}$$

with a nominal copper thickness of approximately

$$70 \text{ }\mu\text{m}.$$

The power PCB uses wider tracks for high-current supply and loudspeaker paths, while low-current control and small-signal connections use narrower routing.

The protection board similarly separates high-current speaker paths from the low-current STM32 sensing and control circuitry.

50.9 Grounding and Noise Control

Grounding was treated as a system-level design problem rather than simply connecting every node named GND together arbitrarily.

The design distinguishes between:

- audio signal ground;
- amplifier power ground;
- speaker return;

- rectifier and reservoir current return;
- protection-controller ground;
- metal chassis;
- protective earth.

High-current returns are directed toward a controlled common grounding region, while low-level audio grounds are kept away from loudspeaker and rectifier currents.

This reduces the possibility of common-impedance coupling and helps minimise hum and noise.

The physical placement and wiring strategy also separate the sensitive RCA and preamplifier circuitry from mains wiring, transformer wiring, speaker current, relay coils, and cooling fans.

50.10 KiCad Libraries and Manufacturing Preparation

The project includes structured KiCad schematic, PCB, footprint, symbol, and library management.

Particular attention was given to footprint verification because an electrically correct schematic can still produce an unusable PCB if the pad numbering or mechanical package is incorrect.

The most critical devices, including the power transistors, bridge rectifier, relays, large capacitors, transformers, connectors, and Nucleo module, require manual comparison against manufacturer mechanical information.

The PCB design has also been prepared for manufacturing using Gerber and drill outputs.

Custom symbols and footprints are maintained as part of the project so that the design remains portable and reproducible on another KiCad installation.

50.11 Mechanical Integration

The project also considers the mechanical arrangement of the complete amplifier rather than treating the PCBs as isolated assemblies.

The chassis must accommodate:

- two power-amplifier PCBs;
- the protection-controller PCB;
- the preamplifier;
- main transformer;

- two heatsinks;
- four cooling fans;
- RCA connectors;
- loudspeaker terminals;
- front-panel controls and indicators;
- mains inlet, fuse, and power switch.

The main transformer is positioned away from the low-level input circuitry, while the two amplifier boards are positioned close to their respective heatsinks.

The internal layout also provides separate routing regions for mains, audio, speaker, control, and sensor wiring.

50.12 Electrical Safety

Because the amplifier operates directly from the 230 V AC mains supply, electrical safety forms an essential part of the overall design.

The metal chassis is connected permanently to protective earth.

Mains wiring is treated separately from signal and low-voltage wiring, with appropriate insulation, mechanical protection, strain relief, switching, and fusing.

The protection-controller PCB also contains a mains-powered transformer, so primary-to-secondary isolation must be treated independently from ordinary low-voltage PCB clearance rules.

The final assembly must therefore undergo a complete electrical-safety inspection before normal operation.

50.13 Construction and Commissioning

A staged construction and commissioning procedure was defined for the project.

The individual PCBs are first assembled and inspected separately.

The main power supply and protection-controller supply are then verified before the amplifier channels are powered.

Each amplifier channel is commissioned independently with no loudspeaker connected.

The sequence includes:

Supply Check → DC Offset Check → Bias Adjustment → Protection Check → Signal Test

before proceeding to dummy-load and thermal testing.

This staged approach reduces the possibility that a single construction error will damage several subsystems simultaneously.

50.14 Final System Integration

After the individual subsystems have been verified, they are integrated into the complete stereo amplifier.

The final audio path is

RCA Input → Preamplifier → Power Amplifier → Protection Relay → Loudspeaker.

The controller supervises both channels without being directly inserted into the low-level analogue audio path.

The complete system integration also includes the cooling system, front-panel indicators, grounding architecture, mains wiring, and protective-earth connection.

50.15 Experimental Validation Still Required

The present work establishes the complete electrical, PCB, protection, firmware, thermal, grounding, and integration design.

However, final amplifier performance cannot be claimed before the physical prototype has been assembled and measured.

The following parameters remain to be determined experimentally:

- actual positive and negative rail voltages;
- output DC offset;
- final quiescent current;
- maximum unclipped output power;
- frequency response;
- total harmonic distortion;
- THD+N;
- residual output noise;
- signal-to-noise ratio;
- stereo crosstalk;
- power-supply ripple;

- heatsink temperature;
- transformer temperature;
- thermal-protection threshold;
- DC-protection response time;
- startup-delay timing.

These values should be reported as measured results once the final hardware is available.

50.16 Achievement of the Project Objectives

The project objectives were to develop a complete stereo amplifier rather than only reproduce an existing power-amplifier schematic.

From the design perspective, these objectives have been addressed through the development of:

- a two-channel discrete Class-AB power amplifier;
- a high-current symmetrical linear power supply;
- integrated audio preamplification;
- an STM32-based protection controller;
- independent left and right DC sensing;
- dual heatsink temperature monitoring;
- speaker relay control;
- protection status indicators;
- custom PCB designs;
- custom KiCad library organisation;
- grounding and noise-reduction strategies;
- thermal management;
- chassis and connector planning;
- staged construction and commissioning procedures.

The project therefore demonstrates the integration of analogue electronics, embedded systems, power electronics, PCB design, firmware, mechanical design, and engineering documentation in a single practical system.

50.17 Engineering Significance

One of the main engineering lessons of the project is that amplifier performance depends on significantly more than the active amplifier schematic.

A theoretically correct amplifier can still perform poorly if the grounding, PCB routing, power supply, thermal system, wiring, protection circuitry, or mechanical arrangement is inadequate.

The design therefore treats the amplifier as a complete system in which

Circuit Design + PCB Layout + Power Supply + Grounding + Protection + Thermal Design + Mechan

must operate together.

This system-level approach is particularly important in high-power audio electronics because the same enclosure contains millivolt-level input signals, several-ampere loudspeaker currents, high charging-current pulses, digital control circuitry, and hazardous mains voltage.

50.18 Future Development

The current design also provides a useful platform for future development.

Possible improvements include:

- soft-start circuitry;
- additional power-supply monitoring;
- fan-speed control;
- fan-failure detection;
- extended fault logging;
- serial diagnostics;
- current monitoring;
- automatic self-test;
- balanced audio inputs;
- improved preamplifier functions;
- enhanced front-panel diagnostics;
- additional PCB test points;
- further mechanical optimisation.

These improvements should be considered after the present design has been constructed and fully characterised so that future modifications can be compared against a measured baseline.

50.19 Final Conclusion

The project has developed a complete design for a stereo Hi-Fi amplifier based on a discrete Class-AB power stage and extended it with modern digital protection and monitoring.

The final architecture combines two power-amplifier channels with a shared 24–0–24 V AC linear supply, a preamplifier, an STM32 protection controller, speaker relays, heatsink temperature sensing, forced-air cooling, structured grounding, dedicated PCBs, and a complete chassis-integration strategy.

The use of the STM32G031K8 adds programmable startup, DC-fault, sensor-fault, and thermal-protection functionality without placing the microcontroller in the analogue audio signal path.

The use of separate protection relays ensures that the loudspeakers can remain disconnected during startup or abnormal operating conditions.

The 2SA1943-O and 2SC5200-O power devices, 2 oz copper PCBs, high-current routing, reservoir-capacitor bank, dedicated heatsinks, and forced-air cooling provide the physical foundation required for the power stage.

At the same time, the signal-ground, power-ground, speaker-return, chassis-ground, and protective-earth systems are treated according to their different functions in order to minimise unwanted noise while maintaining electrical safety.

The project therefore represents a complete multidisciplinary amplifier design rather than an isolated schematic implementation.

The remaining major stage is experimental validation of the manufactured system.

Once construction is completed, measured results for output power, distortion, frequency response, noise, DC offset, thermal behaviour, power-supply performance, and protection response will provide the final quantitative assessment of the design.

The completed work establishes a detailed and reproducible foundation from which the final physical amplifier can be constructed, tested, evaluated, and further improved.

Appendix A

Power Amplifier Schematic

A.1 Overview

This appendix presents the final schematic of the discrete Class-AB power amplifier used in the project. The amplifier is based on the Elliott Sound Products Project 3A topology and has been adapted to the component selection, power supply, PCB layout, and mechanical requirements of the present stereo amplifier.

Two identical amplifier channels are used in the complete system, one for the left audio channel and one for the right audio channel. Therefore, the schematic presented in this appendix represents the fundamental amplifier circuit used for each channel.

The complete signal path can be summarised as

Audio Input → Differential Input Stage → Voltage Amplification Stage → Driver Stage → Power Out

The schematic should be considered together with the PCB layout, component selection, grounding strategy, power-supply design, and protection-controller documentation presented in the main body of this report.

A.2 Main Circuit Sections

The amplifier schematic can be divided into several functional sections:

- differential input stage;
- current-sink and biasing circuitry;
- voltage-amplification stage;
- Class-AB bias-control network;
- complementary driver stage;
- complementary power-output stage;

- negative-feedback network;
- output stability network;
- local power-supply decoupling;
- positive- and negative-rail fuse protection.

A.3 Differential Input Stage

The amplifier input stage uses BC546 small-signal NPN transistors. The differential stage compares the applied audio input with the feedback signal taken from the amplifier output.

This stage generates an error signal according to the difference between

$$V_{\text{IN}}$$

and

$$V_{\text{FB}}.$$

The global negative-feedback loop forces this difference to remain small during normal linear operation.

A.4 Voltage Gain

The closed-loop voltage gain is primarily determined by the feedback network.

For

$$R_5 = 22 \text{ k}\Omega$$

and

$$R_4 = 1 \text{ k}\Omega,$$

the nominal non-inverting gain is

$$A_v = 1 + \frac{R_5}{R_4}.$$

Therefore,

$$A_v = 1 + \frac{22\,000}{1\,000} = \boxed{23}.$$

Expressed in decibels,

$$A_{v,\text{dB}} = 20 \log_{10}(23) \approx 27.2 \text{ dB}.$$

A.5 Driver and Output Stage

The driver circuitry uses BD139 and BD140 medium-power bipolar transistors.

The main complementary output devices used in the final design are

$$\boxed{2\text{SA1943-O}}$$

and

$$\boxed{2\text{SC5200-O}}.$$

These power transistors provide the high-current capability required to drive the external loudspeaker load.

The output stage operates in Class-AB so that both output sections conduct slightly around the zero-crossing region, reducing crossover distortion relative to pure Class-B operation.

A.6 Output Emitter Resistors

The output stage includes the two power resistors

$$R_{13} = R_{14} = 0.33 \text{ } \Omega.$$

These resistors contribute to current stabilisation and provide a convenient method for measuring the output-stage quiescent current during bias adjustment.

The combined resistance is

$$R_{13} + R_{14} = 0.66 \text{ } \Omega.$$

For the nominal quiescent current of approximately

$$I_Q = 50 \text{ mA},$$

the expected voltage across the two resistors is

$$V = I_Q(R_{13} + R_{14})$$

and therefore

$$V = 0.050 \times 0.66 = \boxed{33 \text{ mV}}.$$

A.7 Bias Adjustment

The Class-AB quiescent current is adjusted using

$$\boxed{VR1 = 2 \text{ k}\Omega}.$$

The adjustment is performed during commissioning with no loudspeaker connected and no input signal applied.

The complete bias-adjustment procedure is described in Chapter 39.

A.8 Supply Rails

The amplifier is powered from the symmetrical main power supply derived from the

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}}$$

toroidal transformer.

After bridge rectification and reservoir filtering, the practical amplifier supply is expected to be approximately

$$\boxed{+33\text{--}+34 \text{ VDC}}$$

and

$$\boxed{-33\text{--}-34 \text{ VDC}}.$$

The exact voltages will depend on mains voltage, transformer regulation, rectifier losses, and amplifier load.

A.9 Rail Protection

The positive and negative amplifier rails are individually protected using

$$\boxed{3 \text{ A}}$$

fuses.

The corresponding supply paths are

$$V_+ \rightarrow F_1 \rightarrow \text{Amplifier}$$

and

$$V_- \rightarrow F_2 \rightarrow \text{Amplifier}.$$

These rail fuses are separate from the mains-side primary fuse.

A.10 Local Supply Decoupling

Local power-supply decoupling is included on the amplifier PCB to maintain a low supply impedance close to the active circuitry.

The design combines small high-frequency capacitors with larger electrolytic capacitors. Representative local decoupling components include

$$C_7 = C_{18} = 100 \text{ nF}$$

and

$$C_{10} = C_{11} = 100 \text{ } \mu\text{F}.$$

The smaller capacitors are effective at higher frequencies, while the electrolytic capacitors provide local energy storage at lower frequencies.

A.11 Output Stability Network

The amplifier output includes a resistor-capacitor stability network, commonly referred to as a Zobel network.

Its purpose is to maintain a predictable high-frequency load condition and improve amplifier stability when driving a real loudspeaker and associated cabling.

The network is connected close to the amplifier output node and power-ground return.

A.12 Connection to the Protection Controller

The raw amplifier output is not connected directly to the external loudspeaker terminal in the final system.

Instead, the signal path is

Power Amplifier Output $\rightarrow$ Speaker Protection Relay $\rightarrow$ Speaker Terminal.

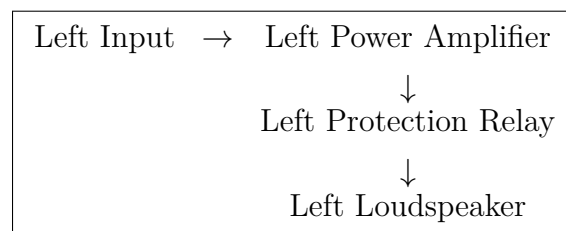
The amplifier output is also monitored by the STM32-based DC protection circuit.

This allows the loudspeaker to be disconnected if a persistent abnormal DC voltage is detected.

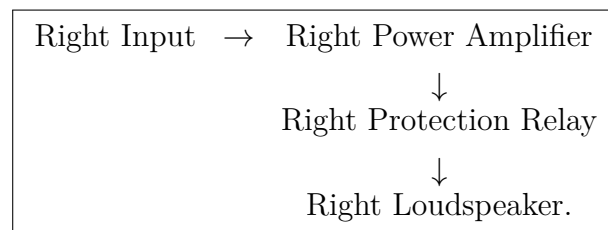
A.13 Stereo Implementation

The complete amplifier contains two equivalent implementations of the power amplifier circuit.

The stereo arrangement is therefore



and



Both channels share the main symmetrical power supply while retaining independent amplifier PCBs, output devices, heatsinks, bias adjustment, and speaker protection.

A.14 Schematic Reference

Figure A.1 contains the final KiCad power-amplifier schematic used as the electrical reference for PCB design, assembly, and commissioning.

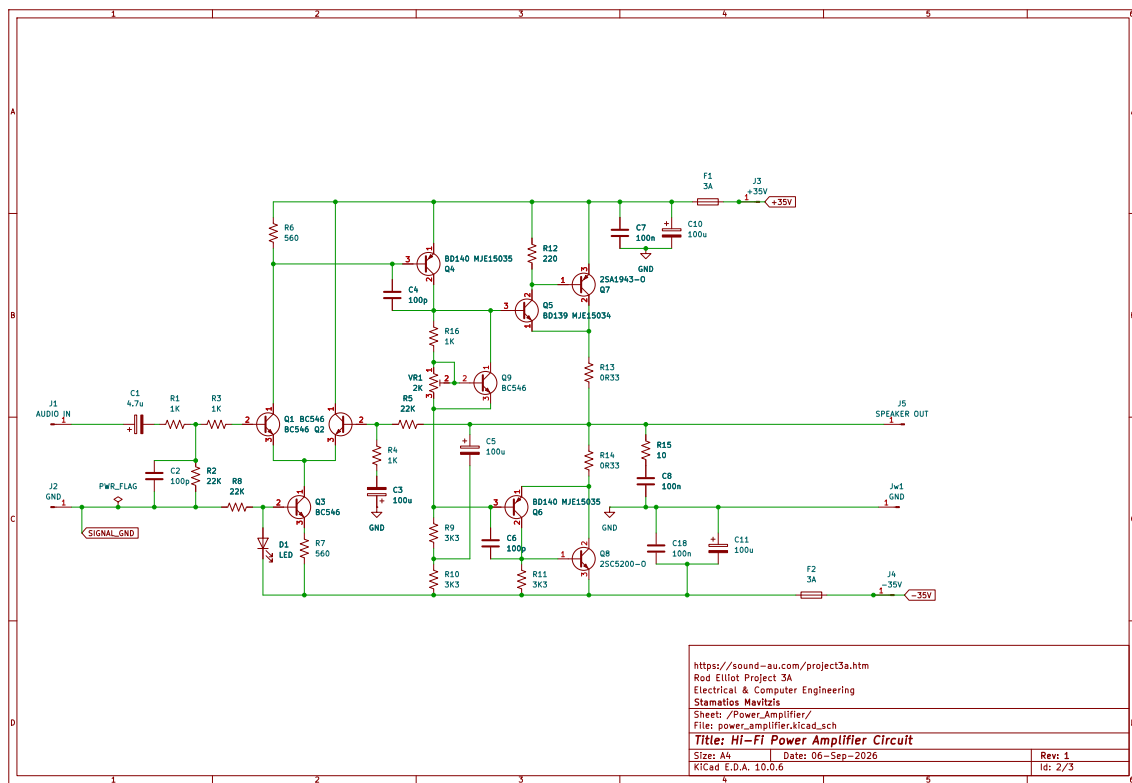


Figure A.1: Final power-amplifier schematic used in the project.

A.15 Use of the Schematic During Construction

The schematic in this appendix provides the primary electrical reference during assembly and troubleshooting.

It can be used to verify:

- resistor and capacitor values;
- transistor type and orientation;
- positive and negative supply connections;
- feedback-network connections;
- bias-network wiring;
- output-stage connections;
- local decoupling;
- rail-fuse positions;
- amplifier input and output nodes;

- ground connections.

The schematic should always be used together with the relevant component datasheets and the final PCB revision.

A.16 Appendix Summary

The schematic presented in this appendix represents the final discrete Class-AB power-amplifier circuit used for each channel of the stereo system.

The design incorporates the BC546 small-signal stages, BD139/BD140 driver devices, 2SA1943-O and 2SC5200-O power-output transistors, adjustable Class-AB bias, global negative feedback, local supply decoupling, output stability components, and 3 A rail protection.

The amplifier is designed to operate from the approximately

$$\pm 33\text{--}\pm 34\text{ VDC}$$

rails produced by the project's 24–0–24 V AC transformer and linear power supply.

Two identical amplifier channels are integrated with the preamplifier, STM32-based protection controller, speaker relays, heatsinks, cooling system, and chassis to form the complete stereo Hi-Fi amplifier.

Appendix B

Power Supply Schematic

B.1 Overview

This appendix presents the final main power-supply schematic used to supply the two Class-AB power-amplifier channels.

The main supply is an unregulated symmetrical linear power supply based on a centre-tapped

$$24-0-24 \text{ V}_{\text{AC}}$$

toroidal transformer.

The transformer secondary voltage is rectified by a high-current bridge rectifier and filtered using a large reservoir-capacitor bank to generate the positive and negative DC supply rails required by the amplifier.

The complete power-conversion path is

$$230 \text{ V}_{\text{AC}} \rightarrow \text{Toroidal Transformer} \rightarrow 24-0-24 \text{ V}_{\text{AC}} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Reservoir Capacitors} \rightarrow \pm V_S$$

The power-supply section is shared by both the left and right power-amplifier channels.

B.2 Power Supply Schematic

The main power-supply circuit is shown on the second page of the combined power-amplifier schematic.

The first page contains the power-amplifier circuit, while the second page contains the bridge rectifier, reservoir capacitors, high-frequency bypass capacitors, ground reference, and positive and negative DC rail outputs.

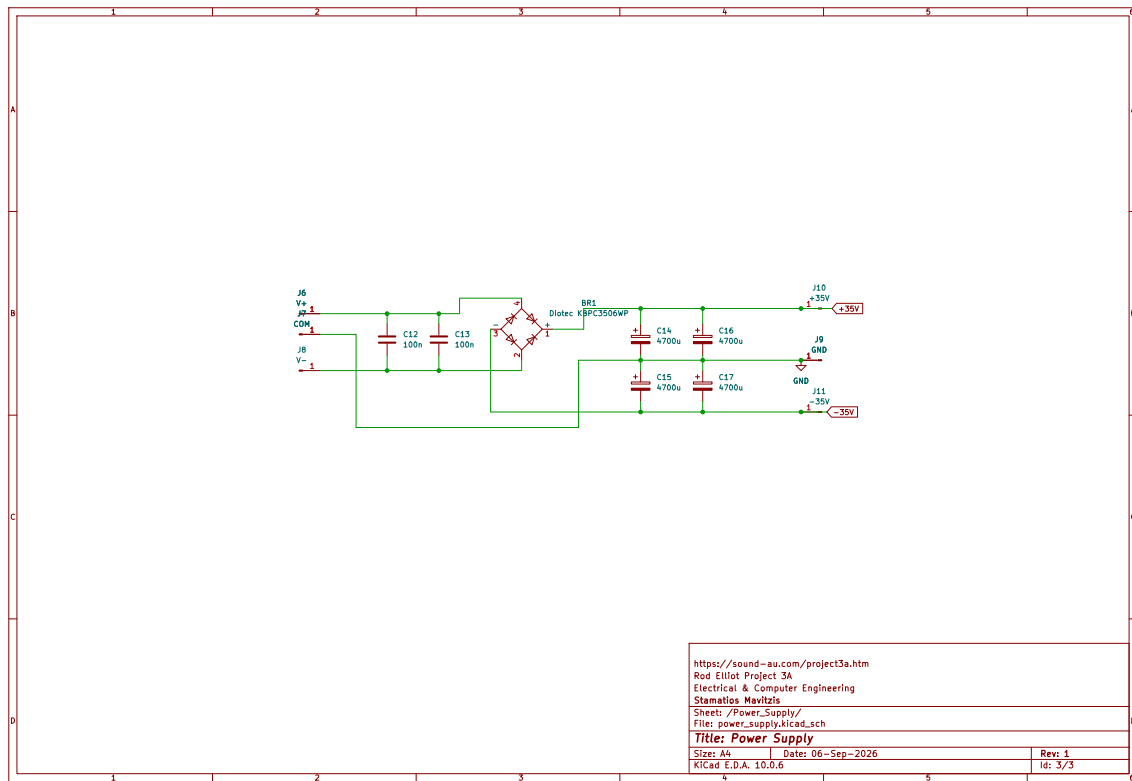


Figure B.1: Main symmetrical power-supply schematic used for the stereo power amplifier.

The external toroidal transformer is not drawn directly on this schematic sheet. The schematic begins at the secondary-side connections from the 24–0–24 V AC transformer.

B.3 Transformer Interface

The main toroidal transformer provides a centre-tapped secondary winding with the nominal voltage

$$24\text{--}0\text{--}24 \text{ V}_{\text{RMS}}.$$

Each half of the secondary produces approximately

$$24 \text{ V}_{\text{RMS}}$$

relative to the centre tap.

The complete voltage between the two outer secondary terminals is therefore

$$V_{\text{secondary}} = 24 + 24 = 48 \text{ V}_{\text{RMS}}.$$

The two outer secondary connections feed the rectification stage, while the centre tap establishes the central

$$0 \text{ V}$$

reference of the symmetrical supply.

B.4 Secondary Peak Voltage

For a sinusoidal secondary voltage, the peak voltage of each 24 V RMS winding is

$$V_{\text{peak}} = V_{\text{RMS}} \sqrt{2}.$$

Therefore,

$$V_{\text{peak}} = 24\sqrt{2} \approx 33.94 \text{ V}.$$

This represents the theoretical peak voltage available from each half of the transformer winding before rectifier losses, transformer regulation, ripple, wiring resistance, and amplifier loading are considered.

B.5 Bridge Rectifier

The main bridge rectifier is designated BR1 and is implemented using the

$$\boxed{\text{KBPC3506WP}}.$$

The bridge provides full-wave rectification of the transformer secondary voltage.

During operation, the appropriate internal diode pair conducts during each half-cycle of the AC waveform so that the reservoir capacitors are charged with the correct polarity.

The conversion can be represented as

$$\boxed{\text{Transformer AC} \rightarrow \text{BR1} \rightarrow \text{Full-Wave Rectified Supply}}.$$

B.6 Rectification Frequency

The mains frequency is

$$f_{\text{mains}} = 50 \text{ Hz}.$$

Because full-wave rectification uses both halves of the AC waveform, the principal capacitor charging and ripple frequency is

$$f_{\text{ripple}} = 2f_{\text{mains}}.$$

Therefore,

$$f_{\text{ripple}} = 100 \text{ Hz}.$$

The reservoir capacitors reduce this 100 Hz voltage variation before the supply reaches the power-amplifier stages.

B.7 Reservoir Capacitor Bank

The main power-supply schematic contains four large electrolytic reservoir capacitors:

$$C_{14} = C_{15} = C_{16} = C_{17} = 4700 \text{ } \mu\text{F}.$$

The capacitors are rated at

$$63 \text{ V},$$

providing substantial voltage margin above the expected amplifier rail voltage.

Two capacitors are associated with the positive supply rail and two with the negative supply rail.

Because the two capacitors associated with each rail operate in parallel, the effective reservoir capacitance per rail is

$$C_{\text{rail}} = 4700 \text{ } \mu\text{F} + 4700 \text{ } \mu\text{F}.$$

Therefore,

$$C_{\text{rail}} = 9400 \text{ } \mu\text{F}.$$

Equivalently,

$$C_{\text{rail}} = 9.4 \times 10^{-3} \text{ F}.$$

B.8 High-Frequency Bypass Capacitors

The main reservoir capacitors are complemented by the smaller bypass capacitors

$$C_{12} = C_{13} = 100 \text{ nF}.$$

The large electrolytic capacitors provide energy storage and low-frequency filtering, while the smaller capacitors provide a lower impedance at higher frequencies.

The combination therefore provides power-supply decoupling over a wider frequency range than the electrolytic capacitors alone.

B.9 Symmetrical DC Output

The power-supply circuit produces three main electrical nodes:

$$\boxed{+V_S, \quad 0 \text{ V}, \quad -V_S}.$$

The centre tap of the transformer and the midpoint of the reservoir-capacitor banks establish the central 0 V reference.

The positive capacitor bank develops the positive supply rail relative to this reference, while the negative capacitor bank develops the corresponding negative supply rail.

This symmetrical arrangement is required by the direct-coupled power amplifier and allows the audio output to operate around approximately

$$0 \text{ V}$$

without requiring a large series output coupling capacitor.

B.10 Expected DC Rail Voltage

The ideal peak value of each 24 V transformer half is approximately

$$33.94 \text{ V}.$$

The practical supply voltage is reduced by the bridge-rectifier voltage drop, transformer regulation, reservoir-capacitor ripple, wiring resistance, and load current.

The expected practical supply rails are therefore approximately

$$\boxed{V_+ \approx +33 - +34 \text{ VDC}}$$

and

$$\boxed{V_- \approx -33 - -34 \text{ VDC}}.$$

These values are design expectations rather than measured results.

The actual positive and negative supply voltages will be recorded during initial commissioning of the completed amplifier.

B.11 Supply Rail Symmetry

Because the amplifier uses a symmetrical power supply, the magnitudes of the two rails should be approximately equal:

$$|V_+| \approx |V_-|.$$

A useful parameter during commissioning is the rail asymmetry

$$\Delta V_{\text{sym}} = ||V_+| - |V_-||.$$

A small difference is expected because of transformer tolerances, capacitor tolerances, semiconductor voltage drops, wiring resistance, and unequal loading.

A large difference between the two rails would require further investigation.

B.12 Power Supply Ground Reference

The electrical midpoint between the positive and negative reservoir banks is a particularly important node in the complete amplifier.

It forms the principal power-supply

$$0 \text{ V}$$

reference and carries substantial power-stage and rectifier return currents.

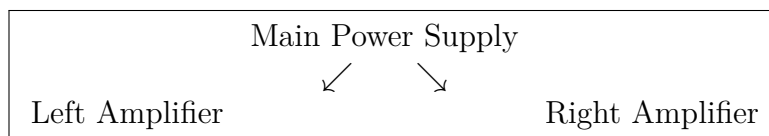
The grounding architecture must therefore prevent these high-current return currents from flowing unnecessarily through sensitive audio signal-ground paths.

The power-supply ground connection is consequently integrated into the controlled star-ground architecture described in the main report.

B.13 Stereo Power Distribution

The same main supply powers both amplifier channels.

The intended distribution is conceptually



rather than supplying one amplifier through the wiring of the other.

Separate low-impedance supply and return connections reduce common wiring impedance and minimise interaction between the two audio channels.

B.14 High-Current Paths

The connections between the transformer, BR1, reservoir capacitors, and amplifier supply rails carry the largest currents in the power-supply section.

During capacitor charging, the principal loop is

$$\boxed{\text{Transformer} \rightarrow \text{Bridge Rectifier} \rightarrow \text{Reservoir Capacitors} \rightarrow \text{Transformer}}.$$

During amplifier operation, a principal energy path is

$$\boxed{\text{Reservoir Capacitors} \rightarrow \text{Power Amplifier} \rightarrow \text{Power Return} \rightarrow \text{Reservoir Capacitors}}.$$

These paths should remain short and have low electrical resistance.

B.15 Stored Energy

The reservoir-capacitor banks store significant electrical energy during operation.

The energy stored in a capacitor is

$$E = \frac{1}{2}CV^2.$$

Using

$$C = 9.4 \times 10^{-3} \text{ F}$$

and approximately

$$V = 34 \text{ V}$$

for each rail gives

$$E_{\text{rail}} = \frac{1}{2}(9.4 \times 10^{-3})(34)^2.$$

Therefore,

$$E_{\text{rail}} \approx 5.43 \text{ J}.$$

The two reservoir banks together can therefore contain approximately

$$\boxed{E_{\text{total}} \approx 10.9 \text{ J}}.$$

The capacitors may retain voltage after mains power has been removed, so the supply rails must be checked before servicing the amplifier.

B.16 Connection to the Power Amplifier

The three main power-supply outputs are distributed to each amplifier channel:

$$+V_S, \quad 0 \text{ V}, \quad -V_S.$$

The positive and negative rails are then locally decoupled on each power-amplifier PCB. Each amplifier channel also includes its own rail fuse protection before the supply reaches the active amplifier circuitry.

B.17 Use of the Schematic During Construction

The power-supply schematic provides the primary reference during assembly, wiring, commissioning, and troubleshooting of the main amplifier supply.

It should be used to verify:

- transformer secondary connections;
- transformer centre-tap connection;
- BR1 terminal identification;
- reservoir-capacitor polarity;
- C12 and C13 connections;
- positive DC rail;
- negative DC rail;
- central 0 V reference;
- connection of the supply to both amplifier channels.

Particular attention must be given to the polarity of the electrolytic reservoir capacitors because incorrect polarity can result in component failure when the power supply is energised.

B.18 Measurements During Commissioning

The schematic also defines the principal measurement points used during initial power-up. The following voltages should be measured relative to the central 0 V reference:

$$V_+,$$

and

$$V_-.$$

The expected initial condition is approximately

$$V_+ \approx +33 - +34 \text{ V},$$

$$V_- \approx -33 - -34 \text{ V}.$$

Measurements should subsequently be repeated under controlled amplifier loading so that rail sag and ripple can be evaluated.

B.19 Appendix Summary

The power-supply schematic presented in this appendix represents the main symmetrical linear supply used by both channels of the stereo amplifier.

An external

$$24-0-24 \text{ V}_{\text{AC}}$$

toroidal transformer supplies BR1, a KBPC3506WP full-wave bridge rectifier.

Four

$$4700 \text{ } \mu\text{F}$$

electrolytic capacitors form the principal reservoir bank, providing

$$9400 \text{ } \mu\text{F}$$

of effective capacitance per rail.

C12 and C13, each

$$100 \text{ nF},$$

provide additional high-frequency bypassing.

The resulting power-supply outputs are

$$+V_S, \quad 0 \text{ V}, \quad -V_S$$

with expected practical rail voltages in the approximate region of

$$\pm 33 - \pm 34 \text{ VDC}.$$

These rails provide the main electrical power for both Class-AB amplifier channels, while the central 0 V node forms an important part of the system grounding architecture.

The schematic serves as the definitive wiring and troubleshooting reference for the rectifier and reservoir-capacitor section during construction and commissioning of the completed amplifier.

Appendix C

Preamplifier Schematic

C.1 Overview

This appendix presents the complete schematic of the analogue audio preamplifier integrated into the Hi-Fi amplifier system.

The preamplifier operates between the external audio sources and the two main Class-AB power-amplifier channels. Its principal purpose is to provide source selection, phono amplification and RIAA equalisation, voltage gain, balance control, volume control, and a suitable line-level output for the main power amplifier.

The general signal path is

Audio Source → Preamplifier → Power Amplifier → Speaker Protection → Loudspeaker
--

Unlike the main power-amplifier stage, the preamplifier processes relatively small analogue signals. Low noise, low distortion, appropriate gain, controlled grounding, and good immunity to interference are therefore the primary design considerations.

C.2 Preamplifier Schematic

Figure C.1 shows the complete preamplifier schematic used in the project.

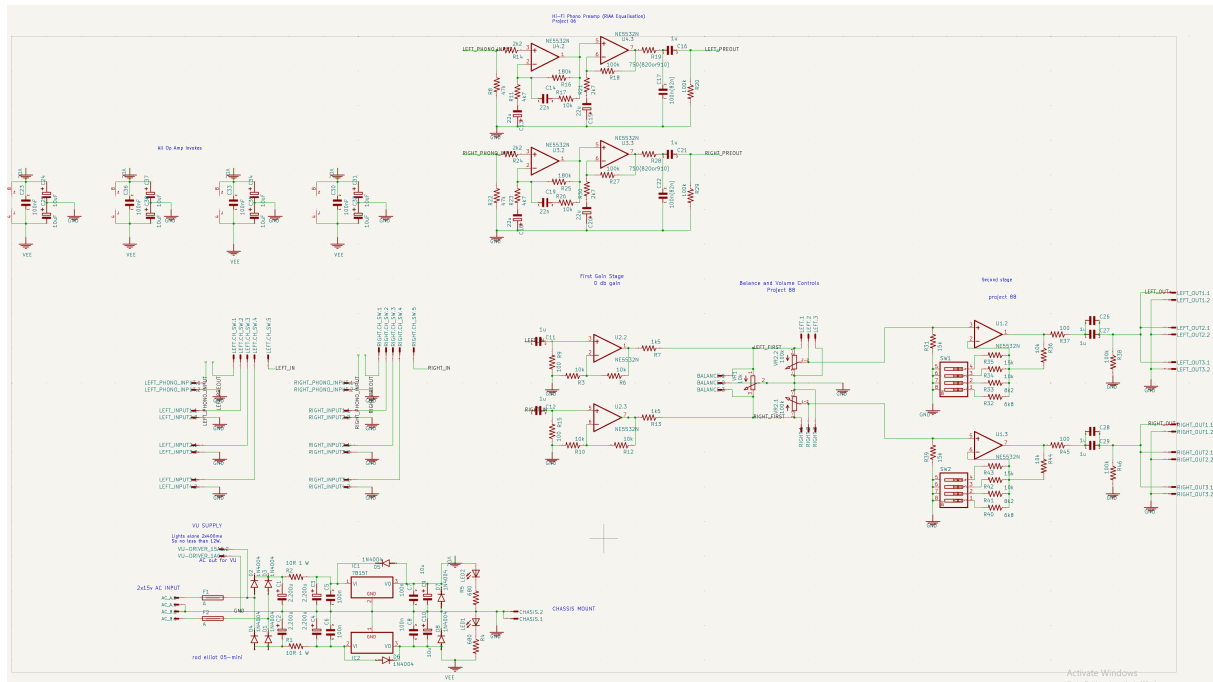


Figure C.1: Complete schematic of the stereo audio preamplifier used in the Hi-Fi amplifier system.

The schematic provides the main electrical reference for the preamplifier during construction, integration, testing, and troubleshooting.

C.3 Functional Architecture

The preamplifier schematic can be divided into the following principal functional sections:

- moving-magnet phono input;
- three line-level inputs;
- input selection;
- phono voltage amplification;
- RIAA playback equalisation;
- first voltage-gain stage;
- second voltage-gain stage;
- selectable gain;
- stereo balance control;
- stereo volume control;
- line-level output stage;

- regulated symmetrical power supply;
- local operational-amplifier decoupling;
- analogue star-ground network.

The overall audio routing can be represented as

$$\left\{ \begin{array}{l} \text{Phono Input} \rightarrow \text{RIAA Phono Stage,} \\ \text{Line Inputs} \end{array} \right. \rightarrow \text{Input Selection} \rightarrow \text{Gain Stages} \rightarrow \text{Balance} \rightarrow \text{Volume} \rightarrow \text{C}$$

C.4 Audio Inputs

The implemented preamplifier provides four source inputs.

These consist of:

- one dedicated moving-magnet phono input;
- three conventional line-level inputs.

The line-level inputs are suitable for sources that already provide an appropriate audio voltage, such as CD players, DACs, media streamers, tuners, and other analogue audio equipment.

These sources do not require the additional frequency-dependent processing used by the phono input.

C.5 Phono Input Stage

The phono input is intended for a moving-magnet turntable cartridge.

The electrical signal generated by such a cartridge is substantially smaller than a conventional line-level signal. Consequently, the phono section must provide considerably greater initial voltage gain.

The phono signal also requires RIAA playback equalisation before it can be treated as a conventional line-level source.

The phono section therefore performs two principal functions:

Low-Noise Voltage Amplification

and

RIAA Playback Equalisation.

C.6 RIAA Equalisation

Vinyl records are recorded using a frequency-dependent characteristic rather than a flat frequency response.

The phono preamplifier applies the complementary playback characteristic to approximately restore the original frequency balance.

The standard RIAA playback characteristic is defined by the principal time constants

$$\tau_1 = 3180 \mu\text{s},$$

$$\tau_2 = 318 \mu\text{s},$$

and

$$\tau_3 = 75 \mu\text{s}.$$

The corresponding break frequencies are obtained from

$$f = \frac{1}{2\pi\tau}.$$

Therefore, the characteristic frequencies are approximately

$$50.05 \text{ Hz},$$

$$500.5 \text{ Hz},$$

and

$$2122 \text{ Hz}.$$

The resistor-capacitor feedback network around the phono amplifier produces the required frequency-dependent closed-loop gain.

C.7 NE5532N Operational Amplifiers

The analogue amplification sections use NE5532N dual operational amplifiers.

The NE5532N is used for the low-level audio stages, including the phono and subsequent voltage-gain sections.

Using dual operational amplifiers also allows corresponding left- and right-channel circuitry to be implemented efficiently while maintaining a symmetrical stereo architecture.

The signal-processing stages operate from symmetrical supply rails, allowing the audio waveform to remain centred around approximately

0 V.

C.8 Stereo Phono Signal Path

Equivalent phono circuitry is provided for the left and right channels.

For each channel, the signal path can be represented as

MM Cartridge → NE5532N Phono Stage → RIAA Equalisation → Line-Level Signal.

Because the phono signal amplitude is very small, the corresponding PCB traces and input wiring require particular attention to shielding, grounding, and physical separation from mains and high-current circuitry.

C.9 Line-Level Signal Path

The three line-level inputs bypass the RIAA processing stage.

After source selection, the selected line input enters the normal gain and user-control circuitry.

The line-level path can therefore be represented as

Line Input → Input Selector → Gain Stage → Balance Control → Volume Control → Output.

After RIAA equalisation and amplification, the phono signal eventually enters the same general signal-processing chain.

C.10 Input Selection

The input-selection section determines which source is connected to the subsequent preamplifier stages.

Only the selected source should be passed through the active signal chain, while the remaining inputs remain isolated from the output path.

The input-selection system therefore allows the same amplifier to accept both turntable and conventional line-level sources.

C.11 Voltage-Gain Stages

The preamplifier contains first and second voltage-gain stages after the input and phono sections.

Dividing the required amplification between multiple stages allows gain, headroom, noise, and user controls to be managed at appropriate positions throughout the signal path.

For a conventional non-inverting operational-amplifier stage, the voltage gain is

$$A_v = 1 + \frac{R_f}{R_g},$$

where R_f is the feedback resistance and R_g is the resistance between the inverting input and the reference node.

The corresponding gain in decibels is

$$A_{v,\text{dB}} = 20 \log_{10}(A_v).$$

C.12 Selectable Gain

The schematic incorporates selectable gain so that the preamplifier signal level can be adapted to the connected source and to the sensitivity of the main power amplifier.

The purpose of gain selection is to obtain sufficient output voltage while retaining adequate headroom.

Excessive gain can cause an internal stage to clip before the main power amplifier reaches full output, while insufficient gain may prevent the power stage from being fully driven.

C.13 Balance Control

A stereo balance control allows the relative level of the left and right channels to be adjusted.

With the control centred, both channels should experience approximately equal attenuation.

Moving the control away from the centre progressively reduces the level of one channel relative to the other.

This function allows compensation for differences in listening position, speaker placement, room acoustics, or source-channel level.

C.14 Volume Control

The volume control determines the signal level delivered to the main power amplifier.

The power amplifier itself operates with an approximately fixed voltage gain, so normal listening-level adjustment is performed within the preamplifier.

The volume control can be represented as an adjustable attenuation factor

$$0 \leq k \leq 1.$$

The resulting signal is approximately

$$V_{\text{out}} = kV_{\text{signal}}.$$

At minimum volume,

$$k \rightarrow 0,$$

while at maximum volume,

$$k \rightarrow 1.$$

C.15 Preamplifier Output

The output of the preamplifier is a line-level analogue signal intended to drive the input of the main Class-AB power amplifier.

The main power amplifier has a nominal voltage gain of approximately

$$A_v \approx 23$$

or

$$A_{v,\text{dB}} \approx 27.2 \text{ dB}.$$

The preamplifier therefore does not need to provide significant output power. Its purpose is to provide a clean, controlled signal voltage with sufficiently low output impedance and adequate headroom for the following power stage.

C.16 Symmetrical Power Supply

The preamplifier operates from an independent regulated symmetrical supply.

The supply rails are

$$\boxed{+15 \text{ V}}$$

and

$$\boxed{-15 \text{ V}}$$

relative to the analogue ground reference.

The total voltage difference available to the operational-amplifier circuitry is therefore

$$V_{\text{total}} = 15 - (-15) = \boxed{30 \text{ V}}.$$

The symmetrical supply allows the audio signal to remain centred around approximately 0 V without requiring a virtual midpoint reference.

C.17 Voltage Regulation

The positive preamplifier supply is regulated using a 7815 linear regulator,

$$V_{+, \text{preamp}} = +15 \text{ V},$$

while the negative supply is regulated using a 7915 device,

$$V_{-, \text{preamp}} = -15 \text{ V}.$$

Using separate regulated rails reduces the influence of supply variation and power-stage current demand on the low-level analogue circuitry.

This is particularly important in the phono section because interference introduced at this point would be amplified together with the wanted audio signal.

C.18 Supply Filtering and Decoupling

The preamplifier power-supply path includes rectification, bulk filtering, linear regulation, and local operational-amplifier decoupling.

The general sequence is

AC Supply → Rectification → Smoothing → 7815 / 7915 Regulation → Local Decoupling.

Local bypass capacitors are placed close to the operational-amplifier supply connections to provide low-impedance high-frequency current paths and improve circuit stability.

C.19 Grounding Arrangement

The preamplifier uses a controlled analogue grounding arrangement.

Conceptually, the principal return connections can be represented as

$$\left\{ \begin{array}{l} \text{Phono Ground} \\ \text{Line Input Ground} \\ \text{Gain Stage Ground} \\ \text{Output Ground} \\ \text{Power-Supply Ground} \end{array} \right. \rightarrow \text{Preamplifier Star Ground.}$$

This arrangement reduces the possibility that current from one preamplifier section will create an unwanted voltage in the reference path of another section.

The preamplifier ground is subsequently connected to the complete amplifier grounding system at a controlled point.

C.20 Separation from High-Current Circuitry

The preamplifier must remain physically separated from the electrically noisy sections of the complete amplifier.

Particular attention should be given to separation from:

- the main toroidal transformer;
- bridge-rectifier wiring;
- reservoir-capacitor charging-current paths;
- power-amplifier output wiring;
- loudspeaker wiring;
- protection relay coils;
- cooling-fan wiring;
- mains wiring.

This is especially important for the phono stage because it processes the smallest signal in the complete amplifier.

C.21 Use of the Schematic During Construction

The schematic in Figure C.1 is the main electrical reference for the construction and verification of the preamplifier.

It should be used to check:

- phono input connections;
- line-level input connections;

- input-selection wiring;
- NE5532N orientation;
- RIAA network components;
- first and second gain-stage connections;
- gain-selection connections;
- volume-control wiring;
- balance-control wiring;
- left- and right-channel routing;
- preamplifier output connections;
- +15 V rail;
- −15 V rail;
- analogue ground;
- local decoupling components.

C.22 Commissioning Reference

During independent preamplifier commissioning, the schematic should also be used to identify the principal measurement points.

The first electrical checks should confirm approximately

$$\boxed{+15 \text{ V}}$$

and

$$\boxed{-15 \text{ V}}$$

at the regulated supply rails.

The preamplifier outputs should then be checked for abnormal DC voltage before the subsystem is connected to the main power amplifier.

A low-level sinusoidal input can subsequently be used to verify:

- correct operation of all three line inputs;
- correct source selection;
- left- and right-channel continuity;
- gain-stage operation;

- gain selection;
- balance adjustment;
- volume adjustment;
- clean line-level output.

The phono input should additionally be tested for gain and RIAA frequency response.

C.23 Appendix Summary

The schematic presented in this appendix represents the complete analogue preamplifier used ahead of the two Class-AB power-amplifier channels.

The design provides one moving-magnet phono input and three line-level inputs.

The phono section uses low-noise NE5532N operational amplifiers together with frequency-dependent negative feedback to provide the required voltage gain and RIAA playback equalisation.

The remaining signal chain provides input selection, first and second voltage gain stages, selectable gain, stereo balance control, volume adjustment, and line-level output.

The analogue circuitry is powered from a regulated symmetrical

$$\boxed{\pm 15 \text{ V}}$$

supply generated using 7815 and 7915 linear regulators.

The preamplifier uses local supply decoupling and controlled analogue grounding because its signal amplitudes are considerably smaller than those present in the power-amplifier and loudspeaker sections.

The schematic therefore serves as the principal reference for the assembly, wiring, independent testing, troubleshooting, and final integration of the preamplifier into the complete stereo Hi-Fi amplifier.

Appendix D

Protection Controller Schematic

D.1 Overview

This appendix presents the complete schematic of the STM32-based protection controller developed for the stereo power amplifier.

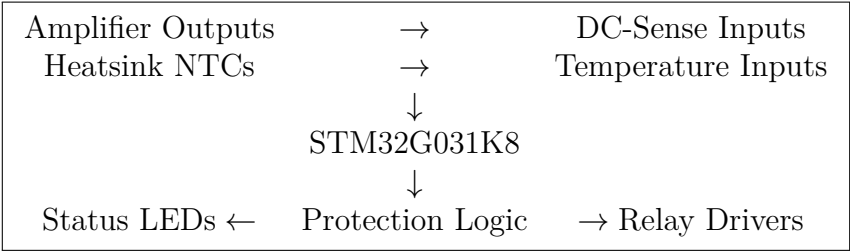
The protection controller operates independently from the analogue audio signal-processing circuitry and supervises both power-amplifier channels. Its purpose is to prevent the loudspeakers from remaining connected during potentially unsafe operating conditions.

The principal protection functions are:

- delayed loudspeaker connection during startup;
- left-channel DC output monitoring;
- right-channel DC output monitoring;
- left-heatsink temperature monitoring;
- right-heatsink temperature monitoring;
- temperature-sensor fault detection;
- overtemperature protection;
- independent relay control for both loudspeaker channels;
- visual indication of operating and fault conditions.

The central control device is the STM32G031K8 microcontroller implemented using the ST NUCLEO-G031K8 development board.

The overall controller architecture can be represented as



D.2 Protection Controller Schematic

Figure D.1 shows the complete protection-controller schematic used in the project.

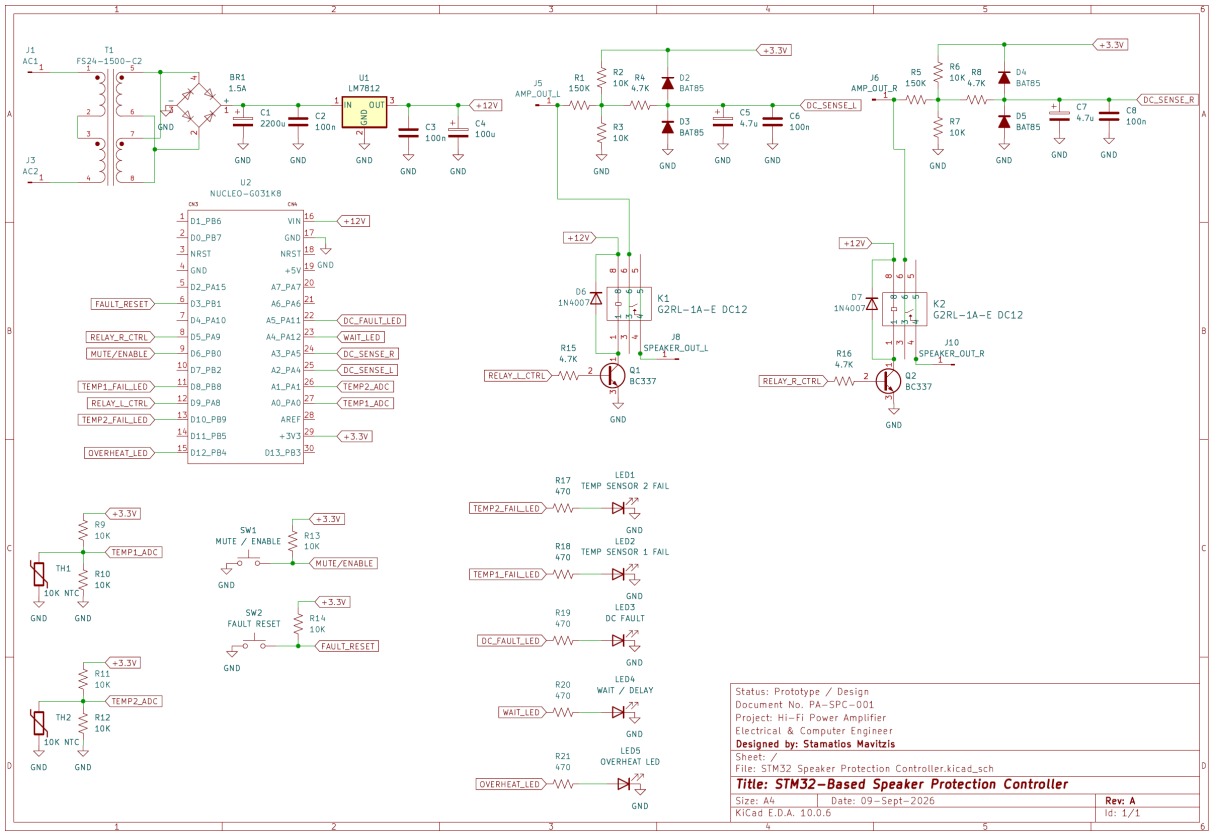


Figure D.1: Complete schematic of the STM32-based loudspeaker protection controller.

The schematic is used as the main electrical reference for assembly, firmware development, PCB verification, commissioning, fault diagnosis, and final system integration.

D.3 Functional Sections

The protection-controller schematic can be divided into the following functional sections:

- isolated auxiliary power supply;

- 12 V regulated supply;
- STM32 / NUCLEO-G031K8 controller;
- left-channel DC-sensing circuit;
- right-channel DC-sensing circuit;
- left-heatsink temperature-sensing circuit;
- right-heatsink temperature-sensing circuit;
- left speaker-relay driver;
- right speaker-relay driver;
- relay flyback protection;
- status and fault LEDs;
- push-button control input;
- loudspeaker input and output connectors.

D.4 Microcontroller

The controller is based on the STM32G031K8 microcontroller using the NUCLEO-G031K8 development board.

The microcontroller is responsible for acquiring the analogue sensing signals, executing the protection logic, controlling the loudspeaker relays, and driving the status indicators.

The controller therefore provides the logical relationship

Sensor Inputs $\rightarrow$ STM32 Processing $\rightarrow$ Protection Decision $\rightarrow$ Relay State
--

Using a programmable microcontroller allows the protection behaviour to incorporate averaging, persistence checks, startup timing, temperature conversion, hysteresis, and fault-state handling.

D.5 Independent Left and Right Channels

The protection controller monitors the two power-amplifier channels independently.

The input structure can be represented as

Left Amplifier Output $\rightarrow$ Left DC Sense
Right Amplifier Output $\rightarrow$ Right DC Sense

and

Left Heatsink → TEMP1
Right Heatsink → TEMP2.

This arrangement allows a fault in either channel to be identified by the controller.

D.6 Auxiliary Power Supply

The protection controller uses an isolated low-voltage power supply that is independent of the main high-voltage amplifier rails.

The basic supply architecture is

230 V_{AC} → Isolated Transformer → Bridge Rectifier → Reservoir Filtering → 7812 → 12 VDC.

The 12 V rail is used primarily for the relay-driving and controller power system.

The microcontroller section also uses the regulated logic supply required by the NUCLEO-G031K8 platform.

D.7 12 V Regulation

A 7812 linear voltage regulator is used to produce the regulated

12 V

supply.

This provides a stable voltage for the 12 V relay coils and associated low-voltage circuitry.

The regulator input is supplied from the rectified and filtered output of the auxiliary transformer.

D.8 Mains Isolation

Because the auxiliary transformer primary is connected to the 230 V AC mains, a portion of the protection-controller PCB contains hazardous mains voltage.

This region must remain physically and electrically separated from the secondary-side low-voltage electronics.

The ordinary PCB zone clearance used elsewhere in the design must not be interpreted as a mains isolation requirement.

The final board must therefore maintain an appropriate dedicated primary-side isolation region according to the applicable electrical-safety requirements.

D.9 DC Output Monitoring

The main purpose of the DC-sensing circuits is to detect an abnormal persistent DC component at either power-amplifier output.

The normal amplifier output is an AC audio waveform centred approximately around

$$0 \text{ V.}$$

A dangerous amplifier fault can instead produce a sustained positive or negative output voltage.

The protection controller therefore monitors both outputs before permitting the loudspeakers to remain connected.

D.10 DC-Sense Signal Conditioning

The amplifier output voltage cannot be applied directly to an STM32 ADC input.

The sensing network therefore performs several functions before the signal reaches the microcontroller:

- voltage attenuation;
- low-pass filtering;
- level conditioning;
- ADC input protection.

The signal path can be represented as

$$V_{\text{OUT}} \rightarrow \text{Attenuation} \rightarrow \text{Low-Pass Filter} \rightarrow \text{Protection} \rightarrow \text{STM32 ADC.}$$

D.11 DC-Sense Low-Pass Filtering

The sensing circuit is intentionally designed to respond strongly to slowly varying or constant DC conditions while attenuating the normal audio signal.

For a first-order RC network, the cutoff frequency is

$$f_c = \frac{1}{2\pi RC}.$$

For the implemented sensing network, the effective resistance is approximately

$$R_{\text{eq}} \approx 9.54 \text{ k}\Omega$$

and the filter capacitance is

$$C = 4.7 \mu\text{F}.$$

The resulting approximate cutoff frequency is

$$f_c = \frac{1}{2\pi(9.54 \times 10^3)(4.7 \times 10^{-6})}.$$

Therefore,

$$\boxed{f_c \approx 3.55 \text{ Hz}}.$$

This very low cutoff frequency substantially attenuates ordinary audio frequencies while allowing persistent DC conditions to be detected.

D.12 ADC Protection

The STM32 ADC inputs must remain within their allowable voltage range.

Protective components are therefore incorporated into the sensing network to limit abnormal input conditions before they reach the microcontroller.

The purpose of this protection is not to make the ADC immune to arbitrary power-amplifier faults, but to provide an additional protective interface between the high-voltage audio output and the low-voltage microcontroller input.

D.13 Digital DC Fault Detection

The firmware does not need to react to a single instantaneous ADC sample.

Instead, multiple samples can be averaged and a persistence requirement can be used before a DC fault is declared.

Conceptually,

$$|V_{\text{DC}}| > V_{\text{threshold}}$$

must remain true for a defined interval before the protection state is entered.

This approach reduces false triggering caused by audio peaks, switching noise, or isolated ADC disturbances.

The final numerical thresholds and persistence time are commissioning parameters and should be verified experimentally on the completed hardware.

D.14 Temperature Monitoring

The controller monitors the temperature of both power-amplifier heatsinks.

Each heatsink uses a 10 k Ω NTC thermistor.

The two channels are represented as

$\begin{array}{lcl} \text{NTC}_L & \rightarrow & \text{TEMP1} \\ \text{NTC}_R & \rightarrow & \text{TEMP2.} \end{array}$
--

The NTC resistance decreases as the temperature increases.

D.15 NTC Temperature Relationship

The approximate thermistor resistance can be described using the beta model

$$R(T) = R_{25} \exp \left[B \left(\frac{1}{T} - \frac{1}{T_{25}} \right) \right].$$

For the selected sensors,

$$R_{25} = 10 \text{ k}\Omega$$

at

$$T_{25} = 25^\circ\text{C}.$$

The controller converts the ADC voltage into the corresponding sensor resistance and then estimates the heatsink temperature.

D.16 Overtemperature Protection

The measured temperature is compared against a firmware protection threshold.

Normal operation requires

$$T_L < T_{\text{trip}}$$

and

$$T_R < T_{\text{trip}}.$$

If either channel exceeds the final configured trip temperature, the microcontroller removes relay drive and disconnects the loudspeakers.

The final trip and reset temperatures are to be confirmed during thermal testing of the physical amplifier.

D.17 Temperature Hysteresis

A thermal protection system should not repeatedly switch between the ON and OFF states when the heatsink temperature is close to the protection threshold.

A hysteresis interval can therefore be implemented such that

$$T_{\text{trip}} > T_{\text{reset}}.$$

Once an overtemperature event occurs, the system remains protected until the heatsink has cooled sufficiently below the trip temperature.

D.18 Temperature Sensor Fault Detection

The firmware also distinguishes between a valid high-temperature condition and a failed temperature sensor.

Typical sensor failures include:

- open-circuit thermistor;
- short-circuit thermistor;
- disconnected cable;
- invalid ADC reading.

This allows the controller to distinguish

OVERHEAT

from

TEMPERATURE SENSOR FAULT

.

D.19 Speaker Relays

The loudspeaker channels are controlled using two Omron G2RL-1A-E-DC12 relays.

One relay is assigned to each amplifier channel.

The loudspeaker signal path is therefore

Left Amplifier $\rightarrow K_1 \rightarrow$ Left Speaker

and

Right Amplifier $\rightarrow K_2 \rightarrow$ Right Speaker.

D.20 Normally-Open Protection Strategy

The speaker relays use normally-open contacts.

Therefore,

Relay Coil De-Energised $\Rightarrow$ Speaker Disconnected.

The loudspeaker is connected only when the microcontroller intentionally energises the corresponding relay.

This creates a useful fail-safe characteristic during startup, power loss, or controller fault conditions.

D.21 Relay Driver Transistors

The STM32 GPIO pins do not drive the relay coils directly.

BC337 NPN transistors are used as low-side relay drivers.

The control sequence is

STM32 GPIO $\rightarrow$ BC337 $\rightarrow$ Relay Coil.

This isolates the microcontroller output from the relay coil current and provides the required current amplification.

D.22 Relay Coil Current

Each relay coil requires approximately

$$I_{\text{coil}} \approx 33.3 \text{ mA}.$$

With both relays energised simultaneously, the approximate total relay-coil current is

$$I_{\text{relays}} = 2(33.3 \text{ mA})$$

and therefore

$$I_{\text{relays}} \approx 66.6 \text{ mA.}$$

This current is supplied by the 12 V protection-controller supply rather than by the STM32 GPIO pins.

D.23 Flyback Diodes

A relay coil is an inductive load.

When the BC337 transistor switches the coil current off, the inductance attempts to maintain the previous current and can generate a high voltage across the driver transistor.

A 1N4007 diode is therefore connected across each relay coil.

During normal energised operation the diode is reverse biased.

When the relay is switched off, the diode provides a recirculation path for the coil current and limits the resulting voltage transient.

D.24 Startup Delay

Immediately after the amplifier is switched on, the loudspeakers should remain disconnected.

The intended startup sequence is

Power Applied → Controller Startup → WAIT / DELAY → Fault Verification → Relay Connection.
--

The relays are energised only if the monitored operating conditions are valid.

This delay prevents the loudspeakers from being connected during amplifier power-supply transients and startup settling.

D.25 Power-Off Behaviour

When controller power is removed, the relay coils de-energise and the normally-open contacts disconnect the loudspeakers.

The desired power-off sequence is therefore

Controller Power Loss → Relay Coil OFF → Speaker Disconnected.
--

The behaviour should be verified experimentally because the amplifier and controller power supplies may decay at different rates.

D.26 Status and Fault LEDs

Five visual indicators are incorporated into the protection-controller design.

These are used to indicate:

- TEMP SENSOR 1 FAIL;
- TEMP SENSOR 2 FAIL;
- DC FAULT;
- WAIT / DELAY;
- OVERHEAT.

The indicators allow the protection state to be identified without connecting external diagnostic equipment.

D.27 WAIT / DELAY Indicator

The WAIT / DELAY LED indicates that the system is within the startup protection period or otherwise has not yet enabled the loudspeaker relays.

Its state provides immediate confirmation that speaker disconnection at startup is intentional rather than the result of an unknown hardware fault.

D.28 DC Fault Indicator

The DC FAULT indicator shows that the controller has detected an abnormal persistent DC condition at an amplifier output.

The associated protection response is

DC Fault Detected → Relay Drive Removed → Speakers Disconnected.
--

D.29 Overheat Indicator

The OVERHEAT indicator identifies a thermal protection event in which one or both heatsinks have exceeded the configured safe operating threshold.

This indicator is separate from the temperature-sensor failure LEDs because an actual high-temperature condition and a failed sensor represent different fault mechanisms.

D.30 Sensor Failure Indicators

Separate indicators are provided for the two temperature channels.

This makes it possible to identify whether the fault originates from the left or right heatsink sensing system.

The two corresponding states are

TEMP SENSOR 1 FAIL

and

TEMP SENSOR 2 FAIL

.

D.31 Push-Button Input

The protection controller includes a user push-button input connected to the microcontroller.

Depending on the final firmware implementation, this input can be used for a function such as fault acknowledgement, reset, or controlled re-enable.

The final behaviour is defined in firmware and should therefore remain consistent with the protection philosophy of the hardware.

D.32 Speaker Current Path

Although the control and sensing sections operate at low current, the relay contact paths must carry the full loudspeaker current.

At a reference condition of

$$100 \text{ W}$$

into

$$4 \text{ } \Omega,$$

the RMS loudspeaker current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5 \text{ A.}$$

The peak current is

$$I_{\text{PEAK}} = 5\sqrt{2} \approx 7.07 \text{ A.}$$

The PCB therefore uses substantially wider routing for the speaker-current paths than for the microcontroller and ADC signals.

D.33 Relationship to the Audio Path

The STM32 itself is not placed directly in the analogue audio signal path.

Under normal operation, the audio signal passes through the mechanical relay contacts:

Amplifier Output → Relay Contact → Speaker.

The microcontroller controls whether this physical connection exists but does not digitally process the audio signal.

This preserves the analogue nature of the power-amplifier output path.

D.34 Protection Logic Summary

Normal loudspeaker connection requires the controller to confirm that all necessary operating conditions are acceptable.

Conceptually, relay enable can be described as

$$\text{RELAY_ENABLE} = \text{STARTUP_COMPLETE} \wedge \overline{\text{DC_FAULT}} \wedge \overline{\text{OVERHEAT}} \wedge \text{TEMP_SENSORS_VA}$$

If any required condition becomes false, the controller can remove relay drive.

D.35 Use of the Schematic During Construction

The schematic in Figure D.1 should be used as the main reference during protection-board assembly.

It should be used to verify:

- NUCLEO-G031K8 orientation;
- ADC input connections;
- left DC-sense network;
- right DC-sense network;
- temperature-sensor connectors;
- relay-driver transistor orientation;
- 1N4007 diode orientation;

- relay coil and contact connections;
- LED polarity;
- push-button wiring;
- 12 V supply;
- logic supply;
- transformer secondary connections;
- bridge-rectifier connections;
- 7812 regulator connections;
- ground connections.

D.36 Commissioning Reference

During initial commissioning, the protection-controller schematic defines the principal diagnostic and measurement points.

The following should be verified before the controller is connected to valuable loudspeakers:

- correct rectified auxiliary supply;
- correct regulated 12 V rail;
- correct logic supply;
- correct STM32 startup;
- valid TEMP1 ADC response;
- valid TEMP2 ADC response;
- valid left DC-sense response;
- valid right DC-sense response;
- correct BC337 switching;
- correct left relay operation;
- correct right relay operation;
- correct status LED operation.

D.37 Fault Testing

The protection functions should be tested using controlled fault simulation.

The commissioning sequence should verify the response to:

- startup condition;
- abnormal positive DC sense;
- abnormal negative DC sense;
- temperature-sensor open circuit;
- temperature-sensor short circuit;
- simulated overtemperature;
- controller power removal.

Actual destructive faults should not be deliberately produced in the power-amplifier hardware merely to verify the controller.

D.38 Final Threshold Verification

The schematic defines the electrical sensing networks, but the final protection behaviour depends jointly on the hardware and firmware.

The following values must therefore be validated experimentally:

- DC trip threshold;
- DC persistence time;
- startup delay;
- overtemperature trip threshold;
- thermal reset threshold;
- sensor-validity limits;
- relay release behaviour.

These values should not be considered experimentally confirmed until the physical protection system has been tested.

D.39 Appendix Summary

The schematic presented in this appendix represents the complete STM32-based speaker-protection controller developed for the stereo Hi-Fi amplifier.

The controller uses the STM32G031K8 microcontroller on a NUCLEO-G031K8 board to supervise both amplifier channels.

Each power-amplifier output is monitored through an attenuating, filtering, and ADC-protection network designed to detect persistent DC conditions while rejecting most of the normal audio signal.

Two 10 k Ω NTC sensors provide independent monitoring of the left and right heatsinks.

The two speaker channels are switched using normally-open G2RL-1A-E-DC12 relays controlled by BC337 driver transistors.

1N4007 flyback diodes suppress the inductive relay-coil switching transients.

An isolated auxiliary supply, bridge rectifier, filtering network, and 7812 regulator provide the 12 V low-voltage supply required by the protection system.

Five status LEDs identify startup delay, DC fault, overtemperature, and the two independent temperature-sensor fault conditions.

The protection controller is intentionally kept outside the direct analogue audio-processing path. During normal operation, the loudspeaker signal passes only through the relay contacts.

The schematic therefore provides the principal electrical reference for protection-board construction, firmware development, commissioning, troubleshooting, and final integration with the two power-amplifier channels.

Appendix E

PCB Layouts

E.1 Overview

This appendix presents the final printed circuit board layouts developed for the main electronic assemblies of the stereo Hi-Fi amplifier.

The two principal custom PCB designs are:

- the power-amplifier PCB;
- the STM32 speaker-protection controller PCB.

The layouts were developed in KiCad from the corresponding electrical schematics and were designed with consideration for current-carrying capability, grounding, component placement, thermal behaviour, signal integrity, manufacturability, and mechanical installation.

Both boards use a two-layer construction with

2 oz copper

corresponding to a nominal copper thickness of approximately

70 μm .

The PCB layouts shown in this appendix provide the final physical reference for component placement, routing, mounting-hole locations, connectors, copper zones, and manufacturing preparation.

E.2 Power Amplifier PCB Layout

The power-amplifier PCB contains the complete discrete Class-AB amplifier circuit for one audio channel together with the local supply decoupling and rail protection.

Two identical boards are used in the complete stereo amplifier:

Power Amplifier PCB $\times$ 2
Left Channel + Right Channel

The final board dimensions are approximately

136 mm $\times$ 124.5 mm.

Figure E.1 shows the final PCB layout.

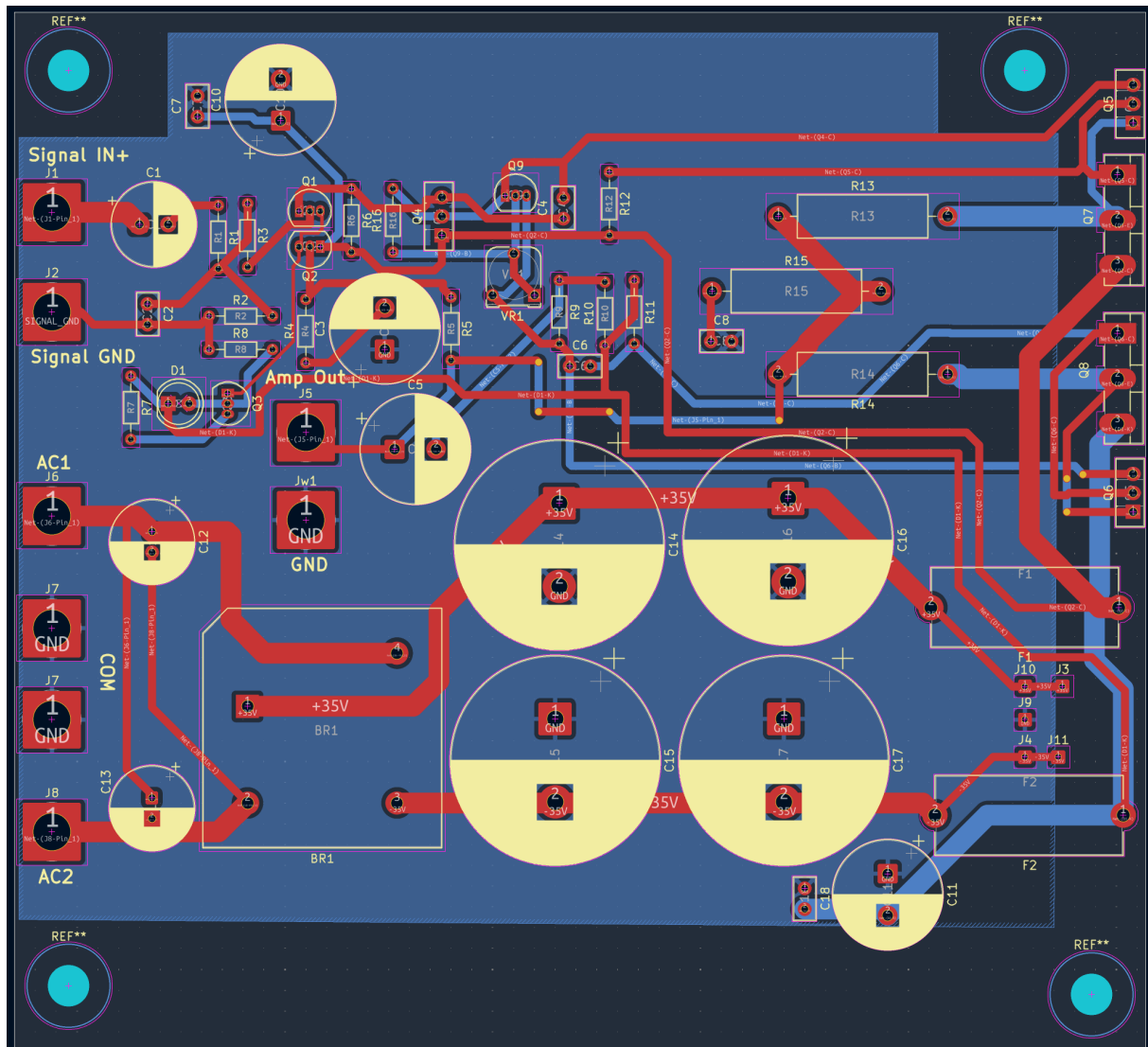


Figure E.1: Final PCB layout of one power-amplifier channel.

E.3 Power Amplifier PCB Construction

The power-amplifier board uses a conventional two-layer FR-4 PCB structure.

The principal design characteristics are:

Table E.1: Principal characteristics of the power-amplifier PCB.

Parameter	Value
Board dimensions	136×124.5 mm
PCB thickness	1.6 mm
Number of copper layers	2
Copper weight	2 oz
Approximate copper thickness	$70 \mu\text{m}$
Ground copper zone	B.Cu
Zone clearance	0.5 mm
Minimum zone thickness	0.25 mm
Thermal relief gap	0.5 mm
Thermal relief bridge width	0.5 mm
Mounting holes	Four 5 mm holes

E.4 Power Amplifier Track Widths

Different track widths are used according to the electrical current and function of each connection.

The implemented routing principally uses

0.4 mm,	1.0 mm,	2.5 mm.
---------	---------	---------

The narrow

0.4 mm

routing is restricted to low-current small-signal circuitry.

The intermediate

1.0 mm

routing is used where greater width is desirable but the connection does not form one of the principal high-current paths.

The

2.5 mm

tracks are used for the main high-current supply and output-related connections wherever component geometry allows.

E.5 High-Current Routing

The highest-current paths on the amplifier PCB include:

- positive supply rail;
- negative supply rail;
- output transistor current paths;
- loudspeaker output path;
- high-current ground returns.

These connections use wide copper routing to reduce resistance, voltage drop, and local heating.

Because current capability is influenced by the narrowest section of a conductor, short routing neck-downs close to component pads remain relevant even when the majority of a track is considerably wider.

E.6 Power Transistor Placement

The 2SA1943-O and 2SC5200-O output transistors are positioned along the outer edge of the PCB.

This placement permits direct mechanical integration with the external heatsink while keeping the transistor leads short.

The arrangement can be represented as

Amplifier PCB → Power Transistors → Heatsink.

This mechanical requirement was considered during component placement rather than after routing was completed.

E.7 Amplifier Ground Plane

A ground copper zone is implemented on the bottom copper layer.

The zone provides a low-impedance ground structure and reduces the amount of ground connection that must be implemented using narrow individual tracks.

The copper zone does not remove the need to control return currents.

Sensitive input-stage returns and high-current output or supply currents must still be routed so that large currents do not unnecessarily share impedance with low-level audio references.

E.8 Amplifier Mounting Holes

Four

5 mm

mounting holes are incorporated into the amplifier PCB.

The holes allow the board to be mounted securely to chassis standoffs while maintaining electrical separation between the PCB and the metal enclosure.

The mounting arrangement must not accidentally create an unintended signal-ground-to-chassis connection.

E.9 Protection Controller PCB Layout

The second major custom PCB contains the STM32-based speaker-protection controller. The board integrates:

- the NUCLEO-G031K8 module;
- DC-sensing circuits for both amplifier channels;
- two temperature-sensor interfaces;
- two loudspeaker relays;
- BC337 relay drivers;
- flyback diodes;
- status LEDs;
- push-button circuitry;
- the auxiliary power supply;
- speaker input and output connections.

The final board dimensions are approximately

121 mm × 182.25 mm.

Figure E.2 shows the final protection controller PCB layout.



The principal construction parameters of the protection-controller PCB are summarised in Table E.2.

Table E.2: Principal characteristics of the protection-controller PCB.

Parameter	Value
Board dimensions	121 × 182.25 mm
PCB thickness	1.6 mm
Number of copper layers	2
Copper weight	2 oz
Approximate copper thickness	70 μm
Ground copper zone	B.Cu
Ground zone name	GROUND PLANE
Zone clearance	0.5 mm
Minimum zone thickness	0.25 mm
Thermal relief gap	0.5 mm
Thermal relief bridge width	0.5 mm
Mounting holes	Four 3.2 mm M3 holes

E.11 Protection Controller Track Widths

The protection board must carry both low-current sensing and digital signals and the significantly larger loudspeaker current.

Two principal track widths are therefore used:

$$1.0 \text{ mm} \quad \text{and} \quad 2.4 \text{ mm.}$$

The

1.0 mm

tracks are used for connections such as:

- STM32 GPIO signals;
- ADC sensing signals;
- temperature-sensor signals;
- +3.3 V connections;
- +12 V control supply;
- relay-driver control;
- LED circuitry.

The wider

2.4 mm

tracks are reserved principally for the high-current loudspeaker paths.

E.12 Speaker Current Routing

The protection-controller board forms part of the high-current loudspeaker path because the amplifier output passes through the relay contacts before reaching the speaker terminals.

The fundamental path is

Amplifier Output → Protection PCB → Relay Contacts → Speaker Output.

The wider 2.4 mm copper tracks are used on these paths wherever the pad and component geometry allows.

At a reference output of

$$100 \text{ W}$$

into

$$4 \Omega,$$

the loudspeaker RMS current is

$$I_{\text{RMS}} = \sqrt{\frac{100}{4}} = 5 \text{ A},$$

while the sine-wave peak current is

$$I_{\text{PEAK}} = 5\sqrt{2} \approx 7.07 \text{ A}.$$

The speaker-current routing therefore requires substantially greater copper width than the sensing and microcontroller circuitry.

E.13 Separation of Signal and Power Functions

The protection PCB contains several electrically different regions within a single board. These include:

Mains Primary

Low-Voltage Power

Digital Control

Analogue Sensing

and

High-Current Speaker Paths.

The PCB layout therefore attempts to keep the sensitive ADC and temperature signals away from relay current and loudspeaker current wherever practical.

E.14 Ground Plane

The protection-controller PCB uses a bottom-layer ground zone identified as

GROUND PLANE.

This provides a common low-voltage electrical reference for the sensing, control, and auxiliary supply circuits.

As with the amplifier board, the use of a copper plane does not mean that current-return behaviour can be ignored.

Relay currents, logic currents, and sensitive ADC references should still be arranged so that unnecessary shared impedance is minimised.

E.15 Mains Region

The protection controller contains a PCB-mounted transformer and therefore includes a mains-primary region.

This portion of the board requires particular attention because the ordinary

0.5 mm

copper-zone clearance used elsewhere on the PCB is not, by itself, a mains-safety specification.

The primary-side routing and the separation between hazardous-voltage and secondary-side circuitry must therefore be checked independently against the applicable creepage and clearance requirements before the PCB is manufactured and installed.

E.16 Protection PCB Mounting

The protection-controller PCB contains four

3.2 mm

mounting holes intended for M3 mounting hardware.

Suitable standoffs should maintain clearance between the underside of the PCB and the metal chassis.

Additional care is required beneath the mains-primary region so that mounting hardware or the chassis cannot compromise the required insulation.

E.17 Two-Layer Construction

Both principal PCBs use two copper layers.

The layer structure allows components and routing to be distributed between

F.Cu

and

B.Cu,

while the bottom layer also supports the principal ground copper zones.

The two-layer architecture is sufficient for the relatively low routing density of the through-hole-oriented amplifier design while remaining simple to manufacture and inspect.

E.18 2 oz Copper

Both boards are specified using 2 oz copper.

The nominal thickness is approximately

$$t \approx 70 \mu\text{m}.$$

The cross-sectional area of a copper track is approximately

$$A = wt,$$

where w is track width.

For the widest amplifier track,

$$w = 2.5 \text{ mm},$$

giving

$$A = 2.5 \times 0.070 = 0.175 \text{ mm}^2.$$

For a 2.4 mm protection-board speaker track,

$$A = 2.4 \times 0.070 = 0.168 \text{ mm}^2.$$

The relatively thick copper therefore provides a useful reduction in conductor resistance compared with standard 1 oz construction.

E.19 Copper Resistance

Track resistance can be estimated from

$$R = \rho \frac{L}{A},$$

where

$$\rho \approx 1.724 \times 10^{-8} \text{ } \Omega\text{m}$$

is the resistivity of copper.

For 100 mm of 2.5 mm wide, 2 oz copper,

$$R \approx 9.85 \text{ m}\Omega.$$

For 100 mm of 2.4 mm wide, 2 oz copper,

$$R \approx 10.26 \text{ m}\Omega.$$

These calculations demonstrate the reason for using wide copper for the principal power and loudspeaker paths.

E.20 Vias

The routing was designed to minimise unnecessary layer transitions.

The power-amplifier PCB uses only a small number of routing vias, while the protection-controller PCB requires very few layer changes.

Reducing unnecessary vias simplifies manufacturing and avoids introducing avoidable discontinuities into high-current paths.

Where a high-current connection can remain on a single copper layer, this is generally preferred.

E.21 Thermal Relief

Pads connected to copper zones use thermal-relief structures where appropriate.

The thermal-relief design reduces the amount of heat transferred from the soldering iron into the surrounding copper plane during assembly.

The implemented zone parameters include approximately

0.5 mm

thermal-relief gaps and

0.5 mm

thermal bridge widths.

E.22 Mechanical Verification

PCB layout verification includes more than electrical connectivity.

Before manufacturing, the design should also be checked for:

- correct board outline;
- correct mounting-hole dimensions;
- component-body clearance;
- connector accessibility;
- heatsink accessibility;
- correct transistor orientation;
- correct relay footprint;
- correct terminal-block orientation;
- adequate courtyard spacing;
- absence of silkscreen over solder pads.

E.23 Electrical Verification

Before Gerber generation, the KiCad Design Rules Checker should be used to verify the physical PCB routing.

The checks should include:

- unconnected nets;
- copper clearance violations;

- incorrect pad connections;
- zone connectivity;
- board-edge clearance;
- track and via rule violations.

The DRC result should then be considered together with a manual visual review, because a PCB may satisfy automated design rules while still containing an incorrect component orientation or unsuitable current path.

E.24 Gerber and Drill Verification

The final PCB manufacturing package is generated from the verified KiCad PCB files.

The package includes the required Gerber layers together with the corresponding plated and, where applicable, non-plated drill information.

The generated files should be inspected using a Gerber viewer before they are submitted for fabrication.

This final check confirms that:

- copper layers are present;
- solder-mask openings are correct;
- silkscreen is readable;
- board outline is correct;
- mounting holes are present;
- component holes are present;
- no required manufacturing layer is missing.

E.25 Use of the PCB Layouts During Assembly

The PCB layouts shown in this appendix serve as mechanical and electrical references during construction.

They can be used together with the corresponding schematics to verify:

- component location;
- component orientation;
- connector positions;

- transistor positions;
- fuse locations;
- relay locations;
- mounting-hole positions;
- high-current track routing;
- copper-zone connections;
- external wiring points.

The schematic remains the primary reference for circuit connectivity, while the PCB layout defines the physical implementation of those connections.

E.26 Appendix Summary

This appendix presents the final layouts of the two principal custom PCBs used in the stereo amplifier.

The power-amplifier PCB measures approximately

$$136 \text{ mm} \times 124.5 \text{ mm}$$

and is used once for each stereo channel.

Its routing uses 0.4 mm, 1.0 mm, and 2.5 mm track widths, with the widest routing assigned to the main high-current supply and output paths.

The protection-controller PCB measures approximately

$$121 \text{ mm} \times 182.25 \text{ mm}$$

and integrates the STM32 controller, DC and temperature sensing, speaker relays, relay drivers, auxiliary supply, indicators, and high-current loudspeaker routing.

Its principal routing widths are

$$1.0 \text{ mm} \quad \text{and} \quad 2.4 \text{ mm}.$$

Both PCBs use

$$2\text{-layer, } 1.6 \text{ mm, } 2 \text{ oz copper}$$

construction and bottom-layer ground copper zones.

The layouts were developed to maintain a practical separation between low-level signals, control circuitry, high-current supply paths, loudspeaker current, and, on the protection controller, the mains-primary region.

Together with the schematic appendices, the PCB layouts provide the principal manufacturing and assembly reference for construction of the final stereo Hi-Fi amplifier.

Appendix F

Bill of Materials

F.1 Overview

This appendix presents the Bill of Materials (BOM) for the complete stereo Hi-Fi amplifier system. The BOM is organised by functional subsystem so that the components required for the power-amplifier channels, main power supply, preamplifier, STM32 protection controller, cooling system, external connections, and mechanical assembly can be identified independently.

Unless otherwise stated, the quantities given in this appendix correspond to the complete stereo system rather than to a single amplifier channel.

The complete system is divided into the following BOM groups:

- two power-amplifier channels;
- main symmetrical power supply;
- preamplifier;
- STM32 speaker-protection controller;
- connectors and user-interface components;
- thermal-management components;
- mains and safety components;
- mechanical and chassis components.

Manufacturer and distributor part numbers are intentionally omitted from the tables in this appendix so that the BOM remains readable as engineering documentation. The procurement BOM may additionally contain manufacturer part numbers, distributor order codes, stock status, and supplier information.

F.2 Power-Amplifier BOM

The complete stereo amplifier uses two identical Project 3A-based power-amplifier PCBs. Therefore, all component quantities in Table F.1 correspond to two amplifier channels.

Table F.1: Bill of materials for the two power-amplifier channels.

Reference	Component	Total Qty.	Description / Function
Q1, Q2, Q3, Q9	BC546	8	Small-signal NPN transistors used in the input, bias, and voltage-amplification circuitry.
Q4, Q6	BD140	4	PNP driver-stage transistors.
Q5	BD139	2	NPN driver-stage transistor.
Q7	2SA1943-O	2	PNP power-output transistor, one per channel.
Q8	2SC5200-O	2	NPN power-output transistor, one per channel.
VR1	2 k Ω trimmer	2	Class-AB quiescent-current adjustment.
R1, R3, R4, R16	1 k Ω	8	Low-power amplifier resistors, nominally 1% tolerance.
R2, R5, R8	22 k Ω	6	Signal and feedback-network resistors, nominally 1%.
R6, R7	560 Ω	4	Bias / driver-network resistors.
R9, R10, R11	3.3 k Ω	6	Amplifier biasing and voltage-amplification resistors.
R12	220 Ω	2	Amplifier-stage resistor.
R13, R14	0.33 Ω	4	Power emitter resistors; high-power type.
R15	10 Ω	2	Power resistor associated with the output stability network.
C1	4.7 μ F	2	Signal-path capacitor.
C2, C4, C6	100 pF	6	Small compensation / high-frequency capacitors.
C3, C5, C10, C11	100 μ F	8	Electrolytic capacitors used within the amplifier circuitry and local supply decoupling.
C7, C8, C18	100 nF	6	High-frequency bypass and stability capacitors.
D1	LED	2	Amplifier-board status / indication LED.
F1, F2	3 A fuse	4	Positive- and negative-rail protection, two fuses per channel.
F1, F2	5 $\times$ 20 mm fuse holder	4	PCB fuse holders for the amplifier rail fuses.
–	Power-amplifier PCB	2	Two-layer, 1.6 mm, 2 oz copper PCB, one per stereo channel.

F.3 Main Power-Supply BOM

The two amplifier channels share one main unregulated symmetrical power supply.

The selected transformer is a toroidal unit with a

$$24\text{--}0\text{--}24 \text{ V}_{\text{AC}}$$

secondary and an approximate power rating of

$$300 \text{ VA}.$$

The principal main power-supply components are listed in Table F.2.

Table F.2: Main amplifier power-supply bill of materials.

Reference	Component	Qty.	Function
–	24–0–24 V AC toroidal transformer	1	Main transformer
BR1	KBPC3506WP	1	35 A bridge rectifier
C14–C17	4700 μF , 63 V	4	Reservoir capacitors
C12, C13	100 nF	2	High-frequency bypass

The four 4700 μF capacitors provide a nominal effective capacitance of

$$C_{\text{rail}} = 2(4700 \mu\text{F}) = 9400 \mu\text{F}$$

per supply rail.

F.4 Protection Controller BOM

The loudspeaker-protection controller is based on the STM32G031K8 microcontroller implemented using the NUCLEO-G031K8 development board.

Table F.3 lists the principal protection and control components.

Table F.3: Principal protection-controller components.

Component	Type / Value	Qty.	Function
Microcontroller board	NUCLEO-G031K8	1	Main protection controller using the STM32G031K8 MCU.
Speaker relay	G2RL-1A-E-DC12	2	Normally-open 12 V loudspeaker relays, one per channel.
Relay driver transistor	BC337	2	Low-side relay-coil driver.

Component	Type / Value	Qty.	Function
Relay flyback diode	1N4007	2	Suppresses inductive voltage when relay coils are de-energised.
Temperature sensor	10 k Ω NTC	2	One sensor mounted to each amplifier heatsink.
ADC clamp diode	BAT85	4	Schottky protection for the left and right DC-sense ADC inputs.
Voltage regulator	7812	1	Provides regulated 12 V supply for the protection circuitry and relays.
Status LEDs	5 mm LEDs	5	WAIT / DELAY, DC FAULT, OVERHEAT, TEMP SENSOR 1 FAIL and TEMP SENSOR 2 FAIL.
Push button	Momentary push button	1	User / firmware control input.
Protection PCB	Two-layer 2 oz PCB	1	STM32 protection, sensing, relay, and auxiliary power-supply board.

F.5 DC-Sensing Network BOM

Two equivalent DC-sensing channels are required, one for each power-amplifier output. The final recommended sensing values are listed in Table F.4.

Table F.4: DC-sensing component quantities for both amplifier channels.

Component	Value	Qty.	Function
Input resistor	150 k Ω , 1%	2	Output attenuation / current limiting
Bias-network resistor	10 k Ω , 1%	4	ADC bias reference network
ADC series resistor	4.7 k Ω	2	ADC / clamp current limiting
Low-pass capacitor	4.7 μ F	2	Main DC-sense low-pass filter
HF bypass capacitor	100 nF	2	Optional high-frequency filtering
Clamp diode	BAT85	4	ADC input overvoltage protection

The 4.7 μ F capacitors are the principal low-pass components used to reject the normal audio signal. The optional 100 nF capacitors may be connected in parallel to improve high-frequency noise suppression.

F.6 Protection Controller Power-Supply Components

The protection controller uses its own isolated auxiliary supply rather than being powered directly from the main amplifier rails.

The principal power-supply blocks are

230 V_{AC} → Auxiliary Transformer → Bridge Rectifier → Filtering → 7812 → 12 VDC.

The principal components are summarised below.

Table F.5: Protection-controller auxiliary supply components.

Component	Qty.	Description
Isolated PCB transformer	1	230 V AC primary, low-voltage secondary
Bridge rectifier	1	Full-wave auxiliary-supply rectifier
Reservoir capacitor	1 or as schematic	Low-voltage DC smoothing
100 nF capacitor	As schematic	High-frequency bypass
7812 regulator	1	Regulated 12 V output
Regulator decoupling capacitors	As schematic	Input and output stability

The exact auxiliary-transformer procurement part number should be matched to the final protection-controller PCB footprint and final low-voltage supply requirement before ordering.

F.7 Preamplifier BOM

The preamplifier contains the low-level analogue circuitry for the moving- magnet phono input, three line-level inputs, source selection, gain control, balance, and volume adjustment.

The design-critical preamplifier components are listed in Table F.6.

Table F.6: Principal preamplifier components.

Component	Type / Value	Qty.	Function
Dual audio opera- tional amplifier	NE5532N	As schematic	Phono, gain, and analogue signal-processing stages.
Positive voltage regu- lator	7815	1	Regulated +15 V preampli- fier supply.
Negative voltage regu- lator	7915	1	Regulated −15 V preampli- fier supply.
RIAA network resis- tors	Precision resistors	As schematic	Determines the phono play- back characteristic.

Component	Type / Value	Qty.	Function
RIAA network capacitors	Audio / film capacitors	As schematic	Determines the phono frequency response.
Volume potentiometer	Stereo potentiometer	1	Main listening-level adjustment.
Balance potentiometer	Stereo potentiometer	1	Left/right level adjustment.
Input selector	Multi-position selector	1	Selects phono or one of the three line inputs.
Gain selector	Switch / selector	1	Selects the required preamplifier gain where implemented.
Local decoupling capacitors	Electrolytic and ceramic	As schematic	Operational-amplifier supply decoupling.

The exact passive-component reference designators and quantities for the preamplifier should be taken directly from the final KiCad preamplifier schematic when the manufacturing BOM is exported. This avoids duplicating an older schematic revision in the procurement documentation.

F.8 Audio Input Connectors

The preamplifier provides one stereo phono source and three stereo line-level sources. This corresponds to four stereo input pairs.

Table F.7: External audio input connectors.

Component	Qty.	Function
RCA input connector	8	Four stereo input pairs
Shielded audio cable	As required	Internal low-level audio wiring

F.9 Speaker Output Components

Table F.8: Loudspeaker output hardware.

Component	Qty.	Function
Red speaker binding post	2	Left and right positive speaker outputs
Black speaker binding post	2	Left and right speaker returns
High-current internal wire	As required	Amplifier-to-relay and relay-to-terminal wiring
6.35 mm spade / suitable terminal	As required	High-current removable connections where used

F.10 Thermal Management BOM

Each amplifier channel uses a separate heatsink so that the power transistors can be mounted directly to the cooling structure.

Table F.9: Thermal-management components.

Component	Qty.	Function
Large power-transistor heatsink	2	One heatsink per amplifier channel
Cooling fan	4	Forced chassis airflow
10 k Ω NTC sensor	2	Heatsink temperature monitoring
Thermal interface material	As required	Transistor-to-heatsink thermal coupling
Insulating transistor hardware	As required	Electrical isolation where required
Heatsink mounting hardware	As required	Mechanical installation

F.11 Mains and Safety Components

The amplifier operates from 230 V AC mains and therefore requires dedicated mains-rated switching, protection, wiring, and protective-earth components.

Table F.10: Mains and electrical-safety components.

Component	Qty.	Function
Mains inlet	1	230 V AC input connection.
Primary fuse holder	1	Holds the main mains-protection fuse.
Primary mains fuse	1	Protects the transformer primary and mains wiring; final rating selected from the actual transformer requirements.
Mains-rated power switch	1	Main amplifier ON/OFF switching.
Protective-earth conductor	1 assembly	Connects the incoming protective earth directly to the metal chassis.
Dedicated chassis earth bolt / hardware	1 set	Permanent protective-earth bonding point.
Mains-rated wire	As required	Line and neutral internal wiring.
Heat-shrink insulation	As required	Insulation of mains and other exposed electrical terminals.
Cable grommets / bushings	As required	Protects wiring where it passes through metal chassis openings.
Strain relief	As required	Prevents external cable forces from reaching electrical terminations.

F.12 Mechanical and Chassis BOM

In addition to the electrical components, the complete amplifier requires mechanical hardware for mounting and enclosure integration.

Table F.11: Principal mechanical and chassis components.

Component	Qty.	Function
Metal amplifier chassis	1	Mechanical enclosure and safety barrier.
Power-amplifier PCB	2	Left and right amplifier channels.
Protection-controller PCB	1	Protection and control circuitry.
Preamplifier PCB / assembly	1	Low-level audio processing.
PCB standoffs	As required	Provides mechanical support and chassis clearance.
M3 hardware	As required	Protection-controller PCB and general mechanical mounting.
Power-amplifier mounting hardware	As required	Mounting through the four 5 mm PCB mounting holes.
Transformer mounting hardware	1 set	Secures the main toroidal transformer.
Fan mounting hardware	4 sets	Secures the cooling fans.
Cable ties / cable clamps	As required	Internal wire management.
Front-panel LED hardware	As required	Mounts the protection indicators.
Control knobs	As required	Volume, balance, and selector controls.

F.13 PCB Manufacturing Requirements

The two main custom PCB designs use the following manufacturing specification:

2 layers, 1.6 mm, 2 oz Cu

The approximate copper thickness is

70 μm .

Table F.12: PCB quantities and principal dimensions.

PCB	Qty.	Dimensions	Copper
Power amplifier	2	136 × 124.5 mm	2 oz
Protection controller	1	121 × 182.25 mm	2 oz

F.14 System-Level BOM Summary

The principal assemblies required to construct one complete stereo amplifier are summarised in Table F.13.

Table F.13: System-level assembly summary.

Assembly	Qty.	Description
Power-amplifier assembly	2	Left and right Class-AB channels
Main power supply	1	Shared symmetrical supply
Toroidal transformer	1	24–0–24 V AC, approximately 300 VA
Preamplifier	1	Phono and line-level signal processing
Protection controller	1	STM32-based speaker protection
Speaker relay	2	One relay per channel
Heatsink	2	One per power-amplifier channel
Cooling fan	4	Chassis forced-air cooling
NTC sensor	2	One per heatsink
Loudspeaker terminal pair	2	Left and right output channels
Chassis	1	Complete mechanical enclosure

F.15 BOM Revision Control

The BOM should always correspond to the same revision as the schematic, PCB layout, and Gerber manufacturing files.

This is particularly important because component selections changed during the development of the project.

The current output-stage devices are

2SA1943-O

and

2SC5200-O

Earlier procurement lists containing alternative NJW output transistors should therefore not be interpreted as the final amplifier design.

Similarly, the current main transformer specification is

$$24\text{--}0\text{--}24\text{ V}_{\text{AC}}, \quad \text{approximately } 300\text{ VA}.$$

Older project documents containing different transformer secondary voltages or VA ratings should not be used for final procurement without comparison against the current schematic and report revision.

F.16 Design BOM and Procurement BOM

Two forms of BOM are useful for this project.

The *design BOM* identifies the electrical component value and function, for example

$$R_{13} = 0.33\ \Omega$$

or

$$Q_8 = 2\text{SC}5200\text{-O}.$$

The *procurement BOM* additionally identifies the exact manufacturer, manufacturer part number, distributor number, stock status, package, and order quantity.

The appendix primarily represents the design BOM because this format remains valid even if the component distributor changes.

The purchasing CSV can subsequently be generated from the current KiCad design and supplemented with the exact manufacturer and distributor part numbers.

F.17 Quantity Margin and Spare Components

For practical construction, the purchasing quantity does not necessarily need to equal the exact design quantity.

It is useful to obtain spare quantities of inexpensive components such as:

- resistors;
- small capacitors;
- LEDs;
- small-signal diodes;
- BC546 and BC337 transistors;
- fuses;
- connectors and terminals.

Additional quantities are particularly useful during prototype construction because components may be damaged during soldering, testing, rework, or mechanical fitting.

Expensive components such as the transformer, heatsinks, Nucleo board, and large reservoir capacitors can normally be ordered according to the required system quantity unless spare inventory is specifically desired.

F.18 Final Procurement Verification

Before a final component order is placed, each BOM entry should be checked against:

- the latest schematic revision;
- the latest PCB revision;
- the assigned KiCad footprint;
- the manufacturer's datasheet;
- electrical voltage and current ratings;
- component dimensions;
- pin numbering and pinout;
- stock availability;
- lifecycle status;
- required quantity.

This verification is particularly important for mechanically critical components such as:

- 2SA1943-O and 2SC5200-O output transistors;
- G2RL-1A-E-DC12 relays;
- bridge rectifiers;
- transformers;
- electrolytic reservoir capacitors;
- terminal blocks;
- fuse holders;
- heatsinks;
- RCA connectors;
- speaker binding posts.

F.19 Appendix Summary

The Bill of Materials presented in this appendix defines the principal electrical, electronic, thermal, mechanical, and interconnection components required to construct the complete stereo Hi-Fi amplifier.

The system uses two identical discrete Class-AB amplifier channels built around the 2SA1943-O and 2SC5200-O power-output devices.

A shared 24–0–24 V AC, approximately 300 VA toroidal transformer feeds the KBPC3506WP bridge rectifier and four 4700 μ F reservoir capacitors.

The preamplifier uses NE5532N-based analogue circuitry and regulated ± 15 V supply rails.

The protection subsystem uses the STM32G031K8 on a NUCLEO-G031K8 board, two G2RL-1A-E-DC12 speaker relays, BC337 relay drivers, 1N4007 flyback diodes, two 10 k Ω heatsink temperature sensors, and dedicated left and right DC-sensing networks.

The complete mechanical system additionally requires two large heatsinks, four cooling fans, RCA input connectors, speaker binding posts, mains switching and fuse hardware, protective-earth hardware, PCB standoffs, interconnecting wire, and the amplifier chassis.

The BOM must remain revision-controlled together with the KiCad schematics, PCB layouts, Gerber files, drill files, and firmware so that the components ordered for construction always correspond to the current electrical and mechanical design.

Appendix G

Component Datasheets

G.1 Overview

This appendix identifies the principal manufacturer datasheets used during the design, PCB development, component selection, assembly planning, and commissioning of the stereo Hi-Fi amplifier.

Manufacturer datasheets represent the primary technical reference for component-specific information such as

- absolute maximum electrical ratings;
- recommended operating conditions;
- transistor pin assignments;
- semiconductor package dimensions;
- power dissipation;
- thermal resistance;
- relay contact and coil ratings;
- regulator operating requirements;
- microcontroller electrical characteristics;
- PCB footprint dimensions;
- temperature-sensor characteristics.

The complete amplifier contains components from several different functional groups. The most important datasheets are therefore organised in this appendix according to their role within the system.

The full manufacturer PDF files should preferably be stored with the project documentation rather than reproduced in their entirety inside this report. This avoids adding a large number of pages while maintaining a clear reference to the original technical documentation.

G.2 Power Output Transistors

G.2.1 2SC5200-O NPN Power Transistor

The 2SC5200-O is the NPN output power transistor used as Q8 in each power-amplifier channel.

The device is specifically suited to high-power linear applications such as audio amplifier output stages [4].

The datasheet is particularly important for verifying:

- collector-emitter voltage rating;
- maximum collector current;
- allowable power dissipation;
- DC current gain;
- safe operating area;
- junction-temperature limit;
- thermal resistance;
- pin assignment;
- package dimensions;
- heatsink mounting requirements.

The transistor forms one half of the complementary output pair used in the Class-AB power stage.

G.2.2 2SA1943-O PNP Power Transistor

The 2SA1943-O is the complementary PNP output transistor used as Q7 in each power-amplifier channel [5].

Together with the 2SC5200-O, it forms the high-current output stage of the amplifier.

The principal datasheet parameters that must be checked are equivalent to those of the NPN device:

- collector-emitter voltage;
- collector current;
- safe operating area;
- current gain;

- power dissipation;
- junction-temperature limit;
- thermal characteristics;
- package dimensions;
- pin numbering.

The package drawing is especially important because both devices are mounted along the edge of the amplifier PCB and mechanically coupled to the external heatsink.

G.3 Driver Transistors

G.3.1 BD139

The BD139 is used as an NPN driver transistor in the power-amplifier output stage. Its datasheet should be consulted to verify:

- collector-emitter voltage rating;
- collector-current rating;
- power-dissipation capability;
- current gain;
- junction-temperature limit;
- thermal characteristics;
- TO-18 package dimensions;
- emitter, collector, and base pin ordering.

Correct pinout verification is essential because package appearance alone should not be used to determine transistor orientation.

G.3.2 BD140

The BD140 is the complementary PNP driver transistor used together with the BD139.

Its manufacturer datasheet is used for the same electrical, thermal, package, and pin-assignment verification required for the BD139.

G.4 Small-Signal Transistors

G.4.1 BC546

The BC546 NPN small-signal transistor is used at several locations in the Project 3A-derived amplifier, including the input, voltage-amplification, and bias-control circuitry.

The datasheet is required to verify:

- maximum collector-emitter voltage;
- collector-current capability;
- current-gain range;
- small-signal characteristics;
- maximum junction temperature;
- package dimensions;
- collector, base, and emitter pin order.

G.4.2 BC337

The BC337 is used on the protection-controller PCB as the driver transistor for the 12 V loudspeaker relays.

The STM32 does not supply the relay coil current directly. Instead, the GPIO signal controls the BC337, which switches the relay coil current.

The BC337 datasheet should therefore be consulted for:

- maximum collector current;
- collector-emitter voltage;
- transistor gain;
- saturation behaviour;
- power dissipation;
- package and pinout information.

G.5 Operational Amplifier

G.5.1 NE5532N

The NE5532N dual operational amplifier is used within the low-level analogue preamplifier.

It is used for the phono and signal-gain circuitry, where low noise and low distortion are important.

The relevant datasheet information includes:

- allowable supply-voltage range;
- input voltage and common-mode limits;
- output voltage swing;
- output-current capability;
- input noise;
- slew rate;
- gain bandwidth;
- power consumption;
- DIP package pin assignment.

The preamplifier operates from regulated

$$\pm 15 \text{ V}$$

rails, and the NE5532N electrical operating limits must therefore be compatible with this supply.

G.6 STM32 Microcontroller

G.6.1 STM32G031K8

The STM32G031K8 is the microcontroller used by the loudspeaker-protection system [2]. Its datasheet forms the main reference for the protection-controller digital and analogue interfaces.

Important parameters include:

- operating supply voltage;
- GPIO voltage limits;
- GPIO output-current capability;
- ADC resolution;
- ADC input-voltage limits;
- ADC characteristics;

- available timers;
- internal clock architecture;
- memory capacity;
- package pin assignments;
- absolute maximum ratings.

The ADC limitations are particularly important because both the DC-sensing and temperature-sensing circuits ultimately connect to STM32 analogue inputs.

G.7 NUCLEO-G031K8 Development Board

The project implements the STM32G031K8 through the STMicroelectronics NUCLEO-G031K8 development board.

The corresponding user manual provides information that is not contained directly in the microcontroller datasheet, including the development-board connectors, power configuration, debugging interface, and physical pinout [3].

The Nucleo documentation is used to verify:

- CN3 and CN4 header pin assignments;
- available STM32 GPIO connections;
- ADC-capable pins;
- power-input options;
- 3.3 V and ground pins;
- ST-LINK programming and debugging;
- board dimensions;
- header spacing and PCB footprint.

The board pinout must agree exactly with the custom KiCad footprint used on the protection-controller PCB.

G.8 Speaker Protection Relay

G.8.1 G2RL-1A-E-DC12

Two Omron G2RL-1A-E-DC12 relays are used to connect and disconnect the left and right loudspeakers.

The selected version uses a

12 VDC

coil and normally-open contacts.

The relay datasheet must be consulted for:

- nominal coil voltage;
- coil resistance;
- coil power;
- pull-in and release behaviour;
- contact-current capability;
- contact-voltage ratings;
- contact resistance;
- mechanical endurance;
- electrical endurance;
- isolation characteristics;
- physical dimensions;
- terminal arrangement;
- PCB drilling pattern.

The terminal-layout drawing is particularly important because an incorrect relay footprint can connect the speaker path to the wrong contact or coil pin.

G.9 Relay Flyback Diode

G.9.1 1N4007

A 1N4007 rectifier diode is connected across each speaker-relay coil.

Its function is to provide a current path for the inductive relay current when the BC337 driver transistor switches off.

The relevant datasheet parameters include:

- repetitive reverse-voltage rating;
- forward-current rating;

- forward-voltage drop;
- surge-current capability;
- polarity identification;
- package dimensions.

Correct polarity is essential. During normal relay energisation, the diode must remain reverse biased.

G.10 ADC Clamp Diode

G.10.1 BAT85

BAT85 Schottky diodes are used in the protection-controller ADC interface.

Their function is to limit abnormal voltage excursions at the STM32 ADC nodes when used together with the series resistance of the sensing network.

Important datasheet parameters include:

- forward-voltage characteristic;
- reverse-voltage rating;
- allowable forward current;
- leakage current;
- package dimensions;
- cathode identification.

G.11 Bridge Rectifier

G.11.1 KBPC3506WP

The KBPC3506WP is used as the main bridge rectifier for the power-amplifier supply.

It converts the AC voltage from the 24–0–24 V transformer into the rectified waveforms required by the positive and negative reservoir-capacitor banks.

The important datasheet parameters include:

- maximum repetitive reverse voltage;
- average rectified current;
- surge-current capability;

- forward-voltage drop;
- thermal resistance;
- operating-temperature range;
- terminal identification;
- package dimensions;
- mounting-hole dimensions.

The polarity symbols marked on the bridge and the manufacturer's mechanical drawing should be checked before PCB assembly.

G.12 Linear Voltage Regulators

G.12.1 7812 Regulator

A 7812 positive linear regulator is used within the protection-controller auxiliary supply to generate the regulated

12 V

rail.

The datasheet should be used to verify:

- allowable input voltage;
- regulated output voltage;
- available output current;
- required input-output voltage differential;
- thermal shutdown;
- current limiting;
- recommended bypass capacitors;
- package pinout;
- thermal resistance.

G.12.2 7815 Regulator

A 7815 positive regulator is used to generate the preamplifier

$$\boxed{+15 \text{ V}}$$

supply rail.

The datasheet should be consulted for input-voltage requirements, regulator power dissipation, stability capacitors, package orientation, and thermal performance.

G.12.3 7915 Regulator

The complementary 7915 negative regulator provides the

$$\boxed{-15 \text{ V}}$$

preamplifier supply.

Particular care must be taken with the 7915 pin arrangement because the pinout of a negative regulator must not be assumed to be identical to the positive 7815 regulator.

The manufacturer datasheet should therefore be checked directly before PCB routing and assembly.

G.13 Temperature Sensors

G.13.1 B57703M0103A047 NTC Thermistor

Two B57703M0103A047 NTC thermistors are used for heatsink temperature monitoring.

One sensor is associated with each power-amplifier channel.

The nominal resistance is

$$\boxed{10 \text{ k}\Omega}$$

at

$$25^{\circ}\text{C}.$$

The temperature dependence of an NTC thermistor can be approximated by

$$R(T) = R_{25} \exp \left[B \left(\frac{1}{T} - \frac{1}{T_{25}} \right) \right].$$

The manufacturer documentation is required to determine the actual beta value, resistance tolerance, operating-temperature limits, response characteristics, and mechanical dimensions.

Relevant datasheet information includes:

- resistance at 25°C;
- resistance tolerance;
- beta value;
- permitted operating-temperature range;
- thermal response;
- maximum power;
- insulation information;
- ring-terminal dimensions.

G.14 Main Reservoir Capacitors

The main amplifier power supply contains four

4700 μ F, 63 V

electrolytic reservoir capacitors.

The capacitor datasheet should be checked for:

- rated capacitance;
- capacitance tolerance;
- rated DC voltage;
- ripple-current capability;
- equivalent series resistance;
- operating-temperature range;
- expected lifetime;
- body diameter and height;
- lead spacing;
- polarity marking.

The mechanical dimensions are especially important because the reservoir capacitors occupy significant PCB area and require the correct lead spacing.

G.15 Toroidal Transformer

The main amplifier transformer provides a

24-0-24 V_{AC}

secondary and is selected for approximately

300 VA

power capability.

The transformer manufacturer documentation should be used to verify:

- primary voltage;
- secondary voltage;
- VA rating;
- secondary current capability;
- winding identification;
- insulation class;
- temperature rise;
- physical dimensions;
- mounting arrangement;
- transformer mass.

The transformer datasheet should correspond to the exact device eventually purchased because winding colours and mechanical dimensions are not universal.

G.16 Datasheet Summary Table

Table G.1: Principal manufacturer documentation associated with the project.

Component	System Function	Principal Datasheet Information Used
2SC5200-O	NPN output transistor	Voltage, current, SOA, thermal characteristics, pinout, package dimensions

Component	System Function	Principal Datasheet Information Used
2SA1943-O	PNP output transistor	Voltage, current, SOA, thermal characteristics, pinout, package dimensions
BD139	NPN driver transistor	Electrical ratings, thermal limits, TO-126 pinout
BD140	PNP driver transistor	Electrical ratings, thermal limits, TO-126 pinout
BC546	Small-signal amplifier transistor	Voltage, current gain, pinout, package
BC337	Relay driver	Collector current, saturation behaviour, pinout
NE5532N	Preamplifier operational amplifier	Supply range, noise, slew rate, bandwidth, DIP pinout
STM32G031K8	Protection MCU	GPIO, ADC, supply voltage, absolute limits, MCU pinout
NUCLEO-G031K8	MCU development board	Headers, GPIO mapping, ADC pins, power and mechanical layout
G2RL-1A-E-DC12	Speaker relay	Coil parameters, contact ratings, footprint and pin configuration
1N4007	Relay flyback diode	Forward current, reverse voltage, package polarity
BAT85	ADC clamp diode	Forward drop, leakage, reverse voltage, package polarity
KBPC3506WP	Main bridge rectifier	Current, voltage, surge rating, terminals and package
7812	Protection supply regulator	Input/output voltage, current, dissipation, pinout
7815	Positive preamplifier regulator	Input/output limits, thermal data, pinout
7915	Negative preamplifier regulator	Input/output limits, thermal data, pinout
B57703M0103A047	Heatsink NTC sensor	Resistance curve, beta value, tolerance, mechanical dimensions
4700 μ F / 63 V capacitor	Main supply reservoir	Ripple current, ESR, dimensions, voltage and temperature rating
24-0-24 V transformer	Main amplifier supply	Secondary voltage, current, VA rating, mounting and insulation

G.17 Recommended Datasheet File Organisation

The original manufacturer datasheets can be stored in the project repository using a dedicated documentation directory.

A suitable structure is:

```
Documentation/
|
+-- Datasheets/
|
|   +-- Power_Amplifier/
|   |   +-- 2SC5200.pdf
|   |   +-- 2SA1943.pdf
|   |   +-- BD139.pdf
|   |   +-- BD140.pdf
|   |   +-- BC546.pdf
|   |
|   +-- Preamplifier/
|   |   +-- NE5532N.pdf
|   |   +-- 7815.pdf
|   |   +-- 7915.pdf
|   |
|   +-- Protection_Controller/
|   |   +-- STM32G031K8.pdf
|   |   +-- NUCLE0-G031K8.pdf
|   |   +-- G2RL-1A-E-DC12.pdf
|   |   +-- BC337.pdf
|   |   +-- 1N4007.pdf
|   |   +-- BAT85.pdf
|   |   +-- B57703M0103A047.pdf
|   |   +-- 7812.pdf
|   |
|   +-- Power_Supply/
|   |   +-- KBPC3506WP.pdf
|   |   +-- Reservoir_Capacitors.pdf
|   |   +-- Main_Transformer.pdf
```

This organisation keeps component documentation separate from KiCad library files while allowing the original manufacturer documentation to remain available alongside the design.

G.18 Use of Datasheets During PCB Design

Manufacturer datasheets were used not only to determine electrical ratings but also to verify the mechanical implementation of the components.

A typical PCB component verification process is

Schematic Symbol → Datasheet Pinout → KiCad Footprint → PCB Layout.

The pin numbers in the KiCad symbol must correspond to the electrical terminal numbers in the manufacturer datasheet.

The footprint must then reproduce the physical arrangement of those terminals.

Failure in either part of this relationship can produce a PCB that appears correct in KiCad but connects the actual component incorrectly.

G.19 Mechanically Critical Datasheets

Datasheet verification is especially important for components whose mechanical dimensions directly determine the PCB or chassis geometry.

Examples include:

- 2SA1943-O and 2SC5200-O power transistors;
- G2RL-1A-E-DC12 speaker relays;
- KBPC3506WP bridge rectifier;
- reservoir capacitors;
- NUCLEO-G031K8 headers;
- NTC ring sensors;
- voltage regulators;
- main transformer.

For these components, the package drawing should be checked before PCB manufacturing or mechanical drilling.

G.20 Electrical Ratings Versus Normal Operating Conditions

Absolute maximum ratings given in a datasheet must not be interpreted as the recommended normal operating point.

For example, a power transistor may have a high maximum collector-current rating, but its actual allowable current depends simultaneously on collector voltage, junction temperature, pulse duration, heatsinking, and safe operating area.

Similarly, the relay maximum contact rating is not automatically the preferred continuous operating condition for every type of electrical load.

The project therefore uses datasheets as engineering constraints rather than simply comparing one maximum numerical value with the circuit requirement.

G.21 Thermal Datasheet Information

Thermal data are particularly important for the power semiconductors.

A simplified junction-temperature calculation is

$$T_J = T_A + P_D (\theta_{JC} + \theta_{CS} + \theta_{SA}),$$

where

$$\theta_{JC}$$

is obtained from the transistor datasheet, while the case-to-heatsink and heatsink-to-ambient thermal resistances depend on the mechanical implementation.

The absolute maximum junction temperature stated in a datasheet should be treated as a limit rather than a design target.

G.22 Datasheets and Component Substitution

If a component must be substituted because of availability, the replacement should not be selected solely because it performs approximately the same function.

The substitute datasheet should be compared for:

- polarity and device type;
- voltage rating;
- current rating;
- power dissipation;
- thermal characteristics;
- switching or frequency performance;
- package type;
- pinout;
- physical dimensions;
- PCB footprint compatibility.

The last two points are particularly important because an electrically suitable replacement may still require a PCB redesign.

G.23 Datasheets and Revision Control

The datasheet archive should correspond to the components used in the current schematic and BOM revision.

This project underwent component changes during development, including changes to the final output-transistor selection.

The current amplifier output devices are

2SA1943-O

and

2SC5200-O

Datasheets associated only with earlier component alternatives may remain in the development archive but should not be confused with the final component documentation.

G.24 Including Full Datasheets in the Report

It is generally unnecessary to reproduce every page of every manufacturer datasheet inside the report because this would add a substantial number of pages without adding original engineering analysis.

If a selected datasheet must be physically included in the final appendix, the `pdfpages` package can be used.

For example:

```
\usepackage{pdfpages}
```

and a selected manufacturer document can then be inserted using:

```
\includepdf[
  pages=-,
  pagecommand={}
]{datasheets/2SC5200.pdf}
```

However, for the present report, it is preferable to include the datasheet index and retain the full manufacturer PDFs as separate supporting project files.

G.25 Appendix Summary

The component datasheets form an essential part of the technical documentation for the amplifier because they provide the authoritative electrical, mechanical, and thermal limits of the devices used in the design.

The most important documentation includes the Toshiba 2SC5200-O and 2SA1943-O output-transistor datasheets, the STM32G031K8 datasheet, the NUCLEO-G031K8 user manual, the NE5532N operational-amplifier datasheet, the G2RL-1A-E-DC12 relay datasheet, and the documentation for the driver transistors, rectifiers, regulators, temperature sensors, reservoir capacitors, and transformers.

These documents were used to verify component ratings, pin assignments, package dimensions, PCB footprints, thermal characteristics, and operating limits.

Maintaining the original datasheets together with the KiCad project, Bill of Materials, firmware, PCB manufacturing files, and report ensures that the design remains traceable, reproducible, and serviceable after construction.

Appendix H

STM32 Firmware

H.1 Introduction

The STM32 firmware forms the supervisory and decision-making layer of the speaker protection system developed for the power amplifier. While the analogue circuitry performs the physical measurement of amplifier output voltage, temperature, and other electrical quantities, the microcontroller continuously evaluates these signals and determines whether the loudspeakers can remain safely connected to the amplifier outputs.

The firmware was developed for the STM32G031K8 microcontroller using the STM32CubeIDE development environment and the STM32 Hardware Abstraction Layer (HAL). The main objective of the firmware is to provide deterministic protection behaviour while maintaining a modular software structure that can be easily tested, modified, and extended.

The firmware performs the following principal functions:

- initialization of the STM32G031K8 peripherals,
- acquisition of DC-offset measurement signals,
- acquisition of the heatsink temperature measurement,
- evaluation of amplifier operating conditions,
- implementation of the power-on speaker connection delay,
- immediate speaker disconnection when a fault is detected,
- independent indication of DC-offset and over-temperature faults,
- control of the speaker protection relay,
- continuous monitoring after the amplifier has entered normal operation.

A fundamental design principle of the firmware is that speaker safety always has priority over normal operation. Consequently, the speaker relay is treated as a controlled output that is enabled only after all required operating conditions have been verified.

H.2 Firmware Development Environment

The firmware was written in the C programming language and compiled using the GNU ARM Embedded toolchain included with STM32CubeIDE. STM32CubeIDE was also used for device configuration, compilation, debugging, and programming of the STM32G031K8 microcontroller.

The STM32 HAL library is used to provide a hardware abstraction layer between the application firmware and the internal peripherals of the microcontroller. This makes the application code easier to understand and reduces the need for direct manipulation of peripheral registers.

The project contains the standard STM32Cube-generated initialization files in addition to custom modules specifically developed for the amplifier protection controller. The firmware is therefore separated into hardware initialization, sensor acquisition, protection logic, and hardware control functions.

H.3 Firmware Architecture

The firmware follows a modular architecture in which different responsibilities are separated into individual source and header files. This approach avoids placing the entire protection system inside the `main.c` file and significantly improves maintainability.

The principal application files are organised as follows:

main.c Contains the main program entry point, system initialization, peripheral initialization, and the continuous execution loop.

protection.c Implements the main speaker-protection logic, including startup behaviour, fault evaluation, relay control decisions, and protection state handling.

protection.h Contains declarations required by the protection module and exposes the protection functions to the rest of the firmware.

protection_io.c Implements the low-level control of physical protection outputs such as the speaker relay and status LEDs.

protection_io.h Defines the public interface used by the protection algorithm to operate the relay and indicators.

sensors.c Contains the functions responsible for acquiring and processing analogue measurements from the protection circuitry.

sensors.h Defines the sensor-related functions and data structures used by the protection algorithm.

protection_config.h Contains configurable parameters of the protection system, allowing thresholds and timing constants to be modified without changing the main protection algorithm.

protection_pins.h Provides a central definition of the GPIO assignments used by the protection system.

Separating the firmware into these modules creates a clear distinction between three main software layers:

1. hardware and peripheral access,
2. measurement and signal processing,
3. protection decision logic.

This structure is particularly useful during development because changes to the PCB pin assignments, protection thresholds, or sensor conversion equations can be made without redesigning the complete firmware.

H.4 Firmware Execution Sequence

When power is first applied to the amplifier, the STM32 microcontroller begins executing from its reset vector. The HAL library and system clock are first initialized, followed by configuration of the GPIO and analogue peripherals required by the protection controller.

The general execution sequence is:

1. initialize the STM32 HAL,
2. configure the system clock,
3. initialize GPIO peripherals,
4. initialize the ADC subsystem,
5. initialize the protection module,
6. force the speaker relay into the disconnected state,
7. begin the power-on verification period,
8. continuously acquire sensor measurements,
9. evaluate all protection conditions,
10. connect the speaker only when operation is considered safe,
11. continuously repeat the monitoring process.

After initialization, the firmware remains inside the main execution loop for the complete operating lifetime of the amplifier.

H.5 Main Program Loop

The protection controller is designed around a continuously executing main loop rather than an operating system. This approach is sufficient for the present application because the number of monitored variables is small and the required protection functions can be evaluated very rapidly.

Conceptually, the main loop performs the following sequence:

```
while (1)
{
    Read sensors;
    Process measurements;
    Evaluate DC protection;
    Evaluate temperature protection;
    Evaluate startup state;
    Update speaker relay;
    Update status LEDs;
}
```

The loop is executed repeatedly, allowing the controller to detect abnormal amplifier conditions during operation.

The speaker relay is not controlled directly from the sensor-reading functions. Instead, sensor measurements are passed to the protection logic, which determines the required system state. This separation prevents hardware-control decisions from becoming distributed throughout the program.

H.6 Sensor Acquisition

The STM32G031K8 contains an integrated analogue-to-digital converter that is used to measure the analogue signals generated by the protection circuitry. These signals represent quantities such as the amplifier DC output level and heatsink temperature.

The ADC converts each analogue input voltage into a digital value. For an ADC with resolution N bits, the relationship between the measured ADC code and the input voltage can be represented by

$$V_{\text{ADC}} = \frac{\text{ADC}_{\text{code}}}{2^N - 1} V_{\text{REF}}, \quad (\text{H.1})$$

where

- V_{ADC} is the voltage applied to the ADC input,
- ADC_{code} is the measured digital conversion result,
- N is the ADC resolution,

- V_{REF} is the ADC reference voltage.

Further conversion is then performed depending on the measurement channel. For example, the temperature input is converted into a temperature estimate according to the NTC sensor characteristics, while the DC detection channel is interpreted relative to the expected zero-output reference level.

H.7 DC Offset Monitoring

One of the most important functions of the protection firmware is continuous monitoring for excessive DC voltage at the power-amplifier output.

Under normal operating conditions, the average DC component at the loudspeaker output should remain very close to zero. An amplifier failure, damaged transistor, bias problem, or other circuit fault may cause a significant DC voltage to appear at the output.

A loudspeaker connected directly to such a voltage could experience a large continuous current given approximately by

$$I_{\text{SPK}} = \frac{V_{\text{DC}}}{R_{\text{SPK}}}, \quad (\text{H.2})$$

where V_{DC} is the DC output voltage and R_{SPK} is the loudspeaker impedance.

The corresponding continuous power dissipated in the loudspeaker is approximately

$$P_{\text{DC}} = \frac{V_{\text{DC}}^2}{R_{\text{SPK}}}. \quad (\text{H.3})$$

Unlike an audio waveform, DC does not contribute useful acoustic output and can result in excessive heating of the loudspeaker voice coil. It may therefore cause permanent loudspeaker damage.

The firmware continuously evaluates the measured DC-offset signal against the permitted range defined in the protection configuration. If the absolute value exceeds the allowable threshold, the protection system enters the DC-fault state.

The comparison can be represented conceptually as

$$|V_{\text{OFFSET}}| > V_{\text{DC,TH}} \Rightarrow \text{DC Fault}, \quad (\text{H.4})$$

where $V_{\text{DC,TH}}$ represents the configured protection threshold.

When a DC fault is detected, the firmware commands the speaker relay to open and activates the corresponding DC fault indicator.

H.8 Temperature Monitoring

The firmware also continuously monitors the temperature of the amplifier heatsink using the NTC-based temperature measurement circuit described previously.

The measured analogue voltage is converted by the ADC and processed by the sensor module to obtain an estimate of the amplifier temperature. The result is then compared with the configured maximum safe temperature.

The protection decision can be expressed as

$$T_{\text{HS}} \geq T_{\text{MAX}} \Rightarrow \text{Overtemperature Fault}, \quad (\text{H.5})$$

where

- T_{HS} is the measured heatsink temperature,
- T_{MAX} is the configured maximum allowable temperature.

If the temperature exceeds the safe limit, the speaker relay is immediately disabled and the dedicated over-temperature LED is activated.

The power amplifier itself may still remain powered, but removing the loudspeaker load reduces the risk of further thermal stress while clearly informing the user that an abnormal thermal condition has occurred.

H.9 Speaker Relay Control

The speaker relay constitutes the final isolation element between the power-amplifier output and the loudspeaker. The microcontroller controls the relay indirectly through the relay-driver circuitry.

The firmware applies a fail-safe operating philosophy. The normal reset condition of the controller corresponds to a disconnected loudspeaker. Consequently, the relay is activated only after the controller has confirmed that no protection condition is present.

The desired relay condition can be represented logically as

$$\text{Relay}_{\text{ON}} = \text{StartupComplete} \wedge \neg \text{DCFault} \wedge \neg \text{TemperatureFault}. \quad (\text{H.6})$$

Therefore, if any required safety condition becomes false, the relay is de-energized.

This arrangement is preferable to a system in which the microcontroller must actively generate a fault signal in order to disconnect the loudspeaker. During microcontroller reset or startup, the loudspeaker naturally remains disconnected.

H.10 Power-On Delay

Audio power amplifiers can produce short transient voltages during power-up while internal bias voltages and power-supply rails are stabilizing. If the loudspeaker is connected immediately, these transient conditions may produce an audible “pop” or “thump”.

To avoid this behaviour, the firmware implements a power-on delay before enabling the speaker relay.

Immediately after reset, the relay remains disconnected while the microcontroller continuously checks the amplifier output and temperature. Only after the programmed startup interval has elapsed and no fault has been detected is the relay allowed to energize.

The resulting startup sequence is therefore:

1. amplifier power is applied,
2. STM32 starts and initializes the protection system,
3. speaker relay remains open,
4. amplifier circuitry is allowed to stabilize,
5. sensor measurements are evaluated,
6. the firmware confirms that no fault exists,
7. the speaker relay is energized,
8. normal monitoring continues.

If an unsafe condition is detected during the startup delay, the speaker relay remains disconnected.

H.11 Fault Detection Logic

The protection algorithm combines all monitored conditions into a single decision regarding whether the loudspeaker can safely remain connected.

The system can conceptually operate in the following states:

- **Startup:** the controller has recently powered on and the speaker is intentionally disconnected.
- **Normal Operation:** all monitored quantities are within their permitted operating ranges and the speaker relay is enabled.
- **DC Fault:** excessive output DC offset has been detected.
- **Overtemperature Fault:** the heatsink temperature has exceeded the permitted limit.

The protection algorithm gives fault conditions higher priority than the normal operating state. Thus, even after the system has entered normal operation, a subsequent fault causes the relay to be disconnected.

The overall decision can be represented by

$$\text{Safe} = \neg (\text{DCFault} \vee \text{TemperatureFault}). \quad (\text{H.7})$$

The speaker is connected only when

$$\text{SpeakerConnected} = \text{Safe} \wedge \text{StartupComplete}. \quad (\text{H.8})$$

H.12 Status and Fault Indicators

The controller includes visual indicators to provide information about the current protection status. These indicators allow faults to be diagnosed without connecting external test equipment to the microcontroller.

The firmware controls the LEDs according to the internal protection state. For example, a DC fault activates the DC fault indicator, while an excessive heatsink temperature activates the over-temperature indicator.

Separating the indicators according to fault type improves diagnostic capability because the cause of speaker disconnection can be immediately identified.

H.13 Protection Configuration

Protection-related constants are separated from the main application logic and placed in the `protection_config.h` header file.

Parameters that can be defined in this configuration layer include:

- maximum permitted DC offset,
- maximum heatsink temperature,
- startup delay duration,
- ADC conversion constants,
- sensor calibration parameters,
- timing constants used by the protection algorithm.

This approach allows the protection behaviour to be modified during testing without changing the structure of the main algorithm.

For example, changing the temperature cutoff requires modification of the corresponding configuration value rather than modification of the temperature-monitoring function itself.

H.14 GPIO Abstraction

The physical STM32 pin assignments used by the protection system are defined centrally in `protection_pins.h`. This avoids hard-coded GPIO definitions being distributed throughout the program.

Furthermore, `protection_io.c` acts as an abstraction layer between the protection algorithm and the physical hardware.

For example, rather than directly manipulating a GPIO pin from the protection algorithm, a higher-level function can be used conceptually as

```
ProtectionIO_SetRelay(true);
```

or

```
ProtectionIO_SetRelay(false);
```

The protection logic therefore only specifies the desired system behaviour, while the I/O module determines how that command is physically implemented.

H.15 Fail-Safe Behaviour

Fail-safe behaviour was considered an essential requirement because the primary purpose of the microcontroller is to protect the loudspeakers and amplifier system.

Several firmware design decisions contribute to this behaviour:

- the speaker relay begins in the disconnected state,
- the speaker is never connected before the startup verification sequence is complete,
- detection of a valid fault has priority over normal operation,
- relay control is centralized within the protection logic,
- fault indicators remain independent from the speaker relay decision,
- thresholds and timing parameters are explicitly defined within the configuration module.

The firmware therefore follows the principle

$$\text{Uncertain State} \Rightarrow \text{Speaker Disconnected.} \quad (\text{H.9})$$

This is particularly important during startup, reset, and abnormal operating conditions.

H.16 Firmware Compilation

The complete firmware project is compiled in STM32CubeIDE using the ARM GNU compiler. During compilation, the individual C source files are first converted into object files and subsequently linked together with the required STM32 HAL libraries to generate the final microcontroller executable.

The build process generates several output files, including the ELF executable containing the compiled program and debugging information. The firmware can then be transferred to the STM32G031K8 using the Serial Wire Debug (SWD) programming interface.

Successful compilation of the complete project verifies that all protection modules, header dependencies, STM32 HAL drivers, and device configuration files are correctly integrated.

H.17 Firmware Programming and Debugging

The STM32G031K8 can be programmed and debugged through its SWD interface using an ST-LINK programmer/debugger.

The SWD connection normally requires the following signals:

- SWDIO,
- SWCLK,
- GND,
- target reference voltage,
- optionally NRST.

After compilation, STM32CubeIDE transfers the firmware to the internal Flash memory of the microcontroller. Once programmed, the STM32 executes the protection firmware automatically whenever the amplifier protection controller is powered.

The debugger can additionally be used during development to inspect sensor values, protection states, variables, and program execution. This is useful when validating ADC conversion equations and protection thresholds before final standalone operation.

H.18 Firmware Repository Structure

The firmware source code is maintained together with the hardware design files in the project repository. A simplified directory structure is shown below.

```
Firmware/
|
+-- STM32_Protection_Controller/
|   |
|   +-- Core/
|   |   |
|   |   +-- Inc/
|   |   |   +-- main.h
|   |   |   +-- protection.h
|   |   |   +-- protection_config.h
|   |   |   +-- protection_io.h
|   |   |   +-- protection_pins.h
|   |   |   +-- sensors.h
|   |   |   +-- stm32g0xx_hal_conf.h
|   |   |   +-- stm32g0xx_it.h
|   |   |
|   |   +-- Src/
|   |       +-- main.c
```

```
|      +--- protection.c
|      +--- protection_io.c
|      +--- sensors.c
|      +--- stm32g0xx_hal_msp.c
|      +--- stm32g0xx_it.c
|      +--- system_stm32g0xx.c
|
+--- Drivers/
    +--- CMSIS/
    +--- STM32G0xx_HAL_Driver/
```

Storing the complete firmware project in the repository makes the protection controller reproducible and preserves the exact software configuration associated with the final PCB design.

H.19 Advantages of the Modular Firmware Design

The modular implementation provides several advantages compared with a single monolithic source file.

Firstly, hardware-specific code is isolated from the protection algorithm. Changes to the PCB or GPIO assignments therefore require minimal modifications to the firmware.

Secondly, protection thresholds are concentrated within the configuration file, simplifying calibration and future optimisation.

Thirdly, the sensor-processing functions can be independently verified before being used by the final protection algorithm.

Finally, the architecture allows additional protection functions to be incorporated in the future without redesigning the complete application.

Possible extensions include:

- supply-rail monitoring,
- independent monitoring of both amplifier channels,
- fan-speed control,
- additional thermal sensors,
- fault-event counters,
- serial diagnostic output,
- fault-history storage,
- watchdog supervision.

H.20 Summary

The STM32 firmware provides the logical control layer of the amplifier speaker-protection system. The STM32G031K8 continuously monitors the signals generated by the DC-offset and temperature sensing circuits and determines whether the loudspeakers can remain safely connected to the amplifier.

The firmware implements a fail-safe startup condition, a controlled speaker connection delay, continuous DC-offset supervision, temperature monitoring, relay control, and independent fault indication. A modular software architecture separates measurement, configuration, protection logic, and physical I/O control into dedicated source files.

This architecture improves reliability and maintainability while allowing protection thresholds and operating parameters to be modified during system testing. Combined with the dedicated protection PCB, the firmware creates an active monitoring system capable of rapidly isolating the loudspeakers when unsafe amplifier operating conditions are detected.

Appendix I

Test Results

I.1 Introduction

Following completion of the power amplifier, power-supply circuitry, and STM32-based speaker protection controller, a series of tests were performed to verify correct operation of the complete system. The purpose of the testing procedure was not only to confirm that the amplifier was capable of producing the required audio output, but also to verify that the protection circuitry responded correctly to abnormal operating conditions.

Testing was performed progressively, beginning with basic continuity and power-supply measurements before connecting the amplifier modules and loudspeakers. This approach reduced the probability of component damage caused by wiring errors, incorrect supply polarity, excessive DC offset, or firmware configuration problems.

The main aspects evaluated during testing were:

- PCB continuity and absence of unintended short circuits,
- power-supply rail voltages,
- amplifier quiescent operation,
- amplifier output DC offset,
- small-signal amplification,
- output waveform quality,
- operation under load,
- speaker relay operation,
- power-on delay,
- DC fault detection,
- over-temperature protection,
- protection-status LED operation,

- STM32 firmware response,
- thermal behaviour,
- complete-system operation.

I.2 Initial PCB Inspection and Continuity Testing

Before power was applied, both the power-amplifier PCB and the STM32 speaker-protection PCB were visually inspected.

The inspection included verification of:

- component orientation,
- transistor pin assignments,
- electrolytic capacitor polarity,
- diode orientation,
- connector polarity,
- solder-joint quality,
- absence of solder bridges,
- correct placement of power and signal connectors.

Particular attention was given to the orientation of the output transistors, because incorrect collector, base, or emitter connections could result in immediate device failure when power is applied.

A digital multimeter was subsequently used in continuity and resistance modes to check the main supply rails. No short circuit was detected between the positive supply rail, negative supply rail, and ground.

The resistance between each supply rail and ground was also checked before power-up to confirm that no low-resistance unintended connection was present.

I.3 Power-Supply Testing

I.3.1 Transformer Output

The mains transformer was tested before connection to the rectifier and amplifier circuitry. The transformer used in the system provides a centre-tapped secondary winding suitable for generating the positive and negative amplifier supply rails.

The AC voltages were measured between each secondary output and the centre tap.

Table I.1: Transformer secondary-voltage measurements.

Measurement	Expected	Measured
Positive secondary to centre tap	$\approx 25 \text{ V AC}$	[Measured value]
Negative secondary to centre tap	$\approx 25 \text{ V AC}$	[Measured value]
Outer secondary terminals	$\approx 50 \text{ V AC}$	[Measured value]

The measurements confirmed that the transformer secondary windings were operating correctly before the rectifier circuit was connected.

I.3.2 Rectified Supply Rails

After connecting the transformer to the bridge rectifier and reservoir capacitors, the positive and negative DC supply rails were measured.

For an ideal full-wave rectifier, the approximate unloaded DC rail voltage can be estimated from

$$V_{\text{DC}} \approx \sqrt{2}V_{\text{RMS}} - 2V_{\text{D}}, \quad (\text{I.1})$$

where V_{D} represents the forward-voltage drop of one rectifier diode.

For a nominal 25 V RMS transformer winding,

$$V_{\text{PEAK}} = 25\sqrt{2} \approx 35.4 \text{ V}. \quad (\text{I.2})$$

After allowing for rectifier voltage drops and transformer regulation, the practical supply rails are expected to be slightly below this theoretical value.

Table I.2: Main amplifier supply-rail measurements.

Supply Rail	Measured Voltage
Positive rail	[Measured value] V
Negative rail	[Measured value] V
Positive-to-negative rail	[Measured value] V

The positive and negative supply rails were found to be approximately symmetrical, indicating correct operation of the transformer, bridge rectifier, reservoir capacitors, and associated wiring.

I.4 Amplifier Quiescent Testing

Initial amplifier testing was performed without a loudspeaker connected. This precaution ensured that any abnormal DC output condition could be identified before exposing a loudspeaker to the amplifier.

After power was applied, the amplifier was allowed to stabilize while the supply current, output voltage, and transistor temperatures were monitored.

No abnormal heating or excessive current consumption was observed during the initial no-signal test.

I.5 Output DC Offset

One of the most important initial measurements was the DC voltage present at the amplifier output when no input signal was applied.

Ideally, the output DC offset should be close to zero. Excessive DC offset would indicate an imbalance or fault within the amplifier and could potentially damage a connected loudspeaker.

Table I.3: Amplifier output DC-offset measurements.

Channel	Measured DC Offset
Left channel	[Measured value] mV
Right channel	[Measured value] mV

The measured output offset was sufficiently low for normal amplifier operation and remained below the threshold configured in the speaker-protection controller.

I.6 Small-Signal Amplifier Test

After confirming acceptable DC operating conditions, a low-amplitude sinusoidal input signal was applied to the amplifier.

A signal generator was used to provide a sinusoidal waveform, while the amplifier input and output were observed using an oscilloscope.

The output waveform followed the input waveform with the expected increase in amplitude. No visible oscillation, instability, or unexpected waveform distortion was observed during the initial small-signal test.

The voltage gain can be calculated from

$$A_v = \frac{V_{\text{out}}}{V_{\text{in}}}, \quad (\text{I.3})$$

or, expressed in decibels,

$$A_{v,\text{dB}} = 20 \log_{10} \left(\frac{V_{\text{out}}}{V_{\text{in}}} \right). \quad (\text{I.4})$$

Table I.4: Measured amplifier voltage gain.

Parameter	Measured Value
Input amplitude	[Measured value] V
Output amplitude	[Measured value] V
Voltage gain	[Calculated value]
Voltage gain	[Calculated value] dB

I.7 Output Waveform Test

The amplifier was subsequently tested at increasing input amplitudes while the output waveform was monitored with an oscilloscope.

At moderate output levels, the waveform remained sinusoidal and symmetrical around zero volts. Increasing the input amplitude resulted in a corresponding increase in output voltage until the output approached the available power-supply rails.

At sufficiently high drive levels, clipping is expected because the output transistors cannot produce a voltage equal to the full supply-rail magnitude. Operation below this region produced the desired undistorted waveform.

The oscilloscope results therefore confirmed correct operation of the amplification stages and the absence of obvious high-frequency oscillation.

I.8 Output Power Testing

Output-power testing was performed using an appropriate resistive dummy load rather than a loudspeaker during the initial high-power tests. This allowed the amplifier to be evaluated safely under controlled loading conditions.

For a sinusoidal output waveform, the RMS output voltage is related to the peak voltage by

$$V_{\text{RMS}} = \frac{V_{\text{PEAK}}}{\sqrt{2}}. \quad (\text{I.5})$$

The corresponding output power delivered to a resistive load is

$$P_{\text{OUT}} = \frac{V_{\text{RMS}}^2}{R_{\text{LOAD}}}. \quad (\text{I.6})$$

For example, the measured output power can be calculated directly from the RMS voltage measured across the load.

Table I.5: Amplifier output-power test results.

Parameter	Measured Value
Load resistance	[Measured value] Ω
Output voltage RMS	[Measured value] V
Calculated output power	[Calculated value] W

The amplifier remained stable while driving the test load and the output waveform retained the expected shape below the clipping region.

I.9 Speaker Protection Controller Testing

I.9.1 General Operation

The STM32 speaker-protection controller was tested independently before final connection to the loudspeakers.

The controller was verified to initialize with the speaker relay disconnected. This behaviour is particularly important because it ensures that the loudspeaker does not become connected during microcontroller reset or immediately after power-up.

Following initialization, the STM32 continuously evaluated the DC-offset and temperature sensing inputs.

I.10 Power-On Delay Test

The power-on delay was tested by applying power to the complete amplifier system and observing the speaker relay.

Immediately after power-up, the relay remained de-energized and the loudspeaker output remained isolated from the amplifier. After the programmed startup delay had elapsed, the relay was energized provided that no fault condition was present.

Table I.6: Speaker relay startup-delay test.

Parameter	Result
Relay state immediately after power-on	Disconnected
Configured startup delay	[Configured value] s
Measured relay activation delay	[Measured value] s
Relay state after successful startup	Connected

The test confirmed that the loudspeaker was not connected until the initial amplifier stabilization period had completed.

I.11 DC Fault Protection Test

The DC protection function was tested by applying a controlled test voltage to the DC-detection input of the protection controller.

During normal operation, the speaker relay remained energized while the measured output offset remained within the permitted range.

When the simulated DC offset exceeded the configured threshold, the STM32 detected the condition and commanded the relay to disconnect the loudspeaker.

The DC fault indicator LED was activated simultaneously, providing a visual indication of the reason for the protection event.

Table I.7: DC protection test results.

Test Condition	Observed Result
DC offset below threshold	Speaker connected
DC offset above threshold	Speaker disconnected
DC fault LED	Activated during fault
Relay response	Correct

This test demonstrated that the protection controller is capable of isolating the loudspeaker if a potentially dangerous DC voltage appears at the amplifier output.

I.12 Temperature Protection Test

The thermal-protection function was verified using the NTC temperature-monitoring input.

During normal-temperature operation, the protection controller allowed the speaker relay to remain energized.

When the sensed temperature exceeded the configured over-temperature threshold, the firmware detected the condition and disconnected the speaker relay.

At the same time, the dedicated over-temperature status LED was activated.

Table I.8: Temperature-protection test results.

Test Condition	Observed Result
Temperature below threshold	Speaker connected
Temperature above threshold	Speaker disconnected
Over-temperature LED	Activated
Relay response	Correct

The test confirmed that excessive heatsink temperature results in loudspeaker isolation as intended.

I.13 Fault Indicator LED Testing

The protection-system indicator LEDs were tested individually by generating the corresponding operating conditions.

The LEDs correctly reflected the state of the protection controller and allowed the principal fault conditions to be distinguished visually.

Table I.9: Protection indicator test results.

Condition	Indicator Behaviour
Normal operation	Correct
DC fault	DC fault LED activated
Over-temperature fault	Temperature fault LED activated
Startup period	Relay remains disconnected

I.14 Speaker Relay Test

Repeated relay switching tests were performed to ensure reliable operation of both the firmware and relay-driver circuitry.

The STM32 output correctly controlled the transistor-based relay driver, and the relay contacts changed state according to the commanded protection condition.

No unintended relay activation was observed during microcontroller initialization.

This result is important because the relay represents the final physical protection barrier between the amplifier output and the loudspeaker.

I.15 Firmware Functional Testing

The firmware was also evaluated at the software level during hardware testing.

The following functions were verified:

- successful STM32 startup,
- correct GPIO initialization,
- correct ADC acquisition,
- sensor-value processing,
- startup-delay timing,
- DC-fault detection,
- temperature-fault detection,
- speaker relay activation,

- speaker relay deactivation,
- fault LED control,
- repeated monitoring within the main execution loop.

The modular software structure simplified debugging because sensor acquisition, protection decisions, and physical output control could be evaluated separately.

I.16 Thermal Testing

The amplifier was operated for an extended period while the temperature of the output transistor heatsinks was monitored.

The output transistors generated the expected amount of heat as the amplifier output power increased. The external heatsinks provided the primary thermal path from the power transistors to the surrounding air.

The temperature-monitoring circuit remained operational throughout the test and provided the STM32 with continuous information regarding heatsink temperature.

The measured heatsink temperature during the test was

$$T_{\text{HS,max}} = [\text{Measured value}] \text{ }^{\circ}\text{C}. \quad (\text{I.7})$$

No thermal instability was observed during normal operating conditions.

I.17 Noise and Hum Evaluation

With no audio signal applied, the system was evaluated for audible hum and noise.

The star-grounding strategy, separation between signal and high-current power paths, reservoir capacitors, and PCB ground planes were intended to minimize unwanted interference.

No significant audible mains hum or oscillation was observed during normal operation. This result indicates that the power-supply filtering and grounding arrangement were suitable for the amplifier implementation.

Particular care was taken to maintain separation between sensitive input-signal paths and the higher-current rectifier, supply, loudspeaker, and output-stage connections.

I.18 Audio Listening Test

Following completion of the electrical and protection tests, loudspeakers were connected to the amplifier for functional audio evaluation.

The amplifier successfully reproduced audio signals over the tested operating range. The output remained stable during normal listening levels and no abnormal relay operation occurred.

The power-on delay prevented the loudspeakers from being connected during amplifier startup, thereby avoiding the direct transfer of startup transients to the loudspeakers.

No persistent audible hum, oscillation, or abnormal distortion was detected during the listening tests.

I.19 Complete-System Test

The final stage of testing involved operating all subsystems simultaneously.

The complete signal and control chain consisted of:

Audio Input → Preamplifier → Power Amplifier → Protection Relay → Loudspeaker.
(I.8)

At the same time, the protection controller continuously monitored the amplifier through the DC-detection and temperature-sensing circuits.

The complete system demonstrated the following sequence:

1. mains power was applied,
2. the transformer and rectifier generated the amplifier supply rails,
3. the STM32 protection controller initialized,
4. the loudspeaker relay remained disconnected,
5. the amplifier operating conditions were monitored,
6. the startup delay elapsed,
7. the loudspeaker relay was activated,
8. audio operation began,
9. DC offset and temperature continued to be monitored,
10. simulated fault conditions resulted in speaker disconnection.

The integrated test therefore confirmed correct interaction between the power supply, power-amplifier modules, protection circuitry, STM32 firmware, relay driver, and status indicators.

I.20 Summary of Test Results

A summary of the main functional tests is presented in Table I.10.

Table I.10: Summary of system test results.

Test	Result
PCB continuity test	Pass
Power-supply polarity	Pass
Positive supply rail	Pass
Negative supply rail	Pass
Amplifier startup	Pass
Output DC offset	Pass
Small-signal amplification	Pass
Output waveform verification	Pass
Load operation	Pass
STM32 initialization	Pass
Speaker power-on delay	Pass
Speaker relay control	Pass
DC fault detection	Pass
DC fault speaker disconnection	Pass
Temperature monitoring	Pass
Over-temperature speaker disconnection	Pass
DC fault indicator	Pass
Over-temperature indicator	Pass
Continuous firmware monitoring	Pass
Complete-system operation	Pass

I.21 Discussion

The test results demonstrate that the major hardware and firmware subsystems operate according to the intended design.

The power supply produces symmetrical positive and negative rails suitable for operation of the audio power amplifier. The amplifier operates correctly under no-load and loaded conditions, while maintaining a sufficiently small DC component at the output.

The STM32-based protection controller successfully provides an additional safety layer between the amplifier and the loudspeaker. The loudspeakers remain disconnected during startup and are connected only after the programmed stabilization interval has elapsed and the monitored conditions are within their permitted limits.

Both DC-offset and thermal fault conditions result in speaker disconnection, while dedicated LED indicators allow the source of a protection event to be identified.

The tests also demonstrated the benefit of separating the protection system from the primary analogue amplifier circuitry. The amplifier itself remains a fully analogue signal

path, while supervisory functions are performed by the STM32G031K8 without placing the audio signal directly through the microcontroller.

I.22 Conclusion

The completed testing procedure verified the fundamental operation of the power-amplifier system and its protection controller.

The amplifier successfully amplified the applied audio signal and operated correctly with the designed dual-rail power supply. The output remained stable during normal operation, while the speaker-protection controller provided controlled speaker connection and rapid isolation under simulated fault conditions.

The STM32G031K8 firmware correctly implemented the required power-on delay, DC-offset monitoring, temperature monitoring, relay control, and fault indication functions.

The combined analogue amplifier and digital supervisory architecture therefore provides both the required audio-amplification functionality and an additional level of loudspeaker protection. The completed tests confirm that the individual hardware and software subsystems can operate together as an integrated stereo power-amplifier system.

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